CAREERS360

PRACTICE Series

JEE Main 2026

January Session

Question Paper with Solution

Physics

Q. 1 A gas based geyser heated water flowing at the rate of 5.0 litres per minute from $27^{\circ}\mathrm{C}$ to $87^{\circ}\mathrm{C}$. The rate of consumption of the gas is _____ g/s.
(Take heat of combustion of gas $= 5.0 \times 10^{4}\ \mathrm{J/g}$, specific heat capacity of water $= 4200$ J/kg. ^{0} C)

Option 1:
2.1

Option 2:
0.21

Option 3:
4.2

Option 4:
0.42

Correct Answer:
0.42

Solution:
Sol: $m_{\omega}C_{\omega}\Delta T = m_{g}C_{g}$

$$\frac{\left(5 \times 10^{-3} \times 10^{3}\right) \times 4200}{60} \times (87-27)$$
$$= m_{g} \times 5 \times 10^{4} \Rightarrow m_{g} = 0.42$$

Q. 2 The point of charge of $10^{-8}\mathrm{C}$ is placed at origin. The work done is moving a point at $2\mu C$ from point $\mathrm{A}(4,4,2)\mathrm{m}$ to $\mathrm{B}(2,2,1)\mathrm{m}$ is _____ J. $\left(\frac{1}{4\pi\epsilon_{0}} = 9 \times 10^{9}\right)$ in SI unit

Option 1:
45×10^{-6}

Option 2:
30×10^{-6}

Option 3:
15×10^{-6}

Option 4:
0

Correct Answer:
30×10^{-6}

Solution:
Sol: Work done in moving a charge in an electrostatic field is:

$$W = q_2 \cdot \Delta V$$

We know the formula for potential difference:

$$\Delta V = Kq_1 \left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

So, work done in moving the charge from one point to another point:

$$W = \Delta V = Kq_1q_2 \left(\frac{1}{r_2} - \frac{1}{r_1}\right) = \left(9 \times 10^9\right) \times \left(10^{-8}\right) \times \left(2 \times 10^{-6}\right) \times \left(\frac{1}{3} - \frac{1}{6}\right) = 30 \times 10^{-6}$$

Hence, the correct answer is option 2.

Q. 3 A light wave described by $E = 60\left[\sin\left(3 \times 10^{15}\right)t + \sin\left(12 \times 10^{15}\right)t\right]$ (in SI units) fall on a metal surface of work function 2.8 eV . The maximum kinetic energy of ejected photoelectron is (approximately) _____ $\text{eV} \cdot \left(h = 6.6 \times 10^{-34}\ \text{J}.s. \text{ and } e = 1.6 \times 10^{-19}\text{C}\right)$

Option 1:
3.8

Option 2:
6.0

Option 3:
7.8

Option 4:
5.1

Correct Answer:
5.1

Solution:
Sol: The given light wave equation has two angular frequencies:

$\omega_1 = 3 \times 10^{15}\text{rad/s}$
$\omega_2 = 12 \times 10^{15}\text{rad/s}$

We can find the frequency using the formula: $\nu = \omega/2\pi$

We are given two frequencies, but the photon with the higher frequency will give maximum kinetic energy. So we use, $\omega_2 = 12 \times 10^{15} rad/s.$

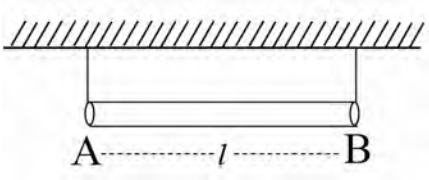
Using the Einstein's Photoelectric Equation:

$$E_{\text{photon}} = h\gamma = \frac{hw}{2\pi} = 7.88\text{eV}$$

$$KE = E_{ph} - \phi = 7.88 - 2.8 = 5.08eV$$

The option closest to 5.08 is 5.1 eV. Hence, the correct answer is option 4.

Q. 4 A uniform rod of mass m and length 1 suspend by means of two identical inextensible light strings as shown in figure. Tension in one string immediately after the other string is cut is _____ (g - acceleration due to gravity)



Option 1:
$\mathrm{mg}/2$

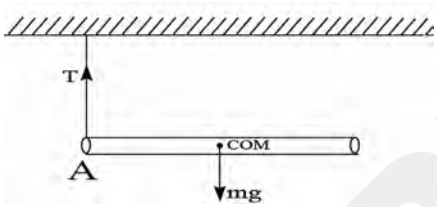
Option 2:
mg

Option 3:
$\mathrm{mg}/3$

Option 4:
$\mathrm{mg}/4$

Correct Answer:
$\mathrm{mg}/4$

Solution:



Sol:

$F_{NET} = ma \Rightarrow mg - T = ma$

$\tau_{COM} = I\alpha \Rightarrow T\frac{l}{2} = \frac{ml^2}{12} \times \alpha \Rightarrow T = \frac{ml\alpha}{6}$

Constraint at $\mathrm{A} \Rightarrow (a_A)_y = 0 \Rightarrow \frac{\alpha l}{2} = a$

On solving $T = \frac{mg}{4}$

Q. 5 A 4 Kg mass moves under the influence of a force $F = \left(4t^3 i - 3tj\right) N$ where t is time in sec . If mass starts from origin $a + t = 0$, the velocity and position after $\mathrm{t} = 2$ s will be

Option 1:
$v = 4i - \frac{3}{2}j; r = \frac{6}{5}i - j$

Option 2:
$v = 4i - \frac{3}{2}\bar{j}; r = \frac{8}{5}i - j$

Option 3:
$v = 4i + \frac{5}{2}j; r = \frac{8}{5}i + 2j$

Option 4:
$v = 3i + \frac{3}{2}j; r = \frac{6}{5}i + j$

Correct Answer:
$v = 4i - \frac{3}{2}\bar{j}; r = \frac{8}{5}i - j$

Solution:

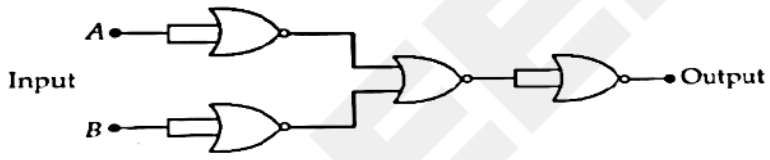
Sol: $F_x = ma_x \Rightarrow F_x = m\left(\frac{dV_x}{dt}\right) \Rightarrow 4t^3 = 4\left(\frac{dV_x}{dt}\right) \Rightarrow \int_0^2 t^3 dt = \int_0^{V_x} dV_x \Rightarrow V_x = 4$

For position: $4t^3 = 4\left(\frac{dV_x}{dt}\right) \Rightarrow \int dV_x = \int t^3 dt \Rightarrow V_x = \frac{t^4}{4}$

$\frac{dx}{dt} = \frac{t^4}{4} \Rightarrow \int_0^x dx = \frac{1}{4}\left(\int_0^t t^4 dt\right) = \frac{1}{4}\left(\frac{t^5}{5}\right) \Rightarrow x = \frac{8}{5}$

$F_y = ma_y \Rightarrow F_y = m\left(\frac{dV_y}{dt}\right) \Rightarrow \int_0^2 (-3t) dt = 4\int_0^{V_y} dV_x \Rightarrow V_y = \frac{-3}{2}$

Q. 6 The given circuit works as:



Option 1:
AND gate

Option 2:
NOR gate

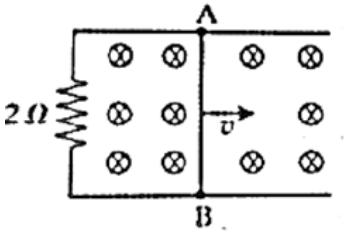
Option 3:
NAND gate

Option 4:
OR gate

Correct Answer:
NAND gate

Solution:
Sol: NAND

Q. 7 A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T . If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of $1.5\ \mathrm{m/s}$ is _____ N.



Option 1:
5.7×10^{-3}

Option 2:
7.5×10^{-3}

Option 3:
7.5×10^{-2}

Option 4:
5.7×10^{-2}

Correct Answer:
7.5×10^{-3}

Solution:
Sol: $F = \frac{B^2 l^2 v}{R} = \frac{(0.1)^2 \times 1^2 \times 1.5}{2} = \frac{3}{400} = 7.5 \times 10^{-3}$

Q. 8 A conducting circular loop of area $1.0m^2$ is placed perpendicular to a magnetic field which varies as $B = \sin(100t)$ Tesla. If the resistance of the loop is 100Ω, Then the average thermal energy dissipated in the loop in one period is _____J.

Option 1:
2π

Option 2:
π

Option 3:
π^2

Option 4:
$\pi/2$

Correct Answer:
π

Solution:
Sol:

$$E = -\frac{d\phi}{dt} = -A \times \frac{dB}{dt} = -A \times (-\cos(100t)) \times 100 = 100A \cdot \cos(100t)$$

$$\text{Heat } = \int_0^T \left(\frac{E^2}{R}\right) dt = \int_0^T \cos^2(100t) dt = \pi$$

Q. 9 A current-carrying solenoid is placed vertically, and a particle of mass m with charge Q is released from rest. The particle moves along the axis of the solenoid. If g is the acceleration due to gravity, then the acceleration (a) of the charged particle will satisfy

Option 1:
$a = g$

Option 2:
$a = 0$

Option 3:
$0 < a < g$

Option 4:
$\mathrm{a} > \mathrm{g}$

Correct Answer:
$a = g$

Solution:
Sol: Magnetic Force $= 0$
Net force = gravity
Acceleration $= \mathrm{g}$

Q. 10 In a double slit experiment, the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slit plan. When one slit is covered with a transparent sheet having thickness t and refractive index $\mathrm{n}(=1.5)$, the central fringe shifts by 0.2 cm . The value of t is

Option 1:
5.6×10^{-4}

Option 2:
5.0×10^{-3}

Option 3:
8×10^{-4}

Option 4:
6.0×10^{-3}

Correct Answer:
8×10^{-4}

Solution:
Sol: Shift $x=\frac{Dt(\mu-v)}{d} \Rightarrow 0.2=\frac{50 \times t \times 0.5}{0.1} \Rightarrow t=8 \times 10^{-4}$ cm

Q. 11 If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is _____ m. (Atomic number of gold $=79$ and $\frac{1}{4\pi\epsilon_0}=9 \times 10^9$ in SI units)

Option 1:
2.95×10^{-14}

Option 2:
2.95×10^{-16}

Option 3:
3.85×10^{-14}

Option 4:
3.85×10^{-16}

Correct Answer:
2.95×10^{-14}

Solution:
Sol: Conservation of mechanical energy

$$\Delta v = -\Delta K$$

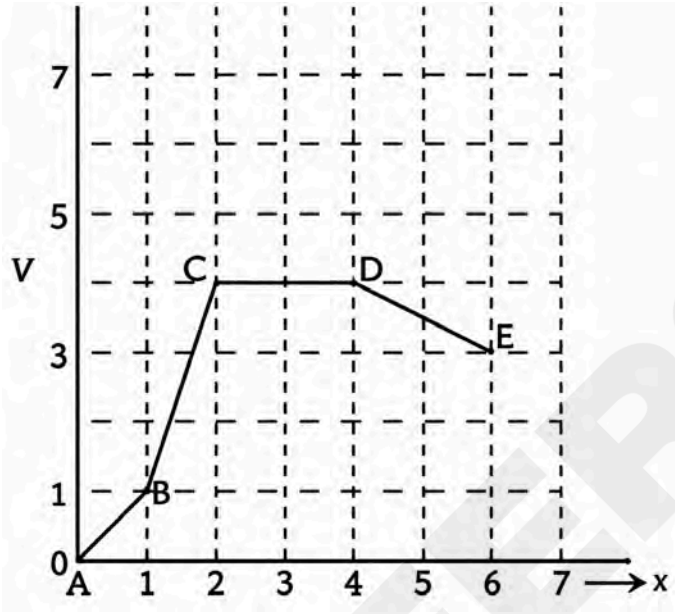
$$\frac{kzq_1q_2}{r} = -\left[0 - \frac{m}{2}v^2\right]\begin{pmatrix}\text{where } q_1 = Ze\\ q_2 = Ze\end{pmatrix}$$

$$\frac{9\times10^9\times79\times\left(1.6\times10^{-19}\right)^2\times2}{r} = 7.7\times10^6\times1.6\times10^{-19}$$

$$r = \frac{9\times10^9\times79\times1.6\times10^{-19}\times2}{7.7\times10^6} = 147.7\times10^{-16}\times2$$

$$= 295.48\times10^{-16} = 2.99548\times10^{-14}$$

Q. 12 Potential energy (V) versus distance (x) is given by the graph. Rank various regions as per the magnitudes of the force. (F) acting on a particle from high to low.



Option 1:
$F_{CD} > F_{AB} > F_{BC} > F_{DE}$

Option 2:
$F_{CD} > F_{DE} > F_{AB} > F_{BC}$

Option 3:
$F_{BC} > F_{CD} > F_{DE} > F_{AB}$

Option 4:
$F_{BC} > F_{AB} > F_{DE} > F_{CD}$

Correct Answer:
$F_{BC} > F_{AB} > F_{DE} > F_{CD}$

Solution:
Sol: $F = -\frac{dU}{dx}$
F a slope

Q. 13 Two strings (A,B) having linear densities $\mu_A = 2 \times 10^{-4}\ \text{kg/m}$ and $\mu_B = 4 \times 10^{-4}\ \text{kg/m}$ and lengths $L_A = 2.5\ \text{m}$ and $L_B = 1.5\ \text{m}$ respectively are joined. Free ends of A and B are tied to two rigid supports C and D , respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time t_1 and t_2 respectively, to reach the joint. The ratio t_1/t_2 is :

Option 1:
1.90

Option 2:
1.67

Option 3:
1.18

Option 4:
1.08

Correct Answer:
1.18

Solution:
Sol:

$$\frac{t_1}{t_2} = \frac{\frac{l_1}{v_1}}{\frac{l_2}{v_2}} = \frac{\frac{2.5}{\sqrt{T/\mu_A}}}{\frac{1.5}{\sqrt{T/\mu_B}}} = \frac{25}{15} \times \sqrt{\frac{\mu_A}{\mu_B}} = \frac{5}{3} \times \sqrt{\frac{2}{4}}$$

$$\frac{5}{3\sqrt{2}} = 1.17$$

Q. 14 Consider a modified Bernoulli equation. $\left(P + \frac{A}{Bt^2}\right) + \rho g(h + Bt) + \frac{1}{2}\rho V^2 =$ Constant If f has the dimension of time then the dimension A and B are _____ , _____ respectively.

Option 1:
$\left[ML^0T^{-2}\right]$ and $\left[M^0LT^{-1}\right]$

Option 2:
$\left[ML^0T^{-1}\right]$ and $\left[M^0LT^{-1}\right]$

Option 3:
$\left[ML^0T^{-2}\right]$ and $\left[M^0LT^{-2}\right]$

Option 4:
$\left[ML^0T^{-1}\right]$ and $\left[M^0LT\right]$

Correct Answer:
$[ML^0T^{-1}]$ and $[M^0LT^{-1}]$

Solution:
Sol: $[h] = [Bt] \Rightarrow [B] = \left[\frac{h}{t}\right] = \left[LT^{-1}\right]$

$[P] = \left[\frac{A}{Bt^2}\right] \Rightarrow [A] = [P][B]\left[t^2\right] = \left[ML^{-1}T^{-2}\right]\left[LT^{-1}\right]\left[T^2\right] = \left[MT^{-1}\right]$

Q. 15 Initially a satellite of 100 kg is in a circular orbit of radius $1.5R_E$. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha \times 10^6 J$ of energy. The value of α is
____ (Take Radius of Earth $R_E = 6 \times 10^6$ m and g $= 10\ \mathrm{m/s^2}$)

Option 1:
100

Option 2:
150

Option 3:
1000

Option 4:
500

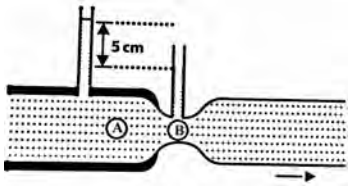
Correct Answer:
1000

Solution:
Sol: Δ Mechanical energy $= \Delta KE + \Delta PE = \frac{m}{2}\left(v_f^2 - v_i^2\right) + \left(\frac{-GMm}{r_f} + \frac{GMm}{r_i}\right)$

$$\frac{m}{2}\left(\frac{GM}{3R} - \frac{GM}{\frac{3}{2}R}\right) + \frac{GMm}{R}\left(-\frac{1}{3} + \frac{1}{\frac{3}{2}}\right) = \frac{-GMm}{6R} + \frac{1}{3}\frac{GMm}{R} = \frac{GMm}{6R} \times \frac{R}{R}$$
$$= \frac{gmR}{6} = \frac{10 \times 100 \times 6 \times 10^6}{6} = 10^9$$

Take $\left[v = \sqrt{\frac{GM}{r}}\right]$ and $\left(\text{ put } g = \frac{GM}{R^2}\right)$

Q. 16 Water flows through a horizontal tube as shown in the figure. The difference in height between the water Columns in vertical tubes is 5 cm and the area of cross-sections at A and B are $6\ \text{cm}^2$ and $3\ \text{cm}^2$ respectively. The rate of flown will be _____ cm^3/s. (take $g = 10\ \text{m/s}^2$)



Option 1:
$200\sqrt{3}$

Option 2:
$\frac{200}{\sqrt{3}}$

Option 3:
$100\sqrt{3}$

Option 4:
$200\sqrt{6}$

Correct Answer:
$200\sqrt{3}$

Solution:
Sol: .

$$Q = \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}}\sqrt{2gH} = \frac{6\times 3\times (10^{-2})^2}{\sqrt{27}} \times \sqrt{2\times 10\times \frac{5}{100}} = \frac{6}{\sqrt{3}} \times 10^{-4} \times 1 = \frac{6}{\sqrt{3}} \times 10^{-4}\,\frac{\text{m}^3}{\text{s}} = \frac{600}{\sqrt{3}}\,\frac{\text{cm}^3}{\text{s}} = 200\sqrt{3}\,\frac{\text{cm}^3}{\text{s}}$$

Q. 17 The electric field in a plane electromagnetic wave is given by: $E_y = 69 \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right] \text{V/m}$
The expansion for magnetic field associated with this electromagnetic wave is _____ T.

Option 1:
$B_z = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x + 1.8 \times 10^{11} t\right]$

Option 2:
$B_z = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

Option 3:
$B_y = 69 \sin\left[0.6 \times 10^3 x + 1.8 \times 10^{11} t\right]$

Option 4:
$B_y = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

Correct Answer:
$B_z = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

Solution:
Sol:

$\hat{B} = \hat{c} \times \hat{E}$

$\Rightarrow \hat{\mathrm{c}} = \hat{\mathrm{i}}$ because phase of electric field is function of x.

$\Rightarrow \hat{\mathrm{E}} = \hat{\mathrm{j}}$ (given)

$\Rightarrow \hat{\mathrm{B}} = \hat{\mathrm{i}} \times \hat{\mathrm{j}} = \hat{\mathrm{k}}$

$|\mathrm{B}| = \frac{|\mathrm{E}|}{\mathrm{c}} = \frac{69 \times 0.6 \times 10^3}{1.8 \times 10^{11}} = \frac{69}{3 \times 10^8} 2$

$|\mathrm{B}| = 2.9 \times 10^{-7}$

$\overrightarrow{\mathrm{B}_2} = 2.9 \times 10^{-7} \sin\left(0.6 \times 10^3 \mathrm{x} - 1.8 \times 10^{11} \mathrm{t}\right)$

(phase is same as that of electric field)

Q. 18 A parallel plate capacitor has capacitance C , when there is vacuum within the parallel plates. A sheet having thickness $\left(\frac{1}{3}\right)^{\text{rd}}$ of the separation between the plates and relative permittivity K is introduced between the plates. The new capacitance of the system is :

Option 1:
$\frac{3KC}{2K+1}$

Option 2:
$\frac{4KC}{3K-1}$

Option 3:
$\frac{CK}{2+K}$

Option 4:
$\frac{3CK^2}{(2K+1)^2}$

Correct Answer:
$\frac{3KC}{2K+1}$

Solution:
Sol: $C^1 = \frac{A\varepsilon_0}{d - t\left(1 - \frac{1}{k}\right)} = \frac{A\varepsilon_0}{d\left(1 - \frac{t}{d}\left(1 - \frac{1}{k}\right)\right)} = \frac{C}{1 - \frac{1}{3}\left(1 - \frac{1}{k}\right)} = \frac{3CK}{2K+1}$

Q. 19 In an experiment the values of two spring constants were measured as $k_1 = (10 \pm 0.2)\mathrm{N/m}$ and $k_2 = (20 \pm 0.3)N/m$. If these spring are connected in parallel, then the percentage error in equivalent spring constant is :

Option 1:
1.67%

Option 2:
1.33%

Option 3:
2.67%

Option 4:
2.33%

Correct Answer:
1.67%

Solution:
Sol:

$K = k_1 + k_2 \quad = 10 + 20 = 30$
$\Delta k = \Delta k_1 + \Delta k_2 = 0.2 + 0.3 = 0.5$
$\% = \dfrac{0.5}{30} \times 100 = 1.67$

Q. 20 An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at $30^\circ\mathrm{C}$. The coefficient of linear expansion of aluminium and steel are $24 \times 10^{-6}/^\circ\mathrm{C}$ and $1.2 \times 10^{-5}/^\circ\mathrm{C}$, respectively. The length of this composite rod when its temperature is raised to $100^\circ\mathrm{C}$, is _____ cm

Option 1:
120.03

Option 2:
120.20

Option 3:
120.06

Option 4:
120.15

Correct Answer:
120.15

Solution:
Sol: $l_f = 120 + \Delta l_{Al} + \Delta l_{\text{sted}} = 120 + 60 \times 24 \times 10^{-6} \times 70 + 60 \times 1.2 \times 10^{-5} \times 70 = 120.1512$

Q. 21 In a microscope, the objective is having focal length $f_o = 2\ \text{cm}$ and eye-piece is having focal length $f_e = 4\ \text{cm}$. The tube length is 32 cm . The magnification produced by this microscope for normal adjustment is _____ .

Correct Answer:
100

Solution:
Sol: Magnification $= \frac{LD}{f_o f_e} = \frac{32 \times 25}{2 \times 4} = 100$

Q. 22 A collimated beam of light diameter 2 mm is propagating along x -axis. The beam is required to be expanded in collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm , then the focal length of second lens is
_____ mm.

Correct Answer:
280

Solution:
Sol: $m = \frac{f_2}{f_1} = \frac{D_2}{D_1} \Rightarrow f_2 = f_1 \left(\frac{D_2}{D_1}\right) = 40 \times \frac{14}{2} = 280$

Q. 23 A 10 mole of oxygen is heated at constant volume from $30^\circ C$ to $40^\circ C$. The change in the internal energy of gas is _____ Cal. (The molar specific heat of oxygen at constant pressure. $C_P = 7\text{Cal/mol}^\circ\text{C}$ and $R = 2\text{Cal/mol}^\circ\text{C}$)

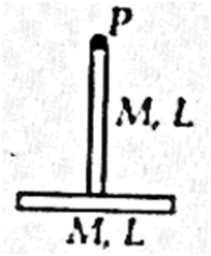
Correct Answer:
500

Solution:

Sol:

$C_P - C_V = R \Rightarrow 7 - C_V = 2 \Rightarrow C_V = 5$
$\Delta U = nC_V\Delta T = 10 \times 5 \times 10 = 500$

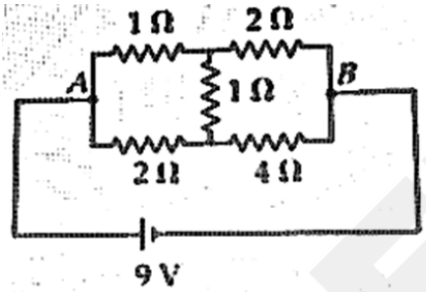
Q. 24 Two identical thin rods of mass M kg and length L m are connected as shown in figure. Moment of inertia of the combined rod system about an axis passing through point P and perpendicular to the plane of the rods is $\frac{x}{12}ML^2kgm^2$. The value of x is _____



Correct Answer:
17

Solution:
Sol: $\frac{ML^2}{3}+\frac{ML^2}{12}+ML^2=\frac{17}{12}ML^2$

Q. 25 The heat generated in 1 minute between points A and B in the given circuit, when a battery of 9 V with internal resistance of 1Ω is connected across these points is _____ J.



Correct Answer:
1080

Solution:
Sol: Wheat stone bridge

$$\mathrm{R}_{eff}=\frac{3\times 6}{3+6}=2(\text{ Between A\& B})$$
$$\therefore R_{\text{total}}=2+1=3$$
$$i=\frac{V}{R_{\text{total}}}=\frac{9}{2+1}=3$$
$$\text{Heat }=i^2Rt=3^2\times 2\times 60=1080$$

Chemistry

Q. 1 MnO_4^{2-} in acidic medium; disproportionates to

Option 1:
MnO_4^- &MnO

Option 2:
Mn_2O_7&MnO_2

Option 3:
Mn_2O_7&MnO

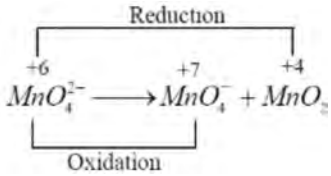
Option 4:
MnO_4^- &MnO_2

Correct Answer:
MnO_4^- &MnO_2

Solution:
Sol:

$$MnO_4^{2-} \xrightarrow[\text{Medium}]{\text{Acidic}} MnO_4^- + MnO_2$$

MnO_4^{2-} : Stable only in strong alkaline medium.



Hence, the correct answer is the option (4).

Q. 2 In Carius method; 0.75 g of an organic compound give 1.2 gm of barium sulphate. Find percentage of sulphur (molar mass $32 gmol^{-1}$). Molar mass of barium sulphate is 233 $gmol^{-1}$.

Option 1:
16.48%

Option 2:
21.97%

Option 3:
10.30%

Option 4:
4.55%

Correct Answer:
21.97%

Solution:
Sol: Carius method:
Weight of organic compound: $0.75\ \text{g}$ O.C. $(\text{s}) \longrightarrow \text{BaSO}_4 \downarrow (1.2\ \text{g})$

$\%S = \frac{1.2}{233} \times \frac{32}{0.75} \times 100 = 21.97\%$

Hence, the correct answer is the option (2).

Q. 3 Given below are two statements:
Statement-I: The number of species among $\text{SF}_4, \text{NH}_4^+, [\text{NiCl}_4], \text{XeF}_4, [\text{PtCl}_4]^{2-}, \text{SeF}_4$ and $[\text{Ni(CN)}_4]^{2-}$ that have tetrahedral geometry is 3 .
Statement - II: In the set $[\text{NO}_2, \text{BeH}_2, \text{BF}_3, \text{AlCl}_3]$; all the molecules have incomplete octet around central atom.

Option 1:
Both statement I and statement II are false

Option 2:
Both statements I and statement II are true

Option 3:
Statement I is false but statement II is true

Option 4:
Statement I is true but statement II is false

Correct Answer:
Statement I is false but statement II is true

Solution:
Sol: Statement I is wrong.

$$\left.\begin{array}{l} \text{SF}_4 \longrightarrow \text{Seesaw} : \text{sp}^3d \\ {}^{+}\text{NH}_4 \longrightarrow \text{Tetrahedral} : sp^3 \\ \text{XeF}_4 \longrightarrow \text{Square planar} : sp^3d^2 \\ [\text{PtCl}_4]^{-2} \longrightarrow \text{Square planar} : dsp^2 \\ \text{SeF}_4 \longrightarrow \text{Seesaw} : sp^3d \\ [\text{Ni(CN)}_4]^{-2} \longrightarrow \text{Square planar} : dsp^2 \end{array}\right\}$$

Tetrahedral:1 only

Statement II is correct.
$\text{NO}_2 \longrightarrow$ Odd e^-molecule
$\text{BeH}_2 \longrightarrow$ Electron deficient molecule
$\text{BF}_3 \longrightarrow$ Electron deficient molecule
$\text{AlCl}_3 \longrightarrow$ Electron deficient molecule
All the molecules have an incomplete octet.

Hence, the correct answer is the option (3).

Q. 4 Identify correct statements:

A: Arginine and Tryptophan are essential amino acids.

B: Histidine does not contain a heterocyclic ring in its structure.

C: Proline is a six-membered ring cyclic amino acid.

D: Glyncine does not have a chiral center.

E: Cysteine has a characteristic feature of side chain as $\mathrm{MeS}-\mathrm{CH}_2-\mathrm{CH}_2-$.

Choose the correct answer from the options given below:

Option 1:
A and D only

Option 2:
B and E only

Option 3:
C and E only

Option 4:
C and D only

Correct Answer:
A and D only

Solution:
Sol: A and D are correct.

Hence, the correct answer is the option (1).

Q. 5 For the reaction $\mathrm{N_2O_4} \quad 2\mathrm{NO_2}$, graph is plotted as shown below.

Identify correct statements.

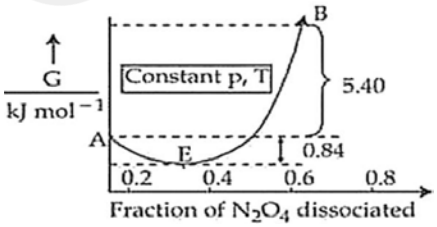
A. Standard free energy change for the reaction is $-5.40\ \mathrm{kJ\ mol^{-1}}$.

B. As $\Delta G^{(-)}$in graph is positive. $\mathrm{N_2O_4}$ Will not dissociate into $\mathrm{NO_2}$ at all.

C. Reverse reaction will go to completion.

D. When 1 mole of $\mathrm{N_2O_4}$ changes into equilibrium mixture, value of $\Delta G^{(-)} = -0.84\mathrm{kJmol^{-1}}$.

E. When 2 mole of NO_2 changes into equilibrium mixture, $\Delta G^{(-)}$for equilibrium mixture is $-6.24\mathrm{kJmol^{-1}}$.



Choose the correct answer from the options given below:

Option 1:
B and C only

Option 2:
A and D only

Option 3:
D and E only

Option 4:
C and E only

Correct Answer:
D and E only

Solution:
Sol: both D and E.

Reaction: $\mathrm{N_2O_4} \rightleftharpoons 2\mathrm{NO_2}$

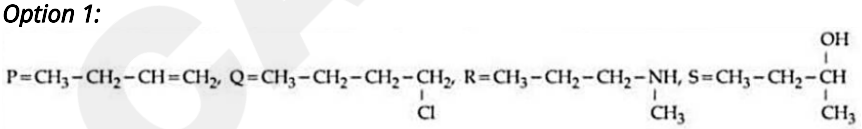
$\Delta \mathrm{G}^{\circ}$ is the standard free energy change $(\sim -5.4\ \mathrm{kJ/mol})$.
$\Delta \mathrm{G}$ depends on the amount of reactants/products, not just $\Delta \mathrm{G}^{\circ}$.

A. $\Delta \mathrm{G}^{\circ} = -5.40\ \mathrm{kJ/mol} \rightarrow$ True for standard $\Delta \mathrm{G}$, not for actual mixture
B. $\Delta \mathrm{G}$ positive → reaction non-spontaneous under that condition, but $\mathrm{N_2O_4}$ still dissociates until equilibrium
C. Reverse reaction goes to completion → Forward reaction is spontaneous, reverse is non-spontaneous
D: When 1 mole of $\mathrm{N_2O_4}$ changes at equilibrium, the value of $\Delta G^{(-)} = -0.84\mathrm{kJmol}^{-1}$ (This is the driving force to equilibrium).
E: when 2 moles of $\Delta G^{(-)} = -0.84\mathrm{kJmol}^{-1}$ changes at equilibrium, the value of $\Delta G^{(-)} = -0.84 - 5.4 = -6.24\mathrm{kJmol}^{-1}$.

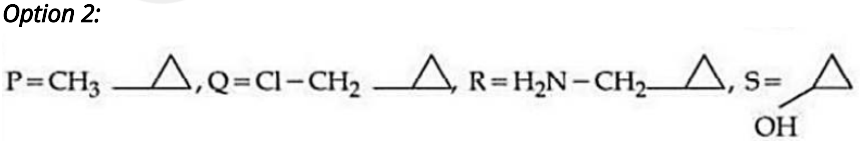
Hence, the correct answer is the option (3).

Q. 6 A hydrocarbon ' P ' $(\mathrm{C_4H_8})$ on reaction with HCl gives an optically active Compound Q ($\mathrm{C_4H_9Cl}$) which on reaction with one mole of ammonia gives compound 'R' $(C_4H_{11}N)$. 'R' on diazotization followed by hydrolysis gives ' S '. Identify $\mathrm{P, Q, R}$ and S .

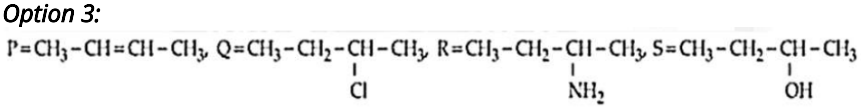
Option 1:



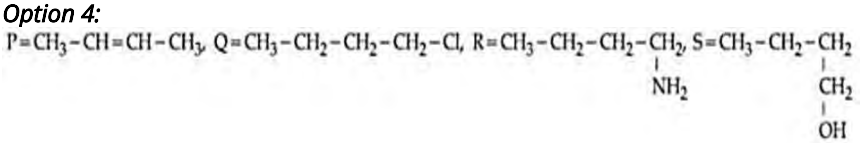
Option 2:



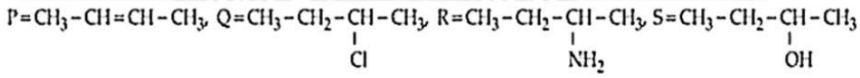
Option 3:



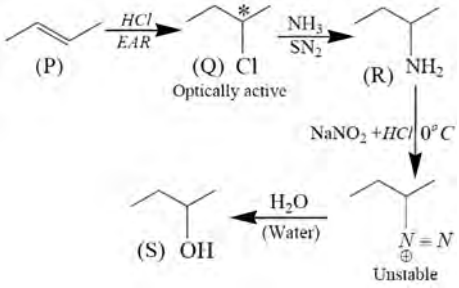
Option 4:



Correct Answer:



Solution:



Hence, the correct answer is the option (3).

Q. 7 An organic compound (P) on treatment with aqueous ammonia under hot conditions forms (Q), which on heating with $\mathrm{Br_2}$ and KOH forms compound (R) having molecular formula $\mathrm{C_6H_7\ N}$. Names of P, Q, and R, respectively, are

Option 1:
Benzoic acid, benzamide, aniline

Option 2:
Toluic acid, methylbenzamide, 2-methylaniline

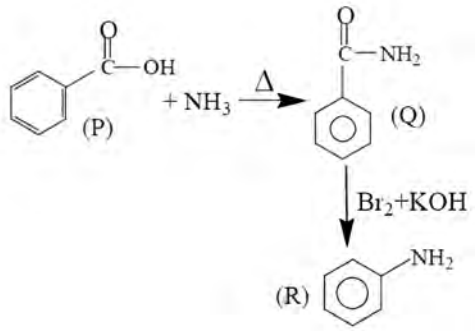
Option 3:
Benzoic acid, 4-methylbenzamide, 4-methylaniline

Option 4:
Phenylethanoic acid, phenylethanamide, benzamine

Correct Answer:
Benzoic acid, benzamide, aniline

Solution:

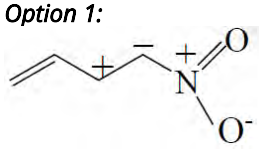
CAREERS360



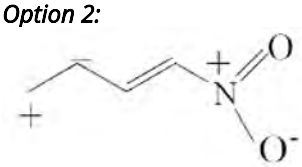
Hence, the correct answer is the option (1).

Q. 8 From the following the least stable structure is

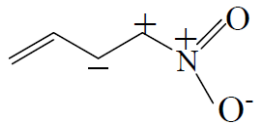
Option 1:



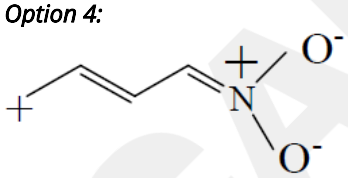
Option 2:



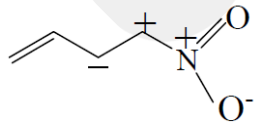
Option 3:



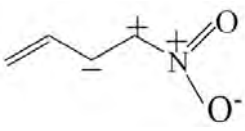
Option 4:



Correct Answer:



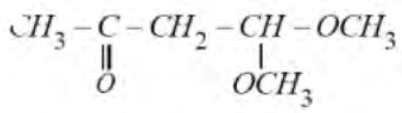
Solution:
unstable resonating structure is



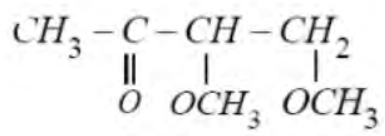
Hence, the correct answer is the option (3).

Q. 9 An organic compound ' P ' of molecular formula $C_6H_{12}O_3$ gives positive iodoform test but negative Tollen's test. When ' P ' is treated with dilute acid, it produces ' Q '. ' Q ' gives positive Tollen's test and also iodoform test. The structure of ' P ' is

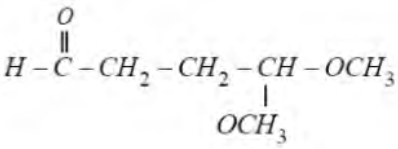
Option 1:



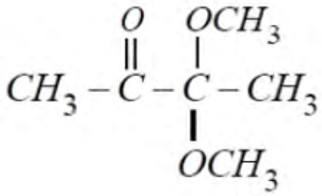
Option 2:



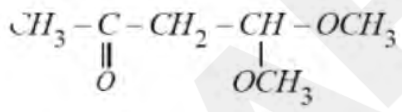
Option 3:



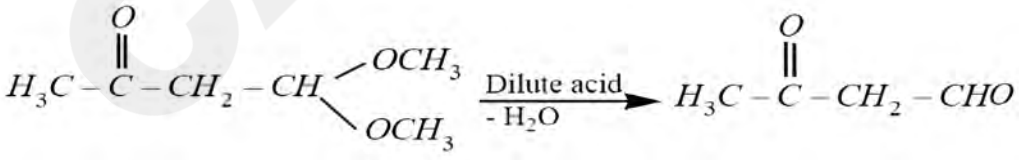
Option 4:



Correct Answer:



Solution:



Hence, the correct answer is the option (1).

Q. 10 Given below are two statements:

Statement-I: Among $[Cu(NH_3)_4]^{2+}$, $[Ni(en)_3]^{2+}$, $[Ni(NH_3)_6]^{2+}$ and $[Mn(H_2O)_6]^{2+}$, $[Mn(H_2O)_6]^{2+}$ has the maximum number of unpaired electrons.

Statement-II: The number of pairs among $\left\{[NiCl_4]^{2-}, [Ni(CO)_4]\right\}$, $\left\{[NiCl_4]^{2-}, [Ni(CN)_4]^{2-}\right\}$ and $\left\{[Ni(CO)_4], [Ni(CN)_4]^{2-}\right\}$ that contain only diamagnetic species is two.

In the light of the above statements, those the correct answer from the given below:

Option 1:
Statement I is false, statement II is true

Option 2:
Both statement I and statement II are false

Option 3:
Statement I is true but statement II is false

Option 4:
Both statement I and statement II are true

Correct Answer:
Statement I is true but statement II is false

Solution:
Sol: Statement–I:

$\left[Mn(H_2O)_6\right]^{2+}$

$Mn^{2+} \rightarrow 3d^5$ ↑ ↑ ↑ ↑ ↑

5 unpaired electrons
Statement–II:

$[NiCl_4]^{2-} \rightarrow$ Para
$[Ni(CO)_4] \rightarrow$ Dia
$[Ni(CN)_4]^{2-} \rightarrow$ Dia

Hence, the correct answer is the option (3).

Q. 11 Which of the following represents the correct trend for the mentioned property?
A: $F > P > S > B$ - First Ionisation Energy.
B: $Cl > F > S > P$-Electron Affinity.
C: $K > Al > Mg > B-$ Metallic Character.
D: $\mathrm{K_2O > Na_2O > MgO > Al_2O_3}$ - Basic character.
Choose the correct answer from the options given below:

Option 1:
A, B and D only

Option 2:
A and B only

Option 3:
B and C only

Option 4:
A, B, C and D

Correct Answer:
A, B and D only

Solution:
Sol: A, B, D are correct.
C is incorrect. $K > Mg > Al > B$ is correct order.

Hence, the correct answer is the option (1).

Q. 12 80 mL of a hydrocarbon on mixing with 264 ml of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL . When the system is treated with KOH solution, the volume decreases to 64 mL . The formula of the hydrocarbon is

Option 1:
$\mathrm{C_2H_4}$

Option 2:
$C_{12}H_6$

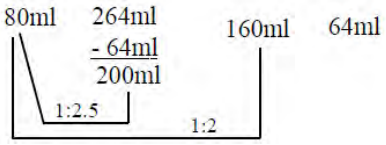
Option 3:
$\mathrm{C_2H_2}$

Option 4:
C_4H_{10}

Correct Answer:
C_2H_2

Solution:
Sol:

$$C_xH_y + O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O$$



Hence the emperical formula $= C_2H_2$

Hence, the correct answer is the option (3).

Q. 13 Given below are two statements:
Statement-I: The number of pairs among $[SiO_2, CO_2]$, $[SnO, SnOO_2]$, $[PbO, PbO_2]$ and $[GeO, GeO_2]$ which contains oxides that are both amphoteric is 2 .
Statement - II: BF_3 is an electron deficient molecule, act as a Lewis acid forms adduct with NH_3 and has a trigonal planar geometry.
In the light of the above statements, Choose the correct answer from the options given below:

Option 1:
Both statements I and Statement II are true

Option 2:
Both statements I and Statement II are false

Option 3:
Statement I is false but statement II is true

Option 4:
Statement I is true but statement II is false

Correct Answer:
Both statements I and Statement II are true

Solution:
Sol: Statement-1: $[SnO; SnO_2]$ and $[PbO; PbO_2]$ are amphoteric.
Statement-1: BF_3 from addict with NH_3, which has trigonal planar geometry.

Hence, the correct answer is the option (1).

Q. 14 Elements ' P ' and ' Q ' from two types of non-volatile; non-ionizable compounds PQ and PQ_2. When 1 g of PQ is dissolved in 50 g of solvent ' A '; ΔT b was 1.176 K while when 1 g of PQ_2 is dissolved in 50 g of of solvent ' A '; ΔTb was 0.689 K . (Kb of A is $5\ \mathrm{K\ kg\ mol^{-1}}$). The molar masses of element P and Q (in $\mathrm{gmol^{-1}}$) respectively are

Option 1:
70,110

Option 2:
25,60

Option 3:
65,145

Option 4:
60,25

Correct Answer:
25,60

Solution:
Sol: i) $M_{PQ} = \frac{K_b \times a \times 100}{\Delta T_b \times b} = \frac{5 \times 1 \times 100}{50 \times 1.176} = 85$
ii) $M_{PQ_2} = \frac{5 \times 1 \times 100}{50 \times 0.689} = 145$

We get, $P + Q = 85 \& P + 2Q = 145$
On solving $P = 25, Q = 60$

Hence, the correct answer is the option (2).

Q. 15 Identify correct statements from the following :
A) Propanal and propanone are functional isomers
B) Ethoxyethane and methoxy propane are metamers
C) But-2-ene shows optical isomerism
D) But-1-ene and But-2-ene are functional isomers
E) Pentane and 2,2-dimethyl propane are chain isomers

Choose the correct answer from the options given below:

Option 1:
A, B and E only

Option 2:
C, D and E only

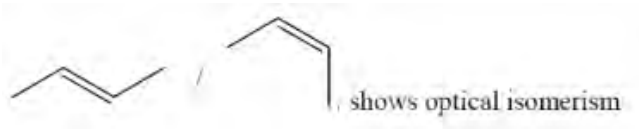
Option 3:
B, C and D only

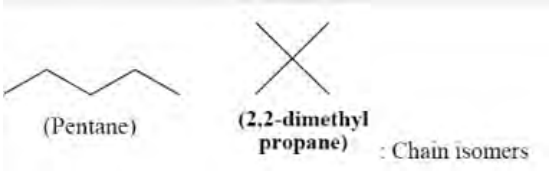
Option 4:
A, B and C only

Correct Answer:
A, B and E only

Solution:
Sol: A: Propanol & propanone is FGI.

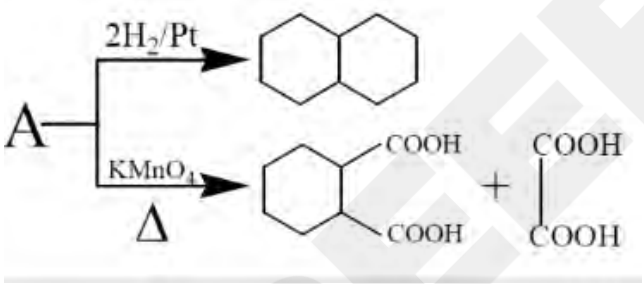
B: $CH_3-CH_2-O-CH_2-CH_3; CH_3-O-CH_2-CH_2-CH_3$ metamers.



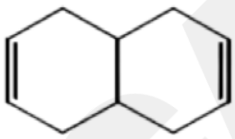


Hence, the correct answer is the option (1).

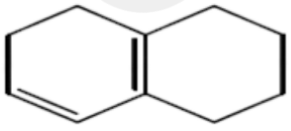
Q. 16 Identify 'A' in the following reaction:



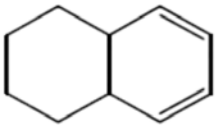
Option 1:



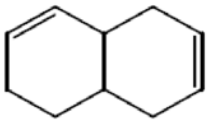
Option 2:



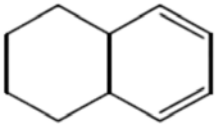
Option 3:



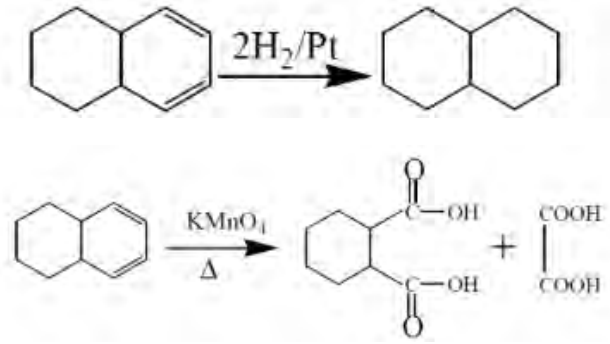
Option 4:



Correct Answer:



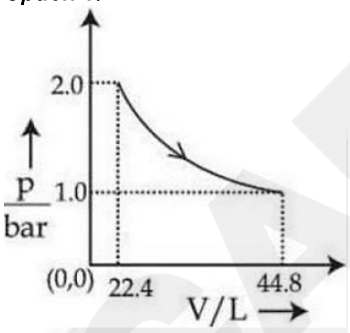
Solution:



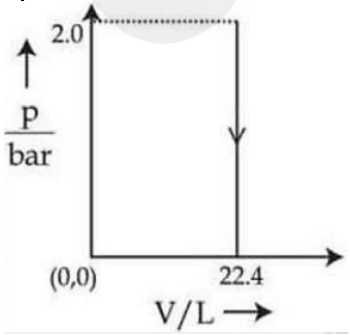
Hence, the correct answer is the option (3).

Q. 17 Which of the following graphs pressure ' P ' versus volume ' V ' represents the maximum workdone?

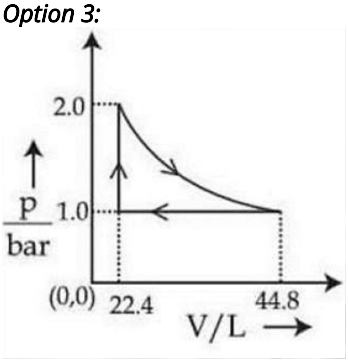
Option 1:



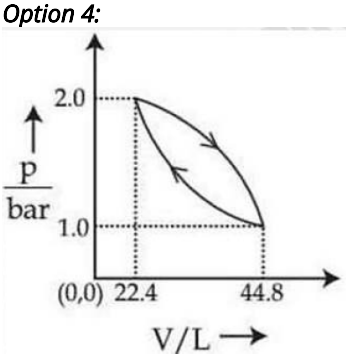
Option 2:



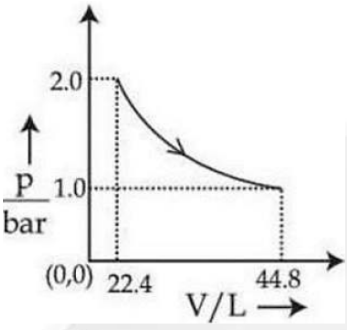
Option 3:



Option 4:



Correct Answer:



Solution:
In graph 1, the work done is positive, and the area under this curve is the largest.

Hence, the correct answer is the option (1).

Q. 18 **Statement – I:** When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.
Statement – II: The frequency of second line Balmer series obtained from He equal to that of first line of Lyman series obtained from hydrogen atom. In the light of the above statements, Choose the correct answer from the options given below:

Option 1:
Both statement I and statement II are false

Option 2:
Both statement I and statement II are true

Option 3:
Statement I is true but statement II is false

Option 4:
Statement I is false but statement II is true

Correct Answer:
Both statement I and statement II are true

Solution:
Sol: 1st line of Hyddrogen atom in Lymen series,

Second line of He^{+}ion in Balmar series,
Third line of Li^{+2} ion in Paschen series Above are having same frequency.

Hence, the correct answer is the option (2).

Q. 19 Consider the following reactions.

$$PbCl_2 + K_2CrO_4 \longrightarrow A + 2KCl$$

(Hot solution)

$$A + NaOH \quad B + Na_2CrO_4$$
$$PbSO_4 + 4CH_3COONH_4 \longrightarrow (NH_4)_2SO_4 + X$$

In the above reactions, A, B and X are respectively.

Option 1:
$PbCrO_4, Na_2[Pb(OH)_4]$ and $[Pb(NH_3)_4]SO_4$

Option 2:
$Na_2[Pb(OH)_4], PbCrO_4$ and $[Pb(NH_3)_4]SO_4$

Option 3:
$PbCrO_4, Na_2[Pb(OH)_4]$ and $(NH_4)_2[Pb(CH_3COO)_4]$

Option 4:
$Na_2[Pb(OH)_4], PbCrO_4$ and $(NH_4)_2[Pb(CH_3COO)_4]$d

Correct Answer:
$PbCrO_4, Na_2[Pb(OH)_4]$ and $(NH_4)_2[Pb(CH_3COO)_4]$

Solution:
Sol:

$$PbCl_2 + K_2CrO_4 \longrightarrow PbCrO_4 + 2KCl$$
$$PbCrO_4 + NaOH \quad Na_2[Pb(OH)_4] + Na_2CrO_4$$
$$PbSO_4 + 4CH_3COONH_4 \longrightarrow (NH_4)2SO_4 + (NH_4)_2[Pb(CH_3COO)_4]$$

Hence, the correct answer is the option (3).

Q. 20 14.0 gm of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K which of the following statements is incorrect?
[Given molar mass in gmol^{-1} of $\mathrm{Ca}-40; \mathrm{Cl}-35.5; \mathrm{H}-1$.

Option 1:
7.84 L of H_2 gas is evolved.

Option 2:
0.35 mol of H_2 gas is evolved.

Option 3:
The limiting reagent is calcium metal.

Option 4:
33.3 g of CaCl_2 is produced.

Correct Answer:
33.3 g of CaCl_2 is produced.

Solution:
Sol: 40 gm of $\mathrm{Ca} \longrightarrow 111 \mathrm{gms} \mathrm{CaCl}_2 \longrightarrow 1$ mole $4 \mathrm{Sol}: 4 \mathrm{H}_2 \longrightarrow 22.4 \mathrm{LH}_2$ STP.
14 gm of $\mathrm{Ca} \longrightarrow ? 38.85 \mathrm{gms} \longrightarrow 0.35$ mole $\mathrm{H}_2 \longrightarrow 7.84 \mathrm{LH}_2$ STP.

Hence, the correct answer is the option (4).

Q. 21 Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by 20 kJmol^{-1}. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K , then in $\frac{k_2}{k_1}$ will be _____ .
(nearest integer) $\left[\mathrm{R}=8.314 \mathrm{]K}^{-1} \mathrm{~mol}^{-1}\right]$

Correct Answer:
8

Solution:
Sol: $\ln\left(\frac{K_2}{K_1}\right)=\frac{Ea_2-Ea_1}{RT}=\frac{20000}{8.314 \times 300}=8$

Hence, the answer is (8).

Q. 22 Consider the following reactions:

$$\mathrm{NaCl} + \mathrm{K_2Cr_2O_7} + \mathrm{H_2SO_4} \longrightarrow \mathrm{A} + \mathrm{KHSO_4} + \mathrm{NaHSO_4} + \mathrm{H_2O}$$
$$\mathrm{A} + \mathrm{NaOH} \longrightarrow \mathrm{B} + \mathrm{NaCl} + \mathrm{H_2O}$$
$$\mathrm{B} + \mathrm{H_2SO_4} + \mathrm{H_2O_2} \longrightarrow \mathrm{C} + \mathrm{Na_2SO_4} + \mathrm{H_2O}$$

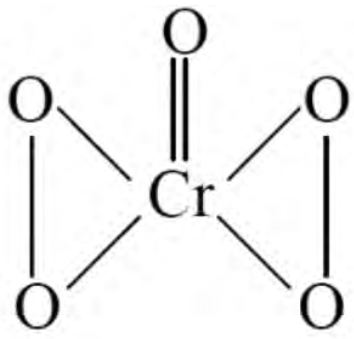
In the product ' C ', ' X ' is the number of $\mathrm{O_2^{2-}}$ units, ' Y ' is the total number oxygen atoms present and ' Z ' is the oxidation state of Cr. The value of $X + Y + Z$ is ____ .

Correct Answer:
13

Solution:
Sol:

$$\mathrm{NaCl} + \mathrm{K_2Cr_2O_7} + \mathrm{H_2SO_4} \longrightarrow \mathrm{CrO_2Cl_2} + \mathrm{KHSO_4} + \mathrm{NaHSO_4} + \mathrm{H_2O}$$
$$\mathrm{CrO_2Cl_2} + \mathrm{NaOH} \longrightarrow \mathrm{Na_2CrO_4} + \mathrm{NaCl} + \mathrm{H_2O}$$

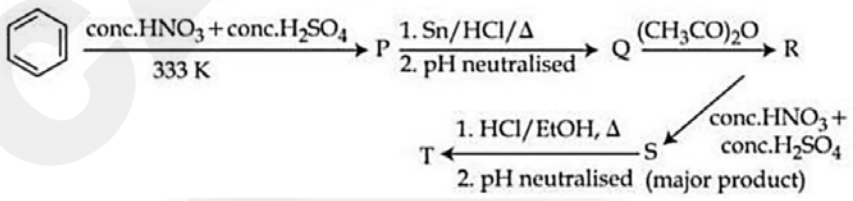
$$\mathrm{Na_2CrO_4} + \mathrm{H_2SO_4} + \mathrm{H_2O_2} \longrightarrow \mathrm{CrO_5} + \mathrm{Na_2SO_4} + \mathrm{H_2O}$$



$x = 2 \quad y = 5 \quad z = 6 \quad x + y + z = 13$

Hence, the answer is (13).

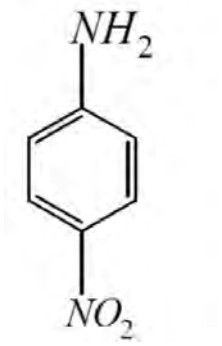
Q. 23 Consider the following reaction sequence



The percentage of nitrogen in product ' T ' formed is ____ %. (Nearest integer)
(Given molar mass in $g\ \mathrm{mol^{-1}}$ $\mathrm{H}: 1, \mathrm{C}: 12,\ \mathrm{N}: 14, \mathrm{O}: 16$)

Correct Answer:
20

Solution:
Sol: The final product T is



% of nitrogen $\frac{28}{138} \times 100 \approx 20\%$

Hence, the answer is (20).

Q. 24 The pH and conductance of a weak acid (HX) was found to be 5 and $4 \times 10^{-5} S$, respectively. The conductance was measured under standard condition using a cell where the electrode plates having a surface area of $1\ \mathrm{cm}^2$ were at a distance of 15 cm apart. The value of the limiting molar conductivity is ____ Sm^2ml^{-1} (nearest integer) (Given: degree of dissociation of the weak acid $(\alpha) \ll 1$)

Correct Answer:
6

Solution:
Sol:

$P^H = 5 \Rightarrow \left[H^+\right] = 10^{-5}\mathrm{M}$

$C = 4 \times 10^{-5} = \frac{1}{R}$

$G^K = \frac{l}{a} = 15\ \mathrm{cm}^{-1} = 1500\ \mathrm{m}^{-1}$

$\therefore K = G \times C = 1500 \times 4 \times 10^{-5} = 6 \times 10^{-2} S.\mathrm{m}^{-1} \Rightarrow m = \frac{K}{1000 \times M} = \frac{6 \times 10^{-2}}{10^3 \times 10^{-5}} = 6$

Hence, the answer is (6).

Q. 25 Use the following date:

Substance	$\frac{\Delta f H^{\ominus}(500K)}{\mathrm{kJmol}^{-1}}$	$\frac{S^{\ominus}(500K)}{\mathrm{JK}^{-1}\ \mathrm{mol}^{-1}}$
$AB(g)$	32	222
$A_2(g)$	6	146
$B_2(g)$	x	280

One mole each of $A_2(g)$ and $B_2(g)$ are taken in a 1 L closed flask and allowed to establish the equilibrium at 500 K .

$A_2(g) + B_2(g) \quad 2AB(g)$

The value of $\mathrm{x}\ (\text{ in } \mathrm{kJmol}^{-1})$ is ____ (Nearest integer)
(Given: $\log K = 2.2 \quad R = 8.3\ \mathrm{J\ K}^{-1}\ \mathrm{mol}^{-1}$)

Correct Answer:
70

Solution:
Sol:

$A_2 + B_2 \quad 2AB$
$\Delta H_f \quad 6 \quad x \quad 2(32)$
$S^0 \quad 146 \quad 280 \quad 222$
$\Delta G^0 = -2.303RT \quad \log K_e^2 = -2.303 \times 8.3 \times 500 \times 2.2 = -21.0264\text{KJ/mole}$
$\Delta H = 64 - 6 - x = (58 - x)\text{KJ/mole}$
$\Delta S = 444 - 426 = 18\ \text{J/k/mole}$
$\therefore \Delta G = \Delta H - T\Delta S \Rightarrow \Delta H = \Delta G + T\Delta S = -21.026 + \dfrac{500 \times 18}{1000} = -12.026\text{KJ}$
$\Rightarrow 58 - x = -12 \Rightarrow x = 70\text{KJ/mole}$

Hence, the answer is (70).

Maths

Q. 1 The value of $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \left(\frac{\pi+4x^{11}}{1-\sin\left(|x|+\frac{\pi}{6}\right)}\right)dx$ is equal to:

Option 1:
8π

Option 2:
2π

Option 3:
6π

Option 4:
4π

Correct Answer:
4π

Solution:
Sol: $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \left(\frac{\pi+4x^{11}}{1-\sin\left(|x|+\frac{\pi}{6}\right)}\right)dx$

$= \int_0^{\frac{\pi}{6}} \frac{2\pi}{(1-\sin(|x|+\frac{\pi}{6}))}dx$

Let $x + \frac{\pi}{6} = \theta$
$\Rightarrow dx = d\theta$

$$= 2\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{d\theta}{(1-\sin\theta)}$$
$$= 2\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{(1+\sin\theta)}{\cos^2\theta d\theta}$$
$$= 2\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\sec^2\theta + \sec\theta\tan\theta\right) d\theta$$
$$= 2\pi[\tan\theta + \sec\theta]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$
$$= 2\pi[(\sqrt{3}+2) - (1\sqrt{3} + \frac{2}{\sqrt{3}})]$$
$$= 2\pi[2]$$
$$= 4\pi$$

Hence, the correct answer is option (4).

Q. 2 The sum of all roots of equation $(x-1)^2 - 5|x-1| + 6 = 0$ is:

Option 1:
4

Option 2:
1

Option 3:
5

Option 4:
3

Correct Answer:
4

Solution:
$(x-1)^2 - 5|x-1| + 6 = 0$
Let $|x-1| = t, t \geq 0$

$t^2 - 5t + 6 = 0$
$t = 2 \text{ or } 3$
$|x-1| = 2 \text{ or } 3$
$\Rightarrow x - 1 = \pm 2 \text{ or } \pm 3$
$\Rightarrow x = 1 \pm 2 \text{ or } 1 \pm 3$
Roots are: $-2, -1, 3, 4$
$\text{Sum} = (-2) + (-1) + 3 + 4 = 4$

Hence, the correct answer is option (1).

Q. 3 The value of $(\text{cosec}\, 10^\circ - \sqrt{3}\sec 10^\circ)$ is equal to:

Option 1:
8

Option 2:
6

Option 3:
4

Option 4:
2

Correct Answer:
4

Solution:
$\operatorname{cosec} 10^{\circ}-\sqrt{3} \sec 10^{\circ}$
$=\frac{1}{\sin 10^{\circ}}-\frac{\sqrt{3}}{\cos 10^{\circ}}$
$=\frac{\cos 10^{\circ}-\sqrt{3} \sin 10^{\circ}}{\sin 10^{\circ} \cdot \cos 10^{\circ}}$
$=\frac{\frac{1}{2} \cdot \cos 10^{\circ}-\frac{\sqrt{3}}{2} \sin 10^{\circ}}{2 \sin 10^{\circ} \cos 10^{\circ}} \times 4$
$=\frac{\sin \left(30^{\circ}-10^{\circ}\right)}{\sin 20^{\circ}} \times 4$
$=4$

Hence, the correct answer is option (3).

Q. 4 If $x^2+x+1=0$,
then the value of $\left(x+\frac{1}{x}\right)^4+\left(x^2+\frac{1}{x^2}\right)^4+\left(x^3+\frac{1}{x^3}\right)^4+\ldots..\left(x^{25}+\frac{1}{x^{25}}\right)^4$ is:

Option 1:
145

Option 2:
162

Option 3:
175

Option 4:
128

Correct Answer:
145

Solution:
Sol: $x^2+x+1=0 \quad \Rightarrow x=w$

$$\left(x+\frac{1}{x}\right)^4+\left(x^2+\frac{1}{x^2}\right)^4+\left(x^3+\frac{1}{x^3}\right)^4+\ldots\ldots+\left(x^{25}+\frac{1}{x^{25}}\right)^4$$
$=(1+1+16)+(1+1+16)+\ldots\ldots\ldots .8 \text{ triplets } +1$
$=8(18)+1=145$

Hence, the correct answer is option (1).

Q. 5 Let $a_1, a_2, a_3 \ldots\ldots$ be a G.P of increasing positive terms such that $a_2 \cdot a_3 \cdot a_4=64$ and $a_1+a_3+a_5=\frac{813}{7}$ then $a_3+a_5+a_7$ is equal to:

Option 1:
3256

Option 2:
3252

Option 3:
3248

Option 4:
3244

Correct Answer:
3252

Solution:

$a_2 \cdot a_3 \cdot a_4 = 64$

$\Rightarrow (ar)\left(ar^2\right)\left(ar^3\right) = 64$

$\Rightarrow a^3r^6 = 64$

$\Rightarrow ar^2 = 4$

$a_1 + a_3 + a_5 = \dfrac{813}{7}$

$\Rightarrow a + ar^2 + ar^4 = \dfrac{813}{7}$

$\Rightarrow a\left(1 + r^4\right) = \dfrac{813}{7} - 4$

$\Rightarrow a\left(1 + r^4\right) = \dfrac{785}{7}$

$\Rightarrow \dfrac{a\left(1 + r^4\right)}{ar^2} = \dfrac{785}{7} \times \dfrac{1}{4}$

$\Rightarrow \dfrac{1 + r^4}{r^2} = \dfrac{785}{28}$

$\Rightarrow 28r^4 + 28 = 785r^2$

$\Rightarrow 28r^4 - 785r^2 + 28 = 0$

$\Rightarrow 28r^4 - 784r^2 - r^2 + 28 = 0$

$\Rightarrow 28r^2\left(r^2 - 28\right) - 1\left(r^2 - 28\right) = 0$

$\Rightarrow \left(r^2 - 28\right)\left(28r^2 - 1\right) = 0$

$\Rightarrow r^2 = 28 \text{ or } r^2 = \dfrac{1}{28} (\text{ Rejected })$

$ar^2 = 4 \Rightarrow a = \dfrac{4}{28} = \dfrac{1}{7}$

$a_3 + a_5 + a_7$
$= ar^2 + ar^4 + ar^6$
$= ar^2\left(1 + r^2 + r^4\right)$
$= 4(1 + 28 + 784)$
$= 4(813)$
$= 3252$

Hence, the correct answer is option (2).

Q. 6 The number of relations, defined on the set $\{a, b, c, d\}$ which are both reflexive & symmetric is equal to:

Option 1:
16

Option 2:
1024

Option 3:
64

Option 4:
256

Correct Answer:
64

Solution:
For a set $S = \{a, b, c, d\}$, a relation is reflexive
⇒ all 4 diagonal pairs $(a, a), (b, b), (c, c), (d, d)$ must be included (1 way).
For symmetry, the 6 off-diagonal pairs $(a, b), (a, c), (a, d), (b, c), (b, d), (c, d)$ can either be included with their symmetric counterparts or excluded (2 choices each).
$\therefore$ Total number of reflexive & symmetric relations:
$1 \times 2^6 = 64$

Hence, the correct answer is option (3).

Q. 7 Let $y = y(x)$ be the solution curve of the differential equation $\left(\left(1 + x^2\right)dy + \left(y - \tan^{-1} x\right)dx = 0\right.$, $y(0) = 1$ then value of $y(1)$ is:

Option 1:
$\frac{4}{e^{\pi/4}} - \pi/2 - 1$

Option 2:
$\frac{2}{e^{\pi/4}} + \pi/4 - 1$

Option 3:
$\frac{2}{e^{\pi/4}} - \pi/4 - 1$

Option 4:
$\frac{4}{e^{\pi/4}} + \pi/2 - 1$

Correct Answer:
$\frac{2}{e^{\pi/4}} + \pi/4 - 1$

Solution:
Sol: $\left(1 + x^2\right)dy + \left(y - \tan^{-1} x\right)dx = 0$

$$\left(1 + x^2\right)\frac{dy}{dx} + y - \tan^{-1} x = 0$$

$$\frac{dy}{dx} + y \cdot \left(\frac{1}{1 + x^2}\right) = \frac{\tan^{-1} x}{1 + x^2}$$

$$\text{I.F} = e^{\int \frac{1}{1+x^2} dx} = e^{\tan^{-1} x}$$

$$\text{y. If } = \int (I.f) \cdot Q(x)dx$$

$$y.e^{\tan^{-1}x} = \int e^{\tan^{-1}x} \cdot \frac{\tan^{-1}x}{1+x^2}dx$$

$$= \int e^t \cdot t \cdot dt \quad \tan^{-1}x = \text{t}$$

$$= t \cdot e^t - \int 1 \cdot e^t \quad \frac{1}{1+x^2}dx = dt = t \cdot e^t - le^t + C$$

$$ye^{\tan^{-1}x} = e^{\tan^{-1}x}\left(\tan^{-1}x - 1\right) + c$$

$$y = \left(\tan^{-1}x - 1\right) + \frac{C}{e^{\tan^{-1}x}}$$

$$x = 0, y = 1, 1 = (0-1) + \frac{c}{1}$$

$$\text{C} = 2$$

$$y = \left(\tan^{-1}x - 1\right) + \frac{2}{e^{\tan^{-1}x}}, \quad x = 1 \Rightarrow y(1) = \frac{\pi}{4} - 1 + \frac{2}{e^{\pi/4}}$$

Hence, the correct answer is option (2).

Q. 8 Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units respectively then the area (in sq units) of equilateral triangle ABC where the Points B and C lie on lines L_1 and L_2 respectively is:

Option 1:
$21\sqrt{3}$

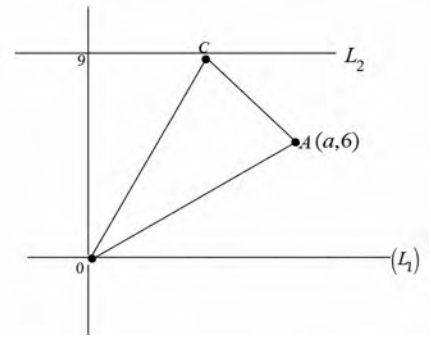
Option 2:
$15\sqrt{6}$

Option 3:
$12\sqrt{2}$

Option 4:
27

Correct Answer:
$21\sqrt{3}$

Solution:



Take $L_1 : y = 0$

$L_2 : y = 9$

Take B is at the origin
Let $A = (a, 6), B(0, 0)$

$C = (b, 9)$

$\triangle ABC$ is equilateral

$$(b, 9) = \frac{x_1 + x_2 \pm \sqrt{3}(y_1 - y_2)}{2}, \frac{y_1 + y_2\sqrt{3}(x_1 - x_2)}{2}$$

$$= \left(\frac{a + 0 \pm \sqrt{3}(6 - 0)}{2}, \frac{6 + 0}{2}\frac{\sqrt{3}(a - 0)}{2}\right) = \left(\frac{a + 6\sqrt{3}}{2}, \frac{6 - \sqrt{3}a}{2}\right) \text{ or } \left(\frac{a - 6\sqrt{3}}{2}, \frac{6 + \sqrt{3}a}{2}\right)$$

$$\text{If } \frac{6 + \sqrt{3}a}{2} = 9 \Rightarrow 6 + \sqrt{3}a = 18$$

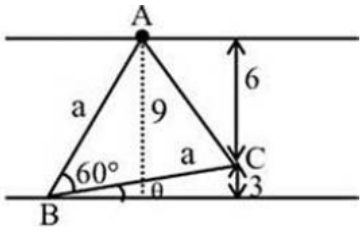
$$+ 12 = \sqrt{3}a$$

$$a = +4\sqrt{3}$$

$$b = \frac{a - 6\sqrt{3}}{2} = -\sqrt{3}$$

$$\text{Area } = \frac{1}{2}|x_1y_2 - x_2y_1| = \frac{1}{2}|9a - 6b| = \frac{1}{2}|36\sqrt{3} + 6\sqrt{3}| = 21\sqrt{3}$$

OR



$$\sin\theta = \frac{3}{a}; \quad \sin(60^\circ + \theta) = \frac{9}{a}$$

$$\Rightarrow \frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta = \frac{9}{a}$$

$$\Rightarrow \sqrt{3}\sqrt{1 - \frac{9}{a^2}} + \frac{3}{a} = \frac{18}{a}$$

$$\Rightarrow a = \sqrt{84}$$

$$\text{Area of } \triangle ABC = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 84 = 21\sqrt{3}$$

Hence, the correct answer is option (1).

Q. 9 Let foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If the eccentricity of the hyperbola is 5, then the length of its latus rectum is:

Option 1:
16

Option 2:
$24\sqrt{5}$

Option 3:
$\frac{96}{\sqrt{5}}$

Option 4:
12

Correct Answer:
$\frac{96}{\sqrt{5}}$

Solution:
Ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$

$$e_E=\sqrt{\frac{36-16}{36}}=\frac{\sqrt{20}}{6}$$

$$\text{Foci }=\left(6\cdot\frac{\sqrt{20}}{6},0\right),\left(-6\cdot\frac{\sqrt{20}}{6},0\right)=(\sqrt{20},0),(-\sqrt{20},0)$$

$$\text{Hyperbola }:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

$$ae_H=\sqrt{20}$$

$$5a=\sqrt{20}=2\sqrt{5}$$

$$a=\frac{2}{\sqrt{5}}$$

$$b^2=a^2\left(e^2-1\right)$$

$$b^2=\frac{4}{5}(25-1)=\frac{96}{5}$$

$$\therefore \text{Length of Latus Rectum}=\frac{2b^2}{a}=2\cdot\frac{96}{5}\cdot\frac{\sqrt{5}}{2}=\frac{96}{\sqrt{5}}$$

Hence, the correct answer is option (3).

Q. 10 Let $f:R\to(0,\infty)$ be a twice differentiable function such that $f(3)=18$, $f^l(3)=0$ and $f(3)=4$, then $\text{Lim}\left[\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right]$ is equal to:

Option 1:
1

Option 2:
18

Option 3:
2

Option 4:
9

Correct Answer:
2

Solution:

Given:

$\lim_{x\to 1} \log\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}$

Since $f(3) = 18$,

$= \lim_{x\to 1} \frac{18}{(x-1)^2} \ln\left(\frac{f(2+x)}{18}\right)$

As $x \to 1$,

$\ln\left(\frac{f(2+x)}{18}\right) \to 0$ and $(x-1)^2 \to 0$

So form is $\frac{0}{0}$.

Applying the L'Hospital rule, we get,

$$= \lim_{x\to 1} \frac{18 \cdot \frac{f'(2+x)}{f(2+x)}}{2(x-1)}$$

$$= \lim_{x\to 1} \frac{9f'(2+x)}{(x-1)f(2+x)}$$

Since $f'(3) = 0$, still $\frac{0}{0}$.

Applying the L'Hospital rule again, we get,

Numerator derivative:

$9f''(2+x)$

Denominator derivative:

$f(2+x) + (x-1)f'(2+x)$

So,

$= \lim_{x\to 1} \frac{9f''(2+x)}{f(2+x)+(x-1)f'(2+x)}$

Substitute $x = 1$:

$= \frac{9f''(3)}{f(3)} = \frac{9\cdot 4}{18} = \frac{36}{18} = 2$

Hence, the correct answer is option (3).

Q. 11 The number of strictly increasing functions f from set $\{1, 2, 3, 4, 5, 6\}$ to the set $\{1, 2, 3, \ldots\ldots\ldots, 9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$ is equal to:

Option 1:
21

Option 2:
27

Option 3:
22

Option 4:
28

Correct Answer:
28

Solution:
No. of functions $= 8_{C_6} = 8_{C_2} = 28$

Hence, the correct answer is option (4).

Q. 12 Let $\bar{a}=-i+2j+2\bar{k}, \bar{b}=8i+7j-3k$ and c be a vector such that $a\times c=\bar{b}$ if $c.\left(i+j+k\right)=4$ then $|a+c|^2$ is equal to:

Option 1:
30

Option 2:
27

Option 3:
33

Option 4:
35

Correct Answer:
27

Solution:
$\bar{a}=-\bar{i}+2\bar{j}+2\bar{k}$

$\bar{b}=8\bar{i}+7\bar{j}-3\bar{k}$

Let $\bar{c}=x\bar{i}+y\bar{j}+z\bar{k}$

$$\bar{a}\times\bar{c}=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\-1&2&2\\x&y&z\end{vmatrix}$$

$\bar{b}=\bar{i}(2z-2y)-\bar{j}(-z-2x)+\bar{k}(-y-2x)$
$2z-2y=8\rightarrow(1),\quad 2x-z=7\rightarrow(2),\quad -y-2x=-3\rightarrow(3)$
Solving $-y+z=4$
$x+y+z=4$
$x+2z=8$
$2x+4z=16$
$2x+z=7$
$3z=9$
$z=3$
$-y+3=4\Rightarrow y=-1$
$2x+3=7$
$x=2$
$\bar{a}+\bar{c}=(-\bar{i}+2\bar{j}+2\bar{k})+(2\bar{i}-\bar{j}+3\bar{k})=\bar{i}+\bar{j}+5\bar{k}$

$|\bar{a}+\bar{c}|=\sqrt{1+1+25}=\sqrt{27}$
$\Rightarrow|a+c|^2=27$

Hence, the correct answer is option (2).

Q. 13 If the domain of the function $f(x)=\cos^{-1}\left(\frac{2x-5}{11-3x}\right)+\sin^{-1}\left(2x^2-3x+1\right)$ is the interval $[\alpha,\beta]$, then $\alpha+2\beta$ is equal to:

Option 1:
5

Option 2:
1

Option 3:
3

Option 4:
2

Correct Answer:
3

Solution:

$$-1 \leq \frac{2x-5}{11-3x} \leq 1, 11-3x \neq 0, -1 \leq 2x^2-3x+1 \leq 1$$

$$\frac{2x-5}{11-3x}+1 \geq 0, \frac{2x-5}{11-3x}-1 \leq 0, x \neq \frac{11}{3}$$

$$2x^2-3x+2 \geq 0, 2x^2-3x \leq 0$$

$$\frac{2x-5+11-3x}{11-3x} \geq 0, \frac{2x-5-11+3x}{11-3x} \leq 0, x \neq \frac{11}{3}, x \in R$$

$$x(2x-3) \leq 0$$

$$\frac{6-x}{11-3x} \geq 0, \frac{5x-16}{11-3x} \leq 0, x \neq \frac{11}{3}, 0 \leq x \leq \frac{3}{2}$$

$$(x-6)(5x-16) \geq 0, x \in \left[0, \frac{3}{2}\right]$$

$$x \in \left[\left(-\infty, \frac{16}{5}\right) \cup (6, \infty)\right] \cap \left[0, \frac{3}{2}\right]$$

$$\text{Domain } = \left[0, \frac{3}{2}\right]$$

$$\alpha = 0, \beta = \frac{3}{2}$$

$$\alpha + 2\beta = 3$$

Hence, the correct answer is option (3).

Q. 14 The area of the region inside the ellipse $x^2+4y^2=4$ and outside the region bounded by the curve $y=|x|-1$ and $y=1-|x|$ is:

Option 1:
$2(\pi-1)$

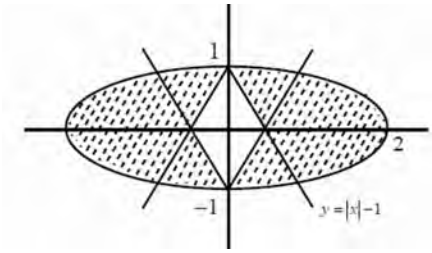
Option 2:
$2\pi-1/2$

Option 3:
$3(\pi-1)$

Option 4:
$2\pi-1$

Correct Answer:
$2(\pi-1)$

Solution:
Ellipse: $\frac{x^2}{4}+\frac{y^2}{1}=1$



$y=|x|-1$

$y=1-|x|$

$\therefore \text{Required area } =\pi(2)(1)-\frac{1}{2}(2)(2)=2\pi-2=2(\pi-1)$

Hence, the correct answer is option (1).

Q. 15 If the coefficient of x in the expansion of $\left(ax^2+bx+c\right)(1-2x)^{26}$ is -56 and the coefficient of x^2 and x^3 are both zero then $\mathrm{a}+\mathrm{b}+\mathrm{c}$ is equal to:

Option 1:
1483

Option 2:
1500

Option 3:
1300

Option 4:
1403

Correct Answer:
1403

Solution:
Given: $\left(ax^2+bx+c\right)(1-2x)^{26}$

$=\left(ax^2+bx+c\right)\left[26_{C_0}-26_{C_1}(2x)+26_{C_2}(2x)-26_{C_3}(2x)^3\right]$
$b\cdot 1+c\cdot(-26_{C_1}\cdot 2)=-56$
$b-52c=-56 \quad ----------(1)$
$a\cdot 1+b\cdot(26_{C_1}\cdot 2)+C\cdot 26_{C_2}(4)=0$
$a-52b+1300c=0 \quad ----------(2)$

And $a.(-26_{C_1}.2)+b.26_{C_2}(4)+c.(-26_{C_3}.8)=0$

$-52a+1300b-20800c=0 \quad ----------(3)$

Solving (1) ,(2) ,(3) ; $a=1300, b=100, c=3$

So, $a+b+c=1403$

Hence, the correct answer is option (4).

Q. 16 Let PQ and MN be two straight lines touching the circle $x^2+y^2-4x-6y-3=0$ at the points A and B, respectively. Let O be the centre of the circle and $\angle AOB=\frac{\pi}{3}$, then the locus of the point of intersection of the lines PQ and MN is:

Option 1:
$x^2+y^2-8x-12y-25=0$

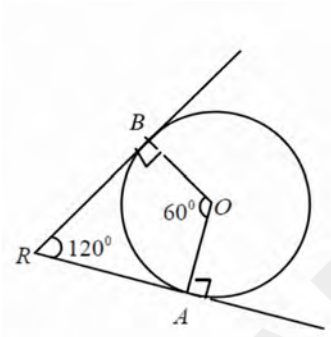
Option 2:
$x^2+y^2-12x-18y-25=0$

Option 3:
$3\left(x^2+y^2\right)-12x-18y-25=0$

Option 4:
$3\left(x^2+y^2\right)-18x-12y+25=0$

Correct Answer:
$3\left(x^2+y^2\right)-12x-18y-25=0$

Solution:
Given: $x^2+y^2-4x-6y-3=0$



Center, 'o' $=(2,3)$

Radius $r=\sqrt{4+9+3}=4$
Let the point of intersection of PQ and MN be $R\left(x_1,y_1\right)$

$\angle ARB=120^\circ$

$\Rightarrow \tan\left(\frac{120^\circ}{2}\right)=\frac{r}{\sqrt{S_{11}}}$

$\Rightarrow \sqrt{3}=\frac{4}{\sqrt{S_{11}}}$

$\Rightarrow 3\left[x_1^2+y_1^2-4x_1-6y_1-3\right]=16$

Locus of R is: $3\left(x^2+y^2\right)-12x-18y-25=0$

Hence, the correct answer is option (3).

Q. 17 Let the mean and variance of 7 observations $2, 4, 10, x, 12, 14, y, x > y$, be 8 and 16, respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability that the smaller number among the two chosen numbers is less than 4 is:

Option 1:
$\frac{4}{5}$

Option 2:
$\frac{1}{3}$

Option 3:
$\frac{3}{5}$

Option 4:
$\frac{2}{5}$

Correct Answer:
$\frac{4}{5}$

Solution:
Given: $\frac{2+4+10+x+12+14+y}{7} = 8$
$\Rightarrow 42 + x + y = 56$
$\Rightarrow x + y = 14 \rightarrow (1)$
$\Rightarrow \sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$
$\Rightarrow 16 = \frac{4 + 16 + 100 + x^2 + 144 + 196 + y^2}{7} - 64$
$\Rightarrow (80)(7) = x^2 + y^2 + 460$
$\Rightarrow x^2 + y^2 = 560 - 460$
$\Rightarrow x^2 + y^2 = 100 \rightarrow (2)$
$\Rightarrow x = 8, y = 6$
Given set $\{1, 2, 3, 4, 6, 5\}$
Let E be the event that both the numbers are greater ar equal to 4

$P(E) = \frac{3_{C_2}}{6_{C_2}} = \frac{3}{15} = \frac{1}{5}$

Hence, Required probability $= 1 - \frac{1}{5} = \frac{4}{5}$

Hence, the correct answer is option (1).

Q. 18 Let c and d be vectors such that $|c + d| = \sqrt{29}$ and $c \times (2i + 3j + 4k) = (2i + 3j + 4k) \times d$. If $\lambda_1, \lambda_2\ (\lambda_1 > \lambda_2)$ are the possible values of $(c + d) \cdot (-7i + 2j + 3k)$, then the equation $K^2x^2 + (K^2 - 5K + \lambda_1)xy + \left(3K + \frac{\lambda_2}{2}\right) \cdot y^2 - 8x + 12y + \lambda_2 = 0$ represents of a circle for K equal to:

Option 1:
4

Option 2:
2

Option 3:
-1

Option 4:
1

Correct Answer:
1

Solution:
$|\bar{c}+\bar{d}|=\sqrt{29}$
Let $\bar{b}=2\bar{i}+3\bar{j}+4\bar{k}$

$\bar{c}\times\bar{b}=\bar{b}\times\bar{d}$
$(\bar{c}\times\bar{b})+(\bar{d}\times\bar{b})=\bar{0}$
$(\bar{c}+\bar{d})\times\bar{b}=\bar{0}$
$\bar{c}+\bar{d}=t\bar{b}$
$|\bar{c}+\bar{d}|=|t|\sqrt{4+9+16}$
$\sqrt{29}=|t|\sqrt{29}$
$t=\pm1$
$\bar{c}+\bar{d}=\pm(2\bar{i}+3\bar{j}+4\bar{k})$
$(\bar{c}+\bar{d})\cdot(-7\bar{i}+2\bar{j}+3\bar{k})=\pm(-14+6+12)=\pm(4)$
$\lambda_1=4,\lambda_2=-4$
$K^2x^2+\left(K^2-5K+\lambda_1\right)xy+\left(3K+\frac{\lambda_2}{2}\right)y^2-8x+12y+\lambda_2=0$

Represents a circle $K^2-5K+4=0\Rightarrow K=1$, or 4

$K^2=3K-2\Rightarrow K^2-3K+2=0\Rightarrow K=1$ or 2

Hence $\mathrm{K}=1$

Hence, the correct answer is option(4).

Q. 19 Let O be the vertex of the parabola $x^2=4y$ and Q be any point on it let the locus of the point P which divides The line segment OQ internally in the ratio $2:3$ be the conic C then the equation of the chord of C which is bisected at point $(1,2)$, is:

Option 1:
$5x-4y+3=0$

Option 2:
$5x-y-3=0$

Option 3:
$4x-5y+6=0$

Option 4:
$x-2y+3=0$

Correct Answer:
$5x-4y+3=0$

Solution:
$x^2=4y$
Let $Q=\left(2t,t^2\right),P(h,k)$
$(h,k)=\left(\frac{2(2t)}{5},\frac{2t^2}{5}\right)$

$h=\frac{4t}{5},K=\frac{2t^2}{5}$

$$\Rightarrow 5K=2\left(\frac{5h}{4}\right)^2$$
$$\Rightarrow 8K=5h^2$$

Locus of P is $5x^2=8y$
Equation of chord having $(1,2)$ as mid point is

$S_1=S_{11}$
$\Rightarrow 5x(1)-4(y+2)=5(1)^2-8(2)$
$\Rightarrow 5x-4y-8=5-16$
$\Rightarrow 5x-4y+3=0$

Hence, the correct answer is option(1).

Q. 20 Let (α,β,γ) be the co-ordinates of the foot of the perpendicular drawn from the point (5,4,2) on the line $r=(-i+3j+k)+\lambda(2i+3j-k)$ then the length of the projection of vector $\alpha i+\beta j+\gamma k$ on the vector $6i+2j+3k$ is:

Option 1:
$\frac{18}{7}$

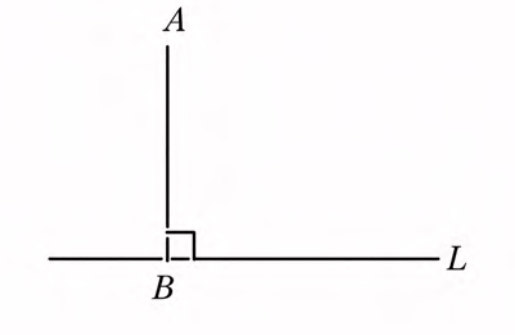
Option 2:
$\frac{15}{7}$

Option 3:
3

Option 4:
4

Correct Answer:
$\frac{18}{7}$

Solution:
Given line : $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$



$A(5,4,2)$
$B(\alpha,\beta,\gamma)$

Let $B=(2\lambda-1,3\lambda+3,-\lambda+1)$
Drs of $\overrightarrow{AB}=(2\lambda-6,3\lambda-1,-\lambda-1)$

$AB\perp L$
$2(2\lambda-b)+3(3\lambda-1)-1(-\lambda-1)=0$
$4\lambda-12+9\lambda-3+\lambda+1=0$
$14\lambda-14=0$
$\lambda=1$
$B=(1,6,0)$

Let $\bar{a}=\alpha\bar{i}+\beta\bar{j}+\gamma,\bar{k}=\bar{i}+b\bar{j}$ and $\bar{b}=b\bar{i}+2\bar{j}+3\bar{k}$

$\bar{a}\cdot\bar{b}=6+12=18$
$|\bar{b}|=\sqrt{36+4+9}=7$

Projection of $\bar{a}$ on $\bar{b}=\frac{\bar{a}.\bar{b}}{|\bar{b}|}\quad=\frac{18}{7}$

Hence, the correct answer is option(1).

Q. 21 Let $a_1=1$ and for $n\geq 1, a_{n+1}=\frac{1}{2}a_n+\frac{n^2-2n-1}{n^2(n+1)^2}$ then $\left|\sum_{n=1}^{\infty}\left(a_n-\frac{2}{n^2}\right)\right|$ is equal to:

Correct Answer:
2

Solution:
$a_{n+1}=\frac{1}{2}a_n+\frac{n^2-2n-1}{n^2(n+1)^2}=\frac{1}{2}a_n+\frac{2n^2-(n+1)^2}{n^2(n+1)^2}$

$a_{n+1}-\frac{2}{(n+1)^2}=\frac{1}{2}a_n-\frac{1}{n^2}=\frac{1}{2}\left[a_n-\frac{2}{n^2}\right]$

Let $b_n=a_n-\frac{2}{n^2}$

$b_{n+1}=\dfrac{1}{2}b_n$
$b_1,b_2,\ldots\ldots b_{n+1}$ are in GP

With $r=\frac{1}{2}$

$b_1=a_1-\frac{2}{1}=1-2=-1$

Now $\left|\sum_{n=1}^{\infty}\left(a_n-\frac{2}{n^2}\right)\right|=\left|\sum_{n=1}^{\infty}b_n\right|=\left|\frac{-1}{1-\frac{1}{2}}\right|=2$

Hence, the correct answer is 2.

Q. 22 $6\int_0^{\pi}|(\sin 3x+\sin 2x+\sin x)|dx$ is equal to:

Correct Answer:
17

Solution:

Sol: $6\int_0^{\pi}|\sin 3x+\sin 2x+\sin x|dx$

$=6\int_0^{\pi}|\sin 2x+2\sin 2x\cos x|dx=6\int_0^{\pi}|\sin 2x(1+2\cos x)|dx$

$=6\int_0^{\pi}|2\cos x(1+2\cos x)|\sin xdx \quad \cos x=t$

$=12\int_1^{-1}|t(1+2t)|-dt \quad -\sin x\cdot dx=dt$

$=12\int_{-1}^{1}|t(1+2t)|dt=12\left[\int_{-1}^{-1/2}\left(t+2t^2\right)dt-\int_{-1/2}^{0}\left(t+2t^2\right)dt+\int_0^{1}\left(t+2t^2\right)dt\right]$

$=12\left[\left(\frac{t^2}{2}+\frac{2t^3}{3}\right)_{-1}^{-1/2}-\left(\frac{t^2}{2}+\frac{2t^3}{3}\right)_{-1/2}+\left(\frac{t^2}{2}+\frac{2t^3}{3}\right)\right]$

$=12\left[\left(\frac{1}{8}-\frac{1}{12}\right)-\left(\frac{1}{2}-\frac{2}{3}\right)+\left(\frac{1}{8}-\frac{1}{12}\right)+\left(\frac{1}{2}+\frac{2}{3}\right)\right]$

$=12\left[\frac{2}{8}-\frac{2}{12}+\frac{4}{3}\right]=12\left[\frac{1}{4}-\frac{1}{6}+\frac{4}{3}\right]=3-2+16=17$

OR,

$6\int_0^{\pi}|2\sin 2x\cos x+\sin 2x|dx$

$6\int_0^{\pi}\left|4\sin x\cos^2 x+2\sin\cos x\right|dx$

$I=12\int_0^{\pi}\sin x\left|2\cos^2 x+\cos x\right|dx \quad \text{Put } \cos x=t,-\sin xdx=dt$

Hence, the correct answer is 17.

Q. 23 For some $\alpha,\beta\in R$, let $A=\begin{bmatrix}\alpha & 2\\ 1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}1 & 1\\ 1 & \beta\end{bmatrix}$ be such that $A^2-4A+2I=B^2-3B+I=0$ then $\left(\det\left(\text{adj}\left(A^3-B^3\right)\right)\right)^2$ is equal to:

Correct Answer:
225

Solution:
Trace of a matrix, $\text{tr}(A)=4$

$\alpha + 2 = 4$
$\alpha = 2$
$\text{tr}(B) = 3$
$1 + \beta = 3$
$\beta = 2$

$$A^2 = \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 4 & 6 \end{pmatrix}$$
$$A^3 = \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 6 & 8 \\ 4 & 6 \end{pmatrix} = \begin{bmatrix} 20 & 28 \\ 14 & 20 \end{bmatrix}$$
$$B^2 = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$
$$B^3 = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 8 & 13 \end{bmatrix}$$
$$A^3 - B^3 = \begin{bmatrix} 15 & 20 \\ 6 & 7 \end{bmatrix}$$
$$\left|A^3 - B^3\right| = 105 - 120 = -15$$

$$\left|\text{adj}\left(A^3 - B^3\right)\right|^2 = \left(\left|A^3 - B^3\right|^{2-1}\right)^2 = (-15)^2 = 225$$

Hence, the correct answer is 225.

Q. 24 Let $S = \{(m, n) : m, n \in (1, 2, 3 \ldots . 50)\}$ if number of elements (m, n) In S such that $6^m + 9^n$ is a multiple of 5 is P and the number of elements (m, n) in S such that $\mathrm{m} + \mathrm{n}$ is a square of a prime number is Q then $\mathrm{P} + \mathrm{Q}$ is equal to:

Correct Answer:
1333

Solution:
$6^m = (5 + 1)^m = 5k + 1$
$9^n = (10 - 1)^n = 10l - 1$ if n is odd
$6^m + 9^n$ is divisible by 5

If m can be any of $\{1, 2, 3, \ldots . . . . . 50\}$
And n is odd from $\{1, 2, 3, \ldots . . . . . 50\}$
No.of ways $= 50\text{X}25$

$\mathrm{P} = 1250$

$m + n = 4 \rightarrow$ no. of ways $= 3$
$m + n = 9 \rightarrow$ no. of ways $= 8$
$m + n = 25 \rightarrow$ no. of ways $= 24$
$m + n = 49 \rightarrow$ no. of ways $= 48$

$Q = 3 + 8 + 24 + 48 = 83$
$\therefore P + Q = 1250 + 83 = 1333$

Hence, the correct answer is 1333.

Q. 25 Let $f : R \to R$ be a twice differentiable function such that the quadratic equation

$$f(x)m^2 - 2f'(x)m + f'(x) = 0$$

In m, has two equal roots for every $x \in R$. If $f(0) = 1, f'(0) = 2$, and (α, β) is the largest interval in which the functions $f\left(\log_e x - x\right)$ is increasing, then $\alpha + \beta$ is equal to:

Correct Answer:
1

Solution:
$f(x) \cdot m^2 - 2f'(x)m + f''(x) = 0$

$\Delta = 0$
$\left[2f'(x)\right]^2 = 4f(x)f''(x)$
$\int \frac{f'(x)}{f(x)}dx = \int \frac{f''(x)}{f'(x)}dx$

$\text{Ln}\, f(x) = \ln f'(x) + \ln C$

$f(x) = Cf'(x)$
$f(0) = 1 \Rightarrow 1 = Cf'(0)$
$1 = \text{ C. } 2$
$C = \frac{1}{2}$
$\therefore 2f(x) = f'(x)$
$\frac{f'(x)}{f(x)} = 2$
$\int \frac{f'(x)}{f(x)}dx = \int 2dx$
$\ln f(x) = 2x + K$
$x = 0 \Rightarrow 0 = 0 + K$

$K = 0$
$\therefore \ln f(x) = 2x$
$f(x) = e^{2x}$

Let $g(x) = f[\ln x - x]$

$g(x) = e^{2(\ln x - x)}$
$g'(x) = e^{2(\ln x - x)}\left[\frac{1}{x} - 1\right]$

$g(x)$ is increasing

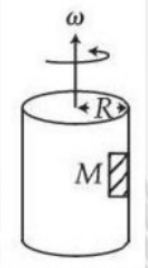
$g'(x) > 0$
$\frac{1}{x} - 1 > 0$
$\frac{1}{x} > 1$
$0 < x < 1$
$x \in (0, 1)$

$\alpha = 0, \beta = 1 \quad \alpha + \beta = 1$

Hence, the correct answer is 1.

Physics

Q. 1 A large drum having radius R is spinning around its axis with angular velocity ω as shown in the figure. The minimum value of ω so that body of mass M remains stuck to the inner wall of the drum taking the coefficient of friction between drum surface and mass M as μ is



Option 1:
$\sqrt{\frac{g}{2\mu R}}$

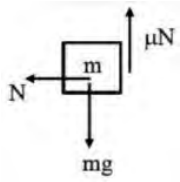
Option 2:
$\sqrt{\frac{2g}{\mu R}}$

Option 3:
$\sqrt{\frac{g}{\mu R}}$

Option 4:
$\sqrt{\frac{\mu g}{R}}$

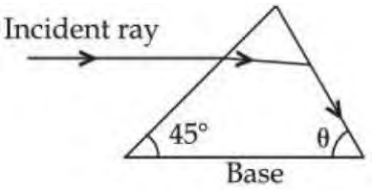
Correct Answer:
$\sqrt{\frac{g}{\mu R}}$

Solution:



$N = M\omega^2 R$ and $Mg = \mu N$
$\mu \times M\omega^2 R = Mg$
$\omega = \sqrt{\frac{g}{\mu R}}$

Q. 2 As shown in the diagram, when the incident ray is parallel to base of the prism the emergent ray grazes along the second surface.



If refractive index of the material of prism is $\sqrt{2}$, the angle θ of prism is

Option 1:
45°

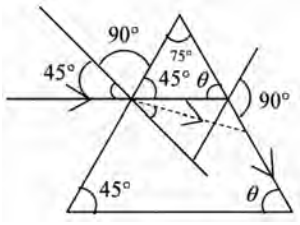
Option 2:
90°

Option 3:
60°

Option 4:
70°

Correct Answer:
60°

Solution:



$\mu = \sqrt{2}$
$i_C = r_2 = 45^\circ$
$r_1 = 30^\circ$
$r_1 + r_2 = A = 75^\circ$
$\therefore \theta = 180^\circ - 75^\circ - 45^\circ = 60^\circ$

Q. 3 Keeping significant figures in view, the sum of the physical quantities $52.01\ \text{m}, 153.2\ \text{m}$ and 0.123 m is

Option 1:
205.33 m

Option 2:
205.333 m

Option 3:
205.3 m

Option 4:
205 m

Correct Answer:
205.3 m

Solution:
52.01+153.2+0.123 = 205.333

Q. 4 The r.m.s. speed of oxygen molecules at 47°C is equal to that of hydrogen molecules kept at _____ $^\circ\text{C}$. (mass of oxygen molecule/mass of hydrogen molecule $= 32/2$)

Option 1:
-235

Option 2:
-20

Option 3:
-100

Option 4:
-253

Correct Answer:
-253

Solution:

$V_{rms} = \sqrt{\frac{3RT}{M}}$

$V_{rms} = V_{rmsH_2}$

$T_{O_2} = 273 + 47 = 320\ \text{K}$

$\sqrt{\frac{3RT_{O_2}}{M_{O_2}}} = \sqrt{\frac{3RT_{H_2}}{M_{H_2}}}$

$\frac{T_2}{M_{O_2}} = \frac{T_{H_2}}{M_{H_2}}$

$\frac{320}{32} = \frac{T_{H_2}}{2}$

$T_{H_2} = 20\ \text{K}$

$T_{\text{H}_2} = -253^{\circ}\text{C}$

Q. 5 A capacitor C is first charged fully with potential difference of V_0 and disconnected from battery. The charged capacitor is connected across an inductor having inductance L . In $\text{t sec}\ 25\%$ of initial energy in the capacitor is transferred to inductor. The value of t is _____ sec.

Option 1:
$\frac{\pi\sqrt{LC}}{2}$

Option 2:
$\pi\sqrt{\frac{\text{LC}}{2}}$

Option 3:
$\frac{\pi\sqrt{LC}}{6}$

Option 4:
$\frac{\pi\sqrt{LC}}{3}$

Correct Answer:
$\frac{\pi\sqrt{LC}}{6}$

Solution:

$U_{c_f} = 75\%U_{c_i}$

$Q_F^2 = \frac{3}{4}Q_I^2$

$Q_i\cos\omega t = \frac{\sqrt{3}}{2}Q_i \Rightarrow t = \frac{T}{12}$

$t = \frac{\pi}{6}\sqrt{LC}$

Q. 6 Consider two identical metallic spheres of radius R each having charge Q and mass m . Their centres have an initial separation of 4 R . Both spheres are given and initial speed u towards each other. The minimum value of u, so that they can just touch each other is (Take $k = \frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where G is the gravitational constant)

Option 1:
$\sqrt{\frac{kQ^2}{2mR}\left(1 - \frac{Gm^2}{kQ^2}\right)}$

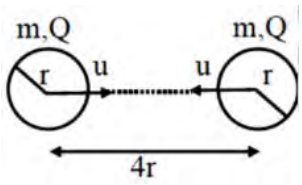
Option 2:
$\sqrt{\frac{kQ^2}{4mR}\left(1 + \frac{Gm^2}{kQ^2}\right)}$

Option 3:
$\sqrt{\frac{kQ^2}{4mR}\left(1-\frac{Gm^2}{kQ^2}\right)}$

Option 4:
$\sqrt{\frac{kQ^2}{2mR}\left(1-\frac{Gm^2}{2kQ^2}\right)}$

Correct Answer:
$\sqrt{\frac{kQ^2}{4mR}\left(1-\frac{Gm^2}{kQ^2}\right)}$

Solution:
Using energy conservation



$$(2)\ \left(\frac{1}{2}mu^2\right)-\frac{Gm^2}{4r}+\frac{KQ^2}{4r}=-\frac{Gm^2}{2r}+\frac{KQ^2}{2r}$$

$$u=\sqrt{\frac{1}{4mr}(KQ^2-Gm^2)}$$

Q. 7 Two cars A and B each of mass 10^3 kg are moving on parallel tracks separate by distance of 10 m , in same direction with speeds $72\ \mathrm{km/h}$ and $36\ \mathrm{km/h}$. The magnitude of angular momentum of car A with respect to car B is _____ Js .

Option 1:
2×10^5

Option 2:
3.6×10^5

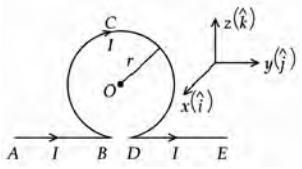
Option 3:
10^5

Option 4:
3×10^5

Correct Answer:
10^5

Solution:
$L=m\cdot V_{rel}r_{\perp}=1000\times\left(36\times\frac{5}{18}\right)\times10=10^5$

Q. 8 An infinitely long straight wire carrying current I is bent in a planer shape as shown in the diagram. The radius of the circular part is r. The magnetic field at the centre O of the circular loop is



Option 1:
$\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

Option 2:
$-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

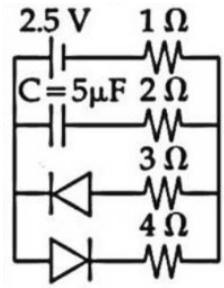
Option 3:
$-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi+1)\hat{i}$

Option 4:
$\frac{\mu_0}{2\pi}\frac{I}{r}(\pi+1)\hat{i}$

Correct Answer:
$-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

Solution:
$B=-\frac{\mu_0 I}{2r}+\frac{\mu_0 I}{2\pi r}=-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

Q. 9 The charge stored by the capacitor C in the given circuit in the steady state is _____ μC.



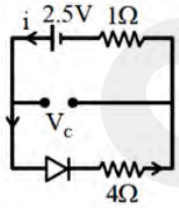
Option 1:
7.5

Option 2:
10

Option 3:
12.5

Option 4:
5

Correct Answer:
10

Solution:


in steady state

$i=2.5/5=0.5\ \text{A}$
$V_c=4\times 0.5$
$V_c=2\ \text{V}$

Charge $Q=CV_C=5\times 2=10\mu C$

Q. 10 The kinetic energy of simple harmonic oscillator is oscillating with angular frequency of $176\text{rad}/\text{sec}$. The frequency of the simple harmonic oscillator is _____ Hz. $(\pi=22/7)$

Option 1:
88

Option 2:
176

Option 3:
28

Option 4:
14

Correct Answer:
14

Solution:
$\omega = 176$

$f' = \frac{\omega}{2} = \frac{176 \times 7}{4 \times 22} = 14$

Q. 11 A river of width 200 m flowing from west to east with a speed $18\ \mathrm{km/h}$. A boat moving with speed of $36\ \mathrm{km/h}$ in still water, is made to travel 1 round trip (bank to bank of the river). Minimum time taken by the boat for this journey and also displacement along the river bank are _____ and _____ respectively.

Option 1:
40 s and 200 m

Option 2:
40 s and 0 m

Option 3:
40 s and 100 m

Option 4:
20 s and 100 m

Correct Answer:
40 s and 200 m

Solution:
$t = \frac{400}{10} = 40\ \mathrm{s}$

$x = 5 \times 40 = 200$

Q. 12 A spherical body of radius r and density σ falls freely through a viscous liquid having density ρ and viscosity η and attains a terminal velocity v_0. Estimated maximum error in the quantity η is (ignore errors associated with σ, ρ and g, gravitational acceleration)

Option 1:
$2\left[\frac{\Delta r}{r} + \frac{\Delta V_0}{V_0}\right]$

Option 2:
$2\left[\frac{\Delta r}{r} - \frac{\Delta V_0}{V_0}\right]$

Option 3:
$2\frac{\Delta r}{r} - \frac{\Delta V_0}{V_0}$

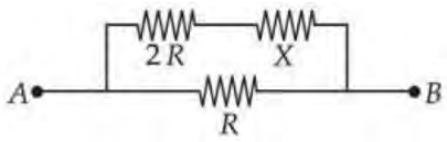
Option 4:
$\frac{2\Delta r}{r} + \frac{\Delta V_0}{V_0}$

Correct Answer:
$\frac{2\Delta r}{r}+\frac{\Delta V_0}{V_0}$

Solution:
$V_0=\frac{2}{9}(\rho_s-\rho_l)\frac{r^2g}{\eta}$ (Here v_0 is terminal velocity) $\Rightarrow \eta=\frac{2}{9}(\rho_s-\rho_l)\frac{r^2g}{\eta}$

$\therefore \frac{\Delta\eta}{\eta}=\frac{2\Delta r}{r}+\frac{\Delta V_0}{V_0}$.

Q. 13 Two known resistance of $\mathrm{R\Omega}$ and $\mathrm{2R\Omega}$ and one unknown resistance $\mathrm{X\Omega}$ are connected in a circuit as shown in the figure. If the equivalent resistance between points A and B in the circuit is $\mathrm{X\Omega}$, then value of X is _____ Ω.



Option 1:
R

Option 2:
$(\sqrt{3}+1)R$

Option 3:
$2(\sqrt{3}-1)R$

Option 4:
$(\sqrt{3}-1)R$

Correct Answer:
$(\sqrt{3}-1)R$

Solution:
$\frac{(2R+x)R}{3R+x}$
$2R^2+xR=3Rx+x^2$
$x^2+2Rx-2R^2=0$
$x=\frac{-2R\pm\sqrt{4R^2+8R^2}}{2}=\frac{-2R\pm2\sqrt{3}R}{2}=R(\sqrt{3}-1)$

Q. 14 The pulley shown in the figure is made using a thin rim and two rods of equal length to the diameter of rim. The rim and each rod have a mass of M. Two blocks of mass M and m are attached to two ends of a light string passing over the pulley, which is hinged to rotate freely in vertical plane about its center. The magnitudes of the acceleration experienced by the block is _____ (Assume no slipping of string on pulley)



Option 1:
$\frac{(M-m)g}{M+m}$

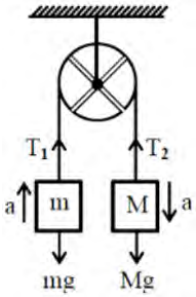
Option 2:
$\frac{(M-m)g}{2M+m}$

Option 3:
$\frac{(M-m)g}{[(\frac{13}{6})M+m]}$

Option 4:
$\frac{(M-m)g}{[(\frac{8}{3})M+m]}$

Correct Answer:
$\frac{(M-m)g}{[(\frac{8}{3})M+m]}$

Solution:



$Mg - T_2 = Ma \ldots\ldots\ldots\ldots (1)$
$T_1 - mg = ma \ldots\ldots\ldots (2)$
$(T_2 - T_1)r = I\frac{a}{r} \ldots\ldots\ldots\ldots (3)$
$(1) + (2) + (3)$
$(M-m)g = \left(M + m + \frac{I}{r^2}\right)a$
$\text{Here } I = Mr^2 + \frac{M \times (2r)^2}{12} \times 2$

On substituting I value in above equation we will get $\frac{(M-m)g}{[(\frac{8}{3})M+m]}$

Q. 15 Given below are two statements:

Statement 1: In Young's double slit experiment, the angular separation of fringes will increase as the screen is moved away from the plane of the slits.

Statement II: In a Young's double slit experiment, the angular separation of fringes will increase when monochromatic source is replaced by another monochromatic source of higher wavelength.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both statement I \& II are true

Option 2:
Statement I false but Statement II is true

Option 3:
Statement I true but Statement II is false

Option 4:
Both statement I \& II are false

Correct Answer:
Statement I false but Statement II is true

Solution:
$\theta = \frac{\lambda}{d}.$

Q. 16 A battery with EMF E and internal resistance r is connected across a resistance R. The power consumption in R will be maximum when:

Option 1:
$R = \sqrt{2}r$

Option 2:
$R = r$

Option 3:
$R = 2r$

Option 4:
$R = \frac{r}{2}$

Correct Answer:
$R = r$

Solution:
$$I = \frac{E}{R + r}$$
$$P = \frac{E^2}{(R + r)}R$$

For $P_{\max} \Rightarrow R = r$

Q. 17 A body of mass 2 kg is moving along x direction such that its displacement as function of time is given by $x(t) = at^2 + \beta t + \gamma m$, where $\alpha = 1\ \mathrm{m/s^2}$, $\beta = 1\ \mathrm{m/s}$ and $\gamma = 1\ \mathrm{m}$. The work done on the body during the time interval $t = 2\ \mathrm{s}$ to $t = 3\ \mathrm{s}$, is ____ J.

Option 1:
49

Option 2:
12

Option 3:
24

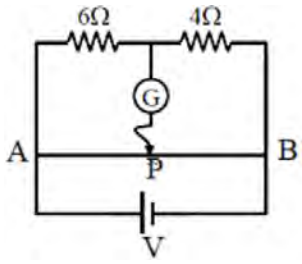
Option 4:
42

Correct Answer:
24

Solution:
$$v = \frac{dx}{dt} = 2t + 1$$
$a = 2; F = ma$
$F = 2 \times 2 = 4N$
$X_{(t=2\text{sec})} = 2^2 + 2 + 1 = 7$
$X_{(t=3\text{sec})} = 3^2 + 3 + 1 = 13$

Displacement $= \Delta x = 13 - 7 = 6$
Work done by force $= F \times x = 4 \times 6 = 24J$

Q. 18 The total length of potentiometer wire AB is 50 cm in the arrangement as shown in the figure. If P is the point where the galvanometer shows Zero reading then length of AP is _____ cm .



Option 1:
20

Option 2:
15

Option 3:
30

Option 4:
25

Correct Answer:
30

Solution:

$$\frac{6}{4}=\frac{x}{50-x}$$
$$\frac{3}{2}=\frac{x}{50-x}$$
$$150-3x=2x$$

$x=\frac{150}{5}=30\ \text{cm}.$

Q. 19 Surface tension of two liquids (having same densities) T_1 and T_2 are measured using capillary rise method utilising two tubes of inner radius r_1 and r_2 where $r_1 > r_2$. The measure liquid heights in these tubes are h_1 and h_2 respectively. (ignore weight of liquid about the lowest point of meniscus). The heights h_1 and h_2 and surface tension T_1 and T_2 satisfy the relation.

Option 1:
$h_1 < h_2$ and $T_1 = T_2$

Option 2:
$h_1 > h_2$ and $T_1 = T_2$

Option 3:
$h_1 = h_2$ and $T_1 = T_2$

Option 4:
$h_1 > h_2$ and $T_1 < T_2$

Correct Answer:
$h_1 < h_2$ and $T_1 = T_2$

Solution:
When a capillary tube is dipped in a liquid, the liquid rises due to surface tension until the upward force of surface tension is balanced by the weight of the liquid column.

The surface tension T acts along the circumference of the meniscus. If θ is the contact angle, the vertical component of the force is :

$$\mathrm{F_{up}} = \mathrm{T}\cos\theta \times (2\pi \mathrm{r})$$

The weight of the liquid column of height h is :

$W = \left(\pi r^2 h\right)\rho g$

At equilibrium :

$\mathrm{T}\cos\theta(2\pi \mathrm{r}) = \pi \mathrm{r}^2\,\mathrm{h}\rho\,\mathrm{g}$
$\Rightarrow\ \mathrm{h} = \dfrac{2\,\mathrm{T}\cos\theta}{\mathrm{r}\rho\,\mathrm{g}}$

If we assume the two liquids refer to the same substance, then:
$\mathrm{T}_1 = \mathrm{T}_2 \ldots$ (i) (Surface tension is a property of the material and temperature).
ρ and g are constant.
θ is constant.
This results the relationship: $\mathrm{h} \propto \frac{1}{\mathrm{r}}$.
We are given that $r_1 > r_2 \Rightarrow \frac{r_1}{r_2} > 1$.
So, the ratio of the heights of capillary rise,

$\frac{\mathrm{h}_1}{\mathrm{h}_2} = \frac{\mathrm{r}_2}{\mathrm{r}_2} < 1 \Rightarrow\ \mathrm{h}_1 < \mathrm{h}_2 \ldots (ii)$
From both the relations, $\mathrm{h}_1 < \mathrm{h}_2$ and $\mathrm{T}_1 = \mathrm{T}_2$.

Q. 20 The energy of an electron in an orbit of Bohr's atom is $-0.04E_0\mathrm{eV}$ where E_0 is the ground state energy. If L is the angular momentum of electron in this orbit and h is planks constant then $\frac{2\pi L}{h}$ is ____ .

Option 1:
2

Option 2:
6

Option 3:
5

Option 4:
4

Correct Answer:
5

Solution:
Angular momentum $L = \frac{nh}{2\pi}$

$n = \frac{2\pi L}{h}$

Energy $E = \frac{-13.6}{n^2} \cdot z^2 = -0.04 \times \left(\frac{+13.6}{1} \cdot z^2\right)$

$n^2 = 25$
$n = 5$

Q. 21 A diatomic gas $(\gamma = 1.4)$ does $100J$ of work when it is expanded isobarically. Then the heat given to the gas is ____ J.

Correct Answer:
350

Solution:
$W = 100J = nR\Delta T$ for isobaric process .

$Q = nC_p\Delta T = \left(\frac{f}{2} + 1\right)nR\Delta T$

$\frac{7}{2}(100) = 350$ Joule

Q. 22 A particle having electric charge $3 \times 10^{-19}\text{C}$ and mass 6×10^{-21} kg is accelerated by applying an electric potential of 1.21 V. Wave length of the matter wave associated with the particle is $\alpha \times 10^{-12}$ m. The value of α is _____. (Take Planck's constant $= 6.6 \times 10^{-34}$ J. s)

Correct Answer:
10

Solution:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 18 \times 10^{-46} \times 1.21}}$$

$$\lambda = 10^{-11}\text{ m} = 10 \times 10^{-12}\text{ m}$$

$$\alpha = 10$$

Q. 23 The terminal velocity of a metallic ball of radius 6 mm in a viscous fluid 20 cm/s. The terminal velocity of another ball of same material and having radius 3 mm in same fluid will be _____ cm/sec.

Correct Answer:
5

Solution:
We know
Terminal velocity α (radius) ^{2}

$$\frac{(v_T)_1}{(v_T)_2} = \left(\frac{6}{3}\right)^2$$

$$(v_T)_2 = \frac{(v_T)_1}{4} = 5\text{ cm/sec}$$

Q. 24 In a Young's double slit experiment setup, the two slits are kept 0.4 mm apart and screen is placed at 1 m from slits. If a thin transparent sheet of thickness 20μ m is introduced in front of one of the slits then center bright fringe shifts by 20 mm on the screen. The refractive index of transparent sheet is given by $\frac{\alpha}{10}$, where α is _____

Correct Answer:
14

Solution:

$$\frac{Y}{D} \cdot t = t(\mu - 1)$$

$$\mu = \left(\frac{Yd}{Dt} + 1\right) = \left(\frac{20 \times 10^{-3} \times 4 \times 10^{-4}}{1 \times 20 \times 10^{-6}}\right) + 1 = \frac{4}{10} + 1 \Rightarrow \mu = \frac{14}{10}$$

Q. 25 An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma = 10\text{mho/m}$. The ratio of maximum conduction current density to maximum displacement current density is _____.

$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2\right]$

Correct Answer:
1800

Solution:

$$i_c = A.\sigma E = A\sigma E_0 \sin(\omega t - kx)$$
$$i_d = A\varepsilon_r\varepsilon_0 \frac{\partial E}{\partial t}$$
$$= A\varepsilon_r\varepsilon_0 E_0 \omega \cos(\omega t - kx)$$
$$= \frac{i_{c\max}}{i_{d\max}} = \frac{\sigma}{\varepsilon_r\varepsilon_0\omega}$$
$$= \frac{10}{8.85 \times 10^{-12} \times 100 \times 10^6} = 1800.$$

Chemistry

Q. 1 Aqueous HCl reacts with $\mathrm{MnO_2}(s)$ to form $\mathrm{MnCl_2}(aq), \mathrm{Cl_2}(g)$ and $\mathrm{H_2O}(l)$.
What is the weight (in g) of $\mathrm{Cl_2}$ liberated when 8.7 g of $\mathrm{MnO_2}(\mathrm{~s})$ reacted with excess aqueous HCl solution? (Given Molar mass in gmol^{-1}

$Mn = 55, Cl = 35.5, O = 16, H = 1)$

Option 1:
7.1

Option 2:
17.1

Option 3:
14.2

Option 4:
3.5

Correct Answer:
7.1

Solution:
$\underset{0.1 \text{ moles}}{\mathrm{MnO_2}} + 4\mathrm{HCl} \rightarrow \underset{0.1 \text{ moles}}{\mathrm{MnCl_2}} + \mathrm{Cl_2} + 2\mathrm{H_2O} \quad \frac{8.7}{87.}$ Excess $= 0.1$ mole

$\underset{0.1 \text{ moles}}{\mathrm{MnO_2}} + 4\mathrm{HCl} \rightarrow \underset{0.1 \text{ moles}}{\mathrm{MnCl_2}} + \mathrm{Cl_2} + 2\mathrm{H_2O} \quad \frac{8.7}{87.}$ Excess $= 0.1$ mole

Hence, the correct answer is the option (1).

Q. 2 Consider the following data:

$\Delta_f{}^{\ominus}$

$(\text{ methane, g}) = -X \text{ kJ mol}^{-1}$

Enthalpy of sublimation of graphite $= Y \text{ kJ mol}^{-1}$
Dissociation enthalpy of $\mathrm{H_2} = \mathrm{Z kJ mol}^{-1}$
The bond enthalpy of $C - H$ bond is given by

Option 1:
$\frac{X+Y+2Z}{4}$

Option 2:
$\frac{X+Y+4Z}{2}$

Option 3:
$X + Y + Z$

Option 4:
$\frac{-X+Y+Z}{4}$

Correct Answer:
$\frac{X+Y+2Z}{4}$

Solution:
$\mathrm{C}(s)+2\mathrm{H}_2(g) \rightarrow \mathrm{CH}_4(g)$

$-x=(\Delta H_{\text{sub}}$ of carbon $)+2\times($ B.E. of H-H $)-4\times($ B.E. of $C-\mathrm{H})$

$-x=y+2z-4(\text{ B.E. of C-H })$

B.E of $\mathrm{C}-\mathrm{H}=\frac{\mathrm{y+2z+x}}{4}$

Hence, the correct answer is the option (1).

Q. 3 Consider the following spectral lines for atomic hydrogen:

A) First line of Paschen series
B) Second line of Balmer series
C) Third line of Paschen series
D) Fourth line of Bracket series

The correct arrangement of the above lines in ascending order of energy is:

Option 1:
$D<A<C<B$

Option 2:
$C<D<B<A$

Option 3:
$A<B<C<D$

Option 4:
$D<C<A<B$

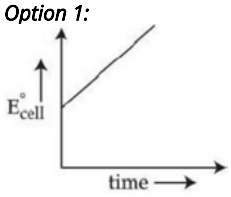
Correct Answer:
$D<A<C<B$

Solution:
$\Delta E=13.6Z^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$

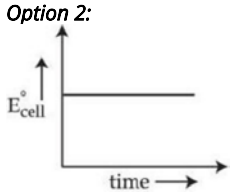
Series	n_1	n_2
1st line of (Paschen)	3	4
2nd line of (Balmer)	2	4
3rd line of (Paschen)	3	6
4th line of (Bracket)	4	8

Hence, the correct answer is the option (1).

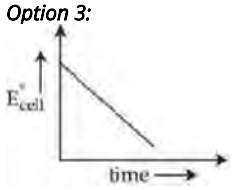
Q. 4 For a closed circuit Daniell cell, which of the following plots is the accurate one at a given temperature?

Option 1:


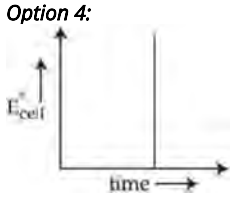
Option 2:



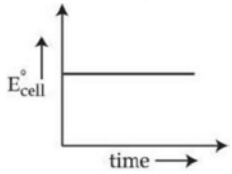
Option 3:



Option 4:



Correct Answer:



Solution:

E^0_{cell} remain constant with time

Hence, the correct answer is the option (2).

Q. 5 Decomposition of A is a first order reaction at $T(K)$ and is given by $A(g) \rightarrow B(g) + C(g)$.

In a closed 1 L vessel, 1 bar $\mathrm{A(g)}$ is allowed to decompose at $\mathrm{T(K)}$. After 100 minutes, the total pressure was 1.5 bar . What is the rate constant (in min^{-1}) of the reaction? $(\log 2 = 0.3)$

Option 1:

6.9×10^{-4}

Option 2:

6.9×10^{-3}

Option 3:

6.9×10^{-1}

Option 4:

6.9×10^{-2}

Correct Answer:

6.9×10^{-3}

Solution:

$A_g + B_g + C_g$

$1 \quad - \quad -$

$1 - P \quad P \quad P$

$P_{\text{total}} = 1 + P$

$1.5 = 1 + P$

$P = 0.5$

$$K = \frac{1}{100} \ln \frac{1}{0.5} = \frac{0.693}{100} = 6.9 \times 10^{-3}\ \mathrm{min}^{-1}$$

Hence, the correct answer is the option (2).

Q. 6 Given below are two statements

Statement-I: The correct order in terms of atomic/ionic radii is $\mathrm{Al} > \mathrm{Mg} > \mathrm{Mg}^{2+} > \mathrm{Al}^{3+}$

Statement -II : The correct order in terms of the magnitude of electron gain enthalpy is $\mathrm{Cl} > \mathrm{Br} > \mathrm{S} > \mathrm{O}$.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both statement I and II are correct.

Option 2:
Both statement I and II are false.

Option 3:
Statement I is correct and II is false.

Option 4:
Statement I is false and II is correct .

Correct Answer:
Statement I is false and II is correct .

Solution:
Correct order of size is $\mathrm{Mg} > \mathrm{Al} > \mathrm{Mg}^{2+} > \mathrm{Al}^{3+}$

Third period electron gain enthalpy will be more than second period.

Hence, the correct answer is the option (4).

Q. 7 Given below are two statements
Statement I : The correct order in terms of bond dissociation energy order is :

$\mathrm{Cl}_2 > \mathrm{Br}_2 > \mathrm{F}_2 > \mathrm{I}_2$

Statement II: The correct trend in the covalent character of the metal halides is

$[\mathrm{SnCl}_4 > \mathrm{SnCl}_2], [\mathrm{PbCl}_4 > \mathrm{PbCl}_2] \text{ and } [\mathrm{UF}_4 > \mathrm{UF}_6]$

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both statement I and II are true

Option 2:
Both statement I and II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Bond energy order is $\mathrm{Cl}_2 > \mathrm{Br}_2 > \mathrm{F}_2 > \mathrm{I}_2$ $[UF_4 < UF_6]$ due to more positive charge on U .

Hence, the correct answer is the option (3).

Q. 8 On heating a mixture of common salt and $K_2Cr_2O_7$ in equal amount along with concentrated H_2SO_4 in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are:

Option 1:
CrO_2Cl_2 and +6

Option 2:
$Cr_2O_2Cl_2$ and +6

Option 3:
$Cr_2O_2Cl_2$ and +3

Option 4:
CrO_2Cl_2 and +5

Correct Answer:
$Cr_2O_2Cl_2$ and +3

Solution:
Chromyl chloride test : Product is deep red vapours of CrO_2Cl_2 in which oxidation state of Cr is +6 .

Hence, the correct answer is the option (3).

Q. 9 The correct increasing order of $C-H(A), C-O(B), C=O(C) \& C\equiv N(D)$ bonds in terms of covalent bond length is:

Option 1:
$A<B<C<D$

Option 2:
$D<C<A<B$

Option 3:
$A<D<C<B$

Option 4:
$D<C<B<A$

Correct Answer:
$A<D<C<B$

Solution:
$C-H \quad 107pm$
$C\equiv N \quad 116pm$
$C\equiv O \quad 143pm$
$C\equiv O \quad 121pm$

Data-based (From NCERT)

Hence, the correct answer is the option (3).

Q. 10 Given below are some of the statements about Mn and Mn_2O_7. Identify the correct statements.

A) Mn forms the oxide Mn_2O_7, in which Mn is in its highest oxidation state.
B) Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn .
C) Mn_2O_7 is an ionic oxide.
D) The structure of Mn_2O_7 consists of one bridged oxygen.

Choose the correct answer from the options given below :

Option 1:
A, B, C and D

Option 2:
A, B and C only

Option 3:
A, C and D only

Option 4:
A, B and D only

Correct Answer:
A, B and D only

Solution:
Mn higher oxidation state will be $+7, \mathrm{Mn}$ is form multiple bond with oxygen, $\mathrm{Mn_2O_7}$ is 1 bridge bond.

Hence, the correct answer is the option (3).

Q. 11 Given below are two statements:

Statement I : Crystal field stabilization Energy (CFSE) of $[\mathrm{Cr(H_2O)_6}]^{2+}$ is greater than that of $[\mathrm{Mn(H_2O)_6}]^{2+}$

Statement II : Potassium ferricyanide has a greater spin - only magnetic moment than sodium ferrocyanide.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II both are true

Option 2:
Statement I is true but Statement II is false.

Option 3:
Statement I is false but statement II is true.

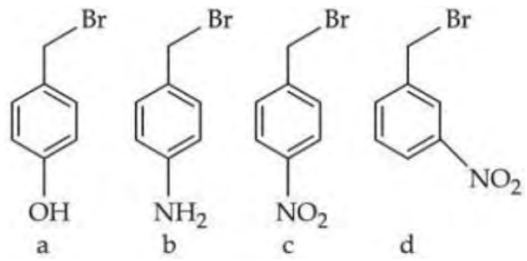
Option 4:
Both statement I and statement II are false.

Correct Answer:
Both Statement I and Statement II both are true

Solution:
$[\mathrm{Mn(H_2O)_6}]^{2+}$ is more stable than $[\mathrm{Cr(H_2O)_6}]^{2+}$
For : $K_3\,[\mathrm{Fe(CN)_6}], \mu = \sqrt{1(1+2)} = \sqrt{3}\ \mathrm{B.M}$
For $K_4\,[\mathrm{Fe(CN)_6}], \mu = \sqrt{0}$ B.M

Hence, the correct answer is the option (1).

Q. 12 The correct order of reactivity of the following benzyl halides towards reaction with KCN is:



Option 1:
$a > b > c > d$

Option 2:
$b > a > c > d$

Option 3:
$a > b > d > c$

Option 4:
$b > a > d > c$

Correct Answer:
$b > a > d > c$

Solution:
This is S_N1 reaction.
Rate of S_N1 reaction α stability of carbocation

Hence, the correct answer is the option (4).

Q. 13 **Match List - I with List - II**

	List -I (Reagent)		**List - II** **Reaction Name (Involving aldehydes)**
A	$H_2, Pd - BaSO_4$	I	Etard reaction
B	$SnCl_2, HCl$	II	Rosenmund reduction
C	CrO_2Cl_2, CS_2	III	Gattermann Koch reaction
D	CO, HCl, Anhyd. $AlCl_3$	IV	Stephen reaction

Choose the correct answer from the options given below

Option 1:
$A - IV, B - III, C - I, D - II$

Option 2:
$A - IV, B - I, C - II, D - III$

Option 3:
$A - II, B - III, C - IV, D - I$

Option 4:
$A - II, B - IV, C - I, D - III$

Correct Answer:
$A - II, B - IV, C - I, D - III$

Solution:
Based on NCERT data

Hence, the correct answer is the option (4).

Q. 14 **Match LIst - I with List - II**

List-I (Pair of Compounds)		List-II (Type of Isomers)	
A.	2-Methylpropene and but-1-ene	I	Stereoisomers
B.	Cis-but-2-ene and trans – but-2-ene	II	Position isomers
C.	2-Butanol and diethyl ether	III	Chain isomers
D.	But-1-ene and but-2-ene	IV	Functional isomers

Option 1:
$A-III, B-I, C-IV, D-II$

Option 2:
$\mathrm{A-II, B-I, C-IV, D-III}$

Option 3:
A - III, B - I, C - II, D - IV

Option 4:
$\mathrm{A-I, B-IV, C-III, D-II}$

Correct Answer:
$A-III, B-I, C-IV, D-II$

Solution:
$\mathrm{2\text{-}Methylpropene\ and\ but\text{-}1\text{-}ene \rightarrow Chain\ isomers}$

Cis-but-2-ene and trans - but-2-ene → Stereoisomers (Geometrical Isomers)

2-Butanol and diethyl ether → Functional isomers

But-1-ene and but-2-ene→ Position isomers

Hence, the correct answer is the option (1).

Q. 15 The correct statements are:

A) Activation energy for enzyme catalysed hydrolysis of sucrose is lower than that of acid catalysed hydrolysis
B) During denaturation, secondary and tertiary structures of a protein are destroyed but primary structure remains intact.
C) Nucleotides are joined together by glycosidic linkage between C_1 and C_4 carbons of the pentose sugar
D) Quaternary structure of proteins represents overall folding of the polypeptide chain

Choose the correct answer from the options given below :

Option 1:
A , B and D only

Option 2:
A and B only

Option 3:
A, C and D only

Option 4:
B and C only

Correct Answer:
A and B only

Solution:
The activation energy (E_a) for enzyme - catalysed hydrolysis of sucrose (using sucrose) is significantly lower - approximately $2.15\ \text{kJ mol}^{-1}$ compared to $6.22\ \text{kJ mol}^{-1}$ for acid catalysed hydrolysis.
The primary structure, which is the sequence of amino acids held together by strong covalent peptide bonds, remain intact.

Hence, the correct answer is the option (2).

Q. 16 The correct order of the rate of the reaction for the following reaction with respect to nuclophiles is:
$\text{CH}_2\text{Br} + \text{Nu}^- \rightarrow \text{CH}_3\text{Nu} + \text{Br}^-$

Option 1:
$^-\text{OH} > \text{PhO}^- > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$

Option 2:
$\text{PhO}^- >^- \text{OH} > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$

Option 3:
$\text{ClO}_4^- > \text{CH}_3\text{COO}^- >^- \text{OH} > \text{PhO}^-$

Option 4:
$\text{CH}_3\text{COO}^- > \text{PhO}^- >^- \text{OH} > \text{ClO}_4$

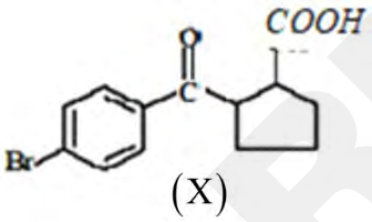
Correct Answer:
$^-\text{OH} > \text{PhO}^- > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$

Solution:
Based on the strength of nucleophile (the more stable an anion is less nucleophilic).

Hence, the correct answer is the option (1).

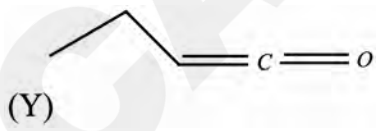
Q. 17 Given below are two statements:

Statement-I : Compound (X), shown below, dissolves in NaHCO_3 solution and has two chiral carbon atoms



(X)

Statement-II : Compound (Y), shown below, has two carbons with sp^3 hybridization, one carbon with sp^2 and one carbon with sp hybridization



(Y)

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Statement I is false but Statement II is true

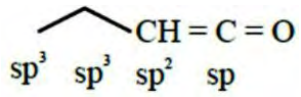
Option 2:
Statement I is true but Statement II is false

Option 3:
Both Statement I and Statement II are false

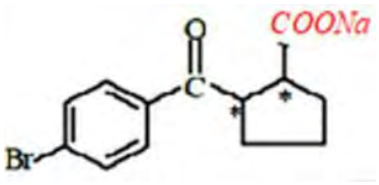
Option 4:
Both Statement I and Statement II are true

Correct Answer:
Both Statement I and Statement II are true

Solution:
Both Statements are correct



So there are $2sp^3, 1sp^2$ and 1 sp hybridised carbon in the given compound.



Dissolves in $NaHCO_3$ → This confirms presence of a carboxylic acid (–COOH) group.

From the given structure of compound (X), it contains:

- One –COOH group
- Two carbon atoms attached to four different groups → two chiral carbons

Hence, the correct answer is the option (4).

Q. 18 By usual analysis 1.00 g of compound (X) gave 1.79 g of magnesium pyrophosphate. The percentage of phosphorus in compound (X) is: (nearest integer) (Give molar mass in gmol^{-1}; $\mathrm{O}=16, \mathrm{Mg}=24, \mathrm{P}=31$)

Option 1:
50

Option 2:
20

Option 3:
30

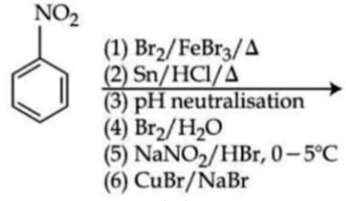
Option 4:
40

Correct Answer:
50

Solution:
$\%$ of $P=\frac{{}^{n}\mathrm{Mg_2P_2O_7}\times 2\times 31}{W_{\text{(unknown compound)}}}\times 100=\frac{\left(\frac{1.79}{222}\right)\times 2\times 31}{1}\times 100=50\%$

Hence, the correct answer is the option (1).

Q. 19



Consider the above sequence of reactions. The number of bromine atom(s) in final product (P) will be?

Option 1:
5

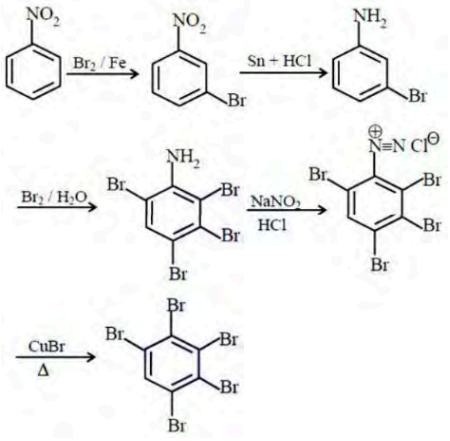
Option 2:
6

Option 3:
3

Option 4:
1

Correct Answer:
5

Solution:



Number of Br atom in major product (P) =5

Hence, the correct answer is the option (1).

Q. 20 Given below are four compounds:

a) n-propyl chloride
b) iso-propyl chloride
c) sec-butyl chloride
d) neo - pentyl chloride

Percentage of carbon in the one which exhibits optical isomerism is :

Option 1:
46

Option 2:
40

Option 3:
52

Option 4:
56

Correct Answer:
52

Solution:
Among the given compounds, only **sec-butyl chloride (2-chlorobutane)** shows optical isomerism because it has one chiral carbon.

Molecular formula $= \mathrm{C_4H_9Cl}$

Molar mass $= (4 \times 12) + (9 \times 1) + 35.5 = 92.5$
Mass of carbon $= 4 \times 12 = 48$

$\mathrm{C_4H_9Cl} = \frac{48}{92.5} \times 100 = 51.89$

Hence, the correct answer is the option (3).

Q. 21 The osmotic pressure of a living cell is 12 atm at 300 K . The strength of sodium chloride solution that is isotonic with the living cell at this temperature is $_gL^{-1}$.
(Nearest integer)
Given: $R = 0.08L\ \mathrm{atm}K^{-1}\ \mathrm{mol}^{-1}$
Assume complete dissociation of NaCl
(Given: Molar mass of $\mathrm{Na\&Cl}$ are $23\&35.5\ \mathrm{g\ mol}^{-1}$ respectively.)

Correct Answer:
15

Solution:
$\pi = iCRT$
$12 = 2 \times C \times 0.08 \times 300$
$C = \frac{1}{4}\ \mathrm{mole/L}$
$\mathrm{Conc.} = \frac{1}{4} \times 58.5\ \mathrm{g/L} = 14.625\ \mathrm{g/L} = 14.625 \times 10^{-2}\ \mathrm{g/L}$

Hence, the answer is (15).

Q. 22 The first and second ionization constants of $\mathrm{H_2X}$ are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} in $0.1MH_2X$ solution is $____ \times 10^{-15}M$. (nearest integer)

Correct Answer:
100

Solution:
$H_2X = \quad H^+ +$
$0.1 - x \quad x + y \quad x - y$

$2.5 \times 10^{-8} = \quad (x+y)(x-y)$
$\qquad\qquad\quad 0.1 - x$
$HX^- \qquad H^+ + X^2$
$x - y \qquad x + y$
$1 \times 10^{-13} = \quad \frac{(x+y)(y)}{x-y}$

Approximate: $K_{a_1} \gg K_{a_2} \Rightarrow$ So $x \gg y$

$x + y \approx x, x - y \approx x$
$10^{-13} = \frac{x \cdot y}{x} \qquad y = 10^{-13}$
$\left[X^{2-}\right] = 10^{-13}$
$\left[X^{2-}\right] = 100 \times 10^{-15}$

Hence, the answer is (100).

Q. 23 A substance "X" (1.5g) dissolved in 150 g of a solvent "Y" (molar mass = $300\ \mathrm{g\ mol}^{-1}$) led to an elevation of the boiling point by 0.5 K . The relative lowering in the vapour pressure of the solvent " Y " is $____ \times 10^{-2}$. (nearest integer) [Given: K_b of the solvent $= 5.0K\ \mathrm{kg\ mol}^{-1}$]
Assume the solution to be dilute and no association or dissociation of X takes place in solution.

Correct Answer:
3

Solution:
: $\Delta T_b = ik_b m$ $\qquad 0.5 = 5 \times im$

$\frac{P_0 - P_s}{P_0} = \frac{im \times M_{\text{solvent}}}{1000} = 0.1 \times \frac{300}{1000} = 0.03 = 3 \times 10^{-2}$

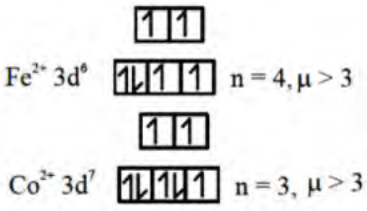
Hence, the answer is (3).

Q. 24 Identify the metal ions among $\mathrm{Co}^{2+}, \mathrm{Ni}^{2+}, \mathrm{Fe}^{2+}, \mathrm{V}^{3+}$ and Ti^{2+} having a spin - only magnetic moment value more than 3.0 BM . The sum of unpaired electrons present in the high spin octahedral complexes formed by those metal ions is :

Correct Answer:
7

Solution:
$V^{3+} 3d^3, \mathrm{Ti}^{2+} 3d^2, \mathrm{Ni}^{2+} 3d^8, \mathrm{Fe}^{2+} 3d^6, \mathrm{Co}^{2+} 3d^7$

Only Fe^{2+} and Co^{2+} can form high spin octahedral complex .



$\therefore$ Number of unpaired electrons $= 4 + 3 + 7$

Hence, the answer is (7).

Q. 25 MX is a sparingly soluble salt that follows the given solubility equilibrium at 298 K .

$MX(s) \rightleftharpoons M^+(aq) + X^-(aq); K_{sp} = 10^{-10}$

If the standard reduction potential for $W = -\int_{\eta}^{r_2} F.dr$. is $\left(E^{\Theta}_{M^+/M}\right) = 0.79V$, then the value of the standard reduction potential for the metal / metal insoluble salt electrode $E^{\Theta}_{X^-/MX(s)/M}$ is ____ mV. (nearest integer)
(Given: $\frac{2.303RT}{F} = 0.059$ V)

Correct Answer:
200

Solution:
$E^0_{CT/AgC/Ag} = E^0_{Ag^+/Ag} - \frac{0.059}{1}\log\frac{1}{K_{sp}} = 0.79 - 0.059 \times \log 10^{10} = 0.79 - 0.59 = 0.2$ volt $= 200$ millivolt.

Hence, the answer is (200).

Maths

Q. 1 Let the line L_1 be parallel to the vector $-3\hat{i} + 2\hat{j} + 4\hat{k}$ and pass through the point $(2, 6, 7)$, and the line L_2 be parallel to the vector $2\hat{i} + \hat{j} + 3\hat{k}$ and pass through the point $(4, 3, 5)$. If the line L_3 is parallel to the vector $-3\hat{i} + 5\hat{j} + 16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $|CD|^2$ is equal to:

Option 1:
89

Option 2:
312

Option 3:
171

Option 4:
290

Correct Answer:
290

Solution:

$L_1 : \frac{x-2}{-3} = \frac{y-6}{2} = \frac{z-7}{4} = t$

$L_2 : \frac{x-4}{2} = \frac{y-3}{1} = \frac{z-5}{3} = s$

Any point on L_1 is $C = (-3t+2, 2t+6, 4t+7)$
Any point on L_2 is $D = (2s+4, s+3, 3s+5)$
D.R.s of $CD = (2s+3t+2, s-2t-3, 3s-4t-2)$

CD is parallel to $-3i+5j+16k$

$\frac{2s+3t+2}{-3} = \frac{s-2t-3}{5} = \frac{3s-4t-2}{16}$

$13s+9t+1=0 \ldots\ldots (1)$
$41s+36t+26=0 \ldots (2)$

Solve (1) and (2) $s=2, t=-3$

$C = (11, 0, -5)$
$D = (8, 5, 11)$

$CD^2 = 290$

Hence, the correct answer is option (4).

Q. 2 If the area of the region $\{(x,y) : 1-2x \le y \le 4-x^2, x \ge 0, y \ge 0\}$ is $\frac{\alpha}{\beta}, \alpha, \beta \in N, \gcd(\alpha,\beta)=1$, then the value of $(\alpha+\beta)$ is :

Option 1:
73

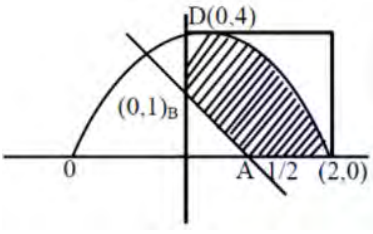
Option 2:
67

Option 3:
91

Option 4:
85

Correct Answer:
67

Solution:



Upper curve: $y = 4-x^2$
Lower curve: $y = 1-2x$

Since $y \ge 0$, the line cuts the x-axis at

$1-2x=0 \Rightarrow x=\frac{1}{2}$

Parabola cuts x-axis at

$4-x^2=0 \Rightarrow x=2$

So required area:

$A=\int_0^2(4-x^2)dx-\int_0^{1/2}(1-2x)dx$

Now compute:

$\int_0^2(4-x^2)dx=\left[4x-\frac{x^3}{3}\right]_0^2=8-\frac{8}{3}=\frac{16}{3}$

$\int_0^{1/2}(1-2x)dx=\left[x-x^2\right]_0^{1/2}=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}$

So, $A=\frac{16}{3}-\frac{1}{4}=\frac{55}{12}$

Thus, $\alpha=55, \beta=12$

$\alpha+\beta=67$

Hence, the correct answer is option (2).

Q. 3 A random variable X takes values $0,1,2,3$ with probabilities $\frac{2a+1}{30}, \frac{8a-1}{30}, \frac{4a+1}{30}$, b respectively, where $a, b \in R$. Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2+\mu^2=2$. Then $\frac{a}{b}$ is equal to:

Option 1:
3

Option 2:
30

Option 3:
12

Option 4:
60

Correct Answer:
60

Solution:

x	0	1	2	3
$p(n)$	$\frac{2a+1}{30}$	$\frac{8a-1}{30}$	$\frac{4a+1}{30}$	b

$\mu=\Sigma \times p(n)=0+\frac{8a-1}{30}+2\times\frac{4a+1}{30}+3b=\frac{16a+90b+1}{30}$

$\sigma^2=\Sigma x^2p(n)-\mu^2 \Rightarrow \sigma^2+\mu^2=\Sigma x^2p(n)=2$

$\frac{8a-1}{30}+\frac{4(4a+1)}{30}+9b=2$

$8a+90b-19=0 \ldots\ldots\ldots (i)$

$\Sigma p(n)=1 \Rightarrow 14a+1+30b=30$

$14a+30b-29=0 \ldots\ldots (ii)$

From (i) and (ii)

$a=2 \quad b=\frac{1}{30} \quad \frac{a}{b}=60$

Hence, the correct answer is option (4).

Q. 4 The positive integer n , for which the solutions of the equation $x(x+2)+(x+2)(x+4)+\ldots+(x+2n-2)(x+2n)=\frac{8n}{3}$ are two consecutive even integers, is:

Option 1:
3

Option 2:
9

Option 3:
6

Option 4:
12

Correct Answer:
3

Solution:

$$nx^2+(2x+6x+10x+\ldots\ldots+(4n-2))+(2.4+4.6+\ldots+(2n-2)2n)=\frac{8n}{3}$$

$$nx^2+2x(1+3+5+\ldots.+2n-1)+\sum_{r=1}^{n}(2r-2)2r=\frac{8n}{3}$$

$$nx^2+2n^2x+4\left(\frac{n(n+1)(2n-2)}{6}\right)=\frac{8n}{3}$$

$$3x^2+6nx+4\left(n^2-1\right)=8$$

For $\mathrm{n}=3$

$$3x^2+18x+24=0$$

$x=-2,-4$ (two consecutive even integers)

Hence, the correct answer is option (1).

Q. 5 Let $f(x)=x^3+x^2f'(1)+2xf''(2)+f'''(3), x\in R$. Then the value of $f'(5)$ is :

Option 1:
$\frac{2}{5}$

Option 2:
$\frac{117}{5}$

Option 3:
$\frac{657}{5}$

Option 4:
$\frac{62}{5}$

Correct Answer:
$\frac{117}{5}$

Solution:
Let $f'(1)=a$

$f'(2) = b$
$f'(3) = c$
$f(x) = x^3 + ax^2 + bx + c$
$f'(x) = 3x^2 + 2ax + b$
$f'(x) = a = 3 + 2a + b \Rightarrow a + b = 3 \ldots\ldots (1)$
$f'(x) = 6x + 2a$
$f''(2) = 12 + 2a = \frac{b}{2} \Rightarrow 4a - b = -24$
$f''(x) = 6$
$f''(3) = c = 6$
$\Rightarrow a = \frac{-27}{5}, b = \frac{12}{5}$
$f'(5) = 75 + 10a + b$
$= 75 - 54 + \frac{12}{5} = 21 + \frac{12}{5} = \frac{117}{5}$

Hence, the correct answer is option (2).

Q. 6 Let $y = y(x)$ be the solution of the differential equation $\sec x \frac{dy}{dx} - 2y = 2 + 3\sin x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

Option 1:
$-3\sqrt{3} - 7$

Option 2:
$-\frac{5}{2}$

Option 3:
$-3\sqrt{2} - 7$

Option 4:
$-\frac{5}{4}$

Correct Answer:
$-\frac{5}{2}$

Solution:
$\frac{dy}{dx} - 2\cos x \cdot y = \cos x(2 + 3\sin x)$
$I.F = e^{-2\sin x}$
General Solution: $y \cdot e^{-2\sin x} = \int e^{-2\sin x} \cos x(2 + 3\sin x)dx$
Put $\begin{matrix}\sin x = t \\ \cos x \cdot dx = dt\end{matrix}$
$y = e^{-2\sin x}\left(\frac{2 + 3\sin x}{-2} - \frac{3}{4}\right) + c$
When $x = 0, c = 0$
When $x = \frac{\pi}{6}, y = -\frac{5}{2}$

Hence, the correct answer is option (2).

Q. 7 If the line $\alpha x + 4y = \sqrt{7}$, where $\alpha \in R$, touches the ellipse $3x^2 + 4y^2 = 1$ at the point P in the first quadrant, then one of the focal distances of P is :

Option 1:
$\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{5}}$

Option 2:
$\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{7}}$

Option 3:
$\frac{1}{\sqrt{3}}-\frac{1}{2\sqrt{11}}$

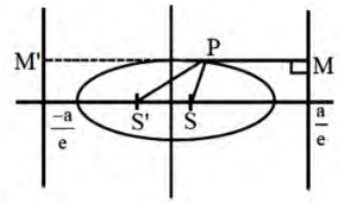
Option 4:
$\frac{1}{\sqrt{3}}-\frac{1}{2\sqrt{5}}$

Correct Answer:
$\frac{1}{\sqrt{3}}+\frac{1}{2\sqrt{7}}$

Solution:
$\alpha x+4y-\sqrt{7}=0$ touches $3x^2+4y^2=1$
$\therefore c^2=a^2m^2+b^2$
$\frac{7}{16}=\frac{1}{3}\times\frac{\alpha^2}{16}+\frac{1}{4}\Rightarrow\alpha=3,-3$

Tangent is $3x+47-\sqrt{7}=0$
Let the point of contact is $P(x_1,y_1)$

$\therefore\frac{3x_1}{3}=\frac{4y_1}{4}=\frac{1}{\sqrt{7}}$
$\therefore P\left(\frac{1}{\sqrt{7}},\frac{1}{\sqrt{7}}\right)$
$\text{eccentricity }=\sqrt{1-\frac{3}{4}}=\frac{1}{2}$



$PS=e(PM)=e\left(\frac{a}{e}-\frac{1}{\sqrt{7}}\right)=\frac{1}{2}\left(\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{7}}\right)=\frac{1}{\sqrt{3}}-\frac{1}{2\sqrt{7}}$
$PS=e\left(PM'\right)=\frac{1}{2}\left(\frac{a}{e}+\frac{1}{\sqrt{7}}\right)=\frac{1}{2}\left(\frac{1}{\sqrt{7}}+\frac{2}{\sqrt{3}}\right)=\frac{1}{\sqrt{3}}+\frac{1}{2\sqrt{7}}$

Hence, the correct answer is option (2).

Q. 8 For the matrices $A=\begin{bmatrix}3&-4\\1&-1\end{bmatrix}$ and $B=\begin{bmatrix}-29&49\\-13&18\end{bmatrix}$, if $\left(A^{15}+B\right)\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$, then among the following which one is true ?

Option 1:
$x=16,y=3$

Option 2:
$x=11,y=2$

Option 3:
$x=18,y=11$

Option 4:
$x=5,y=7$

Correct Answer:
$x=11,y=2$

Solution:
$A=\begin{bmatrix}3&-4\\1&-1\end{bmatrix},B=\begin{bmatrix}23&49\\45&21\end{bmatrix}$

Characteristic equation of A ,

$(3-\lambda)(-1-\lambda)+4=0$
$(\lambda-3)(\lambda+1)+4=\lambda^2-2\lambda+=0$
$\Rightarrow A^2-2A+I=0$
$A^2=2A-I, A^3=2A^2-A=2(2A-I)-A=3A-2I$
$A^4=4A^2+I-4A=4(2A-I)-4A+I$
$=4(2A-I)-3A=5A-4I$
$\left(A^5\right)^3=(5A-4I)^3$
$=125A^3-3\times 25A^2(4)+3(5A)\left(4^2\right)-64I$
$=125(3A-2I)-300(2A-I)+240A-64I$
$=A(375-600+240)+I(-250+300-64)$
$=15A-14I$
$\Rightarrow A^{15}+B=15A-14I+B$
$=\begin{bmatrix}2 & -11\\ 1 & -11\end{bmatrix}\quad \left(A^{15}+B\right)\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}\Rightarrow 2x=11y$

Hence, the correct answer is option (2).

Q. 9 If the system of equations

$3x+y+4z=3$
$2x+\alpha y-z=-3$
$x+2y+z=4$

has no solution, then the value of α is equal to :

Option 1:
23

Option 2:
4

Option 3:
13

Option 4:
19

Correct Answer:
19

Solution:
$\Delta=\begin{vmatrix}3 & 1 & 4\\ 2 & \alpha & -1\\ 1 & 2 & 1\end{vmatrix}$

$\det(\Delta)=0$
$\alpha=19$

Hence, the correct answer is option (4).

Q. 10 For a triangle ABC, let $\vec{p}=\vec{BC},\ \vec{q}=\vec{CA}$ and $\vec{r}=\vec{BA}$. If $|\vec{p}|=2\sqrt{3},\ |\vec{q}|=2$ and $\cos\theta=\frac{1}{\sqrt{3}}$ where θ is the angle between $\vec{p}$ and $\vec{q}$, then $|\vec{p}\times(\vec{q}-3\vec{r})|^2+3|\vec{r}|^2$ is equal to :

Option 1:
340

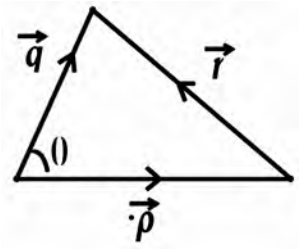
Option 2:
410

Option 3:
220

Option 4:
200

Correct Answer:
200

Solution:



$\Rightarrow r = q + p$

$|r| = \sqrt{q^2 + p^2 + 2pq\cos\theta}$

$|r|^2 = 4 + 12 + 2 \times \sqrt{3} \times 2 \times \frac{1}{\sqrt{3}}$

$|r|^2 = 24$

$\left|p \times \left(q - 3(q-p)^2\right)\right|^2 + 3|r|^2 = |-2p \times q|^2 + 3|r|^2$

$= 4 \times |p||q|^2 \sin^2\theta + 3 \times 24 = 4 \times 12 \times 4 \times \left(\frac{2}{3}\right) + 72 = 128 + 72 = 200$

Hence, the correct answer is option (4).

Q. 11 Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \ldots, \frac{a_{10}}{2^9}$ be a G.P . of common ratio $\frac{1}{\sqrt{2}}$. If $a_1 + a_2 + \ldots + a_{10} = 62$, then a_1 is equal to:

Option 1:
$2 - \sqrt{2}$

Option 2:
$2(2 - \sqrt{2})$

Option 3:
$\sqrt{2} - 1$

Option 4:
$2(\sqrt{2} - 1)$

Correct Answer:
$2(\sqrt{2} - 1)$

Solution:

$\frac{a_2}{2a_1} = \frac{a_3}{2a_2} = \frac{a_4}{2a_3} = \ldots = \frac{a_{10}}{2a_9} = \frac{1}{\sqrt{2}}$

$\therefore a_1, a_2, a_2, \ldots a_{10}$ are in G.P. With common ratio $\sqrt{2}$

$\therefore \sum_{i=1}^{10} a_i = \frac{a_1\left((\sqrt{2})^{10} - 1\right)}{\sqrt{2} - 1} = 62 \Leftrightarrow a_{1i} = 2(\sqrt{2} - 1)$

Hence, the correct answer is option (4).

Q. 12 Let the line L pass through the point $(-3, 5, 2)$ and make equal angles with the positive coordinate axes. If the distance of L from the point $(-2, r, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

Option 1:
6

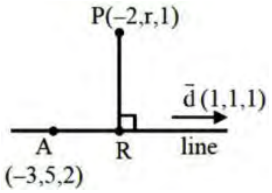
Option 2:
10

Option 3:
12

Option 4:
16

Correct Answer:
10

Solution:
Equation line is : $\frac{x+3}{1}=\frac{y-5}{1}=\frac{z-2}{1}=\lambda$
$\therefore$ General point R on line is $r(\lambda-3,\lambda+5,\lambda+2)$



$PR=(\lambda-1,\lambda+5-r,\lambda+1)$

Now $PR=.d=0$

$\Rightarrow(\lambda-1)1+(\lambda+5-r)1+(\lambda+1)1=0$
$\Rightarrow 3\lambda+5=0$
$\Rightarrow\lambda=\frac{r-5}{3}$
$\therefore R\equiv\left(\frac{r-5}{3}-3,\frac{r-5}{3}+5,\frac{r-5}{3}+2\right)$
$\therefore R\equiv\left(\frac{r-14}{3},\frac{r+10}{3},\frac{r+1}{3}\right)$

Now $PR=\sqrt{\frac{14}{3}}\Rightarrow(PR)^2=\frac{14}{3}$

$\Rightarrow\left(\frac{r-14}{3}+2\right)+\left(\frac{r+10}{3}-r\right)^2+\left(\frac{r+1}{3}-1\right)^2=\frac{14}{3}$
$\Rightarrow\frac{(r-8)^2}{9}+\frac{(10-2r)^2}{9}+\frac{(r-2)^2}{9}=\frac{14}{3}$
$\Rightarrow\left(r^2-16r+64\right)+\left(100+4r^2-40r\right)+\left(r^2-4r+4\right)=42$
$\Rightarrow 6r^2-60r+126=0$
$\Rightarrow r^2-10r+21=0$
$\Rightarrow r=3,7$

Sum of possible value of r is $=10$

Hence, the correct answer is option 2).

Q. 13 Let α and β be the roots of the equation $x^2+2ax+(3a+10)=0$ such that $\alpha<1<\beta$. Then the set of all possible values of a is :

Option 1:
$(-\infty,-2)\cup(5,\infty)$

Option 2:
$(-\infty,-3)$

Option 3:
$\left(-\infty, \frac{-11}{5}\right)$

Option 4:
$\left(-\infty, \frac{-11}{5}\right) \cup (5, \infty)$

Correct Answer:
$\left(-\infty, \frac{-11}{5}\right)$

Solution:
$\alpha < 1 < \beta$
$f(1) < 0$

$\Rightarrow 1 + 2a + (3a + 10) < 0$
$\Rightarrow 5a + 11 < 0$
$a < \dfrac{-11}{5}$
$\therefore a \in \left(-\infty, \dfrac{-11}{5}\right)$

Hence, the correct answer is option 3).

Q. 14 The largest $n \in N$, for which 7^n divides 101! is:

Option 1:
18

Option 2:
19

Option 3:
16

Option 4:
15

Correct Answer:
16

Solution:
Exponent of 7 in $101! = \left[\frac{101}{7}\right] + \left[\frac{101}{7^2}\right] + \ldots..$ (Where [.] is G.I.F.) $= 14 + 12 = 16$

Hence, the correct answer is option (3).

Q. 15 Let $y^2 = 12x$ be the parabola with its vertex at O . Let P be point on the parabola and A be a point on the x -axis such that $\angle OPA = 90^\circ$. Then the locus of the centroid of such triangles OPA is:

Option 1:
$y^2 - 9x + 6 = 0$

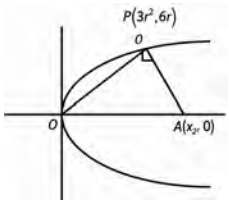
Option 2:
$y^2 - 6x + 4 = 0$

Option 3:
$y^2 - 4x + 8 = 0$

Option 4:
$y^2 - 2x + 8 = 0$

Correct Answer:
$y^2 - 2x + 8 = 0$

Solution:



$m_{op}m_{PA} = -1\frac{6t}{3t^2} \times \frac{6t}{3t^2 - x} = -1$

$12 = -3t^2 + x$

$x = 12 + 3t^2$

$h = \frac{x + 3t^2 + 0}{3}; k = \frac{0 + 0 + 6t}{3}$

$3h = 12 + 6th2; k = 2t$

$h = 4 + 2t^2; k = 2t$

Substitute t& replace h by x&k by y

$y^2 - 2x + 8 = 0$

Hence, the correct answer is option (4).

Q. 16 Let $A = \{2, 3, 5, 7, 9\}$. Let R be the relation on A defined by xRy if and only if $2x \leq 3y$. Let l be the number of elements in R , and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l + m$ is equal to:

Option 1:
23

Option 2:
21

Option 3:
25

Option 4:
27

Correct Answer:
25

Solution:
For $x = 2$

$4 \leq 3y \quad y \in \{2, 3, 5, 7, 9\}$
$(2, 2)(2, 3)(2, 5)(2, 7)(2, 5)$

For $x = 3$

$6 \leq 3yy \in \{2, 3, 5, 7, 9\}$
$(3, 2)(3, 3)(3, 5)(3, 7)(3, 5)$

For $x = 5$

$10 \leq 3yy \in \{5, 7, 9\}$
$(5, 5)(5, 7)(5, 5)$

For $x = 7$

$14 \leq 3yy \in \{5, 7, 9\}$
$(7, 5)(7, 7)(7, 9)$

For $x = 9$

$18 \leq 3yy \in \{7, 9\}$
$(9, 7)(9, 9)$

$R = 5 + 5 + 3 + 3 + 2 = 18$
$(2,3) \in R$ but $(3,2) \in R$(symmetric)
$(2,5) \in R$ but $(5,2) \notin R$ Add$(5,2)$
$(2,7) \in R$ but $(7,2) \notin R$ Add$(7,2)$
$(2,9) \in R$ but $(9,2) \notin R$ Add$(9,2)$
$(3,5) \in R$ but $(5,3) \notin R$ Add$(5,3)$
$(3,7) \in R$ but $(7,3) \notin R$ Add$(7,3)$
$(3,9) \in R$ but $(9,3) \notin R$ Add$(9,3)$
$(5,7) \in R$ but $(7,5) \in R$
$(5,9) \in R$ but $(9,5) \notin R$ Add$(9,5)$
$(7,9) \in R$ but $(9,7) \in R$
$18 + 7 = 25$

Hence, the correct answer is option 3).

Q. 17 Let one end of a focal chord of the parabola $y^2 = 16x$ be $(16,16)$. If $P(\alpha,\beta)$ divides this focal chord internally in the ratio $5:2$, then the minimum value of $\alpha+\beta$ is equal to:

Option 1:
7

Option 2:
5

Option 3:
22

Option 4:
16

Correct Answer:
7

Solution:
If $P(\alpha,\beta)$ divides focal chord internally in $5:2$
If $A(t_1)\&B(t_2)$ are ends of focal chord $A(16,16)$

$t_1t_2 = -1$
$2t_2 = -1$
$t_2 = -\frac{1}{2}$
$B = (1,-4)$

$P(\alpha,\beta) = \left(\frac{37}{7},\frac{12}{7}\right)$ or $\left(\frac{82}{7},\frac{72}{7}\right)$

Minimum of $\alpha+\beta = \frac{37}{7}+\frac{12}{7} = 7$

Hence, the correct answer is option (1).

Q. 18 Let $f: R \to R$ be a twice differentiable function such that $f''(x) > 0$ for all $x \in R$ and $f'(a-1) = 0$, where a is a real number.
Let $g(x) = f\left(\tan^2 x - 2\tan x + a\right), 0 < x < \frac{\pi}{2}$.
Consider the following two statements :

(I) g is increasing in $\left(0, \frac{\pi}{4}\right)$
(II) g is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Then,

Option 1:
Only (I) is True

Option 2:
Neither (I) nor (II) is True

Option 3:
Only (II) is True

Option 4:
Both (I) and (II) are True

Correct Answer:
Neither (I) nor (II) is True

Solution:
$g(x) = f\left((\tan x - 1)^2 + a - 1\right)$
$g'(x) = f'\left((\tan x - 1)^2 + a - 1\right) \cdot 2(\tan x - 1) \cdot \sec^2 x.$

Given $f''(x) > 0 \quad \forall x \in R$
$\therefore f'(x)$ is increasing function
Case (i) $0 < x < \frac{\pi}{4}$

$0 < \tan x > 1$
$-1 < \tan x - 1 < 0$
$0 < (\tan x - 1)^2 < 1$
$a - 1 < (\tan x - 1)^2 + (a - 1) < a$
$\therefore f'\left((\tan x - 1)^2 + a - 1\right) > f'(a - 1) = 0$

From (i) $g'(x) = (+)(-)(+) = -ve$
$g(x)$ is decreasing in $\left(0, \frac{\pi}{4}\right)$
Similarly we can prove that $g(x)$ is increasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Hence, the correct answer is option (2).

Q. 19 Let Z be the complex number satisfying $|z - 5| \leq 3$ and having maximum positive principal argument . Then $34\left|\frac{5z-12}{5iz+16}\right|^2$ is equal to;

Option 1:
16

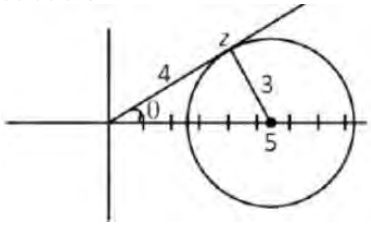
Option 2:
12

Option 3:
20

Option 4:
26

Correct Answer:
20

Solution:



$\Rightarrow \arg(z) = \sin^{-1}\left(\frac{3}{5}\right) \tan^{-1}\left(\frac{3}{4}\right) \Rightarrow z = |z|^{i\theta}$

$= 4 \times (\cos\theta + i\sin\theta) = 4 \times \left(\frac{4}{5} + i\frac{3}{5}\right)$

$= \frac{16}{5} + \frac{12i}{5} \Rightarrow 5z = 16 + 12i$

$\left(\frac{5z-12}{5zi+16}\right) = \frac{(4+12i)}{(4+16i)} = \frac{(1+3i)}{(1+4i)}$

$\Rightarrow \left(\frac{5z-12}{5zi+16}\right) = \frac{\sqrt{10}}{\sqrt{17}}$

$34\left(\frac{5z-12}{5zi+16}\right)^2 = 34 \times \frac{10}{17} = 20$

Hence, the correct answer is option 3).

Q. 20 Let $A = \{x : |x^2 - 10| \leq 6\}$ and $B = \{x : |x-2| > 1\}$. Then

Option 1:
$A \cap B = [-4, -2] \cup [3, 4]$

Option 2:
$A - B = [2, 3)$

Option 3:
$B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$

Option 4:
$A \cup B = (-\infty, 1] \cup (2, \infty)$

Correct Answer:
$B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$

Solution:
$A = x : |x^2 - 10| \leq 6$

$-6 \leq x^2 - 10 \leq 6$

$4 \leq x^2 \leq 16$

$2 \leq |x| \leq 4$

$A = [-4, -2] \cup [2, 4]$

$B = x : |x - 2| > 1$

$x - 2 > 1$ or $x - 2 < -1$

$x > 3$ or $x < 1$

$B = (-\infty, 1) \cup (3, \infty)$

$A \cap B$

$([-4, -2] \cup [2, 4]) \cap ((-\infty, 1) \cup (3, \infty))$

$= [-4, -2] \cup (3, 4]$

So statement 1 is incorrect.

$A - B$

$([-4, -2] \cup [2, 4]) - ((-\infty, 1) \cup (3, \infty))$

$= [2, 3]$

So statement 2 is incorrect.

$B - A$

$((-\infty, 1) \cup (3, \infty)) - ([-4, -2] \cup [2, 4])$

$= (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$

So statement 3 is correct.

$A \cup B$

$([-4, -2] \cup [2, 4]) \cup ((-\infty, 1) \cup (3, \infty))$

$= (-\infty, 1) \cup [2, \infty)$

So statement 4 is incorrect.

Hence, the correct answer is option **3**.

Q. 21 If P is a point on the circle $x^2 + y^2 = 4$, Q is a point on the straight line $5x + y + 2 = 0$ and $x - y + 1 = 0$ is the perpendicular bisector of PQ , then 13 times the sum of abscissa of all such points P is _____

Correct Answer:
2

Solution:
Let $P(x_1, y_1)$ be a point on the circle and $Q(x_2, y_2)$ be a point on the line.

Given circle $x_1^2 + y_1^2 = 4$

Given line for $Q : 5x_2 + y_2 + 2 = 0$

The perpendicular bisector of PQ is $x - y + 1 = 0 \Rightarrow y = x + 1$

Slope of this line is 1.

Since it is perpendicular to PQ,

$$\frac{y_2 - y_1}{x_2 - x_1} = -1$$

So $y_2 - y_1 = -(x_2 - x_1)$

$y_2 = -x_2 + x_1 + y_1$

Substitute in $5x_2 + y_2 + 2 = 0$

$5x_2 + (-x_2 + x_1 + y_1) + 2 = 0$

$4x_2 + x_1 + y_1 + 2 = 0$

$$x_2 = -\frac{x_1 + y_1 + 2}{4}$$

Now $y_2 = -x_2 + x_1 + y_1$

$$y_2 = \frac{x_1 + y_1 + 2}{4} + x_1 + y_1$$

$$y_2 = \frac{5x_1 + 5y_1 + 2}{4}$$

Since the midpoint of PQ lies on $x - y + 1 = 0$

$$\frac{x_1 + x_2}{2} - \frac{y_1 + y_2}{2} + 1 = 0$$

$x_1 + x_2 - y_1 - y_2 + 2 = 0$

Substitute x_2, y_2

$$x_1 - \frac{x_1 + y_1 + 2}{4} - y_1 - \frac{5x_1 + 5y_1 + 2}{4} + 2 = 0$$

Multiplying by 4

$$4x_1 - (x_1 + y_1 + 2) - 4y_1 - (5x_1 + 5y_1 + 2) + 8 = 0$$

$$-2x_1 - 10y_1 + 4 = 0$$

$$x_1 + 5y_1 - 2 = 0$$

$$x_1 = 2 - 5y_1$$

Using circle equation $(2 - 5y_1)^2 + y_1^2 = 4$

$$4 - 20y_1 + 25y_1^2 + y_1^2 = 4$$

$$26y_1^2 - 20y_1 = 0$$

$$2y_1(13y_1 - 10) = 0$$

$y_1 = 0$ or $y_1 = \frac{10}{13}$

Corresponding x_1

$y_1 = 0 \Rightarrow x_1 = 2$

$y_1 = \frac{10}{13} \Rightarrow x_1 = 2 - \frac{50}{13} = -\frac{24}{13}$

Sum of abscissae $2 - \frac{24}{13} = \frac{2}{13}$

Therefore $13 \times \left(\frac{2}{13}\right) = 2$

Hence, the correct answer is 2.

Q. 22 If $\left(\frac{1}{{}^{15}C_0} + \frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1} + \frac{1}{{}^{15}C_2}\right)\ldots\left(\frac{1}{{}^{15}C_{12}} + \frac{1}{{}^{15}C_{13}}\right) = \frac{\alpha^{13}}{{}^{14}C_0\,{}^{14}C_1\ldots{}^{14}C_{12}}$, then 30α is equal to ____

Correct Answer:
32

Solution:

$\frac{1}{{}^nC_r} + \frac{1}{{}^nC_{r+1}} = \frac{{}^nC_{r+1} + {}^nC_r}{{}^nC_r \cdot {}^nC_{r+1}} = \frac{{}^{n+1}C_{r+1}}{{}^nC_r \cdot {}^nC_{r+1}} = \frac{\frac{n+1}{r+1} + {}^nC_r}{{}^nC_r \cdot {}^nC_{r+1}} = \frac{(n+1)}{(r+1)\cdot\frac{n}{r+1}\cdot{}^{n-1}C_r} = \frac{n+1}{n\cdot{}^{n+1}C_r}$

$$\therefore \left(\frac{1}{15C_0} + \frac{1}{15C_1}\right)\left(\frac{1}{15C_1} + \frac{1}{15C_2}\right)\cdots\left(\frac{1}{15C_{12}} + \frac{1}{15C_{13}}\right) = \frac{16}{15\cdot{}^{14}C_0}\cdot\frac{16}{15\cdot{}^{14}C_1}\cdot\frac{16}{15\cdot{}^{14}C_{12}} = \frac{\left(\frac{16}{15}\right)^{13}}{{}^{14}C_0\cdot{}^{14}C_1\cdot{}^{14}C_2\cdot\ldots\cdot{}^{14}C_{12}}$$

$\therefore \alpha = \frac{16}{15}$

$\therefore 30\alpha = 32$

Hence, the correct answer is 32.

Q. 23 Let [ ] denote the greatest integer function and $f(x) = \lim_{n\to\infty}\frac{1}{n^3}\sum_{k=1}^{n}\left[\frac{k^2}{3^x}\right]$. Then $12\sum_{j=1}^{\infty} f(j)$ is equal to

Correct Answer:
2

Solution:

$$f(x)=\lim_{x\to\infty}\left(\frac{1}{n^3}\cdot\sum_{k=1}^{n}\left[\frac{k^2}{3^x}\right]\right)=\lim_{n\to\infty}\frac{1}{n^3}\cdot\sum_{k=1}^{n}\left(\frac{k^2}{3^x}\right)-\lim_{n\to\infty}\frac{1}{n^3}\cdot\sum_{k=1}^{n}\left\{\frac{k^2}{3^x}\right\}$$

$$f(x)=\lim_{n\to\infty}\frac{1}{n^3}\times\frac{n(n+1)(2n+1)}{6\times 3^x}-0\Rightarrow f(x)=\frac{1}{3^{x+1}}$$

$$12\sum_{j=1}^{\infty}f(j)=12\left(\frac{1}{3^2}+\frac{1}{3^3}+\frac{1}{3^4}+\ldots..\right)=12\left(\frac{\frac{1}{9}}{1-\frac{1}{3}}\right)=\frac{12}{9-3}=2$$

Hence, the correct answer is 2.

Q. 24 If $\int_0^1 4\cot^{-1}\left(1-2x+4x^2\right)dx=a\tan^{-1}(2)-b\log_e(5)$, where $a\cdot b\in N$, then $(2a+b)$ is equal to...

Correct Answer:
9

Solution:

$$4\int_0^1 \mathrm{Tan}^{-1}\left(\frac{1}{1-2x+4x^2}\right)dx$$

$$4\int_0^1 \mathrm{Tan}^{-1}\left(\frac{2x-(2x-1)}{1+2x+(2x-1)}\right)dx$$

$$I=4\int_0^1 \mathrm{Tan}^{-1}2x.\,dx-4\int_0^1 \mathrm{Tan}^{-1}(2x-1)dx\ldots\ldots..(i)$$

$$4\int_0^1 \mathrm{Tan}^{-1}2x.\,dx-4\int_0^1 \mathrm{Tan}^{-1}(2(1-x)-1)dx$$

$$I=4\int_0^1 \mathrm{Tan}^{-1}2x.\,dx+4\int_0^1 \mathrm{Tan}^{-1}(2x-1)dx\ldots..(ii)$$

$$\text{(i)}\ +\ \text{(ii)}\ 2I=8\int_0^1 \mathrm{Tan}^{-1}2x.\,dx$$

By integration by parts $I=4\,\mathrm{Tan}^1 2-\log 5$

$\therefore 2a+b=8+1=9$

Hence, the correct answer is 9.

Q. 25 Let the maximum value of $\left(\sin^{-1}x\right)^2+\left(\cos^{-1}x\right)^2$ for $x\in\left[-\frac{\sqrt{3}}{2},\frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n}\pi^2$, where $\gcd(m,n)=1$. Then $m+n$ is equal to ____

Correct Answer:
65

Solution:

$$\frac{-\sqrt{3}}{2}\le x\le\frac{1}{\sqrt{2}}$$

$$-\frac{\pi}{3}\le\sin^{-1}x\le\frac{\pi}{4}$$

$$-\frac{\pi}{3}-\frac{\pi}{4}\le\sin^{-1}x-\frac{\pi}{4}\le 0$$

$$-\frac{7\pi}{12}\le\sin^{-1}x-\frac{\pi}{4}\le 0$$

$$0\le\left(\sin^{-1}x-\frac{\pi}{4}\right)^2\le\frac{49\pi^2}{144}$$

$$\frac{\pi^2}{16}\le\left(\sin^{-1}x-\frac{\pi}{4}\right)^2+\frac{\pi^2}{16}\le\frac{49\pi^2}{144}+\frac{\pi^2}{16}$$

$$\frac{\pi^2}{16}\le\left(\sin^{-1}x-\frac{\pi}{4}\right)^2+\frac{\pi^2}{16}\le\frac{58\pi^2}{144}$$

$$\frac{\pi^2}{8}\le 2\left(\sin^{-1}x-\frac{\pi}{4}\right)^2+\frac{\pi^2}{8}\le\frac{29\pi^2}{36}$$

Hence, the correct answer is 65.

Physics

Q. 1 The volume of an ideal gas increases 8 times and temperature becomes $(1/4)^{th}$ of initial temperature during a reversible change. If there is no exchange of heat in this process $(\Delta Q = 0)$ then identify the gas from the following options (Assuming the gases given in the options are ideal gases):

Option 1:
$\mathrm{NH_3}$

Option 2:
He

Option 3:
$\mathrm{O_2}$

Option 4:
$\mathrm{CO_2}$

Correct Answer:
He

Solution:
It is given that there is no exchange of heat in this process, which means $\Delta Q = 0$, i.e., this is an adiabatic process.
For an ideal gas undergoing a reversible adiabatic process, the relationship between Temperature (T) and Volume (V) is given by the equation:
$\mathrm{T \cdot V^{\gamma-1}} =$ constant
Where γ is the adiabatic index or the ratio of specific heats $\left(\frac{\mathrm{C_p}}{\mathrm{C_v}}\right)$. This γ value is different depending on the atomicity of the gases (monoatomic, diatomic, polyatomic).

$\mathrm{T_1 \cdot V_1^{\gamma-1} = T_2 \cdot V_2^{\gamma-1}}$
Final volume (V_2) is 8 times the initial volume (V_1) $\Rightarrow V_2 = 8V_1$
Final temperature (T_2) is $\left(\frac{1}{4}\right)^{\mathrm{th}}$ the initial temperature $(T_1) \Rightarrow T_2 = \frac{T_1}{4}$

$$T_1 \cdot V_1^{\gamma-1} = \left(\frac{T_1}{4}\right) \cdot (8\,V_1)^{\gamma-1}$$
$$\Rightarrow V_1^{\gamma-1} = \frac{1}{4} \cdot 8^{\gamma-1} \cdot V_1^{\gamma-1}$$
$$\Rightarrow 1 = \frac{1}{4} \cdot 8^{\gamma-1}$$
$$\Rightarrow 4 = 8^{\gamma-1}$$
$$\Rightarrow 2^2 = 2^{3(\gamma-1)}$$
$$\Rightarrow 2 = 3(\gamma-1)2 = 3\gamma - 3$$
$$\Rightarrow 5 = 3\gamma$$
$$\Rightarrow \gamma = \frac{5}{3}$$

The value of γ is directly related to the degrees of freedom (f) of the gas molecules by the formula :

$$\gamma = 1 + \frac{2}{f}$$
$$\Rightarrow \frac{5}{3} = 1 + \frac{2}{f}$$
$$\Rightarrow \frac{5}{3} - 1 = \frac{2}{f}$$
$$\Rightarrow \frac{2}{3} = \frac{2}{f}$$
$$\Rightarrow f = 3$$

A gas with 3 degrees of freedom (all translational, no rotational or vibrational modes activated) is a monoatomic gas.
NH_3 (Ammonia) is a polyatomic molecule (non-linear).
O_2 (Oxygen) is a diatomic molecule.
CO_2 (Carbon Dioxide) is a polyatomic molecule (linear).
He (Helium) is a noble gas that exists as single atoms. It is a monoatomic gas. Therefore, the given gas is a monoatomic gas (He).

Q. 2 A solid sphere of mass 5 kg and radius 10 cm is kept in contact with another solid sphere of mass 10 kg and radius 20 cm . The moment of inertia of this pair of spheres about the tangent passing through the point of contact is _____ kgm^2.

Option 1:
0.18

Option 2:
0.36

Option 3:
0.63

Option 4:
0.72

Correct Answer:
0.63

Solution:

$I = \frac{7}{5}\left[m_1 R_1^2 + m_2 R_2^2\right]$

$= \frac{7}{5}\left[5(10)^2 + 10 \times (20)^2\right] \times 10^{-4}$

$I = 63 \times 10^{-2}\ \text{kg m}^2$

$I = 0.63\ \text{kg m}^2$

Q. 3 A Projectile is thrown upward at an angle 60° with the horizontal. The speed of the projectile is $20\ \text{m/s}$ when its direction of motion is 45° with the horizontal. The initial speed of the projectile is _____ m/s

Option 1:
$20\sqrt{3}$

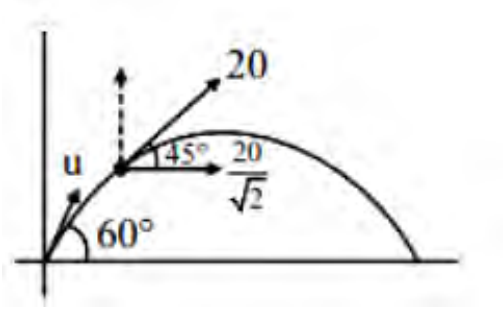
Option 2:
$40\sqrt{2}$

Option 3:
40

Option 4:
$20\sqrt{2}$

Correct Answer:
$20\sqrt{2}$

Solution:



$$u\cos 60^\circ = \frac{20}{\sqrt{2}}$$

$$\frac{u}{2} = \frac{20}{\sqrt{2}}$$

$$u = \frac{40}{\sqrt{2}}$$

$$u = 20\sqrt{2}\ m/s$$

Q. 4 Given below are two statements:

Statement I: Pressure of fluid is exerted only on a solid surface in contact as the fluid- Pressure does not exist everywhere in a still fluid .

Statement II: Excess potential energy of the molecules on the surface of a liquid. When compared to interior, results in surface tension.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Statement I is true but Statement II is false

Option 3:
Statement I is false but Statement II is true

Option 4:
Both Statement I and Statement II are false

Correct Answer:
Statement I is false but Statement II is true

Solution:

According to Pascal's law, pressure at any point in a liquid at rest is the same in all directions. It exists at every point in the liquid, not just at boundaries. So statement (1) is false. For the interior molecule net cohesive forces are zero statement (2) is correct.

Q. 5 Match the LIST-I with LIST-II

List -I	List-II
A Spring constant	I. $ML^2T^{-2}K^{-1}$
B Thermal conductivity	II. $ML^{\circ}T^{-2}$
C Boltzmann constant	III. $ML^2T^{-3}A^{-2}$
D. Inductive reactance	IV. $MLT^{-3}K^{-1}$

Choose the correct answer from the options given below:

Option 1:
A-II,B-IV,C-I,D-III

Option 2:
A-III,B-II,C-IV,D-I

Option 3:
A-II, B-I,C-IV,D-III

Option 4:
A-I,B-IV,C-II,D-III

Correct Answer:
A-II,B-IV,C-I,D-III

Solution:
$(A) F = Kx$
$\left[\mathrm{MLT}^{-2}\right] = [\mathrm{K}][\mathrm{L}]$
$[\mathrm{K}] = \mathrm{ML}^{0}\,\mathrm{T}^{-2}$

(B) Thermal conductivity

$$\frac{dQ}{dt} = \frac{kA}{\ell}\Delta T$$
$$ML^2 T^{-3} = \frac{[k]L^2 K}{L}$$
$$[K^1] = MLT^{-3} K^{-1}$$

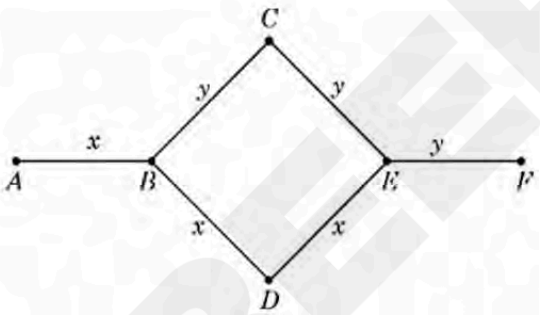
(C) Boltzmann constant

$$[K] = ML^2 T^{-2} K^{-1}$$

(D) Inductive reactance

$$\frac{[V]}{[I]} = \frac{ML^2 T^{-3} A^{-1}}{A}$$
$$= ML^2 T^{-3} A^{-2}$$

Q. 6 Rods x and y of equal dimensions but of different materials are joined as shown in figure, Temperatures of end points A and F are maintained at $100^\circ C$ and $40^\circ C$ respectively. Given the thermal conductivity of $\text{rod}\, x$ is three times of that of $\text{rod}\, y$, the temperature at junction points B and E are (close to):



Option 1:
$80^\circ C$ and $60^\circ C$ respectively

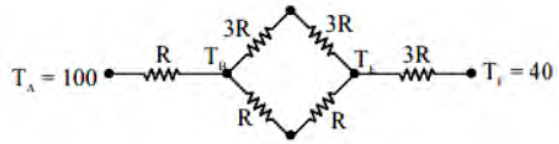
Option 2:
$80^\circ C$ and $70^\circ C$ respectively

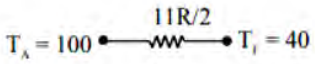
Option 3:
$89^\circ C$ and $73^\circ C$ respectively

Option 4:
$60^\circ C$ and $45^\circ C$ respectively

Correct Answer:
$89^{\circ}\mathrm{C}$ and $73^{\circ}\mathrm{C}$ respectively

Solution:
Let $\left[\mathrm{R}=\frac{\ell}{3\mathrm{KA}}\right]$





$$\left[\mathrm{H}=\frac{100-40}{\frac{11\mathrm{R}}{2}}\right] \ldots (1)$$

$$H=\frac{100-T_B}{R} \ldots\ldots (2)$$

$$H=\frac{100-T_B}{R} \ldots\ldots (3)$$

using (1) and (2)

$$120=1100-11\ \mathrm{T_A}$$

$$\mathrm{T_B}=89^{\circ}\mathrm{C}$$

using (1) and (3)

$$\mathrm{T_E}=73^{\circ}\mathrm{C}$$

Q. 7 $7.9\mathrm{MeV}$ α-particle scatters from a target material of atomic number 79. From the given data the estimated diameter of nuclei of the target material is (approximately) _____ m.
$\left|\frac{1}{4x\epsilon_0}=9\times10^{9}\mathrm{Nm}^2/\mathrm{C}^2\right.$ and electron charge $e=1.6\times10^{-19}\mathrm{C}\ |$

Option 1:
1.69×10^{-12}

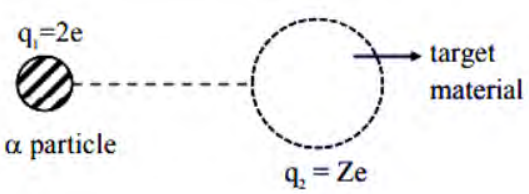
Option 2:
2.88×10^{-14}

Option 3:
5.76×10^{-14}

Option 4:
1.44×10^{-13}

Correct Answer:
2.88×10^{-14}

Solution:
By mechanical energy conservation



$(Me)_i = (Me)_f$

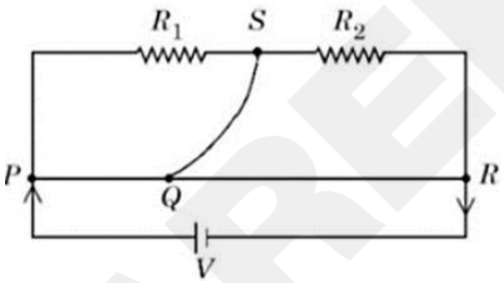
$PE_i + KE_i = PE_f + KE_f$

$$0 + 7.9 \times 10^6 \times 1.6 \times 10^{-19} = \frac{k(2e)(Ze)}{r} + 0$$

$$r = \frac{9 \times 10^9 \times 2 \times \left(1.6 \times 10^{-19}\right)^2 \times 79}{7.9 \times 10^6 \times 1.6 \times 10^{-19}} = 2.88 \times 10^{-14}\ \text{m}$$

$$\text{For diameter } \Rightarrow D = 2r = 5.76 \times 10^{-14}\ \text{m}$$

Q. 8 A meter bridge with two resistance R_1 and R_2 as shown in figure was balanced (null point) At 40 cm from the point P. The null point changed to 50 cm from the point P, when 16Ω resistance is connected in parallel to R_2. The values of resistances R_1 and R_2 are _____



Option 1:
$R_2 = 4\Omega, R_1 = \frac{4}{3}\Omega$

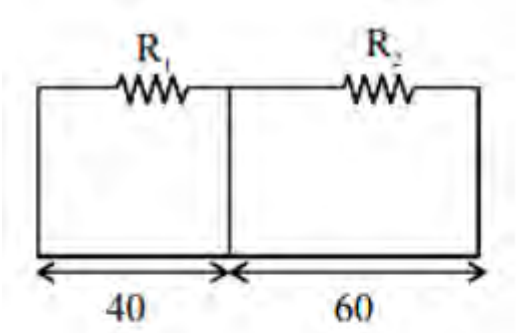
Option 2:
$R_2 = 8\Omega, R_1 = \frac{16}{3}\Omega$

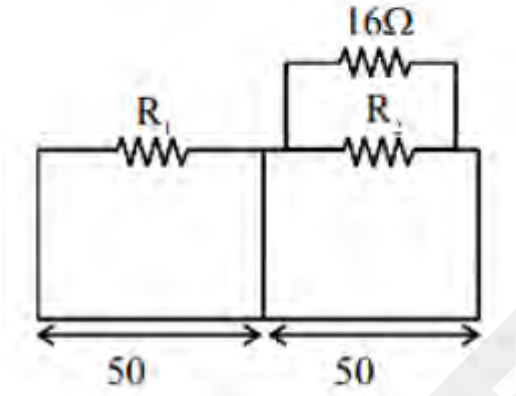
Option 3:
$R_2 = 12\Omega, R_1 = \frac{12}{3}\Omega$

Option 4:
$R_2 = 16\Omega, R_1 = \frac{16}{3}\Omega$Key : 2

Correct Answer:
$R_2 = 8\Omega, R_1 = \frac{16}{3}\Omega$

Solution:





$$\frac{R_1}{R_2} = \frac{40}{60} = \frac{2}{3} \ldots (1)$$

$$\frac{R_1}{\left(\frac{R_2 \times 16}{R_2+16}\right)} = \frac{50}{50} \Rightarrow R_1 = \frac{16R_2}{16+R_2} \ldots\ldots$$

$$\frac{2}{3}R_2 = \frac{16R_2}{16+R_2}$$

$$\frac{32}{3} + \frac{2R_2}{3} = 16$$

$$\frac{2R_2}{3} = 16 - \frac{32}{3} = \frac{16}{3}$$

$R_2 = 8\Omega$

By equation (1)

$R_1 = \frac{2}{3}R_2 = \frac{16}{3}\Omega$

Q. 9 The escape velocity from a spherical planet A is $10\ \mathrm{km/s}$. The escape velocity from another planet B whose density and radius are 10% of those of planet A is _____ $\mathrm{m/s}$.

Option 1:
$1000\sqrt{2}$

Option 2:
$100\sqrt{10}$

Option 3:
$200\sqrt{5}$

Option 4:
1000

Correct Answer:
$100\sqrt{10}$

Solution:

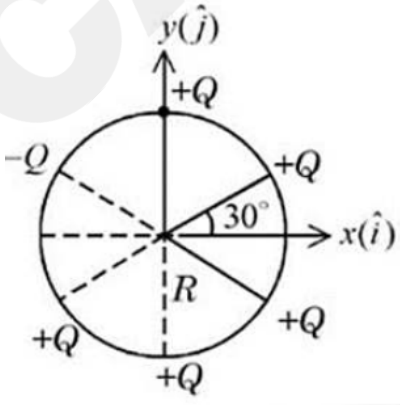
$$V_e=\sqrt{\frac{2GM}{R}}=\sqrt{\frac{2G\times\rho\times\frac{4\pi R^3}{3}}{R}}\Rightarrow V_e\propto\sqrt{\rho}\times R$$

$$\frac{(V_e)_B}{(V_e)_A}=\sqrt{\frac{\rho_B}{\rho_A}}\times\frac{R_B}{R_A}=\sqrt{\frac{0.1\rho_A}{\rho_A}}\times\left(\frac{0.1R_A}{R_A}\right)$$

$$\frac{(V_e)_B}{(V_e)_A}=\frac{1}{10}\times\frac{1}{\sqrt{10}}$$

$$(V_e)_B=\frac{10\times1000}{10\sqrt{10}}=100\sqrt{10}\ m/sec$$

Q. 10 Six point charges are kept 60° apart from each other on the circumference of a circle of radius R as shown in figure. The net electric field at the center of the circle is _____



(ϵ_0 is permittivity of free space)

Option 1:
$-\frac{Q}{4\pi\in_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$

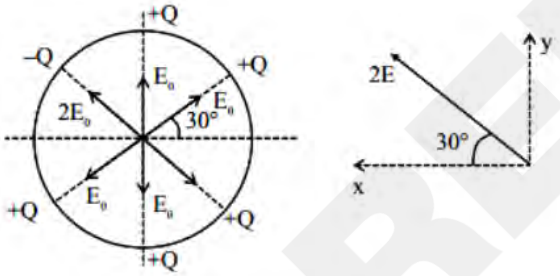
Option 2:
$\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$

Option 3:
$-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i}+\sqrt{3}\hat{j})$

Option 4:
$-\left(\frac{5Q}{8\pi\epsilon_\circ R^2}\right)(\hat{i}-3\hat{j})$

Correct Answer:
$\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$

Solution:
Let $\frac{\mathrm{KQ}}{\mathrm{r}^2}=\mathrm{E}_0$



$\overrightarrow{\mathrm{E}}_{\mathrm{net}}=2\mathrm{E}_0\cos 30^\circ(-\hat{\mathrm{i}})+2\mathrm{E}_0\sin 30^\circ(\hat{\mathrm{j}})$

$=\frac{2\mathrm{kQ}}{\mathrm{r}^2}\left[\frac{\sqrt{3}}{2}(-\hat{\mathrm{i}})+\frac{1}{2}\hat{\mathrm{j}}\right]$

$=\frac{-1Q}{4\pi\varepsilon_0 r^2}(\sqrt{3}\hat{i}-\hat{j})$

Q. 11 Three identical coils C_1, C_2 and C_3 are closely placed such that they share a common axis, C_2 is exactly midway. C_1 carries current I in anti-clockwise direction while C_3 carries current I in clockwise direction . An induced current flows though C_2 will be in clockwise direction when

Option 1:
C_1 moves away from C_2 and C_3 moves towards C_2

Option 2:
C_1 moves towards C_2 and C_3 moves away from C_2

Option 3:
C_1 and C_3 move with equal speeds away from C_2

Option 4:
C_1 and C_3 move with equal speeds towards C_2

Correct Answer:
C_1 moves towards C_2 and C_3 moves away from C_2

Solution:
C_1 carries current in an anti-clockwise direction, so its magnetic field at C_2 is out of the plane.
C_3 carries current in a clockwise direction, so its magnetic field at C_2 is also out of the plane.

Thus, both coils produce magnetic flux in the same direction through C_2.

According to Lenz's law, the induced current opposes the change in magnetic flux.

C_1 moves towards C_2 : magnetic flux through C_2 increases.
C_3 moves away from C_2 : magnetic flux through C_2 decreases.

Net effect: decrease in outward magnetic flux through C_2.

To oppose the decrease in outward flux, the induced current in C_2 must produce outward magnetic field. This corresponds to a clockwise induced current.

Q. 12 The minimum frequency of photon required to break a particle of mas 15.348 amu into 4α particles is _____ kHz
[mass of He nucleus $= 4.002$ amu, 1 amu $= 1.66 \times 10^{-27}$ kg. h $= 6.6 \times 10^{-34}$ J. s and c $= 3 \times 10^{g}$ m/s]

Option 1:
14.94×10^{19}

Option 2:
9×10^{19}

Option 3:
14.94×10^{20}

Option 4:
9×10^{20}

Correct Answer:
14.94×10^{20}

Solution:
Einstein's equation tells us that mass and energy are interchangeable.

$$\mathrm{E} = \mathrm{mc}^2$$

When a nuclear reaction occurs, the total mass of the products is often slightly different from the initial mass of the reactants.

If the products are lighter, the missing mass is released as energy.
If the products are heavier, energy must be supplied to create that extra mass.
It is given that a particle breaks into 4α particles.
Initial mass of the single particle. $\mathrm{M}_{\text{initial}} = 15.348\text{amu}$
Final mass of 4α particles $= \mathrm{M}_{\text{final}} = 4\times$ mass of He nucleus

$$\mathrm{M}_{\text{final}} = 4 \times 4.002\text{amu} = 16.008\text{amu}$$

The mass defect ($\Delta \mathrm{m}$) during this nuclear reaction is,

$$\Delta \mathrm{m} = \mathrm{M}_{\text{final}} - \mathrm{M}_{\text{initial}}$$
$$\Rightarrow \Delta \mathrm{m} = 16.008\text{amu} - 15.348\text{amu}$$
$$\Rightarrow \Delta\ \mathrm{m} = 0.66\text{amu}$$

It is given that $1\text{amu} = 1.66 \times 10^{-27}\ \text{kg}$.

$$\Delta \mathrm{m} = 0.66 \times \left(1.66 \times 10^{-27}\right)\text{kg}$$
$$\Rightarrow \Delta\ \mathrm{m} = 1.0956 \times 10^{-27}\ \text{kg}$$

Now using Einstein's equation,

$$\mathrm{E} = \Delta \mathrm{m} \cdot \mathrm{c}^2$$
$$\Rightarrow E = \left(1.0956 \times 10^{-27}\right) \times \left(3 \times 10^8\right)^2$$
$$\Rightarrow E = \left(1.0956 \times 10^{-27}\right) \times \left(9 \times 10^{16}\right)$$
$$\Rightarrow E = 9.8604 \times 10^{-11} \text{ Joules}$$

This energy is provided by a single photon. The energy of a photon is directly proportional to its frequency (v), given by Planck's equation:

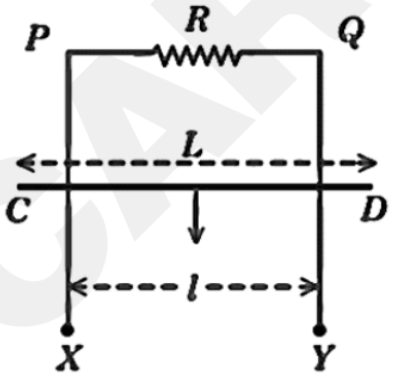
$$\mathrm{E} = \mathrm{h}\nu$$

Where h is Planck's constant $\left(6.6 \times 10^{-34} \text{ J} \cdot \text{ s}\right)$.

$$v = \frac{E}{h}$$
$$\Rightarrow v = \frac{9.8604 \times 10^{-11}}{6.6 \times 10^{-34}}$$
$$\Rightarrow v = 1.494 \times 10^{23} \text{ Hz}$$
$$\Rightarrow v_{\mathrm{kHz}} = 1.494 \times 10^{23} \times 10^{-3} \mathrm{kHz}$$

$$\Rightarrow v_{\mathrm{kHz}} = 14.94 \times 10^{19} \mathrm{kHz}$$

Q. 13 $XPQY$ is a vertical smooth long loop having a total resistance R where PX is parallel to QY and separation between them is l. A constant magnetic field B perpendicular to the plane of the loop exists in the entire space.
A rod CD of length $L(L > l)$ and mass m is made to slide down from rest under the gravity as shown in figure. The terminal speed acquired by the rod is _____ $\mathrm{m/s}$. (g = acceleration due to gravity)



Option 1:
$\frac{8mgR}{B^2l^2}$

Option 2:
$\frac{mgR}{B^2l^2}$

Option 3:
$\frac{2mgR}{B^2l^2}$

Option 4:
$\frac{2mgR}{B^2l^2}$

Correct Answer:
$\frac{mgR}{B^2l^2}$

Solution:
Sol : $Mg = Bil$

Q. 14 A thin convex lens of focal length 5 cm and a thin concave lens of focal length 4 cm are combined together (without any gap) and this combination has magnification m_1 when an object is placed 10 cm before the convex lens. Keeping the positions of convex lens and object undisturbed a gap of 1 cm is introduced between the lenses by moving the concave lens away. Which lead to a change in magnification of total lens system to m_2.The value of $\left|\frac{m_1}{m_2}\right|$ is_____

Option 1:
$\frac{5}{27}$

Option 2:
$\frac{25}{27}$

Option 3:
$\frac{3}{2}$

Option 4:
$\frac{5}{9}$

Correct Answer:
$\frac{5}{27}$

Solution:
Key : $\frac{5}{6}$ not available in options

Sol : $m_2{}' = \frac{f_1}{u_2+f_1}$

$$m_2'' = \frac{f_2}{u_2' + f_2}$$
$$m_2 = m_2' \times m_2''$$
$$u_2 = -10 \text{ cm}, f_1 = +5 \text{ cm}, u_2' = 9 \text{ cm}, f_2 = -4 \text{ cm}$$

Q. 15 Net gravitational force at the center of a square is found to be F_1 when four particles having mass $\mathrm{M}, 2\mathrm{M}, 3\mathrm{M}$ and 4 M are placed at the four corners of the square as shown in figure and it is F_2 when the positions of 3 M and 4 M are interchanged. The ratio $\frac{F_1}{F_2}$ is $\frac{a}{\sqrt{5}}$, The value of a is _____

Option 1:
3

Option 2:
2

Option 3:
1

Option 4:
$\frac{2}{\sqrt{5}}$

Correct Answer:
2

Solution:
Sol :

$$F_2 = \sqrt{10}$$
$$\frac{F_1}{F_2} = \frac{2}{\sqrt{5}}$$

Q. 16 Consider an equilateral prism (refractive index $\sqrt{2}$) A ray of light is incident on its one surface at a certain angle i. If the emergent ray is found to graze along the other surface then the angle of refraction at the incident surface is close to

Option 1:
20°

Option 2:
40°

Option 3:
15°

Option 4:
30°

Correct Answer:
15°

Solution:
Sol: $r_1 + r_2 = A$

Q. 17 A simple pendulum has a bob with mass m and charge q. The pendulum string has negligible mass. When a uniform and horizontal electric field E is applied. The tension the string changes. The final tension in the string when pendulum attains an equilibrium position is _____
(g acceleration due to gravity)

Option 1:
$\sqrt{m^2g^2 + q^2E^2}$

Option 2:
$mg + qE$

Option 3:
$mg - qE$

Option 4:
$\sqrt{m^2g^2 - q^2E^2}$

Correct Answer:
$\sqrt{m^2g^2+q^2E^2}$

Solution:
Sol: $T=\sqrt{q^2E^2+m^2g^2}$

Q. 18 Electric field in a region is given by $E=Ax\hat{i}+By\hat{j}$, where $A=10\ \mathrm{V/m^2}$ and $B=5\ \mathrm{V/m^2}$ If the electric potential at a point $(10,20)$ is 500 V , then the electric potential at origin is _____ V.

Option 1:
500

Option 2:
0

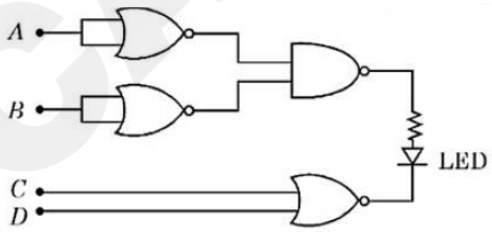
Option 3:
1000

Option 4:
2000

Correct Answer:
500

Solution:
Sol : $V_p-V_0=-\int_0^p E.dl$

Q. 19 Find the correct combination of $\mathrm{A},\mathrm{B},\mathrm{C}$ and D inputs which can cause the LED to glow.



Option 1:
0011

Option 2:
1000

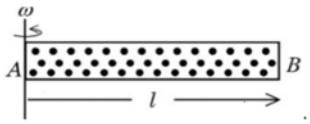
Option 3:
1101

Option 4:
0100

Correct Answer:
1101

Solution:
Conceptual

Q. 20 A cylindrical tube AB of length l, closed at both ends contains an ideal gas of 1 mol having molecular weight M . The tube is rotated in a horizontal plane with constant angular velocity ω about an axis perpendicular AB and passing through the edge at end A as shown in the figure. If P_A and P_B are the pressures at A and B respectively. Then (Consider the temperature is same at all points in the tube)



Option 1:
$P_B = P_A \exp\left(M\omega^2 l^2/2RT\right)$

Option 2:
$P_B = P_A \exp\left(M\omega^2 l^2/RT\right)$

Option 3:
$P_B = P_A$

Option 4:
$P_B = P_A \exp\left(M\omega^2 l^2/3RT\right)$

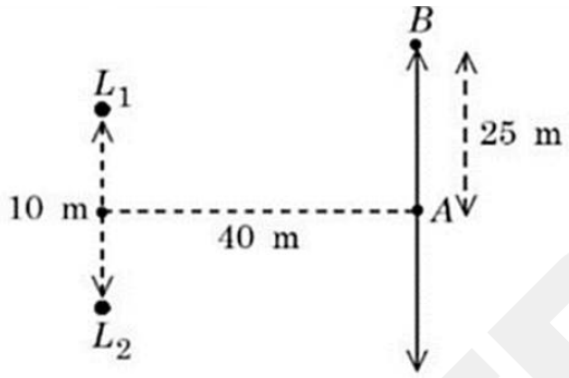
Correct Answer:
$P_B = P_A \exp\left(M\omega^2 l^2/2RT\right)$

Solution:
Sol :

$$\int_{P_A}^{P_B} \frac{dP}{P} = \frac{M}{RT}\omega^2 \int_0^L x dx$$

$$P_B = P_A \exp\left(M\omega^2 l^2/2RT\right)$$

Q. 21 Two loudspeakers $(L_1$ and $L_2)$ are placed with a separation of 10 m . as shown in figure. Both speakers are fed with an audio input signal of same frequency with constant volume A voice recorder, initially at point A at equidistance to both loud speakers is moved by 25 m along the line AB while monitoring the audio signal The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is _____ Hz (Speed of sound in air is $324\ \mathrm{m/s}$ and $\sqrt{5} = 2.23$)



Correct Answer:
348

Solution:
Sol : path difference $= L_2B - L_1B = 10\lambda = 10\frac{\mathrm{C}}{\mathrm{f}}$

Q. 22 A circular disc has radius R_1 and thickness T_1. Another circular disc made of the same material has radius R_2 and thickness T_2 If the moment of inertia of both discs are same and $\frac{R_1}{R_2} = 2$ then $\frac{T_1}{T_2} = \frac{1}{\alpha}$, The value of α is _____

Correct Answer:
16

Solution:
Sol : $I_1 = I_2$

$$\left(\frac{R_2}{R_1}\right)^4 = \frac{T_1}{T_2}$$

Q. 23 Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω. The energy density in the inductor when the current reaches $\left(\frac{1}{e}\right)$ of its maximum value is $\alpha\pi \times \frac{1}{e^2}\ \mathrm{J/m^3}$. The value of α is $\left(\mu_0 = 4\pi \times 10^{-7}\mathrm{Tm/A}\right)$

Correct Answer:
-

Solution:
Sol : If $n = \frac{N}{l} = 10^4$ then energy density $\frac{B^2}{2\mu_0}$

$B = \mu_0 ni$

Q. 24 The electric field of a plane electromagnetic wave, travelling in an unknown nonmagnetic medium is given by . $E_y = 20\sin\left(3 \times 10^6 x - 4.5 \times 10^{14} t\right)\mathrm{V/m}$ (where x, t and other values have S.I. units). The dielectric constant of the medium is _____
(Speed of light in free space is $3 \times 10^8\ \mathrm{m/s}$)

Correct Answer:
4

Solution:
Sol :

$$C = \frac{\omega}{k} = \frac{C_0}{\sqrt{\mu_r \epsilon_r}}$$

$\mu_r = 1$

Q. 25 A parallel beam of light travelling in air (refractive index 1.0) is incident on a convex spherical glass surface of radius of curvature 50 cm . Refractive index of glass is 1.5 . The rays converge to a point at a distance x cm from the centre of the curvature of the spherical surface . The value of x is _____ cm.

Correct Answer:
100

Solution:

Sol : $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

Chemistry

Q. 1 As compared with chlorocyclohexane, which of the following statements correctly apply to chlorobenzene?

A. The magnitude of negative charge is more on chlorine atom

B. The $\mathrm{C} - \mathrm{C}l$ bond has partial double bond character

C. $\mathrm{C} - \mathrm{Cl}$ bond is less polar

D. $\mathrm{C} - \mathrm{C}l$ bond is longer due to repulsion between delocalised electrons of the aromatic ring and lone pairs of electrons of chlorine.

E. The $\mathrm{C} - \mathrm{Cl}$ bond is formed using sp^2 hybridized orbital of carbon

Choose the correct answer from the options given below :

Option 1:
A, D and E Only

Option 2:
B, C and D Only

Option 3:
B, C and E Only

Option 4:
A, C and E Only

Correct Answer:
B, C and E Only

Solution:
Sol : correct statements: **B, C, E**

A: In chlorobenzene, the lone pair on Cl partially overlaps with the π-system of the benzene ring (resonance). This delocalizes some electron density away from Cl, so the negative charge on Cl is less than in chlorocyclohexane. So, A is false.

B: is True. The resonance C–Cl → C–C π-system gives partial double bond character, restricting rotation.

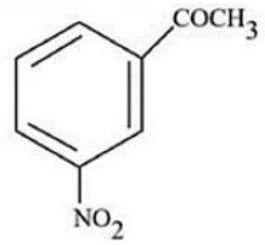
C: In chlorobenzene, carbon is sp^2 hybridized, more electronegative than sp^3 carbon of chlorocyclohexane. Polarity depends on the difference in electronegativity: C(sp^2)–Cl bond is actually slightly less polar than C(sp^3)–Cl. So, C is correct.

D: The C–Cl bond in chlorobenzene is slightly shorter than normal (due to partial double bond character), not longer. So, D is false.

E: In chlorobenzene, carbon is sp^2 hybridized. So the σ-bond uses sp^2 orbital of carbon.
So, E is correct.

Hence, the correct answer is the option (3).

Q. 2 Given below are two statements :
Statement I: Benzene is nitrated to give nitrobenzene, which on further treatment with $CH_3COCl/AlCl_3$ will give



Statement II : $-NO_2$ group in a m-directing, and deactivating group
In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:
Statement I is correct but Statement II is incorrect

Option 2:
Statement I is incorrect but Statement II is correct

Option 3:
Both Statement I and Statement II are incorrect

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Statement I is incorrect but Statement II is correct

Solution:
Statement I: Nitrobenzene is strongly deactivated due to $–NO_2$ (electron-withdrawing). Friedel–Crafts acylation does not occur efficiently on nitrobenzene. So Statement I is actually wrong.

Statement II: $-NO_2$ is meta-directing and deactivating. This is correct.

Hence, the correct answer is the option (2).

Q. 3 Given below are two statements:

Statement I: The Henry's law constant $\mathrm{K_H}$ is constant with respect to variations in solution's concentration over the range for which the solution is ideally dilute.
Statement II: $\mathrm{K_H}$ does not differ for the same solute in different solvents. In the light of the above statements, choose the correct answer from the options given below

Option 1:
Statement I is true and Statement II are false

Option 2:
Statement I is false but Statement II is true

Option 3:
Both Statement I and Statement II are false

Option 4:
Both Statement I and Statement II are true

Correct Answer:
Statement I is true and Statement II are false

Solution:
Sol :

Statement I: This is true.
because Henry's law applies to ideally dilute solutions, and in that range, the solubility of a gas is proportional to its partial pressure, so K_H remains effectively constant.

Statement II: This is false.
K_H depends not only on the solute but also strongly on the nature of the solvent.

Hence, the correct answer is the option (1).

Q. 4 In the reaction

$$2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$$

Option 1:
$67.2\ LH_2(g)$ at STP is produced for every mole of Al that reacts

Option 2:
$12\ LHCl(aq)$ is consumed for every $6\ L_2(g)$ produced.

Option 3:
$33.6\ LH_2(g)$ is produced regardless of temperature and pressure for every mole of Al that reacts

Option 4:
$11.2\ LH_2(g)$ at STP is produced for every mole of HCl consumed

Correct Answer:
$11.2\ LH_2(g)$ at STP is produced for every mole of HCl consumed

Solution:
Sol :

6 mole $HCl \Leftrightarrow 3$ mole H_2
1 mole $HCl \Leftrightarrow ? \quad = 11.2$ lit

Hence, the correct answer is the option (4).

Q. 5 Given below are two statements :
Statement-I : The halogen that makes longest bond with hydrogen in HX, has the smallest covalent radius in its group.
Statement-II : A group 15 elements hybride EH_3 has the lowest boiling point among corresponding hybrids of other group 15 elements. The maximum covalency of that element E is 4
In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

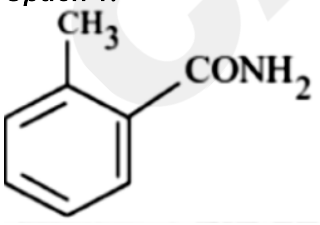
Correct Answer:
Both Statement I and Statement II are false

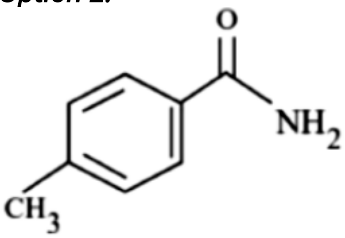
Solution:
Boiling point order

$$\mathrm{PH_3} < \mathrm{AsH_3} < \mathrm{NH_3} < \mathrm{SbH_3}$$
$$\mathrm{P} = \text{ max valency } = 6$$

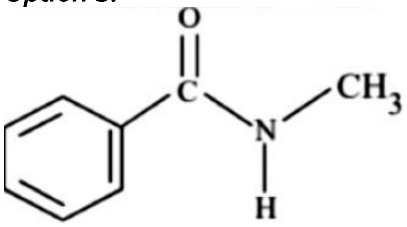
Hence, the correct answer is the option (2).

Q. 6 ' A ' is a neutral organic compound ($M.F: \mathrm{C_8H_9ON}$). On treatment with aqueous $\mathrm{Br_2/HO^{(-)}}$, 'A' forms a compound 'B' which is soluble in dilute acid. 'B' on treatment with aqueous $\mathrm{NaNO_2/HCl}\,(0-5^{\circ}\mathrm{C})$ produces a compound ' C ' which on treatment with $\mathrm{CuCN/NaCN}$ produces 'D'. Hydrolysis of 'D' produces ' E ' which is also obtainable from the hydrolysis of ' A '. ' E ' on treatment with acidified $\mathrm{KMnO_4}$ produces ' F '. ' F ' contains two different types of hydrogen atoms. The structure of ' A ' is

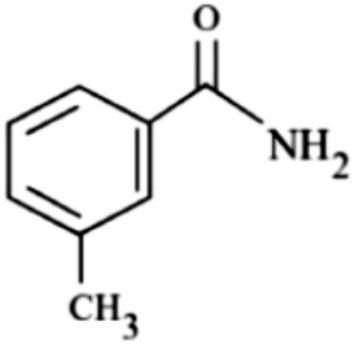
Option 1:


Option 2:


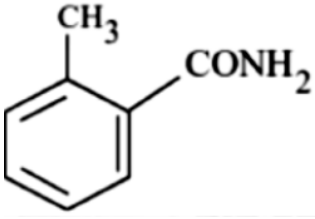
Option 3:



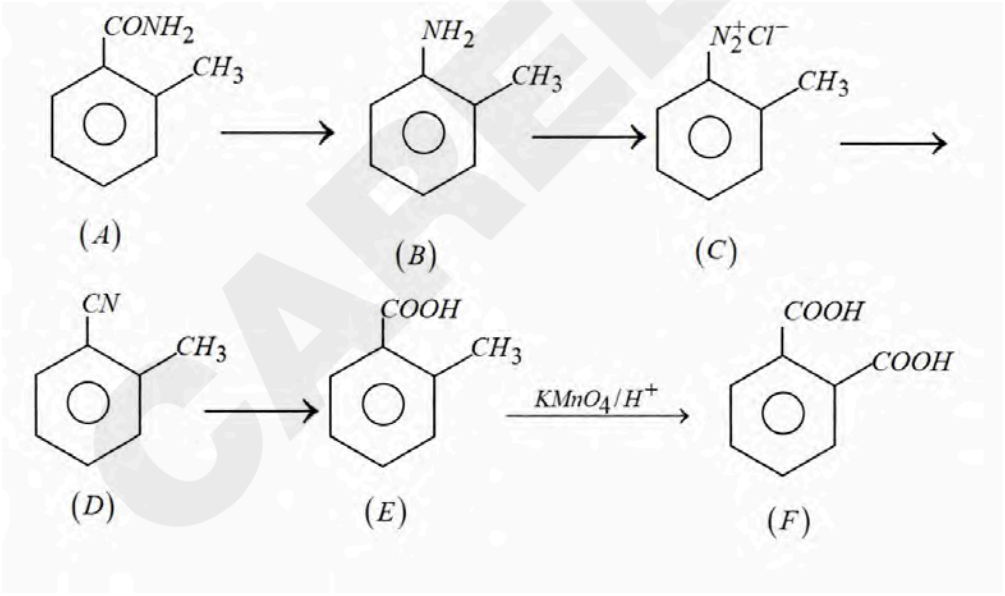
Option 4:



Correct Answer:



Solution:



Hence, the correct answer is the option (2).

Q. 7 Two p - block elements X and Y form fluorides of the type EF_3. The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridizations of the central atoms of XF_3 and YF_3 respectively are

Option 1:
sp^2 and sp^3

Option 2:
Both sp^3

Option 3:
Both sp^2

Option 4:
sp^3 and sp^2

Correct Answer:
sp^2 and sp^3

Solution:
Sol :

Lewis acid XF_3:

- Electron-deficient → no lone pair on central atom.
- Fluorides (F) are highly electronegative → central atom has only 3 bonding pairs.
- Geometry: Trigonal planar
- Hybridization: sp^2

Lewis base YF_3:

- Has a lone pair → can donate electrons → tetrahedral electron domain, 1 lone pair + 3 bonds
- Geometry: Trigonal pyramidal
- Hybridization: sp^3

Hence, the correct answer is the option (1).

Q. 8 Match the List-I with List – II

List-I (Reagents)	List-II (Name of Reaction involving carbonyl compounds)
A. $NH_2 - NH_2, KOH$	I. Tollen's Test
B. $Ag(NH_3)_2OH$	II. Clemmensen Reduction
C. Aq. $CuSO_4$, Sodium Potassium tartarate, KOH	III. Wolff - Kishner Reduction
D. $Zn - Hg, HCl$	IV. Fehling's Test

Choose the correct answer from the options given below L

Option 1:
A-III, B-IV, C-I, D-II

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
As per NCERT

A. $NH_2 - NH_2, KOH \rightarrow$ Wolff-Kishner Reduction

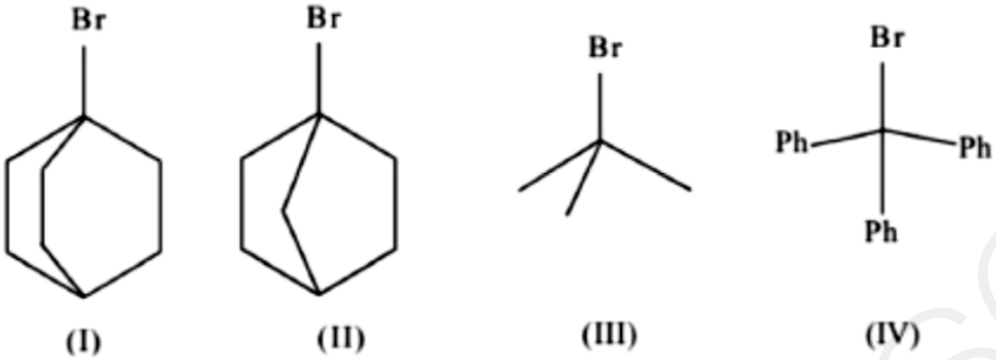
B. $Ag(NH_3)_2OH \rightarrow$ Tollen's Test

C. Aq. $CuSO_4$, Sodium Potassium tartarate, KOH $\rightarrow$ Fehling's Test

D. $Zn - Hg, HCl \rightarrow$ Clemmensen Reduction

Hence, the correct answer is the option (3).

Q. 9 The correct order of the rate of reaction of the following reactants with nucleophile by S_N1 mechanism is
(Given : Structures I and II are rigid)



(I) (II) (III) (IV)

Option 1:
I < II < III < IV

Option 2:
III < I < II < IV

Option 3:
II < I < III < IV

Option 4:
IV < III < II < I

Correct Answer:
II < I < III < IV

Solution:
Sol: The SN1 reaction involves the formation of a carbocation. The more stable the carbocation, the faster the reaction rate.

Structure IV (Triphenylmethyl bromide): This forms the triphenylmethyl carbocation. It is the most stable because the positive charge is extensively delocalized through resonance across three phenyl rings.
Structure III (tert-Butyl bromide): This forms the tert-butyl carbocation, a tertiary (3°) carbocation stabilized by hyperconjugation and the inductive effect of three methyl groups. While very stable, it is less stable than the resonance-stabilized triphenylmethyl cation.
Structure I (1-bromobicyclooctane): This is a bridgehead halide. According to Bredt's Rule, bridgehead carbons in small bicyclic systems cannot easily achieve the planar,
hybridized geometry required for a stable carbocation. However, the system is slightly more flexible than the system, allowing for a very slow but measurable SN1 reaction.

Structure II (1-bromobicycloheptane): This is a more rigid and strained bridgehead system than structure I. The small rings prevent the bridgehead carbon from becoming even partially planar, making the carbocation formation extremely difficult and the reactant virtually unreactive in SN1 mechanisms.

Hence, the correct answer is the option (3).

Q. 10 Given below are two statements :

Statement I : Phenol on treatment with $\mathrm{CHCl_3}$ / aq. KOH under refluxing condition, followed by acidification produces p - hydroxy benzaldehyde as the major product and oohydroxy benzaldehyde as the minor product.

Statement II : The mixture of p-hydroxybenzaldehyde and o- hydroxybenzaldehyde can be easily separated through steam distillation.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are false

Option 2:
Statement I is true but Statement II is false

Option 3:
Statement I is false but Statement II is true

Option 4:
Both Statement I and Statement II are true

Correct Answer:
Both Statement I and Statement II are true

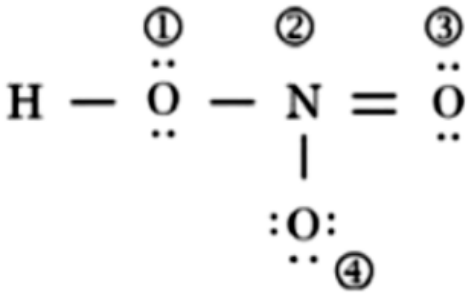
Solution:
Sol: Due to steric factor of K^+para product is major

Statement I: True — Phenol + $\mathrm{CHCl_3}$/aq. KOH → Reimer-Tiemann reaction, gives p-hydroxybenzaldehyde (major) and o-hydroxybenzaldehyde (minor).

Statement II: True — The two isomers can be separated by steam distillation.

Hence, the correct answer is the option (4).

Q. 11 The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are



Option 1:
$0, 0, -1, +1$

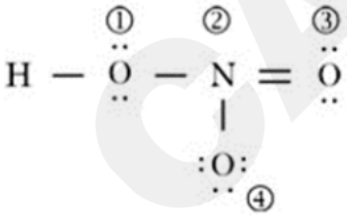
Option 2:
$0, -1, 0, +1$

Option 3:
$0, +1, 0, -1$

Option 4:
$+1, 0, 0, -1$

Correct Answer:
$0, +1, 0, -1$

Solution:



$1 = 6 - 4 - \frac{1}{2}[4] = 0$

$2. 5 - 0 - \frac{1}{2}[8] = +1$
$3. 6 - 4 - \frac{1}{2}[4] = 0$
$4. 6 - 6 - \frac{1}{2}[2] = -1$

Hence, the correct answer is the option (3).

Q. 12 The energy required by electrons, present in the first Bohr orbit of hydrogen atom to be excited to second Bohr orbit is _____ $\mathrm{Jmol^{-1}}$
Given: $\mathrm{R_H} = 2.18 \times 10^{-11}\mathrm{ergs}$

Option 1:
9.835×10^{5}

Option 2:
9.835×10^{12}

Option 3:
1.635×10^{-11}

Option 4:
1.635×10^{-18}

Correct Answer:
9.835×10^{5}

Solution:
Sol :

$$E_2 - E_1 = R_H\left[\frac{1}{N_1^2} - \frac{1}{N_2^2}\right] \times 6 \times 10^{23} = 2.18 \times 10^{-18}\left[\frac{3}{4}\right] \times 6 \times 10^{23} = 9.81 \times 10^{5}\ \mathrm{J/mol}$$

Hence, the correct answer is the option (1).

Q. 13 $\mathrm{A} \rightarrow$ product (First order reaction)
Three sets of experiment were performed for a reaction under similar experimental conditions :

Run $1 \Rightarrow 100\ \mathrm{mL}$ of 10 M solution of reactant A

Run $2 \Rightarrow 200\ \mathrm{mL}$ of 10 M solution of reactant A

Run $3 \Rightarrow 100\ \mathrm{mL}$ of 10 M solution of reactant $\mathrm{A} + 100\ \mathrm{mL}$ of $\mathrm{H_2O}$ added.

The correct variation of rate of reaction is

Option 1:
$\text{Run } 1 < \text{Run } 2 < \text{Run } 3$

Option 2:
$\text{Run } 3 < \text{Run } 1 = \text{Run } 2$

Option 3:
$\text{Run } 1 = \text{Run } 2 = \text{Run } 3$

Option 4:
$\text{Run } 3 < \text{Run } 1 < \text{Run } 2$

Correct Answer:
$\text{Run } 3 < \text{Run } 1 = \text{Run } 2$

Solution:
Sol : $r = K(C)^1$

Run 1: 100 mL of 10 M solution
Concentration = **10 M**

$\text{Rate}_1 = k(10)$

Run 2: 200 mL of 10 M solution
Concentration = still 10 M (Even though volume doubled, concentration did not change.)

$\text{Rate}_2 = k(10)$

Run 3: 100 mL of 10 M solution + 100 mL water

$$\text{Now total volume } = 200 \text{ mL}$$
$$\text{Initial moles of } A = 10 \times 0.1 = 1 \text{ mole}$$
$$\text{New concentration } = \frac{1}{0.2} = 5\text{M}$$

$\text{Rate}_3 = k(5)$

Final conclusion : $\text{Rate}_1 = \text{Rate}_2 > \text{Rate}_3$

Hence, the correct answer is the option (2).

Q. 14 A first row transition metal (M) does not liberate H_2 gas from dilute HCl .1 mol of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then $\mathrm{H}_2\ \mathrm{S}(\ \mathrm{g})$ is passed through the solution. The amount of MS (metal sulphide) formed from the above reaction is _____ mol

Option 1:
3

Option 2:
0

Option 3:
1

Option 4:
2

Correct Answer:
0

Solution:
Sol : $\mathrm{M}=\mathrm{Cu} \rightarrow \mathrm{Cu}^{+1}$ in complex

$$\underset{1 \text{ mole}}{\mathrm{CuSO}_4}+\mathrm{KCN} \underset{1 \text{ mole}}{\mathrm{Cu}}\ (\mathrm{CN})_2 \xrightarrow{\mathrm{KCN}} \underset{0.5 \text{ mole}}{\mathrm{Cu}_2}\ (\mathrm{CN})_2$$

$$\mathrm{Cu}_2(\mathrm{CN})_2 \xrightarrow{\mathrm{KCN}} 2\ \mathrm{K}_3\left[\mathrm{Cu}(\mathrm{CN})_4\right]+(\mathrm{CN})_2$$

NO, free Cu^{+}ions are available

Hence, the correct answer is the option (2).

Q. 15 Consider the transition metal ions $\mathrm{Mn}^{3+}, \mathrm{Cr}^{3+}, \mathrm{Fe}^{3+}$ and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

Option 1:
$\mathrm{Mn}^{3+}>\mathrm{Fe}^{3+}>\mathrm{Co}^{3+}>\mathrm{Cr}^{3+}$

Option 2:
$Fe^{3+} > Co^{3+} > Mn^{3+} > Cr^{3+}$

Option 3:
$Cr^{3+} > Fe^{3+} > Co^{3+} > Mn^{3+}$

Option 4:
$Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$

Correct Answer:
$Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$

Solution:

Mn^{+3}	Cr^{+3}	Fe^{+3}	Co^{+3}
d^4	d^3	d^5	d^6
$t_{2g^3}^3 eg^1$	t_{2g}^3	$t^{25}e^0$	${}^t2g^6{}_{eg}{}^0$
$n=2$	$n=3$	$n=1$	$n=0$

$Cr^{+3} > Mn^{+3} > Fe^{+3} > Co^{+3}$

Hence, the correct answer is the option (4).

Q. 16 The correct order of reactivity of CH_3Br in methanol with the following nucleophiles is $F^-, I^-, C_2H_5O^-$ and $C_6H_5O^-$

Option 1:
$I^- > C_2H_5O^- > F^- > C_6H_5O^-$

Option 2:
$I^- > C_2H_5O^- > C_6H_5O^- > F^-$

Option 3:
$I^- > C_6H_5O^- > F^- > C_2H_5O^-$

Option 4:
$I^- > F^- > C_6H_5O^- > C_2H_5O^-$

Correct Answer:
$I^- > C_2H_5O^- > C_6H_5O^- > F^-$

Solution:
Sol : $I^- > EtO^- > PhO^- > F^-$

Hence, the correct answer is the option (2).

Q. 17 A ' p ' block element (E) and hydrogen form a binary cation $(EH_x)^+$, while EH_3 on treatment with K_2HgI_4 in alkaline medium gives a precipitate of basic mercury (II) amido - iodine. Given below are first ionisation enthalpy values $(kJmol^{-1})$ for first element each from group 13, 14, 15 and 16. Identify the correct first ionisation enthalpy value for element E .

Option 1:
1086

Option 2:
1312

Option 3:
1402

Option 4:
801

Correct Answer:
1402

Solution:
Sol : I.P order $II_A < IV_A < VI_A < V_A$

Hence, the correct answer is the option (3).

Q. 18 Given below are two statements :

Statement I : Sucrose is dextrorotatory. However, sucrose upon hydrolysis gives a solution having mixture of products. This solution shows laevorotation.

Statement II : Hydrolysis of sucrose gives glucose and fructose. Since the laevorotation of glucose is more than the dextrorotation of fructose, the resulting solution becomes laevorotatory.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are true

Option 2:
Statement I true but Statement II is false

Option 3:
Statement I is false but Statement II are true

Option 4:
Both Statement I and Statement II are fasle

Correct Answer:
Statement I true but Statement II is false

Solution:
Sol : laevo rotation of fructose is more than dextro rotation of glucose

Hence, the correct answer is the option (2).

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Q. 19 Match the List-I with List - II

List-I	List-II
Thermodynamic Process	**Magnitude in kJ**
A. Work done in reversible, isothermal expansion of 2 mol of ideal gas from $2\mathrm{dm}^3$ to $20\mathrm{dm}^3$ at 300 K	I. 4
B. Work done in irreversible isothermal expansion of 1 mol ideal gas form $1\ \mathrm{m}^3$ to $3\ \mathrm{m}^3$ at 300 K against a Constant pressure of 3 kPa	II. 11.5
C. Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature $= 320\ \mathrm{K}$ and $\overline{\mathrm{C}}_{\mathrm{v}} = \frac{3}{2}\mathrm{R}$	III. 6
D. Change in enthalpy at constant pressure of 1 mol ideal gas With change of temperature $= 337\ \mathrm{K}$ and $\overline{\mathrm{C}}_{\mathrm{p}} = \frac{5}{2}\mathrm{R}$	IV. 7

Choose the correct answer from the options given below :

Option 1:
A-II,B- III, C-I, D-IV

Option 2:
A-I,B- II, C-III, D-IV

Option 3:
A-II,B-I, C-III, D-IV

Option 4:
A-III,B-II, C-IV, D-I

Correct Answer:
A-II,B- III, C-I, D-IV

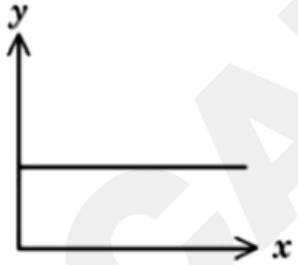
Solution:
Sol : $W_{it,\text{ rev}} = 2.303 \times 2 \times 8.314 \times 10^{-3} \times 300 \log \frac{20}{2} = 11.5$
$(W) = P\Delta V = 3 \times 10^3(3-1) = 6 \times 10^3 T$

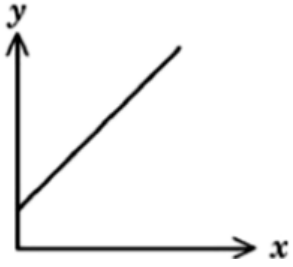
$\Delta U = {}^{n}C_r dt = 1 \times \frac{3R}{2} \times 320 = 4$

Hence, the correct answer is the option (1).

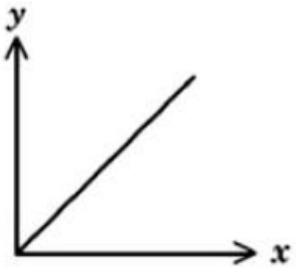
Q. 20 Consider a solution $\mathrm{CO_2(\ g)}$ dissolved in water in a closed container Which one of the following plots correctly represents variation of log (partial pressure of $\mathrm{CO_2}$ in vapour phase above water) [y-axis] with log (mole fraction of $\mathrm{CO_2}$ in water)

[x-axis] at $25^\circ\mathrm{C}$?

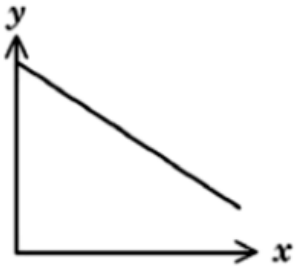
Option 1:


Option 2:


Option 3:



Option 4:



Correct Answer:



Solution:

Sol: : $P_{CO_2} = K_H \cdot X_{CO_2}$

$\log P_{CO_2} = \log K_H + \log X_{CO_2}$

Hence, the correct answer is the option (2).

Q. 21 Consider the following electrochemical cell at 298 K

$$Pt\left|HSnO_2^-(aq)\right|Sn(OH)_6^{2-}(aq)\left|OH^-(aq)\right|Bi_2O_3(s)\mid Bi(s)$$

If the reaction quotient at a given time is 10^6, then the cell EMF (E_{cell}) is ____ $\times 10^{-1}$ V (Nearest integer).

Given the standard half - cell reduction potential as

$$E^\circ_{Bi_2O_3//Bi,OH^-} = -0.44\ V^\circ \text{ and } E^\circ_{Sn(OH)_6^{2-}/HSnO_2^-,OH^-} = -0.90\ V$$

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Correct Answer:
4

Solution:
Sol : $\mathrm{HSnO_2^-} \rightarrow \mathrm{Sn(OH)_6^{-2}} + 2e^-$

$E = (-0.44 + 0.9) - \frac{0.06}{6}\log 10^6 = 0.4 = 4 \times 10^{-1}$

Hence, the answer is (4).

Q. 22 Dissociation of a gas $\mathrm{A_2}$ takes place according to the following chemical reaction. At equilibrium, the total pressure is 1 bar at 300 K

$\mathrm{A_2(\ g)} \quad 2\ \mathrm{A(\ g)}$

The standard Gibbs energy of formation of the involved substances has been provided below :

Substance	$\mathbf{\Delta G_f^\circ / kJ\ mol^{-1}}$
$\mathbf{A_2}$	-100.00
A	-50.832

The degree of dissociation of $\mathrm{A_2(\ g)}$ given by $\left(\mathrm{x} \times 10^{-2}\right)^{1/2}$ where $\mathrm{x} =$ ____ (Nearest integer)
[Given : $\mathrm{R} = 8\ \mathrm{J\ mol^{-1}\ K^{-1}}, \log 2 = 0.3010, \log 3 = 0.48$]
Assume degree of dissociation is not negligible

Correct Answer:
33

Solution:
Sol :

$A_2 \rightleftarrows 2A$
$\Delta G = (-50.832)2 - (-100)$
r
$= 100 - 101.664$
$-1.664 = -2.303 \times 8 \times 10^{-3} \times 300 \log K$
$\log K = \log 2 \Rightarrow K = 2$
$A_2 \rightarrow 2A$
$k_P = \left(\frac{2\alpha}{1+\alpha} \cdot P\right)^2$
$\left(\frac{1-\alpha}{1+\alpha} \cdot P\right) = P = 1$
$3\alpha^2 = 1$

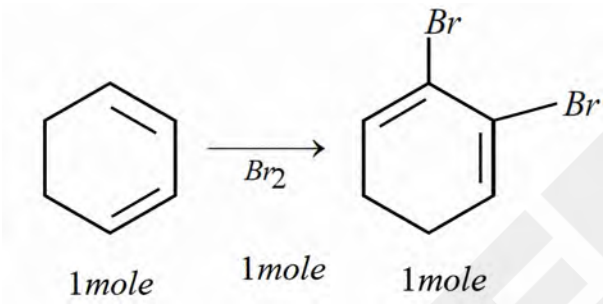
$\alpha = \left(\frac{1}{3}\right)^2 = \left(33.3 \times 10^{-2}\right)^{1/2}$

Hence, the answer is (33).

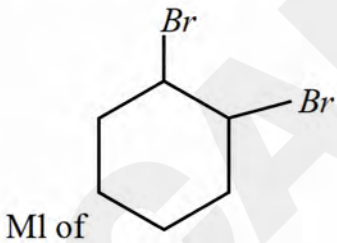
Q. 23 The cycloalkene (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which $\mathrm{C}:\mathrm{Br}$ ratio is $3:1$. The percentage of bromine in the product (Y) is _____ \% (Nearest integer)
(Given : molar mass in $\mathrm{gmol}^{-1}\mathrm{H}:1, \mathrm{C}:12, \mathrm{O}:16, \mathrm{Br}:80$)

Correct Answer:
66

Solution:
Sol : $\mathrm{C}:\mathrm{Br} \Rightarrow 3:2$



$\mathrm{C{:}Br} \Rightarrow 6{:}2 \Rightarrow 3{:}1$



Ml of
$C_6H_{10}Br_2 \Rightarrow 66.11$

Hence, the answer is (66).

Q. 24 The temperature at which rate constants of the given below two gaseous reactions become equal is _____ K (Nearest integer)

$$X \longrightarrow Y \qquad k_1 = 10^6 e^{\frac{-30000}{T}}$$

$$P \longrightarrow Q \qquad k_2 = 10^4 e^{\frac{-24000}{T}}$$

Given : $\ln 10 = 2.303)$

Correct Answer:
1303

Solution:
Sol:
$K_1 = K_2$

$10^6 \cdot e^{-\frac{3\times10^4}{T}} = 10^4 \cdot e^{-\frac{2.4\times10^4}{T}}$

$2 \times 2.303 - \dfrac{3 \times 10^4}{T} = \dfrac{-2.4 \times 10^4}{T}$

$T = \dfrac{6 \times 10^3}{4.606}$

Hence, the answer is (1303).

Q. 25 Sodium fusion extract of an organic compound (Y) with $CHCl_3$ and chlorine water gives violet colour to the $CHCl_3$ layer. 0.15 g of (Y) gave 0.12 g of the silver halie precipitate in Carius method. Percentage of halogen in the compound (Y) is _____ (Nearest integer)
(Given : molar mass $gmol^{-1} C : 12, H : 1, Cl : 35.5, Br : 80, I : 127$)

Correct Answer:
43

Solution:
Sol : $\frac{127}{235} \times \frac{0.12}{0.15} \times 100 = 43.23$

$\frac{\text{Awof } I}{\text{Mw of } AgI} \times \frac{\text{wof } AgI}{\text{wof } O.C} \times 100$

Hence, the answer is (43).

Maths

Q. 1 Two distinct numbers a and b are selected at random from $1, 2, 3, \ldots\ldots, 50$. The probability that their product ab is divisible by 3 is:

Option 1:
$\frac{664}{1225}$

Option 2:
$\frac{272}{1225}$

Option 3:
$\frac{561}{1225}$

Option 4:
$\frac{8}{25}$

Correct Answer:
$\frac{664}{1225}$

Solution:
Numbers divisible by $3: 3, 6, 9, \ldots, 48 \rightarrow$ total 16 numbers
Numbers not divisible by 3: remaining numbers → total $50 - 16 = 34$ numbers

A product is divisible by 3 if at least one of its factors is divisible by 3.
Count this using complementary cases:
Pairs where exactly one number is divisible by 3: $16 \times 34 = 544$
Pairs where both numbers divisible by 3 : $\binom{16}{2} = 120$
Total favourable pairs: $544 + 120 = 664$
Total possible pairs: $\binom{50}{2} = 1225$

So, Probability: $P(E) = \frac{664}{1225}$

Hence, the correct answer is option (1).

Q. 2 Let the solution curve of the differential equation
$x dy - y dx = \sqrt{x^2 + y^2} dx, x > 0, y(1) = 0, be, y = y(x)$. Then $y(3)$ is equal to:

Option 1:
1

Option 2:
2

Option 3:
4

Option 4:
6

Correct Answer:
4

Solution:

$xdy - ydx = \sqrt{x^2+y^2}dx$

$\Rightarrow \frac{xdy-ydx}{x^2} = \frac{\sqrt{x^2+y^2}}{x^2}dx$

$\Rightarrow \int \frac{d(y/x)}{\sqrt{1+(y/x)^2}} = \int \frac{1}{x}dx$

$\Rightarrow \log\left(\frac{y}{x} + \sqrt{1+\left(\frac{y}{x}\right)^2}\right) = \log x + \log C$

$\Rightarrow y + \sqrt{x^2+y^2} = x^2$

$\Rightarrow y - \sqrt{x^2+y^2} = -1$

$\Rightarrow 2y = x^2 - 1$

$\Rightarrow y = \frac{x^2-1}{2}$

Now,

$y(3) = \frac{9-1}{2} = \frac{8}{2} = 4$

Hence, the correct answer is option (3).

Q. 3 The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{[x]+4}\right)dx$, where [.] denotes the greatest integer function, is:

Option 1:
$\frac{1}{60}(\pi - 7)$

Option 2:
$\frac{7}{60}(3\pi - 1)$

Option 3:
$\frac{7}{60}(\pi - 3)$

Option 4:
$\frac{1}{60}(21\pi - 1)$

Correct Answer:
$\frac{7}{60}(3\pi - 1)$

Solution:

$I = \int_{-\frac{\pi}{2}}^{-1} \frac{1}{-2+4}dx + \int_{-1}^{0} \frac{1}{-1+4}dx + \int_{0}^{1} \frac{1}{0+4}dx + \int_{1}^{\frac{\pi}{2}} \frac{1}{1+4}dx$

$I = \frac{1}{2}\left(-1+\frac{\pi}{2}\right) + \frac{1}{3}(0-(-1)) + \frac{1}{4}(1-0) + \frac{1}{5}\left(\frac{\pi}{2}-1\right)$

$I = \frac{\pi}{4} - \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{\pi}{10} - \frac{1}{5}$

$I = \frac{7\pi}{20} - \frac{7}{60}$

$I = \frac{7}{60}(3\pi - 1)$

Hence, the correct answer is option (2).

Q. 4 Let $P(\alpha, \beta, \gamma)$ be the point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at distance $4\sqrt{14}$ from the point $(1, -1, 0)$ and nearer to the origin. Then the shortest distance, between the line $\frac{x-\alpha}{1} = \frac{y-\beta}{2} = \frac{z-\gamma}{3}$ and $\frac{x+5}{2} = \frac{y-10}{1} = \frac{z-3}{1}$, is equal to:

Option 1:

$4\sqrt{\frac{5}{7}}$

Option 2:

$7\sqrt{\frac{5}{4}}$

Option 3:

$2\sqrt{\frac{7}{4}}$

Option 4:

$4\sqrt{\frac{7}{5}}$

Correct Answer:

$4\sqrt{\frac{7}{5}}$

Solution:

$\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z}{1} = t$

$(1+2t, -1-3t, t)$

$4t^2 + 9t^2 + t^2 = 16 \times 14$

$t = \pm 4$

Points are $(9, -13, 4), (-7, 11, -4)$

$(\alpha, \beta, \gamma) = (-7, 11, -4)$

$\bar{a} = (-7, 11, -4), \bar{c} = (-5, 10, 3)$

$\bar{c} - \bar{a} = (2, -1, 7)$

$b \times d = (-1, 5, -3)$

$$S.D = \left| \frac{-2 - 5 - 21}{\sqrt{1 + 25 + 9}} \right| = \frac{28}{\sqrt{35}}$$

$$\frac{28}{\sqrt{35}} \times \frac{\sqrt{35}}{\sqrt{35}}$$

$$= \frac{28\sqrt{35}}{35}$$

$$= \frac{4\sqrt{35}}{5}$$

Now rewrite $\sqrt{35}$:

$\sqrt{35} = \sqrt{7 \times 5} = \sqrt{7}\sqrt{5}$

$$\frac{4\sqrt{35}}{5} = \frac{4\sqrt{7}\sqrt{5}}{5}$$

$$= 4\sqrt{7} \cdot \frac{\sqrt{5}}{5}$$

Since

$$\frac{\sqrt{5}}{5} = \frac{1}{\sqrt{5}}$$

$$= 4\frac{\sqrt{7}}{\sqrt{5}}$$

$$= 4\sqrt{\frac{7}{5}}$$

Hence, the correct answer is option (4).

Q. 5 Let $f(x) = x^{2025} - x^{2000}, x \in [0, 1]$ and the minimum value of the function $f(x)$ in the interval $[0, 1]$ be $(80)^{80}(n)^{-81}$. Then n is equal to:

Option 1:
-80

Option 2:
-40

Option 3:
-81

Option 4:
-41

Correct Answer:
-81

Solution:
$f(x) = x^{2025} - x^{2000}; x \in [0,1]$
$f'(x) = 2025x^{2024} - 2000x^{1999}$
$= x^{1999}\left(2025x^5 - 2000\right)$
$f'(x) = 0 \Rightarrow x^5 = \dfrac{80}{81}$
$f(x) = \left(x^5\right)^{81} - \left(x^5\right)^{80}$
$= \left(\dfrac{80}{81}\right)^{81} - \left(\dfrac{80}{81}\right)^{80}$
$= (80)^{80}(-81)^{-81}$

Hence, the correct answer is option (3).

Q. 6 The number of distinct real solutions of the equation $x|x+4| + 3|x+2| + 10 = 0$ is:

Option 1:
1

Option 2:
0

Option 3:
2

Option 4:
3

Correct Answer:
1

Solution:
Critical points from modulus terms:
$x + 4 = 0 \Rightarrow x = -4, \quad x + 2 = 0 \Rightarrow x = -2$
So consider intervals:
1. $x < -4$
2. $-4 \leq x < -2$
3. $x \geq -2$

Case 1: $x < -4$

$|x+4| = -(x+4), \quad |x+2| = -(x+2)$

$\Rightarrow x(-x-4) + 3(-x-2) + 10 = 0$

$\Rightarrow -x^2 - 4x - 3x - 6 + 10 = 0$

$\Rightarrow -x^2 - 7x + 4 = 0$

$\Rightarrow x^2 + 7x - 4 = 0$

Roots:

$x = \frac{-7 \pm \sqrt{65}}{2}$

Valid root for $x < -4$:

$x = \frac{-7 - \sqrt{65}}{2}$

Hence, 1 solution.

Case 2: $-4 \leq x < -2$

$|x+4| = x+4, \quad |x+2| = -(x+2)$

$x(x+4) + 3(-x-2) + 10 = 0$

$x^2 + 4x - 3x - 6 + 10 = 0$

$x^2 + x + 4 = 0$

Discriminant:

$1 - 16 = -15 < 0$

Hence, No real solution

Case 3: $x \geq -2$

$|x+4| = x+4, \quad |x+2| = x+2$

$x(x+4) + 3(x+2) + 10 = 0$

$x^2 + 4x + 3x + 6 + 10 = 0$

$x^2 + 7x + 16 = 0$

Discriminant:

$49 - 64 = -15 < 0$

Hence, No real solution.

Hence, the correct answer is option (1).

Q. 7 The coefficient of x^{48} in $(1+x) + 2(1+x)^2 + 3(1+x)^3 + \ldots + 100(1+x)^{100}$ is equal to:

Option 1:

${}^{100}C_{C_{50}} + {}^{101}C_{49}$

Option 2:

$100.\ {}^{100}C_{49} - {}^{101}C_{50}$

Option 3:

$100.\ {}^{101}C_{49} - {}^{100}C_{50}$

Option 4:
$100 \cdot {}^{100}C_{49} - {}^{100}C_{48}$

Correct Answer:
$100. {}^{101}C_{49} - {}^{100}C_{50}$

Solution:
$S = (1+x) + 2(1+x)^2 + 3(1+x)^3 + \ldots + 100(1+x)^{100}$

$(1+x)S = (1+x)^2 + \ldots + 99(1+x)^{100} + 100(1+x)^{101}$

$-xS = (1+x)\left(\frac{(1+x)^{100}-1}{x}\right) - 100(1+x)^{101}$

$S = \left(\frac{(1+x)-(1+x)^{101}}{x^2}\right) + \frac{100(1+x)^{101}}{x}$

CoeffIcient of $x^{48} = -{}^{101}C_{50} + 100 \times {}^{(101)}C_{49}$

Hence, the correct answer is option (3).

Q. 8 If the image of the points $P(1,2,a)$ in the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{7-z}{2}$ is $Q(5,b,c)$, then $a^2+b^2+c^2$ is equal to:

Option 1:
264

Option 2:
298

Option 3:
283

Option 4:
293

Correct Answer:
298

Solution:
The given line is $\frac{x-6}{3} = \frac{y-7}{2} = \frac{7-z}{2}$. To find the direction ratios (d_1, d_2, d_3), we rewrite the z term:
$L: \frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$
The direction vector of the line is $v = (3,2,-2)$.

$\overrightarrow{PQ} = (5-1, b-2, c-a) = (4, b-2, c-a)$
Since $PQ \perp$ line,

$$(4, b-2, c-a)\cdot(3,2,-2)=0$$
$$3(4)+2(b-2)-2(c-a)=0$$
$$12+2b-4-2c+2a=0$$
$$a+b-c=-4$$

Midpoint of P and Q :

$$M\left(\frac{1+5}{2}, \frac{2+b}{2}, \frac{a+c}{2}\right)$$
$$M\left(3, \frac{2+b}{2}, \frac{a+c}{2}\right)$$

Since midpoint lies on the line,

$$\frac{3-6}{3}=\frac{\frac{2+b}{2}-7}{2}=\frac{7-\frac{a+c}{2}}{2}$$
$$-1=\frac{b-12}{4}$$
$$b=8$$
$$-1=\frac{7-\frac{a+c}{2}}{2}$$
$$7-\frac{a+c}{2}=-2$$
$$a+c=18$$

Solving,

$$a+b-c=-4$$
$$a+8-c=-4$$
$$a-c=-12$$
$$a+c=18$$
$$a=3, \quad c=15$$
$$a^2+b^2+c^2=3^2+8^2+15^2$$
$$=9+64+225$$
$$=298$$

Hence, the correct answer is option (2).

Q. 9 Let relation R on the set $M=\{1,2,3,\ldots,16\}$ be given by $R=\{(x,y): 4y=5x-3, x, y\in M\}$ Then the minimum number of elements required to be added in R, in order to make the relation symmetric, is equal to:

Option 1:
1

Option 2:
3

Option 3:
2

Option 4:
4

Correct Answer:
2

Solution:
Given:

$$M = \{1, 2, 3, \ldots, 16\}$$

$$R = \{(x, y) : 4y = 5x - 3, x, y \in M\}$$

$$4y = 5x - 3$$

$$y = \frac{5x - 3}{4}$$

For y to be an integer,

$$5x - 3 \equiv 0 \pmod 4$$

Since $5 \equiv 1 \pmod 4$,

$$x - 3 \equiv 0 \pmod 4$$

$$x \equiv 3 \pmod 4$$

Thus

$$x = 3, 7, 11, 15$$

Now compute y :

$$x = 3 \Rightarrow y = 3$$

$$x = 7 \Rightarrow y = 8$$

$$x = 11 \Rightarrow y = 13$$

$$x = 15 \Rightarrow y = 18 \notin M$$

So relation R :

$$R = \{(3, 3), (7, 8), (11, 13)\}$$

For symmetry:
If $(a, b) \in R$, then (b, a) must also belong to R.
$(3, 3)$
Already symmetric.
But

$$(7, 8) \Rightarrow (8, 7) \notin R$$

$$(11, 13) \Rightarrow (13, 11) \notin R$$

Thus, two pairs must be added:
$(8, 7), (13, 11)$
Minimum number of elements to be added $= 2$

Hence, the correct answer is option (3).

Q. 10 If the line $\alpha x + 2y = 1$, where $\alpha \in R$, does not meet the hyperbola $x^2 - 9y^2 = 9$, then a possible value of α is:

Option 1:
0.6

Option 2:
0.7

Option 3:
0.5

Option 4:
0.8

Correct Answer:
0.8

Solution:
$\alpha x+2y=1, \alpha \in R$
$x^2-9y^2=9$

$y=\dfrac{1-\alpha x}{2}$
$4x^2-9\left(1+\alpha^2x^2-2\alpha x\right)=36$
$\Delta<0$
$1620\alpha^2-720>0$
$\alpha^2>0.55$
$\alpha=0.8$

Hence, the correct answer is option (4).

Q. 11 If the domain of the function $f(x)=\sin^{-1}\left[\frac{5-x}{3+2x}\right]+\frac{1}{\log_e(10-x)}$ is $(-\infty,\alpha]\cup[\beta,\gamma)-\{\delta\}$, then $6(\alpha+\beta+\gamma+\delta)$ is equal to:

Option 1:
66

Option 2:
67

Option 3:
68

Option 4:
70

Correct Answer:
70

Solution:

$$f(x)=\sin^{-1}\left[\frac{5-x}{3+2x}\right]+\frac{1}{\log_e(10-x)}$$

$$-1\leq\frac{5-x}{3+2x}\leq 1$$

$$\frac{2-3x}{3+2x}\leq 0$$

$$(3x-2)(2x+3)\geq 0$$

$$(3x-2)(2x+3)\geq 0x\in\left(-\infty,-\frac{3}{2}\right]\cup\left[\frac{2}{3},\infty\right)$$

$$\frac{8+x}{3+2x}\geq 0$$

$$x\in(-\infty,-8]\cup\left[\frac{-3}{2},\infty\right)$$

$$x\in(-\infty,-8]\cup\left[\frac{2}{3},10\right)-\{9\}=6(\alpha+\beta+\gamma+\delta)=70$$

Hence, the correct answer is option (4).

Q. 12 Let the line $x=-1$ divided the area of the region $\left\{(x,y):1+x^2\leq y\leq 3-x\right\}$ in the ratio $m:n,\gcd(m,n)=1$. Then $m+n$ is equal to:

Option 1:
28

Option 2:
25

Option 3:
26

Option 4:
27

Correct Answer:
27

Solution:

$m=\int_{-2}^{-1}(3-x)-\left(1+x^2\right)dx$

$$m=2+\frac{3}{2}-\frac{7}{3}=\frac{7}{6}$$

$$M=\int_{-1}^{1}(3-x)-\left(1+x^2\right)dx=4-\frac{1}{2}(0)-\frac{1}{3}(1+1)=4-\frac{2}{3}=\frac{10}{3}$$

$$m:M=\frac{7}{6}:\frac{10}{3}=\frac{7}{20}$$

$m+n=27$

Hence, the correct answer is option (4).

Q. 13 Let $\overline{AB} = 2i + 4j - 5k$ and $\overline{AD} = i + 2j + \lambda k, \lambda \in R$. Let the projection of the vector $\bar{v} = i + j + k$ on the diagonal $\overline{AC}$ of the parallelogram $ABCD$ be of length one unit .If α, β, where $\alpha > \beta$, be the roots of equation $\lambda^2 x^2 - 6\lambda x + 5 = 0$ then $2\alpha - \beta$ is equal to:

Option 1:
4

Option 2:
3

Option 3:
6

Option 4:
1

Correct Answer:
3

Solution:
$\overline{AB} = 2\bar{i} + 4\bar{j} - 5\bar{k}$
$\overline{AD} = \bar{i} + 2\bar{j} + \lambda\bar{k}$
$\bar{V} = \bar{i} + \bar{j} + \bar{k}$
$\bar{V} \cdot \overline{AC}$
$|\overline{AC}|$
$\overline{AC} = 1$
$18\lambda = 54$
$18\lambda = 54$
$\lambda = 3$

Hence, the correct answer is option (2).

Q. 14 If the sum of the first term of an A.P. is 6 and the sum of its first six terms is 4 , then the sum of its first twelve terms is:

Option 1:
-26

Option 2:
-24

Option 3:
-20

Option 4:
-22

Correct Answer:
-22

Solution:
$a+(a+d)+(a+2d)+(d+3d)=6$

$2a+3d=3$
$6a+15d=4$
$d=-5/6$
$a=11/4$
$S_{12}=\frac{12}{2}\left[2\left(\frac{11}{4}\right)+11\times(-5/6)\right]=-22$

Hence, the correct answer is option (4).

Q. 15 If the random variable x has the probability distribution:

x	0	1	2	3	4	5	6	7
$P(x)$	0	$2K$	K	$3K$	$2K^2$	$2K$	K^2+k	$7K^2$

Then $P(3<x\leq 6)$ is equal to:

Option 1:
0.22

Option 2:
0.33

Option 3:
0.64

Option 4:
0.34

Correct Answer:
0.33

Solution:

$2K + K + 3K + 2K^2 + 2K + K^2 + K + 7K^2 = 1$
$(K + 1)(10K - 1) = 0$
$K = -1 \text{ or } K = \frac{1}{10}$

$$P(3 < x \leq 6) = \frac{2}{100} + \frac{2}{10} + \frac{1}{100} + \frac{1}{10} = 0.33$$

Hence, the correct answer is option (2).

Q. 16 Let the set of all values of r, for which the circle $(x+1)^2 + (y+4)^2 = r^2$ and $x^2 + y^2 - 4y - 2y - 4 = 0$ intersect at two distinct points be the interval (α, β). Then $\alpha\beta$ is equal to:

Option 1:
20

Option 2:
25

Option 3:
24

Option 4:
21

Correct Answer:
25

Solution:
Circle 1: $(x+1)^2 + (y+4)^2 = r^2$
Center $C_1 = (-1, -4)$, Radius $r_1 = r$
Circle 2: $x^2 + y^2 - 4x - 2y - 4 = 0 \implies (x-2)^2 + (y-1)^2 = 9$
Center $C_2 = (2, 1)$, Radius $r_2 = 3$
Distance Between Centers, $d = \sqrt{(2-(-1))^2 + (1-(-4))^2} = \sqrt{3^2 + 5^2} = \sqrt{34}$
For two distinct intersection points:
$|r_1 - r_2| < d < r_1 + r_2$
$|r - 3| < \sqrt{34} < r + 3$
From $\sqrt{34} < r + 3$:
$r > \sqrt{34} - 3 \quad (\alpha)$
From $|r - 3| < \sqrt{34}$:
$r - 3 < \sqrt{34} \implies r < \sqrt{34} + 3$
the interval is $(\sqrt{34} - 3, \sqrt{34} + 3)$.
$\alpha\beta = (\sqrt{34} - 3)(\sqrt{34} + 3)$
$\alpha\beta = (\sqrt{34})^2 - 3^2$
$\alpha\beta = 34 - 9 = 25$

Hence, the correct answer is option (2).

Q. 17 If the chord joining the points $P_1\left(x_1, y_1\right)$ and $P_2\left(x_2, y_2\right)$ on the parabola $y^2=12x$ subtends a right angle at the vertex of the parabola, then $\left(x_1x_2-y_1y_2\right)$ is equal to:

Option 1:
284

Option 2:
280

Option 3:
292

Option 4:
288

Correct Answer:
288

Solution:
For the parabola $y^2=12x$, we have $a=3$.
Let the points be $P_1\left(3t_1^2, 6t_1\right)$ and $P_2\left(3t_2^2, 6t_2\right)$.
Since P_1 and P_2 subtend a right angle at the vertex $(0,0)$,
the product of the slopes of OP_1 and OP_2 is -1 .

$m_1 \cdot m_2=\left(\frac{2}{t_1}\right)\left(\frac{2}{t_2}\right)=-1 \Longrightarrow t_1t_2=-4$

$x_1x_2=\left(3t_1^2\right)\left(3t_2^2\right)=9\left(t_1t_2\right)^2=9(-4)^2=144$

$y_1y_2=\left(6t_1\right)\left(6t_2\right)=36\left(t_1t_2\right)=36(-4)=-144$

$\therefore x_1x_2-y_1y_2=144-(-144)=288$

Hence, the correct answer is option (4).

Q. 18 The number of solutions of $\tan^{-1} 4x+\tan^{-1} 6x=\frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}}<x<\frac{1}{2\sqrt{6}}$, equal to:

Option 1:
1

Option 2:
3

Option 3:
0

Option 4:
2

Correct Answer:
1

Solution:
$\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$

$$\tan^{-1}\left(\frac{10x}{1-24x^2}\right) = \frac{\pi}{6}$$

$$24x^2 + 10\sqrt{3}x - 1 = 0$$

$$x = \frac{-5\sqrt{3} \pm 3\sqrt{11}}{24}$$

The given interval is $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, which is approximately $-0.204 < x < 0.204$.
Case $1(x > 0) : x = \frac{3\sqrt{11}-5\sqrt{3}}{24} \approx \frac{9.95-8.66}{24} \approx 0.053$. (Valid)
Case $2(x < 0) : x = \frac{-5\sqrt{3}-3\sqrt{11}}{24} \approx \frac{-8.66-9.95}{24} \approx -0.775$. (Invalid, outside the range)
The number of solutions is 1.

Hence, the correct answer is option (1).

Q. 19 If $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$ then the determinant of the matrix $\left(A^{2025} - 3A^{2024} + A^{2023}\right)$ is:

Option 1:
28

Option 2:
12

Option 3:
16

Option 4:
24

Correct Answer:
16

Solution:
$A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}, |A| = 1$

$$\left|A^{2025}-3A^{2024}+A^{2023}\right|=\left|A^{2023}\right|\left|A^{2}-3A+I\right|=16$$

Hence, the correct answer is option (3).

Q. 20 Let $f:[1,\infty)\to R$ be a differentiable function. If $6\int_1^x f(t)dt=3xf(x)+x^3-4$ For all $x\geq 1$, then the value of $f(2)-f(3)$ is:

Option 1:
-4

Option 2:
-3

Option 3:
3

Option 4:
4

Correct Answer:
3

Solution:
$6\int_1^x f(t)dt=3xf(x)+x^2-4$
By using leibnz rule

$6f(x)=3f(x)+3xf'(x)+3x^2$

$$\frac{f(x)-xf'(x)}{x^2}=1$$

$$\frac{d}{dx}\left(\frac{f(x)}{x}\right)=-1f(x)=-x^2+kx$$

$1=-1+k(\text{ since } f(1)=1)$
$f(x)=-x^2+2x$
$f(2)-f(3)=3$

Hence, the correct answer is option (3).

Q. 21 Let $\alpha=\frac{-1+i\sqrt{3}}{2}$ and $\beta=\frac{-1-i\sqrt{3}}{2}, i=\sqrt{-1}$. If

$(7-7\alpha+9\beta)^{20}+(9+7\alpha-7\beta)^{20}+(-7+9\alpha+7\beta)^{20}+(14+7\alpha+7\beta)^{20}=m^{10}$,
Then m is _____

Correct Answer:
49

Solution:

$Z = (7 - 7\alpha + 9\beta)^{20} + (9 + 7\alpha - 7\beta)^{20} + (-7 - 9\alpha + 7\beta)^{20} + (14 + 7\alpha + 7\beta)^{20}$

$= \alpha^{20}(7\beta - 7 + 9\alpha)^{20} + \beta^{20}(-7 + 9\alpha + 7\beta)^{20} + (-7 + 9\alpha + 7\beta)^{20} + 7^{20}$

$= (-7 + 9\alpha + 7\beta)^{20}(0) + 7^{20}$

$m^{10} = 7^{20}$

$m = 49$

Hence, the correct answer is 49.

Q. 22 Let A be a 3×3 matrix such that $A + A^T = 0$. If $A = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}, A^2 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 19 \\ 24 \end{bmatrix}$ and $\det(\text{adj}(2\,\text{adj}(A + I))) = (2)^{\alpha} \cdot (3)^{\beta} \cdot (11)^{\gamma}, \alpha, \beta, \gamma$ are nonnegative integers, then $\alpha + \beta + \gamma$ is equal to _____

Correct Answer:
18

Solution:

$A = -A^T$

$A = \begin{bmatrix} 0 & \alpha & \beta \\ -\alpha & 0 & \gamma \\ -\beta & -\gamma & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & \alpha & \beta \\ -\alpha & 0 & \gamma \\ -\beta & -\gamma & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$

$\alpha = -3, -\beta + \gamma = 2$

$\begin{bmatrix} 0 & \alpha & \beta \\ -\alpha & 0 & \gamma \\ -\beta & -\gamma & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 19 \\ -24 \end{bmatrix}$

$\beta = 3$

$|A + I| = \begin{vmatrix} 1 & -3 & 3 \\ 3 & 1 & 5 \\ -3 & -5 & 1 \end{vmatrix} = 44$

$\det(\text{adj}(2\,\text{adj}(A + I))) = 2^6 \det(\text{adj}(\text{adj}(A + I))) = 2^6(44)^4 = 2^{14} \times 11^4$

Comparing $2^{14} \cdot 3^0 \cdot 11^4$ with $2^{\alpha} \cdot 3^{\beta} \cdot 11^{\gamma}$:

$\alpha = 14$

$\beta = 0$

$\gamma = 4$

Sum: $\alpha + \beta + \gamma = 14 + 0 + 4 = 18$

Hence, the correct answer is 18.

Q. 23 Let ABC be a triangle Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC, None of these points is a vertex of the triangle ABC. Then the total number of pentagons that can be formed by taking all the vertices from the points $p_1, p_2, \ldots, p_{13}$, is_____

Correct Answer:
660

Solution:
Case 1: 2 points from AB, 2 points from BC, 1 point from AC
$^4C_2 \times {}^5C_2 \times {}^4C_1 = 6 \times 10 \times 4 = 240$

Case 2: 2 points from BC, 2 points from AC, 1 point from AB
$^5C_2 \times {}^4C_2 \times {}^4C_1 = 10 \times 6 \times 4 = 240$

Case 3: 2 points from AB, 2 points from AC, 1 point from BC
$^4C_2 \times {}^4C_2 \times {}^5C_1 = 6 \times 6 \times 5 = 180$

$\therefore$ Total Pentagons $= 240 + 240 + 180 = 660$

Hence, the correct answer is 660.

Q. 24 If $\frac{\cos^2 48^\circ - \sin^2 12^\circ}{\sin^2 24^\circ - \sin^2 6^\circ} = \frac{\alpha + \beta\sqrt{5}}{2}$, where $\alpha, \beta \in N$, then $\alpha + \beta$ is equal to_____

Correct Answer:
4

Solution:
$\frac{\cos 60^\circ \cos 36^\circ}{\sin 30^\circ \cdot \sin 18^\circ} = \frac{\sqrt{5}+1}{4} \cdot \frac{4}{\sqrt{5}-1} = \frac{3+\sqrt{5}}{2}$

$\therefore \alpha + \beta = 4$

Hence, the correct answer is 4.

Q. 25 If $\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = -\frac{p_1}{q_1}(\cot x)^{\frac{9}{2}} - \frac{p_2}{q_2}(\cot x)^{\frac{5}{2}} - \frac{p_3}{q_3}(\cot x)^{\frac{1}{2}} + \frac{p_4}{q_4}(\cot x)^{\frac{-3}{2}} + C$, where p_i and q_j are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15 p_1 p_2 p_3 p_4}{q_1 q_2 q_3 q_4}$ is equal to_____

Correct Answer:
16

Solution:

$$\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = \int \frac{1}{\mathrm{Tan}^{11/2} x} \cdot \sec^8 x dx = \int \frac{\left(1+\mathrm{Tan}^2 x\right)^3 \sec^2 x}{\mathrm{Tan}^{11/2} x} dx$$

$\text{put } \mathrm{Tan}\, x = t$

$$\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4} = \frac{15(2 \times 6 \times 6 \times 2)}{9 \times 5 \times 1 \times 3} = 16$$

Hence, the correct answer is 16.

CAREERS360

Physics

Q. 1 Given below are two statements:

Statement-I: A satellite is moving around earth in the orbit very close to the earth surface. The time period of revolution of satellite depends upon the density of earth

Statement-II: The time period of revolution of the satellite is $\mathrm{T}=2\pi\sqrt{\frac{R_e}{g}}$ (for satellite very close to the earth surface), where R_e radius of earth and g acceleration due to gravity.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Statement I is false but Statement II is true

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II are true

Solution:
Statement - 1 is correct

Explanation: $T=2\pi\sqrt{\frac{\mathrm{R_e}}{g}}$

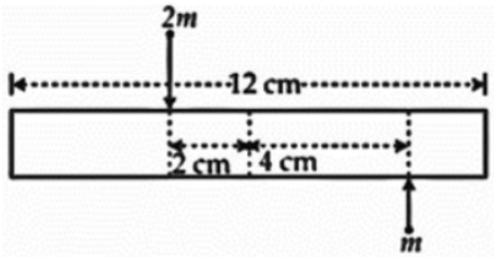
$$T=2\pi\sqrt{\frac{\mathrm{R_e}}{\frac{4}{3}\pi G\rho \mathrm{Re}}}$$

$$T=Z\pi\sqrt{\frac{3}{4\pi g\rho}}$$

Statement -2 is correct

$$T=2\pi\sqrt{\frac{\mathrm{Re}}{g}}$$

Q. 2 A uniform bar of length 12 cm and mass 20 m lies on a smooth horizontal table. Two point masses m and $2m$ are moving in opposite direction with same speed of v and in the same plane as the bar, as showing figure. These masses strike the bar simultaneously and got stuck to it. After collision the entire system is rotating with the angular frequency ω. The ratio of v and
ω is:



Option 1:
33

Option 2:
32

Option 3:
$2\sqrt{88}$

Option 4:
66

Correct Answer:
33

Solution:
Using angular momentum conservation about com of rod $L_i = L_f$

$$m \times v \times 4 + 2m \times v \times 2 = \left(\frac{20m(12)^2}{12} + m \times 4^2 + 2m \times 2^2\right)\omega$$

$$8mv = (240m + 24m)\omega$$

$$8v = 264\omega$$

$$\frac{v}{\omega} = 33$$

Q. 3 A Laser beam has intensity of $4.0 \times 10^{14}\ \mathrm{W/m^2}$. The amplitude of magnetic field associated with beam is ..... T. (Take $\epsilon_0 = 8.85 \times 10^{-12}\mathrm{C^2/Nm^2}$ and $c = 3 \times 10^8\ \mathrm{m/s}$)

Option 1:
5.5

Option 2:
18.3

Option 3:
1.83

Option 4:
2.0

Correct Answer:
1.83

Solution:
$I = \left(\frac{B^2}{2\mu 0}C\right) = 4 \times 10^{14}$

$$B^2 = \frac{2\mu_0}{c} \times 4 \times 10^{14}$$
$$B^2 = \frac{2 \times 4\pi \times 10^{-7} \times 4 \times 10^{14}}{3 \times 10^8}$$
$$B^2 = \frac{3.2\pi}{3} \quad B^2 = \frac{32\pi}{3} \times 10^{-1}$$
$$B = 1.83T$$

Q. 4 The smallest wavelength of Lyman series is 91 nm . The difference between the largest wavelengths of Paschen and Balmer series is nearly _____ nm.

Option 1:
1784

Option 2:
1875

Option 3:
1217

Option 4:
1550

Correct Answer:
1217

Solution:
For Lyman Series $\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) \Rightarrow \lambda = \frac{1}{R} = 91 \text{ nm}$
For Balmer services → Largest Wavelength

$(n_1 = 2 \& n_2 = 3)$

$$\frac{1}{\lambda_B} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$$

$$\lambda_B = \frac{36}{5R} = \frac{36}{5} \times 91 = 655.2$$

$$\therefore \lambda_B = 655.2 \text{ nm}$$

For Paschen series → Largest Wavelength

$(n_1 = 3 \& n_2 = 4)$

$$\frac{1}{\lambda_P} = R\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = \frac{7R}{144}$$

$$\lambda_P = \frac{144}{7R} = \frac{144}{7} \times 91$$

$$\lambda_P = 1872 \text{ nm}$$

$$\lambda_P - \lambda_B = 1216.8 \text{ nm}$$

$$\lambda_P - \lambda_B = 1217 \text{ nm}(\text{ nearly })$$

Q. 5 In an open organ pipe v_3 and v_6 are 3rd and 6th harmonic frequencies, respectively. If $v_6 - v_3 = 2200$ Hz then length of the pipe is _____ mm . (Take velocity of sound in air is 330 m/s).

Option 1:
225

Option 2:
275

Option 3:
250

Option 4:
200

Correct Answer:
275

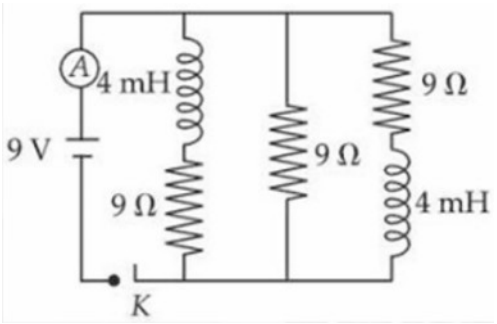
Solution:

$$f = n\frac{V_0}{2L}(\text{ For open pipe })$$

$$\frac{6V_0}{2L} - \frac{3V_0}{2L} = 2200$$

$$\frac{3 \times 330}{2L} = 2200$$
$$\frac{3}{2L} \times 3 = 20 \Rightarrow L = \frac{9}{40}\ \text{m}$$
$$\therefore L = \frac{9 \times 1000}{40} = \frac{900}{4}\ \text{mm}$$
$$L = 225\ \text{mm}$$

Q. 6 Figure shows the circuit that contains three resistances (9Ω each) and two inductors (4 mH each). The reading of ammeter at the moment switch K is turned ON , is _____ A.



Option 1:
1

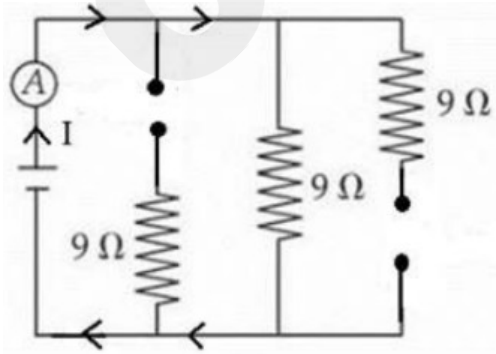
Option 2:
3

Option 3:
zero

Option 4:
2

Correct Answer:
1

Solution:



At $t = 0$ all inductors act as open circuit $I = \frac{9}{9} = 1\ \text{A}$

Q. 7 The wavelength of light, while it is passing through the water of 540 nm . The refractive index of water is $4/3$. The wavelength of the same light when it is passing through a transparent medium having refractive index of $3/2$ is _____ nm

Option 1:
380

Option 2:
840

Option 3:
480

Option 4:
540

Correct Answer:
480

Solution:

$$\lambda \alpha \frac{1}{\mu} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\mu_2}{\mu_1}$$

$$\therefore \frac{\lambda_1}{\lambda_2} = \frac{3/2}{4/3} = \frac{9}{8}$$

$$\lambda_2 = \frac{8}{9}\lambda_1$$

$$\therefore \lambda_2 = \frac{8}{9} \times 540$$

$$\lambda_2 = 480 \text{ nm}$$

Q. 8 An electric power line having total resistance of 2Ω, delivers 1 kW of power at 250 V . The percentage efficiency of transmission line is _____

Option 1:
96.9

Option 2:
100

Option 3:
86.5

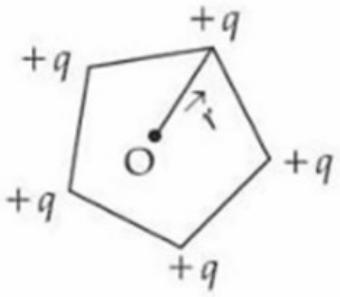
Option 4:
92.5

Correct Answer:
96.9

Solution:
Total current $\mathrm{i}=\frac{1000}{250}=4\ \mathrm{A}$
$\mathrm{i}^2\mathrm{RW}=4^2\times 2=32\ \mathrm{W}$
(power dissipated in 2Ω)

$$\therefore \%\ \text{wastage}\ =\frac{32}{1000}\times 100=3.2\%$$
$$\therefore \eta=(100-3.2)\%=96.8\%$$

Q. 9 Five positive charges each having charge q are placed at the vertices of a pentagon as shown in the figure. The electric potential (V) and the electric field (E) at the centre O of the pentagon due to these five positive charges are:



Option 1:
$\mathrm{V}=\frac{5q}{4\pi\varepsilon_0 r}$ and $E=0$

Option 2:
$\mathrm{V}=\frac{5q}{4\pi\varepsilon_0 r}$ and $E=\frac{5\sqrt{3}q}{8\pi\varepsilon_0 r^2}\hat{r}$

Option 3:
$\mathrm{V}=0$ and $E=0$

Option 4:
$\mathrm{V}=\frac{5q}{4\pi\varepsilon_0 r}$ and $E=\frac{5q}{4\pi\varepsilon_0 r^2}\hat{r}$

Correct Answer:
$\mathrm{V}=\frac{5q}{4\pi\varepsilon_0 r}$ and $E=0$

Solution:
$V=\left(\frac{5kq}{r}\right)=\frac{5q}{4\pi\epsilon_0 R}$

$|\overrightarrow{\mathrm{E}}|=0$ (Asit is a regular pentagon)

Q. 10 Three small identical bubbles of water having same charge on each coalesce to form a bigger bubble. Then the ratio of the potentials on one initial bubble and that on the resultant bigger bubble is:

Option 1:
$1:3^{1/3}$

Option 2:
$3^{2/3}:1$

Option 3:
$1:2^{2/3}$

Option 4:
$1:3^{2/3}$

Correct Answer:
$1:3^{2/3}$

Solution:
Let each water bubble has radius r.

Then $\mathrm{V_i}=\frac{\mathrm{Kq}}{\mathrm{r}}$ (on one bubble)

$V_f=\frac{K3q}{R}=\frac{3Kq}{R}$

From volume conservation, it yields, $3\left(\frac{4}{3}\pi \mathrm{r}^3\right)=\frac{4}{3}\pi \mathrm{R}^3$

$R=(3)^{\frac{1}{3}}$

$\therefore \frac{V_i}{V_f}=\frac{\mathrm{Kq/r}}{\mathrm{k3q/R}}=\frac{1}{3^{2/3}}$

$V_i:V_f=1:3^{2/3}$

Q. 11 In parallax method for the determination of focal length of a concave mirror, the object should always be placed:

Option 1:
Between the focus (F) and the centre of curvature (C) of the mirror ONLY

Option 2:
Between the pole (P) and the focus (F) of the concave mirror ONLY

Option 3:
Beyond the centre of curvature (C) of the mirror ONLY

Option 4:
At any point beyond the focus (F) of the mirror

Correct Answer:
Between the focus (F) and the centre of curvature (C) of the mirror ONLY

Solution:
Between the focus (f) & the center of curvature of the mirror(c) For more magnification (3) Beyond the centre of curvature of the mirror.

Q. 12 Given below are two statements:

Statement I: An object moves from position r_1 to position r_2 under a conservative force field F. The work done by the force is $W = -\int_{r_1}^{r_2} F.dr$.

Statement II: Any object moving from one location to another location can follow infinite number of paths. Therefore, the amount of work done by the object changes with the path it follows for a conservative force.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both statement I and statement II are false

Option 2:
Both statement I and statement II are true

Option 3:
Statement I is false but statement II is true

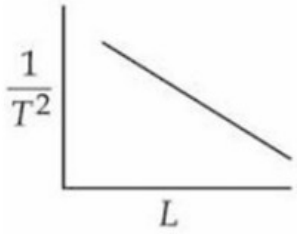
Option 4:
Statement I is true but statement II is false

Correct Answer:
Both statement I and statement II are false

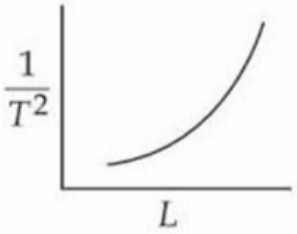
Solution:
Both statement-1 \& statement-2 are false work done in a closed path if it's
O , the work force associated with this force is conservative \& work done is $\mathrm{W} = \int_{\mathrm{r}_1}^{\mathrm{r}_2} \mathrm{F}.\mathrm{dr}$ which is independent of path.

Q. 13 Using a simple pendulum experiment g is determined by measuring its time period T . Which of the following plots represent the correct relation between the pendulum length L $\&$ time period T ?

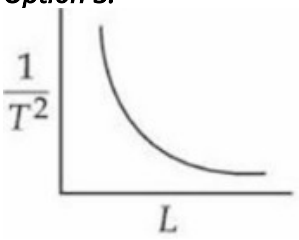
Option 1:



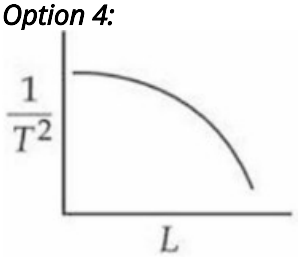
Option 2:



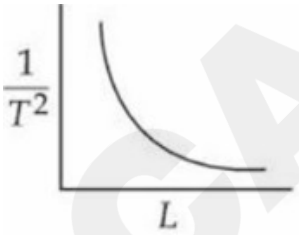
Option 3:



Option 4:



Correct Answer:



Solution:

$$T = 2\pi\sqrt{\frac{1}{g}} \Rightarrow T^2 = 4\pi^2\frac{l}{g}$$

$$\frac{1}{g} = \frac{T^2}{4\pi^2} \Rightarrow \frac{1}{T^2} = \frac{g}{4\pi^2}$$

$$\left(\frac{1}{T^2}\right)1 = \text{CONSTANT (RECTANGULAR HYPERBOLA)}$$

Q. 14 When a part of a straight capillary tube is placed vertically in liquid, the liquid raises upto certain height h. If the inner radius of the capillary tube, density of the liquid and surface tension of the liquid decrease by 1% each, then the height of the liquid in the tube will change by _____

Option 1:
-3

Option 2:
+1

Option 3:
+3

Option 4:
-1

Correct Answer:
+1

Solution:
$H = \frac{2T\cos\theta}{\rho g r} \Rightarrow \frac{dh}{h} = \frac{dT}{T} - \frac{d\rho}{\rho} - \frac{dr}{r} = -1\% + 1\% + 1\% = +1\%$

Q. 15 The correct truth table for the given input data of the following logic gate is:



Option 1:

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	0
1	0	1	0	1
1	1	1	1	1

Option 2:

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	0
1	1	1	1	1

Option 3:

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	1
1	1	1	1	0

Option 4:

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	1
1	0	1	0	1
1	1	1	1	1

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Correct Answer:

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	0
1	1	1	1	1

Solution:

$Y = (\overline{A \cdot B}) \cdot (C + D)$

$Y = A \cdot B + \overline{C + D}$

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	0
1	1	1	1	1

Q. 16 Consider two boxes containing ideal gases A and B such that their temperatures, pressures and number densities are same. The molecular size of A is half of that of B and mass of molecules A is four times that of B. If the collision frequency in gas B is $32 \times 10^{18}/\text{s}$ then collision frequency in gas A is _____ /s.

Option 1:
4×10^8

Option 2:
8×10^8

Option 3:
2×10^8

Option 4:
32×10^8

Correct Answer:
4×10^8

Solution:

$\mathrm{P, T, N} =$ same

collision frequency

$$z \propto \frac{d^2}{\sqrt{m}} =$$

$$Z_B = 32 \times 10^{8/3}$$

$$\frac{Z_A}{Z_B} = \frac{{d_A}^2/\sqrt{m_A}}{{d_B}^2/\sqrt{m_B}} = \left(\frac{d_A}{d_B}\right)^2 = \sqrt{\frac{m_B}{m_A}}$$

$$\frac{Z_A}{32 \times 10^8} = (1/2)^2 \times \sqrt{\frac{1}{4}\frac{m_A}{m_B}} = 4 \times 10^8/s$$

Q. 17 If $\in, \mathrm{E}$ and t represent the free space permittivity, electric field and time respectively, then the unit of $\frac{\in E}{t}$ will be :

Option 1:
Am^2

Option 2:
$\mathrm{A/m}^2$

Option 3:
Am

Option 4:
$\mathrm{A/m}$

Correct Answer:
$\mathrm{A/m}^2$

Solution:

$$\frac{\epsilon \cdot E}{t} = \frac{F/m \times N/C}{S}$$

$$\epsilon \cdot E = \frac{C}{m^2}$$

$$m^2 = A$$

Q. 18 Given below are two statements:

Statement-I: For a mechanical system of many particles total kinetic energy is the sum of kinetic energies of all the particles.

Statement-II: The total kinetic energy can be the sum of kinetic energy of the center of mass w.r.t. to the origin and the kinetic energy of the all the particles w.r.t. the center of mass as the reference.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement-I and Statement-II are true

Option 2:
Statement-I is true but Statement-II is false

Option 3:
Statement-I is false but Statement-II is true

Option 4:
Both Statement-I and Statement-II are false

Correct Answer:
Both Statement-I and Statement-II are true

Solution:
Both statements are true (1&2) correct explanation

Q. 19 Which of the following are true for a single slit diffraction?

A. Width of central maxima increases with increasing in wavelength keeping slit width constant
B. Width of central maxima increases with decrease in wavelength keeping slit width constant
C. Width of central maxima increases with decrease in slit width at constant wavelength.
D. Width of central maxima increases with increase in slit width at constant wavelength.
E. Brightness of central maxima increases for decrease in wavelength at constant slit width

Option 1:
B, D Only

Option 2:
B, C Only

Option 3:
A,D,E Only

Option 4:
A, D Only

Correct Answer:
A,D,E Only

Solution:
$\beta = \frac{\lambda D_2}{a}$

A, C are true
(A) $\beta \propto \lambda$
(C) $\beta \propto \frac{1}{d}$
(E) Shorter wave length's distance length.

Q. 20 Light is incident on a metallic plate having work function 110×10^{-20} J. If the produced photoelectrons have zero kinetic energy then the angular frequency of the incident light is ...rad/s. $\left(h = 6.63 \times 10^{-30}\ \mathrm{J/s}\right)$

Option 1:
1.04×10^{16}

Option 2:
1.66×10^{15}

Option 3:
1.04×10^{13}

Option 4:
1.66×10^{16}

Correct Answer:
1.04×10^{13}

Solution:

$$\mathrm{w} = \frac{2\pi \mathrm{E}}{\mathrm{h}} \quad \mathrm{KE} = \mathrm{W} = 110 \times 10^{-20}\ \mathrm{J}$$

$$= \frac{2 \times 3.14 \times 110 \times 10^{-20}}{6.6 \times 10^{-34}} = 104 \times 10^{14} = 1.04 \times 10^{16} \mathrm{rad/sec}$$

Q. 21 A conducting circular loop is rotated about is diameter at a constant angular speed of 100 $\mathrm{rad/s}$ in a magnetic field of 0.5 T perpendicular to the axis of rotation. When the loop is rotated by 30^0 from the horizontal position, the induced EMF is 15.4 mV . The radius of the loop is _____ mm.
(Take $\pi = \frac{22}{7}$)

Correct Answer:
14

Solution:

$$E(t) = BA\omega \sin(\omega t) = 15.4 \times 10^{-3} V$$

$$0.5 \times \frac{22}{7} x r^2 \times 100 \times \frac{1}{2} = 15.4 \times 10^{-3} V$$

$$\Rightarrow r = 14\ \mathrm{mm}$$

Q. 22 A capacitor P with capacitance $10 \times 10^{-6}\ \mathrm{F}$ is fully charged with a potential difference of 6.0 V and disconnected from the battery. The charged capacitor P is connected across another capacitor Q with capacitance $20 \times 10^{-6}\ \mathrm{F}$. The charge on capacitor Q when equilibrium is established will be $\alpha \times 10^{-5}\mathrm{C}$ (assume capacitor Q does not have any charge initially), the value of α is _____

Correct Answer:
40

Solution:
$C_P = 10\mu F \quad Q = 60\mu c$
$C_Q = 20\mu F$
$$Q \propto C \Longrightarrow \frac{Q_P}{Q_Q} = \frac{C_P}{C_Q} = \frac{10}{20}$$
$$\frac{Q_P}{Q_Q} = \frac{1}{2} \Rightarrow Q_Q = 40\mu C$$

Q. 23 An insulated cylinder of volume $60\ \mathrm{cm}^3$ is filled with a gas at $27^{\circ}\mathrm{C}$ and 2 atmospheric pressure. Then the gas is compressed making the final volume as $20\ \mathrm{cm}^3$ while allowing the temperature to rise to $77^{\circ}\mathrm{C}$. The final pressure is _____ atmospheric pressure.

Correct Answer:
7

Solution:
$\frac{\mathrm{P_1\,V_1}}{\mathrm{T_1}} = \frac{\mathrm{P_2\,V_2}}{\mathrm{T_2}}$
$$\mathrm{P_1} = 2,\ \mathrm{T_1} = 300,\ \mathrm{V_1} = 60$$
$$\mathrm{P_2} = ?,\ \mathrm{T_2} = 350,\ \mathrm{V_2} = 20$$
$$\Rightarrow \mathrm{P_2} = \frac{\mathrm{P_1\,V_1}}{\mathrm{T_1}} \cdot \frac{\mathrm{T_2}}{\mathrm{V_2}} = 7$$

Q. 24 A cylindrical conductor of length 2 m and area of cross-section $0.2\ \mathrm{mm}^2$ carries an electric current of 1.6 A when its ends are connected to a 2 V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3}\ \mathrm{m}^2/\mathrm{V.s}$. The value of α is: (electron concentration $= 5 \times 10^{28}/\mathrm{m}^3$ and electron charge $= 1.6 \times 10^{-19}\mathrm{C}$)

Correct Answer:
1

Solution:

$l = 2\ m$

$A = 0.2\ mm^2$

$i = 1.6\ A$

$V = 2v$

$n = 5 \times 10^{28}/m^3$

$$V_d = \frac{i}{neA}$$

$$\frac{V_d}{E} = \frac{V_d \cdot 1}{V} = \frac{i}{neA} \cdot \left(\frac{1}{v}\right) = \mu = 1 \times 10^{-3}\ m^2/v \cdot s$$

Q. 25 Two masses m and $2m$ are connected by a light string going over a pulley (disc) of mass 30 m with radius $r = 0.1$ m. The pulley is mounted in a vertical plane and it is free to rotate about its axis. The 2 m mass is released from rest and its speed when it has descended through a height of 3.6 m is _____ m/s. (Assume string does not slip and $g = 10\ m/s^2$)

Correct Answer:
2

Solution:
Loss of Gravitational potential energy of 2 m

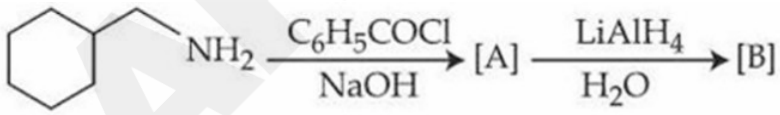
$$2\ m \cdot 10.36 = \frac{1}{2} \times 2\ m \times v^2 + \frac{1}{2} \times m \times v^2 + m \times 10(3.6) + \frac{1}{2} \times 30\ m \cdot \frac{r^2}{2} \cdot \frac{v^2}{r^2}$$

$$36\ m = mv^2 + \frac{mv^2}{2} + \frac{1}{2} \times 30\ m \times \frac{v^2}{2}$$

$$36 = 9v^2 \Rightarrow v = 2\ m/s$$

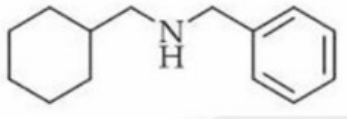
Chemistry

Q. 1

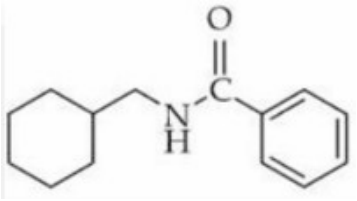


The final product [B] is:

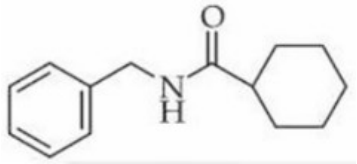
Option 1:



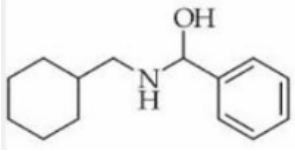
Option 2:



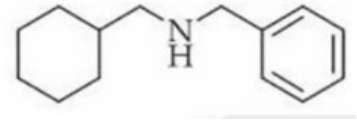
Option 3:



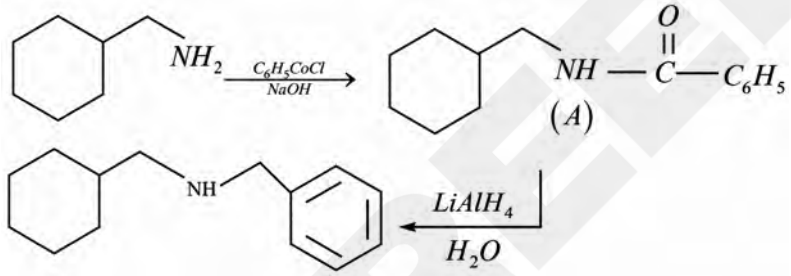
Option 4:



Correct Answer:



Solution:



Hence, the correct answer is the option (1).

Q. 2 When 1 g of compound (X) is subjected to Kjeldahl's method for estimation of nitrogen, $15\ \mathrm{mL}1\mathrm{M}\mathrm{H}_2\mathrm{SO}_4$ was neutralized by ammonia evolved. The percentage of nitrogen in compound (X) is:

Option 1:
42

Option 2:
0.21

Option 3:
0.42

Option 4:
21

Correct Answer:
42

Solution:
Equivalents of H_2SO_4 = Equivalents of ammonia

$$= \frac{1.5 \times 1 \times 2}{1000} = \text{ moles of ammonia } \times 1$$
$$= \text{ moles of Ammonia } = \text{ moles of Nitrogen}$$
$$= \text{ weight of nitrogen } = \frac{15 \times 1 \times 2}{1000} \times 14 = 0.42$$
$$\% \text{ of Nitrogen } = \frac{\text{weight of nitrogen}}{\text{weight of organic compound}} \times 100 = \frac{0.42}{1} \times 100 = 42\%$$

Hence, the correct answer is the option (1).

Q. 3 Which of the following mixture gives a buffer solution with $\mathrm{pH} = 9.25$?
Given: $\mathrm{pK}_b\ (\mathrm{NH_4OH}) = 4.75$

Correct Answer:
$0.2\mathrm{M}\mathrm{NH_4OH}(0.5\ \mathrm{L}) + 0.1\mathrm{M}\mathrm{HCl}(0.5\ \mathrm{L})$

Solution:
$\mathrm{NH_4OH} + \mathrm{HCl} \rightarrow \mathrm{NH_4Cl} + \mathrm{H_2O}$
$0.2\mathrm{M} \times 0.5$ lit 0.1×0.5

100 milli moles 50 milli moles 50 milli moles $100 - 50 = 50$ millimoles

$$POH = pkb + \log \frac{[\text{ salt }]}{[\text{ Base }]}$$

$$PH = 14 - 4.75 = 9.25$$

Hence, the correct answer is the option (3).

Q. 4 At $\mathrm{T(K)}, 100$ g of $98\%\mathrm{H_2SO_4(w/w)}$ aqueous solution is mixed with 100 g of $49\%\mathrm{H_2SO_4(w/w)}$ aqueous solution. What is the mole fraction of $\mathrm{H_2SO_4}$ in the resultant solution? (Given: Atomic mass $\mathrm{H} = 1\mathrm{u}; \mathrm{S} = 32\mathrm{u}; \mathrm{O} = 16\mathrm{u}$) (Assume that temperature after mixing remains constant)

Option 1:
0.663

Option 2:
0.9

Option 3:
0.337

Option 4:
0.1

Correct Answer:
0.337

Solution:
$98\%\mathrm{H_2SO_4}$ means 98 gm of $\mathrm{H_2SO_4}\&2\mathrm{gm}$ of $\mathrm{H_2O}$
$49\%\mathrm{H_2SO_4}$ means 49 gm of $\mathrm{H_2SO_4}\&51\mathrm{gm}$ of $\mathrm{H_2O}$
Total mass of solute $= 98 + 49 = 147\mathrm{gm}$
Total mass of solvent $= 2 + 51 = 53\mathrm{gm}$
No. of moles of $\mathrm{H_2SO_4} = \frac{147}{98} = 1.5$ moles of $\mathrm{H_2SO_4}$
No. of moles of $\mathrm{H_2O} = \frac{53}{18} = 2.94$ moles of $\mathrm{H_2O}$

$$X_{\mathrm{H_2SO_4}} = \frac{1.5}{1.5+2.94} = \frac{1.5}{4.44} = 0.337$$

Hence, the correct answer is the option (3).

Q. 5 Given below are two statements:
Statement I: Elements ' X ' and ' Y ' are the most and least electronegative elements, respectively among $\mathrm{N}, \mathrm{As}, \mathrm{Sb}$ and P . The nature of the oxides $\mathrm{X_2O_3}$ and $\mathrm{Y_2O_3}$ is acidic and amphoteric, respectively
Statement II: $\mathrm{BCl_3}$ is covalent in nature and gets hydrolysed in water. It produces $[\mathrm{B(OH)_4}]^-$ and $[\mathrm{B(H_2O)_6}]^{3+}$ in aqueous medium.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Statement I is true but statement II is false

Option 2:
Both statement I and statement II are false

Option 3:
Both statement I and statement II are true

Option 4:
Statement I is false but statement II is true

Correct Answer:
Statement I is true but statement II is false

Solution:
Statement-1
Electronegativity of 15th group

$$\underset{(3.0)}{N} > \underset{(2.1)}{p} > \underset{(2.0)}{As} > \underset{(1.9)}{Sb} = \underset{(1.9)}{Bi}$$

Most Electronegative - nitrogen - $\mathrm{N_2O_3}$
Least Electronegative - Sb \& Bi
$\mathrm{Sb_2O_3}$-Amphoteric Nature
$\mathrm{Bi_2O_3}$ - Basic Nature
Statement-2

$$\mathrm{BCl_3 + 3H_2O \rightarrow \underset{\text{BoricAcid}}{H_3BO_3} + 3HCl}$$

It produces Boric Acid
∴ Statement -2 is incorrect.

Hence, the correct answer is the option (1).

Q. 6 Consider the following reduction processes:

$$\mathrm{Al^{3+} + 3e^- \rightarrow Al(s), E^0 = -1.66\ V}$$
$$\mathrm{Fe^{3+} + e^- \rightarrow Fe^{2+}, E^0 = +0.77\ V}$$
$$\mathrm{Co^{3+} + e^- \rightarrow Co^{2+}, E^0 = +1.81\ V}$$
$$\mathrm{Cr^{3+} + 3e^- \rightarrow Cr(\ s), E^0 = -0.74\ V}$$

The tendency to acts as reducing agent decreases in the order:

Option 1:
$\mathrm{Al > Cr > Fe^{2+} \geq Co^{2+}}$

Option 2:
$\mathrm{Al > Cr > Co^{2+} > Fe^{2+}}$

Option 3:
$\mathrm{Al > Fe^{2+} > Cr > Co^{2+}}$

Option 4:
$\mathrm{Cr > Fe^{2+} > Al > Co^{2+}}$

Correct Answer:
$\mathrm{Al > Cr > Fe^{2+} \geq Co^{2+}}$

Solution:
More Negative the reduction potential values, that act as strong reducing agent

$Al^{+3}/Al = -1.60\ V$ $Fe^{3+}/Fe^{2+} = +0.77\ V$

$Co^{+3}/Co^{2+} = +1.81\ V$ $Cr^{+3}/Cr = -0.7\ V$

Decreasing order of reducing nature $Al > Cr > Fe^{+2} > Co^{+2}$

Hence, the correct answer is the option (1).

Q. 7 $[Ni(PPh_3)_2Cl_2]$ is a paramagnetic complex. Identify the INCORRECT statements about this complex.

A. The complex exhibits geometrical isomerism
B. The complex is white in colour
C. The calculated spin-only magnetic moment of the complex is 2.84 BM
D. The calculated CFSE (Crystal Field Stabilization Energy) of Ni in this complex is $-0.8\Delta_0$
E. The geometrical arrangement of ligands in this complex is similar to that in $Ni(CO)_4$.
Choose the correct answer from the options given below:

Option 1:
C and D only

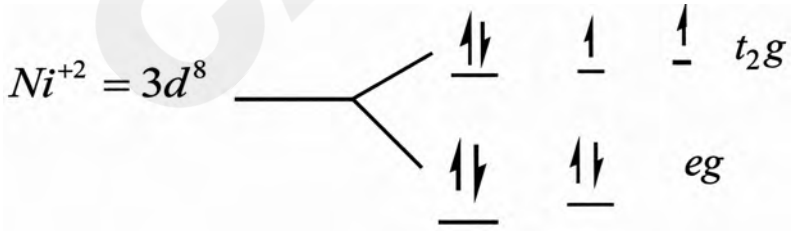
Option 2:
C,D and E only

Option 3:
A and B only

Option 4:
A,B and D only

Correct Answer:
A,B and D only

Solution:
$[Ni(pph_3)_2Cl_2] \Rightarrow$ oxidations state of Ni is +2



It is tetrahedral complex, paramagnetic nature
It is blue colour complex
$CFSE$ for d^8 is $-1.2\Delta_0$ irrespective of ligand
Tetrahedral complexes do not show geometrical isomerism
A, B, D incorrect

Hence, the correct answer is the option (4).

Q. 8 Correct statements regarding Arrhenius equation among the following are:

A. Factor $\mathrm{e}^{-Ea/RT}$ corresponds to fraction of molecules having kinetic energy less than Ea.
B. At a given temperature, lower the Ea , faster is the reaction
C. Increase in temperature by about $10^{\circ}\mathrm{C}$ doubles the rate of reaction
D. Plot of $\log \mathrm{k}$ vs $\frac{1}{\mathrm{T}}$ gives a straight line with slope $= -\frac{\mathrm{Ea}}{\mathrm{R}}$

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
B and C only

Option 3:
A and C only

Option 4:
B and D only

Correct Answer:
B and C only

Solution:
* The factor $e^{-Ea/RT}$ in the Arrhenius equation represents the fraction of molecules with kinetic energy greater than activation Energy

* Plot of $\log KVs\frac{1}{T}$ gives a straight line with slope $\frac{-Ea}{2.303R}$ B,C are correct

Hence, the correct answer is the option (2).

Q. 9 $\mathbf{A + 2B \rightarrow AB_2}$

36.0 g of 'A' (Molar mass : $60\ \mathrm{g\ mol^{-1}}$) and 56.0 g of 'B' (Molar mass : $80\ \mathrm{g\ mol^{-1}}$) are allowed to react. Which of the following statements are correct?
A. 'A' is the limiting reagent
B. 77.0 g of $\mathrm{AB_2}$ is formed
C. Molar mass of AB_2 is $140\ \mathrm{g\ mol^{-1}}$
D. 15.0 g of A is left unreacted after the completion of reaction Choose the correct answer from the options given below:

Option 1:
C and D only

Option 2:
A and B only

Option 3:
B and D only

Option 4:
A and C only

Correct Answer:
B and D only

Solution:
$A + 2B \rightarrow AB_2$

(A) Moles of $(\mathrm{A})\&(\ \mathrm{B}) = \frac{36}{60}\&\frac{56}{80} = 0.6$ mole $(\mathrm{A})\&0.7$ mole (B)

Baked on stoichiometric ration ' B ' is limiting reagent
(B) 2 moles of $\mathrm{B} \rightarrow$ produces 1 mole of AB_2
$0.7 \rightarrow (x)$
$x = \frac{0.7}{2} = 0.35$ mole of AB_2
Weight of formed $AB_2 = 0.35 \times$ mol. wt of AB_2

$0.35 \times 220 = 77\ \mathrm{g}$

(C) Molar mass of AB_2 is A-60, B-80

$\therefore 220$

(D) 1 mole of $\mathrm{A} \rightarrow$ reacts with 2 moles of B
$(x) \leftarrow 0.7$ moles of B

$x = \frac{0.7}{2} = 0.35(\ \text{Required of A})$

$\therefore 0.6 - 0.35 = 0.25$ mole of A is left
$\therefore 0.25 \times 60 = 15\ \mathrm{g}$ of ' n ' is left
$\therefore \mathrm{B}, \mathrm{D}$ Correct

Hence, the correct answer is the option (3).

Q. 10 The compound $\mathrm{A}, \mathrm{C_8H_8O_2}$ reacts with acetophenone to form a single product via crossaldol condensation. The compound A on reaction with conc. NaOH forms a substituted benzyl alcohol as one of the two products. The compound A is:

Option 1:
4-methoxy benzaldehyde

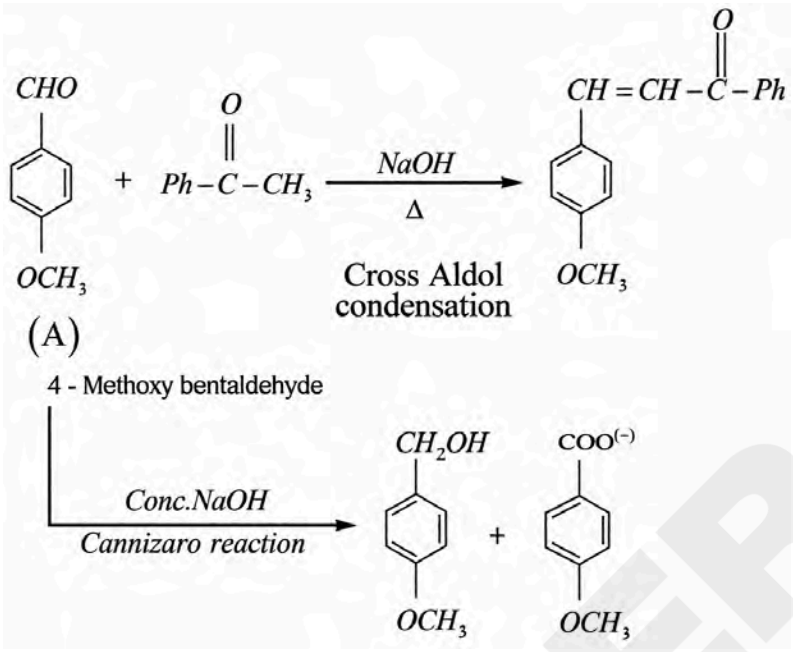
Option 2:
4-hydroxy benzylaldehyde

Option 3:
4-methyl benzoic acid

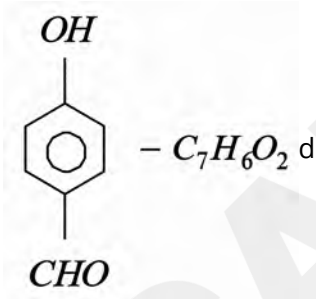
Option 4:
2-hydoxy acetophenone

Correct Answer:
4-methoxy benzaldehyde

Solution:

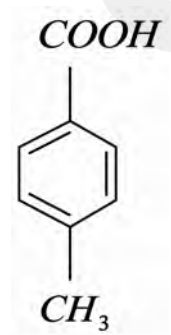


(1)



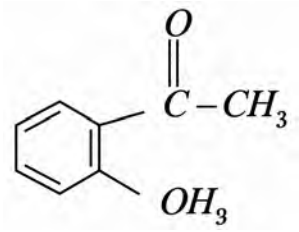
$-C_7H_6O_2$ does not match the molecular formula

(2)



Not Aldehyde & Ketone does not give Aldol (or) Cannizaro reactions

(3)



(4) do not undergoes canizzaro due to presence of α-Hydrogen

Hence, the correct answer is the option (1).

Q. 11 The energy of first (lowest) Balmer line of H atom is x J . The energy (in J) of second Balmer line of H atom is:

Option 1:
x^2

Option 2:
$\frac{x}{1.35}$

Option 3:
$1.35x$

Option 4:
$2x$

Correct Answer:
$1.35x$

Solution:
First line of Balmer $\Delta E_1 = RHhc\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = RHhc\left[\frac{5}{36}\right]$

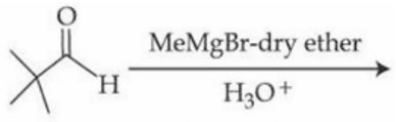
Second line of Balmer $\Delta E_2 = RHhc\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = RHhc\left[\frac{3}{16}\right]$

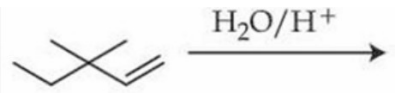
$\frac{\Delta E_2}{\Delta E_1} = \frac{\frac{3}{11}}{\frac{5}{36}} = \frac{3\times 36}{16\times 5} = \frac{108}{80} = \frac{27}{20}$

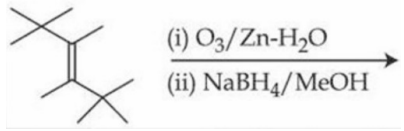
The energy of 2nd line $= \frac{27}{20}x = 1.35x$

Hence, the correct answer is the option (3).

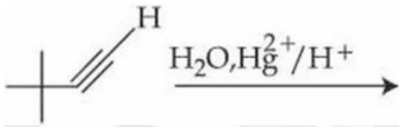
Q. 12 3, 3-Dimethyl-2-butanol cannot be prepared by:

A. 

B. 

C. 

D. 

E. 

Choose the correct answer from the options given below:

Option 1:
B and C only

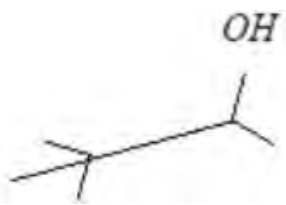
Option 2:
B and E only

Option 3:
B only

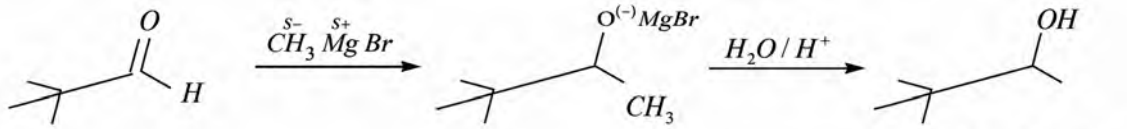
Option 4:
B,C and E only

Correct Answer:
B and E only

Solution:
3,3-dimethyl-2-butanol

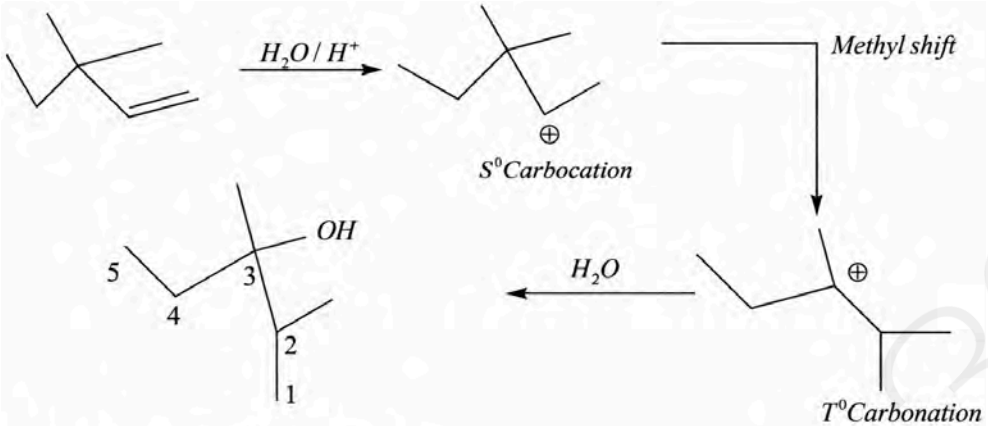


(A)



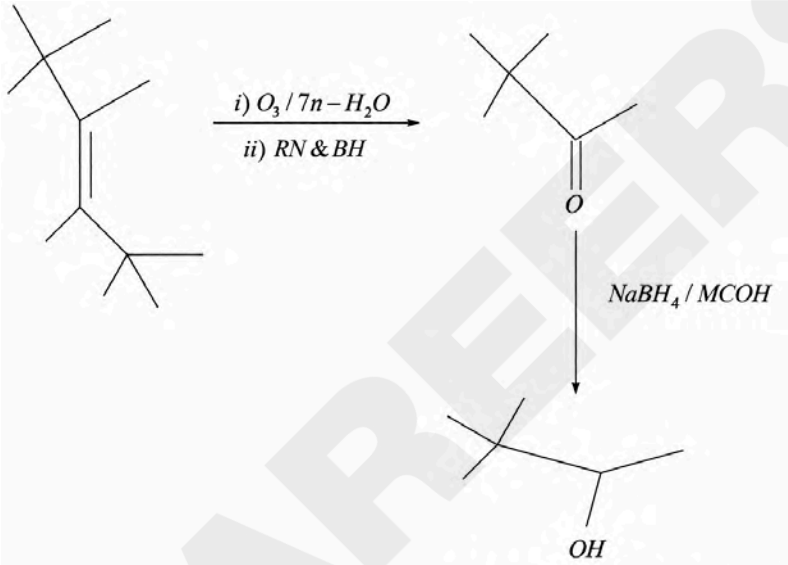
3,3 – dimethyl – 2 – butanol

(B)

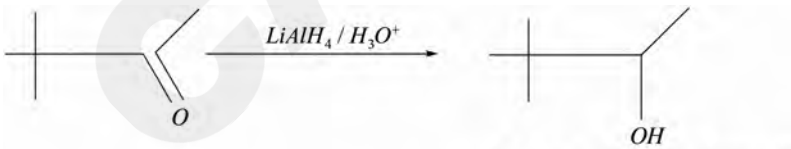


2,3 – dimethyl pentan – 3 – ol

(C)

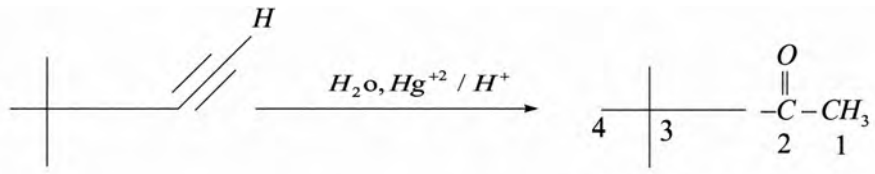


3,3 – dimethyl – 2 – butanol



3,3 – dimethyl – 2 – butanol

(D)



3,3 –dimethyl Butan – 2 – one

(E)

$\therefore \mathrm{B}, \mathrm{E}$ are correct

Hence, the correct answer is option (2)

Q. 13 Among $\mathrm{H}_2\ \mathrm{S}, \mathrm{H}_2\mathrm{O}, \mathrm{NF}_3, \mathrm{NH}_3$ and CHCl_3, identify the molecule (X) with lowest dipole moment value. The number of lone pairs of electrons present on the central atom of the molecule (X) is:

Option 1:
2

Option 2:
3

Option 3:
0

Option 4:
1

Correct Answer:
1

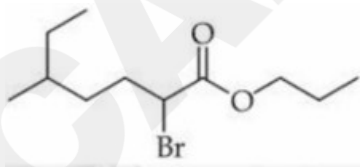
Solution:

Molecule	$\mathrm{H}_2\ \mathrm{S}$	$\mathrm{H}_2\mathrm{O}$	NF_3	NH_3	CHCl_3
Dipole moment	0.95	1.85	0.23	1.47	1.04

$\therefore NF_3$ has least dipole moment \& has one lone pair of electrons

Hence, the correct answer is the option (4).

Q. 14 The IUPAC name of the following compound is:



Option 1:
n-propyl-1-bromo-4-methylhexanoate

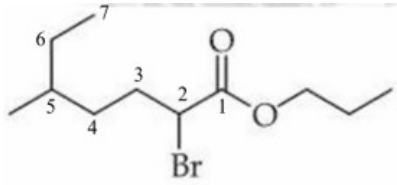
Option 2:
n-propyl-2-bromo-5-methylheptanoate

Option 3:
2-bromo-5-methylpropanoate

Option 4:
2-bromo-5-methylhexylpropanoate

Correct Answer:
n-propyl-2-bromo-5-methylheptanoate

Solution:



IUPAC name of ethers - Alkyl Alkanoate

n-propyl-2-bromo-5-methylheptanoate

Hence, the correct answer is the option (2).

Q. 15 Given below are two statements:

Statement I: $\mathrm{C} < \mathrm{O} < \mathrm{N} < \mathrm{F}$ is the correct order in terms of first ionization enthalpy values.

Statement II: $\mathrm{S} > \mathrm{Se} > \mathrm{Te} > \mathrm{Po} > \mathrm{O}$ is the correct order in terms of the magnitude of electron gain enthalpy values.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Statement I is true but statement II is false

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
Statement-I: Ionization energy of 2nd period

	C	N	O	F
$\mathrm{I.E/K.J/mol}$	1086	1402	1314	1681

$\therefore \mathrm{C} < \mathrm{O} < \mathrm{N} < \mathrm{F}$ IS CORRECT

Statement - II: Magnitude of electron gain enthalpy

$\text{Group} - 16:$	O	S	Se	Te	Po
$\Delta_{eg}/\text{K. J/mol}$	-141	-200	-195	-190	-174

$\therefore$ Both statements are correct

Hence, the correct answer is the option (2).

Q. 16 Match LIST-I with LIST-II.

	List-I		**List-II**
	Reaction of glucose with		Product Formed
A	Hydroxylamine	I	Gluconic acid
B	Br_2 water	II	Glucose pentacetate
C	Excess acetic anhydride	III	Saccharic acid
D	Concentrated HNO_3	IV	glucoxime

Choose the correct answer from the options given below:

Option 1:
$A - I, B - III, C - IV, D - II$

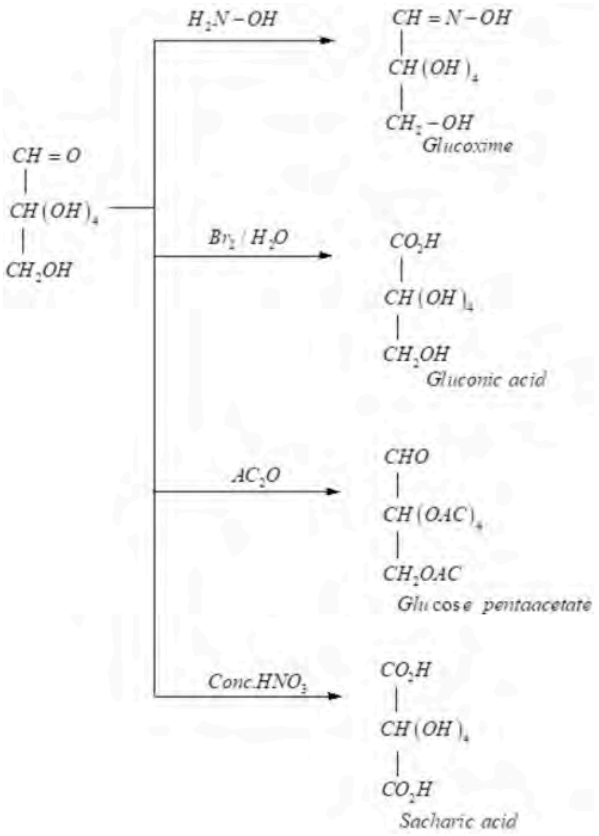
Option 2:
$A - III, B - I, C - IV, D - II$

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-IV, B-I, C-II, D-III

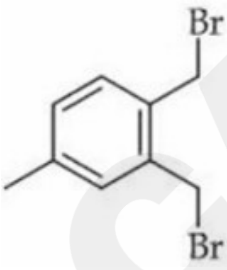
Solution:



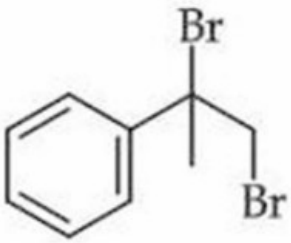
Hence, the correct answer is the option (3).

Q. 17 The dibromo compound $[\mathrm{P}]$ (molecular formula : $\mathrm{C_9H_{10}Br_2}$) when heated with excess sodamide followed by treatment with dilute HCl gives $[\mathrm{Q}]$. On warming $[\mathrm{Q}]$ with mercuric sulphate and dilute sulphuric acid yield $[R]$ which gives positive Iodoform test but negative Tollen's test. The compound $[\mathrm{P}]$ is:

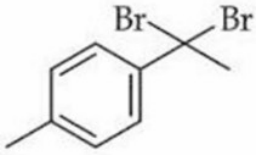
Option 1:



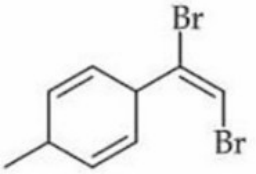
Option 2:



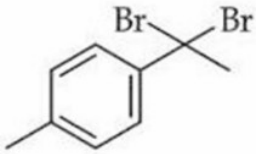
Option 3:



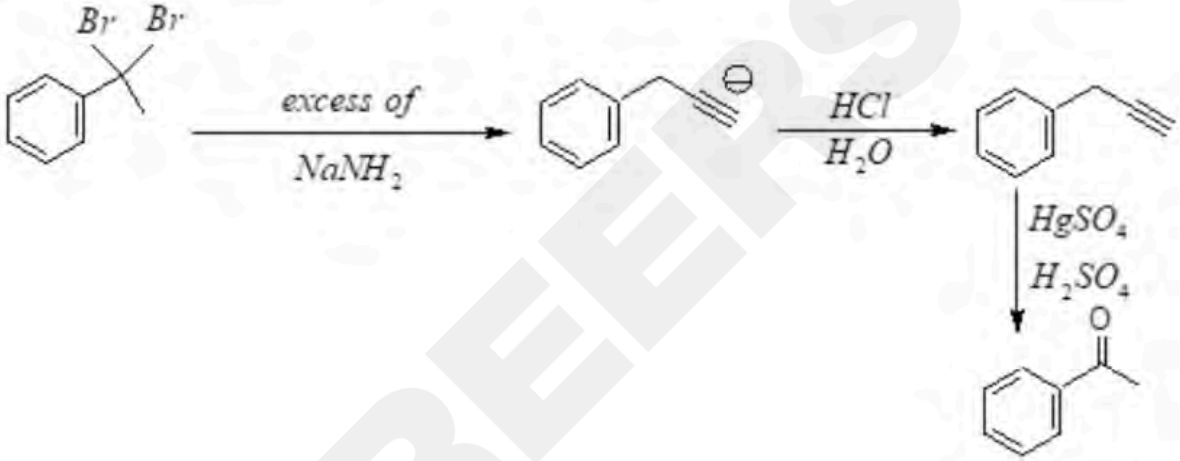
Option 4:



Correct Answer:



Solution:



Hence, the correct answer is the option (3).

Q. 18 Given below are two statements:

Statement I: The first ionization enthalpy of Cr is lower than that of Mn

Statement II: The second and third ionization enthalpies of Cr are higher than those of Mn.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both statement I and statement II are false

Option 2:
Statement I is true but statement II is false

Option 3:
Both statement I and statement II are true

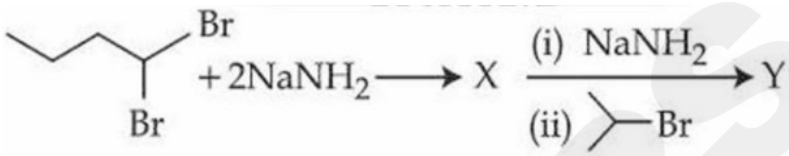
Option 4:
Statement I is false but statement II is true

Correct Answer:
Statement I is true but statement II is false

Solution:
$IE_1(Cr) = 653\text{KJ/mol}$
$IE_2(Cr) = 1592\text{KJ/mol}$
$IE_3(Cr) = 2990$
$IE_1(\text{Mn}) = 717$
$IE_2(\text{Mn}) = 1509$
$IE_3(\text{Mn}) = 3260$

Hence, the correct answer is the option (2).

Q. 19 Consider the following reaction:



The product Y formed is:

Option 1:
2-methylhex-2-yne

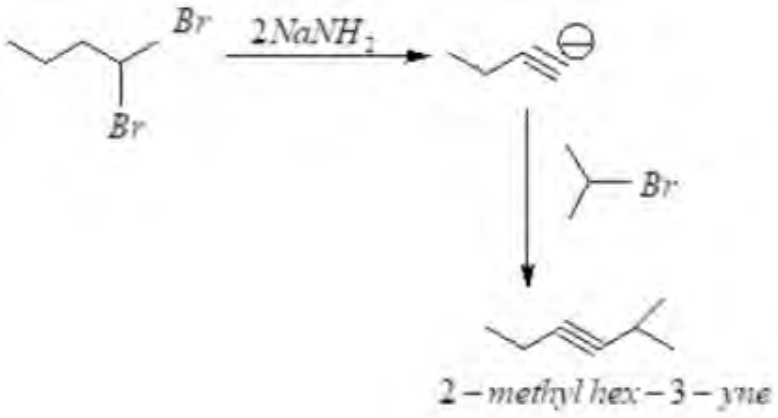
Option 2:
Isopropylbut-1-yne

Option 3:
5-methylhex-2-yne

Option 4:
2-methylhex-3-yne

Correct Answer:
2-methylhex-3-yne

Solution:



Hence, the correct answer is the option (4).

Q. 20 Identify the correct statements:

A. Hydrated salts can be used as primary standard
B. Primary standard should not undergo any reaction with air
C. Reactions of primary standard with another substance should be instantaneous and stoichiometric.
D. Primary standard should not be soluble in water
E. Primary standard should have low relative molar mass

Choose the correct answer from the options given below:

Option 1:
A,B and E only

Option 2:
A,B and C only

Option 3:
A,B,C and E only

Option 4:
D and E only

Correct Answer:
A,B and E only

Solution:
A) Hydrated Salt can be primary standard
B) Primary standard should not react with air
E) Primary standard should have low relative molar mass are correct

Hence, the correct answer is the option (1).

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Q. 21 Among the following oxides of 3d elements, the number of mixed oxides are _____

$Ti_2O_3,\ V_2O_4, Cr_2O_3, Mn_3O_4, Fe_3O_4, Fe_2O_3, Co_3O_4$

Correct Answer:
3

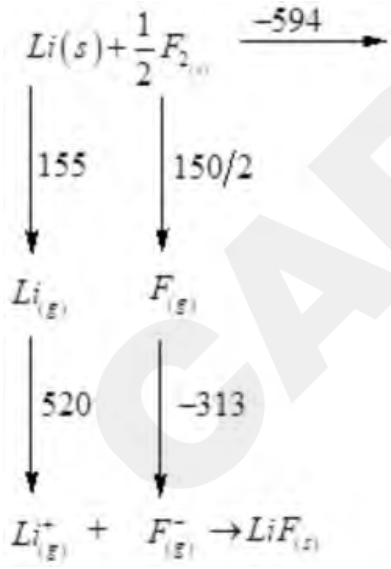
Solution:
$Mn_3O_4 \Rightarrow 2MnO, MnO_2$
$Fe_3O_4 \Rightarrow FeO, Fe_2O_3$
$Co_3O_4 \Rightarrow CoO, Co_2O_3$

Hence, the answer is (3).

Q. 22 If the enthalpy of sublimation of Li is $155\ \text{kJ mol}^{-1}$, enthalpy dissociation of F_2 is $150\ \text{kJ mol}^{-1}$, ionization enthalpy of Li is $520\ \text{kJ mol}^{-1}$, electron gain enthalpy of F is $-313\ \text{kJ mol}^{-1}$, standard enthalpy of formation of LiF is $-594\ \text{kJ mol}^{-1}$. The magnitude of lattice enthalpy of LiF is _____ kJmol^{-1} (Nearest integer).

Correct Answer:
1031

Solution:



$-594 = 155 + 520 + \frac{150}{2} - 313 + L \cdot E \Rightarrow L \cdot E = -1031$

Hence, the answer is (1031).

Q. 23 Consider the following electrochemical cell:

$$\mathrm{Pt} \mid \mathrm{O}_2(\mathrm{~g})(1 \text{ bar }) \left|\mathrm{HCl}(\mathrm{aq})\right\| \mathrm{M}^{2+}(\mathrm{aq} \cdot 1.0 \mathrm{M}) \left|\mathrm{M}(\mathrm{~s})\right.$$

The pH above which, oxygen gas would start to evolve at anode is _____ (nearest integer).

Given: $E^0_{M^{2+}/M} = 0.994$ V, $E^0_{O_2/H_2O} = 1.23$ V } standard reduction potential

and $\frac{RT}{F}(2.303) = 0.059$ V at the given condition

Correct Answer:
4

Solution:
For spontaneity $E_{\text{cell}} > 0$

$$E_{O \text{ id}}(\text{ Anode }) = -E_{\text{Red}}(\text{ Cathode })$$

$$\mathrm{H_2O} \rightarrow 2\mathrm{H^+} + 1/2\mathrm{O_2} + 2e^-$$

$$E_{\text{cell}} = E^0_{\text{cell}} - \frac{0.059}{2}\log\left[\frac{[\mathrm{H^+}]^2 \times \mathrm{PO_2^{1/2}}}{1}\right]$$

$$-0.994 = -1.23 + 0.059 \times p^H$$

$$p^H = 4$$

Hence, the answer is (4).

Q. 24 Consider $\mathrm{A} \xrightarrow{k_1} \mathrm{B}$ and $\mathrm{C} \xrightarrow{k_2} \mathrm{D}$ are two reactions. If the rate constant (k_1) of the $\mathrm{A} \rightarrow \mathrm{B}$ reaction can be expressed by the following equation $\log_{10} \mathrm{k} = 14.34 - \frac{1.5 \times 10^4}{\mathrm{T/K}}$ and activation energy of $\mathrm{C} \rightarrow \mathrm{D}$ reaction (Ea_2) is $\frac{1}{5}$ th of the $\mathrm{A} \rightarrow \mathrm{B}$ reaction (Ea_1), then the value of (Ea_2) is _____ $\mathrm{kJmol^{-1}}$ (Nearest Integer)

Correct Answer:
57

Solution:
For $A \rightarrow B \quad \log k = \log A - \frac{Ea_1}{2.303RT} \rightarrow (1)$

Given $\log_{10}^K = 14.34 - \frac{105 \times 10^4}{T/k} \rightarrow (2)$

Compare (1) \& (2)

$\frac{Ea_1}{2.303R} = 1.5 \times 10^4 \Rightarrow Ea_1 = 1.5 \times 10^4 \times 2.303 \times 8.314$
$Ea_1 = 287.207\text{KJ}$

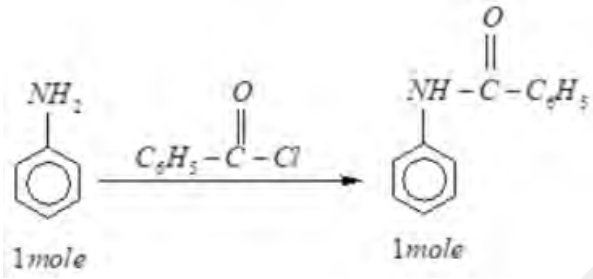
For $C \rightarrow D \Rightarrow Ea_2 = 1/5 \times Ea_1 \Rightarrow Ea_2 = 57.44\text{KJ} \Rightarrow Ea_2 = 57$ (Nearest integer)

Hence, the answer is (57).

Q. 25 The mass of benzanilide obtained from the benzoylation reaction of 5.8 g of aniline, if yield of product is 82%, is _____ g (nearest integer).
(Given molar mass in $\text{gmol}^{-1}\text{H}: 1, \text{C}: 12,\ \text{N}: 14, \text{O}: 16$)

Correct Answer:
10

Solution:



Given 5.8 grams (0.0623 mole)
Product is 82% yield then mole of

$\text{product} = \frac{0.0623 \times 82}{100} \Rightarrow 0.051 \text{ mole}$

Mass of product $= 0.051 \times 197 = 10.047$
Ans: 10 (Nearest integer)

Hence, the answer is (10).

Maths

Q. 1 If $\text{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations $\text{AX} = \text{B}$, Where

$\text{Adj } A = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$ then $|x + y + z|$ is equal to:

Option 1:
$\frac{3}{2}$

Option 2:
3

Option 3:
2

Option 4:
1

Correct Answer:
2

Solution:

$$X = A^{-1}B = \left(\frac{adjA}{|A|}\right)B = \pm\frac{1}{10}\begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix}\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} = \pm\frac{1}{10}\begin{pmatrix} 20 \\ -10 \\ 10 \end{pmatrix} = \pm\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$\therefore |x + y + z| = 2$

Hence the correct answer is option (3)

Q. 2 The area of the region $A = \{(x, y) : 4x^2 + y^2 \leq 8 \text{ and } y^2 \leq 4x\}$ is:

Option 1:
$\frac{\pi}{2} + 2$

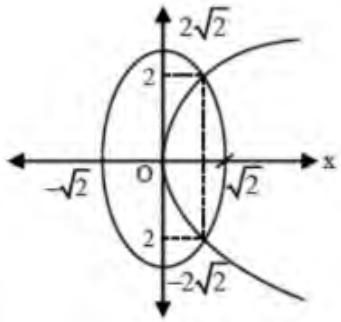
Option 2:
$\frac{\pi}{2} + \frac{1}{3}$

Option 3:
$\pi + 4$

Option 4:
$\pi + \frac{2}{3}$

Correct Answer:
$\pi + \frac{2}{3}$

Solution:



$$A = 2\int_0^1 2\sqrt{x}dx + 2\int_1^{\sqrt{2}} \sqrt{8-4x^2}dx = \frac{8}{3}\left(X^{\frac{3}{2}}\right)_0^1 + 4\int_1^{\sqrt{2}} \sqrt{2-x^2}dx$$

$$\frac{8}{3} + 4 \times \frac{1}{2}\left[x\sqrt{2-x^2} + 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\right]_1^{\sqrt{2}}$$

$$\frac{8}{3} + 2\left[2 \times \frac{\pi}{2} - 1 - 2 \times \frac{\pi}{4}\right] = \frac{8}{3} + 2\pi - 2 - \pi = \pi + \frac{2}{3} \text{ Sq. unit}$$

Hence the correct answer is option (4)

Q. 3 Let $\mathbf{P(10, 2\sqrt{15})}$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, whose foci are S and S'. If the length of its latus rectum is 8 , then the square of the area of $\triangle PSS'$ is equal to:

Option 1:
1462

Option 2:
2700

Option 3:
4200

Option 4:
900

Correct Answer:
2700

Solution:

$P(10, 2\sqrt{15})$ lies on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\therefore \frac{100}{a^2} - \frac{60}{b^2} = 1 \ldots\ldots\ldots\ldots (1)$

Length of latus rectum $= 8$

$\frac{2 \cdot b^2}{a} = 8 \Rightarrow \frac{b^2}{a} = 4 \ldots\ldots\ldots\ldots$

From (1) & (2)

$$\frac{100}{a^2}-\frac{60}{4a}=1$$
$$400-60a=4a^2$$
$$4a^2+60a-400=0$$
$$a^2+15a-100=0$$
$$a=5\&-20(\text{ rejected })$$
$$\Rightarrow b=\sqrt{20}$$

$\therefore$ Hyperbola is $\frac{x^2}{25}-\frac{y^2}{20}=1$

$$\therefore \text{ Focal length } SS'=2ae=2\cdot 5\cdot\left(\sqrt{1+\frac{4}{5}}\right)=6\sqrt{5}$$

$$\therefore \text{ Area of } \triangle PSS'=\frac{1}{2}\cdot 6\sqrt{5}\cdot 2\sqrt{15}=30\sqrt{3}=A$$

$$\therefore A^2=2700$$

Hence the correct answer is option (2).

Q. 4 Let $S=\left\{z\in:4z^2+\bar{z}=0\right\}$, then $\sum_{z\in S}|z|^2$ is equal to

Option 1:
$\frac{7}{64}$

Option 2:
$\frac{5}{64}$

Option 3:
$\frac{3}{16}$

Option 4:
$\frac{1}{16}$

Correct Answer:
$\frac{5}{64}$

Solution:
$$4z^2=\bar{z}=0$$
Let $z=x+iy$
$$4(x+iy)^2+x-iy=0$$
$$4x^2-4y^2+8xyi+x-iy=0$$
$$4x^2-4y^2+x=0\&y(8x-1)=0$$
$$\Rightarrow y=0 \text{ or } x=\frac{1}{8}$$

If $y=0, 4x^2+x=0$

$x = 0, \frac{-1}{4}$

$\therefore z_1 = 0 + 0i \quad |z_1|^2 = 0$

$z_2 = -\frac{1}{4} + 0i \quad |z_2|^2 = \frac{1}{16}$

If $x = \frac{1}{8}$

$4 \times \frac{1}{64} - 4y^2 + \frac{1}{8} = 0 \Rightarrow 4y^2 = \frac{3}{16} \Rightarrow y = \pm\frac{\sqrt{3}}{8}$

$\therefore z_3 = \frac{1}{8} + \frac{\sqrt{3}}{8}i \qquad |z_3|^2 = \frac{1}{64} + \frac{3}{64} = \frac{1}{16}$

$z_4 = \frac{1}{8} - \frac{\sqrt{3}}{8}i \qquad |z_4|^2 = \frac{1}{64} + \frac{3}{64} = \frac{1}{16}$

$\therefore \sum_{i=1}^{n} |z_i|^2 = 0 + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{3}{16}$

Hence the correct answer is option (2).

Q. 5 Let C_r denote the coefficient of x^r in the binomial expansion of $(1+x)^n, n \in N, 0 \le r \le n$. If $P_n = C_0 - C_1 + \frac{2^2}{3}C_2 - \frac{2^3}{4}C_3 + \ldots + \frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25} \frac{1}{P_{2n}}$ equals:

Option 1:
675

Option 2:
650

Option 3:
525

Option 4:
580

Correct Answer:
675

Solution:

$P_n = \sum_{r=0}^{n} \frac{{}^nC_r(-2)^r}{r+1} = \sum_{r=0}^{n} \frac{1}{(n+1)}{}^{n+1}C_{r+1}(-2)^r$

$= \frac{-1}{2(n+1)} \sum_{r=0}^{n} {}^{n+1}C_{r+1}(-2)^{r+1} = \frac{-1}{2(n+1)}\left[(1-2)^{n+1} - 1\right]$

$P_n = \frac{1}{2(n+1)}\left[1 - (-1)^{n+1}\right]$

$P_{2n} = \frac{1}{2(2n+1)}\left[1 - (-1)^{2n+1}\right]$

$P_{2n} = \frac{1}{2n+1}$

$\sum_{n=0} \frac{1}{P_{2n}} = \sum_{n=1}^{25}(2n+1) = 3 + 5 + \ldots\ldots + 51 = \frac{25}{2}[51+3] = 25 \times 27 = 675$

Hence the correct answer is option (1).

Q. 6 Let f and g be functions satisfying $f(x+y)=f(x)f(y), f(1)=7$ and $g(x+y)=g(xy), g(1)=1$, for all $x, y \in N$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right)=19607$, then n is equal to:

Option 1:
6

Option 2:
5

Option 3:
4

Option 4:
7

Correct Answer:
5

Solution:
$f(x+y)=f(x)\cdot f(y) \Rightarrow f(x)=a^x$
$\left(\quad f(1)=7 \Rightarrow a^1=7\right)$

So $f(x)=7^x$
Now

$g(x+y)=g(xy)(\text{ put } y=1) \Rightarrow g(x+1)=g(x)$

So $g(1)=g(2)=g(3)=\ldots\ldots\ldots=g(n)=1$

$$\text{Given } \sum_{x=1}^{n}\frac{f(x)}{g(x)}=19607$$

$$\sum_{x=1}^{n}\frac{7^x}{1}=19607 \Rightarrow 7\left(\frac{7^n-1}{7-1}\right)=19607$$

$$7^n-1=\frac{6}{7}\times 19607$$

$$7^n=16807 \Rightarrow n=5$$

Hence the correct answer is option (2).

Q. 7 Let $f(x)=[x]^2-[x+3]-3, x \in R$, where [ ] is the greatest integer function. Then

Option 1:
$f(x)>0$ only for $x \in [4,\infty)$

Option 2:
$f(x) = 0$ for finitely many values of x

Option 3:
$f(x) < 0$ only for $x \in [-1, 3)$

Option 4:
$\int_0^2 f(x)dx = -6$

Correct Answer:
$f(x) < 0$ only for $x \in [-1, 3)$

Solution:
$(3) \Rightarrow f(x) = 0 \Rightarrow [x] = 3 \text{ or } [x] = -2$
$\Rightarrow x \in [3, 4) \cup [-2, -1) \Rightarrow f(x) = 0 \text{ has infinite solutions}$
$f(x) = [x]^2 - [x + 3] - 3$
$= [x^2] - [x] - 6 = ([x] - 3)([x] + 2)$
$(2) \Rightarrow \int_0^2 f(x)dx = \int_0^2 \left([x]^2 - [x] - 6\right)dx$
$\int_0^2 f(x)dx = \int_0^2 \left([x]^2 - [x] - 6\right)dx = \int_0^2 \left([x]^2 - [x]\right)dx - 12 = -12$
$(4) \Rightarrow f(x) < 0 \Rightarrow ([x] - 3)([x] + 2) < 0 \Rightarrow -2 < [x] < 3 \Rightarrow -1 \leq [x] < 3$
$(1) \Rightarrow f(x) > 0 \Rightarrow [x] < -2 \text{ or } [x] > 3 \Rightarrow x < -2 \text{ or } x \geq 4 \Rightarrow x \in (-x, -2)0[4, \infty]$

Hence the correct answer is option (3).

Q. 8 Let α, β be the roots of the quadratic equation $12x^2 - 20x + 3\lambda = 0, \lambda \in \mathbf{Z}$. If $\frac{1}{2} \leq |\beta - \alpha| \leq \frac{3}{2}$, then the sum of all possible values of λ is:

Option 1:
3

Option 2:
4

Option 3:
1

Option 4:
6

Correct Answer:
3

Solution:

$$\frac{1}{2} \leq |\alpha-\beta| \leq \frac{3}{2}$$
$$\frac{1}{4} \leq |\alpha-\beta|^2 \leq \frac{9}{4}$$
$$\frac{1}{4} \leq (\alpha+\beta)^2 - 4\alpha\beta \leq \frac{9}{4}$$
$$\frac{1}{4} \leq \frac{25}{9} - 4 \times \frac{\lambda}{4} \leq \frac{9}{4}$$
$$-\frac{91}{36} \leq -\lambda \leq \frac{-19}{36}$$
$$\frac{19}{36} \leq \lambda \leq \frac{91}{36}$$
$$\lambda = 1, 2$$
$$\text{Sum } = 3$$

Hence the correct answer is option (1).

Q. 9 If the mean deviation about the median of the numbers $\mathrm{k}, 2\mathrm{k}, 3\mathrm{k} \ldots 1000\mathrm{k}$ is 500 , then $\mathbf{k}^2$ is equal to:

Option 1:
1

Option 2:
4

Option 3:
16

Option 4:
9

Correct Answer:
4

Solution:
Median $= \frac{1001k}{2} = X_M$

$$\therefore \text{ mean deviation about median } = \frac{\sum |X_i - X_M|}{n}$$
$$= \frac{2\left(\frac{k}{2} + \frac{3k}{2} + \frac{5k}{2} + \ldots\ldots 500 \text{ terms }\right)}{1000} = \frac{2 \cdot \frac{k}{2}(500)^2}{1000} = \frac{500k}{2} = 500(\text{ given })$$
$$\therefore k = 2$$
$$\therefore k^2 = 4$$

Hence the correct answer is option (2).

Q. 10 If $\lim_{x\to 0}\frac{e^{(a-1)x}+2\cos bx+(c-2)e^{-x}}{x\cos x-\log_e(1+x)}=2$, then $a^2+b^2+c^2$ is equal to:

Option 1:
9

Option 2:
7

Option 3:
3

Option 4:
5

Correct Answer:
7

Solution:

$$\lim_{x\to 0}\left[1+\frac{(a-1)}{1!}x+\frac{(a-1)^2}{2!}x^2+\ldots\ldots\ldots\infty\right]$$
$$+2\left[1-\frac{(bx)^2}{2!}+\ldots\ldots\infty\right]+(c-2)\left[1-\frac{x^1}{1!}+\frac{x^2}{2!}-\ldots\ldots\infty\right]$$
$$x\left(1-\frac{x^2}{2!}+\ldots\ldots\infty\right)-\left(x-\frac{x^2}{2}+\ldots\ldots\ldots\infty\right)$$
$$\Rightarrow\lim_{x\to 0}\frac{(c+1)+(a-c+1)x+\left[\frac{(a-1)^2}{2}-b^2+\frac{(c-2)}{2}\right]x^2+\ldots\ldots\ldots\infty}{\left(\frac{x^2}{2}-\frac{x^3}{2}+\ldots\ldots\ldots\ldots\right)}=2$$

$c+1=0\Rightarrow c=-1$

$a-c+1=0\Rightarrow a+1+1=0\Rightarrow a=-2$

$$\frac{\left(\frac{(a-1)^2}{2}-b^2+\frac{(c-2)}{2}\right)}{\left(\frac{1}{2}\right)}=2$$

$\Rightarrow 9-2b^2-3=2\Rightarrow 2b^2=4\Rightarrow b^2=2$

$\therefore a^2+b^2+c^2=4+2+1=7$

Hence the correct answer is option (2).

Q. 11 Let n be the number obtained on rolling a fair die. If the probability that the system

$$x - ny + z = 6$$
$$x + (n-2)y + (n+1)z = 8$$
$$(n-1)y + z = 1$$

has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is:

Option 1:
24

Option 2:
20

Option 3:
21

Option 4:
22

Correct Answer:
22

Solution:

$$x - ny + z = 6$$
$$x - (n-2)y + (n+1)z = 8$$
$$(n-1)y + z = 1$$

$$\begin{vmatrix} 1 & -n & 1 \\ 1 & (n-2) & n+1 \\ 0 & n-1 & 1 \end{vmatrix} = 0 \Rightarrow n = 1, 2 \text{ or } n = -1 \text{ (rejected)}$$

For unique solution $\mathrm{n} = 3, 4, 5, 6$
Now P (Probability when system of equations has unique solution) $= \frac{4}{6}$
So $\mathrm{k} = 4$
Now required sum $= 4 + (3 + 4 + 5 + 6) = 22$

Hence the correct answer is option (4).

Q. 12 Let the domain of the function

$$f(x) = \log_3 \log_5 \left(7 - \log_2 \left(x^2 - 10x + 85\right)\right) + \sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right) \text{ be } (\alpha, \beta].$$

Then $\alpha + \beta$ is equal to:

Option 1:
8

Option 2:
9

Option 3:
10

Option 4:
12

Correct Answer:
9

Solution:
let $x^2-10x+85=\lambda$
$\therefore$ Domain for first term $\lambda>0$.
& $7-\log_2\lambda>0\Rightarrow\lambda<2^7$.
& $\log_5(7-\log_2\lambda)>0\Rightarrow\lambda<2^6$.
$\therefore$ from (1),(2) & (3)
$0<\lambda<2^6$
$0<x^2-10x+85<64\Rightarrow x\in(3,7)$
& domain for second term $-1\leq\frac{3x-7}{x-17}\leq 1\Rightarrow x\in[-5,6]$

From (A) & (B), domain of function will be (3,6]
$\Rightarrow\alpha=3,\beta=6\Rightarrow\alpha+\beta=9$

Hence the correct answer is option (2)

Q. 13 Let L be the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z+3}{6}$ and let S be set of all points $(\mathrm{a},\mathrm{b},\mathrm{c})$ on L , whose distance from the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z-9}{0}$ along the line L is 7 . Then $\sum_{(\mathrm{a},\mathrm{b},\mathrm{c})\in\mathrm{s}}(a+b+c)$ is equal to

Option 1:
34

Option 2:
40

Option 3:
28

Option 4:
6

Correct Answer:
34

Solution:
M is the point of intersection of $L_1 \& L_2$

$\Rightarrow 2\lambda - 1 = 2\mu - 1, 3\lambda - 1 = 3\mu - 1, 6\lambda - 3 = 9 \Rightarrow \lambda = 2 = \mu$
$M(3, 5, 9)$

Now let point P be $(2K - 1, 3K - 1, 6K - 3)$ on L_2
Such that $\mathrm{PM} = 7$

$\Rightarrow \sqrt{(2K-4)^2 + (3K-6) + (6K-12)^2} = 7$
$\Rightarrow 49K^2 + 196 - 196K = 49$
$\Rightarrow K^2 + 4 - 4K = 1 \Rightarrow K^2 - 4K + 3 = 0 \Rightarrow K = 1, 3$

So points $P \& Q$ are $(1, 2, 3) \& (5, 8, 15)$ So sum of all co-ordinates of $P \& Q = 34$

Hence the correct answer is option (1)

Q. 14 Among the statements,
(S1): If $\mathrm{A}(5, -1)$ and $\mathrm{B}(-2, 3)$ are two vertices of a triangle, whose orthocentre is $(0, 0)$ then its third vertex is $(-4, -7)$ and
(S2): If positive numbers $2\mathrm{a}, \mathrm{b}, \mathrm{c}$ are three consecutive terms of an A.P., then the lines $ax + by + c = 0$ are concurrent at $(2, -2)$

Option 1:
Only (S1) is correct

Option 2:
Both are correct

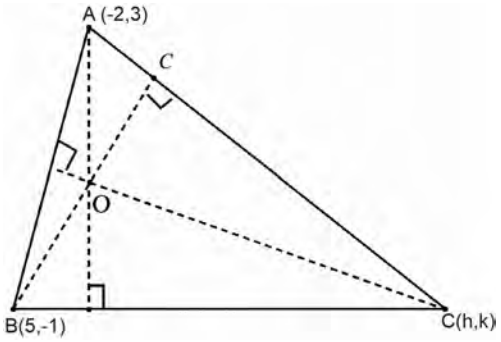
Option 3:
Only (S2) is correct

Option 4:
Both are incorrect

Correct Answer:
Both are incorrect

Solution:
Solution of statement-I

$m_{AO} m_{BC} = -1$



$\Rightarrow 2h - 3k = 13 \ldots\ldots\ldots (1)$

$\&m_{AB}m_{OC} = -1$
$\Rightarrow 4k = 7h \ldots\ldots\ldots$

⇒ third vertex is $(-4, -7)$
Solution of statement -2

$2a, b, c \rightarrow A \cdot P$
$b = \dfrac{2a + c}{2} \Rightarrow 2a - 2b + c = 0$

∴ lines $ax + by + c = 0$ are concurrent then

$\dfrac{x}{2} = \dfrac{y}{-2} = \dfrac{1}{1}$
$x = 2 \text{ and } y = -2$

∴ Pt. Of concurrency is $(2, -2)$
∴ Statement 2 is correct.

Hence the correct answer is option (4)

Q. 15 Let [.] denote the greatest integer function, and let $f(x) = \min\left\{\sqrt{2}x, x^2\right\}$. Let $S = \{x \in (-2, 2) : \text{the function } g(x) = |x|\left[x^2\right] \text{ is discontinuous at } x\}$. Then $\sum_{x \in S} f(x)$ equals:

Option 1:
$2\sqrt{6} - 3\sqrt{2}$

Option 2:
$2 - \sqrt{2}$

Option 3:
$1 - \sqrt{2}$

Option 4:
$\sqrt{6} - 2\sqrt{2}$

Correct Answer:
$\sqrt{6}-2\sqrt{2}$

Solution:
$g(x)=|x|\left[x^2\right]$

Points of discontinuity of g(x) in $(-2,2)$ are $(\pm 1,\pm\sqrt{2},\pm\sqrt{3})$

$$\therefore S=\{-1,1,-\sqrt{2},\sqrt{2}\cdot-\sqrt{3},\sqrt{3}\}$$
$$f(x)=\min\left\{\sqrt{2x},x^2\right\}$$
$$\therefore \sum_{x\in S} f(x)=-\sqrt{2}+1-2+2-\sqrt{6}+\sqrt{6}=1-\sqrt{2}$$

Hence the correct answer is option (4).

Q. 16 If $y=y(x)$ satisfies the differential equation

$16(\sqrt{x+9\sqrt{x}})(4+\sqrt{9+\sqrt{x}})\cos y dy=(1+2\sin y)dx, x>0$ and $y(256)=\frac{\pi}{2}, y(49)=\alpha$

then $2\,\mathrm{Sin}\,\alpha$ is equal to :

Option 1:
$3(\sqrt{2}-1)$

Option 2:
$\sqrt{2}-1$

Option 3:
$2(\sqrt{2}-1)$

Option 4:
$2\sqrt{2}-1$

Correct Answer:
$3(\sqrt{2}-1)$

Solution:
$\int \frac{\cos y}{1+2\sin y}dy=\int \frac{1}{16(\sqrt{x+9\sqrt{x}})(4+\sqrt{9+\sqrt{x}})}dx$

$\Rightarrow \frac{1}{2}\int \frac{2\cos y}{1+2\sin y}dy=\int \frac{1}{4t}dt$

Put $4+\sqrt{9+\sqrt{x}}=t$

$$\Rightarrow \frac{1}{2\sqrt{9+\sqrt{x}}} \cdot \frac{1}{2\sqrt{x}} dx = dt \Rightarrow \frac{1}{4\sqrt{x+9\sqrt{x}}} dx = dt$$

$$\Rightarrow \frac{1}{2}\log|1+2\sin y| = \frac{1}{4}\log|t| + \log c \Rightarrow \sqrt{1+2\sin y} = ct^{1/4} \Rightarrow \frac{1}{4}\log|t| + \log c$$

$$\Rightarrow \sqrt{1+2\sin y} = ct^{1/4} \Rightarrow \sqrt{1+2\sin y} = c(4+\sqrt{9+\sqrt{x}})^{1/4}$$

$$\left(256, \frac{\pi}{2}\right) \Rightarrow \sqrt{3} = c(\sqrt{3}) \Rightarrow c = 1$$

$$(49, \alpha) \Rightarrow \sqrt{1+2\sin\alpha} = (1)8^{1/4}$$

$$\Rightarrow 1+2\sin\alpha = 2\sqrt{2} \Rightarrow 2\sin\alpha = 2\sqrt{2}-1$$

Hence, the correct answer is option (1).

Q. 17 The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y \in \mathbf{Z}\}$ is

Option 1:
89

Option 2:
86

Option 3:
77

Option 4:
67

Correct Answer:
67

Solution:
$4x^2 + y^2 < 52, x, y \in Z$

$x = 0 \Rightarrow y = 0, \pm1, \pm2, \pm3, \pm4, \pm5, \pm6, \pm7 \rightarrow 1 \times 15 = 15$

$x = \pm1 \Rightarrow y = 0, \pm1, \pm2, \pm3, \pm4, \pm5, \pm6 \rightarrow 2 \times 13 = 26$

$x = \pm2 \Rightarrow y = 0, \pm1, \pm2, \pm3, \pm4, \pm5 \rightarrow 2 \times 11 = 22$

$x = \pm3 \Rightarrow y = 0, \pm1, \pm2, \pm3 \rightarrow 2 \times 7 = 14$

$15 + 26 + 22 + 14 = 77$

Number of elements $= 77$

Hence the correct answer is option (4)

Q. 18 Let $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\lambda\hat{j}+2\hat{k}, \lambda \in Z$ be two vectors, Let $\vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}|=\sqrt{53}$, then the maximum possible value of $(\vec{c}.\vec{d})^2$ is equal to:

Option 1:
52

Option 2:
104

Option 3:
26

Option 4:
208

Correct Answer:
208

Solution:

$$\vec{c}=\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&1\\0&\lambda&2\end{vmatrix}=(2-\lambda)\vec{i}-\hat{j}(4)+(2\lambda)\hat{k}$$

$$|\vec{c}|^2=53\Rightarrow(2-\lambda)^2+16+(2\lambda)^2=53=5\lambda^2-4\lambda-33=0$$

$$\Rightarrow 5\lambda^2-15\lambda+11\lambda-33=0$$

$$\Rightarrow 5\lambda(\lambda-3)+11(\lambda-3)=0$$

$$\Rightarrow(\lambda-3)(5\lambda+11)=0$$

$$\Rightarrow\lambda=3,\lambda\neq-\frac{11}{5}\quad(\lambda\in Z)$$

$$\therefore\vec{c}=(-1,-4,6)$$

Let $\therefore\vec{d}=(0,y,z)$

Given $|\vec{d}|=2\Rightarrow y^2+z^2=4$

$$\vec{c}.\vec{d}=0-4y+6z$$

$$(\vec{c}.\vec{d})^2=(-4y+6z)^2$$

$$=16y^2+36z^2-48yz$$

$$=16y^2+36\left(4-y^2\right)-48y\cdot\sqrt{4-y^2}$$

Let $\mathrm{f(y)}=-20y^2+144-48y\sqrt{4-y^2}$

$$f'(y)=-40y-48\left(\sqrt{4-y^2}\right)-48y\left(\frac{1}{2\sqrt{4-y^2}}\right)(-2y)$$

$$= -40y - 48\left(\frac{4-y^2-y^2}{\sqrt{4-y^2}}\right)$$

$$f'(y) = 0 \Leftarrow 40y = 96\left(\frac{y^2-2}{\sqrt{4-y^2}}\right)$$

$$\Rightarrow 5y\sqrt{4-y^2} = 12\left(y^2-2\right)$$

$$\Rightarrow 25y^2\left(4-y^2\right) = 144\left(y^4+4-4y^2\right)$$

$$= 169y^4 - 676y^2 + 576 = 0$$

$$= 169y^4 - 208y^2 - 468y^2 + 576 = 0$$

$$\Rightarrow 13y^2\left(13y^2-16\right) - 36\left(13y^2-16\right)$$

$$\Rightarrow y^2 = \frac{16}{13}(\text{ or })\frac{36}{13}$$

$$\Rightarrow y = \pm\frac{4}{\sqrt{13}}(\text{ or }) \pm \frac{6}{\sqrt{13}}$$

$$\therefore \text{ Maximum of } (\vec{c}.\vec{d})^2 = 208$$

Hence the correct answer is option (4)

Q. 19 Let the locus of the mid-point of the chord through the origin O of the parabola $y^2 = 4x$ be curve S . Let P be any point on S . Then the locus of the point, which internally divides OP in ratio $3:1$, is:

Option 1:
$3y^2 = 2x$

Option 2:
$2y^2 = 3x$

Option 3:
$2x^2 = 3y$

Option 4:
$3x^2 = 2y$

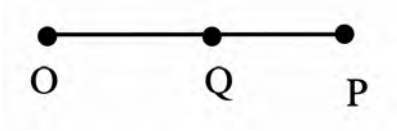
Correct Answer:
$2y^2 = 3x$

Solution:
Let $M\left(x_1, y_1\right)$ be the locus of the midpoint of the chord of the parabola $y^2 = 4x$
$\therefore$ Equation of the chord is $S_1 = S_{11} \Rightarrow yy_1 - 2\left(x+x_1\right) = y_1^2 - 4x_1$
It passes through $0(0,0)$

$$\therefore -2x_1 = y_1^2 - 4x_1$$

Locus of M is $y^2 = 2x$
Let $P\left(\frac{1}{2}t^2, t\right)$ be a point on this locus

$3:1$



$$Q(x,y)=\left(\frac{\frac{3t^2}{2}+0}{4},\frac{3t+0}{4}\right)\Rightarrow\left(\frac{3t^2}{8},\frac{3t}{4}\right)$$

$$\Rightarrow t^2=\frac{8x}{3},t=\frac{4y}{3}\Rightarrow\frac{16y^2}{9}=\frac{8x}{3}\Rightarrow 2y^2=3x$$

Hence the correct answer is option (2).

Q. 20 Let S and S' be the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ and $\mathrm{P}(\alpha,\beta)$ be a point on the ellipse in the first quadrant. If $(SP)^2+(S'P)^2-SP.S'P=37$, then $\alpha^2+\beta^2$ is equal to:

Option 1:
17

Option 2:
11

Option 3:
13

Option 4:
15

Correct Answer:
15

Solution:

$$\mathrm{P\ lies\ on\ ellipse}\ \Rightarrow\frac{\alpha^2}{25}+\frac{\beta^2}{9}=1$$

$$PS+PS'=2a\Rightarrow PS+PS'=10$$

$$(PS)^2+\left(PS'\right)^2-PS.PS'=37$$

$$\left(PS+PS'\right)-3PS.PS'=37$$

$$100-3PS.PS'=37$$

$$3PS.PS'=63\Rightarrow PS.PS'=21$$

$PS\&PS'$ are $\left(5 \pm \frac{4}{5} \cdot \alpha\right)$

$\therefore PS \cdot PS' = 25 - \frac{16}{25}\alpha^2 = 21$

$\frac{16}{25}\alpha^2 = 4$

$\alpha = \frac{5}{2} \Rightarrow \alpha^2 = \frac{25}{4}$

$\therefore \beta^2 = \frac{27}{4}$

$\therefore \alpha^2 + \beta^2 = \frac{52}{4} = 13$

Hence the correct answer is option (4).

Q. 21 Let [.] be the greatest integer function. If $\alpha = \int_0^{64} \left(x^{1/3} - \left[x^{1/3}\right]\right) dx$, then $\frac{1}{\pi}\int_0^{\alpha\pi} \left(\frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta}\right) \mathrm{d}\theta$ is equal to _____

Correct Answer:
36

Solution:

$$\int_0^{64} X^{\frac{1}{3}} dx = \frac{3}{4} \cdot \left[X^{\frac{4}{3}}\right]_0^{64} = 192\&$$

$$\int_0^{64} \left[X^{1/3}\right] dx = \int_0^1 \left[X^{1/3}\right] dx + \int_1^8 \left[X^{1/3}\right] dx + \int_8^{27} \left[X^{1/3}\right] dx \int_{27}^{64} \left[X^{1/3}\right] dx = 156$$

So $\alpha = 192 - 156 = 36$

Now $E = \frac{1}{\pi}\int_0^{36\pi} \frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta} d\theta = \frac{36}{\pi} \cdot 2\int_0^{\pi} \frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta} d\theta$

$\Rightarrow E = \frac{36}{\pi}\int_0^{\pi/2} \frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta} d\theta$

Let $J = \int_0^{\pi/2} \frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta} d\theta \ldots\ldots\ldots$

Applying King $J = \int_0^{\pi/2} \frac{\cos^2\theta}{\sin^6\theta+\cos^6\theta} d\theta \ldots\ldots\ldots\ldots (2).$

Now $2J = \int_0^{\pi/2} \frac{1}{\sin^6\theta+\cos^6\theta} d\theta (\text{add}(1)\&(2)) = \int_0^{\pi/2} \frac{\sec^6\theta}{\tan^6\theta+1} d\theta$

Put $\tan\theta = \lambda$

$$\int_0^\infty \frac{\left(1+\lambda^2\right)^2}{\lambda^4-\lambda^2+1}d\lambda$$

$$\int_0^\infty \frac{1+\frac{1}{\lambda^2}}{\lambda^2-1+\frac{1}{\lambda^2}}d\lambda=\pi$$

$$\Rightarrow J=\frac{\pi}{2}$$

$$E=\frac{36}{\pi}\cdot 2J=36$$

Hence, the correct answer is 36.

Q. 22 Let a vector $a=\sqrt{2}\hat{i}-\hat{j}+\lambda\hat{k},\lambda>0$, make an obtuse angle with the vector $b=-\lambda^2\hat{i}+4\sqrt{2}\hat{j}+4\sqrt{2}\hat{k}$ and an angle θ, $\frac{\pi}{6}<\theta<\frac{\pi}{2}$, with positive z -axis. If the set of all possible values of λ is $(\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to _____

Correct Answer:
5

Solution:
$a=\sqrt{2}\hat{i}-\hat{j}+\lambda\hat{k},\lambda>0$

$b=-\lambda^2\hat{i}+4\sqrt{2}\hat{j}+4\sqrt{2}\hat{k}$

Obtuse angle condition

$a\cdot b<0$

$a\cdot b=(\sqrt{2})(-\lambda^2)+(-1)(4\sqrt{2})+\lambda(4\sqrt{2})$

$=\sqrt{2}(-\lambda^2+4\lambda-4)$

$=\sqrt{2}[-(\lambda-2)^2]<0$

$\lambda\neq 2$

Angle with positive z-axis

$$\cos\theta=\frac{a\cdot\hat{k}}{|a|}=\frac{\lambda}{\sqrt{\lambda^2+3}}$$

$$\frac{\pi}{6}<\theta<\frac{\pi}{2}\Rightarrow 0<\cos\theta<\frac{\sqrt{3}}{2}$$

$$\frac{\lambda}{\sqrt{\lambda^2+3}}<\frac{\sqrt{3}}{2}$$

$$\frac{\lambda^2}{\lambda^2+3}<\frac{3}{4}$$

$4\lambda^2 < 3(\lambda^2 + 3)$

$\lambda^2 < 9$

$0 < \lambda < 3$

$(0, 3) - 2$

$\alpha = 0, \beta = 3, \gamma = 2$

$\alpha + \beta + \gamma = 5$

Hence, the correct answer is 5.

Q. 23 Let S be the set of first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2$ and the product of all elements of B is even $\}$ is _____

Correct Answer:
1979

Solution:
$A = \{1, 2, 3 \ldots\ldots\ldots 11\}$

$\therefore n(B) \geq 2$ \& product of all elements in B is even

Case (i): $n(\text{ B}) = 2 \Rightarrow {}^{11}C_2 - {}^{6}C_2$
$n(B) = 3 \Rightarrow {}^{11}C_3 - {}^{6}C_3$
$n(B) = 4 \Rightarrow {}^{11}C_4 - {}^{6}C_4$
$n(B) = 5 \Rightarrow {}^{11}C_5 - {}^{6}C_5$
$n(B) = 6 \Rightarrow {}^{11}C_6 - {}^{6}C_6$
$n(B) = 7 \Rightarrow {}^{11}C_7$
:
$n(B) = 11 \Rightarrow {}^{11}C_{11}$

$\therefore$ number of sets $B = \sum_{r=2}^{11} {}^{11}C_r - \sum_{r=2}^{6} {}^{6}C_r = 2^{11} - (12) - \left(2^6 - 7\right) = 2048 - 64 - 5 = 1979$

Alternate sol.
Total subsets $= 2^{11}$
No. of subsets having odd terms only $= 2^6$
No. Of subsets having one term only \& also having even terms $= 5$
Req. Ways $= 2^{11} - 2^6 - 5 = 1979$

Hence the correct answer is 1979

Q. 24 Let $\cos(\alpha + \beta) = -\frac{1}{10}$ and $\sin(\alpha - \beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$. If $\tan 2\alpha = \frac{3(1 - r\sqrt{5})}{\sqrt{11}(s + \sqrt{5})}, r, s \in \mathbf{N}$, then $r + s$ is equal to _____

Correct Answer:
20

Solution:
$\tan 2\alpha = \tan[(\alpha+\beta)+(\alpha-\beta)]$

$$\tan 2\alpha = \frac{\tan(\alpha+\beta)+\tan(\alpha-\beta)}{1-\tan(\alpha+\beta)\cdot\tan(\alpha-\beta)}$$

$$\tan 2\alpha = \frac{\left(-\sqrt{99}+\frac{3}{\sqrt{55}}\right)}{1-(\sqrt{99})\left(\frac{3}{\sqrt{55}}\right)}$$

$$\tan 2\alpha = \frac{-3\sqrt{11}+\frac{3}{\sqrt{5}\times\sqrt{11}}}{1+\frac{9\sqrt{11}}{\sqrt{5}\times\sqrt{11}}}$$

$$\tan 2\alpha = \frac{3(1-11\sqrt{5})}{\sqrt{11}(9+\sqrt{5})}$$

$r = 11, s = 9$

$r + s = 20$

Hence the correct answer is 20.

Q. 25 Suppose a, b, c are in A.P. and $a^2, 2b^2, c^2$ are in G.P. If $a < b < c$ and $a + b + c = 1$, then $9\left(a^2+b^2+c^2\right)$ is equal to ____

Correct Answer:
9

Solution:
$a = b - d, c = b + d, \Rightarrow b = \frac{1}{3} \Rightarrow 4b^4 = a^2c^2$

$4b^4 = [(b-d)(b+d)]^2$

$$\frac{4}{81} = \left(\frac{1}{9}-d^2\right)^2 \Rightarrow \frac{4}{81} = \frac{1}{81} - \frac{2d^2}{9} + d^4 \Rightarrow d^4 - \frac{2d^2}{9} - \frac{1}{27} = 0 \Rightarrow 27d^4 - 6d^2 - 1 = 0$$

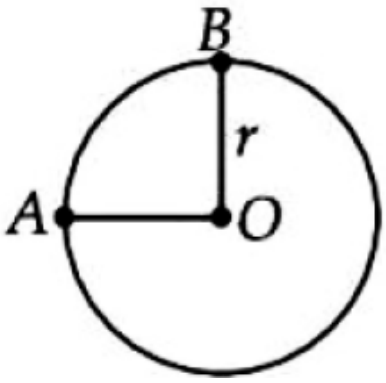
$$d^2 = 1/3 \Rightarrow d = +\frac{1}{\sqrt{3}} (\text{ as a} > \text{b} > \text{c})$$

$$9\left(a^2+b^2+c^2\right) = 9\left[\left(\frac{1}{3}-\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}+\frac{1}{\sqrt{3}}\right)^2\right] = 9\left[\frac{1}{3}+\frac{2}{3}\right] = 3+6 = 9$$

Hence the correct answer is 9.

Physics

Q. 1 A Wire of uniform resistance $\lambda\Omega/\mathrm{m}$ is bent into a circle of radius r and another piece of wire with length 2 r is connected between points A and $\mathrm{B}(\mathrm{AOB})$ as shown in figure. The equivalent resistance between point A and B is _____ Ω



Option 1:
$2\pi\lambda r$

Option 2:
$\frac{3\pi\lambda r}{8}$

Option 3:
$\frac{6\pi\lambda r}{3\pi+16}$

Option 4:
$(\pi+1)2r\lambda$

Correct Answer:
$\frac{6\pi\lambda r}{3\pi+16}$

Solution:
Sol : $R_1 = \lambda\frac{3}{4} \times 2\pi r = \frac{3\lambda\pi r}{2}$

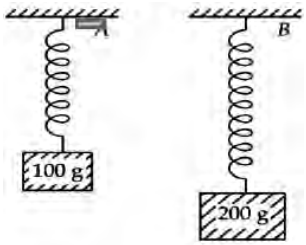
$R_2 = \lambda\frac{2\pi r}{4} = \frac{\lambda\pi r}{2}$

$$R_3 = \lambda 2r \quad R_{CH} = \frac{6\pi\lambda r}{16+3\pi}$$

$$\frac{1}{\mathrm{Re}} = \frac{1}{\lambda 2r} + \frac{2}{\lambda\pi r} + \frac{2}{2\lambda\pi r} = \frac{1}{\lambda r}\left[\frac{1}{2} + \frac{2}{\pi} + \frac{2}{3\pi}\right]$$

$$= \frac{\lambda}{\lambda r}\left(\frac{3\pi+12+4}{6\pi}\right) \quad R_{eff} = \frac{6\pi\lambda r}{3\pi+16}$$

Q. 2 Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B as shown in figure. The energy stored in A is E . The energy stored in B , when spring constants $k_A, k_B\mathrm{fA}$ and B respectively satisfy the relation $4k_A = 3k_B$ is



Option 1:
4 E

Option 2:
3E

Option 3:
2 E

Option 4:
$\frac{4}{3}E$

Correct Answer:
3E

Solution:
Sol:

Assume they are released from natural length

$$E = \frac{1}{2}mA^2w^2 \quad KA = mg = \frac{1}{2}KA^2$$
$$E = \frac{1}{2}K \cdot \left(\frac{mg}{k}\right)^2 = \frac{1}{2}\frac{(mg)^2}{K}$$
$$\frac{E_1}{E_2} = \frac{m_1^2}{m_2^2} \times \frac{k_2}{k_1} = \left(\frac{100}{200}\right)^2 \times \frac{K_B}{K_A} = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$$
$$E_2 = 3E$$

Q. 3 Four persons measure the length of a rod as $20.00\ \text{cm}, 19.75\ \text{cm}, 17.01\ \text{cm}$ and $18.25\ \text{cm}$. The relative error in the measurement of average length of the rod is:

Option 1:
0.06

Option 2:
0.08

Option 3:
0.24

Option 4:
0.18

Correct Answer:
0.06

Solution:
Sol : $20.00 + 19.75 + 17.01 + 18.25 = 75.01$

$x_m = 18.75$
$\Delta x_1 = -1.2475, \quad \Delta x_2 = -0.9975, \quad \Delta x_3 = 1.7425$
$\Delta x_2 = 0.5025$
$\Delta x_m = 1.1225$
$\frac{\Delta x}{x} = 00598 \quad = 0.06$

Q. 4 Given Below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R) Consider a ferromagnetic material:
Assertion (A) : The individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact with one another in such a way that they spontaneously align themselves forming domains.
Reason (R): At high enough temperature, the domain structure of ferromagnetic material disintegrate. Thus magnetization will disappear at high enough temperature known as Curie temperature.
In the light of the above statement choose the correct answer from the options given below:

Option 1:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Option 2:
(A) is true but (R) is false

Option 3:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Option 4:
(A) is false but (R) is true

Correct Answer:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Solution:
In ferromagnetic materials magnetic domains exist above curve temperature domains disappear

Q. 5 In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg moving in opposite directions with speeds of $10\ \mathrm{m/s}$ and $30\ \mathrm{m/s}$ respectively, strike each other and stick together. The rise in temperature (in C), if all the heat produced during the collision is retained by these spheres is:
(specific heat of sphere material $31\mathrm{cal/kg.}^{0}\mathrm{C}$ and $1\mathrm{cal} = 4.2\ \mathrm{J}$)

Option 1:
1.95

Option 2:
1.44

Option 3:
1.75

Option 4:
1.15

Correct Answer:
1.44

Solution:

$15 \times 10 - 25 \times 30 = 40 \times v$ (linear momentum conservation)

$150 - 750 = 40v$

$\Delta E = \frac{1}{2}mu_1^2 + \frac{1}{2}m_2u^2 - \frac{1}{2}(m_2 + m_2)w$

$\Delta E = 12000 - (20 \times 225) = 12000 - 4500$

$\Delta E = 7500$

$\Delta E = ms(\Delta T)$

$\Delta T = \frac{7500}{40 \times 31 \times 4.2} = 1.44$

Q. 6 Two small balls with masses m and 2 m are attached to both ends of a rigid rod of length d and negligible mass. If angular momentum of this system is L about an axis (A) passing through its centre of mass and perpendicular to the rod then angular velocity of the system about A is:

Option 1:
$\frac{3}{2}\frac{L}{md^2}$

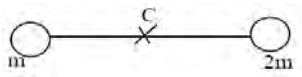
Option 2:
$\frac{4}{3}\frac{L}{md^2}$

Option 3:
$\frac{2L}{5md^2}$

Option 4:
$\frac{2L}{md^2}$

Correct Answer:
$\frac{3}{2}\frac{L}{md^2}$

Solution:
Sol : $x_n = \frac{m(0)+2md}{3m} = \frac{2d}{3}$



$I = m\left(\frac{2d}{3}\right)^2 + 2m \times \left(\frac{d}{3}\right)^2$

$md^2\frac{4}{9} + 2\frac{md^2}{9} = md^2\left(\frac{6}{9}\right) = \frac{2md^2}{3}$

$L = \frac{2md^2}{3}\omega$

Q. 7 A 20 m long uniform copper wire held horizontally is allowed to fall under the gravity $\left(g = 10\ \mathrm{m/s^2}\right)$ trough a uniform horizontal magnetic field of 0.5 Gauss perpendicular to the length of the wire. The induced EMF across the wire when it travels a vertical distance of 200 m is _____ mV

Option 1:
$2\sqrt{10}$

Option 2:
$0.2\sqrt{10}$

Option 3:
$20\sqrt{10}$

Option 4:
$200\sqrt{10}$

Correct Answer:
$20\sqrt{10}$

Solution:

$v = \sqrt{2gh}; e = Bvl$

$$\mathrm{e} = 0.5 \times 10^{-4} \times \sqrt{2 \times 10 \times 200} \times 20 = \frac{1}{2} \times 10 \times \frac{20\sqrt{10} \times 20}{10000} \text{ volts}$$
$$= \frac{2\sqrt{10}}{100} = \frac{\sqrt{10}}{100} \times \frac{10^3}{10^3} = 20\sqrt{10}\mathrm{mV}$$

Q. 8 A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produced dispersion without deviation. The angle of second prism is ____

Option 1:
5^0

Option 2:
6°

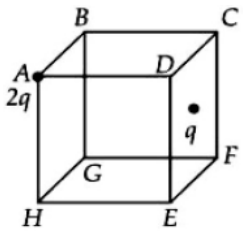
Option 3:
4^0

Option 4:
4.5°

Correct Answer:
4^0

Solution:
$A_2 = \frac{(\mu_1 - 1)A_1}{(\mu_2 - 1)} = \frac{5(0.72)}{0.9} = 4^0$

Q. 9 Two point charges $2q$ and q are placed at vertex A and centre of face CDEF of the cube as shown in figure. The electric flux passing through the cube is:



Option 1:
$\frac{3q}{2\epsilon_0}$

Option 2:
$\frac{3q}{\epsilon_0}$

Option 3:
$\frac{3q}{4\epsilon_0}$

Option 4:
$\frac{q}{\epsilon_0}$

Correct Answer:
$\frac{3q}{4\epsilon_0}$

Solution:
Sol: $\phi = \frac{1}{\in 0}\left(\frac{2q}{8} + \frac{q}{2}\right) = \frac{3q}{4\epsilon_0}$

Q. 10 In a screw gauge, the zero of the circular scale lies 3 divisions above the horizontal pitch line when their metallic studs are brought in contact. Using this instrument thickness of a sheet is measure If pitch scale reading is 1 mm and the circular scale reading is 51 then the correct thickness of the sheet is _____ mm.
[Assume least count is 0.01 mm]

Option 1:
1.50

Option 2:
1.48

Option 3:
1.51

Option 4:
1.54

Correct Answer:
1.54

Solution:
Error is negative $e = -3 \times L.C$

$$\text{Reducing } = (1 + 51 \times 0.01 + 3 \times 0.01)\text{mm} = 1.54 \text{ mm}$$

Q. 11 The moment of inertia of a square loop made of four uniform solid cylinders, each having radius R and length $\mathrm{L}(\mathrm{R} < \mathrm{L})$ about an axis passing through the mid points of opposite sides is (Take the mass of the entire loop as M)

Option 1:
$\frac{3}{8}MR^2 + \frac{7}{12}ML^2$

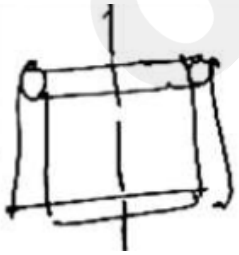
Option 2:
$\frac{3}{8}MR^2 + \frac{1}{6}ML^2$

Option 3:
$\frac{3}{4}MR^2 + \frac{7}{12}ML^2$

Option 4:
$\frac{3}{4}MR^2 + \frac{1}{6}ML^2$

Correct Answer:
$\frac{3}{8}MR^2 + \frac{1}{6}ML^2$

Solution:
$I = \left[\frac{m_1L^2}{12} + \frac{m_1R^2}{4}\right] \times 2 + \left[\frac{m_1R^2}{2} + m_1\left(\frac{L}{2}\right)^2\right] \times 2 \quad m_1 = \frac{M}{4}$

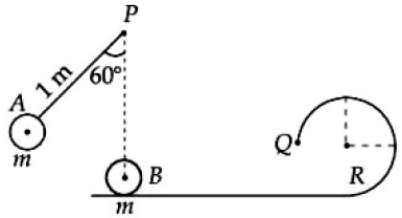


$$\frac{m_1L^2}{12}\times 2+\frac{m_1R^2}{4}+\frac{m_1R^2}{4}\times 2+\frac{m_1R^2}{2}\times 2m_1L^2\left[\frac{1}{6}+\frac{1}{2}\right]+m_1R^2\left[\frac{1}{2}+1\right]$$

$$m_1L^2\left(\frac{2+6}{12}\right)+\frac{3}{2}m_1R^2$$

$$\frac{ML^2}{6}+\frac{3MR^2}{8}$$

Q. 12 A small bob A of mass m is attached to a massless rigid rod of length 1 m pivoted at point P and kept at an angle of 60° with vertical as shown in figure. At distance of 1 m below point P , an identical bob B is kept at rest on a smooth horizontal surface that extends to a circular track of radius R as shown in figure. If bob B just manages to complete the circular path of radius R upto a point Q after being hit elastically by bob A, then radius R is _____ m.



Option 1:
$\frac{1}{5}$

Option 2:
$\frac{2-\sqrt{3}}{5}$

Option 3:
$\frac{2+\sqrt{3}}{5}$

Option 4:
$\frac{3}{5}$

Correct Answer:
$\frac{1}{5}$

Solution:

$u = \sqrt{2gl(1-\cos 60)}$

$\sqrt{10 \times 1} = \sqrt{10}$

$\sqrt{10} = \sqrt{5gR}$

$10 = 5gR$

$R = \frac{10}{5g} = \frac{1}{5}$

Q. 13 Match List - I with List-II

List-I	List-II
A) $\int E \cdot dl = -\frac{d}{dt}\int B \cdot da$	I) Ampere's circuital law
B) $\int B.dl = \mu_0\left(I + \epsilon_0 \frac{d\phi_E}{dt_0}\right)$	II) Faraday's law of electromagnetic induction
C) $\int E.da = \frac{1}{\epsilon_0}\int_v \rho dv$	III) Ampere – Maxwell law
D) $\int B.dl = \mu_0 I$	IV) Gauss's law of electrostatics

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-I, B-IV, C-III, D-II

Option 3:
A-II, B-III, C-I, D-IV

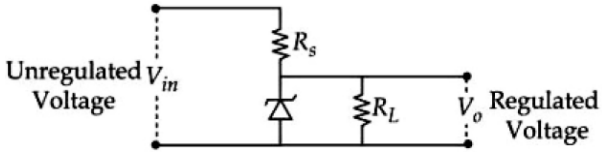
Option 4:
A-IV, B-I, C-II,D-III

Correct Answer:
A-I, B-IV, C-III, D-II

Solution:
Key :1 or 2

Sol:Conceptual

Q. 14 The following diagram shows a Zener diode as a voltage regulator. The Zener diode is rated at $V_z = 5\ \text{V}$ and the desired current in load is 5 mA . The unregulated voltage source can supply upto 25 V . Considering the Zener diode can withstand four times of the load current, the value of resistor R_S (Shown in circuit) should be _____ Ω.



Option 1:
4000

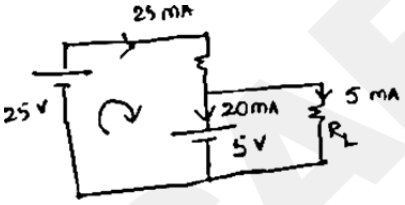
Option 2:
1000

Option 3:
100

Option 4:
10

Correct Answer:
4000

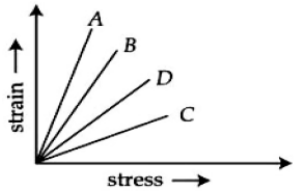
Solution:



$$25 - 25R_5 - 5 = 0$$

$$R_5 = \frac{20}{25}k\Omega = \frac{4}{5} = 0.8k\Omega = 800\Omega; R_s > 800\Omega$$

Q. 15 The strain-stress plot for materials A,B,C and D is shown in the figure. Which material has the largest Young's modulus?



Option 1:
C

Option 2:
B

Option 3:
D

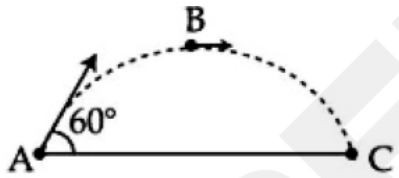
Option 4:
A

Correct Answer:
C

Solution:
$Y = \frac{\text{Sless}}{\text{Stain}} = \frac{1}{\text{slope}}$

C-least slope

Q. 16 An object is projected with kinetic energy K from a point A at an angle 60° with the horizontal. The ratio of the difference in kinetic energies at point B and C to that at point A (See figure), in the absence of air friction is:



Option 1:
1:2

Option 2:
1:4

Option 3:
3:4

Option 4:
2:3

Correct Answer:
3:4

Solution:
Sol:

At A $KE_A = \frac{1}{2}mv^2 = \mathrm{K}$
At B $KE_B = \frac{K}{4}$
At C $KE_C = \frac{1}{2}mv^2 = \mathrm{K}$

$\frac{K_B - K_C}{K_A} = \frac{\frac{K}{4} - K}{K} = \frac{3}{4}$

Q. 17 A simple pendulum of straight length 30 cm performs 20 oscillations in 10 s . The length of the string required for the pendulum to perform 40 oscillations in the same time duration is _____ cm [Assume that the mass of the pendulum remains same]

Option 1:
15

Option 2:
120

Option 3:
7.5

Option 4:
0.75

Correct Answer:
7.5

Solution:
Sol:

$T = 2\pi\sqrt{\frac{l}{g}}$

$f\alpha\frac{1}{\sqrt{l}}$

$\frac{f_2}{f_1} = \sqrt{\frac{l_1}{l_2}}$

$\frac{4}{2} = \sqrt{\frac{30}{l_2}}$

$l_2 = 7.5\ \mathrm{cm}$

Q. 18 Consider light travelling from a medium A to medium B separated by a plane interface, If the light undergoes total internal reflection during its travel from medium A to B and the speed of light in media A and B are $2.4 \times 10^8\ \mathrm{m/s}$ and $27 \times 10^8\ \mathrm{m/s}$, respectively then the value of critical angle is:

Option 1:
$\sin^{-1}\left(\frac{9}{8}\right)$

Option 2:
$\cot^{-1}\left(\frac{3}{\sqrt{13}}\right)$

Option 3:
$\cos^{-1}\left(\frac{8}{9}\right)$

Option 4:
$\tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$

Correct Answer:
$\tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$

Solution:
Sol:

$$\mu_A = \frac{c}{V_A} = \frac{3\times10^8}{2.4\times10^8} = \frac{10}{8}$$
$$\mu_B = \frac{c}{V_B} = \frac{3\times10^8}{2.7\times10^8} = \frac{10}{9}$$
$$\mu_A > \mu_B$$
$$\mu_A \sin i = \mu_B \sin r$$
$$\frac{10}{8}\sin i = \frac{10}{9}\sin 90^0$$
$$\sin i = \frac{8}{9}$$
$$i = \tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$$

Q. 19 In hydrogen atom spectrum ($R \rightarrow$ Rydberg's constant)
A) The maximum wavelength of the radiation of Lyman series is $\frac{4}{3R}$
B) The Balmer series lies in the visible region of the spectrum
C) The minimum wavelength of the radiation of Paschen series $\frac{9}{R}$
D) The minimum wavelength of Lyman series is $\frac{52}{4R}$

Choose the correct answer from the options given below:

Option 1:
A, B and C Only

Option 2:
A, B Only

Option 3:
B, D Only

Option 4:
A, B and D Only

Correct Answer:
A, B and C Only

Solution:
Sol:

$$\frac{1}{\lambda} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$\mathrm{A}: n_1 = 1; n_2 = 2$

$\lambda_{\max} = \dfrac{4}{3R}$

$\mathrm{B}:$ Correct

$\mathrm{C}: n_1 = 3, n_2 = \alpha$

$\lambda \text{ minimum } = \dfrac{9}{R}$

$\mathrm{D}: n_1 = 1; n_2 = \alpha$

$\lambda \text{ minimum } = \mathrm{R}$

Q. 20 The de Broglie wavelength of an oxygen molecule at $27^{\circ}\mathrm{C}$ is $x \times 10^{-12}$ m. The value of x is (take Planck's constant $= 6.63 \times 10^{-34}$ J. s Boltzmann constant $= 1.38 \times 10^{-23}$ J/K. mass of oxygen molecule $= 5.31 \times 10^{-26}$ kg)

Option 1:
20

Option 2:
24

Option 3:
26

Option 4:
30

Correct Answer:
26

Solution:
$\lambda = \frac{h}{p} = \frac{h}{\sqrt{3mkT}}$
$\lambda = 26 \times 10^{-12}$
$x = 26$

Q. 21 The equation the electric field of an electromagnetic wave propagating through free space is given by : $E = \sqrt{377}\sin\left(6.27 \times 10^3 t - 2.09 \times 10^{-5} x\right) N/C$

The average power of the electromagnetic wave is $\left(\frac{1}{\alpha}\right)\mathrm{W/m^2}$. The value of α is _____

$\left(\text{Take } \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \text{ in SI units}\right)$

Correct Answer:
2

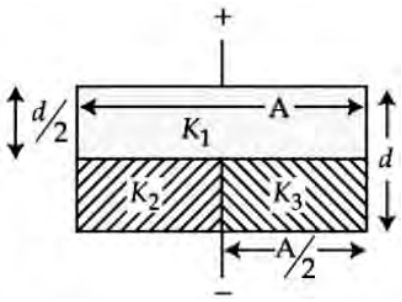
Solution:
$$\mathrm{I_{ave}} = \frac{E_0^2}{2Z} = \frac{(\sqrt{377})^2}{2 \times 377} = \frac{1}{2}$$
$\alpha = 2$

Q. 22 In two separate Young's double - slit experimental set-ups and two monochromatic light sources of different wavelength are used to get fringes of equal width. The ratios of the slits separations and that of the wavelengths of light used are $2:1$ and $1:2$ respectively. The corresponding ratio of the distance between the slits and the respective screens (D_1/D_2) is _____

Correct Answer:
4

Solution:
$\beta_1 = \beta_2$
$$\frac{D_1\lambda_1}{d_1} = \frac{D_2\lambda_2}{d_2}$$
$$\frac{D_1}{D_2} = \frac{d_1}{d_2} \times \frac{\lambda_2}{\lambda_1} = 4$$

Q. 23 The space between the plates of a parallel plate capacitor of capacitance C (without any dielectric) is now filled with three dielectric slabs of dielectric constants $K_1 = 2, K_2 = 3$ and $K_3 = 5$ (as shown in figure). If new capacitance is $\frac{n}{3}C$ then the value of n is _____



Correct Answer:
8

Solution:

$C = \frac{\varepsilon_0 A}{d}$

$C_1 = \frac{2\varepsilon_0 A}{d/2} = 4C$

$C_2 = 3C$

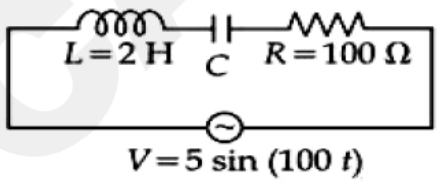
$C_3 = 5C$

$C_P = 3C + 5C = 8C$

$C_{eff} = \frac{4C \times 8C}{12C} = \frac{8C}{3} = \frac{n}{3}C$

$n = 8$

Q. 24 Using a variable frequency a.c. voltage source the maximum current measured in the given LCR circuit is 50 mA for $V = 5\sin(100t)$ The values of L and R are shown in the figure. The capacitance of the capacitor (C) used is _____ μF.



Correct Answer:
50

Solution:
Maximum current is possible in resonance condition

$x_L = x_C$

$L\omega = \frac{1}{c\omega} = c = \frac{1}{\omega^2 L}$

$c = \frac{1}{(100)^2(2)} = 50\mu F$

Q. 25 A Simple Pendulum made of mass 10 g and a metallic wire of length 10 cm is suspended vertically in a uniform magnetic field of 2 T . The magnetic field direction is perpendicular to the plane of oscillations of the pendulum. If the pendulum is released from an angle of 60° with vertical, then maximum induced EMF between the point of suspension and point of oscillations is _____ mV . (Take $g = 10\ \mathrm{m/s^2}$)

Correct Answer:
100

Solution:

$mgh = \frac{1}{2}mv^2$

$mgl\left[1 - \cos 60^\circ\right] = \frac{1}{2}mv^2$

$V = 1m/s$

$\mathrm{Emf} = \frac{1}{2}Bl^2\omega$

$\frac{1}{2}Blv = \frac{1}{2} \times 2 \times 0.1 \times 1 = 100\mathrm{mV\ s}$

Chemistry

Q. 1 The statements that are incorrect about the nickel (II) complex of dimethylglyoxime are:
A) It is red in colour.
B) It has a high solubility in water at $\mathrm{pH} = 9$.
C) The Ni ion has two unpaired d-electrons.
D) The $\mathrm{N - Ni - N}$ bond angle is almost close to 90°
E) The complex contains four five-membered metallacycles (metal containing rings)

Choose the correct answer from the options given below :

Option 1:
A, D and B only

Option 2:
C and D only

Option 3:
C and E only

Option 4:
B, C and E only

Correct Answer:
B, C and E only

Solution:
Sol: $\mathrm{NiDMG_2}$ shows dsp hybridization, dia magnetic and produces red coloured ppt. in ammonical solution $\mathrm{P^H9}$. It shows square planar structure and bond angle is 90°.

It consist of two five member rings and two six member rings.

Hence, the correct answer is the option (4).

Q. 2 Identify the molecule (X) with maximum number of lone pairs of electrons (obtained using Lewis dot structure) among $\mathrm{HNO_3}, \mathrm{H_2SO_4}, \mathrm{NF_3}$ and $\mathrm{O_3}$. Choose the correct bond angle made by the central atom of the molecule (X).

Option 1:
120°

Option 2:
102°

Option 3:
107^0

Option 4:
116°

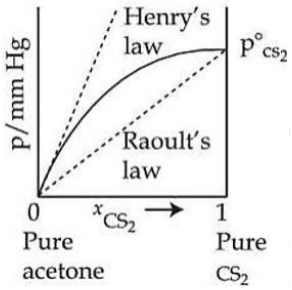
Correct Answer:
102°

Solution:
Sol : HNO_3 contains 7 lone pairs
H_2SO_4 contains 8 lone pairs
NF_3 contains 10 lone pairs
O_3 contains 6 lone pairs
Based on above data NF_3 contains more no. of lone pairs and bond angle nearly equals to 102^0.

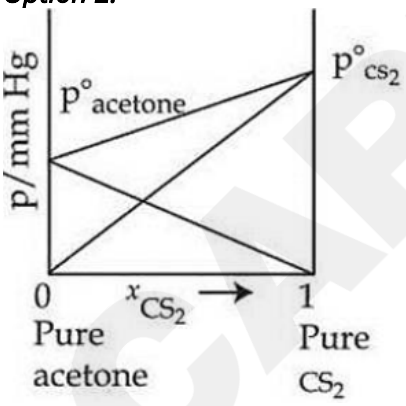
Hence, the correct answer is the option (2).

Q. 3 Which of the following graphs accurately represents the plot of partial pressure of CS_2 vs its mole fraction in a mixture of acetone and CS_2 at constant temperature?

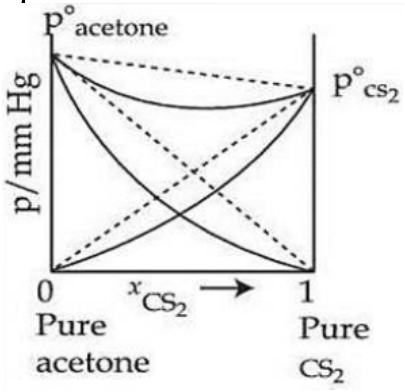
Option 1:



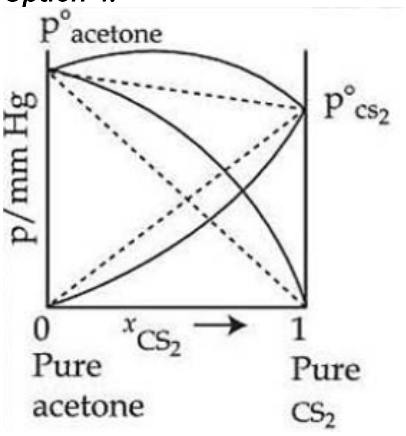
Option 2:



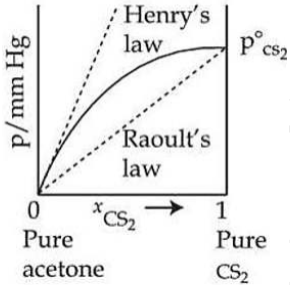
Option 3:



Option 4:



Correct Answer:



Solution:

CS_2 CH_3 CO CH_3

(non polar) (polar)

non ideal solution

Henry's law behaviour at low mole fraction.

Hence, the correct answer is the option (1).

Q. 4 Given below are two statements:

Statement – I : Sublimation is used for the separation and purification of compounds with low melting point.

Statement – II : The boiling point of a liquid increases as the external pressure is reduced.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:

Both Statement – I and Statement – II are false

Option 2:

Both Statement – I and Statement – II are true

Option 3:
Statement – I is false but Statement – II is true

Option 4:
Statement – I is true but Statement – II is false

Correct Answer:
Both Statement – I and Statement – II are true

Solution:
Sol: Both Statement – I and Statement – II are true
Sublimation is used for the separation and purification of compounds

Hence, the correct answer is the option (2).

Q. 5 Match List – I wish List – II:

List – I	List – II
Functional group (detection)	**Change observed during detection**
A) Unsaturation (Baeyer's test)	I) Red colour appears
B) Alcoholic group (Ceric ammonium nitrite test)	II) Silver mirror appears
C) Aldehyde group (Tollen's reagent)	III) Violet colour appears
D) Phenolic group$FeCl_3$ test	IV) Discharge of pink colour

Choose the correct answer from the options given below:

Option 1:
A – IV; B – III; C – II; D – I

Option 2:
A – III; B – IV; C – I; D – II

Option 3:
A – III; B – IV; C – II; D – I

Option 4:
A – IV; B – I; C – II; D – III

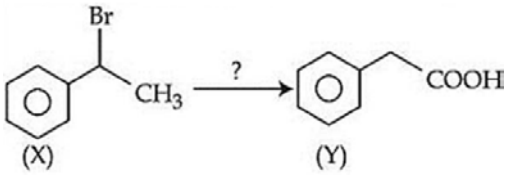
Correct Answer:
A – IV; B – I; C – II; D – III

Solution:
Sol : A) Baeyer's reagent is test for unsaturation (pink colour discharge)
B) CAN – red colour complex
C) Aldehyde group - Silver mirror appears

D) Phenolic group $\mathrm{FeCl_3}$ test - Violet colour appears

Hence, the correct answer is the option (4).

Q. 6



The correct sequence of reagents for the above conversion of X to Y is :

Option 1:
(i) $\mathrm{B_2H_6/H_2O_2}$ (ii) NaOEt (iii) Jones reagent

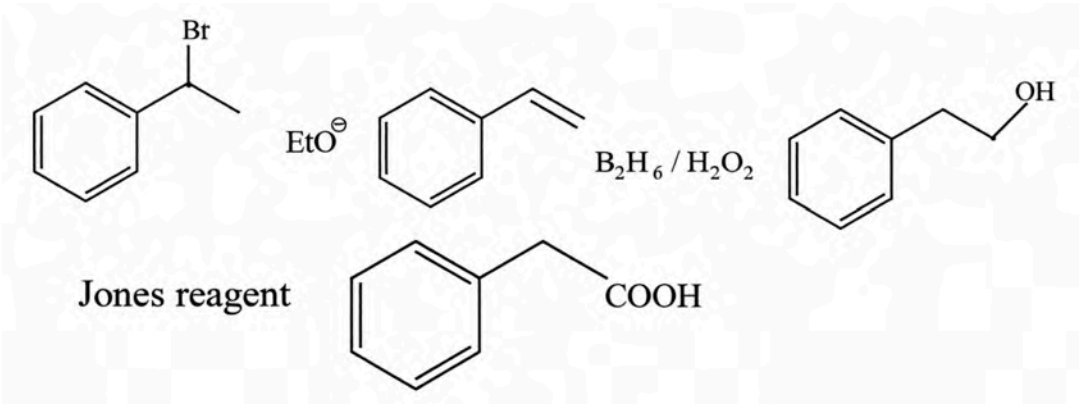
Option 2:
(i) $\mathrm{NaOH(aq)}$ (ii) Jones reagent (iii) $\mathrm{H_3O}$

Option 3:
(i) NaOEt (ii) $\mathrm{B_2H_6/H_2O_2}$ (iii) Jones reagent

Option 4:
(i) Jones reagent (ii) NaOEt (iii) $\mathrm{HotKMnO_4/KOH}$

Correct Answer:
(i) NaOEt (ii) $\mathrm{B_2H_6/H_2O_2}$ (iii) Jones reagent

Solution:



Hence, the correct answer is the option (3).

Q. 7 Given :

(A) $\mathrm{n} \quad 5,\ \mathrm{m}_1 \quad 1$

(B) $\mathrm{n} \quad 3, l \quad 2,\ \mathrm{m}_1 \quad 1,\ \mathrm{m}_\mathrm{s} \quad \frac{1}{2}$

The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

Option 1:
26 and 1

Option 2:
2 and 4

Option 3:
8 and 1

Option 4:
4 and 1

Correct Answer:
8 and 1

Solution:
Sol:

Part $-\mathrm{A} : \mathrm{n} = 5,\ \mathrm{m} = -1$

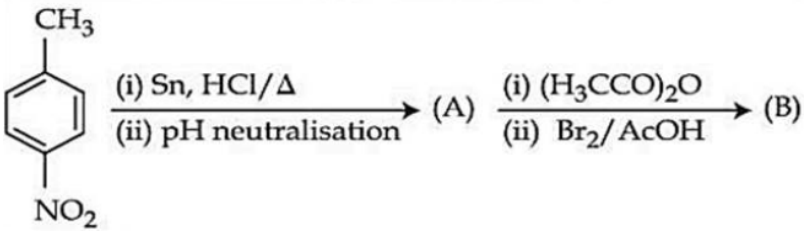
$\mathrm{m} = -1$ allowed l values are $l = 1, 2, 3, 4$

No. of orbitals $= 4$

Maximum of no. of electrons $= 4 \quad 2 \quad 8$
Part $-\mathrm{B}: \mathrm{n}=3, l=2, \ \mathrm{m}=-1, \ \mathrm{s}=+\frac{1}{2}$
Maximum no. of electrons $= 1$

Hence, the correct answer is the option (3).

Q. 8 Consider the following sequ3ences of reactions:



4- nitrotoluene
Assuming that the reaction proceeds to completion then 137 mg of 4-nitrotoluene will produce _____ mg of B .
(Given molar mass in $\mathrm{gmol}^{1}\mathrm{H}: 1, \mathrm{C}: 12, \ \mathrm{N}: 14, \mathrm{O}: 16, \mathrm{Br}: 80$)

Option 1:
208

Option 2:
301

Option 3:
228

Option 4:
146

Correct Answer:
228

Solution:

$$\text{mole } \frac{137 \quad 10^{3}}{137} \quad 0.001 \text{ mol}$$

$$\text{Mole of product } = 0.001 \text{ mole.}$$

$$\text{Mass of product } = 0.001 \quad 228\text{gr} \quad = 0.228 = 228\text{gr.}$$

Hence, the correct answer is the option (3).

Q. 9 Consider the general reaction given below at 400 K

$$xA_g \quad yB_g$$

The value of K_p and K_c are studied under the same condition of temperature but variation in x and y .

I) $K_p = 85.87$ and $K_c \quad 2.586$ appropriate units

II) $K_p \quad 0.862$ and $K_c \quad 28.62$ appropriate units

The values of x and y in (i) and (ii) respectively are :

Option 1:
(i) 3,1 (ii) 3,1

Option 2:
(i) 4,1 (ii) 4,1

Option 3:
(i) 1,3 (ii) 2,1

Option 4:
(i) 1,2 (ii) 2,1

Correct Answer:
(i) 1,2 (ii) 2,1

Solution:
A B
$K_P \quad K_C \quad RT$
At 400 k RT 0.082400
32.8 nearly 33

Case (i) $\frac{K_P}{K_C} \quad \frac{85.87}{2.586} \quad 33$
n 1
Case 2 : $\frac{K_P}{K_C} \quad \frac{0.862}{28.62} \quad 0.03$
$33^1 \quad 0.03$
$n \quad 1$

Hence, the correct answer is the option (4).

Q. 10 The correct statements from the following are:
A) Ionic radii of trivalent cations of group 13 elements decreases down the group.
B) Electro negativity of group 13 elements decreases down the group
C) Among the group 13 elements, Boron has highest first ionisation enthalpy
D) The trichloride and triiodide of group 13 elements are covalent in nature.
Choose the correct answer from the options given below :

Option 1:
C and D only

Option 2:
B and D only

Option 3:
A and D only

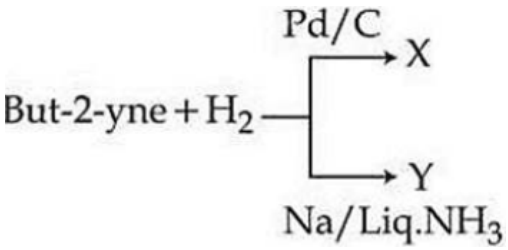
Option 4:
A and C only

Correct Answer:
C and D only

Solution:
Order of ionic radii: $B^3 \quad Al^3 \quad Ga^3 \quad In^3 \quad Tl^3$
Electro negativity: $B > Tl > In > Ga > Al$
Ionization enthalpy: $B > Tl > Ga > Al > In$
MX_3 shows covalent nature.

Hence, the correct answer is the option (1).

Q. 11 But-2-yne and hydrogen (one mole each) are separately treated with i Pd/C and ii $Na/$ liq. NH_3 to give the products X and Y respectively.



Identify the incorrect statements :
A) X and Y are stereo isomers
B) Dipole moment of X is zero
C) Boiling point of X is higher than Y
D) X and Y react with O_3/Zn H_2O to give different products

Choose the correct answer from the options given below :

Option 1:
A and C only

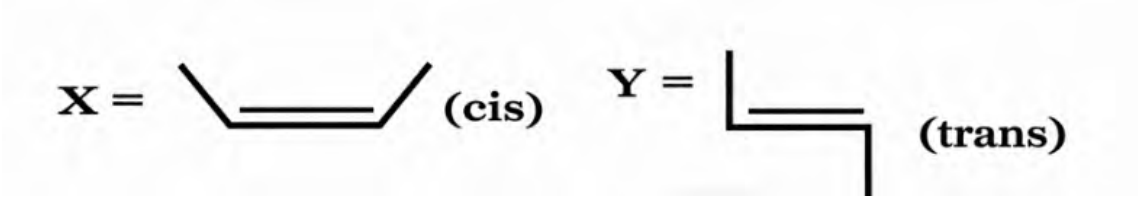
Option 2:
B and C only

Option 3:
B and D only

Option 4:
A and B only

Correct Answer:
B and D only

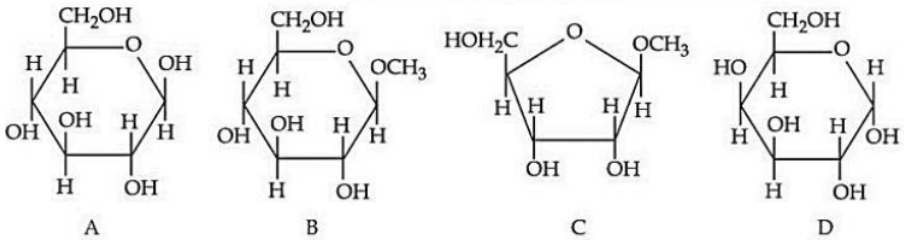
Solution:



X is having higher dipole moment than Y .
X, Y O_3/Zn gives same product ethanol

Hence, the correct answer is the option (3).

Q. 12 From the given following (A to D) cyclic structures, those which will not react with Tollen's reagent are :



Option 1:
B and D

Option 2:
A and B

Option 3:
B and C

Option 4:
A and D

Correct Answer:
B and C

Solution:
Tollens' Test gives a positive result with aldehydes or hemiacetals (because they can open to an aldehyde).
But acetals do NOT react because they lack the free anomeric –OH and cannot form an aldehyde.

In the given structures

- B and C are acetals (protected carbonyl, no free –OH).
- Hence they cannot open to aldehyde → no Tollens reaction.

Hence, the correct answer is the option (3).

Q. 13 ' x ' is the product which is obtained from propanenitrile and stannous chloride in the presence of hydrochloric acid, followed by hydrolysis. ' y ' is the product which is obtained from the but-2-ene by the ozonolysis, followed by hydrolysis. From the following, which product is not obtained when one mole of ' x ' and one mole of ' y ' react with each other in the presence of alkali, followed by heating?

Option 1:
Pent-2-enal

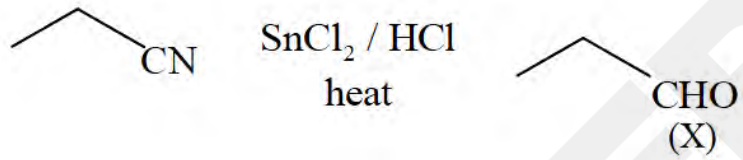
Option 2:
3-Methylbut-2-enal

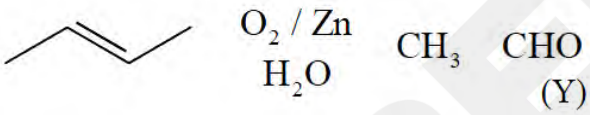
Option 3:
2-Methylpent-2-enal

Option 4:
2-methylbut-2-enal

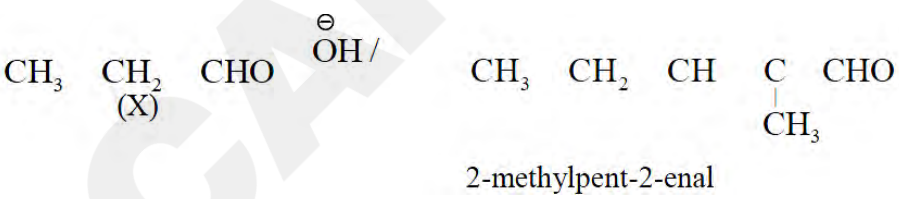
Correct Answer:
3-Methylbut-2-enal

Solution:



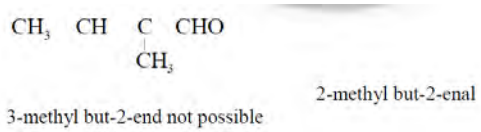


X Y $\overset{\ominus}{OH}$ / possible products



$CH_3\underset{(Y)}{CHO} \overset{\ominus}{OH}/= CH_3 \underset{C}{} CHCH \underset{\text{but-2-enal}}{} CHO$

$CH_3\ CH_2\ CHO\ CH_3\ CHO\ CH_3\ CH_2\ CH\ CH\ CHO$ pent-2-enal



Hence, the correct answer is option (2)

Q. 14 In the given electrochemical cell,
$\mathrm{Ag\ \ s|AgCls|FeCl_2aq, FeCl_3aq\ |\ Pt\ \ s}$ at 298 K , the cell potential $\mathrm{E_{cell}}$ will increase when :
A) Concentration of $\mathrm{Fe^2}$ is increased
B) Concentration of $\mathrm{Fe^3}$ is decreased
C) Concentration of $\mathrm{Fe^2}$ is decreased
D) Concentration of $\mathrm{Fe^3}$ is increased
E) Concentration of Cl is increased

Choose the correct answer from the options given below :

Option 1:
A and E only

Option 2:
B only

Option 3:
C, D and E only

Option 4:
A and B only

Correct Answer:
C, D and E only

Solution:
Overall reaction:

$$\mathrm{Ag + Cl^- + Fe^{3+} \rightarrow AgCl + Fe^{2+}}$$

Reaction quotient:

$$Q = \frac{\left[\mathrm{Fe^{2+}}\right]}{\left[\mathrm{Fe^{3+}}\right]\left[\mathrm{Cl^-}\right]}$$

From the Nernst Equation

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - 0.059 \log Q$$

So E_{cell} increases when Q decreases.

Check options:
A) $\uparrow Fe^{2+} \rightarrow Q \uparrow \rightarrow E \downarrow$
B) $\downarrow Fe^{3+} \rightarrow Q \uparrow \rightarrow E \downarrow$
C) $\downarrow Fe^{2+} \rightarrow Q \downarrow \rightarrow E \uparrow$
D) $\uparrow Fe^{3+} \rightarrow Q \downarrow \rightarrow E \uparrow$
E) $\uparrow Cl^{-} \rightarrow Q \downarrow \rightarrow E \uparrow$

Hence, the correct answer is the option (3).

Q. 15 Given below are two statements :
Statement I: $\mathrm{CoBr_4}^{2}$ ions will absorb light of lower energy than $\mathrm{CoCl_4}^{2}$ ion.
Statement - II : In $\mathrm{CoI_4}^{2}$ ion, the energy separation between the two set of d-orbitals is more than $\left[\mathrm{CoCl}^{4}\right]^{2-}$ ion.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Statement - I is true but Statement - II is false

Option 2:
Both Statement - I and Statement - II are true

Option 3:
Both Statement - I and Statement - II are false

Option 4:
Statement - I is false but Statement - II is true

Correct Answer:
Statement - I is true but Statement - II is false

Solution:
Statement I
$\mathrm{CoBr_4^{2-}}$ vs $\mathrm{CoCl_4^{2-}}$
Br^{-} is weaker $\rightarrow$ smaller $\Delta \rightarrow$ absorbs lower energy light.
Hence, True

Statement II
$\mathrm{CoI_4^{2-}}$ vs $\mathrm{CoCl_4^{2-}}$
I^{-} is weaker $\rightarrow$ smaller Δ than Cl^{-}.

But statement says more separation.
Hence, False

Hence, the correct answer is the option (1).

Q. 16 The correct trend in the first ionization enthalpies of the elements in the 3rd periodic table is :

Option 1:
$S < Si < Al < P < Cl$

Option 2:
$Al < Si < S < P < Cl$

Option 3:
$Al < S < P < Si < Cl$

Option 4:
$Si < S < Al < P > Cl$

Correct Answer:
$Al < Si < S < P < Cl$

Solution:
Sol:: Order of ionization energy is $Al < Si < S < P < Cl$
Phosphorus contains half filled P sub shell

Hence, the correct answer is the option (2).

Q. 17 A cup of water at $5^{\circ}C$ (system) is placed in a microwave oven and the oven is turned on for one minute during which the water begins to boil. Which of the following option is true?

Option 1:
q ve, w ve, U ve

Option 2:
q ve, w 0, U ve

Option 3:
q ve, w ve,U ve

Option 4:
q ve, w ve, U ve

Correct Answer:
q ve, w ve, U ve

Solution:
Cup of water $5^{\circ}C$
Heated in microwave oven. After 1 min . water begins to boil.
System = water

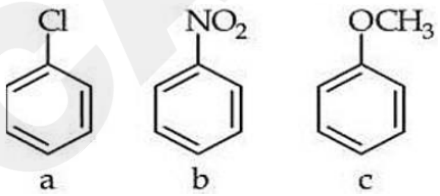
$q = +ve$

- System: water
- Energy absorbed: yes, from microwave → ΔU > 0
- Temperature: rises from 5°C → 100°C
- Entropy: increases (more molecular disorder)

Conclusion: Internal energy and entropy of water increase.

Hence, the correct answer is the option (1).

Q. 18 Consider the following compounds



Arrange these compounds in the increasing order of reactivity with nitrating mixture.

Option 1:
c < b < a

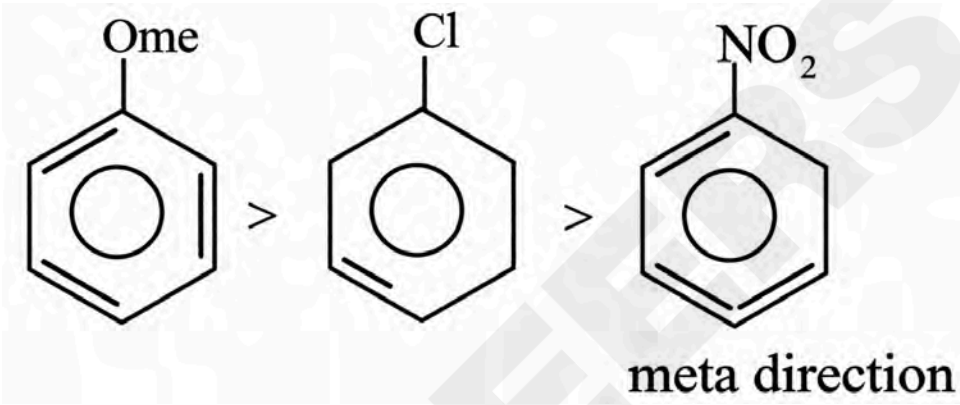
Option 2:
b < c < a

Option 3:
b < a < c

Option 4:
c < a < b

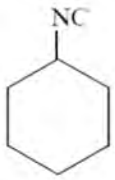
Correct Answer:
b < a < c

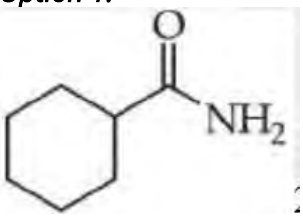
Solution:
Sol: Rate of EAS order



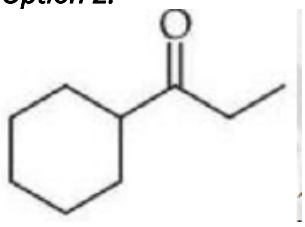
Hence, the correct answer is the option (3).

Q. 19 Compound ' P ' undergoes the following sequence of reactions:

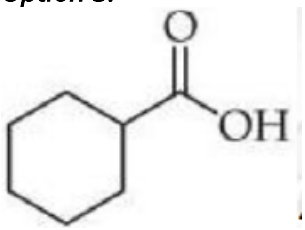
$\mathrm{P}\ \overset{\mathrm{i}\ \mathrm{NH_3}}{\underset{\mathrm{ii}\ \mathrm{Q}}{\mathrm{i}\ \mathrm{KOH}, \mathrm{Br_2CHCl_3}, \mathrm{KOH}\ \text{alc,}}}$  P is :

Option 1:


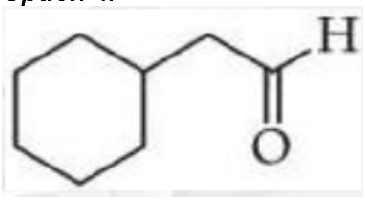
Option 2:



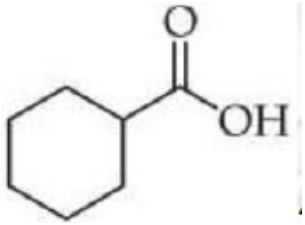
Option 3:



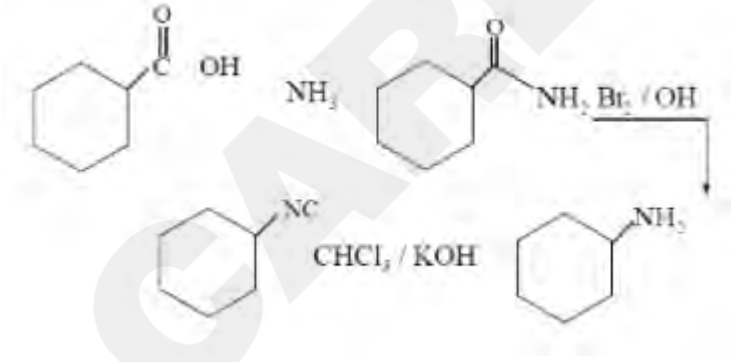
Option 4:



Correct Answer:



Solution:



Hence, the correct answer is the option (3).

Q. 20 Which of the following statements regarding the energy of the stationary state is true the following one-electron systems?

Option 1:
$2.18 \quad 10^{18}$ J for second orbit of He ion

Option 2:
$1.09 \quad 10^{18}$ J for sec ond orbit of H atom.

Option 3:
$2.18 \quad 10^{18}$ J for third orbit of Li^2 ion.

Option 4:
$8.72 \quad 10^{18}$ J for first orbit of He ion.

Correct Answer:
$2.18 \quad 10^{18}$ J for third orbit of Li^2 ion.

Solution:
Using the Bohr Model formula:

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2}$$

For Li^{2+} :

$$Z = 3, n = 3$$

$$E = -2.18 \times 10^{-18} \times \frac{3^2}{3^2}$$
$$E = -2.18 \times 10^{-18} J$$

Hence, the correct answer is the option (3).

Q. 21 $x$mg of pure HCl was used to make an aqueous solution 25.0 ml . of $0.1\mathrm{MBaOH}_2$ solution is used when the HCl solution was titrated against it. The numerical value of ' x ' is _____ 10^1 (nearest integer)Given : molar mass of HCl and BaOH_2 are 36.5 and 171.0 g mol^1 respectively).

Correct Answer:
1825

Solution:
Reaction:

$$\mathrm{Ba(OH)_2 + 2HCl \rightarrow BaCl_2 + 2H_2O}$$

Let's use the titration relation:

$$\frac{W}{M_{\mathrm{HCl}}} \times 1000 = N_2 V_2 \times \text{ stoichiometric factor}$$

Where:
- W = mass of HCl in grams
- $M_{\mathrm{HCl}} = 36.5\ \mathrm{g/mol}$
- $N_2 = 0.1\mathrm{M}$ (molarity of $\mathrm{Ba(OH)_2}$)
- $V_2 = 25\ \mathrm{mL}$ (volume of $\mathrm{Ba(OH)_2}$)
- Stoichiometric factor $= 2$ (because 1 mole $\mathrm{Ba(OH)_2}$ reacts with 2 moles HCl)

$$\frac{W}{36.5} \times 1000 = 0.1 \times 25 \times 2$$

$$\frac{W}{36.5} \times 1000 = 5$$

$$W = \frac{5 \times 36.5}{1000} = 0.1825\ \mathrm{g} = 182.5\mathrm{mg}$$

Expressed in $\times 10^1 \mathrm{mg}$ form:
$182.5\mathrm{mg} = 1825 \times 10^{-1}\mathrm{mg}$ or simply 1825×10^1 as in your reference.

Hence, the answer is (1825).

Q. 22 For the following gas phase equilibrium reaction at constant temperature.

$$\mathrm{NH_3\ g} \quad \tfrac{1}{2}\ \mathrm{N_2\ g} \quad \tfrac{3}{2}\mathrm{H_2\ g}$$

If the total pressure is $\sqrt{3}\ \mathrm{atm}$ and the pressure equilibrium constant K_p is 9 atm , then the degree of dissociation is given as $\mathrm{x} \quad 10^2 1/2$. The value of ' x ' is .....
(nearest integer)

Correct Answer:
125

Solution:
Sol:

$$\begin{array}{llll} NH_3 & \frac{1}{2}N_2 & \frac{3}{2}H_2 \\ 1 & 0 & 0 \\ 1 & -\frac{3}{2} & \frac{3}{2} \end{array}$$

$$\text{Total moles at equilibrium} \quad 1 \quad \overline{2} \quad \frac{3}{2} \quad 1$$

$$PP_{NH_3} \quad \frac{1}{1}\sqrt{3}; PP_{N_2} \quad \overline{21}\sqrt{3}$$

$$PP_{H_2} \quad \frac{3}{21}\sqrt{3}$$

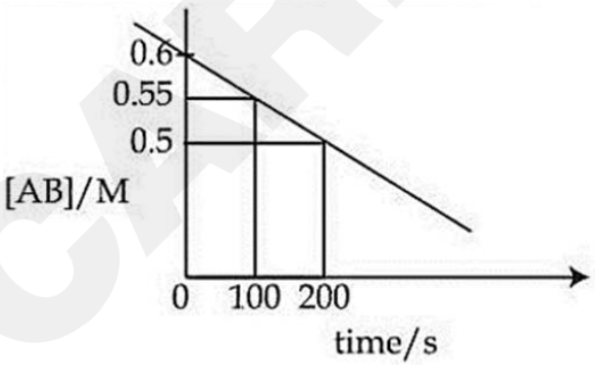
$$K_P \frac{P_{N_2}^{\frac{1}{2}} P_{H_2}^{3/2}}{P_{NH_3}}$$

After substitution and simplification

$$\begin{array}{ll} K_P & \frac{3\sqrt{3}}{4}\frac{P}{1}2 \\ 9 & \frac{3\sqrt{3}}{4}\frac{^2\sqrt{3}}{1} \\ & \frac{4}{5} \quad \sqrt{\frac{4}{5}} \\ & \times 10^2 \\ 2 & \frac{1}{\times 1/2} \\ \frac{4}{5} & \frac{1}{\times 10^2} \\ \times & 125 \end{array}$$

Hence, the answer is (125).

Q. 23 For the thermal decomposition of reactant $AB(g)$, the following plot is constructed.



The half life of the reaction is ' x ' $\min$.

$x =$ _____ min. (nearest integer)

Correct Answer:

10

Solution:

Sol: Given graph zero order

$\text{At } t=0(AB)=0.6$
$t=200,(AB)=0.5$
$\text{Change } =0.1$
$\text{K} \quad \frac{0.1}{200} \quad 510^4$
$\text{t}_{\frac{1}{2}} \quad \frac{\text{a}_0}{2\text{k}} \quad \frac{0.6}{2 \quad 5 \quad 10^4} \quad \frac{0.6}{10^3} \quad 600\text{sec.} \quad 10 \text{ min.}$

Hence, the answer is (10).

Q. 24 The crystal field splitting energy of Co oxalate $_3{}^3$ complex is ' n ' times that of the Cr oxalate $_3{}^3$ complex. Here ' n ' is _____ (Assume 0 P)

Correct Answer:
2

Solution:
For octahedral complexes:

$$\Delta \propto \frac{Z^2}{r^5}$$

- Co^{3+} radius $\approx 0.545, \text{Cr}^{3+}$ radius ≈ 0.615

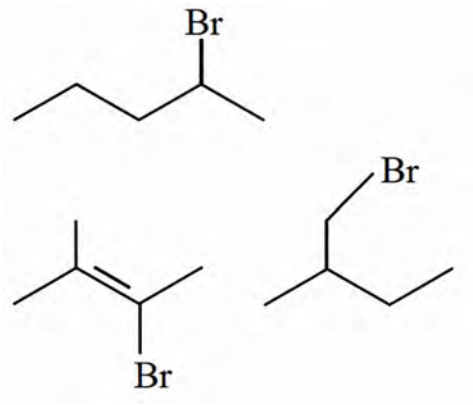
$$n=\frac{\Delta_{\text{Co}}}{\Delta_{\text{Cr}}}=\left(\frac{0.615}{0.545}\right)^5 \approx 1.8$$

Hence, the answer is (2).

Q. 25 Consider all the structural isomers with molecular formula $\text{C}_5\text{H}_{11}\text{Br}$ are separately treated with KOH(aq) to give respective substitution products, without any rearrangement. The number of products which can exhibit optical isomerism from these is _____

Correct Answer:
6

Solution:



Hence, the answer is (6).

Maths

Q. 1 A rectangle is formed by the lines $x = 0, y = 0, x = 3$ and $y = 4$. Let the line L be perpendicular to $3x + y + 6 = 0$ and divide the area of the rectangle into two equal parts. Then the distance of the point $\left(\frac{1}{2}, -5\right)$ from the line L is equal to :

Option 1:
$2\sqrt{10}$

Option 2:
$3\sqrt{10}$

Option 3:
$\sqrt{10}$

Option 4:
$2\sqrt{5}$

Correct Answer:
$2\sqrt{10}$

Solution:
The given line is
$3x + y + 6 = 0$
Convert to slope form:
$y = -3x - 6$
Hence, slope $m_1 = -3$.
For a line perpendicular to it:
$m_1 m_2 = -1$
$(-3)m_2 = -1$
$m_2 = \frac{1}{3}$
The rectangle is bounded by
$x = 0, y = 0, x = 3, y = 4$

Vertices:
$(0,0),(3,0),(3,4),(0,4)$
A line dividing a rectangle into two equal areas passes through its centre.
Centre of rectangle:
$\left(\frac{0+3}{2},\frac{0+4}{2}\right)=\left(\frac{3}{2},2\right)$

Equation of line L with slope $\frac{1}{3}$ passing through $\left(\frac{3}{2},2\right)$:

$$y-2=\frac{1}{3}\left(x-\frac{3}{2}\right)$$
$$6y-12=2x-3$$
$$2x-6y+9=0$$

Now distance of point $\left(\frac{1}{2},-5\right)$ from the line.
Distance formula:

$$d=\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$$
$$d=\frac{\left|2\left(\frac{1}{2}\right)-6(-5)+9\right|}{\sqrt{2^2+(-6)^2}}$$
$$=\frac{|1+30+9|}{\sqrt{40}}$$
$$=\frac{40}{2\sqrt{10}}$$
$$=2\sqrt{10}$$

Hence, the correct answer is option (1).

Q. 2 The value of $\frac{{}^{100}C_{50}}{51}+\frac{{}^{100}C_{51}}{52}+\ldots.+\frac{{}^{100}C_{100}}{101}$ is :

Option 1:
$\frac{2^{101}}{101}$

Option 2:
$\frac{2^{100}}{101}$

Option 3:
$\frac{2^{100}}{100}$

Option 4:
$\frac{2^{101}}{100}$

Correct Answer:
$\frac{2^{100}}{101}$

Solution:

We know that,
$\frac{^nC_r}{r+1} = \frac{^{n+1}C_{r+1}}{n+1}$
This follows from the factorial definition of combinations.

$$^nC_r = \frac{n!}{r!(n-r)!}$$

$$\frac{^nC_r}{r+1} = \frac{n!}{(r+1)!(n-r)!}$$

$$= \frac{1}{n+1} \cdot \frac{(n+1)!}{(r+1)!(n-r)!}$$

$$= \frac{^{n+1}C_{r+1}}{n+1}$$

Now,
$\frac{^{100}C_{50}}{51} + \frac{^{100}C_{51}}{52} + \ldots + \frac{^{100}C_{100}}{101}$
Using the identity:
$\frac{^{100}C_r}{r+1} = \frac{^{101}C_{r+1}}{101}$
So the sum becomes
$\frac{1}{101}\left(^{101}C_{51} + ^{101}C_{52} + \ldots + ^{101}C_{101}\right)$
Let $k = r + 1$.
Then the sum becomes
$\frac{1}{101}\sum_{k=51}^{101} ^{101}C_k$
Using binomial symmetry:
$\sum_{k=0}^{101} ^{101}C_k = 2^{101}$
And
$^{101}C_k = ^{101}C_{101-k}$
Hence, half of the total sum is
$\sum_{k=51}^{101} ^{101}C_k = 2^{100}$
Therefore, $\frac{1}{101} \times 2^{100} = \frac{2^{100}}{101}$

Hence, the correct answer is option (2).

Q. 3 Let α and β respectively be the maximum and the minimum values of the function

$$f(\theta) = 4\left(\sin^4\left(\frac{7\pi}{2} - \theta\right) + \sin^4(11\pi + \theta)\right) - 2\left(\sin^6\left(\frac{3\pi}{2} - \theta\right) + \sin^6(9\pi - \theta)\right), \theta \in \mathbf{R}.$$

Then $\alpha + 2\beta$ is equal to :

Option 1:
5

Option 2:
6

Option 3:
3

Option 4:
4

Correct Answer:
5

Solution:
$\sin\left(\frac{7\pi}{2}-\theta\right)=\sin\left(3\pi+\frac{\pi}{2}-\theta\right)=-\sin\left(\frac{\pi}{2}-\theta\right)=-\cos\theta$
$\sin(11\pi+\theta)=-\sin\theta$
$\sin\left(\frac{3\pi}{2}-\theta\right)=-\cos\theta$
$\sin(9\pi-\theta)=\sin\theta$
Substituting these into $f(\theta)$ and noting that even powers eliminate the negative signs:
$f(\theta)=4\left(\cos^4\theta+\sin^4\theta\right)-2\left(\cos^6\theta+\sin^6\theta\right)$
Using the identities for the sum of powers:
$\sin^4\theta+\cos^4\theta=\left(\sin^2\theta+\cos^2\theta\right)^2-2\sin^2\theta\cos^2\theta=1-2\sin^2\theta\cos^2\theta$
$\sin^6\theta+\cos^6\theta=\left(\sin^2\theta+\cos^2\theta\right)\left(\sin^4\theta-\sin^2\theta\cos^2\theta+\cos^4\theta\right)$
$=1\cdot\left(1-2\sin^2\theta\cos^2\theta-\sin^2\theta\cos^2\theta\right)=1-3\sin^2\theta\cos^2\theta$
Now,
$f(\theta)=4\left(1-2\sin^2\theta\cos^2\theta\right)-2\left(1-3\sin^2\theta\cos^2\theta\right)$
$f(\theta)=4-8\sin^2\theta\cos^2\theta-2+6\sin^2\theta\cos^2\theta$
$f(\theta)=2-2\sin^2\theta\cos^2\theta$
Using the double angle formula
$\sin 2\theta=2\sin\theta\cos\theta$
$\Rightarrow\sin^2 2\theta=4\sin^2\theta\cos^2\theta$:
$f(\theta)=2-\frac{1}{2}\left(4\sin^2\theta\cos^2\theta\right)=2-\frac{1}{2}\sin^2 2\theta$
Since $0\le\sin^2 2\theta\le 1$:
The maximum value α occurs when $\sin^2 2\theta=0$:
$\alpha=2-0=2$
The minimum value β occurs when $\sin^2 2\theta=1$:
$\beta=2-\frac{1}{2}=\frac{3}{2}$
Calculating the required value $\alpha+2\beta$:
$\alpha+2\beta=2+2\left(\frac{3}{2}\right)=2+3=5$
Hence, the correct answer is option (1).

Q. 4 Let $\vec{a}=-\hat{i}+\hat{j}+2\hat{k},\ \vec{b}=\hat{i}-\hat{j}-3\hat{k},\vec{c}=\vec{a}\times\vec{b}$ and $\vec{d}=\vec{c}\times\vec{a}$. Then $(\vec{a}-\vec{b})\cdot\vec{d}$ is equal to:

Option 1:
2

Option 2:
-4

Option 3:
4

Option 4:
-2

Correct Answer:
-2

Solution:
Let us define d using the given relations $c = a \times b$ and $d = c \times a$:
$d = (a \times b) \times a$
Using the Vector Triple Product identity $(u \times v) \times w = (u \cdot w)v - (v \cdot w)u$:

$d = (a \cdot a)b - (b \cdot a)a$
Now, we need to find $(a - b) \cdot d$:

$(a - b) \cdot [(a \cdot a)b - (b \cdot a)a]$
$= (a \cdot a)(a \cdot b) - (b \cdot a)(a \cdot a) - (a \cdot a)(b \cdot b) + (b \cdot a)(b \cdot a)$
$= -(a \cdot a)(b \cdot b) + (a \cdot b)^2$
Now, the required dot products from $a = -\hat{i} + \hat{j} + 2\hat{k}$ and $b = \hat{i} - \hat{j} - 3\hat{k}$:
$a \cdot a = (-1)^2 + (1)^2 + (2)^2 = 1 + 1 + 4 = 6$
$b \cdot b = (1)^2 + (-1)^2 + (-3)^2 = 1 + 1 + 9 = 11$
$a \cdot b = (-1)(1) + (1)(-1) + (2)(-3) = -1 - 1 - 6 = -8$
So,
$(a - b) \cdot d = -(6)(11) + (-8)^2$
$= -66 + 64$
$= -2$

Hence, the correct answer is option (4).

Q. 5 Let $S = \{z : 3 \leqslant |2z - 3(1 + i)| \leqslant 7\}$ be a set of complex numbers. Then $\text{Min}_{z \in S} \left|\left(z + \frac{1}{2}(5 + 3i)\right)\right|$ is equal to :

Option 1:
$\frac{3}{2}$

Option 2:
$\frac{5}{2}$

Option 3:
2

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{3}{2}$

Solution:
Let us rewrite the condition for the set S in the standard form $|z-z_0|$:
$3 \leqslant |2z-3(1+i)| \leqslant 7$
$\frac{3}{2} \leqslant \left|z-\frac{3}{2}(1+i)\right| \leqslant \frac{7}{2}$
$\frac{3}{2} \leqslant \left|z-\left(\frac{3}{2}+\frac{3}{2}i\right)\right| \leqslant \frac{7}{2}$
This describes an annulus (a ring-shaped region) centred at $C=\left(\frac{3}{2},\frac{3}{2}\right)$ with an inner radius $r_1=1.5$ and an outer radius $r_2=3.5$.

Let P be the point corresponding to:
$-\frac{1}{2}(5+3i)=-\frac{5}{2}-\frac{3}{2}i$
So, the point P is $(-2.5,-1.5)$. We want to find the minimum value of the distance $d=\mid z-P \mid$.

Calculate the distance between the center $C(1.5,1.5)$ and the point $P(-2.5,-1.5)$:

$CP=\sqrt{(1.5-(-2.5))^2+(1.5-(-1.5))^2}$
$CP=\sqrt{(4)^2+(3)^2}=\sqrt{16+9}=5$
Since the distance from the point P to the centre C is 5, which is greater than the outer radius $r_2=3.5$, the point P lies outside the shaded region S.

The minimum distance from a point P to the annulus is:
Min distance $=CP-$ (outer radius r_2)
Min distance $=5-3.5=1.5$
Min distance $=\frac{3}{2}$

Hence, the correct answer is option (1).

Q. 6 If α and $\beta(\alpha<\beta)$ are the roots of the equation $(-2+\sqrt{3})(|\sqrt{x}-3|)+(x-6\sqrt{x})+(9-2\sqrt{3})=0, x \geqslant 0$, then $\sqrt{\frac{\beta}{\alpha}}+\sqrt{\alpha\beta}$ is equal to :

Option 1:
8

Option 2:
11

Option 3:
9

Option 4:
10

Correct Answer:
10

Solution:
Let $t = \sqrt{x} - 3$. Then $x = (\sqrt{x})^2$, and we can rewrite the middle term:

$x - 6\sqrt{x} = (\sqrt{x})^2 - 6\sqrt{x} + 9 - 9 = (\sqrt{x} - 3)^2 - 9$
Substituting this into the original equation:

$(-2 + \sqrt{3})|t| + \left(t^2 - 9\right) + (9 - 2\sqrt{3}) = 0$
$t^2 + (\sqrt{3} - 2)|t| - 2\sqrt{3} = 0$
Since $t^2 = |t|^2$, let $y = |t|$, where $y \geqslant 0$:

$y^2 + (\sqrt{3} - 2)y - 2\sqrt{3} = 0$
This is a quadratic equation in y. We can factorize it by splitting the middle term:

$y^2 + \sqrt{3}y - 2y - 2\sqrt{3} = 0$
$y(y + \sqrt{3}) - 2(y + \sqrt{3}) = 0$
$(y - 2)(y + \sqrt{3}) = 0$
Since $y = |t| \geqslant 0$, the only valid solution is:

$y = 2 \Rightarrow |\sqrt{x} - 3| = 2$
This gives us two cases:
Case 1: $\sqrt{x} - 3 = 2 \Rightarrow \sqrt{x} = 5 \Rightarrow x = 25$
Case 2: $\sqrt{x} - 3 = -2 \Rightarrow \sqrt{x} = 1 \Rightarrow x = 1$
Given $\alpha < \beta$, we have:

$\alpha = 1 \text{ and } \beta = 25$

Now, calculate the required value $\sqrt{\frac{\beta}{\alpha}} + \sqrt{\alpha\beta}$:

$\sqrt{\frac{25}{1}} + \sqrt{1 \times 25} = \sqrt{25} + \sqrt{25}$
$= 5 + 5 = 10$

Hence, the correct answer is option (4).

Q. 7 Let the mean and variance of 8 numbers $-10, -7, -1, x, y, 9, 2, 16$ be $\frac{7}{2}$ and $\frac{293}{4}$, respectively. Then the mean of 4 numbers $x, y, x + y + 1, |x - y|$ is :

Option 1:
12

Option 2:
10

Option 3:
9

Option 4:
11

Correct Answer:
11

Solution:

$$\text{Mean } = \frac{7}{2}$$

$$\frac{-10-7-1+x+y+16+2+9}{8} = \frac{7}{2}$$

$$\Rightarrow x+y+9 = 28$$

$$\Rightarrow x+y = 19$$

$$\text{Var } = \frac{293}{4}$$

$$\frac{100+49+1+x^2+y^2+256+4+8}{8} - \frac{49}{4} = \frac{293}{4}$$

$$\frac{491+x^2+y^2}{8} = \frac{342}{4} \times 2$$

$$x^2+y^2 = 193$$

$$\Rightarrow (x,y) = (7,12) \text{ or } (12,7)$$

$$\text{Mean } = \frac{x+y+x+y+1+(x-y)}{4}$$

$$= \frac{19+20+5}{4}$$

$$= 11$$

Hence, the correct answer is option (4).

Q. 8 The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are $(1, 6, 3)$ and $(4, 9, \alpha)$ respectively and C is at a distance of 10 units from B. The area (in sq. units) of $\triangle ABC$ is :

Option 1:
$5\sqrt{13}$

Option 2:
$10\sqrt{13}$

Option 3:
$15\sqrt{13}$

Option 4:
$20\sqrt{13}$

Correct Answer:
$5\sqrt{13}$

Solution:
The line equation is given by:

$\frac{x}{1}=\frac{1-y}{-2}=\frac{z-2}{3}\Rightarrow\frac{z}{1}=\frac{y-1}{2}=\frac{z-2}{3}$
Since $B(4,9,\alpha)$ lies on this line:

$$\frac{4}{1}=\frac{9-1}{2}=\frac{\alpha-2}{3}$$
$$4=4=\frac{\alpha-2}{3}\Rightarrow\alpha=14$$
Here, $A=(1,6,3)$ and $B=(4,9,14)$:

$$\overrightarrow{AB}=(4-1)\hat{\imath}+(9-6)\hat{\jmath}+(14-3)\hat{k}=3\hat{\imath}+3\hat{\jmath}+11\hat{k}$$
The vector $\overrightarrow{BC}$ has magnitude 10 and is parallel to the line direction $\hat{v}=\frac{\hat{\imath}+2\hat{\jmath}+3\hat{k}}{\sqrt{14}}$:

$$\overrightarrow{BC}=\frac{10}{\sqrt{14}}(\hat{\imath}+2\hat{\jmath}+3\hat{k})$$
Now,
$$\overrightarrow{AB}\times\overrightarrow{BC}=\frac{10}{\sqrt{14}}\begin{vmatrix}\hat{\imath}&\hat{\jmath}&\hat{k}\\3&3&11\\1&2&3\end{vmatrix}$$
$$\overrightarrow{AB}\times\overrightarrow{BC}=\frac{10}{\sqrt{14}}[\hat{\imath}(9-22)-\hat{\jmath}(9-11)+\hat{k}(6-3)]$$
$$\overrightarrow{AB}\times\overrightarrow{BC}=\frac{10}{\sqrt{14}}(-13\hat{\imath}+2\hat{\jmath}+3\hat{k})$$
Calculate the magnitude:

$$|\overrightarrow{AB}\times\overrightarrow{BC}|=\frac{10}{\sqrt{14}}\sqrt{(-13)^2+2^2+3^2}=\frac{10\sqrt{182}}{\sqrt{14}}$$
$$|\overrightarrow{AB}\times\overrightarrow{BC}|=10\sqrt{13}$$
$$\text{Area }=\frac{1}{2}\times10\sqrt{13}$$
$$=5\sqrt{13}$$

Hence, the correct answer is option (1).

Q. 9 Let $\mathrm{A}=\{-2,-1,0,1,2,3,4\}$. Let R be a relation on A defined by $x \mathrm{R} y$ if and only if $2x+y \leqslant 2$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it a reflexive and symmetric relations respectively. Then $l+\mathrm{m}+\mathrm{n}$ is equal to :

Option 1:
33

Option 2:
32

Option 3:
34

Option 4:
35

Correct Answer:
33

Solution:
A relation R is defined on $A=\{-2,-1,0,1,2,3,4\}$ by $xRy \iff y \leqslant 2-2x$.

We count pairs (x,y) such that $y \leqslant 2-2x$:
For $x=-2, y \leqslant 6 \Rightarrow$ all 7 values of y
For $x=-1, y \leqslant 4 \Rightarrow$ all 7 values of y
For $x=0, y \leqslant 2 \Rightarrow y \in \{-2,-1,0,1,2\}$ (5 values)
For $x=1, y \leqslant 0 \Rightarrow y \in \{-2,-1,0\}$ (3 values)
For $x=2, y \leqslant -2 \Rightarrow y \in \{-2\}$ (1 value)
For $x=3, y \leqslant -4 \Rightarrow 0$ values
For $x=4, y \leqslant -6 \Rightarrow 0$ values

$l=7+7+5+3+1=23$

A relation is reflexive if $(x,x) \in R$ for all $x \in A$.
Check $2x+x \leqslant 2 \Rightarrow 3x \leqslant 2$:
This holds for $x \in \{-2,-1,0\}$.
It fails for $x \in \{1,2,3,4\}$.
$m=4$ (Pairs to add: $(1,1),(2,2),(3,3),(4,4)$)

A relation is symmetric if $(x,y) \in R \Longrightarrow (y,x) \in R$.
Identify $(x,y) \in R$ where $(y,x) \notin R$ (i.e., $2y+x>2$):
Pairs are: $(-2,3),(-2,4),(-1,2),(-1,3),(-1,4),(0,2)$
$n=6$ (Pairs to add: $(3,-2),(4,-2),(2,-1),(3,-1),(4,-1),(2,0)$)
$l+m+n=23+4+6=33$

Hence, the correct answer is option (1).

Q. 10 Among the statements :

1: If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$,

then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

II: If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = px + q$, then $p^2 = 196q^2$,

Option 1:
Both are false

Option 2:
Only II is true

Option 3:
Only I is true

Option 4:
Both are true

Correct Answer:
Both are false

Solution:
Statement I
Let $\cos\alpha = a, \cos\beta = b, \cos\gamma = c$:
The first determinant is:

$$\begin{vmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{vmatrix} = 1\left(1-c^2\right) - a(a-bc) + b(ac-b)$$

$$= 1 - a^2 - b^2 - c^2 + 2abc$$

The second determinant is:

$$\begin{vmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 0 \end{vmatrix} = 0 - a(0-bc) + b(ac-0) = 2abc$$

Equating the two:

$$1 - \left(a^2+b^2+c^2\right) + 2abc = 2abc$$

$$\Rightarrow a^2 + b^2 + c^2 = 1$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Hence, the statement $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$ is False.

Statement II

To find q in $px + q$, substitute $x = 0$ into the determinant:

$$q = \begin{vmatrix} 0 & 1 & -2 \\ -1 & 0 & -3 \\ 3 & -1 & -1 \end{vmatrix}$$

$q = -1(1 - (-9)) - 2(1 - 0) = -10 - 2 = -12$

To find p, we evaluate the coefficient of x (by taking $x = 1$ or differentiating):

The full expansion yields $24x - 12$.

Here, $p = 24$ and $q = -12$.

Checking $p^2 = 196q^2$:

$p^2 = (24)^2 = 576$

$196q^2 = 196(-12)^2 = 196 \times 144 = 28224$

$576 \neq 28224$

Hence, the statement $p^2 = 196q^2$ is False.

So, Both Statement I and Statement II are False.

Hence, the correct answer is option (1).

Q. 11 Let the domain of the function $f(x) = \log_3 \log_5 \log_7 \left(9x - x^2 - 13\right)$ be the interval (m, n). Let the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ have eccentricity $\frac{n}{3}$ and the length of the latus rectum $\frac{8m}{3}$. Then $b^2 - a^2$ is equal to :

Option 1:
7

Option 2:
11

Option 3:
5

Option 4:
9

Correct Answer:
7

Solution:

For $f(x)=\log_3\log_5\log_7\left(9x-x^2-13\right)$ to be defined:

$\log_5\log_7\left(9x-x^2-13\right)>0$

$\log_7\left(9x-x^2-13\right)>5^0$

$\log_7\left(9x-x^2-13\right)>1$

$9x-x^2-13>7^1$

$x^2-9x+20<0$

$(x-4)(x-5)<0$

$x\in(4,5)\Rightarrow m=4, n=5$

Eccentricity $e=\frac{n}{3}=\frac{5}{3}$

$$e^2=1+\frac{b^2}{a^2}$$

$$\frac{25}{9}=1+\frac{b^2}{a^2}\Rightarrow\frac{b^2}{a^2}=\frac{16}{9}$$

$$b^2=\frac{16}{9}a^2$$

Length of latus rectum $LR=\frac{8m}{3}=\frac{32}{3}$

$$\frac{2b^2}{a}=\frac{32}{3}$$

$$\frac{2}{a}\left(\frac{16a^2}{9}\right)=\frac{32}{3}$$

$$\frac{32a}{9}=\frac{32}{3}\Rightarrow a=3$$

$$a^2=9$$

$$b^2=\frac{16}{9}(9)=16$$

$$b^2-a^2=16-9=7$$

Hence, the correct answer is option (1).

Q. 12 Let $f(x)=\int\frac{\left(2-x^2\right)\cdot e^x}{(\sqrt{1+x})(1-x)^{3/2}}\,dx$. If $f(0)=0$, then $f\left(\frac{1}{2}\right)$ is equal to:

Option 1:
$\sqrt{3e}+1$

Option 2:
$\sqrt{2e}+1$

Option 3:
$\sqrt{2e}-1$

Option 4:
$\sqrt{3e}-1$

Correct Answer:
$\sqrt{3e}-1$

Solution:
$(\sqrt{1+x})(1-x)^{3/2}=\sqrt{1+x}\cdot(1-x)\cdot\sqrt{1-x}$
$=(1-x)\sqrt{(1+x)(1-x)}$
$=(1-x)\sqrt{1-x^2}$
Now, rewriting the integrand, we get:

$I=\int e^x\left[\frac{2-x^2}{(1-x)\sqrt{1-x^2}}\right]dx$
We can split the numerator $2-x^2$ as $(1)+\left(1-x^2\right)$:

$$I=\int e^x\left[\frac{1}{(1-x)\sqrt{1-x^2}}+\frac{1-x^2}{(1-x)\sqrt{1-x^2}}\right]dx$$
$$I=\int e^x\left[\frac{1}{(1-x)\sqrt{1-x^2}}+\frac{\sqrt{1-x^2}}{1-x}\right]dx$$
Let $g(x)=\frac{\sqrt{1-x^2}}{1-x}=\sqrt{\frac{(1-x)(1+x)}{(1-x)^2}}=\sqrt{\frac{1+x}{1-x}}$
Now,
$$g'(x)=\frac{d}{dx}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$
$$g'(x)=\frac{1}{2}\left(\frac{1+x}{1-x}\right)^{-\frac{1}{2}}\cdot\frac{(1-x)(1)-(1+x)(-1)}{(1-x)^2}$$
$$g'(x)=\frac{1}{2}\sqrt{\frac{1-x}{1+x}}\cdot\frac{2}{(1-x)^2}$$
$$g'(x)=\frac{1}{\sqrt{1+x\cdot(1-x)^{3/2}}}$$
$$g'(x)=\frac{1}{(1-x)\sqrt{1-x^2}}$$
Since the integral is in the form $\int e^x\left[g'(x)+g(x)\right]dx$:
$f(x)=e^x\cdot g(x)+C$
$$f(x)=e^x\sqrt{\frac{1+x}{1-x}}+C$$

Given $f(0)=0$:

$0=e^0\sqrt{\frac{1+0}{1-0}}+C\Rightarrow 0=1+C\Rightarrow C=-1$
So, $f(x)=e^x\sqrt{\frac{1+x}{1-x}}-1$
Now, find $f(\frac{1}{2})$:

$$f(\frac{1}{2}) = e^{\frac{1}{2}}\sqrt{\frac{1+\frac{1}{2}}{1-\frac{1}{2}}} - 1$$

$$f(\frac{1}{2}) = \sqrt{e}\sqrt{\frac{3/2}{\frac{1}{2}}} - 1$$

$$f(\frac{1}{2}) = \sqrt{e}\cdot\sqrt{3} - 1$$

$$f(\frac{1}{2}) = \sqrt{3e} - 1$$

Hence, the correct answer is option (4).

Q. 13 The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is:

Option 1:
$\frac{\pi}{18}$

Option 2:
$\frac{\pi}{6}$

Option 3:
$\frac{\pi}{3}$

Option 4:
$\frac{\pi}{12}$

Correct Answer:
$\frac{\pi}{12}$

Solution:
Let the given integral be I :

$$I = \int_{\frac{\pi}{2\pi}}^{\frac{\frac{5\pi}{24}}{3\pi}} \frac{dx}{1+\sqrt{\tan 2x}}$$

The sum of the limits is $a+b = \frac{\pi}{24} + \frac{5\pi}{24} = \frac{6\pi}{24} = \frac{\pi}{4}$.
Using the property $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$, replace x with $\left(\frac{\pi}{4} - x\right)$:

$$I = \int_{\frac{\pi}{2\pi}}^{\frac{\sin}{2\pi}} \frac{dx}{1+\sqrt[3]{\tan 2\left(\frac{\pi}{4} - x\right)}}$$

$$I = \int_{\frac{\pi}{2\pi}}^{\frac{\sin}{2\pi}} \frac{dx}{1+\sqrt[3]{\tan\left(\frac{\pi}{2} - 2x\right)}}$$

Since $\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$:

$$I=\int_{\frac{\pi}{2\pi}}^{\frac{\pi}{2\pi}}\frac{dx}{1+\sqrt[3]{\cot 2x}}$$

$$I=\int_{\frac{\pi}{2\pi}}^{\frac{\pi}{2\pi}}\frac{dx}{1+\frac{1}{\sqrt[3]{\tan 2x}}}$$

$$I=\int_{\frac{\pi}{2\pi}}^{\frac{\pi}{2\pi}}\frac{\sqrt[3]{\tan 2x}}{1+\sqrt{\tan 2x}}dx$$

Now, add the two expressions for I :

$$2I=\int\frac{\frac{\pi}{\pi}}{\frac{\pi}{2\pi}}\left(\frac{1}{1+\sqrt[3]{\tan 2x}}+\frac{\sqrt[3]{\tan 2x}}{1+\sqrt[3]{\tan 2x}}\right)dx$$

$$2I=\int\frac{\frac{\pi}{2}}{\frac{1+\sqrt[3]{\tan 2x}}{1+\sqrt[3]{\tan 2x}}}dx$$

$$2I=\int\frac{\frac{\pi}{2\pi}}{\frac{\pi}{24}}1dx$$

$$2I=[x]\frac{\frac{\pi x}{\pi}}{24}$$

$$2I=\frac{5\pi}{24}-\frac{\pi}{24}=\frac{4\pi}{24}=\frac{\pi}{6}$$

$$I=\frac{\pi}{12}$$

Hence, the correct answer is option (4).

Q. 14 Let $f(x)=\begin{cases}\frac{ax^2+2ax+3}{4x^2+4x-3} & ,x\neq-\frac{3}{2},\frac{1}{2}\\ b & ,x=-\frac{3}{2},\frac{1}{2}\end{cases}$ be continuous at $x=-\frac{3}{2}$. If $f\circ f(x)=\frac{7}{5}$, then x is equal to:

Option 1:
1.4

Option 2:
0

Option 3:
1

Option 4:
2

Correct Answer:
1

Solution:

The denominator $4x^2 + 4x - 3$ vanishes at $x = -\frac{3}{2}$.
For continuity, the numerator must also be zero:

$$a\left(-\frac{3}{2}\right)^2 + 2a\left(-\frac{3}{2}\right) + 3 = 0$$

$$\frac{9a}{4} - 3a + 3 = 0 \Rightarrow -\frac{3a}{4} = -3 \Rightarrow a = 4$$

The function becomes:

$f(x) = \frac{4x^2+8x+3}{4x^2+4x-3} = \frac{(2x+3)(2x+1)}{(2x+3)(2x-1)}$
For $x \neq -\frac{3}{2}, \frac{1}{2}$:

$f(x) = \frac{2x+1}{2x-1}$
Since $f(x)$ is continuous at $x = -\frac{3}{2}$:

$b = \lim_{x\to-\frac{3}{2}} \frac{2x+1}{2x-1} = \frac{-3+1}{-3-1} = \frac{1}{2}$
Given the composite function equation:

$f(f(x)) = \frac{7}{5}$
Let $f(x) = y$. Then $f(y) = \frac{7}{5}$:

$$\frac{2y+1}{2y-1} = \frac{7}{5}$$

$5(2y + 1) = 7(2y - 1)$
$10y + 5 = 14y - 7 \Rightarrow 4y = 12 \Rightarrow y = 3$
Now, substitute $f(x) = 3$ to find x :

$$\frac{2x+1}{2x-1} = 3$$

$2x + 1 = 3(2x - 1)$
$2x + 1 = 6x - 3$
$4x = 4 \Rightarrow x = 1$
Hence, the correct answer is option (3).

Q. 15 Let the line $y - x = 1$ intersect the ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$ at the points A and B. Then the angle made by the line segment AB at the centre of the ellipse is :

Option 1:
$\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$

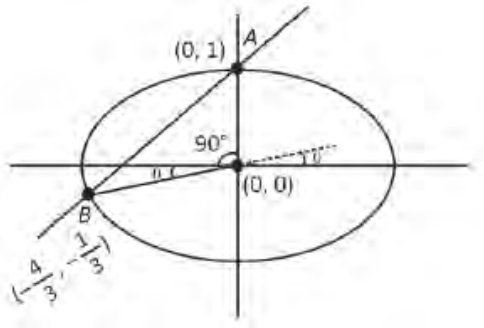
Option 2:
$\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$

Option 3:
$\frac{\pi}{2} + 2\tan^{-1}\left(\frac{1}{4}\right)$

Option 4:
$\frac{\pi}{4}+\tan^{-1}\left(\frac{1}{4}\right)$

Correct Answer:
$\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

Solution:



$y=x+1$ intersects $\frac{x^2}{2}+y^2=1$

$\Rightarrow \frac{x^2}{2}+(x+1)^2=1 \Rightarrow x^2+2x^2+4x=0$

$\Rightarrow x=0,\frac{-4}{3}$

⇒ Points A and B are

$(0,1),\left(\frac{-4}{3},\frac{-1}{3}\right)$

$\theta=\tan^{-1}\left(\frac{\frac{-1}{3}}{\frac{-4}{3}}\right)=\tan^{-1}\left(\frac{1}{4}\right)$

$\Rightarrow \angle AOB=\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

Hence, the correct answer is option (1).

Q. 16 Let $y=y(x)$ be the solution of the differential equation $x^4\,dy+\left(4x^3y+2\sin x\right)dx=0, x>0, y\left(\frac{\pi}{2}\right)=0$. Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

Option 1:
81

Option 2:
72

Option 3:
92

Option 4:
64

Correct Answer:
81

Solution:
The given equation is:

$x^4dy + \left(4x^3y + 2\sin x\right)dx = 0$
$\Rightarrow x^4dy + 4x^3ydx = -2\sin xdx$
Notice that the left side is the exact derivative of the product $x^4 \cdot y$:

$\frac{d}{dx}\left(x^4y\right) = x^4\frac{dy}{dx} + 4x^3y$
So, the equation can be written as:

$d\left(x^4y\right) = -2\sin xdx$

$\int d\left(x^4y\right) = \int -2\sin xdx$
$x^4y = 2\cos x + C$

2. Use the initial condition $y\left(\frac{\pi}{2}\right) = 0$ to find C:

$\left(\frac{\pi}{2}\right)^4(0) = 2\cos\left(\frac{\pi}{2}\right) + C$
$0 = 2(0) + C \Rightarrow C = 0$
The particular solution is:

$x^4y = 2\cos x \Rightarrow y = \frac{2\cos x}{x^4}$

$y\left(\frac{\pi}{3}\right) = \frac{2\cos\left(\frac{\pi}{3}\right)}{\left(\frac{\pi}{3}\right)^4}$

$y\left(\frac{\pi}{3}\right) = \frac{2\left(\frac{\pi}{3}\right)}{\frac{\pi^4}{31}} = \frac{1}{\frac{\pi^4}{31}} = \frac{81}{\pi^4}$

$\pi^4\left(\frac{81}{\pi^4}\right) = 81$

Hence, the correct answer is option (1).

Q. 17 A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. The mason A along will complete the construction work in :

Option 1:
24 days

Option 2:
36 days

Option 3:
42 days

Option 4:
30 days

Correct Answer:
36 days

Solution:
Let the time taken by mason A alone be x days.
According to the problem, mason A is faster, so mason B takes $x + 24$ days.
The combined rate is given as 22.5 days, which is $\frac{45}{2}$ days.
The combined rate per day is $\frac{2}{45}$.

$\frac{1}{x} + \frac{1}{x+24} = \frac{2}{45}$

$$\frac{(x+24)+x}{x(x+24)} = \frac{2}{45}$$

$$\frac{2x+24}{x^2+24x} = \frac{2}{45}$$

$\frac{x+12}{x^2+24x} = \frac{1}{45}$

$x^2 + 24x = 45(x + 12)$

$x^2 + 24x = 45x + 540$

$x^2 - 21x - 540 = 0$

$x^2 - 36x + 15x - 540 = 0$

$x(x - 36) + 15(x - 36) = 0$

$(x - 36)(x + 15) = 0$

$x = 36$ or $x = -15.$

Since time cannot be negative, $x = 36$.

Hence, the correct answer is option (2).

Q. 18 Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0, \theta \in [-3\pi, 2\pi]$ is :

Option 1:
4

Option 2:
3

Option 3:
0

Option 4:
5

Correct Answer:
5

Solution:
Substitute $\cos 2\theta = 2\cos^2\theta - 1$ into the given equation:

$$\sqrt{3}\left(2\cos^2\theta - 1\right) + 8\cos\theta + 3\sqrt{3} = 0$$
$$2\sqrt{3}\cos^2\theta - \sqrt{3} + 8\cos\theta + 3\sqrt{3} = 0$$
$$2\sqrt{3}\cos^2\theta + 8\cos\theta + 2\sqrt{3} = 0$$
$$\sqrt{3}\cos^2\theta + 4\cos\theta + \sqrt{3} = 0$$
$$\sqrt{3}\cos^2\theta + 3\cos\theta + \cos\theta + \sqrt{3} = 0$$
$$\sqrt{3}\cos\theta(\cos\theta + \sqrt{3}) + 1(\cos\theta + \sqrt{3}) = 0$$
$$(\sqrt{3}\cos\theta + 1)(\cos\theta + \sqrt{3}) = 0$$
This gives two possible values for $\cos\theta$:

$$\cos\theta = -\frac{1}{\sqrt{3}} \text{ or } \cos\theta = -\sqrt{3}$$
Since $-\sqrt{3} < -1$, the second case $\cos\theta = -\sqrt{3}$ has no real solutions.
We need to find the number of solutions for $\cos\theta = -\frac{1}{\sqrt{3}}$ in $\theta \in [-3\pi, 2\pi]$:
In $[0, 2\pi]$, $\cos\theta$ is negative in the 2 nd and 3rd quadrants $\rightarrow 2$ solutions.
In $[-2\pi, 0]$, by periodicity, there are $\rightarrow 2$ solutions.
In $[-3\pi, -2\pi]$, the graph covers the 2 nd and 3rd quadrants of the previous cycle $\rightarrow 1$ solution (at the point where it crosses $-\frac{1}{\sqrt{3}}$).
Total solutions $= 2 + 2 + 1 = 5$.

Hence, the correct answer is option (4).

Q. 19 Let the direction cosines of two lines satisfy the equations: $4l + m - n = 0$ and $2mn + 10nl + 3lm = 0$. Then the cosine of the acute angle between these lines is :

Option 1:
$\frac{10}{3\sqrt{38}}$

Option 2:
$\frac{20}{3\sqrt{38}}$

Option 3:
$\frac{10}{\sqrt{38}}$

Option 4:
$\frac{10}{7\sqrt{38}}$

Correct Answer:
$\frac{10}{3\sqrt{38}}$

Solution:
From the first equation: $n = 4l + m$.

$2m(4l + m) + 10(4l + m)l + 3lm = 0$
$8lm + 2m^2 + 40l^2 + 10lm + 3lm = 0$
$40l^2 + 21lm + 2m^2 = 0$

Case 1: $m = -8l$
From $(i), n = 4l + (-8l) = -4l.$
Direction ratios $(l_1, m_1, n_1) \propto (l, -8l, -4l) \Rightarrow (1, -8, -4).$

Case 2: $2m = -5l \Rightarrow m = -2.5l$
From $(i), n = 4l - 2.5l = 1.5l.$
Direction ratios $(l_2, m_2, n_2) \propto (l, -2.5l, 1.5l) \Rightarrow (2, -5, 3).$
$\cos\theta = \frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$
$\cos\theta = \frac{|(1)(2)+(-8)(-5)+(-4)(3)|}{\sqrt{1+64+16}\sqrt{4+25+9}}$
$\cos\theta = \frac{|2+40-12|}{\sqrt{81}\sqrt{38}}$
$\cos\theta = \frac{30}{9\sqrt{38}} = \frac{10}{3\sqrt{38}}$

Hence, the correct answer is option (1).

Q. 20 The sum of all possible values of $\mathrm{n} \in \mathbf{N}$, so that the coefficients of x, x^2 and x^3 in the expansion of $\left(1 + x^2\right)^2(1 + x)^{\mathrm{n}}$, are in arithmetic progression is :

Option 1:
9

Option 2:
7

Option 3:
12

Option 4:
3

Correct Answer:
9

Solution:
The total expansion is $\left(1+2x^2+x^4\right)\left(1+\binom{n}{1}x+\binom{n}{2}x^2+\binom{n}{3}x^3+\ldots\right)$.
Coefficient of x (c_1) : Only from $(1)\times\left(\binom{n}{1}x\right)\cdot c_1=\binom{n}{1}=n$
Coefficient of x^2 (c_2) : From $(1)\times\left(\binom{n}{2}x^2\right)$ and $\left(2x^2\right)\times(1).\, c_2=\binom{n}{2}+2=\frac{n(n-1)}{2}+2$
Coefficient of x^3 (c_3) : From (1) $\times\left(\binom{n}{3}x^3\right)$ and
$\left(2x^2\right)\times\left(\binom{n}{1}x\right)\cdot c_3=\binom{n}{3}+2\binom{n}{1}=\frac{n(n-1)(n-2)}{6}+2n$

Since c_1, c_2, c_3 are in AP:
$2c_2=c_1+c_32\left[\frac{n(n-1)}{2}+2\right]$
$=n+\left[\frac{n(n-1)(n-2)}{6}+2n\right]n^2-n+4$
$=3n+\frac{n(n-1)(n-2)}{6}n^2-4n+4$
$=\frac{n(n-1)(n-2)}{6}(n-2)^2$
$=\frac{n(n-1)(n-2)}{6}$

One obvious solution is $n-2=0\Rightarrow\mathbf{n=2}$.
If $n\neq 2$, we can divide both sides by
$(n-2):(n-2)=\frac{n(n-1)}{6}6n-12=n^2-nn^2-7n+12=0(n-3)(n-4)=0$
So, $\mathbf{n=3}$ and $\mathbf{n}=4$.

All values of $n(2, 3,$ and $4)$ are natural numbers.
Sum $=2+3+4=9$

Hence, the correct answer is option (1).

Q. 21 Let $|A|=6$, where A is a 3×3 matrix. If $\left|\operatorname{adj}\left(3\operatorname{adj}\left(A^2\cdot\operatorname{adj}(2A)\right)\right)\right|=2^m\cdot 3^n, m, n\in N$, then $m+n$ is equal to_____

Correct Answer:
62

Solution:
Given $|A|=6$ and $n=3$.
We need to simplify the expression: $X=\left|\operatorname{adj}\left(3\operatorname{adj}\left(A^2\cdot\operatorname{adj}(2A)\right)\right)\right|$
Let $B=A^2\cdot\operatorname{adj}(2A)$.
Using the product rule: $|B|=\left|A^2\right|\cdot|\operatorname{adj}(2A)|$
Since $n=3,|\operatorname{adj}(2A)|=|2A|^{3-1}=|2A|^2$.
Also, $|2A|=2^3|A|=8|A|$.
Substituting these in: $|B|=|A|^2\cdot(8|A|)^2=|A|^2\cdot 64|A|^2=64|A|^4$
Given $|A|=6=2\cdot 3:|B|=2^6\cdot(2\cdot 3)^4=2^6\cdot 2^4\cdot 3^4=2^{10}\cdot 3^4$
Let $C=3\operatorname{adj}(B)\cdot|C|=|3\operatorname{adj}(B)|=3^3|\operatorname{adj}(B)|$
Using the adjoint property again: $|C|=3^3\cdot|B|^{3-1}=3^3\cdot|B|^2$
Substitute $|B|=2^{10}\cdot 3^4:|C|=3^3\cdot\left(2^{10}\cdot 3^4\right)^2=3^3\cdot 2^{20}\cdot 3^8=2^{20}\cdot 3^{11}$
We need $|\operatorname{adj}(C)|:|\operatorname{adj}(C)|=|C|^{n-1}=|C|^2$
Substitute $|C|=2^{20}\cdot 3^{11}:|\operatorname{adj}(C)|=\left(2^{20}\cdot 3^{11}\right)^2=2^{40}\cdot 3^{22}$

Comparing with $2^m \cdot 3^n : m = 40 n = 22$
$m + n = 40 + 22 = 62$

Hence, the correct answer is 62.

Q. 22 The number of 4 -letter words, with or without meaning, which can be formed using the letters PQRPQRSTUVP, is _____.

Correct Answer:
1422

Solution:
The given letters are P, Q, R, P, Q, R, S, T, U, V, P.
- P:3
- Q:2
- R:2
- S, T, U, V: 1 each

Total distinct letters available: $7\{P, Q, R, S, T, U, V\}$.

Case 1: All 4 letters are distinct
Select 4 distinct letters from 7: ${}^7C_4 = 35$
Arrange them: $4! = 24$
Total: $35 \times 24 = 840$

Case 2: 2 letters are alike, and 2 are distinct
Select 1 pair from $\{P, Q, R\}$ and 2 distinct letters from the remaining 6: ${}^3C_1 \times {}^6C_2 = 3 \times 15 = 45$
Arrange them: $\frac{41}{2!} = 12$
Total: $45 \times 12 = 540$

Case 3: 2 letters are alike (Type 1), and 2 are alike (Type 2)
Select 2 pairs from the 3 available pairs $\{P, Q, R\} : {}^3C_2 = 3$
Arrange them: $\frac{41}{212!} = 6$
Total: $3 \times 6 = 18$

Case 4:3 letters are alike, and 1 is distinct
Select the triplet (only P is available) and 1 distinct letter from the remaining 6: ${}^1C_1 \times {}^6C_1 = 6$
Arrange them: $\frac{41}{31} = 4$
Total: $6 \times 4 = 24$

Total number of 4 -letter words $= 840 + 540 + 18 + 24 = 1422$
Hence, the correct answer is 1422.

Q. 23 From the first 100 natural numbers, two numbers, first a and then b are selected randomly without replacement. If the probability that $\mathrm{a}-\mathrm{b} \geqslant 10$ is $\frac{\mathrm{m}}{\mathrm{n}}, \gcd(\mathrm{m}, \mathrm{n})=1$, then $\mathrm{m}+\mathrm{n}$ is equal to

____ .

Correct Answer:
311

Solution:
Since two numbers are selected from the first 100 natural numbers without replacement:

$n(S) = 100 \times 99 = 9900$

We need $a - b \geq 10$, which can be rewritten as $a \geq b + 10$.
If $b = 1$, then $a \in \{11, 12, \ldots, 100\} \rightarrow 90$ values.
If $b = 2$, then $a \in \{12, 13, \ldots, 100\} \rightarrow 89$ values.
If $b = 3$, then $a \in \{13, 14, \ldots, 100\} \rightarrow 88$ values.
...
If $b = 90$, then $a \in \{100\} \rightarrow 1$ value.
If $b = 91$, then a has no values because a must be ≥ 101.

The total number of favourable outcomes is the sum:

$n(E) = 90 + 89 + 88 + \cdots + 1$

$$n(E) = \frac{90 \times (90+1)}{2} = \frac{90 \times 91}{2} = 45 \times 91 = 4095$$

$P = \frac{n(E)}{n(S)} = \frac{4095}{9500}$

$P = \frac{819}{1980}$

$P = \frac{91}{220}$
Since $\gcd(91, 220) = 1$, we have:

$m = 91$
$n = 220$

$m + n = 91 + 220 = 311$
Hence, the correct answer is 311.

Q. 24 Let the area of the region bounded by the curve $y = \max\{\sin x, \cos x\}$, lines $x = 0, x = \frac{3\pi}{2}$, and the x-axis be A . Then, $\mathrm{A} + \mathrm{A}^2$ is equal to ____ .

Correct Answer:
3

Solution:

Required area $= \int_0^{\pi/4} \cos x dx + \int_{\pi/4}^{\pi} \sin x dx$

$+ \int_{\pi}^{5\pi/4} (-\sin x) dx + \int_{5\pi/4}^{3\pi/2} (-\cos x) dx$

$= [\sin x]_0^{\pi/4} + [-\cos x]_{\pi/4}^{\pi} + [\cos x]_{\pi}^{5\pi/4} - [\sin x]_{5\pi/4}^{3\pi/2}$

$= \frac{1}{\sqrt{2}}\left[1 + \frac{1}{\sqrt{2}}\right] + \left[-\frac{1}{\sqrt{2}} + 1\right] - \left[-1 + \frac{1}{\sqrt{2}}\right]$

$= \sqrt{2} + 1 - \frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}$

$= 3$

Hence, the correct answer is 3.

Q. 25 Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x \left((f(t))^2 + (f'(t))^2\right) \mathrm{dt}$. Then the mean of $f(\log_e(1)), f(\log_e(2)), \ldots\ldots, f(\log_e(625))$ is equal to ____

Correct Answer:
1565

Solution:

Given: $(f(x))^2 = 25 + \int_0^x \left[(f(t))^2 + (f'(t))^2\right] dt$

Differentiate both sides with respect to x :

$2f(x)f'(x) = 0 + \left[(f(x))^2 + (f'(x))^2\right]$

$(f'(x))^2 - 2f(x)f'(x) + (f(x))^2 = 0$

This is a perfect square:

$\left(f'(x) - f(x)\right)^2 = 0$

$\Rightarrow f'(x) = f(x)$

$\Rightarrow \frac{f'(x)}{f(x)} = 1$

$\int \frac{f'(x)}{f(x)} dx = \int 1 dx$

$\ln|f(x)| = x + C$

$f(x) = e^{x+C} = ke^x$

Substitute $x = 0$ into the original equation:

$(f(0))^2 = 25 + \int_0^0 [\ldots] dt$

$(f(0))^2 = 25 \Rightarrow f(0) = 5$ (since f is non-negative).

Substitute $x = 0$ into $f(x) = ke^x$:

$5 = ke^0 \Rightarrow k = 5$
Thus, $f(x) = 5e^x$

The terms are $f(\ln 1), f(\ln 2), \ldots, f(\ln 625)$.
Since $f(\ln r) = 5e^{\ln r} = 5r$:
The values are $5(1), 5(2), 5(3), \ldots, 5(625)$.

$$\text{Mean } = \frac{\sum_{x=1}^{625} 5r}{625}$$

$$\text{Mean } = \frac{5 \cdot \frac{625(0.5+1)}{2}}{625}$$

$$\text{Mean } = \frac{5 \cdot 626}{2}$$

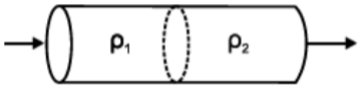
$$\text{Mean } = 5 \cdot 313 = 1565$$

Hence, the correct answer is 1565.

CAREERS360

Physics

Q. 1 Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is:



Option 1:
Zero

Option 2:
$\frac{(\rho_1+\rho_2)I\varepsilon_0}{2}$

Option 3:
$\varepsilon_0 I\,|\rho_1-\rho_2|$

Option 4:
$\varepsilon_0 I\,|\rho_1+\rho_2|$

Correct Answer:
$\varepsilon_0 I\,|\rho_1-\rho_2|$

Solution:

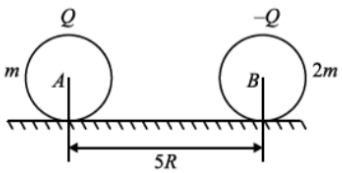
Apply Gauss's law: $\frac{q_{\text{in}}}{\varepsilon_0}$

(Outgoing flux - incoming flux)

$$V = IR = \frac{I\rho\ell}{A} \Rightarrow \frac{V}{\ell}A = I\rho \Rightarrow EA = I\rho$$

$$\Rightarrow \frac{q_{\text{in}}}{\varepsilon_0} = I\,|\rho_1-\rho_2| \Rightarrow q_{\text{in}} = I\varepsilon_0\,|\rho_1-\rho_2|$$

Q. 2 Two smooth spherical non conducting shells each of radius R having uniformly distributed charge Q and $-Q$ on their surfaces are released on a smooth non conducting surface when the distance between their centres is $5R$. The mass of A is m and that of B is $2m$. The speed of A just before A and B collide is [Neglect gravitational interaction] $\left(K=\frac{1}{4\pi\varepsilon_0}\right)$



Option 1:
$\sqrt{\frac{2KQ^2}{5mR}}$

Option 2:
$\sqrt{\frac{4KQ^2}{5mR}}$

Option 3:
$\sqrt{\frac{8KQ^2}{5mR}}$

Option 4:
$\sqrt{\frac{16KQ^2}{5mR}}$

Correct Answer:
$\sqrt{\frac{2KQ^2}{5mR}}$

Solution:
$mv = 2m\,(v')$

$$\frac{KQ(-Q)}{5R} = \frac{KQ(-Q)}{2R} + \frac{1}{2}mv^2 + \frac{1}{2}(2m)\left(v'\right)^2$$

$$\Rightarrow v = \sqrt{\frac{2KQ^2}{5mR}}$$

Q. 3 Consider a conductor with a spherical cavity in it. A point charge q_0 is placed at the centre of cavity and a point charge Q is placed outside conductor.
Statement-1: Total charge induced on cavity wall is equal and opposite to the charge inside.
Statement-2: If cavity is surrounded by a Gaussian surface, where all parts of Gaussian surface are located inside the conductor, $\oint \vec{E} \cdot \vec{ds} := 0$; hence $q_{\text{induced}} = -q_0$

Option 1:
(1) Statement-1 is true, statement -2 is true and statement-2 is correct explanation for statement- 1 .

Option 2:
(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

Option 3:
(3) Statement-1 is true, statement-2 is false.

Option 4:
(4) Statement- 1 is false, statement- 2 is true.

Correct Answer:

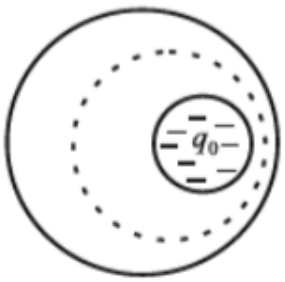
(1) Statement-1 is true, statement -2 is true and statement-2 is correct explanation for statement- 1 .

Solution:

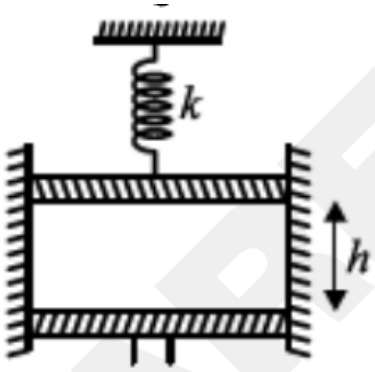
Since $\mathrm{E}_{\text{inside}}$ conductor $= 0$

$\oint \vec{E} \cdot \overrightarrow{ds} = \frac{q_{\text{in}}}{\varepsilon_0} = \frac{q_0 + q_{\text{in on surface}}}{\varepsilon_0}$

$\Rightarrow q_{\text{in}}$ on surface $= -q_0$



Q. 4 An ideal gas at temperature 273 K is enclosed in an adiabatic vertical cylinder having area of cross section $A = 27\ \mathrm{cm}^2$, between two light movable pistons as shown in the figure. Spring with force constant $k = 3700\ \mathrm{N/m}$ is in a relaxed state initially. Now the lower piston is moved upwards by a height $h/2$ and h being the initial length of gas column. It is observed that the upper piston moves up by a distance $h/16$. Taking λ for the gas to be 1.5 , find the final temperature of the gas.



Option 1:

273 K

Option 2:

640 K

Option 3:

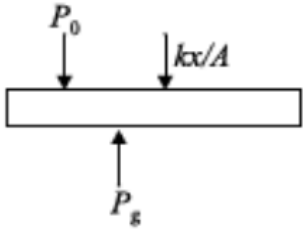
364 K

Option 4:

527 K

Correct Answer:
364 K

Solution:



$$P_g = P_0 + \frac{kx}{A} = P_0 + \frac{kh}{16A}$$

For adiabatic process $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$$P_0(hA)^{\gamma} = P_g\left[\left(\frac{h}{2} + \frac{h}{16}\right)A\right]^{\gamma}$$

$$10^5(hA)^{\gamma} = \left[10^5 + \frac{3700 \times h}{16 \times 27 \times 10^{-4}}\right]\left[\frac{9h}{16}A\right]^{\gamma}$$

$$\frac{64}{27} = 1 + \frac{370 \times h}{16 \times 27}$$

$$\frac{37}{27} = \frac{370 \times h}{16 \times 27} \Rightarrow h = \frac{16}{10}$$

$$PV = nRT$$

Initially, $P_0hA = nR^{\times}273 \ldots$ (i)
Finally, $\left\{P_0 + \frac{kh}{16A}\right\}\left\{\frac{Ph}{16}A\right\} = nRT$
(ii) ÷ (i) gives

$$\left\{1 + \frac{kh}{16A \times P_0}\right\}\frac{9}{16} = \frac{T}{273}$$

$$T = 273 \times \frac{9}{16}\left[1 + \frac{370}{270}\right]$$

$$T = 273 \times \frac{4}{3} = 364K$$

Q. 5 In an industrial process 10 kg of water per hour is to be heated from $20^{\circ}\mathrm{C}$ to $80^{\circ}\mathrm{C}$. To do this, steam at $150^{\circ}\mathrm{C}$ is passed from a boiler into a copper coil immersed in water. The steam condenses in the coil and is returned to the boiler as water at $90^{\circ}\mathrm{C}$. How many kg of steam is required per hour. (Specific heat of steam $= 1\mathrm{cal/g^{\circ}C}$, Latent heat of vaporisation $= 540\mathrm{cal/g}$, specific heat of water $= 1\mathrm{cal/g^{\circ}C}$)

Option 1:
1 g

Option 2:
1 kg

Option 3:
10 g

Option 4:
10 kg

Correct Answer:
1 kg

Solution:
Suppose m kg steam is required per hour Heat is released by steam in following three steps
(i) When $150^{\circ}\mathrm{C}$ steam $\xrightarrow[Q_1]{}$ $100^{\circ}\mathrm{C}$ steam

$$Q_1 = mc_{\text{steam}}\,\Delta\theta = m \times 1(150-100) = 50\text{mcal}$$

(i) When $100^{\circ}\mathrm{C}$ steam $\xrightarrow[Q_2]{}$ $100^{\circ}\mathrm{C}$

Water
$Q_2 = mL_{\text{v}} = m \times 540 = 540\text{mcal}$
$Q_2 = mLv = m540 = 540m\text{cal}$
(iii) When $100^{\circ}\mathrm{C}$ water $\xrightarrow[Q_2]{}$ $90^{\circ}\mathrm{C}$ water

$$Q_3 = mC_{\text{w}}\Delta\theta = m \times 1 \times (100-90) = 10\text{mcal}$$

Hence
total heat given by the steam

$$Q = Q_1 + Q_2 + Q_3 = 600\ \text{mcal}$$

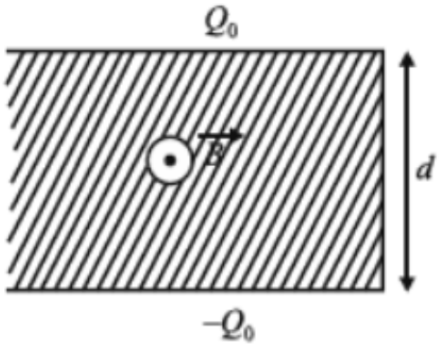
Heat taken by 10 kg water

$$Q' = mC_w\Delta\theta = 10 \times 10^3 \times 1 \times (80-20) = 600 \times 10^3\text{cal}$$

Hence

$$Q = Q' \Rightarrow 600\ \text{m} = 600 \times 10^3 \Rightarrow m = 10^3\text{gm} = 1\ \text{kg}$$

Q. 6 A slab is introduced to the parallel plate capacitor having charge Q_0. The capacitor is discharged through the slab. A magnetic field is present throughout and is out of the plane of the paper. The total momentum given to the slab after complete discharge is



Option 1:
BQ_0d

Option 2:
$2BQ_0d$

Option 3:
$\frac{3}{2}BQ_0d$

Option 4:
$\frac{4}{3}BQ_0d$

Correct Answer:
BQ_0d

Solution:

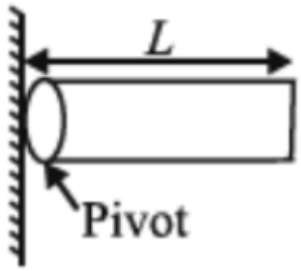
$$dB(Idt) = mdv$$

$$\Rightarrow dB\int_0^{Q_0} Idt = m\int_0^{v} dv$$

$$\Rightarrow dBQ_0 = mv$$

CAREERS360

Q. 7 A uniform rod of length L and mass M is free to rotate about a frictionless pivot at one end as shown in the figure. The rod is released from rest in the horizontal position.



Option 1:
Initial linear acceleration of right end of the rod is g and is common to all points on the rod.

Option 2:
Initial linear acceleration of the right end of the rod is $3g/2$ and is common to all points on the rod.

Option 3:
Initial linear acceleration of the right end of the rod is $3g/2$ and its value is different at different points.

Option 4:
No points on the rod have a linear acceleration greater than g.

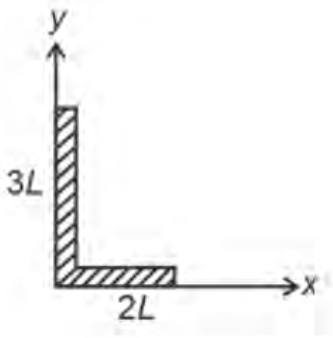
Correct Answer:
Initial linear acceleration of the right end of the rod is $3g/2$ and its value is different at different points.

Solution:
$(\mathrm{A})\tau_{mg} = I\alpha$

$$Mg \times \frac{L}{2} = \frac{ML^2}{3}\alpha \text{ i.e, } \alpha = 3g/2L$$

$(\mathrm{B})\ a = r\alpha$

Q. 8 A rod of length 5 L is bent right angle keeping one side length as 2 L



The position of the centre of mass of the system: (Consider L=10 cm)

Option 1:
$3\hat{i}+7\hat{j}$

Option 2:
$4\hat{i}+9\hat{j}$

Option 3:
$2\hat{i}+3\hat{j}$

Option 4:
$5\hat{i}+8\hat{j}$

Correct Answer:
$4\hat{i}+9\hat{j}$

Solution:

$m_1=(2m); x_1=L, y_1=0$

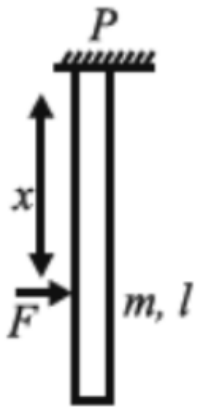
$m_2=(3m); x_2=0, y_2=\dfrac{3L}{2}$

$\text{So, } x_{\text{cm}}=\dfrac{2m(L)+0}{5m}=\dfrac{2L}{5}=4\hat{i}(\text{ cm})$

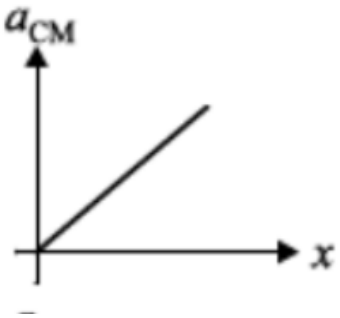
$y_{\text{cm}}=\dfrac{(0)+\frac{9mL}{2}}{5m}=\dfrac{9L}{10}=9\hat{j}(\text{ cm})$

$\text{So } \overrightarrow{r_{\text{cm}}}=(4\hat{i}+9\hat{j})\text{cm}$

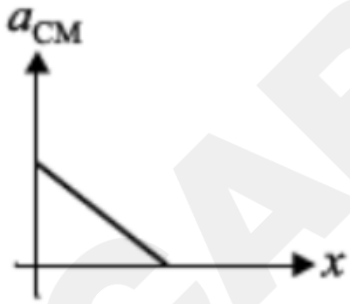
Q. 9 A rod of mass m and length ℓ is hinged at one of its end P. A Force F is applied at a distance x from P as shown in the figure. The acceleration of centre of mass (a_{CM}) varies with x as:



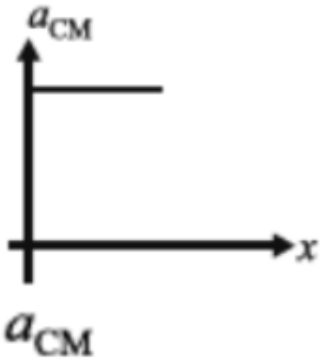
Option 1:



Option 2:

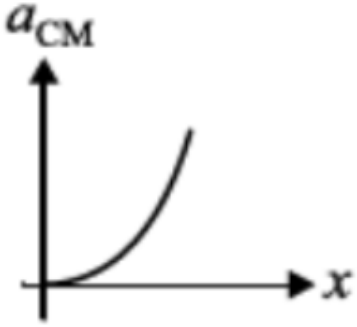


Option 3:

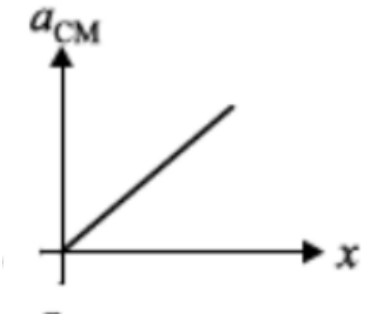


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Option 4:



Correct Answer:

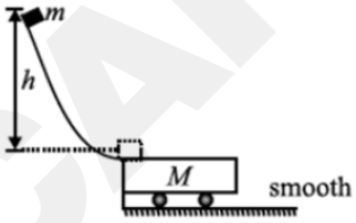


Solution:

$$F \cdot x = \frac{m\ell^2}{3}\alpha \Rightarrow \alpha = \frac{3Fx}{m\ell^2}$$

$$\therefore a_{CM} = \frac{\alpha\ell}{2} = \frac{3Fx}{m\ell^2} \cdot \frac{\ell}{2} = \frac{3Fx}{2m\ell}$$

Q. 10 A carriage of mass M and length ℓ is joined to the end of a slope as shown in the figure. A block of mass m is released from the slope from height h. It slides till end of the carriage. The friction between the mass and the slope and also friction between carriage and horizontal floor is negligible. Coefficient of friction between block and carriage is μ. Find h in terms of given parameters.



Option 1:

$\mu\left(1+\frac{M}{m}\right)\ell$

Option 2:

$2\mu\left(1+\frac{M}{m}\right)\ell$

Option 3:
$\mu\left(2+\frac{m}{M}\right)\ell$

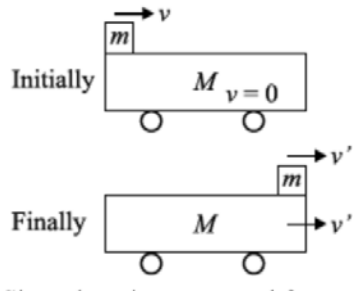
Option 4:
$\mu\left(1+\frac{m}{M}\right)\ell$

Correct Answer:
$\mu\left(1+\frac{m}{M}\right)\ell$

Solution:
Suppose velocity of block at the end of slop $= v$
Then by work energy theorem

$$mgh = \frac{1}{2}mv^2$$

Now,



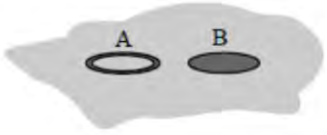
Since there is no external force on (cart + block) system

$$P_i = P_f$$
$$mv = (m+M)v'$$
$$v' = \frac{mv}{m+m}$$

Now work done by friction $=$ change in kinetic energy of system.

$$-\mu mg\ell = \frac{1}{2}(m+M)v'^2 - \frac{1}{2}mv^2$$
$$\mu mg\ell = \frac{1}{2}mv^2 - \frac{1}{2}(m+M)\frac{m^2v^2}{(m+M)^2}$$
$$h = \mu\left(\frac{m+M}{M}\right)\ell$$

Q. 11 A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses?



Option 1:
Both have the same mass.

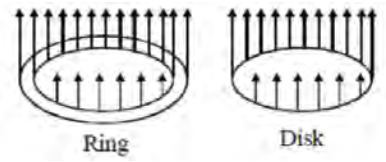
Option 2:
Mass of the ring is half of that of the disk.

Option 3:
Mass of the ring is double to that of the disk.

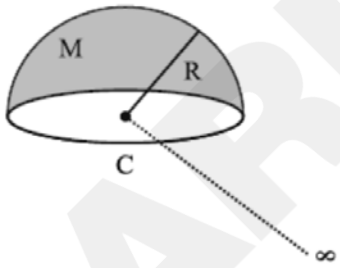
Option 4:
More information is needed to decide.

Correct Answer:
More information is needed to decide.

Solution:
Ring have double surface than that of disk.



Q. 12 A particle of mass m was transferred from infinity into the centre of the base of a uniform hemisphere of mass M and radius R. What work was performed in the process by the gravitational force exerted on the particle by the hemisphere?



Option 1:
$\frac{GmM}{2R}$

Option 2:
$\frac{3GmM}{2R}$

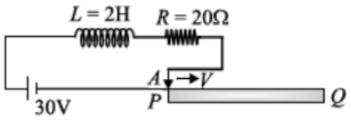
Option 3:
$\frac{GmM}{R}$

Option 4:
$\frac{3GmM}{R}$

Correct Answer:
$\frac{3GmM}{2R}$

Solution:
$W_g = \Delta U_g$
$= m\left(\Delta V_g\right) = m\left(V_{gf} - V_{gi}\right) = m\left(V_\infty - V_c\right)$
$W_g = m\left(0 - \left(\frac{-3GM}{2R}\right)\right) = \frac{3GmM}{2R}$

Q. 13 In the circuit shown in figure, A is a sliding contact which can move over a smooth rod PQ. Resistance per unit length of the rod PQ is 1ohm/m. Initially slider is just left to the point P and circuit is in the steady state. At $t = 0$ slider starts moving with constant velocity $v = 5\text{ m/s}$ towards right. Current in the circuit at $t = 2\text{sec}$ is



Option 1:
1 amp

Option 2:
less than 0.5 amp

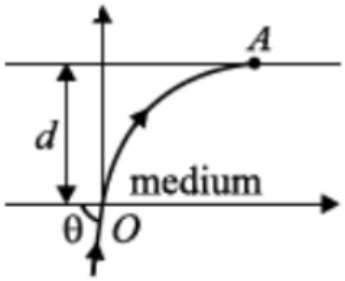
Option 3:
more than 1 amp

Option 4:
in between 0.5 to 1.0 amp

Correct Answer:
more than 1 amp

Solution:
$30 - L\frac{di}{dt} = i(20 + 5t)$
$\frac{di}{dt} = -ve \Rightarrow 30 - L\frac{di}{dt} > 30$
$\Rightarrow \text{ at } t = 2\text{sec}; i > 1\text{amp}$

Q. 14 A long rectangular slab of transparent medium of thickness d is placed on table with length parallel to x-axis and width parallel to the y-axis. A ray of light travelling along y-axis enters at origin such that angle of incidence $\theta\,(\theta \to 90^\circ)$. The refractive index of the medium varies as $\mu = \sqrt{1 + e^{x/d}}$. The refractive index of the air is 1 . Find the x-coordinate of the point A, where the ray intersects the upper surface of the slab air boundary.



Option 1:
$x = 2d^2$

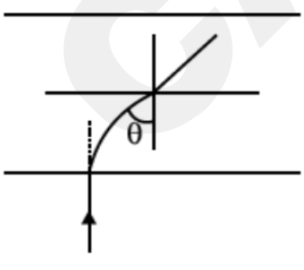
Option 2:
$x = 4d^2$

Option 3:
$x = 4d \ln 2$

Option 4:
$x = 2d \ln 2$

Correct Answer:
$x = 2d \ln 2$

Solution:
Apply Snell's law $\mu \sin\theta =$ constant,



From Snell's law

$$1 = \mu \sin\theta$$

$$\sin\theta = \frac{1}{\mu}$$

$$\frac{dy}{dx} = \frac{1/\mu}{\frac{(\mu^2-1)^{1/2}}{\mu}}$$

$$\frac{dy}{dx} = e^{\frac{-x}{2d}}$$

$$\int dy = \int e^{\frac{-x}{2d}} dx$$

$$y = -2dx^{\frac{-x}{2d}} + C$$

$$\text{at } = x = 0, y = 0, C = 2d$$

$$\text{at } y = 0$$

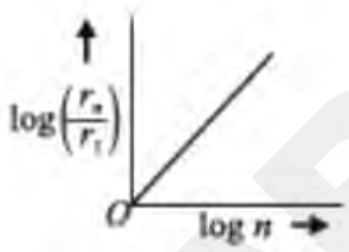
$$y = -2dx^{\frac{-x}{2d}} + 2d; -d = -2de^{\frac{-x}{2d}}; \frac{1}{2} = e\frac{-x}{2d}$$

$$\ln 2 = \frac{x}{2d}$$

$$2d \ln 2 = x$$

Q. 15 In hydrogen atom, the radius of nth Bohr's orbit is r_n. The graph between $\log\left(\frac{r_n}{r_1}\right)_{\text{and}} \log(n)$ will be

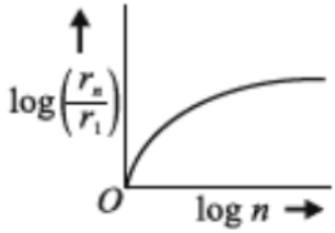
Option 1:



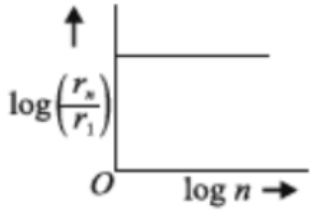
Option 2:



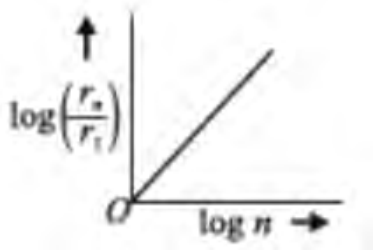
Option 3:



Option 4:



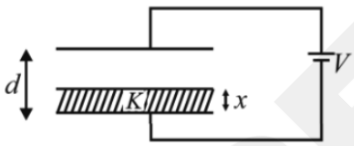
Correct Answer:



Solution:

$r_n \propto n^2; \log \frac{r_n}{r_1} = 2 \log n$

Q. 16 In Two conducting plates each having area A are kept at a separation d parallel to each other. The two plates are connected to a battery of emf V. The space between the plates is filled with a liquid of dielectric constant K. The height (x) of the liquid between the plates increases at a uniform rate from zero to d in a time interval t_0. Write the current through the cell at time t.



Option 1:

$\frac{K\varepsilon_0 A}{(K-1)d} \times \frac{dx}{dt} \times V$

Option 2:

$\frac{VK\varepsilon_0 A(K-1)dx}{(Kd-(K-1)x)^2 dt}$

Option 3:

$\frac{VK\varepsilon_0 A(K-1)dx}{(Kd-(K-1)x)dt}$

Option 4:

None of these

Correct Answer:

$\frac{VK\varepsilon_0 A(K-1)dx}{(Kd-(K-1)x)^2 dt}$

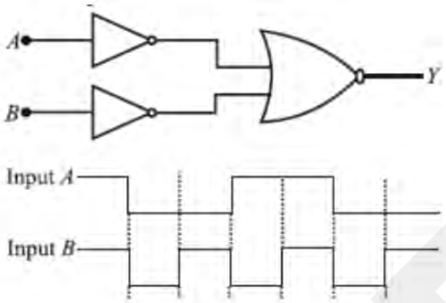
Solution:
Height of liquid at time t is

$x = \frac{d}{t_0}t$

We can take the system as two capacitors in series

$$C_1 = \frac{K \in_0 A}{x} \text{ and } C_2 = \frac{\in_0 A}{d-x}$$
$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{x}{K \in_0 A} = \frac{Kd-(K-1)x}{K \in_0 A}$$
$$\Rightarrow C = \frac{K \in_0 A}{Kd(K-1)x}$$

Q. 17 The logic circuit shown below has the input waveforms ' A ' and ' B ' as shown. Pick out the correct output waveform



Option 1:



Option 2:



Option 3:



Option 4:

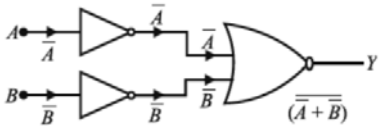


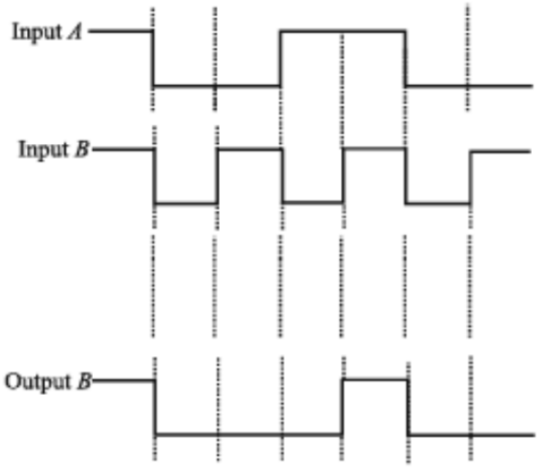
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Correct Answer:



Solution:

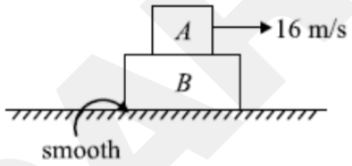




$Y = \overline{(\bar{A} + \bar{B})}$

$Y = A \cdot B$

Q. 18 A block ' A ' of mass 1 Kg is kept on a block ' B ' of mass 2 Kg . Block ' A ' is imparted a velocity of $16\ \mathrm{m/s}$ at $t = 0$. If the coefficient of friction between A and B is 0.2 , find the time (in sec) after which momentum of A and B relative to ground will be equal. (Take $g = 10\ \mathrm{m/s^2}$).



Option 1:
2 sec

Option 2:
1 sec

Option 3:
3 sec

Option 4:
4 sec

Correct Answer:
4 sec

Solution:
Momentum of A at time ' t ' is

$$P_A = m_A v_A = m_A(16 - \mu g t)$$

$$\text{And } P_B = m_B v_B = m_B \left(\frac{\mu m A g}{m_B}\right) t$$

$$P_A = P_B$$
$$\Rightarrow 16 - \mu g t = \mu g t$$
$$\Rightarrow \mu g t = 8$$
$$\therefore t = \frac{8}{(0.2)(10)} = 4\text{sec}$$
$$\therefore t = 4\text{sec}$$

Q. 19 A block A standing wave is maintained in a homogeneous string of cross-sectional area s and density ρ. It is formed by the superposition of two waves travelling in opposite directions given by the equation $y_1 = a\sin(\omega t - kx)$ and $y_2 = 2a\sin(\omega t + kx)$ The total mechanical energy confined between the sections corresponding to the adjacent antinodes is

Option 1:
$\frac{3\pi s\rho\omega^2 a^2}{2k}$

Option 2:
$\frac{\pi s\rho\omega^2 a^2}{2k}$

Option 3:
$\frac{5\pi s\rho\omega^2 a^2}{2k}$

Option 4:
$\frac{2\pi s\rho\omega^2 a^2}{2k}$

Correct Answer:
$\frac{5\pi s\rho\omega^2 a^2}{2k}$

Solution:

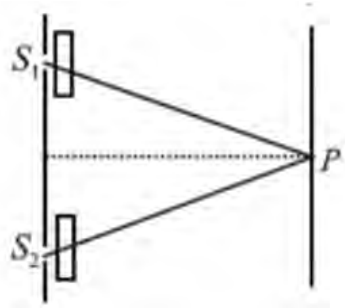
Distance between two adjacent
antinodes is $\frac{\lambda}{2}$ or $\frac{\pi}{2}$
volume of string between two adjacent antinodes

$V=\frac{\pi}{k}aE=(V)\left(u_1+u_2\right)$

Here, u is the energy density (unity per unit volume)
which is equal to $\frac{1}{2}\rho A^2\omega^2$

$\Rightarrow E=\left(\frac{\pi}{k}\right)s\left[\frac{1}{2}\rho a^2\omega^2+\frac{1}{2}(2a)^2\omega^2\right]=\frac{5\pi s\rho a^2\omega^2}{2k}$

Q. 20 One slit in a YDSE set up is covered by a glass plate (Refractive index $=\mu_1$) and other by another glass plate (Refractive index $=\mu_2$) of same thickness. If I_0 is the intensity of light through each slit and is the wavelength of light, then intensity at point P is (P is symmetrical w.r.t. slits S_1 and S_2)



Option 1:
$4I_0\cos^2\frac{\pi}{\lambda}(\mu_1-\mu_2)t$

Option 2:
$4I_0\cos^2\frac{\pi}{\lambda}(\mu_1+\mu_2)t$

Option 3:
$4I_0\sin^2\frac{\pi}{\lambda}(\mu_1-\mu_2)t$

Option 4:
$4I_0\sin^2\frac{\pi}{\lambda}(\mu_1+\mu_2)t$

Correct Answer:
$4I_0\cos^2\frac{\pi}{\lambda}(\mu_1-\mu_2)t$

Solution:

$$\Delta x = (\mu_1 - \mu_2)t$$

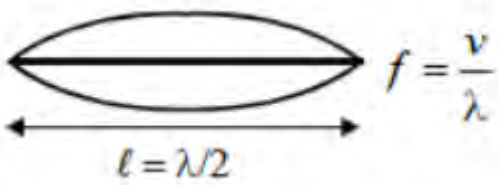
$$I_P = 2I_0\left[1 + \cos\left(\frac{2\pi}{\lambda}\right)(\mu_1 - \mu_2)t\right]$$

$$= 4I_0\cos^2\frac{\pi}{\lambda}(\mu_1 - \mu_2)t$$

Q. 21 A string will break apart if it is placed under too much tensile stress. One type of steel has density $\rho_{\text{steel}} = 10^4\ \text{kg/m}^3$ and breaking stress $\sigma = 9 \times 10^8\ \text{N/m}^2$. We make a guitar string from (4π) gram of this type of steel. It should be able to withstand $(900\pi)\text{N}$ without breaking. What is the highest possible fundamental frequency (in Hz) of standing waves on the string if the entire length of the string vibrates?

Correct Answer:
375

Solution:



Given
- Density of steel:$\rho = 10^4\ \text{kg m}^{-3}$

- Breaking stress:$\sigma = 9 \times 10^8\ \text{N m}^{-2}$

- Mass of string:$m = 4\pi\ \text{g} = 0.004\pi\ \text{kg}$

- Maximum safe tension:$T_{\max} = 900\pi\ \text{N}$

Cross-sectional area (from breaking stress)

$\sigma = \frac{T}{A} \quad \Rightarrow \quad A = \frac{T}{\sigma} = \frac{900\pi}{9\times10^8} = \pi \times 10^{-6}\ \text{m}^2$
Volume of the string:

$V = \frac{m}{\rho} = \frac{0.004\pi}{10^4} = 4 \times 10^{-7}\pi\ \text{m}^3$

Length of the string
Since $V = AL$,

$L = \frac{V}{A} = \frac{4\times10^{-7}\pi}{\pi\times10^{-6}} = 0.4\ \text{m}$
Linear mass density

$\mu = \frac{m}{L} = \frac{0.004\pi}{0.4} = 0.01\pi\ \text{kg m}^{-1}$

Fundamental frequency of a stretched string

$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

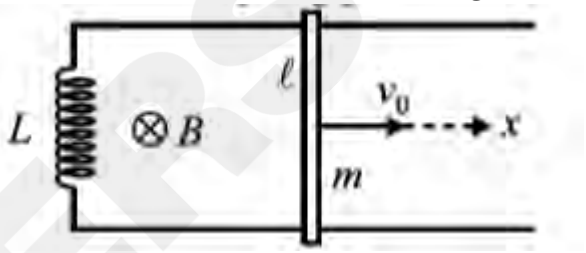
Substitute values:

$f_1 = \frac{1}{0.8}\sqrt{\frac{900\pi}{0.01\pi}} = 1.25 \times \sqrt{90000} = 1.25 \times 300$

Final Answer

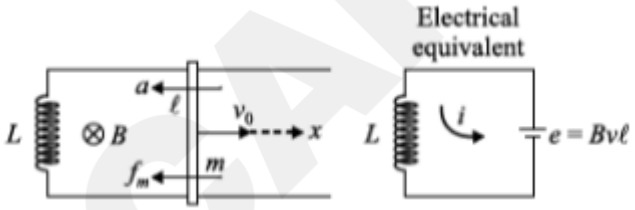
$f_{\max} = 375 \text{ Hz}$

Q. 22 A loop is formed by two parallel conductors connected by a solenoid with inductance $L = 2\text{H}$ and a conducting rod of mass $m = 80$ g , which can freely (without friction) slide over the conductors. The conductors are located in a horizontal plane in a uniform vertical magnetic field with induction $B = 4\text{T}$. The distance between the conductors is equal to $\ell = 4$ cm. At the moment $t = 0$ the rod is imparted an initial velocity $v_0 = 2$ cm/s directed to the right. Find the amplitude (in cm) for the oscillations of conducting rod, if the electric resistance of the loop is negligible.



Correct Answer:
5

Solution:



According to Len's law, magnetic force, will oppose the motion of conducting rod.

$F_m = i\ell B; ma = i\ell B \text{ and } \left(a = -\frac{dv}{dt}\right)$

So,

$m \times \left(-\frac{L}{B\ell}\frac{d^2i}{dt^2}\right) = i\ell B \text{ or } \frac{d^2i}{dt^2} = -\left(\frac{B^2\ell^2}{mL}\right)i$

And

$\varepsilon = L\frac{di}{dt}; Bv\ell = L\frac{di}{dt}; (B\ell)\left(\frac{dv}{dt}\right) = L\frac{d^2i}{dt^2}$

This is a second order differential equation;

So, $i = i_0 \sin(\omega t)$

Where $\frac{1}{2}mv_0^2 = \frac{1}{2}Li_0^2$

$\Rightarrow \left[i_0 = v_0\sqrt{\frac{m}{L}}\right]$ and $\left[\omega = \sqrt{\frac{B^2\ell^2}{mL}}\right]$

We have,

$B v \ell = L\frac{di}{dt}$ or $B(vdt)\ell = L(di)$
Or $B\ell(dx) = L(di)$

$$\text{Or } x = \frac{Li}{B\ell} = \frac{Li_0\sin\omega t}{B\ell} = \frac{Lv_0\sqrt{\frac{m}{L}}}{B\ell}\sin\omega t$$
$$\Rightarrow x = \frac{v_0}{\omega}\sin\omega t$$
$$\text{So, amplitude } = \frac{v_0}{\omega} = \frac{v_0}{B\ell}\sqrt{mL} = 5cm$$

Q. 23 Hydrogen gas in the atomic state is excited to an energy level such that the electrostatic potential energy of hydrogen atom becomes -1.7 eV . Now a photoelectric plate having work function 2.3 eV is exposed to the emission spectra of this gas. Assuming all the transitions to be possible, find the minimum de-Broglie wavelength of the ejected photoelectrons in $\mathring{A}$ to the nearest integer.

Correct Answer:
4

Solution:
$U = 1.7\text{eV}$

$$\therefore E = \frac{U}{2} = -0.85\text{eV}$$
$$-0.85\text{eV} = \frac{-13.6}{n^2}$$
$N = 4$
Energy of the electron transition from $n = 4$ to $n = 1$ is

$$\Delta E = -0.85 - (-13.6) = 12.75\text{eV}$$
$K_{\max} = \Delta E - \phi = 10.45\text{eV}$
Wavelength of electron is

$$\lambda = \frac{h}{\sqrt{2mK_{\max}}} = 3.8\mathring{A} \approx 4\mathring{A}$$

Q. 24 An infinite hollow cylindrical of radius R having uniform surface charge density σ is rotated with constant angular velocity ω about its axis. Magnetic field at the centre of cylinder is $B = \mu_0\sigma R^x\omega$. Find x

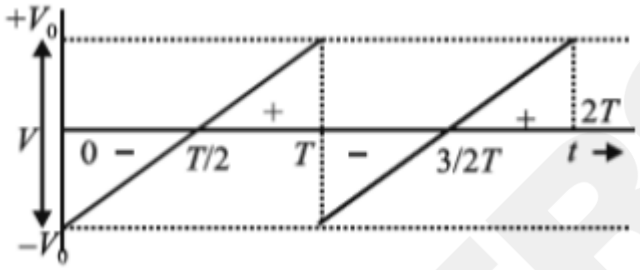
Correct Answer:
1

Solution:
Consider unit length of cylinder charge $= \sigma 2\pi R \times 1$

$$\text{time } = 2\pi/\omega$$
$$\text{So } ni = \frac{2\pi R\sigma\omega}{2\pi} = \sigma R\omega$$
Now $B = \mu_0 ni$

$$B = \mu_0 \sigma R\omega$$

Q. 25 The rms value for the saw-tooth voltage of peak value V_0 from $t = 0$ to $t = 2$ T as shown in figure is $\frac{V_0}{\sqrt{x}}$. Find the value of x



Correct Answer:
3

Solution:
General equation of V
$$V = \frac{V_0}{\frac{T}{2}}t - V_0 = \frac{2V_0}{T}t - V_0$$
$$V_{rms} = \left[\frac{\int_0^T V^2 dt}{T}\right]^{\frac{1}{2}} = \left[\frac{\int_0^T \left(\frac{2V_0}{T}t - V_0\right)^2 dt}{T}\right] = \frac{V_0}{\sqrt{3}}$$
$$x = 3$$

Chemistry

Q. 1 For the following conversion, which set of reagents are required:

$$\text{But-2-ene} \longrightarrow \text{2-hydroxy-2-methyl butanoic acid}$$

Select the correct set of reagent:

Option 1:
$HBr; {}^{\ominus}CN, HCN, H_3O^+$

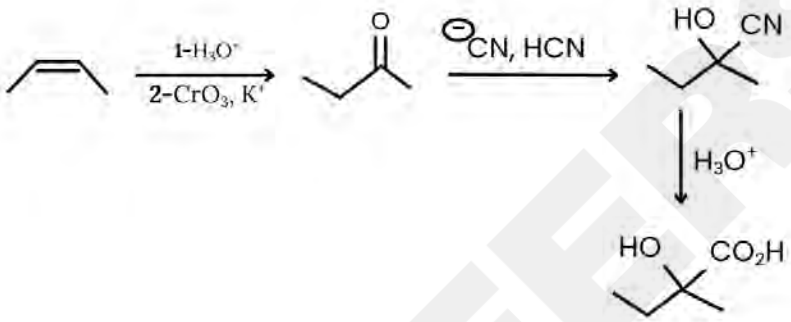
Option 2:
$H_3O^+; CrO_3, H^+; {}^{\ominus}CN, HCN; LiAlH_4$

Option 3:
$H_3O^{\oplus}; CrO_3, H^{\oplus}; {}^{-}CN, HCN; H_3O^{\oplus}$

Option 4:
$HBr; PCC; {}^{\ominus}CN, HCN; H_2/Pd$

Correct Answer:
$H_3O^{\oplus}; CrO_3, H^{\oplus}; {}^{-}CN, HCN; H_3O^{\oplus}$

Solution:



Q. 2 Which of the below quantity will increase on dilution of the solution?

Option 1:
Specific conductance (k) of strong electrolyte

Option 2:
Specific conductance (k) of weak electrolyte

Option 3:
Molar conductance (Λ_m) of a strong electrolyte

Option 4:
Limiting molar conductance (Λ_m^{∞}) of a strong electrolyte

Correct Answer:
Molar conductance (Λ_m) of a strong electrolyte

Solution:
The molar conductance of a strong electrolyte. Increases due to a decrease in interionic attraction forces on dissolution of the solution.

Q. 3 In which of the following compound all the bonds are of equal length?

Option 1:
PCl_5

Option 2:
PF_2Cl_3

Option 3:
ClF_3

Option 4:
XeF_4.

Correct Answer:
XeF_4.

Solution:
In TBP geometry, axial bonds are longer than equatorial bonds.

Q. 4 From the below given options, select the option having the correct order of radius as well as electron affinity.

Option 1:
$Cl > Br$

Option 2:
$Na^+ < K^+$

Option 3:
$Cl > F$

Option 4:
$Li^+ < Na^+$

Correct Answer:
$Cl > F$

Solution:

Because of the larger size of Cl, it can easily accommodate the extra added electron.

Q. 5 An. aqueous solution of $10^{-3}M$ K_2SO_4 and $10^{-3}M$ KI present. Now pure $Pb(NO_3)_2$ (s) is added slowly in the mixture. Calculate concentration of $SO_4{}^{2-}$ ion when PbI_2 starts precipitating in the solution.

$$\text{Ksp}\,(PbI_2) = 10^{-10} \quad \& \quad \text{Ksp}\,(PbSO_4) = 10^{-9}$$

Option 1:
10^{-5}

Option 2:
10^{-6}

Option 3:
10^{-4}

Option 4:
10^{-7}

Correct Answer:
10^{-5}

Solution:

$$[Pb^{2+}][I^-]^2 = 10^{-10}$$
$$[Pb^{2+}](10^{-3})^2 = 10^{-10}$$
$$[Pb^{2+}] = 10^{-4}$$

$$[Pb^{2+}]\,[SO_4^{2-}] = 10^{-9}$$
$$10^{-4}\,[SO_4^{2-}] = 10^{-9}$$
$$[SO_4^{2-}] = 10^{-5}$$

Q. 6 Which of the following Lanthanoids does not have any electrons in the 5d subshell, while writing the electronic configuration in atomic form?

Option 1:
Cerium $(_{58}Ce)$

Option 2:
Europium ($_{63}$Eu)

Option 3:
Gadolinium $(_{64}Cd)$

Option 4:
Lutetium ($_{71}\mathrm{Lu}$)

Correct Answer:
Europium ($_{63}$Eu)

Solution:
Electronic configuration.

$Eu : 4f^{7}6s^{2}.$

Q. 7 Consider the following cases:
(A) $\mathrm{O} + e^{-} \longrightarrow \mathrm{O}^{-}$
(B) $\mathrm{O}^{-} + e^{-} \longrightarrow \mathrm{O}^{2-}$
(C) $\mathrm{F} + \mathrm{e}^{-} \longrightarrow \mathrm{F}^{-}$
(D) $\mathrm{He} + e^{-} \longrightarrow \mathrm{He}^{-}$
(E) $\mathrm{Na} \longrightarrow \mathrm{Na}^{+} + e^{-}$

In which of the above mentioned cases, $\triangle H$ for the process is negative

Option 1:
A,B,C,D

Option 2:
A,C,D

Option 3:
A,C

Option 4:
C, D, E

Correct Answer:
A,C

Solution:
Only A and C cases are exothermic, that's why $\triangle H$ is negative, and the rest of the cases are endothermic.

Q. 8 The correct order of atomic radius is :

Option 1:
$\mathrm{Li} > \mathrm{Na} > K > \mathrm{Rb} > \mathrm{Cs}$

Option 2:
$\mathrm{Cl} < \mathrm{F} < \mathrm{Br} < \mathrm{I}$

Option 3:
$\mathrm{Li} > \mathrm{Be} > \mathrm{B} > \mathrm{Al}$

Option 4:
$\mathrm{Cl} < \mathrm{S} < \mathrm{Li} < \mathrm{Na}$

Correct Answer:
$\mathrm{Cl} < \mathrm{S} < \mathrm{Li} < \mathrm{Na}$

Solution:
Atomic Radius:

Na	Li	S	Cl
186ppm	152ppm	104ppm	99ppm

Q. 9 Which of the following combinations can not act as a buffer?

Option 1:
$\mathrm{CH_3COOH} + \mathrm{CH_3COONa}$

Option 2:
$\mathrm{NH_4OH} + \mathrm{NH_4Cl}$

Option 3:
$\mathrm{NaH_2PO_4} + \mathrm{Na_2HPO_4}$

Option 4:
$\mathrm{NH_4OH} + \mathrm{NaOH}.$

Correct Answer:
$\mathrm{NH_4OH} + \mathrm{NaOH}.$

Solution:
A solution of a weak base and a strong base cannot act a buffer.

Q. 10 1 mole of $\mathrm{O_2}$ gas is adiabatically and irreversibly expanded in one step from 27 atm & 300 K to 3 atm . Calculate the final temperature of the system after the expansion -
(to the nearest integer).

Option 1:
224K

Option 2:
390K

Option 3:
412K

Option 4:
500K

Correct Answer:
224K

Solution:
(1) diabatic psocess $(q=0)$ and Irreversible expansion.
(2) O_2 is diatomic gas.
(3) Molar heat capacity $(C_v)=\frac{5}{2}R$

$P_1=27\text{ atm}\quad T_1=300\text{ K}$
$P_{\text{ext}}=3atin\quad P_2(\text{ final pressure })=3atin$

(4) Apply FLOT:- $\Delta U=q+\omega$.

$q=0$
$\therefore\quad \Delta U=w.$
Change in Internal evergy $\Delta U=nC_V\left(T_2-T_1\right)$

$\omega=-\text{Pext}\left(V_2-V_1\right)$
$nC_V\left(T_2-T_1\right)=-\text{Pext}\left(V_2-V_1\right)$
$nC_V\left(T_2-T_1\right)=-\text{Pext}\left(\frac{nRT_2}{2}-\frac{nRT_1}{P_1}\right)$

$C_v=2.5R$
$2.5\left(T_2-300\right)=-3\left(\frac{T_2}{3}-\frac{300}{27}\right)$

simplify it.

$T_g=223.8\text{ K}\simeq 224\text{ K}$

Q. 11 Consider the following electrolysis reaction:

(A) An aqueous solution of $\mathrm{HgNO_3}$ using silver electrodes
(B) An aqueous solution of $\mathrm{AgNO_3}$ using Pt electrodes

(C) An aqueous solution of $\mathrm{CuSO_4}$ using Pt electrodes

(D)A Concentrated solution of $\mathrm{NH_4HSO_4}$ using Pt electrodes

(E)An aqueous solution of NaCl using Pt electrodes

In which of the above cases,O_2 is obtained as a product of electrolysis at the anode.

Option 1:
A,B,D,E

Option 2:
B,C

Option 3:
B,C,D

Option 4:
A,B,E

Correct Answer:
B,C

Solution:
(A) Anode: $\mathrm{Ag^+}$
(B) Anode: $\mathrm{O_2(\ g)}$
(C) Anode: $\mathrm{O_2(\ g)}$
(D) Anode: $\mathrm{S_2O_8^{2-}}$
(E) Anode: $\mathrm{Cl_2}$ (g)

Q. 12 Q. If the rate of disappearance of hydrogen is $12\ \mathrm{g\ L^{-1}\ s^{-1}}$.

$-\frac{d[H_2]}{dt} = 12\ \mathrm{g}L^{-1}\ \mathrm{s}^{-1}.$

Then, calculate the rate of disappearance of $\mathrm{N_2}$ gas for the reaction $\mathrm{N_2 + 3H_2 \longrightarrow 2NH_3}$.

Option 1:
$4\mathrm{gL^{-1}\ s^{-1}}$

Option 2:
$28\ \mathrm{g\ L^{-1}\ s^{-1}}$

Option 3:
$56\ \mathrm{g\ L^{-1}\ s^{-1}}$

Option 4:
$2\ \mathrm{g\ L^{-1}\ s^{-1}}$

Correct Answer:
$56\ \mathrm{g\ L^{-1}\ s^{-1}}$

Solution:

$$-\frac{d\left[H_2\right]}{dt}=12gL^{-1}s^{-1}=6\,\mathrm{mol}\,L^{-1}s^{-1}.$$

$$\frac{-\frac{1}{3}\frac{d[H_2]}{dt}}{3}=-\frac{d\left[N_2\right]}{dt}=\frac{6}{3}=2\,\mathrm{mol}\,L^{-1}s^{-1}.$$

$$\frac{-d\left[N_2\right]}{dt}=2\ \mathrm{mol}L^{-1}s^{-1}$$

$$\frac{-d\left[N_2\right]}{dt}=2\times 28\mathrm{gL^{-1}}s^{-1}=56\mathrm{gL^{-1}}s^{-1}$$

Q. 13 One H atom, One $\mathrm{He^{+}}$ion and one $\mathrm{Li^{2+}}$ ion taken. Electrons in all the atom & ions are present in 4th Excited state. What is the maximum number of spectral lines obtained?

Option 1:
12

Option 2:
10

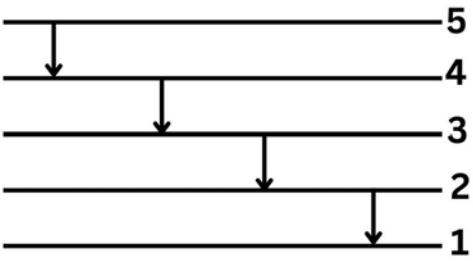
Option 3:
30

Option 4:
29

Correct Answer:
12

Solution:
4 spectral lines each form H, $\mathrm{He^{+}}$and $\mathrm{Li^{2+}}$.

$4+4+4=12$



Q. 14 Which of the following statements is incorrect regarding the Crystal Field Theory?

Option 1:
CFT is successful in explaining the color and magnetic properties of coordination compounds.

Option 2:
CFT is successful in explaining why anionic ligands lying at the lower end of the spectrochemical series.

Option 3:
CFT does not take into account the covalent character of bonding between the ligand and the central atom.

Option 4:
Ligands are treated as point charges in case of anions or point dipoles in case of neutral molecules.

Correct Answer:
CFT is successful in explaining why anionic ligands lying at the lower end of the spectrochemical series.

Solution:
This is a limitation of CFT as it can't explain any anionic ligands lying at the lower end of the spectrochemical series.

Q. 15 In Maltose, the type of glycosidic linkage present is

Option 1:
$(1,4)\alpha$-glycosidic linkage.

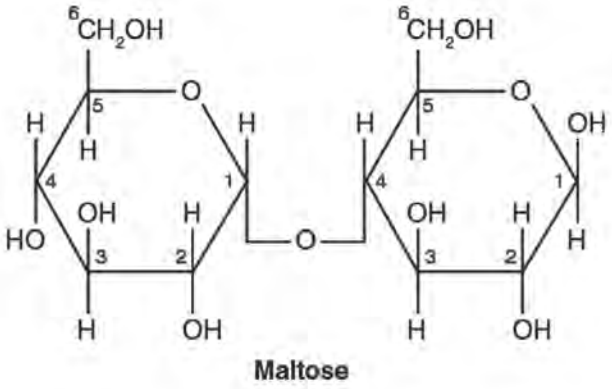
Option 2:
$(1,4)\beta$-glycosidic linkage.

Option 3:
$(1,2)\alpha$-glycosidic linkage.

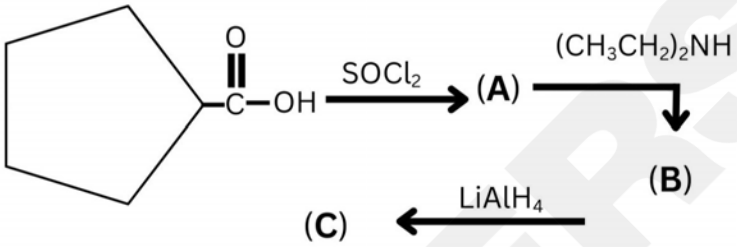
Option 4:
$(1,2)\beta$-glycosidic linkage.

Correct Answer:
$(1,4)\alpha$-glycosidic linkage.

Solution:

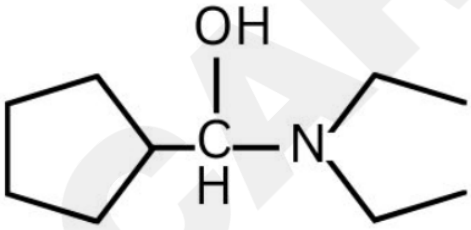


Q. 16

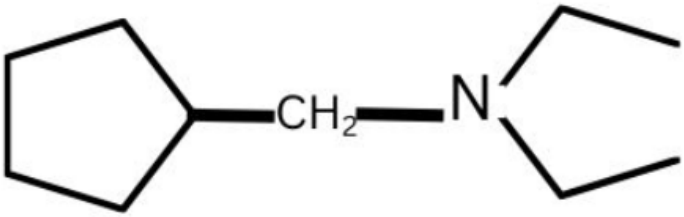


The major product (C) in the following series of reactions is:

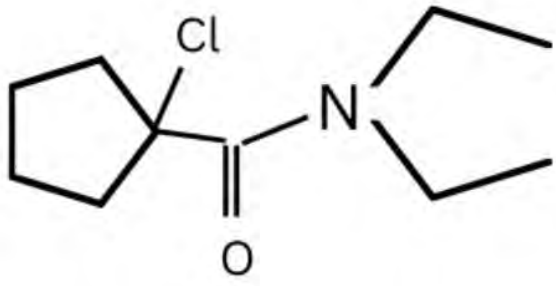
Option 1:



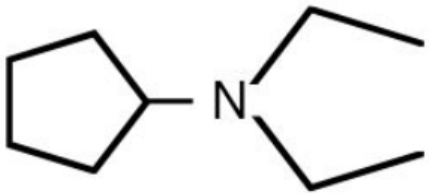
Option 2:



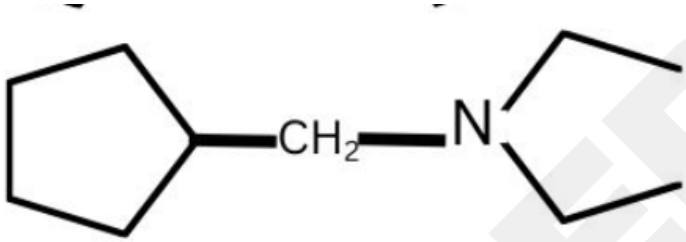
Option 3:



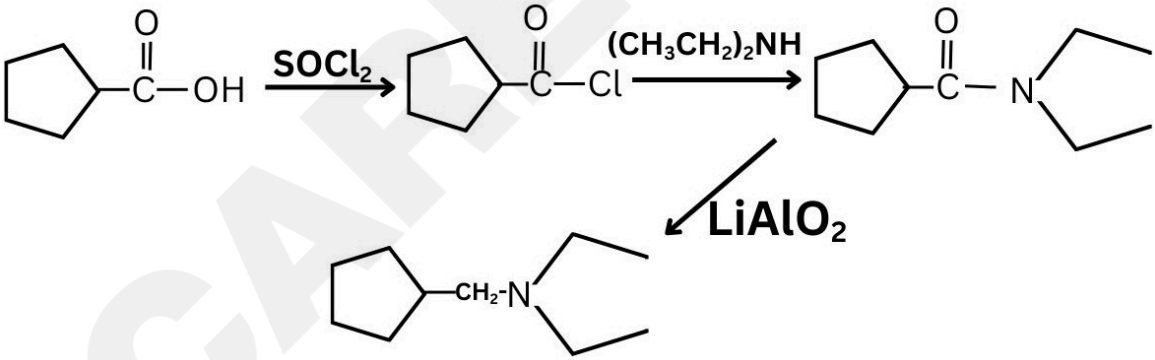
Option 4:



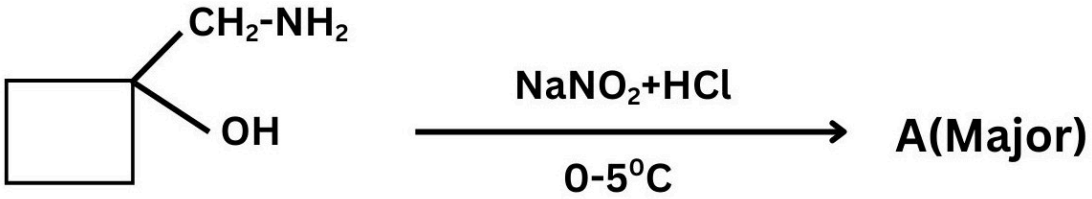
Correct Answer:



Solution:

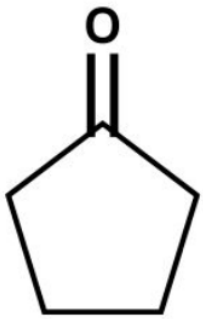


Q. 17

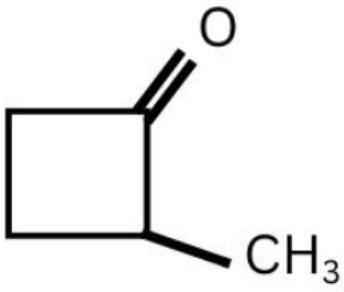


Major Product of reaction is:

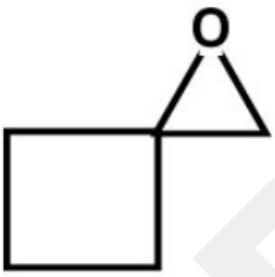
Option 1:



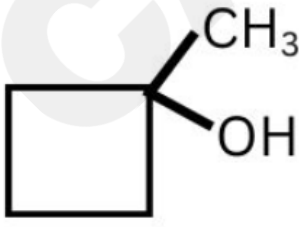
Option 2:



Option 3:

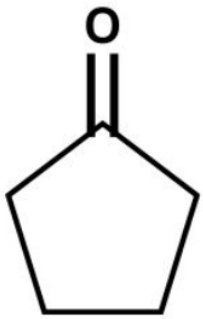


Option 4:

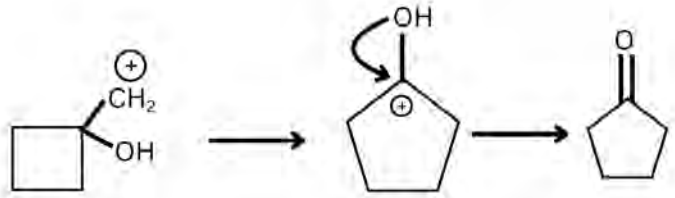


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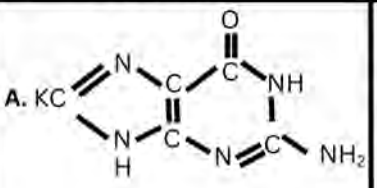
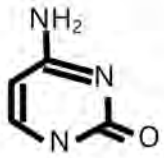
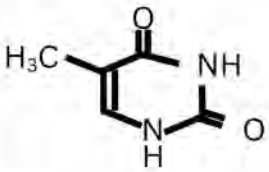
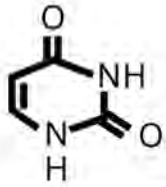
Correct Answer:



Solution:



Q. 18 Column (1) contains structure of DNA and RNA bases whereas column II contains their names. Match the structure with its corresponding name:

Structure	Name
A. 	P. Uracil (U)
B. 	Q. Guanine (G)
C. 	R. Thymine (T)
D. 	S. Cytosine (C)

Option 1:
$(A \rightarrow R); (B \rightarrow S); (C \rightarrow P); (D \rightarrow Q).$

Option 2:
$(A \rightarrow Q); (B \rightarrow S), (C \rightarrow R); (D \rightarrow P)$

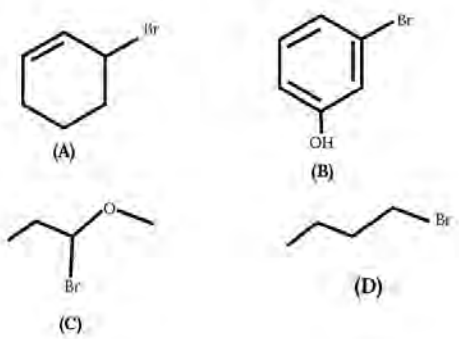
Option 3:
$(A \rightarrow S); (B \rightarrow P); (C \rightarrow Q); (D \rightarrow R)$

Option 4:
$(A \rightarrow P); (B \rightarrow R); (C \rightarrow Q); (D \rightarrow S)$

Correct Answer:
$(A \rightarrow Q); (B \rightarrow S), (C \rightarrow R); (D \rightarrow P)$

Solution:
$(A \rightarrow Q); (B \rightarrow S), (C \rightarrow R); (D \rightarrow P)$

Q. 19 Arrange the following in order of ease of formation of white ppt on reaction with $AgNO_3$.



Option 1:
$C > A > D > B$

Option 2:

$A > C > B > D$

Option 3:
$C > A > B > D$

Option 4:
$A > D > C > B$

Correct Answer:
$C > A > D > B$

Solution:
Stabilizing of carbocation after loss of Br^-

$C > A > D > B$

Q. 20 Which of the following statement(s) is/are correct

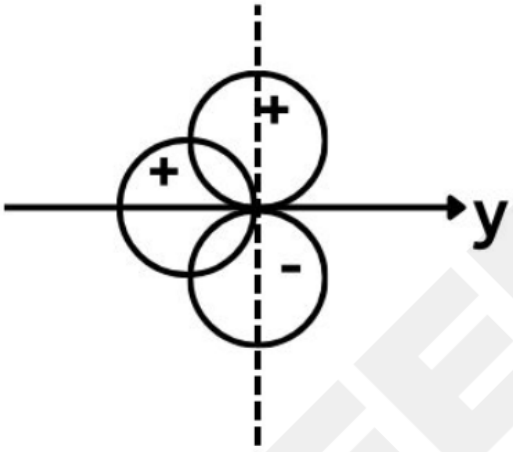
(i) Irespective of internuclear axis, addition of two in-phase atomic orbitals having wave fanctions $\psi_{100} = \left(\frac{1}{2\pi}\right)^{1/2}\left(\frac{z}{a_0}\right)^{3/2} e^{-\left(\frac{2zr}{a_0}\right)}$ gives antibonding molecular orbitals.

(ii) Addition of in-plane atomic orbitals with wave functions

$\psi_{200} \propto \left(\frac{z}{a_0}\right)^{3/2}\left(4 - \frac{z\gamma}{a_0}\right)e^{-\frac{z\gamma}{2a_0}}$ and

$\psi_{210} \propto \left(\frac{z}{2a_0}\right)^{5/2} re^{-\left(\frac{z\gamma}{2a_0}\right)}\cos\theta$ gives bonding molecular orbital, z-axis being internuclear axis.

(iii) Following combination gives zero overlap along y-axis as internuclear axis.



(iv) Number of nodal planes in σ_{2pz} antibonding molecular orbital is 3 .

Option 1:
(i), (ii) (iv) only.

Option 2:
(ii) (iii) only.

Option 3:
(i), (ii), (iii) only.

Option 4:
(ii), (iii) & (iv) only.

Correct Answer:
(ii), (iii) & (iv) only.

Solution:

(i) $\rightarrow$ denotes is orbital

(ii) $\psi_{200} \rightarrow 2\ \mathrm{S}$

$\psi_{210} \rightarrow 2p_2$

Q. 21 Consider the following species. Among them, x are planar and y are non-planar. Then find the value of the product of x and y.

$\mathrm{H_2O}, \mathrm{BrF_3}, \mathrm{B_2H_6}, \mathrm{PCl_3}, \mathrm{SO_3}, \mathrm{XeF_4}, \mathrm{XeO_3}$.

Correct Answer:

12

Solution:

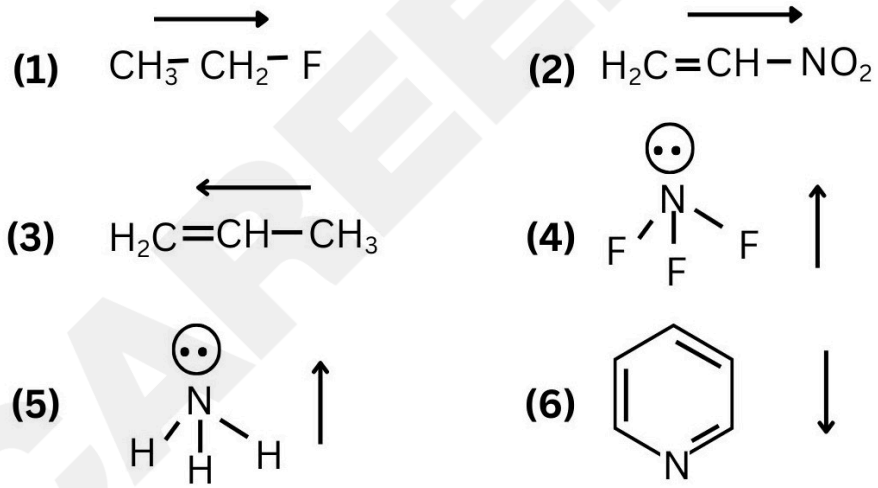
Planar $\rightarrow \mathrm{H_2O}, \mathrm{BrF_3}, \mathrm{XeF_4}, \mathrm{SO_3}$

Non-Planar$\rightarrow \mathrm{B_2H_6}, \mathrm{PCl_3}, \mathrm{XeO_3}$

$x = 4 \quad , y = 3$

X*Y = 12

Q. 22 How many of the below representations are correct for the net dipole vector.



Correct Answer:

5

Solution:

option (1) ,(2) , (3) .(5) (6) are correct.

Q. 23 Among the following compounds, how many compounds will give +ve iodoform test

a) HCHO

b) MeCHO

c) Butanone

d) 2 -Propanol

e) 1-Propanol

f) Ethylene Chloride

g) Ethanoic Acid

h) Methanol

Correct Answer:
3

Solution:
b,c,d gives positive iodoform test.

Q. 24 How many of the following are correct match for the given vitamin and their deficiency diseases.

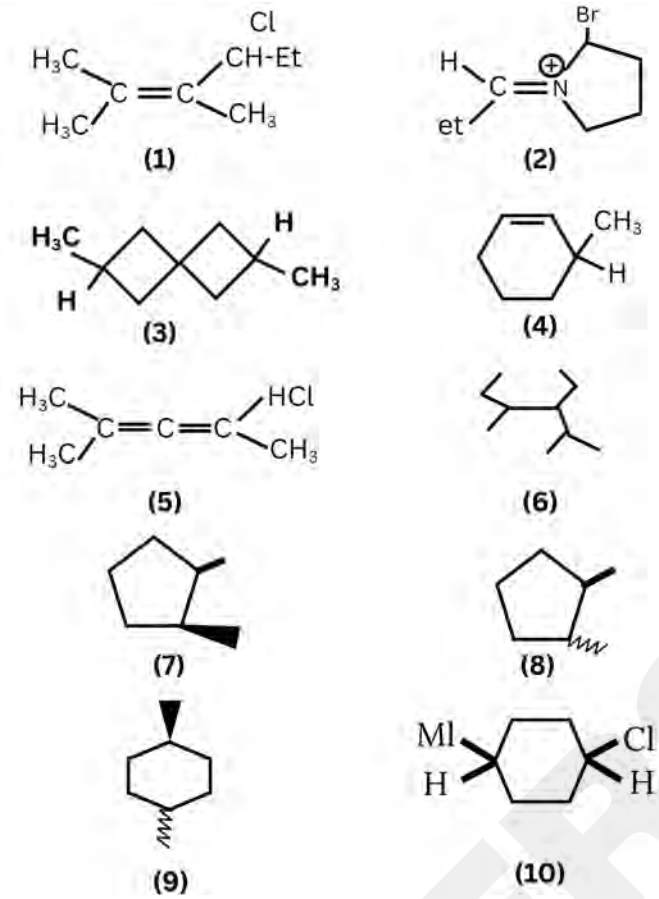
Vitamin $A \rightarrow$ hardening of cornea of eye.
Vitamin $\mathrm{B}_{12} \rightarrow$. Pernicious anaemia.
Vitamin C $\rightarrow$ Scurvy.
Vitamin B1 $\rightarrow$ Cheilosis
Vitamin $\mathrm{B}_2 \rightarrow$ Beri-Beri
Vitamin D $\rightarrow$Osteomalacia.

Correct Answer:
4

Solution:
Vitamin B_1 - Beri -Beri

Vitamin B_2 - Cheilosis

Q. 25 How many (maximum) of the given compounds can be optically active?



Correct Answer:
7

Solution:
(1), (2), (3), (4), (5),(6), (8) are optically active

Maths

Q. 1 Let $a, b \in R$ such that $a^2 + b^2 = 9$ and define $E = \frac{b-\sqrt{2}}{a-4}$. If M and m denotes the maximum and minimum values of E then the value of M.m is:

Option 1:
3

Option 2:
-3

Option 3:
9

Option 4:
-1

Correct Answer:
-1

Solution:
$a^2+b^2=9 \Rightarrow$ Point $P(a,b)$ lies on the circle $c: x^2+y^2=9$ and point $Q(4,\sqrt{2})$ lies on it's director circle. So E denotes the slope of line PQ, and the extreme values of E will occur in the case of two tangents, which. are perpendicular), hence $M \cdot m=-1$

Q. 2 Consider the sequence $\langle a_n \rangle$ such that $a_1=1, a_2=2$ and $a_{n+2}=4a_{n+1}-3a_n+2\ \forall$ $n \in N$; the value of a_{10} is

Option 1:
3^9-9

Option 2:
$4^{10}-8$

Option 3:
$3^{10}-10$

Option 4:
4^9-10

Correct Answer:
3^9-9

Solution:
$a_{n+2}-a_{n+1}+1=3\left(a_{n+1}-a_n+1\right)$
Let $b_n=a_{n+1}-a_n+1$, then $\langle b_n \rangle$ is a G.P. with common ratio 3 ; So $b_n=2 \cdot 3^{n-1}$.

$a_{n+1}-a_n+1=2 \cdot 3^{n-1}$

$\Rightarrow \sum_{n=1}^{9}\left(a_{n+1}-a_n+1\right)=\sum_{n=1}^{9} 2 \cdot 3^{n-1}$

$\Rightarrow a_{10}-a_1+9=3^9-1$

$\Rightarrow a_{10}=3^9-9$

Q. 3 Let $y=y(x)$ be the solution of differential equation $x\frac{dy}{dx}+y\ln\left(\frac{x}{y}\right)=0$ and $y(1)=e^2$, then $y(\ln 2)$ is equal to:

Option 1:
1

Option 2:
$2e \ln 2$

Option 3:
$2 \ln 2$

Option 4:
e^3

Correct Answer:
$2e \ln 2$

Solution:
Put $y = xt$, then $\frac{dy}{dx} = x\frac{dt}{dx} + t$

$$\Rightarrow x\frac{dt}{dx} + t = t\ln t$$
$$\Rightarrow \frac{dt}{t(\ln t - 1)} = \frac{dx}{x}$$
$$\Rightarrow \ln(\ln t - 1) = \ln x + \ln c$$
$$\Rightarrow \ln t - 1 = cx$$
$$\Rightarrow y = xe^{1+cx}$$

Now $y(1) = e^{1+c} = e^2 \Rightarrow c = 1$ and $y(\ln 2) = \ln 2 \cdot e^{1+\ln 2}$

$$= 2e \cdot \ln 2$$

Q. 4 The distance of point $(2, -1, 6)$ from the line $\frac{x-2}{3} = \frac{y}{7} = \frac{z-1}{2}$ measured parallel to the line $\frac{x+1}{6} = \frac{y-3}{15} = \frac{5-z}{1}$ is :

Option 1:
$\sqrt{35}$

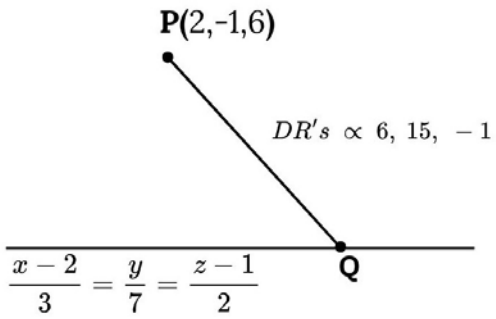
Option 2:
$\sqrt{262}$

Option 3:
$\sqrt{126}$

Option 4:
12

Correct Answer:
$\sqrt{262}$

Solution:
Equation of PQ is $\frac{x-2}{6} = \frac{y+1}{15} = \frac{z-6}{-1}$



So let $Q(6\lambda+2, 15\lambda-1, -\lambda+6)$ As Q lies on given line, so

$\frac{6\lambda}{3} = \frac{15\lambda-1}{7} = \frac{-\lambda+5}{2} \Rightarrow \lambda = 1$

Hence Q is $(8, 14, 5)$ and $PQ = \sqrt{6^2+15^2+1^2} = \sqrt{262}$

Q. 5 If a function $f(x)$ satisfies a condition $f(x) = \cos x + \int_0^{\pi/2} \sin x \cdot \sin y f(y) dy$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

Option 1:
$\frac{1}{3\sqrt{2}}\left(\frac{2+\pi}{2-\pi}\right)$

Option 2:
$\frac{4-\pi}{3\sqrt{2}}$

Option 3:
$\frac{2-\pi}{2\sqrt{2}}$

Option 4:
$\frac{1}{\sqrt{2}}\left(\frac{6-\pi}{4-\pi}\right)$

Correct Answer:
$\frac{1}{\sqrt{2}}\left(\frac{6-\pi}{4-\pi}\right)$

Solution:

Let, $f(x) = \cos x + k \sin x$
where $k = \int_0^{\pi/2} \sin y \cdot f(y) dy$

$$\Rightarrow k = \int_0^{\pi/2} \left(\sin y \cdot \cos y + k \sin^2 y\right) dy = -\frac{\cos 2y}{4} + \frac{k}{2}\left(y - \frac{\sin 2y}{2}\right)\Big|_0^{\pi/2}$$

$$\Rightarrow k = \frac{1}{2} + \frac{k\pi}{4} \Rightarrow k = \frac{2}{4-\pi}$$

Hence $f(x) = \cos x + \left(\frac{2}{4-\pi}\right) \sin x$

Q. 6 If the latus rectum of a hyperbola subtends a right angle at the farther end of the transverse axis, then the eccentricity of the hyperbola is:

Option 1:
2

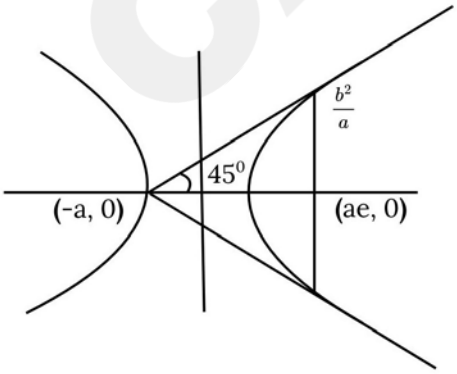
Option 2:
$\sqrt{3}$

Option 3:
$\frac{4}{\sqrt{3}}$

Option 4:
$\frac{2}{\sqrt{3}}$

Correct Answer:
2

Solution:



$\text{Clearly , } \frac{b^2}{a} = a + ae$
$\Rightarrow a\left(e^2 - 1\right) = a + ae$
$\Rightarrow e - 1 = 1$
$\Rightarrow e = 2$

Q. 7 The number of equivalence relations on the set $A = \{\alpha, \beta, \gamma\}$ that contains the ordered pair (α, β) is:

Option 1:
2

Option 2:
3

Option 3:
4

Option 4:
16

Correct Answer:
2

Solution:
$A \times A$ contains 9 ordered pairs, out of which 5 are already included in the relation. Only two such relations exist.

$R_1 = \{(\alpha, \alpha), (\beta, \beta), (\gamma, \gamma), (\alpha, \beta), (\beta, \alpha)\}$
$R_2 = \{(\alpha, \alpha), (\beta, \beta), (\gamma, \gamma), (\alpha, \beta), (\beta, \alpha), (\beta, \gamma), (\gamma, \beta), (\alpha, \gamma), (\gamma, \alpha)\}$

Q. 8 Using the principle values of inverse trigonometric functions, the sum of the greatest and the least value of the function :

$$f(x) = 2\sin^{-1} x + \tan^{-1} x + 3\sec^{-1} x$$

Option 1:
$\frac{5\pi}{2}$

Option 2:
$\frac{5\pi}{4}$

Option 3:
3π

Option 4:
$\frac{7\pi}{4}$

Correct Answer:
3π

Solution:
Domain of $f(x)$ is $\{-1, 1\}$.
$f(1) = \frac{5\pi}{4}$ and $f(-1) = \frac{7\pi}{4}$

Required sum is 3π.

Q. 9 Let z_1, z_2, z_3 be the vertices of a triangle of maximum area and satisfy the condition $\text{Im}\left(\frac{1}{Z_i}\right) = \frac{1}{2}$ for $i = 1, 2, 3$. Then the perimeter of the triangle is:

Option 1:
$\frac{3}{2}$

Option 2:
$3\sqrt{3}$

Option 3:
$\frac{9}{4}$

Option 4:
$\frac{3\sqrt{3}}{4}$

Correct Answer:
$3\sqrt{3}$

Solution:
$\text{Let } z = x + iy$

$$\text{Im}\left(\frac{1}{z}\right) = \frac{1}{2} \Rightarrow \text{Im}\left(\frac{x - iy}{x^2 + y^2}\right) = \frac{1}{2} \Rightarrow x^2 + y^2 + 2y = 0$$

which is a circle, so the triangle of maximum area will be an equilateral triangle inscribed in this circle of radius 1.

$$\text{Side of triangle } a = \sqrt{3}R$$

Here $R = 1$, so

$a = \sqrt{3}.$

Perimeter = 3a = $3\sqrt{3}$

Q. 10 A biased coin is tossed n times. If expectation and variance of the number of heads obtained are 2 and $\frac{4}{3}$ respectively, then the probability of getting exactly 4 heads is:

Option 1:
$\frac{40}{243}$

Option 2:
$\frac{160}{729}$

Option 3:
$\frac{10}{81}$

Option 4:
$\frac{20}{243}$

Correct Answer:
$\frac{20}{243}$

Solution:
Let probability of getting head is p.

$np = 2, npq = \frac{4}{3} \Rightarrow q = \frac{2}{3}$ so $p = \frac{1}{3}$ and $n = 6$.

$$\text{Required probability } = {}^6C_4 \cdot \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 = \frac{15 \times 4}{3^6} = \frac{20}{243}$$

Q. 11 If $7x - y = 0$ and $x + y = 0$ are two sides of rhombus having a vertex $(6, 2)$ then equation of one of it's diagonal is:

Option 1:
$3x + y = 0$

Option 2:
$x - 3y = 8$

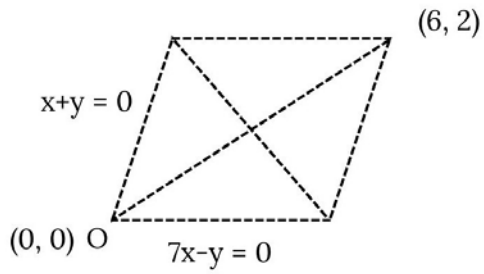
Option 3:
$3x + y = 10$

Option 4:
$x - 3y = 2$

Correct Answer:
$3x + y = 10$

Solution:
Equation of angle bisectors:

$$\frac{x+y}{\sqrt{2}} = \pm\frac{7x-y}{5\sqrt{2}}$$
$$\Rightarrow x - 3y = 0 \text{ and } 3x + y = 0$$



So one diagonal has an equation, $x - 3y = 0$, and the other will be parallel to $3x + y = 0$ and passing through $(3, 1)$, so its equation will be $3x + y = 10$.

Q. 12 Let $f(x)$ be a twice differentiable function such that $f(x + y) = f(x) + f(y) - xy \quad \forall x, y \in R$. If $f(2) = 2$, then area of the region bounded by $y = f(x)$ and x-axis is:

Option 1:
$\frac{8}{3}$

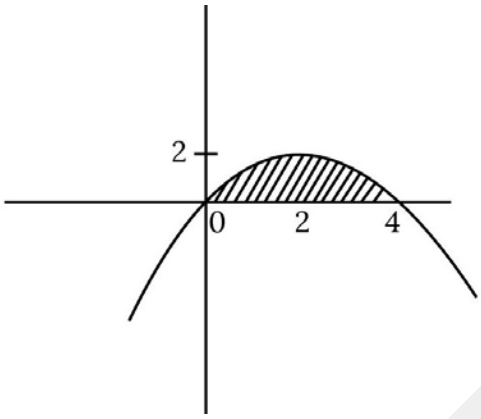
Option 2:
$\frac{4}{3}$

Option 3:
$\frac{10}{3}$

Option 4:
$\frac{16}{3}$

Correct Answer:
$\frac{16}{3}$

Solution:
$f'(x+y) = f'(x) - y$; put $x = 0$
$\Rightarrow f'(y) = f'(0) - y$
$\Rightarrow f(y) = f'(0)y - \frac{y^2}{2} + c$
$\because f(0) = 0 \Rightarrow c = 0$ and $f(2) = 2 \Rightarrow f'(0) = 2$
So $f(x) = 2x - \frac{x^2}{2}$
Required area $= \frac{16}{3}$



Q. 13 Consider a two degree polynomial $P(x)$ such that $|P(-1)| = |P(0)| = |P(3)| = |P(4)| = 6$, then value of $|P(5)|$ is:

Option 1:
24

Option 2:
6

Option 3:
$\frac{95}{4}$

Option 4:
$\frac{75}{4}$

Correct Answer:
24

Solution:
Clearly vertex of parabola $y = P(x)$ lies on $x = \frac{3}{2}$.

$$\text{Let } P(x) = a\left(x - \tfrac{3}{2}\right)^2 + b$$
$$\because |P(0)| = 6 \Rightarrow \left|\tfrac{9}{4}a + b\right| = 6 \ldots\ldots\ldots (1)$$
$$\because |P(-1)| = 6 \Rightarrow \left|\tfrac{25}{4}a + b\right| = 6 \ldots\ldots\ldots\ldots (2)$$

From (1) and (2) we get $(a, b) = \left(3, -\frac{51}{4}\right)$ or $\left(-3, \frac{51}{4}\right)$
Now $|P(5)| = \left|\frac{49a}{4} + b\right| = 24$

Q. 14 A and B be two square matrices of order 3 such that $\det\left(2\,\text{adj}\left(A^2B\right)\right) = 2^7 \cdot 3^4$ and $\det(3\,\text{adj}(A \cdot \text{adj}\,B)) = 2^8 \cdot 3^5$. The value of $\det(\det(A) \cdot \text{adj}(B))$ is :

Option 1:
$2^2 \cdot 3^5$

Option 2:
$2 \cdot 3^2$

Option 3:
$2^4 \cdot 3$

Option 4:
$2^4 \cdot 3^3$

Correct Answer:
$2^4 \cdot 3^3$

Solution:
$\det\left(2\,\text{adj}\left(A^2B\right)\right) = 2^7 \cdot 3^4 \Rightarrow 2^3 \cdot \left|A^2B\right|^2 = 2^7 \cdot 3^4$

$\Rightarrow |A|^2 \cdot |B| = \pm 2^2 \cdot 3^2$

Similarily; $\det(3\,\text{adj}(A \cdot \text{adj}\,B)) = 2^8 \cdot 3^5$

$\Rightarrow 3^3 \cdot |A \cdot \text{adj}(B)|^2 = 2^8 \cdot 3^5 \Rightarrow |A|^2|B|^4 = 2^8 \cdot 3^2.$

From (1) and (2) we get $|A| = \pm 3$ and $|B| = \pm 4$
Now,

$||A|\operatorname{adj} B| = |A|^3|B|^2 = 2^4 \cdot 3^3$

Q. 15 Let $\bar{a}$ and $\bar{b}$ are the components of vector $\hat{i} - 2\hat{j} - \hat{k}$ along and perpendicular to the vector $\hat{i} + \hat{j} + 5\hat{k}$. If $\bar{c}$ be a unit vector, then the maximum volume of a tetrahedron determined by vectors $\bar{a}, \bar{b}$ and $\bar{c}$ is:

Option 1:
6

Option 2:
$\frac{\sqrt{14}}{9}$

Option 3:
$\frac{2\sqrt{14}}{3}$

Option 4:
1

Correct Answer:
$\frac{\sqrt{14}}{9}$

Solution:
Volume of tetrahedron formed by vectors $\vec{a}, \vec{b}, \vec{c}$ is:
$V = \frac{1}{6}|\vec{a} \cdot (\vec{b} \times \vec{c})|$
To maximize V with fixed $\vec{a}$ and $\vec{b}$, and $\vec{c}$ a unit vector:
$|\vec{a} \cdot (\vec{b} \times \vec{c})|$ is maximized when $\vec{c}$ is parallel to $\vec{a} \times \vec{b}$
Maximum volume of tetrahedron $= \frac{1}{6}|\bar{a} \times \bar{b}| = \frac{1}{6}|\bar{a}||\bar{b}| \quad (\text{as } \bar{a} \perp \bar{b})$

Let $\bar{a} = \lambda(\hat{i} + \hat{j} + 5\hat{k})$
and $\hat{i} - 2\hat{j} - \hat{k} = \bar{a} + \bar{b}$
Taking the dot product with $\hat{i} + \hat{j} + 5\hat{k}$, we get,

$-6 = \lambda(27) + 0 \Rightarrow \lambda = -\frac{2}{9}$

So $\bar{a} = -\frac{2}{9}(\hat{i} + \hat{j} + 5\hat{k})$ and $\bar{b} = \hat{i} - 2\hat{j} - \hat{k} + \frac{2}{9}(\hat{i} + \hat{j} + 5\hat{k})$

$\Rightarrow \bar{b} = \frac{1}{9}(11\hat{i} - 16\hat{j} + \hat{k})$
So the maximum volume of a tetrahedron
$= \frac{1}{6} \cdot \frac{2}{9}\sqrt{27} \cdot \frac{1}{9}\sqrt{378}$

$= \frac{\sqrt{14}}{9}$

Q. 16 The number of distinct real roots of the equation $e^{2x} + 1 - x^2 = \frac{\sqrt{3}\cos 20^\circ - \sin 40^\circ}{\cos 10^\circ}$ is :

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
4

Correct Answer:
1

Solution:
$\because \sqrt{3}\cos 20^\circ - \sin 40^\circ = 2\sin 60^\circ \cdot \cos 20^\circ - \sin 40^\circ$
$= \sin 80^\circ = \cos 10^\circ$

So the equation becomes $e^x = \pm x$; which gives only one solution.

Q. 17 Consider a point $P(x_0, y_0)$ on the parabola $y^2 = 4x$ having vertex O. Tangent at P meets the tangent at vertex O. Tangent at P meets the tangent at the vertex at T. Let area of $\triangle OPT$ is $\Delta_1(x_0)$ and area bounded by the two tangents and parabola is $\Delta_2(x_0)$; then $\lim_{x_0 \to \infty} \frac{\Delta_1(x_0)}{\Delta_2(x_0)}$ is :

Option 1:
$\sqrt{3}$

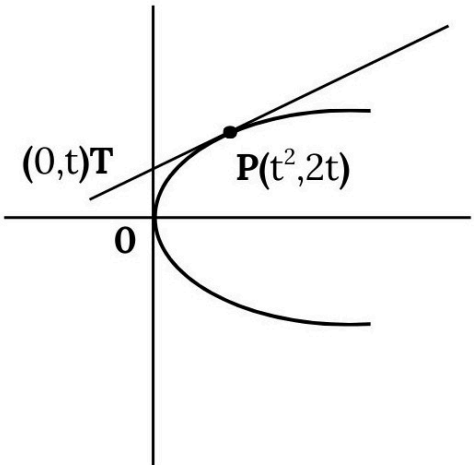
Option 2:
1

Option 3:
3

Option 4:
$\sqrt{6}$

Correct Answer:
3

Solution:



$\Delta_1(x_0) = \frac{1}{2}t \cdot t^2 = \frac{1}{2}t^3$
and

$$\Delta_2(x_0) = \frac{1}{3}(2t \cdot t^2) - \frac{1}{2}t^3$$
$$= \frac{1}{6}t^3$$

So $\frac{\Delta_1(x_0)}{\Delta_2(x_0)} = 3$ (constant)

Q. 18 Let $f : (0, \infty) \to (0, \infty), f(x) = x \cdot e^{1/x}$. Then which of the following option is INCORRECT?

Option 1:
$f(x)$ is not monotonic

Option 2:
$f(x)$ is not bounded

Option 3:
There exist atleast one $c \in (1, \infty)$ for which $f'(c) > e$

Option 4:
$f(x) = \frac{1}{e}$ has no solution in x.

Correct Answer:
There exist atleast one $c \in (1, \infty)$ for which $f'(c) > e$

Solution:
$f(x) = x \cdot e^{1/x}$

$f'(x) = e^{\frac{1}{x}} \left(1 - \frac{1}{x}\right)$

and $f''(x) = \frac{e^{\frac{1}{x}}}{x^3}$
So, $f(x)$ is decreasing in $(0, 1)$ and increasing is $(1, \infty)$

Range of $f(x)$ is $[e, \infty)$.
$f'(x)$ is also increasing in $(1, \infty)$ and $f'(x) < \lim_{x \to \infty} f'(x)$

$$\Rightarrow f'(x) < 1$$

Q. 19

For $n \geqslant 3, A^n = A^{n-2} + A^2 - I$, where $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, A^{49} =$

Option 1:
$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$

Option 2:
$\begin{bmatrix} 1 & 0 & 0 \\ 24 & 0 & 1 \\ 25 & 1 & 0 \end{bmatrix}$

Option 3:
$\begin{bmatrix} 1 & 0 & 0 \\ 24 & 0 & 1 \\ 24 & 1 & 0 \end{bmatrix}$

Option 4:
$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 0 & 1 \\ 24 & 1 & 0 \end{bmatrix}$

Correct Answer:
$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 0 & 1 \\ 24 & 1 & 0 \end{bmatrix}$

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Solution:
We have $A^n = A^{n-2} + A^2 - 1$

$\Rightarrow A^n - A^{n-2} = A^2 - I.$

For $\mathrm{n} = 49$, this gives

$A^{49} - A^{47} = A^2 - 1.$

Further, $\mathrm{A}^{47} - \mathrm{A}^{45} = \mathrm{A}^2 - 1$

$A^5 - A^3 = A^2 - 1$
$A^3 - A = A^2 - 1$

Adding the above equations, we get

$A^{49} = A + 24A^2 - 24I.$

Here $A^2 = \begin{bmatrix}1 & 0 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}$ so that

$$A^{49} = \begin{bmatrix}1 & 0 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix} + \begin{bmatrix}24 & 0 & 0\\ 24 & 24 & 0\\ 24 & 0 & 24\end{bmatrix} - \begin{bmatrix}24 & 0 & 0\\ 0 & 24 & 0\\ 0 & 0 & 24\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 25 & 0 & 1\\ 24 & 1 & 0\end{bmatrix}.$$

Hence, the correct answer is option (4).

Q. 20 The sum of infinite series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \ldots..$ is equal to:

Option 1:
$\int_0^1 \frac{dx}{x^2+x+1}$

Option 2:
$\int_0^1 \frac{dx}{x^2-x+1}$

Option 3:
$\int_0^1 \frac{dx}{1+x+x^2+x^3}$

Option 4:
$\int_0^1 x\ln(1+x)dx$

Correct Answer:
$\int_0^1 \frac{dx}{x^2+x+1}$

Solution:

$$\left(1+\frac{1}{4}+\frac{1}{7}+\cdots\right)-\left(\frac{1}{2}+\frac{1}{5}+\frac{1}{8}+\cdots\right)$$

$$=\sum_{r=0}^{\infty}\frac{1}{3r+1}-\frac{1}{3r+2}=\sum_{r=0}^{\infty}\int_0^1\left(x^{3r}-x^{3r+1}\right)dx$$

$$=\int_0^1\frac{1-x}{1-x^3}dx=\int_0^1\frac{dx}{1+x+x^2}$$

Q. 21 Let $\theta_r=\frac{2\cdot 3^r}{1+3^n}\pi$ then the value of
$\lim\limits_{n\to\infty}(1+2\cos\theta_1)(1+2\cos\theta_2)(1+2\cos\theta_3)\ldots.(1+2\cos\theta_n)$ is :

Correct Answer:
1

Solution:
Let

$$T_r=1+2\cos\left(\frac{2\cdot 3^r\pi}{1+3^n}\right)=\frac{\sin\left(\frac{3^{r+1}\pi}{1+3^n}\right)}{\sin\left(\frac{3^r\pi}{1+3^n}\right)}$$

$$\left(\because\frac{\sin 3\theta}{\sin\theta}=1+2\cos 2\theta\right)$$

So $\prod_{r=1}^{n}T_r=\frac{\sin\left(\frac{3^{n+1}\pi}{1+3^n}\right)}{\sin\left(\frac{3\pi}{1+3^n}\right)}=1$ as $\frac{3\pi}{1+3^n}+\frac{3^{n+1}\pi}{1+3^n}=3\pi$

Q. 22 The number of values of x satisfying $\sec\left(\sqrt{x^3-x+1}\pi\right)\cdot\sec\left(\sqrt{25-x^2}\pi\right)=-1$ is
____

Correct Answer:
3

Solution:

$\sqrt{x^3-x+1}$ and $\sqrt{25-x^2}$ are positive integers of opposite parity. $\sqrt{25-x^2}$ will be integer if $x=0,\pm 3,\pm 4,\pm\sqrt{21},\pm\sqrt{24},\pm 5$ Out of which only $x=0,3,5$ are allowed.

Q. 23 An insect is moving on a surface with condition that in every step it can move to point $(i, j+1)$ or $(i+1, j)$ or $(i+1, j+1)$or $(i+1, j-1)$ from the point (i, j). If initially the insect is at the origin and it is not allowed to move below x-axis, then the number of distinct paths to reach $(7, 0)$ in seven steps is _____.

Correct Answer:
127

Solution:
In each step insect must move to the right side (either straight or diagonally). Also, diagonal steps will be even in numbers (diagonal up and diagonal down).

Case I: No diagonal steps.

$$\text{Number of paths } = 1$$

Case II: Two diagonal steps.

$$\text{Number of paths } = {}^7C_2 = 21$$

Case III: Four diagonal steps.

$$\text{Number of paths } = {}^7C_4 \cdot \left(\tfrac{1}{3} \cdot {}^4C_2\right) = 70$$

Case IV: Six diagonal steps.
Number of paths $= {}^7C_6 \cdot \left(\tfrac{1}{4} \cdot {}^6C_3\right) = 35$
Total number of distinct paths $= 127$

Q. 24 $\left(1+x+x^2\right)^{20} = a_0 + a_1x + a_2x^2 + \ldots . + a_{40}x^{40}$ then the value of $\frac{10a_{10}+31a_9}{a_{11}}$ is _____

Correct Answer:
11

Solution:
$\because \left(1+x+x^2\right)^{20} = a_0 + a_1x + a_2x^2 + \ldots + a_{40}x^{40}$
Differentiating w.r.t. x

$$20(1+2x)\left(1+x+x^2\right)^{19} = 1 \cdot a_1 + 2a_2x + 3a_3x^2 + \ldots + 40a_{40}x^{39}$$
$$\Rightarrow 20(1+2x)\left(1+x+x^2\right)^{20} = \left(1+x+x^2\right)\left(1 \cdot a_1 + 2a_2x + \ldots + 40a_{40}x^{39}\right)$$

Comparing the coefficient of x^{10} both sides, we get
$20\left[a_{10} + 2a_9\right] = 11 \cdot a_{11} + 10a_{10} + 9a_9$
$\Rightarrow 10a_{10} + 31a_9 = 11 \cdot a_{11}$

Q. 25 For $i = 1, 2, 3, \ldots, 30$ the mean of $(x_i + 2)$ is 12 and variance of $(x_1 + i)$ is 200 . If $x_1 + 2x_2 + 3x_3 + \ldots + 30x_{30} = \frac{325}{4}$, then the variance of $x_1, x_2, x_3, \ldots, x_3$ is:

Correct Answer:
130

Solution:
Given $\frac{\sum x_i}{30} = 10$ and $\frac{\sum (x_i+i)^2}{30} - \left(\frac{\sum (x_i+i)}{30}\right)^2 = 200$

$$\Rightarrow \frac{\sum x_i^2}{30} + \frac{31 \times 61}{6} + \frac{2\sum i x_i}{30} - \left(\frac{1}{3} + \frac{31}{2}\right)^2 = 200$$
$$\Rightarrow \frac{\sum x_i^2}{30} + \frac{31 \times 61}{6} + \frac{325}{60} - \frac{9025}{36} = 200$$
$$\Rightarrow \frac{\sum x_i^2}{30} = \frac{1171}{9}$$

So variance $= \frac{\sum x_i^2}{30} - \left(\frac{\sum x_i}{30}\right)^2 = \frac{1171}{9} - \frac{1}{9} = 130$

Physics

Q. 1 Density of water $4^{\circ}\mathrm{c}$ and $20^{\circ}\mathrm{c}$ are $1000\ \mathrm{kg/m^3}$ and $998\ \mathrm{kg/m^3}$ respectively the increase in internal energy of 4 kg of water when it is heated from $4^{\circ}\mathrm{c}$ to $20^{\circ}\mathrm{c}$ is _____ J.
(specific heat capacity of water $= 4.2\ \mathrm{J/kg}$ and 1 atmospheric pressure $= 10^5\ \mathrm{Pa}$)

Option 1:
268799.2

Option 2:
315826.2

Option 3:
258700.8

Option 4:
234699.2

Correct Answer:
268799.2

Solution:
Sol:

$Q = msdt = 4 \times 4200 \times 16 = 268800$
$W = pdv = p\left[v_2 - v_1\right]$
$= P\left[\frac{m}{\rho_2} - \frac{m}{\rho_1}\right]$
$= 10^5\left[\frac{4}{998} - \frac{4}{1000}\right]$
$= \frac{10^5 \times 4 \times 2}{998 \times 1000}$
$= 0.8016$
$\Delta u = dQ - dw$
$= 268800 - 0.8016$
$268799.19J$

Q. 2 In a microscope of tube length 10 cm , two convex len's are arranged with focal length of 2 cm and 5 cm Total magnification obtained with this system for normal advistament is $(5)^{k}$ the value of K is

Option 1:
4

Option 2:
5

Option 3:
2

Option 4:
3.5

Correct Answer:
2

Solution:
Sol:

$\mathrm{f_o} = 2\ \mathrm{cm}, \mathrm{f_e} = 5\ \mathrm{cm},\ \mathrm{L} = 10\ \mathrm{cm}$

Magnification $m = m_o me = \frac{L}{f_o}\left(\frac{D}{fe}\right)$

$\therefore \mathrm{m} = \frac{10}{2}\left(\frac{25}{2}\right) \quad = 5(5)$

$\therefore K = 2$

Q. 3 Two electrons are moving in orbits of two hydrogen like atoms with speed $3 \times 10^5\ \mathrm{m/s}$ and $2.5 \times 10^5\ \mathrm{m/s}$ respectively. If the radii of these orbits are nearly same then, the possible order of energy states are _____ respectively

Option 1:
9 and 8

Option 2:
10 and 12

Option 3:
8 and 10

Option 4:
6 and 5

Correct Answer:
6 and 5

Solution:
Sol:

$\mathrm{E}\alpha\frac{\mathrm{z}^2}{\mathrm{n}^2}\&\ \mathrm{V}\alpha\frac{\mathrm{z}}{\mathrm{n}}$

For hydrogen like atom $\mathrm{E}\alpha\frac{1}{\mathrm{n}^2}\&\ \mathrm{V}\alpha\frac{1}{\mathrm{n}}$

$\Rightarrow \mathrm{E}\alpha\mathrm{V}^2 \Rightarrow \frac{\mathrm{E}_1}{\mathrm{E}_2}=\left(\frac{\mathrm{V}_1}{\mathrm{V}_2}\right)^2=\left(\frac{3\times10^{7}}{2.5\times10^{7}}\right)^2 \Rightarrow \frac{\mathrm{E}_1}{\mathrm{E}_2}=\left(\frac{6}{5}\right)^2=\frac{36}{25}$

But $\frac{\mathrm{E}_1}{\mathrm{E}_2}=\left(\frac{\mathrm{n}_2}{\mathrm{n}_1}\right)^2=\left(\frac{6}{5}\right)^2$

$\therefore$ Possible states are $6\&5$

Q. 4 There are three co-centric conducing spherical shell A, B and C radii a, b and c respectively ($c < b < a$) and they are charged with charge q_1, q_2 and q_3 respectively.
The potentials of the spheres A, B and C respectively, are:

Option 1:
$\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{a}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2}{b}+\frac{q_3}{c}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right]$

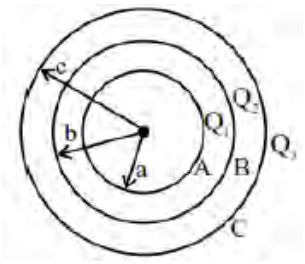
Option 2:
$\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{b}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{c}\right]$

Option 3:
$\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{a}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{b}\right] \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{c}\right]$

Option 4:
$\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{e}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2}{b}+\frac{q_3}{c}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{c}\right]$

Correct Answer:
$\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{e}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2}{b}+\frac{q_3}{c}\right], \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1+q_2+q_3}{c}\right]$

Solution:



$v_A = \frac{1}{4\pi c_0}\left[\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c}\right]$

$v_B = \frac{1}{4\pi c_0}\left[\frac{q_1+q_2}{b} + \frac{q_3}{c}\right]$ $\quad$ $v_c = \frac{1}{4\pi c_0}\left[\frac{q_1+q_2+q_3}{c}\right]$

Q. 5 An unpolarised light is incident at an interface of two dielectric media having refractive indices of 2 and $2\sqrt{3}$ respectively to satisfy the condition that reflected and refracted rays are perpendicular to each other, the angle of incidence is _____

Option 1:
30°

Option 2:
60°

Option 3:
45°

Option 4:
10°

Correct Answer:
60°

Solution:
Sol: From Brewster law

$$\text{Tan}\,\theta_P = \frac{\mu_2}{\mu_1} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$\theta_P = 60^\circ$

Q. 6 Match the list -I with list-II

	List-I		List-II
(A)	Magnetic induction	(I)	$\mathrm{MLT^{-2}\,A^{-2}}$
(B)	Magnetic flues	(II)	$\mathrm{ML^2\,T^{-2}\,A^{-2}}$
(C)	Magnetic permeability	(III)	$\mathrm{ML^{\circ}T^{-2}\,A^{-1}}$
(D)	Self induction	(IV)	$\mathrm{ML^2\,T^{-2}\,A^{-1}}$

Option 1:
$A-III, B-IV, C-II, D-I$

Option 2:
$A-I, B-III, C-IV, D-II$

Option 3:
$A-IV, B-III, C-I, D-II$

Option 4:
$A-III, B-IV, C-I, D-II$

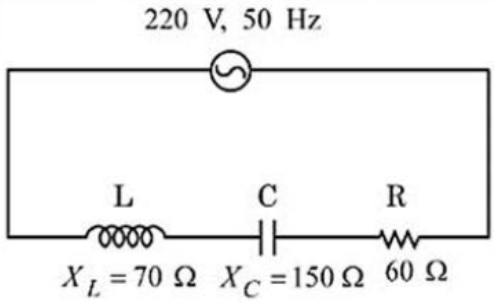
Correct Answer:
$A-III, B-IV, C-I, D-II$

Solution:
Sol:

$$\mathrm{B=F/m=\left[mL^{\circ}T^{-2}\,A^{-1}\right]}, \phi=\mathrm{BA=\left[mL^2\,T^{-2}\,A^{-1}\right]}$$

$$\mu_o=\frac{\mathrm{Fr^2}}{\mathrm{m_1\,m_2}}=\mathrm{\left[mLT^{-2}\,A^{-2}\right]L}=\frac{\mathrm{U}}{\mathrm{i2}}=\mathrm{\left[mL^2\,T^2\,A^{-2}\right]}$$

Q. 7 For a series LCR circuit connected with 220 V 50 Hz a.c source as shown in the figure The power factory is $\frac{\alpha}{10}$ the valve of α is_____



Option 1:
10

Option 2:
4

Option 3:
6

Option 4:
8

Correct Answer:
6

Solution:
Sol:

$$\text{Cos}\,\phi = \frac{\text{R}}{z} = \frac{\text{R}}{\sqrt{(\text{x}_\text{c} - \text{x}_\text{c})^2 + \text{R}^2}} = \frac{60}{\sqrt{(150-70)^2 + (60)^2}}$$

$$\text{Cos}\,\phi = \frac{60}{100} = \frac{\alpha}{10}$$

$$\alpha = 6$$

Q. 8 Match the list -I with list-II

	List-I		List-II
A)	Ratio wave	I)	Is produced by magnetron value
B)	Micro wave	II)	Due to change in vibrational modes of atoms
C)	Infrared – wave	III)	Due to inner shell electron moving from higher energy level to lower energy level
D)	X-ray	IV)	Due to rapid acceleration of chares

Choose the correct answer from the options given below:

Option 1:
$\mathrm{A} \rightarrow \mathrm{IV}, \mathrm{B} \rightarrow \mathrm{I}, \mathrm{C} \rightarrow \mathrm{II}, \mathrm{D} \rightarrow \mathrm{III}$

Option 2:
$\mathrm{A} \rightarrow \mathrm{II}, \mathrm{B} \rightarrow \mathrm{IV}, \mathrm{C} \rightarrow \mathrm{III}, \mathrm{D} \rightarrow \mathrm{I}$

Option 3:
$\mathrm{A} \rightarrow \mathrm{IV}, \mathrm{B} \rightarrow \mathrm{II}, \mathrm{C} \rightarrow \mathrm{I}, \mathrm{D} \rightarrow \mathrm{III}$

Option 4:
$\mathrm{A} \rightarrow \mathrm{IV}, \mathrm{B} \rightarrow \mathrm{III}, \mathrm{C} \rightarrow \mathrm{I}, \mathrm{D} \rightarrow \mathrm{II}$

Correct Answer:
$\mathrm{A} \rightarrow \mathrm{IV}, \mathrm{B} \rightarrow \mathrm{I}, \mathrm{C} \rightarrow \mathrm{II}, \mathrm{D} \rightarrow \mathrm{III}$

Solution:
Sol: Radio waves are produced by acceleration of charges, micro waves are produced by magnetron value IR waves are produced by vibration modern of atoms, x rays are produced by electrons moving from high to low

Q. 9 A cylindrical block of mass M and area of cross section A is floating in a liquid of density ρ and with x -axis vertical when depressed a little an released the block starts oscillating the period of oscillation is_____

Option 1:
$2\pi\sqrt{\frac{\mathrm{A}}{\mathrm{Mg}}}$

Option 2:
$2\pi\sqrt{\frac{M}{\rho Ag}}$

Option 3:
$\pi\sqrt{\frac{\mathrm{A}}{\mathrm{Mg}}}$

Option 4:
$\pi\sqrt{\frac{2\mathrm{M}}{\mathrm{Ag}}}$

Correct Answer:
$2\pi\sqrt{\frac{M}{\rho Ag}}$

Solution:
In equilibrium
When it is depressed

$$\mathrm{F_{net}} = \mathrm{W_{app}} - \mathrm{F_B(x+y)}$$

$$\Rightarrow \mathrm{ma} = \mathrm{Wapp} - F_{Bx} - F_{By} \Rightarrow \mathrm{ma} = -V_{\mathrm{in}}\,\mathrm{e}$$

$$\Rightarrow \mathrm{ma} = -(\mathrm{Ay})\mathrm{e} - \mathrm{g} \Rightarrow \mathrm{a} = -\frac{(\mathrm{Aeg})}{\mathrm{M}}\mathrm{y}$$

Compare with

$$\mathrm{a} = -\mathrm{w}^2\mathrm{y}$$

$$\therefore \mathrm{w}^2 = \frac{\mathrm{Aeg}}{\mathrm{M}} \quad \Rightarrow \mathrm{w} = \sqrt{\frac{\mathrm{Aeg}}{\mathrm{M}}}$$

$$\mathrm{T} = \frac{2\pi}{\omega} \quad \Rightarrow \mathrm{T} = 2\pi\sqrt{\frac{\mathrm{M}}{\mathrm{Aeg}}}$$

Q. 10 Given below are two statements:
Statement –I :- For all elements, grater then mass of the nucleus, greater is the binding energy per nucleon
Statement –II :-For all elements, nuclei with less binding energy per nucleon transforms to nuclei with grater bonding energy per nucleon
In the light of the above statements, choose the correct answer from the option given below

Option 1:
Statement –I is true but Statement –II are false

Option 2:
Both Statement –I and Statement –II are false

Option 3:
Both Statement –I and Statement –II are true

Option 4:
Statement –I is false but Statement –II is true

Correct Answer:
Statement –I is false but Statement –II is true

Solution:
Sol:

Statement-I is false because the binding energy per nucleon generally increases with mass number upto 56 and then it gradually decreases for heavier nuclei

Statement-II is true because less $\mathrm{BE/A}$. nuclei transform into nuclei with greater $\mathrm{BE/A}$

Q. 11 The electro static potential in a charged spherical region of radius r varies as $V = ar^3 + b$. where a and b are constants. The total charge in the sphere of unit radius is $\alpha \times \pi \mathrm{a} \varepsilon_0$. The value of α is _____ .
(permittivity of vacuum is ε_0)

Option 1:
-8

Option 2:
-9

Option 3:
-12

Option 4:
-6

Correct Answer:
-12

Solution:
Sol:

We know $\mathrm{E}=-\frac{\mathrm{dv}}{\mathrm{dr}} \Rightarrow \mathrm{E}=-\mathrm{a}\left(3 \mathrm{r}^2\right)+0 \quad \Rightarrow \mathrm{E}-3 \mathrm{ar}^2$
From gauss's law $\phi=\overline{\mathrm{E}} . \overline{\mathrm{d}} s=\frac{\mathrm{q}}{\varepsilon_0} \Rightarrow-3 \mathrm{ar}^2\left(4 \pi \mathrm{r}^2\right)=\frac{\mathrm{q}}{\varepsilon_0} \Rightarrow-12 \pi \mathrm{ar}^4=\frac{\mathrm{q}}{\varepsilon_0}$
Given $\mathrm{r}=1 \Rightarrow \mathrm{q}=-12 \pi \mathrm{a} \varepsilon_0$

$\therefore \alpha=-12$

Q. 12 Three charges $+2q, +3q$ and $-4q$ are satiated at $(0,-3a), (2a,0)$ and $(-2a,0)$ respectively in the xy plane. The resultant dipole moment about origin is _____ .

Option 1:
$2qa[3\hat{i}-7\hat{j}]$

Option 2:
$2qa[3\hat{j}-7\hat{i}]$

Option 3:
$2qa(7\hat{i}-3\hat{j})$

Option 4:
$2qa(3\hat{j}-\hat{i})$

Correct Answer:
$2qa(7\hat{i}-3\hat{j})$

Solution:
Sol:

$q_1 = +2qr_1 = -3a\hat{j}$

$q_1 = +3qr_2 = 2a\hat{i}$

$q_3 = +4qr_3 = -2a\hat{i}$

$P = \varepsilon qr = q_1r_1 + q_2r_2 + q_3r_3$

$P = (2q)(-3a\hat{j}) + 3q(2a)\hat{i} + (-4q)(-2a)\hat{i}$

$P = 2qa[7\hat{i} - 3\hat{j}]$

Q. 13 Two resistor's of 100Ω each are connected in series with a 9 V battery a voltmeter of 400Ω resistance is connected two measure the voltage drop a cross one of the resistance the voltmeter reading is _____ V

Option 1:
4.5

Option 2:
4

Option 3:
3

Option 4:
2

Correct Answer:
4

Solution:
Sol:

Req $= 100 + 800 = 180$

$I = \frac{9}{180}$

$\text{Voltmeter reading } = iR = \frac{9}{180} \times 80 = 4v$

Q. 14 Two masses 400 g and 350 g are suspended from the ends of a light string passing over a heavy pulley of radius 2 cm when released from rest the heavier mass is observed to fall 81 cm is 9 s . The rotational inertia of the pulley is _____ $kg.m^2.\left(g = 9.8\ m/s^2\right)$

Option 1:
4.75×10^{-3}

Option 2:
1.86×10^{-2}

Option 3:
9.5×10^{-3}

Option 4:
8.3×10^{-3}

Correct Answer:
9.5×10^{-3}

Solution:
Sol:

$a = \frac{(m_2 - m_1)g}{\left[m_1 + m_2 + \frac{I}{R^2}\right]} \Rightarrow S = 21t + \frac{1}{2}at^2$

$$\frac{81}{100} = 0 + \frac{1}{2}\left[\frac{m_2 - m_1}{m_1 + m_2 + \frac{I}{R^2}}\right] g \, (()^{\chi}$$

$$\rightarrow \left[m_1 + m_2 + \frac{I}{R^2}\right] = \frac{1}{2}\frac{(m_2 - m_1)(g)}{2}100$$

$$\left[\frac{750}{1000} + \frac{I}{R^2}\right] = \frac{1}{2}\left[\frac{50}{1000}\right](9.8)100$$

$$I = 9.5 \times 10^{-4} \cdot \text{ kg m}^2$$

Q. 15 A boy throw's a ball in to air at 45° from the horizontal to land on a roof of a building of height H . If the ball attains maximum height in 2 s and land on the building in 3 s after launch then the value of H is _____ $m \left(g = 10 \text{ m/s}^2\right)$

Option 1:
15

Option 2:
20

Option 3:
10

Option 4:
25

Correct Answer:
15

Solution:
Sol:

$t_a = 2 = \frac{u \sin\theta}{g}$ $\quad 2 = \frac{u}{g} \cdot \left(\frac{1}{\sqrt{2}}\right)$

$u = 2\sqrt{2}g$ $\quad Sg = (u \sin\theta)t - \frac{1}{2}gt^2$

$H = 2\sqrt{2}g\left(\frac{1}{\sqrt{2}}\right)(3) - \frac{1}{2}g(3)^2$

$H = 15m$

Q. 16 A brass wire of length 2 m and radius 1 mm at $27^{\circ}\mathrm{C}$ is held taut between two rigid supports initially it was cooled to temperature of $-43^{\circ}\mathrm{c}$ creating a tension T in the wire.The temperature to which the wire has to be cooled in order to increase the tension in it to 1.4 T is _____ $^{\circ}\mathrm{c}$

Option 1:
-80

Option 2:
-65

Option 3:
-71

Option 4:
-86

Correct Answer:
-71

Solution:
Thermal stress $\mathrm{F} = \mathrm{yA}\alpha\Delta\mathrm{t} = \mathrm{yA}\alpha\left[\mathrm{t}_2 - \mathrm{t}_1\right]$

$\frac{\mathrm{T}}{1.4\,\mathrm{T}} = \frac{\mathrm{yA}\alpha[-43-27]}{\mathrm{yA}\alpha[\mathrm{t}-27]} \Rightarrow \frac{1}{1.4} = \frac{-70}{\mathrm{t}-27} \Rightarrow \mathrm{t} - 27 = -70(1.4) \Rightarrow \mathrm{t} = -98 + 27 \Rightarrow \mathrm{t} = -71^{\circ}\mathrm{c}$

Q. 17 Three masses $200\ \mathrm{kg}, 300\ \mathrm{kg}$ and 400 kg are placed at the vertices of an equilateral triangle with sides 20 m . They are rearranged on the vertices of a bigger triangle of side 25 m and with same centre. The work done in this process is _____ J.
(Gravitational constant $G = 6.7 \times 10^{-11}\mathrm{Nm}^2/\mathrm{kg}^2$)

Option 1:
2.85×10^{-7}

Option 2:
4.77×10^{-7}

Option 3:
9.86×10^{-6}

Option 4:
1.74×10^{-7}

Correct Answer:
1.74×10^{-7}

Solution:
Sol:

$$\mathrm{w} = \Delta \mathrm{U}$$

$$U_i = --\left[\frac{Gm_1m_2}{r_{12}} + \frac{m_2m_3}{r_{23}} + \frac{Gm_3m_1}{r_{13}}\right]$$

$$= -\frac{6.67 \times 10^{-11}}{20}[200 \times 300 + 300 \times 400 + 400 \times 200]$$

$$= -\frac{6.67 \times 10^{11}}{20}[6 + 12 + 8] \times 10^4$$

$$= -\frac{6.67 \times 10^{-12}}{\not{2}}[26] \times 10^4$$

$$= -86.71 \times 10^{-8}$$

$$U_f = -\frac{6.67 \times 10^{-11}}{25}\left[26 \times 10^4\right]$$

$$= -6.93 \times 10^{-7}$$

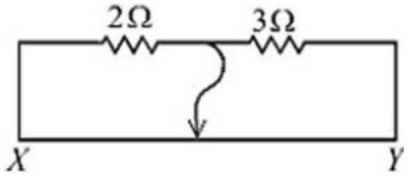
$$= -69.3 \times 10^{-8}$$

$$W = U_f - U_i = [-69.3 + 86.71]10^{-8}$$

$$= 17.41 \times 10^{-8}$$

$$W = 1.741 \times 10^{-7}\ \mathrm{J}$$

Q. 18 Two resistors 2Ω and 3Ω are connected in the gap of bridge as shown in fig the null point is obtained with the contact of jockey at some point on wire XY . When an unknown resistor is connected in parallel with 3Ω resistor , the null point is shifted by 22.5 cm towards Y. The resistance of unknown resistor is _____ Ω



Option 1:
4

Option 2:
3

Option 3:
1

Option 4:
2

Correct Answer:
2

Solution:
Sol:

$$\frac{P}{Q} = \frac{}{100-}$$

$$\frac{2}{3} = \frac{}{100-}$$

$$200 - 2 = 3$$

$$200 = 5 \Rightarrow = \frac{200}{5} = 40 \text{ cm}$$

If unknown resistor (R) is connected in parallel with $3\Omega \mathrm{R}, 3$ are in $\mathrm{P} \rightarrow \frac{3\mathrm{R}}{\mathrm{R}+3}$

$$\mathrm{R}, 3 \text{ are in } \mathrm{P} \rightarrow \frac{3\mathrm{R}}{\mathrm{R}+3}$$

$$^{1} = +22.5 = 40 + 22.5$$

$$= 62.5$$

From $\frac{P}{Q} = \frac{1}{100-1}$

$$\frac{2(R+3)}{3R}=\frac{62.5}{37.5}$$

$$\frac{2(R+3)}{\not{3}R}=\frac{5}{\not{\beta}}\Rightarrow 2R+3(2)=5R\Rightarrow 6=5R-2R$$

$$3R=6$$

$$R=\frac{6}{3}=2\Omega$$

Q. 19 A spring of force constant $15\ N/m$ is cut into two pieces If the ratio of their lengths is $1:3$, then force constant of smaller pieces is _____ N/m

Option 1:
45

Option 2:
15

Option 3:
60

Option 4:
20

Correct Answer:
60

Solution:
Sol:

$K\alpha \xrightarrow{1}\Rightarrow\ K=$ Constant

$K=k_1$ Given $\frac{1}{2}=\frac{1}{3}\Rightarrow {}_2=3_1$

But $={}_1+{}_2={}_1+3_1=4_1\Rightarrow {}_1=\frac{}{4}$

$\therefore 15=K_1\left(\frac{}{4}\right)$

$K_1=15\times 4\Rightarrow\ K_1=60\ N/m$

Q. 20 The exit surface of prism with Refractive index n is coated with material having Refractive index $n/2$ when this prism is set for minimum angle of deviation It Exactly meet's the condition of critical angle, the prism angle is _____

Option 1:
15°

Option 2:
60°

Option 3:
45°

Option 4:
30°

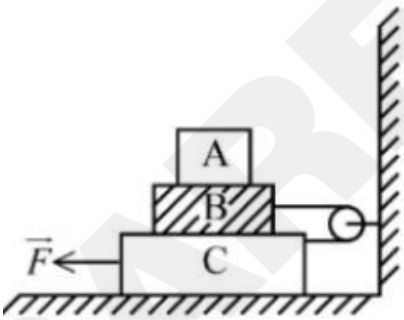
Correct Answer:
60°

Solution:
$\sin c = \frac{n/2}{n} = \frac{1}{2}$

At minimum deviation
A $2r = 60^\circ$

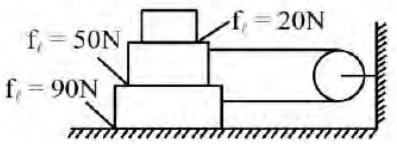
$r_1 = r_2 = c = 30^\circ$

Q. 21 In the given figure the blocks A, B and C weigh $4\ kg, 6\ kg$ and 8 kg respectively. The co-efficient of sliding friction between any two surfaces is 0.5 . The force F required to slide the block C with constant speed is _____ N. (Use $g = 10\ m/s^2$)



Correct Answer:
210

Solution:
For 8 kg to move with constant velocity $F_{net} = 0$.



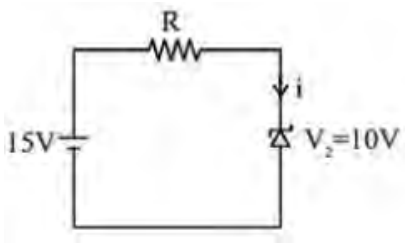
$\therefore F = 90 + T + 50$ (for 8 kg block)
$T = 20 + 50$ (for 6 kg block)

$\therefore \mathrm{F}=210 \mathrm{~N}$.

Q. 22 A voltage regulating circuit consisting of Zener diode, having break-down voltage of 10 V and maximum power dissipation of 0.4 W , is operated at 15 V . The approximate value of protective resistance in this circuit is _____ Ω

Correct Answer:
125

Solution:
Sol:



$\mathrm{P}_{\mathrm{D}}=0.4 \mathrm{w}=10 \mathrm{i}$
$\mathrm{i}=0.04 \mathrm{~A}$
$\mathrm{R}=\frac{15-10}{0.04}=\frac{5}{0.04}=125 \Omega$

Q. 23 A gas of certain mass filled in a closed cylinder at a pressure of 3.23 kPa has temperature $50^{\circ} \mathrm{C}$. The gas is now heated to double its temperature. The modified pressure is _____ Pa .

Correct Answer:
3730

Solution:
$\frac{P_1}{P_2}=\frac{T_1}{T_2} \quad \frac{3.23 \times 10^3}{P_2}=\frac{273+50}{273+100}=\frac{323}{373} P_2=3730 P$

Q. 24 A short bar magnet placed with its axis at 30° with an external field of 800 Gauss, experiences a torque of $0.016 \mathrm{~N} . \mathrm{m}$. The work done in moving it from most stable to most unstable position is $\alpha \times 10^{-3} \mathrm{~J}$. The value of α is _____

Correct Answer:
64

Solution:
Sol:

$$\tau = \mu B \sin\theta \Rightarrow 0.016 = \mu \times B \times \frac{1}{2}$$

$$\Rightarrow \mu = \frac{0.032}{B}$$

$$W_{eu} = U_f - U_i = \mu B - (\mu B) = 2\mu B$$

$$= 2 \times \frac{0.032}{B} \times B$$

$$= 0.064 \text{ J}$$

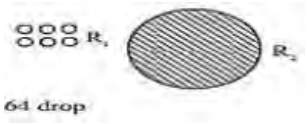
Q. 25 Sixty four rain drops of radius 1 mm each falling down with a terminal velocity of 10 cm/s coalesce to form a bigger drop. The terminal velocity of bigger drop is _____ cm/s.

Correct Answer:
160

Solution:
Sol:

$$\mathrm{V}_r - \frac{2\mathrm{r}^2\,\mathrm{g}}{9\mathrm{n}}[\sigma - \rho]$$

$$\mathrm{V}_r \propto \mathrm{r}^2$$



$$64\left(\frac{4}{3}\pi R_i\right) = \frac{4}{3}\pi R_2$$

$$R_2 - 4R_1$$

$$\frac{(V_T)_1}{(V_T)_2} = \left(\frac{R_2}{R_3}\right)^2 = \left(\frac{1}{4}\right)^2$$

$$\frac{10}{(V_T)_2} = \frac{1}{16}$$

$$(V_T)_2 = 160 \text{ cm/scc}$$

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Chemistry

Q. 1 Given below are two statements:

Statement-I: Hybridisation, shape and spin only magnetic moment of $K_3\left[Co(CO_3)_3\right]$ is sp^3d^2, octahedral and 4.9 BM respectively.

Statement-II: Geometry, hybridisation and spin only magnetic moment values (BM) of the ions $[Ni(CN)_4]^{2-}$, $[MnBr_4]^{2-}$ and $[CoF_6]^{2-}$ respectively are square planar, tetrahedral, octahedral; dsp^2, sp^3, sp^3d^2 and 0, 5.9, 4.9.

In light of the above statements, choose the correct answer from the options given below

Option 1:
S-I is true but S-II is false

Option 2:
Both S-I and S-II are true

Option 3:
S-I is false but S-II is true

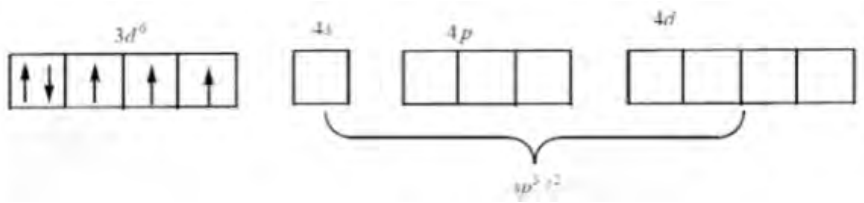
Option 4:
Both S-I and S-II are false

Correct Answer:
Both S-I and S-II are true

Solution:
S-I $K_3\left[Co(CO_3)_3\right]$

$Co^{+3} - 3d^6 4s^0 4p^0$

Co_3^{-2}-Weak field ligand, no pairing of electrons



No. of unpaired electrons $\mu = 4$

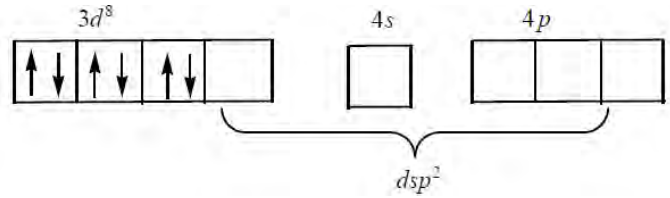
Magnetic moment $\mu = \sqrt{n(n+2)}BM$

$\mu = \sqrt{4(4+2)} = \sqrt{24} = 4.9\text{BM}$

S-II : $[\text{Ni(CN)}_4]^{2-}$

$\text{Ni}^{+} - 3d^8$

CN^{-}- strong field ligand, pairing takes place

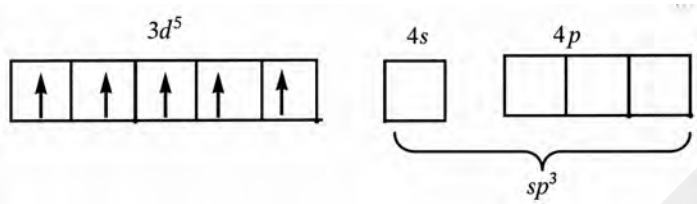


$\text{N} = 0, \mu = 0$

$[\text{MnBr}_4]^{2-}$

$\text{Mn}^{+2} - 3d^5$

Br^{-}-weak field ligand, no pairing



$\text{n} = 5$

$\mu = \sqrt{5(5+2)} \quad = \sqrt{35} \quad = 5.9\text{BM}$

$[\text{CoF}_6]^{3-}$

$\text{Co}^{+3} - 3d^6$

F^{-}-weak field ligand, no pairing

$\text{N} = 4, \mu = \sqrt{4(4+2)} = \sqrt{24} = 4.9\text{BM}$

Hence, the correct answer is the option (2).

Q. 2 Given below are statements about some molecules/ions.
Identify the CORRECT statements.
A. The dipole moment value of NF_3 is higher than that of NH_3.
B. The dipole moment value of BeH_2 is zero.
C. The bond order of O_2^{2-} and F_2 is same
D. The formal charge on the central oxygen atom of ozone is -1 .
E. In NO_2, all the three atoms satisfy the octet rule, hence it is very stable.

Choose the correct answer from the options given below:

Option 1:
B & C only

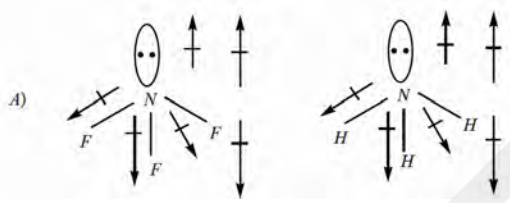
Option 2:
B, C & D only

Option 3:
A, C & D only

Option 4:
A, B, C, D & E

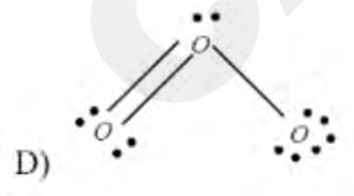
Correct Answer:
B & C only

Solution:
Sol:

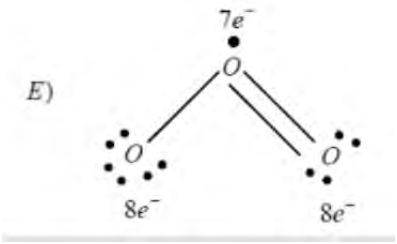


B) $\mathrm{H-Be-H}$

Linear $\mu = 0$
C) $\mathrm{O_2^{-2}},\ \mathrm{F_2} = 18e^-$
$\sigma \quad \overset{*}{\sigma} \quad \sigma \quad \overset{*}{\sigma} \quad \sigma$
$1s^2 \quad 1s^2 \quad 2s^2 \quad 2s^2 \quad 2s_2^2 \quad \pi_{2px}^2 = \pi_{2py}^2 \pi_{2px}^{*2} = \pi_{2py}^{*2}$
B. O. $= \frac{1}{2}[N_b - N_a] = \frac{1}{2}[10-8] = \frac{2}{2} = 1$



Formal charge $N_A = N_{l.p} - \frac{1}{2}N_{B.P}$
Central oxygen $= 6 - 2 - \frac{1}{2}(6) = 6 - 5 = +1$



Hence, the correct answer is the option (1).

Q. 3 Arrange the following combinations in the decreasing order of stability

I. $p - Br - C_6H_4 - \overset{\ominus}{\underset{\cdot\cdot}{C}}H_2$

II. $C_6H_5 - \overset{\ominus}{\underset{\cdot\cdot}{C}}H_2$

III. $p - CH_3O - C_6H_4 - \overset{\ominus}{\underset{\cdot\cdot}{C}}H_2$

IV. $p - CHO - C_6H_4 - \overset{\ominus}{\underset{\cdot\cdot}{C}}H_2$

V. $p - CH_3 - C_6H_4 - \overset{\ominus}{\underset{\cdot\cdot}{C}}H_2$

Choose the correct answer from the options given below:

Option 1:
IV > II > I > III > V

Option 2:
I > IV > II> V > III

Option 3:
I > II > IV > V > III

Option 4:
IV > I > II > V > III

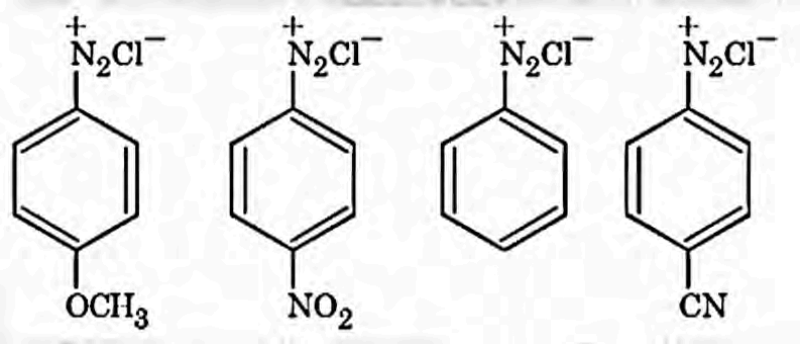
Correct Answer:
IV > I > II > V > III

Solution:
Sol: Stability of carbanion directly proportional to -j groups (E.W.G)

$$P - HO - C_6H_4 - CH_2 > P - Br - C_6H_4 - CH_2 > C_6H_5 - CH_2 > P - CH_3 - C_6H_4 - CH_2 > P - CH_3O - C_6H_4 - CH_2$$

Hence, the correct answer is the option (4).

Q. 4 The correct stability order of the following diazonium salts is



Option 1:
$C > A > D > B$

Option 2:
A > B > C > D

Option 3:
$A > C > D > B$

Option 4:
C > D > B > A

Correct Answer:
$A > C > D > B$

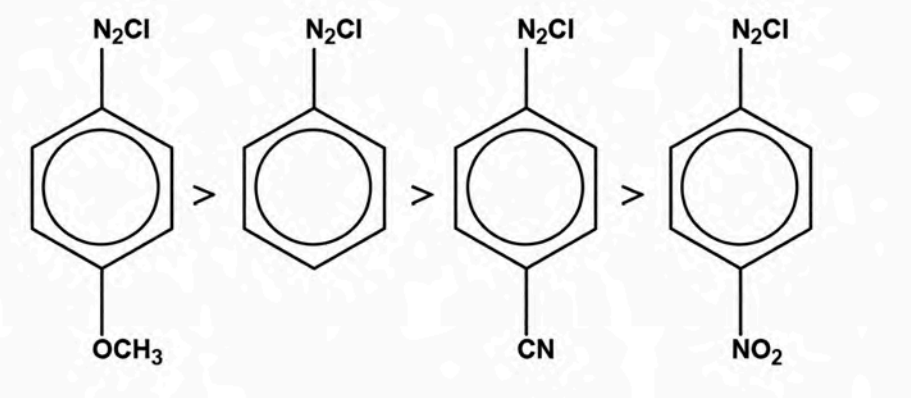
Solution:
Sol: Diazonium salt stabilized by E.D.G.

Effect of given groups

- $-OCH_3$ → strong +M (electron donating) → most stabilizing
- –H → neutral
- –CN → –M, –I withdrawing
- $-NO_2$ → very strong –M, –I withdrawing → most destabilizing

Stability order

A > C > D > B



Hence, the correct answer is the option (3).

Q. 5 Among the following, the correct combinations are
A. $IF_3 \rightarrow T-$ shaped $\left(sp^3d\right)$
B. $IF_5 \rightarrow$ Square pyramidal $\left(sp^3d^2\right)$
C. $IF_7 \rightarrow$ Pentagonal bipyramidal $\left(sp^3d^3\right)$
D. $\mathrm{ClO}_4^- \rightarrow$ Square planar $\left(sp^2d\right)$ Choose the correct answer from the options given below:

Option 1:
A, B and C only

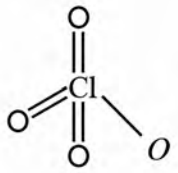
Option 2:
A and B only

Option 3:
A, B, C and D

Option 4:
B, C and D only

Correct Answer:
A, B and C only

Solution:
A) $IF_3 - AB_3E_2$ type - T -shape - Sp^3d
B) $IF_5 - AB_5E$ type - Square pyramidal $-Sp^3d^2$
C) $IF_7 - AB_7$ type - Pentagonal bipyramidal $\left(Sp^3d^3\right)$

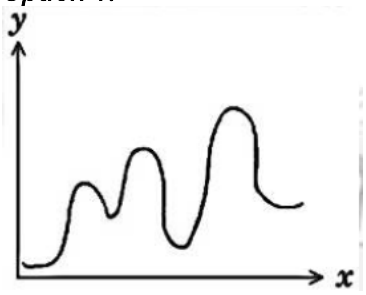
d) ClO_4^- –  $-Sp^3$ – Tetrahedral

Hence, the correct answer is the option (1).

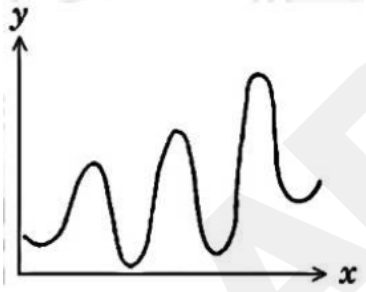
Q. 6 $\mathrm{A} \rightarrow \mathrm{D}$ is an endothermic reaction occurring in three steps (elementary).
(i) $A \rightarrow B \Delta H_i = +ve$
(ii) $B \rightarrow C \Delta H_{ii} = -ve$
(iii) $C \rightarrow D \Delta H_{iii} = -ve$

Which of the following graphs between potential energy (y-axis) vs reaction coordinate (x-axis) correctly represents the reaction profile of $\mathrm{A} \rightarrow \mathrm{D}$?

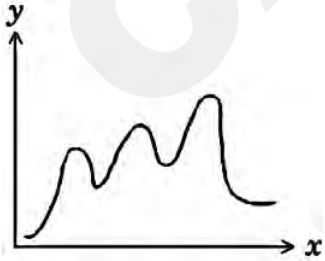
Option 1:



Option 2:

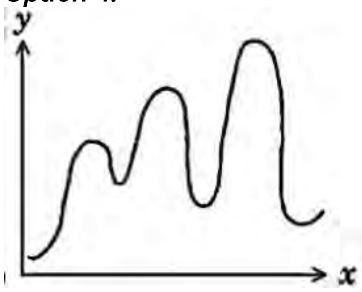


Option 3:

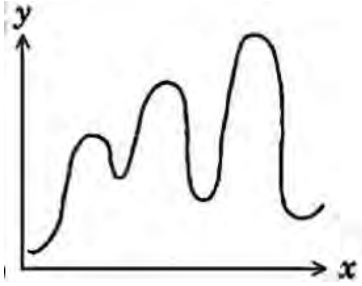


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Option 4:

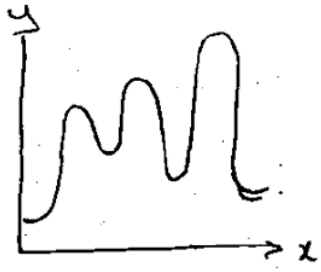


Correct Answer:



Solution:
Sol:

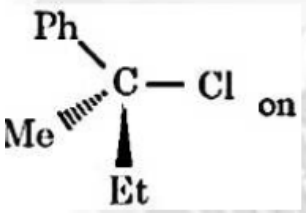
$\mathrm{A} \rightarrow \mathrm{D}\ \Delta H = +ve \quad \Delta H = H_P - H_R\,(H_p > H_R)$



(i) $\mathrm{A} \rightarrow \mathrm{B} \quad \Delta H_i = +ve\,(H_p > H_R)$
(ii) $\mathrm{B} \rightarrow \mathrm{C} \quad \Delta H_{ii} = -\text{ ve}\,(H_p < H_R)$
(iii) $\mathrm{C} \rightarrow \mathrm{D} \quad \Delta H_{iii} = -ve\,(H_p < H_R)$

Hence, the correct answer is the option (4).

Q. 7 Given below are two statements
Statement-I: ' $\mathrm{C} - \mathrm{Cl}'$ bond is strong in $\mathrm{CH_2 = CH - Cl}$ than $\mathrm{CH_3 - CH_2 - Cl}$
Statement-II: The given optically active molecule,



on hydrolysis gives a solution that can rotate the plane polarized light.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
S-I is true but S-II is false

Option 2:
Both S-I and S-II are false

Option 3:
Both S-I and S-II are true

Option 4:
S-I is false but S-II is true

Correct Answer:
Both S-I and S-II are true

Solution:
Statement-I: True

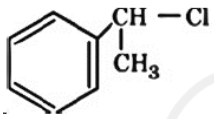
In $\mathrm{CH_2=CHCl}$, Cl lone pair participates in resonance with the double bond → partial double bond character in $\mathrm{C-Cl}$ → bond becomes stronger than in $\mathrm{CH_3CH_2Cl}$.

Statement-2: True

Hydrolysis forms a carbocation, nucleophile attacks from both sides but not equally, so the product mixture is still optically active.

Hence, the correct answer is the option (3).

Q. 8 Match List-I with List-II

List-I Chloro derivative		List-II Example	
(A)	Vinyl Chloride	I.	$CH_2 = CH - CH_2Cl$
(B)	Benzyl Chloride	II.	$CH_3 - CH(Cl)CH_3$
(C)	Alkyl Chloride	III.	$\mathrm{CH}_2=\mathrm{CHCl}$
(D)	Allyl Chloride	IV.	

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-I, D-II

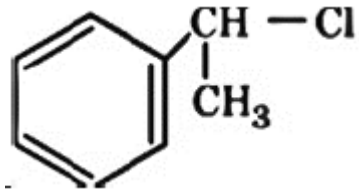
Option 2:
A-IV, B-I, C-III, D-II

Option 3:
A-I, B-II, C-IV, D-III

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:
Sol: Vinyl chloride $CH_2 = CH - Cl$

Benzyl chloride 

Alkyl chloride $CH_3 - CH(Cl) - CH_3$
Allyl chloride $CH_2 = CH - CH_2 - Cl$

Hence, the correct answer is the option (4).

Q. 9 Consider a mixture ' X ' which is made by dissolving 0.4 mol of $[Co(NH_3)_5SO_4]Br$ and 0.4 mol of $[Co(NH_3)_5Br]SO_4$ in water to make 4 L of solution. When 2 L of mixture ' X ' is allowed to react with excess of $AgNO_3$, it forms precipitate ' Y '. The rest 2 L of mixture ' X ' reacts with excel $BaCl_2$ to form precipitate ' Z '. which of the following statements is CORRECT?

Option 1:
0.1 mol of ' Y ' is formed

Option 2:
0.2 mol of ' Z ' is formed

Option 3:
0.4 mol of ' Z ' is formed

Option 4:
' Y ' is $BaSO_4$ and ' Z ' is AgBr

Correct Answer:
0.2 mol of ' Z ' is formed

Solution:
Sol:

$$[CO(NH_3)_5SO_4]Br \longrightarrow \underset{0.4\ mol}{[CO(NH_3)_5SO_4]^+} + \underset{0.4\ mol}{Br}$$

$$\left[CO(NH_3)_5BrSO_4 \underset{a}{\rightarrow} [CO(NH_3)_5Br]SO_4[CO(NH_3)_5Br]^{+2} + \underset{0.4\ mol}{SO_4^{-2}}\right.$$

$X\&Y$ mixed in 4 L

2 L solution contain $0.2\ mol Br^-\ \&\ 0.2\ mol SO_4^{-2}$

2 L Mixture $+AgNO_3 \longrightarrow \underset{0.2\ mol}{AgBr}$

2 L mixture $+BaCl_2 \longrightarrow \underset{0.2\ mol}{BaSO_4}$

Hence, the correct answer is the option (2).

Q. 10 Consider three metal chlorides x, y and z , where x is water soluble at room temperature, y is sparingly soluble in water at room temperature and z is soluble in hot water. x, y and z re respectively

Option 1:
$\mathrm{MgCl_2}, \mathrm{AgCl}$ and $\mathrm{AlCl_3}$

Option 2:
$\mathrm{CuCl_2}, \mathrm{AgCl}$ and $\mathrm{PbCl_2}$

Option 3:
$\mathrm{AgCl}, \mathrm{Hg_2Cl_2}$ and $\mathrm{PbCl_2}$

Option 4:
$\mathrm{AlCl_3}, \mathrm{PbCl_2}$ and $\mathrm{BaCl_2}$

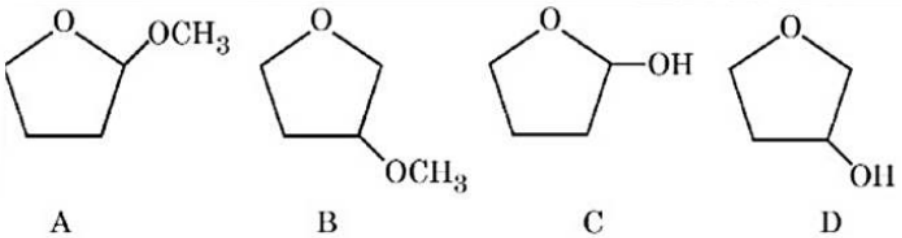
Correct Answer:
$\mathrm{CuCl_2}, \mathrm{AgCl}$ and $\mathrm{PbCl_2}$

Solution:
Sol:

$\mathrm{X} = \mathrm{CuCl_2}$ soluble in water at room temperature
$Y = AgCl$ sparingly soluble in water at room temperature.
$Z = \mathrm{PbCl_2}$ soluble in hot water.

Hence, the correct answer is the option (2).

Q. 11 A student is given one compound among the following compounds that gives positive test with Tollen's reagent.



A B C D

The compound is

Option 1:
A

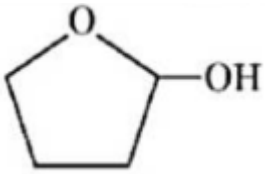
Option 2:
D

Option 3:
C

Option 4:
B

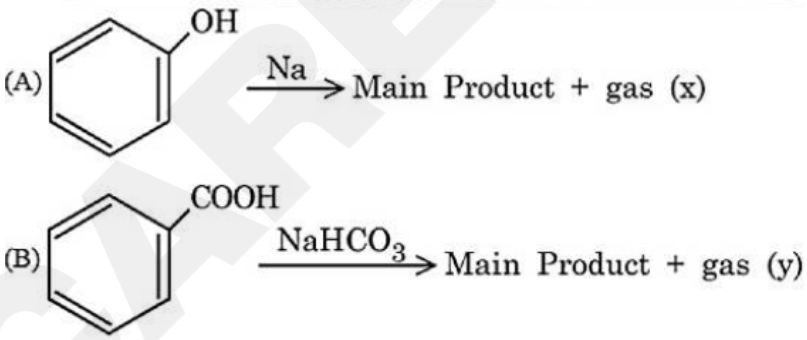
Correct Answer:
C

Solution:
Sol:Only hemiacetals givesTollen's test



Hence, the correct answer is the option (3).

Q. 12 Consider the following two reactions A and B



Numerical value of [molar mass of x + molar mass of y] is ...

Option 1:
4

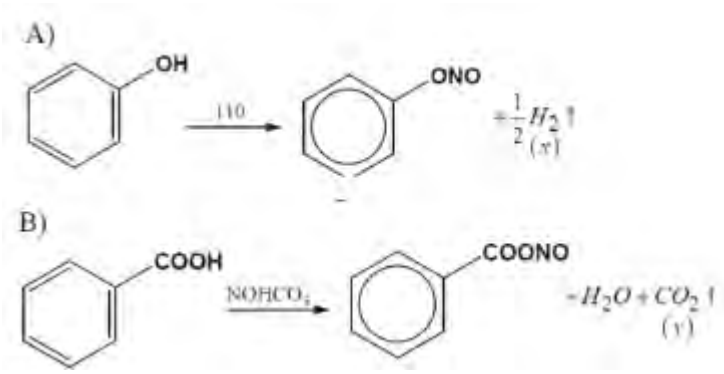
Option 2:
46

Option 3:
88

Option 4:
160

Correct Answer:
46

Solution:
Sol:



Molar mass of $H_2(x) = 2$
Molar mass of $\mathrm{CO}_2(y) = 44$

$x + y = 46$

Hence, the correct answer is the option (2).

Q. 13 Given below are two statements;
Statement-I: The number of paramagnetic species among $[\mathrm{CoF}_6]^{3-}, [\mathrm{TiF}_6]^{3-}, V_2\mathrm{O}_5$ and $[\mathrm{Fe(CN)}_6]^{3-}$ is 3.
Statement-II:

$$\mathrm{K}_4\,[\mathrm{Fe(CN)}_6] < \mathrm{K}_3\,[\mathrm{Fe(CN)}_6] < [\mathrm{Fe(H_2O)}_6]\mathrm{SO}_4 \cdot \mathrm{H_2O} < [\mathrm{Fe(H_2O)}_6]\mathrm{Cl}_3$$

$[\mathrm{Fe(H_2O)}_6]\mathrm{Cl}_3$ is the correct order in terms of number of unpaired element(s) present in the complexes.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both S-I and S-II are true

Option 2:
S-I is false but S-II is true

Option 3:
S-I is true but S-II is false

Option 4:
Both S-I and S-II are false

Correct Answer:
Both S-I and S-II are true

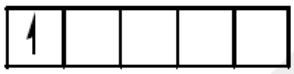
Solution:
Sol:

$S-I$

$[\mathrm{COF_6}]^{3-}\ \mathrm{CO}^{+3}-3d^6$ 

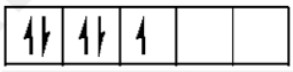
F^-- weatfield ligand, no pairing.

No. of unpaired $e's(n)=4$
$\therefore$ Paramagnetic

$[TiF_6]^{3-}\quad Ti^{+3}-3d^1$ 

$n=1$, paramagnetic
$\mathrm{V_2O_5}\ \mathrm{V}^{+5}-3d^0, n=O$, diamagnetic

$[\mathrm{Fe(CN)_6}]^{3-}\mathrm{Fe}^{+3}-3d^5$ 

CN^-- Strong field ligand, pairing of $e's$ 

S-II

$K_4\,[\mathrm{Fe(CN)_6}]$
$\mathrm{Fe}^{+2}-3d^6$
$\mathrm{CN}-\mathrm{SFL}$



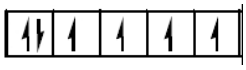
$n=0$

$K_3\,[\mathrm{Fe(CN)_6}]$
$\mathrm{Fe}^{+3}-3d^5$
$\mathrm{CN}-\mathrm{SFL}$



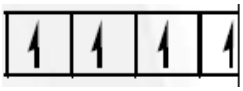
$n = 1$

$\left[\mathrm{Fe}_{(\mathrm{H_2O})_6}\right]\mathrm{SO_4} \cdot \mathrm{H_2O}$
$\mathrm{Fe}^{+2} - 3d^6$
$\mathrm{H_2O} -$ W.F.L.



$n = 4$

$[\mathrm{Fe(H_2O)_6}]\mathrm{Cl_3}$
$\mathrm{Fe}^{+3} - 3d^5$
$\mathrm{H_2O} -$ W.F.L.



$n = 5$

Hence, the correct answer is the option (1).

Q. 14 Given below are two statements:
Statement-I: $\mathrm{K} > \mathrm{Mg} > \mathrm{Al} > \mathrm{B}$ is the correct order in terms of metallic character.
Statement-II: Atomic radius is always greater than the ionic radius for any element.
In the light of the above statements, the correct answer from the options given below

Option 1:
Both S-I and S-II are false

Option 2:
Both S-I and S-II are true

Option 3:
S-I is false but S-II is true

Option 4:
S-I is true but S-II is false

Correct Answer:
S-I is true but S-II is false

Solution:
Sol:

Statement-I: True → Metallic character: $K > Mg > Al > B$ (down group ↑, across period ↓)
Statement-II: False → Atomic radius not always greater than ionic radius (cations shrink, anions expand)

Hence, the correct answer is the option (4).

Q. 15 A solution is prepared by dissolving 0.3 g of non-volatile non-electrolyte solute ' A ' of molar mass $60\ \text{g mol}^{-1}$ and 0.9 g of non-volatile non-electrolyte solute ' B ' of molar mass $180\ \text{g mol}^{-1}$ in $100\ \text{mL} H_2O$ at 27°C. Osmatic pressure of the solution will be

[Given: $\text{R} = 0.082\ \text{L atm K}^{-1}\ \text{mol}^{-1}$]

Option 1:
1.23 atm

Option 2:
1.47 atm

Option 3:
2.46 atm

Option 4:
0.82 atm

Correct Answer:
2.46 atm

Solution:
Sol:

$\pi = C_1 + C_2 = \left(\frac{0.3\times1000}{60\times100} + \frac{0.9\times1000}{180\times100}\right) \times 0.821 \times 300 = 2.46\ \text{atm}$

Hence, the correct answer is the option (3).

Q. 16 The hydroxy compound (X) with molar mass $122\ \mathrm{g\ mol^{-1}}$ is acetylated with acetic anhydride, using a large excess of the reagent ensuring complete acetylation of all hydroxyl groups. The product obtained has a molar mass of $290\ \mathrm{g\ mol^{-1}}$. The number of hydroxyl groups present in compound (X) is:

Option 1:
4

Option 2:
3

Option 3:
2

Option 4:
5

Correct Answer:
4

Solution:
When a hydroxy compound reacts with acetic anhydride, each –OH group is converted into an acetate ester (–$OCOCH_3$).

- Molar mass of X = 122 g/mol
- Molar mass of acetylated product = 290 g/mol

No. of hydroxyl groups $= \frac{\text{molar mass of acetylated product} - \text{molar mass of original compound}}{42}$

$$= \frac{290 - 122}{42} = \frac{168}{142} = 4$$

Hence, the correct answer is the option (1).

Q. 17 At $27^{\circ}\mathrm{C}$ in presence of a catalyst, activation energy of a reaction is lowered by 10 kJ $\mathrm{mol^{-1}}$. The logarithm of ratio of $\frac{\mathrm{k(\ catalysed\ )}}{\mathrm{k(\ uncatalysed\ )}}$ is ....
(Consider that the frequency factor for both reactions is same)

Option 1:
3.482

Option 2:
1.741

Option 3:
17.41

Option 4:
0.1741

Correct Answer:
1.741

Solution:
Sol:

$$\frac{\mathrm{K}_{\text{catalyst}}}{K_{\text{uncatalyst}}} = e^{-Eq/RT}$$

$$\log \frac{\mathrm{K}_c}{K_u} = \frac{\Delta E}{2.303RT} = \frac{10 \times 1000}{2.303 \times 8.314 \times 300} = 1.741$$

Hence, the correct answer is the option (2).

Q. 18 Arrange the following alkenes in decreasing order of stability



Choose the correct answer from the options given below:

Option 1:
I > III > II > IV

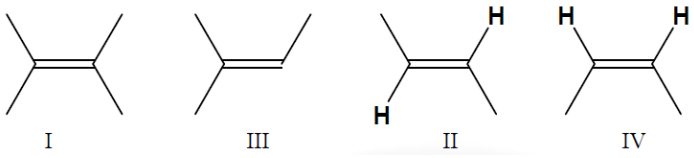
Option 2:
III > II > I > IV

Option 3:
I > III > IV > II

Option 4:
III > I > II > IV

Correct Answer:
I > III > II > IV

Solution:
Sol:More substituted alkeny are more stable by hyper conjugation



Hence, the correct answer is the option (1).

Q. 19 ' W ' g of non-volatile electrolyte solid solute of molar mass ' M ' gmol^{-1} when dissolved in 100 mL water, decreases vapour pressure of water from 640 mm Hg to 600 mm Hg . If aqueous solution of the electrolyte boils at $375\ \mathrm{K}^{\circ}$ and $\mathrm{K_b}$ for water is $0.52\ \mathrm{K\ kg\ mol}^{-1}$, then the mole fraction of the electrolyte solution $\left(\mathrm{x_2}\right)$ in the solution can be expressed as (Given: Density of water $=1\ \mathrm{g/mL}$ and boiling point of water $=373\ \mathrm{K}$)

Option 1:
$\frac{16}{2.6} \times \frac{\mathrm{W}}{\mathrm{M}}$

Option 2:
$\frac{1.3}{8} \times \frac{M}{W}$

Option 3:
$\frac{2.6}{16} \times \frac{M}{W}$

Option 4:
$\frac{1.3}{8} \times \frac{\mathrm{W}}{\mathrm{M}}$

Correct Answer:
$\frac{1.3}{8} \times \frac{\mathrm{W}}{\mathrm{M}}$

Solution:
Sol:

$$\Delta T_b = iK_bm$$

$$2 = i \times 0.52 \times \frac{\frac{w}{m}}{\frac{100}{1000}}$$

$$i \times \frac{w}{m} = 2 \times \frac{100}{1000} \times \frac{1}{52}$$

$$i = \frac{1}{2.6} \times \frac{m}{w}$$

$$RLVP = \frac{P^o - Ps}{P^o} = i \times X_{\text{solute}}$$
$$\frac{640 - 600}{640} = i \times X_{\text{solute}}$$
$$\frac{1w}{16} = \frac{1}{2.6} \times \frac{m}{w} \times X_{\text{solute}}$$
$$X_{\text{solute}} = \frac{1.3}{8} \times \frac{w}{m}$$

Hence, the correct answer is the option (4).

Q. 20 Match List-I with List-II

List-I Isothermal process for ideal gas system		**List-II Work done** $(V_f > V_i)$	
A.	Reversible expansion	I	$w = 0$
B.	Free expansion	II	$w = -nRT \ln \frac{V_f}{V_i}$
C.	Irreversible expansion	III	$w = -P_{ex}(V_f - V_i)$
D.	Irreversible compression	IV	$w = -P_{ex}(V_i - V_f)$

Choose the correct answer from the options given below:

Option 1:
A-IV, B-II, C-III, D-I

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-I, B-III, C-II, D-IV

Option 4:
A-IV, B-I, C-III, D-II

Correct Answer:
A-II, B-I, C-III, D-IV

Solution:
Sol:

Reversible expansion $w=-nRT\ln\frac{V_f}{V_i}$

Free expansion $\mathrm{w}=0$
Irreversible expansion$w=-P_{ex}\left(V_f-V_i\right)$

Irreversible compression$w=-P_{ex}\left(V_i-V_f\right)$

Hence, the correct answer is the option (2).

Q. 21 X and Y are the number of electrons involved, respectively during the oxidation of I^- to I_2 and S^{2-} to S by acidified $\mathrm{K}_2\mathrm{Cr}_2\mathrm{O}_7$. The value of $\mathrm{X}+\mathrm{Y}$ is _____

Correct Answer:
12

Solution:
a) $\mathrm{Cr}_2\mathrm{O}_7^{-2}+14\mathrm{H}^++6e^-\rightarrow 2\mathrm{Cr}^{+3}+7\mathrm{H}_2\mathrm{O}$
b) $\left(2I^-\rightarrow I_2+2e^-\right)\times 3$

$\mathrm{Cr}_2\mathrm{O}_7^{-2}+6\mathrm{I}^-+14\mathrm{H}^+\xrightarrow{6e(x)}2\mathrm{Cr}^{+3}+3\mathrm{I}_2+7\mathrm{H}_2\mathrm{O}$

c) $\left(S^{-2}\rightarrow S+2e^-\right)\times 3$

$\mathrm{Cr}_2\mathrm{O}_7^{-2}+3\ \mathrm{S}^{-2}+14\mathrm{H}^+\xrightarrow{6e(y)}2\mathrm{Cr}^{+3}+7\mathrm{H}_2\mathrm{O}+3\ \mathrm{S}$
$x+y=6+6=12$

Hence, the answer is (12).

Q. 22 Electricity is passed through an acidic solution of Cu^{2+} till all the Cu^{2+} was exhausted, leading to the deposition of 300 mg of Cu metal. However, a current of 600 mA was continued to pass through the same solution for another 28 minutes by keeping the total volume of the solution fixed at 200 mL . The total volume of oxygen evolved at STP during the entire process is _____ mL. (nearest integer)

$\mathrm{Cu}^{2+}(\mathrm{aq})+2\mathrm{e}^-\rightarrow\mathrm{Cu}(\mathrm{s})\mathrm{E}^{\circ}_{\mathrm{red}}=+0.34\ \mathrm{V}$
$\mathrm{O}_2(\mathrm{g})+4\mathrm{H}^++4\mathrm{e}^-\rightarrow 2\mathrm{H}_2\mathrm{O}\mathrm{E}^{\circ}_{\mathrm{red}}=+1.23\ \mathrm{V}$

Molar mass of $\mathrm{Cu}=63.54\ \mathrm{g\ mol}^{-1}$
Molar mass of $\mathrm{O}_2=32\ \mathrm{g\ mol}^{-1}$
Faraday Constant $=96500\mathrm{Cmol}^{-1}$
Molar volume at $\mathrm{STP}=22.4\ \mathrm{L}]$

Correct Answer:
111

Solution:
Sol:

Equation of Cu = Equation of $\mathrm{O_2}$

$$\frac{300 \times 10^{-3} \times 2}{63.54} = {}^{n}O_2 \times 4$$
$${}^{n}O_2 = 2.36 \times 10^{-3}$$

When current is further passed ${}^{n}O_2 \times 4 = \frac{600 \times 28 \times 60}{96500 \times 1000}$

$n_{o_2} = 2.611 \times 10^{-3}$

Total O_2 released $= \left[10^{-3} \times (2.36 + 2.611)\right] \times 22400 = 111.35\mathrm{ml}$

Hence, the answer is (111).

Q. 23 Consider two group IV metal ions X^{2+} and Y^{2+} A solution containing $0.01\mathrm{M}\mathrm{X}^{2+}$ and $0.01\mathrm{M}\mathrm{Y}^{2+}$ is saturated with $\mathrm{H_2\ S}$. The pH at which the metal sulphide YS will form as a precipitate is ... (nearest integer)

(Given: $\mathrm{K_{sp}(XS)} = 1 \times 10^{-22}$ at $25^{\circ}\mathrm{C}, \mathrm{K_{sp}(YS)} = 4 \times 10^{-16}$ at $25^{\circ}\mathrm{C}$, $[\mathrm{H_2\ S}] = 0.1\mathrm{M}$ in solution, $\mathrm{K_{a1} \times K_{a2}\ (H_2\ S)} = 1.0 \times 10^{-21}, \log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.70$)

Correct Answer:
4

Solution:
Sol:

ppt of YS begins when I.P. of $\left(Y^{+2}\right)\left(S^{-2}\right) > K_{SP}$ of YS

$\left[Y^{+2}\right] = 0.01M \& K_{SP}(YS) = 4 \times 10^{-16}$

min sulphide ion conc. Required

$$\left[S^{-2}\right]_{\min} = \frac{K_{SP}(YS)}{Y^{+2}} = \frac{4 \times 10^{-16}}{10^{-2}} = 4 \times 10^{-14}$$

For the dissociation of $\mathrm{H_2\ S}$ in solution. The overall equation constant K_a, K_{a_2} is given by

$\mathrm{H_2\ S} \rightleftarrows 2\mathrm{H}^{+}\mathrm{S}^{-2}$

$$K_{a_1} \cdot K_{a_2} = \frac{\left[H^+\right]^2 \left[S^{-2}\right]}{[H_2S]}$$
$$K_{a_1} \cdot K_{a_2} = 1 \times 10^{-21} \& \left[H_2S\right] = 0.1M$$
$$\left[H^+\right]^2 = \frac{K_{a_1} \cdot K_{a_2} [H_2S]}{S^{-2}} = \frac{1 \times 10^{-21} \times 10^{-1}}{4 \times 10^{-14}}$$
$$\left[H^+\right] = 0.5 \times 10^{-4} M \quad pH = 4.3$$

Hence, the answer is (4).

Q. 24 The hydrogen spectrum consists of several spectral lines in Lyman series $(\mathrm{L}_1,\ \mathrm{L}_2,\ \mathrm{L}_3 \ldots\ldots .; \mathrm{L}_1$ has lowest energy among Lyman series). Similarly it consists of several spectral lines in Balmer series ($\mathrm{B}_1,\ \mathrm{B}_2,\ \mathrm{B}_3 \ldots ; \mathrm{B}_1$ has lowest energy among Balmer lines). The energy of L_1 is x times the energy of B_1. The value of x is $\ldots \times 10^{-1}$ (Nearest integer)

Correct Answer:
54

Solution:
Sol:

$$\Delta E_L = 13.6 \times Z^2 \left[\frac{1}{1^2} - \frac{1}{2^2}\right] = 13.6 \times Z^2 \times \frac{3}{4}$$
$$\Delta E_B = 13.6 \times Z^2 \left[\frac{1}{2^2} - \frac{1}{3^2}\right] = 13.6 \times Z^2 \times \frac{5}{4 \times 9}$$
$$\frac{\Delta E_L}{\Delta E_B} = \frac{3}{5} \times 9 = \frac{27}{5} = 5.4 = 54 \times 10^{-1}$$
$$\therefore x = 54$$

Hence, the answer is (54).

Q. 25 In Dumas method for estimation of nitrogen, 0.50 g of an organic compound gave 70 mL of nitrogen collected at 300 K and 715 mm pressure. The percentage of nitrogen in the organic compound is .... $\%$ (Aqueous tension at 300 K is 15 mm).

Correct Answer:
15

Solution:
Sol:

$$\%N = \frac{28}{22400} \times \frac{\text{vol. no. } N_2 \text{ at STP}}{\text{wt of O.C.}} \times 100$$
$$= \frac{28}{22400} \times \frac{69.16}{0.5} \times 100$$
$$= 14.65\%$$
$$\approx 15$$

STP given

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$
$$\frac{760 \times 70}{300} = \frac{(715-15)V_2}{273}$$
$$V_2 = \frac{760 \times 70}{300} \times \frac{273}{700} = 69.16$$

Hence, the answer is (15).

Maths

Q. 1 Let $S = \left\{Z \in C : \left|\frac{Z-6i}{Z-2i}\right| = 1 \text{ and } \left|\frac{z-8+2i}{z+2i}\right| = \frac{3}{5}\right\}$ then $\sum_{Z \in S} |Z|^2$ is equal to:

Option 1:
385

Option 2:
413

Option 3:
398

Option 4:
423

Correct Answer:
385

Solution:
$$\left|\frac{Z-6i}{Z-2i}\right| = 1 \Rightarrow y = 4$$
$$25|z-8+2i|^2 = 9|z+2i|^2$$
$$\Rightarrow x^2 + y^2 - 25x + 4y + 104 = 0$$
Substituting $y = 4$ into the equation:
$$x^2 + (4)^2 - 25x + 4(4) + 104 = 0$$
$$x^2 + 16 - 25x + 16 + 104 = 0$$
$$x^2 - 25x + 136 = 0$$

$$x^2 - 17x - 8x + 136 = 0$$
$$x(x-17) - 8(x-17) = 0$$
$$(x-8)(x-17) = 0$$

This gives $x = 8$ and $x = 17$.

Since $Z = x + iy$, the points in S are:
$Z_1 = 8 + 4i$
$Z_2 = 17 + 4i$

We need the sum of the squares of the moduli:

$$|Z_1|^2 = 8^2 + 4^2 = 64 + 16 = 80$$
$$|Z_2|^2 = 17^2 + 4^2 = 289 + 16 = 305$$
$$\sum |Z|^2 = 80 + 305 = 385$$

Hence, the correct answer is option (1).

Q. 2 If the function $f(x) = \frac{e^x(e^{\tan x - x} - 1) + \log_e(\sec x + \tan x) - x}{\tan x - x}$ is continues at $x = 0$ then the value of $f(0)$ is equal to:

Option 1:
$\frac{3}{2}$

Option 2:
2

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{2}{3}$

Correct Answer:
$\frac{3}{2}$

Solution:

$$f(0) = \text{lt}_{x\to 0}\, e^x \frac{(e^{\tan x - x} - 1)}{\tan x - x} + \text{lt}_{x\to 0} \frac{\log(\sec x + \tan x) - x}{\tan x - x}$$

$$\text{lt}_{x\to 0}\, e^x \frac{[e^{\tan x - x} - 1]}{\tan x - x} - 1\, \text{lt}_{x\to 0} \frac{\log(\sec x + \tan x) - x}{\tan x - x}$$

$$e^{(0)}(1) + \text{lt}_{x\to 0} \frac{\frac{1}{\sec x + \tan x} \cdot (\sec x \tan x + \sec^2 x) - 1}{\sec^2 x - 1} \quad \text{LHospital rule}$$

$$= 1 + \text{lt}_{x\to 0} \frac{\sec x - 1}{(\sec x - 1)(\sec x) + 1)} = 1 + \frac{1}{2} = \frac{3}{2}$$

Hence, the correct answer is option (1).

Q. 3 Let a circle of radius 4 pass through the origin O. The Points $A(-\sqrt{3}a, 0)$ and $B(0, -\sqrt{2}b)$ where a and b are real parameters $\mathrm{ab} \neq 0$. Then the radius of the locus of the centroid of $\triangle OAB$ is:

Option 1:
$\frac{7}{3}$

Option 2:
$\frac{11}{3}$

Option 3:
$\frac{8}{3}$

Option 4:
$\frac{5}{3}$

Correct Answer:
$\frac{8}{3}$

Solution:

$\mathrm{R} = 4 \Rightarrow \mathrm{R}^2 = 16 \Rightarrow 2\,\mathrm{b}^2 + 3\mathrm{a}^2 = 64$

$\mathrm{b} = \dfrac{-3\mathrm{x}}{\sqrt{2}}, \mathrm{a} = \dfrac{-3\mathrm{y}}{\sqrt{3}}$

$2\,\mathrm{b}^2 + 3\mathrm{a}^2 = 64$

$\Rightarrow \mathrm{x}^2 + \mathrm{y}^2 = \left(\dfrac{8}{3}\right)^2$

$\therefore \mathrm{r} = \dfrac{8}{3}$

Hence, the correct answer is option (3).

Q. 4 Consider an A.P : $\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3 \ldots\ldots, \mathrm{a}_\mathrm{n}; a_1 > 0$ If $a_2 - a_1 = \frac{-3}{4}, a_n = \frac{1}{4}a_1$ and $\sum_{i=1}^{n} a_i = \frac{525}{2}$ then $\sum_{i=1}^{17} a_i$ is equal to:

Option 1:
476

Option 2:
136

Option 3:
952

Option 4:
238

Correct Answer:
238

Solution:

$a_2 - a_1 = -\frac{3}{4} \Rightarrow a_1 + d - a_1 = -\frac{3}{4} \Rightarrow d = -\frac{3}{4}$

$a_n = \frac{1}{4}a_1 \Rightarrow a_1 + (n-1)\cdot\left(-\frac{3}{4}\right) = \frac{1}{4}a_1$

$\Rightarrow 4a_1 3n + 3 = a_1 \Rightarrow 3a_1 - 3n + 3 = 0 \Rightarrow a_1 - n + 1 = 0$

$\sum_{i=l}^{n} a_i = \frac{525}{2} \Rightarrow \frac{n}{2}\left[2a_1 + (n-1)\cdot\left(-\frac{3}{4}\right)\right] = \frac{525}{2} \Rightarrow \frac{n}{2}[a_1 + a_n] = \frac{525}{2}$

$8na_1 - 3n^2 + 3n = 525 \Rightarrow \frac{n}{2}\left[a_1 + \frac{a_1}{4}\right] = \frac{525}{2}$

$\Rightarrow a_n n = 420$

$\Rightarrow a_1\left[a_1 + 1\right] = 420 = 20 \times 21$

$\Rightarrow a_1 = 20$

Using the sum formula for the first 17 terms (S_{17}) :

$S_{17} = \frac{17}{2}[2a_1 + (17-1)d]$

$S_{17} = \frac{17}{2}\left[2(20) + 16\left(-\frac{3}{4}\right)\right]$

$S_{17} = \frac{17}{2}[40 - 12]$

$S_{17} = \frac{17}{2} \times 28 = 17 \times 14 = 238$

Hence, the correct answer is option (4).

Q. 5 Let $\alpha, \beta \in R$ such that the function $f(x) = \begin{cases} 2\alpha\left(x^2 - 2\right) + 2\beta x & , x < 1 \\ (\alpha + 3)x + \alpha - \beta & x \geq 1 \end{cases}$ be differentiable at all $x \in R$ then $34(\alpha + \beta) = ?$

Option 1:
24

Option 2:
84

Option 3:
48

Option 4:
36

Correct Answer:
48

Solution:
Evaluating the function from the left ($x < 1$) and the right ($x \geq 1$):
$f\left(1^{-}\right) = 2\alpha\left(1^2 - 2\right) + 2\beta(1) = -2\alpha + 2\beta$
$f\left(1^{+}\right) = (\alpha + 3)(1) + \alpha - \beta = 2\alpha - \beta + 3$

After equating them, we get,

$$-2\alpha + 2\beta = 2\alpha - \beta + 3$$
$$4\alpha - 3\beta = -3 \quad \ldots(\text{Equation } 1)$$
Differentiate the piecewise parts:
$f'(x) = 4\alpha x + 2\beta(\text{ for } x < 1)$
$f'(x) = \alpha + 3(\text{ for } x \geq 1)$
Equating the derivatives at $x = 1$:
$$4\alpha(1) + 2\beta = \alpha + 3$$
$$3\alpha + 2\beta = 3 \quad \ldots(\text{Equation } 2)$$
We have:
1. $4\alpha - 3\beta = -3$
2. $3\alpha + 2\beta = 3$
Multiply Eq (1) by 2 and Eq (2) by 3:
$8\alpha - 6\beta = -6$
$9\alpha + 6\beta = 9$
Adding the two:
$17\alpha = 3 \Rightarrow \alpha = \frac{3}{17}$
Substitute $\alpha = \frac{3}{17}$ into Eq (2):
$$3\left(\frac{3}{17}\right) + 2\beta = 3$$
$$\frac{9}{17} + 2\beta = 3 \Rightarrow 2\beta = \frac{51 - 9}{17} = \frac{42}{17}$$
$$\beta = \frac{21}{17}$$
$$\alpha + \beta = \frac{3}{17} + \frac{21}{17} = \frac{24}{17}$$
$$34(\alpha + \beta) = 34 \times \frac{24}{17} = 2 \times 24 = 48$$

Hence, the correct answer is option (3).

Q. 6 Let $f(t)=\int\left(\frac{(1-\sin(\log_e t)}{1-\cos(\log_e t)}\right)dt, t>1$. if. $f\left(e^{\pi/2}\right)=-e^{\pi/2}$ and $f\left(e^{\pi/4}\right)=\alpha e^{\pi/4}$ then α equals:

Option 1:
$-1-2\sqrt{2}$

Option 2:
$-1+\sqrt{2}$

Option 3:
$-1-\sqrt{2}$

Option 4:
$1+\sqrt{2}$

Correct Answer:
$-1-\sqrt{2}$

Solution:
Let $\cos \mathrm{t}=\mathrm{x} \Rightarrow \mathrm{t}=\mathrm{e}^{\mathrm{x}} \Rightarrow \mathrm{dt}=\mathrm{e}^{\mathrm{x}}\mathrm{dx}$

$$G.I=\int\frac{1-\sin x}{2\sin^2\frac{x}{2}}e^x dx=\frac{1}{2}\int\left(\cos e^2\frac{x}{2}-2\cot\frac{x}{2}\right)e^x dx=\frac{1}{2}e^x\left(-2\cot\frac{x}{2}\right)+c$$

$$\Rightarrow -t\cot\left(\frac{\log t}{2}\right)+c$$

$$f\left(e^{-\pi/2}\right)=-e^{\pi/2} \quad \Rightarrow c=0$$

$$\therefore d\left(e^{\pi/4}\right)=-e^{-\pi/4}(\sqrt{2}+1)$$

$$\therefore \alpha=-(\sqrt{2}+1)$$

Hence, the correct answer is option (3).

Q. 7 Let $S=\frac{1}{25!}+\frac{1}{3!23!}+\frac{1}{5!21!}+\ldots$. Up to 13 terms. If $13\ \mathrm{S}=\frac{2^k}{n!}, k\in N$ then $\mathrm{n}+\mathrm{k}$ is equal to:

Option 1:
49

Option 2:
51

Option 3:
52

Option 4:
50

Correct Answer:
49

Solution:
The given series is:

$S = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \ldots \text{ (up to 13 terms)}$
The general term can be written as $\frac{1}{r!(25-r)!}$ where r takes the values $1, 3, 5, \ldots, 25$.
There are exactly 13 such odd values between 1 and 25.
To convert these into binomial coefficients $\binom{n}{r} = \frac{n!}{r!(n-r)!}$, we multiply and divide the entire sum by 25!:

$$S = \frac{1}{25!}\left[\frac{25!}{1124!} + \frac{25!}{3122!} + \frac{25!}{5120!} + \cdots + \frac{25!}{2510!}\right]$$
$$\Rightarrow S = \frac{1}{25!}\left[\binom{25}{1} + \binom{25}{3} + \binom{25}{5} + \cdots + \binom{25}{25}\right]$$
We know from the properties of binomial coefficients that the sum of odd-indexed terms is:

$\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots = 2^{n-1}$
For $n = 25$:

$\sum \text{ Odd terms } = 2^{25-1} = 2^{24}$
Given: $S = \frac{2^k}{n!}$
From our calculation: $S = \frac{2^{24}}{25!}$
Comparing the two:

$k = 24$
$n = 25$
So, $n + k = 24 + 25 = 49$

Hence, the correct answer is option (1).

Q. 8 The number of real solutions of a the equation $x|x+3| + |x-1| - 2 = 0$ is:

Option 1:
3

Option 2:
5

Option 3:
2

Option 4:
4

Correct Answer:
3

Solution:
The given equation is $x|x+3|+|x-1|-2=0$.
The critical points are $x=-3$ and $x=1$. We solve the quadratic equations formed in each interval.

Case 1: $x<-3$
The equation becomes: $x[-(x+3)]-(x-1)-2=0$
$\Rightarrow -x^2-3x-x+1-2=0$
$\Rightarrow -x^2-4x-1=0$
$\Rightarrow x^2+4x+1=0$
$\Rightarrow x=\dfrac{-4\pm\sqrt{16-4}}{2}=-2\pm\sqrt{3}$
Comparing with $x<-3$:

$-2-\sqrt{3}\approx -3.732$ (Valid)
$-2+\sqrt{3}\approx -0.268$ (Invalid)
One solution: $x=-2-\sqrt{3}$

Case 2: $-3\le x<1$
The equation becomes: $x(x+3)-(x-1)-2=0$

$\Rightarrow x^2+3x-x+1-2=0$
$\Rightarrow x^2+2x-1=0$
$\Rightarrow x=\dfrac{-2\pm\sqrt{4+4}}{2}=-1\pm\sqrt{2}$
Comparing with $-3\le x<1$:

$-1-\sqrt{2}\approx -2.414$ (Valid)
$-1+\sqrt{2}\approx 0.414$ (Valid)
Two solutions: $x=-1-\sqrt{2}$ and $x=-1+\sqrt{2}$

Case 3: $x\ge 1$
The equation becomes: $x(x+3)+(x-1)-2=0$
$\Rightarrow x^2+3x+x-1-2=0$
$\Rightarrow x^2+4x-3=0$
$\Rightarrow x=\dfrac{-4\pm\sqrt{16+12}}{2}=-2\pm\sqrt{7}$
Comparing with $x\ge 1$:

$-2-\sqrt{7}\approx -4.645$ (Invalid)
$-2+\sqrt{7}\approx 0.645$ (Invalid)
No solutions in this interval.
Comparing all cases:
The solutions are $x=-2-\sqrt{3},-1-\sqrt{2},-1+\sqrt{2}$.
Total number of real solutions $=3$.

Hence, the correct answer is option (1).

Q. 9 Let the lines $L_1 : \bar{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}), \lambda \in R$, and $L_2 : \bar{r} = (4\hat{i} + \hat{j}) + \mu(5i + 2j + k)\mu \in R$ intersect at the point R.
Let P and Q be the points lying on the lines L_1 and L_2 respectively such that $|PR| = \sqrt{29}$ and $|PQ| = \sqrt{\frac{47}{3}}$ and $|PQ| = \sqrt{\frac{47}{3}}$ if the Point P lies in the 1st octant then $27(\mathrm{QR})^2$ is equal to:

Option 1:
340

Option 2:
348

Option 3:
360

Option 4:
320

Correct Answer:
360

Solution:
Finding point $R(-1, -1, -1)$

$P(2\lambda + 1, 3\lambda + 2, 4\lambda + 3)Q(5\mu + 4, 2\mu + 1, \mu)$

$|PQ| = \sqrt{\frac{47}{3}} \Rightarrow (5\mu + 3)^2 + (2\mu - 1)^2 + (\mu - 3)^2 = \frac{47}{3} \Rightarrow \mu = \frac{1}{3}$

$PQ = \sqrt{29} \Rightarrow (2\lambda + 2)^2 + (3\lambda + 3)^2 + (4\lambda + 4)^2 = 29$

$\therefore Q = \left(\frac{7}{3}, \frac{1}{3}, -\frac{1}{3}\right)$

$\therefore (QR)^2 = \frac{120}{9} \Rightarrow 27(QR)^2 = 27 \times \frac{120}{9} = 360$

Hence, the correct answer is option (3).

Q. 10 From a lot containing 10 defective and 90 and defective bulbs, 8 bulbs are selected one by one with replacement then the probability of getting at least 7 defective bulbs is:

Option 1:
$\frac{7}{10^7}$

Option 2:
$\frac{81}{10^8}$

Option 3:
$\frac{73}{10^8}$

Option 4:
$\frac{67}{10^8}$

Correct Answer:
$\frac{73}{10^8}$

Solution:
Required probability

$$= \mathrm{P(X=7)} + \mathrm{P(X=8)} = \mathrm{P(7 \text{ defective, } 1 \text{ fair }) + P(8 \text{ defective })}$$
$$= \frac{10^7 \times 90 \times 8 + 10^8}{100^8}$$
$$= \frac{10^8[72+1]}{(100)^8}$$
$$= \frac{73 \times 10^8}{\left(10^2\right)^8}$$
$$= \frac{73 \times 10^8}{10^{16}}$$
$$= \frac{73}{10^8}$$

Hence, the correct answer is option (3).

Q. 11 Let $\mathrm{A}(1,0)\mathrm{B}(2,-1)$ and $\mathrm{C} = \left(\frac{7}{3}, \frac{4}{3}\right)$ be three points. If the equation of bisector of the angle ABC is $\alpha x + \beta y = 5$ then the value of $\alpha^2 + \beta^2$ is:

Option 1:
8

Option 2:
5

Option 3:
10

Option 4:
13

Correct Answer:
10

Solution:
Let the coordinates be $A(1,0), B(2,-1)$, and $C\left(\frac{7}{3}, \frac{4}{3}\right)$.
Find the slope of line $AB: m_{AB} = \frac{-1-0}{2-1} = -1$
Equation of AB: $(y-0) = -1(x-1) \Rightarrow x+y-1=0$
Find the slope of line $BC: m_{BC} = \frac{4/3-(-1)}{7/3-2} = \frac{7/3}{1/3} = 7$
Equation of $BC: (y+1) = 7(x-2) \Rightarrow 7x-y-15=0$
The distance from a point (x,y) on the bisector to both lines must be equal: |
$\frac{|x+y-1|}{\sqrt{1^2+1^2}} = \frac{|7x-y-15|}{\sqrt{7^2+(-1)^2}}$
$\Rightarrow \frac{|x+y-1|}{\sqrt{2}} = \frac{|7x-y-15|}{\sqrt{50}}$
$\Rightarrow \frac{|x+y-1|}{\sqrt{2}} = \frac{|7x-y-15|}{5\sqrt{2}}$
$\Rightarrow 5|x+y-1| = |7x-y-15|$
Case (Positive)
$5(x+y-1) = 7x-y-15$
$\Rightarrow 5x+5y-5 = 7x-y-15$
$\Rightarrow 2x-6y = 10$
$\Rightarrow x-3y = 5$

Case 2 (Negative):
$5(x+y-1) = -(7x-y-15)$
$\Rightarrow 5x+5y-5 = -7x+y+15$
$\Rightarrow 12x+4y = 20$
$\Rightarrow 3x+y = 5$
The bisector of the angle ABC must pass through the vertex $B(2,-1)$.
Both equations satisfy this.
By checking the position of points A and C relative to the lines, or calculating the slopes:
$m_1 = \frac{1}{3}$ and $m_2 = -3$.
The product of slopes of AB and BC is $(-1)(7) = -7$.
The angle bisector of $\angle ABC$ is $3x+y=5$.
Comparing with the given form $\alpha x+\beta y = 5: \alpha = 3\beta = 1$
Calculating $\alpha^2+\beta^2: \alpha^2+\beta^2 = 3^2+1^2 \Rightarrow 9+1=10$

Hence, the correct answer is option (3).

Q. 12 The value of $\frac{\sqrt{3}\operatorname{cosec} 20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to:

Option 1:
16

Option 2:
12

Option 3:
64

Option 4:
32

Correct Answer:
64

Solution:
$$\frac{2\left(\frac{\sqrt{3}}{2}\cos 20^\circ-\frac{1}{2}\sec 20^\circ\right)}{\frac{1}{2}\times\frac{1}{4}\times\frac{1}{2}}=64$$

Hence, the correct answer is option (3).

Q. 13 If the Domain of the function $f(x)=\log_{(10x^2-17x+7)}\left(18x^2-11x+1\right)$ is $(-\infty,a)\cup(b,c)\cup(d,\infty)-\{e\}$ then $90(\mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}+\mathrm{e})$ equals:

Option 1:
307

Option 2:
177

Option 3:
316

Option 4:
170

Correct Answer:
316

Solution:
For the function $f(x)=\log_{g(x)}h(x)$ to be defined, the following conditions must be met:
1. $h(x)>0$ (Argument must be positive)
2. $g(x)>0$ (Base must be positive)
3. $g(x)\neq 1$ (Base cannot be equal to 1)

Condition 1: $18x^2-11x+1>0$
$\Rightarrow 18x^2-9x-2x+1>0$
$\Rightarrow 9x(2x-1)-1(2x-1)>0$
$\Rightarrow (2x-1)(9x-1)>0$
Roots are $x=\frac{1}{2}$ and $x=\frac{1}{9}$.
By wavy curve method: $x\in\left(-\infty,\frac{1}{9}\right)\cup\left(\frac{1}{2},\infty\right)$... (i)

Condition 2: $10x^2 - 17x + 7 > 0$
$\Rightarrow 10x^2 - 10x - 7x + 7 > 0$
$\Rightarrow 10x(x-1) - 7(x-1) > 0$
$\Rightarrow (10x-7)(x-1) > 0$
Roots are $x = \frac{7}{10}$ and $x = 1$.
By wavy curve method: $x \in \left(-\infty, \frac{7}{10}\right) \cup (1, \infty)$ . . .. (ii)

Condition 3: $10x^2 - 17x + 7 \neq 1$
$\Rightarrow 10x^2 - 17x + 6 \neq 0$
$\Rightarrow 2x(5x-6) - 1(5x-6) \neq 0$
$\Rightarrow (2x-1)(5x-6) \neq 0$
$\Rightarrow x \neq \frac{1}{2}$ and $x \neq \frac{6}{5} \ldots$ (iii)
Intersection of (i), (ii), and (fii):
Intervals are $\left(-\infty, \frac{1}{9}\right), \left(\frac{1}{2}, \frac{7}{10}\right), (1, \infty)$.
From (iii), we must exclude $x = \frac{6}{5}$ (Note: $\frac{1}{2}$ is already an endpoint, so it is naturally excluded).
Domain: $\left(-\infty, \frac{1}{9}\right) \cup \left(\frac{1}{2}, \frac{7}{10}\right) \cup (1, \infty) - \left\{\frac{6}{5}\right\}$.
Comparing with $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$:
$a = \frac{1}{9}$
$b = \frac{1}{2}$
$c = \frac{7}{10}$
$d = 1$
$e = \frac{6}{5}$
So,

$$\text{Sum} = \frac{1}{9} + \frac{1}{2} + \frac{7}{10} + 1 + \frac{6}{5}$$
$$\text{Sum} = \frac{10 + 45 + 63 + 90 + 108}{90} = \frac{316}{90}$$
$$90 \times \left(\frac{316}{90}\right) = 316$$

Hence, the correct answer is option (3).

Q. 14 The mean and variance of data of 10 observations are 10 and 2 respectively. If an observation α in this data is replaced by β, then the mean and variance become 10.1 and 1.99 respectively then $\alpha + \beta$ equals:

Option 1:
10

Option 2:
15

Option 3:
20

Option 4:
5

Correct Answer:
20

Solution:
Let be data $\mathrm{x}_1, \mathrm{x}_2, \ldots\ldots\ldots\ldots \mathrm{x}_\mathrm{a}, \alpha$

$$\sum_{i=1}^{9} x_i = 100 - \alpha$$
$$\sum x_i 2 = 1020 - \alpha^2$$
$$100 - \alpha + \beta = 101\alpha - \beta + 1$$
$$\frac{\sum_{i=1}^{9} xi^2 + \beta^2}{10} - (101)^2 = 1.99$$
$$\beta^2 \cdot \alpha^2 = 20 \Rightarrow \alpha = \frac{19}{2}, \beta = \frac{21}{2}$$
$$\alpha + \beta = 20$$

Hence, the correct answer is option (3).

Q. 15 If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $\sin 7x \left(\cos \frac{13x}{2} + \sin \frac{13x}{2}\right) + \cos 7x \left(\cos \frac{13x}{2} - \sin \frac{13x}{2}\right)$ is equal to:

Option 1:
$\frac{1}{\sqrt{13}}$

Option 2:
$\frac{4}{\sqrt{26}}$

Option 3:
$\frac{5}{\sqrt{13}}$

Option 4:
$\frac{6}{\sqrt{26}}$

Correct Answer:
$\frac{1}{\sqrt{13}}$

Solution:
Let the given expression be E.

$E = \sin 7x \cos \frac{13x}{2} + \sin 7x \sin \frac{13x}{2} + \cos 7x \cos \frac{13x}{2} - \cos 7x \sin \frac{13x}{2}$

$E = \left(\cos 7x \cos \frac{13x}{2} + \sin 7x \sin \frac{13x}{2}\right) - \left(\cos 7x \sin \frac{13x}{2} - \sin 7x \cos \frac{13x}{2}\right)$

Using the identities:

$\cos A \cos B + \sin A \sin B = \cos(A - B)$

$\sin A \cos B - \cos A \sin B = \sin(A - B)$

Here, $A = 7x$ and $B = \frac{13x}{2}$:

$$\cos\left(7x - \frac{13x}{2}\right) = \cos\left(\frac{x}{2}\right)$$

$$\sin\left(\frac{13x}{2} - 7x\right) = \sin\left(-\frac{x}{2}\right) = -\sin\left(\frac{x}{2}\right)$$

$\left(\cos 7x \sin \frac{13x}{2} - \sin 7x \cos \frac{13x}{2}\right) = \sin\left(\frac{13x}{2} - 7x\right) = \sin\left(-\frac{x}{2}\right) = -\sin\frac{x}{2}$

So, the expression becomes:

$E = \cos\left(\frac{x}{2}\right) - \left(-\sin\frac{x}{2}\right) = \cos\frac{x}{2} + \sin\frac{x}{2}$

Given $\cot x = \frac{5}{12}$ and $x \in \left(\pi, \frac{3\pi}{2}\right)$ (3rd Quadrant).

In the 3rd quadrant, $\sin x$ and $\cos x$ are negative.

Using the Pythagorean triplet (5, 12, 13):

$\cos x = -\frac{5}{13}$

We use the half-angle formulae:

$$\cos^2\frac{x}{2} = \frac{1+\cos x}{2} = \frac{1-5/13}{2} = \frac{8}{26} = \frac{4}{13} \Rightarrow \cos\frac{x}{2} = \pm\frac{2}{\sqrt{13}}$$

$$\sin^2\frac{x}{2} = \frac{1-\cos x}{2} = \frac{1+5/13}{2} = \frac{18}{26} = \frac{9}{13} \Rightarrow \sin\frac{x}{2} = \pm\frac{3}{\sqrt{13}}$$

Since $\pi < x < \frac{3\pi}{2}$, then $\frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4}$.

In the 2nd quadrant:

$$\sin\frac{x}{2} \text{ is positive } \Rightarrow \sin\frac{x}{2} = \frac{3}{\sqrt{13}}$$

$$\cos\frac{x}{2} \text{ is negative } \Rightarrow \cos\frac{x}{2} = -\frac{2}{\sqrt{13}}$$

$$E = \cos\frac{x}{2} + \sin\frac{x}{2}$$

$$\Rightarrow E = -\frac{2}{\sqrt{13}} + \frac{3}{\sqrt{13}}$$

$$\Rightarrow E = \frac{1}{\sqrt{13}}$$

Hence, the correct answer is option (1).

Q. 16 Let each of the two ellipses $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1 (A < B)$ have eccentricity $\frac{4}{5}$. Let the Length of the latus rectum of E_1 and E_2 be $_1, _2$ respectively such that $2_1^2 = 9_2$. If the distance between foci of E_1 is 8, then the difference between foci of E_2 is:

Option 1:
$\frac{8}{5}$

Option 2:
$\frac{16}{5}$

Option 3:
$\frac{32}{5}$

Option 4:
18

Correct Answer:
$\frac{32}{5}$

Solution:
Ellipse E_1 (Horizontal Ellipse, $a > b$):
Equation: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
Given eccentricity $e_1 = \frac{4}{5}$ and distance between foci $2ae_1 = 8$.

$$\Rightarrow 2a\left(\frac{4}{5}\right) = 8$$

$$\Rightarrow a = 5$$

Now, calculate b^2 using $b^2 = a^2\left(1 - e^2\right)$:

$$\Rightarrow b^2 = 25\left(1 - \frac{16}{25}\right) = 25\left(\frac{9}{25}\right) = 9$$

Length of latus rectum $(l_1) = \frac{2b^2}{a}$:

$$\Rightarrow l_1 = \frac{2(9)}{5} = \frac{18}{5}$$

Ellipse E_2 (Vertical Ellipse, $A < B$):
Equation: $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$
Given eccentricity $e_2 = \frac{4}{5}$. Since $B > A, A^2 = B^2\left(1 - e_2^2\right)$.

$$\Rightarrow A^2 = B^2\left(1 - \frac{16}{25}\right) = \frac{9B^2}{25}$$

Length of latus rectum $(l_2) = \frac{2A^2}{B}$:

$$\Rightarrow l_2 = \frac{2\left(9B^2/25\right)}{B} = \frac{18B}{25}$$

Use the given relation $2l_1^2 = 9l_2$:

$$\Rightarrow 2\left(\frac{18}{5}\right)^2 = 9\left(\frac{18B}{25}\right)$$
$$\Rightarrow 2\left(\frac{324}{25}\right) = \frac{162B}{25}$$
$$\Rightarrow \frac{648}{25} = \frac{162B}{25}$$
$$\Rightarrow 162B = 648 \Rightarrow B = 4$$

For a vertical ellipse, the distance between foci is $2Be_2$.

$\Rightarrow 2 \times 4 \times \frac{4}{5} = \frac{32}{5}$

Hence, the correct answer is option (3).

Q. 17 Let R be relation defined on the set $\{1,2,3,4\} \times \{1,2,3,4\}$ by $\mathrm{R} = \{((\mathrm{a},\mathrm{b})(\mathrm{c},\mathrm{d})) : 2\mathrm{a} + 3\ \mathrm{b} = 3\mathrm{c} + 4\ \mathrm{d}\}$ then the number of elements in R is:

Option 1:
6

Option 2:
15

Option 3:
12

Option 4:
13

Correct Answer:
12

Solution:
$(\mathrm{ab}) \rightarrow (\mathrm{c},\mathrm{d})$

$(1,1) \rightarrow$ No values
$(1,2) \rightarrow$ No values
$(1,3) \rightarrow (1,2)$
$(1,4) \rightarrow (2,2)$
$(2,1) \rightarrow (1,1)$
$(2,2) \rightarrow (2,1)$
$(2,3) \rightarrow (3,1)$
$(2,4) \rightarrow (4,1)$
$(3,1) \rightarrow$ No values
$(3,2) \rightarrow$ No values

$(3,3) \rightarrow (1,3)$
$(3,4) \rightarrow (2,3)$
$(4,1) \rightarrow (1,2)$
$(4,2) \rightarrow (2,2)$
$(4,3) \rightarrow (3,2)$
$(4,4) \rightarrow (2,2)$

Number of elements $= 12$

Hence, the correct answer is option (3).

Q. 18 Let $729, 81, 9, 1 \ldots\ldots$ be a sequence and $\mathrm{P_n}$ denote the product of the first n terms of the sequence. If $2\sum_{n=1}^{40}(P_n)^{\frac{1}{n}} = \frac{3^{\alpha}-1}{3^{\beta}}$ and $\gcd$ of $(\alpha, \beta) = 1$ then $\alpha + \beta$ is equal to:

Option 1:
79

Option 2:
73

Option 3:
76

Option 4:
75

Correct Answer:
73

Solution:
The given sequence is $729, 81, 9, 1, \ldots$
This is a Geometric Progression (G.P.) where:
First term $(a) = 729 = 3^6$
Common ratio $(r) = \frac{81}{729} = \frac{1}{9} = 3^{-2}$

The nth term $a_n = a \cdot r^{n-1} = 3^6 \cdot \left(3^{-2}\right)^{n-1} = 3^{6-2n+2} = 3^{8-2n}$.

$P_n = a_1 \cdot a_2 \cdot a_3 \cdot \cdots\cdots a_n$
$P_n = 3^6 \cdot 3^4 \cdot 3^2 \cdots\cdots 3^{8-2n}$
$P_n = 3^{(6+4+2+\cdots+(8-2n))}$
The exponent is an Arithmetic Progression with n terms.
Sum of exponent $= \frac{n}{2}[2(6) + (n-1)(-2)] = \frac{n}{2}[12 - 2n + 2] = \frac{n}{2}[14 - 2n] = n(7-n)$.
So, $P_n = 3^{n(7-n)}$.

$(P_n)^{1/n} = \left(3^{n(7-n)}\right)^{1/n} = 3^{7-n}$.
Let $S = \sum_{n=1}^{40} (P_n)^{1/n} = \sum_{n=1}^{40} 3^{7-n}$
This is a G.P. with 40 terms: $3^6, 3^5, 3^4, \ldots, 3^{-33}$.
First term $(A) = 3^6$, Common ratio $(R) = \frac{1}{3}$, Number of terms $= 40$.

$$S = \frac{3^6\left[1-(1/3)^{40}\right]}{1-1/3} = \frac{3^6\left[1-3^{-40}\right]}{2/3} = \frac{3}{2}\cdot 3^6 \cdot \frac{3^{60}-1}{3^{60}}$$

$$S = \frac{3^7\left(3^{60}-1\right)}{2\cdot 3^{30}} = \frac{3^{40}-1}{2\cdot 3^{30}}.$$

Given: $2\sum_{n=1}^{40} (P_n)^{1/n} = \frac{3^a-1}{3^s}$

$2\left[\frac{3^{60}-1}{2\cdot 3^{33}}\right] = \frac{3^{60}-1}{3^{33}}$
Comparing this with $\frac{3^\alpha-1}{3^\beta}$:

$\alpha = 40$
$\beta = 33$

$\alpha + \beta = 40 + 33 = 73.$

Hence, the correct answer is option (2).

Q. 19 Let A_1 be the bounded area enclosed by the curves $y = x^2 + 2, x + y = 8$ and the y-axis that lies in the first quadrant. Let A_2 be bonded area enclosed by the curves $y = x^2 + 2, y^2 = x, x = 2$ and y axis that lies in the first quadrant then $\mathrm{A}_1 - \mathrm{A}_2$ is equal to:

Option 1:
$\frac{2}{3}(2\sqrt{2}+1)$

Option 2:
$\frac{2}{3}(\sqrt{2}+1)$

Option 3:
$\frac{2}{3}(3\sqrt{2}+1)$

Option 4:
$\frac{2}{3}(4\sqrt{2}+1)$

Correct Answer:
$\frac{2}{3}(2\sqrt{2}+1)$

Solution:

$y = x^2 + 2$

$8 - x = x^2 + 2$

$\Rightarrow x^2 + x - 6 = 0 \quad \Rightarrow (x-2)(x+3) = 0 \quad \Rightarrow x = 2, x = -3$

$A_1 = \int_0^2 \left[(8-x)\left(x^2+2\right)\right] dx$

$= \int_0^2 \left(6 - x - x^2\right) dx = \left[6x - \frac{x^2}{2} - \frac{x^3}{3}\right]_0^2 = \left[\left(12 - 2 - \frac{8}{3}\right) - 0\right] \quad = \frac{22}{3}$

$A_2 \int_0^2 \left(x^2 + 2 - \sqrt{x}\right)$

$= \left[\frac{x^3}{3} + 2x - \frac{x^{\frac{3}{3}}}{\frac{3}{2}}\right]^2 \quad = \left[\left(\frac{8}{3} + 4 - \frac{2}{3}2\sqrt{2}\right) - 0\right] = \frac{20}{3} - \frac{2}{3}2\sqrt{2}$

Hence, the correct answer is option (1).

Q. 20 Let $\bar{a} = 2\hat{i} + \hat{j} - 2\hat{k}, \bar{b} = \hat{i} + \hat{j}$ and $\bar{c} = \bar{a} \times \bar{b}$. Let $\bar{d}$ be a vector such that $|\bar{d} - \bar{a}| = \sqrt{11}, |\bar{c} \times \bar{d}| = 3$ and angle between $\bar{c}$ and $\bar{d}$ is $\frac{\pi}{4}$. Then $\bar{a}.\bar{d}$ is equal to:

Option 1:
11

Option 2:
0

Option 3:
1

Option 4:
3

Correct Answer:
0

Solution:

$$\varepsilon = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i}(2) - \hat{j}(2) + \hat{k}(1)$$

$|\bar{c} \times \bar{d}| = 3 \Rightarrow |\bar{c}||\bar{d}| \sin(\bar{c}, \bar{d}) = 3 \Rightarrow 3|\bar{d}|\frac{1}{\sqrt{2}} = 3$

$|\bar{d}| = \sqrt{2}$

$|\bar{d} - \bar{a}| = \sqrt{11} \Rightarrow \bar{d}^2 + \bar{a}^2 - 2\bar{a} \cdot \bar{d} = 1\bar{1} \Rightarrow \bar{d}^2 + 9 - 2\bar{a} \cdot \bar{d} = 11 \Rightarrow \bar{a} \cdot \bar{d} = 0$

Hence, the correct answer is option (2).

Q. 21 The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is___

Correct Answer:
42

Solution:
The number N must satisfy $5000 < N < 9000$. Using the digits $\{0, 1, 2, 5, 9\}$:
The thousands digit d_1 must be 5 or 9.
If $d_1 = 9$, the smallest number is 9000, which is not less than 9000.
Therefore, the thousands digit is fixed as $d_1 = 5$.
The number is of the form $5d_2d_3d_4$.
A number is divisible by 3 if the sum of its digits is a multiple of 3.

$\text{Sum} = 5 + d_2 + d_3 + d_4$
For the sum to be divisible by $3, d_2 + d_3 + d_4$ must leave a remainder of 1 when divided by 3 (since $5 \equiv 2(\bmod 3)$).

Remainder 0 (Type 0): $\{0, 9\}$ - 2 digits
Remainder 1 (Type 1): \{1\} - 1 digit
Remainder 2 (Type 2): $\{2, 5\}$ - 2 digits
We need the sum of the remainders of d_2, d_3, d_4 to be $1(\bmod 3)$ (i.e., $1, 4, 7$).

Case	Remainder Set	Ways to Arrange	Calculation	Total
1	$(0, 0, 1)$	$\frac{3!}{2!}$	$3 \times (2 \times 2 \times 1)$	12
2	$(1, 1, 2)$	$\frac{3!}{2!}$	$3 \times (1 \times 1 \times 2)$	6
3	$(2, 2, 0)$	$\frac{3!}{2!}$	$3 \times (2 \times 2 \times 2)$	24

Total numbers $= 12 + 6 + 24 = 42$.
Since the repetition is allowed, all numbers formed (like 5001) are strictly greater than 5000 and less than 9000.

Hence, the correct answer is 42.

Q. 22 Let a differentiable function f satisfy the equation $\int_0^{36} \mathrm{f}\left(\frac{\mathrm{tx}}{36}\right)\mathrm{dt} = 4\alpha\mathrm{f}(\mathrm{x})$. If $y = f(x)$ is a standard parabola passing through the points $(2, 1)$ and $(-4, \beta)$ then β^{α} is equal to _____

Correct Answer:
64

Solution:

$$\int_0^{36} f\left(\frac{tx}{36}\right)dt \cdot \frac{tx}{36} = ydt = \frac{36}{x}dy$$

$$\therefore \int_0^x \frac{f(y)36dy}{x} = 4\alpha + f(x)$$

$$\int_0^x f(y)dy = \frac{\alpha \cdot x \cdot f(x)}{9} \Rightarrow f(x) = \frac{\alpha}{9}(f + xfi)$$

$$\frac{f^1(x)}{f(x)} = \left(\frac{9}{\alpha} - 1\right)\frac{1}{x} \Rightarrow f(x) = \mathrm{cx}^{\left(\frac{9}{\alpha}-1\right)}(2,1)$$

$$\alpha = 3, c = \frac{1}{4}\left[\therefore \frac{9}{\alpha} - 1 = 2\right]$$

$$y = \frac{x^2}{4}\hat{i}(-4, \beta) \text{ parabola } \Rightarrow \beta = 4$$

$$\beta^{\alpha} = 64$$

Hence, the correct answer is 64.

Q. 23 The number of 3×2 matrices A . which can be formed using the elements of the set $\{-2, -1, 0, 1, 2\}$ such that the sum of all the diagonal elements of $\mathrm{A}^{\mathrm{T}}\mathrm{A}$ is 5, is _____

Correct Answer:
312

Solution:
Let the 3×2 matrix A be:

$$A = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$$

Then the transpose A^T is a 2×3 matrix:

$$A^T = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}$$

The product A^TA is a 2×2 matrix:

$$A^TA = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix} \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = \begin{bmatrix} a^2 + c^2 + e^2 & ab + cd + ef \\ ab + cd + ef & b^2 + d^2 + f^2 \end{bmatrix}$$

The sum of all diagonal elements (Trace) of A^TA is:

$$\text{Sum} = \left(a^2 + c^2 + e^2\right) + \left(b^2 + d^2 + f^2\right)$$

We are given that this sum is 5.
So, $a^2 + b^2 + c^2 + d^2 + e^2 + f^2 = 5$, where each element $\in \{-2, -1, 0, 1, 2\}$.
Step 3: Find possible values for the elements
The squares of the elements in the set $\{-2, -1, 0, 1, 2\}$ are $\{4, 1, 0, 1, 4\}$.
Basically, the squares can be 0, 1, or 4.
We need to solve $x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2 = 5$.

Possible combinations of squares:

Case 1: One '4' and one '1' (and four '0's)
Example: $4+1+0+0+0+0=5$
Ways to choose the position for '4': ${}^6C_1 = 6$
Ways to choose the element whose square is $4 : \{-2, 2\} \rightarrow 2$ choices.
Ways to choose the position for ' 1: ${}^5C_1 = 5$
Ways to choose the element whose square is $1 : \{-1, 1\} \rightarrow 2$ choices.
Total for Case 1: $6 \times 2 \times 5 \times 2 = 120$

Case 2: Five '1's (and one '0')
Example: $1+1+1+1+1+0=5$
Ways to choose the position for ' 0 ': ${}^6C_1 = 6$
Ways to choose the element whose square is $0 : \{0\} \rightarrow 1$ choice.
The remaining 5 positions must be '1's.
For each of the 5 positions, we have 2 choices $(\{-1, 1\})$.
Total for Case 2: $6 \times 1 \times 2^5 = 6 \times 32 = 192$
Total $= 120 + 192 = 312$.

Hence, the correct answer is 312.

Q. 24 Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t) = \frac{|t+1|}{t^2}, t < 0$ is strictly decreasing. Then the local maximum value of the function $g(x) = 2\log_e(x-2) + ax^2 + 4x - \alpha, x > 2$, is _____

Correct Answer:
4

Solution:
Analyze the function $f(t)$ for $t < 0$
The function is $f(t) = \frac{|t+1|}{t^2}$. Since $t < 0$, we check the behavior around the critical point of the modulus, $t = -1$.

Case 1: $-1 \leq t < 0$

$$f(t) = \frac{t+1}{t^2} = \frac{1}{t} + \frac{1}{t^2}$$

$$f'(t) = -\frac{1}{t^2} - \frac{2}{t^3} = \frac{-(t+2)}{t^3}$$

For $-1 \leq t < 0, t+2$ is positive, and t^3 is negative. Thus, $f'(t) > 0$ (Increasing).

Case 2: $t < -1$

$$f(t) = \frac{-(t+1)}{t^2} = -\frac{1}{t} - \frac{1}{t^2}$$

$$f'(t) = \frac{1}{t^2} + \frac{2}{t^3} = \frac{t+2}{t^3}$$

For the function to be strictly decreasing, $f'(t) < 0$:

$\frac{t+2}{t^3} < 0 \Rightarrow t \in (-2, -1)$
The largest interval is $(-2, -1)$. Comparing this with $(2\alpha, \alpha)$, we get:

$\alpha = -1$
Substitute $\alpha = -1$ into $g(x)$:
$g(x) = 2\ln(x-2) + ax^2 + 4x + 1$

$g(x) = 2\ln(x-2) - x^2 + 4x + 1, \quad x > 2$

$g'(x) = \frac{2}{x-2} - 2x + 4$
Set $g'(x) = 0$:

$$\frac{2}{x-2} - 2(x-2) = 0$$
$$\frac{2 - 2(x-2)^2}{x-2} = 0 \Rightarrow (x-2)^2 = 1$$
$x - 2 = 1$ or $x - 2 = -1$
$x = 3$ or $x = 1$
Since $x > 2$, we take $x = 3$.
$g''(x) = -\frac{2}{(x-2)^2} - 2$
$g''(3) = -2 - 2 = -4 < 0.$
Thus, $x = 3$ is a point of local maximum.

$g(3) = 2\ln(3-2) - (3)^2 + 4(3) + 1$
$g(3) = 2\ln(1) - 9 + 12 + 1$
$g(3) = 0 + 4 = 4$

Hence, the correct answer is 4.

Q. 25 Let a line L passing through the point $\mathrm{P}(1, 1, 1)$ be perpendicular to the lines $\frac{x-4}{4} = \frac{y-1}{1} = \frac{z-1}{1}$ and $\frac{x-17}{1} = \frac{y-71}{1} = \frac{z}{0}$. Let the line L intersect the $yz-$ plane at the point Q. Another line parallel to L and passing through the point $\mathrm{S}(1, 0, -1)$ intersects the yz plane at point R. Then the square of the area of the parallelogram PQRS is equal to _____

Correct Answer:
6

Solution:
Line L is perpendicular to lines L_1 and L_2.
Direction vector of L_1 is $b_1 = 4\hat{i} + \hat{j} + \hat{k}$
Direction vector of L_2 is $b_2 = \hat{i} + \hat{j} + 0\hat{k}$

The direction vector v of line L is given by $b_1 \times b_2$:

$$v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i}(0-1) - \hat{j}(0-1) + \hat{k}(4-1) = -\hat{i} + \hat{j} + 3\hat{k}$$

Direction ratios are $(-1, 1, 3)$.

Equation of line L passing through $P(1, 1, 1)$:

$\frac{x-1}{-1} = \frac{y-1}{1} = \frac{z-1}{3} = \lambda$

For intersection with the yz-plane, $x = 0$:

$\frac{0-1}{-1} = \lambda \Rightarrow \lambda = 1$

Coordinates of Q : $y = 1 + 1 = 2, z = 1 + 3 = 4$.

So, $Q(0, 2, 4)$.

The second line passes through $S(1, 0, -1)$ and is parallel to L :

$\frac{x-1}{-1} = \frac{y-0}{1} = \frac{z+1}{3} = \mu$

For intersection with the yz-plane, $x = 0$:

$\frac{0-1}{-1} = \mu \Rightarrow \mu = 1$

Coordinates of R : $y = 1, z = -1 + 3 = 2$. So, $R(0, 1, 2)$.

The area of a parallelogram with adjacent sides PQ and PS is $|PQ \times PS|$.

$PQ = Q - P = (0-1, 2-1, 4-1) = -\hat{i} + \hat{j} + 3\hat{k}$

$PS = S - P = (1-1, 0-1, -1-1) = 0\hat{i} - \hat{j} - 2\hat{k}$

$$PQ \times PS = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 3 \\ 0 & -1 & -2 \end{vmatrix} = \hat{i}(-2-(-3)) - \hat{j}(2-0) + \hat{k}(1-0) = \hat{i} - 2\hat{j} + \hat{k}$$

Area $= \sqrt{1^2 + (-2)^2 + 1^2} = \sqrt{6}$

$(\text{ Area })^2 = (\sqrt{6})^2 = 6$

Hence, the square of the area of the parallelogram PQRS is 6.

Physics

Q. 1 A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30 . Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is _____

Option 1:
$\sqrt{3}mg$

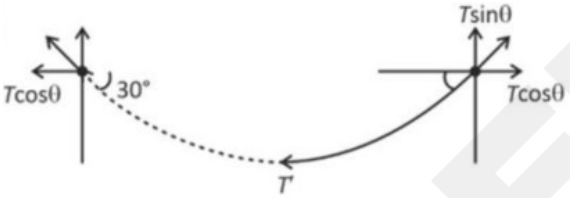
Option 2:
mg

Option 3:
$\frac{\sqrt{3}}{2}mg$

Option 4:
$\frac{1}{2}mg$

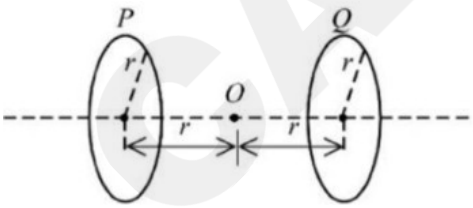
Correct Answer:
$\frac{\sqrt{3}}{2}mg$

Solution:



$$T\frac{\sqrt{3}}{2} = T^1$$

$$T^1 = \frac{mg\sqrt{3}}{2}$$



Q. 2 Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O. The net magnetic field at O is:

Option 1:
$\frac{3\mu_O I}{4\sqrt{2}r}$ towards P

Option 2:
$\frac{3\mu_O I}{4\sqrt{2}r}$ towards Q

Option 3:
$\frac{\mu_O I}{4\sqrt{2}r}$ towards Q

Option 4:
$\frac{\mu_O I}{4\sqrt{2}r}$ towards P

Correct Answer:
$\frac{3\mu_O I}{4\sqrt{2}r}$ towards Q

Solution:

$$B_P = \frac{\mu_o i r^2}{2\,(r^2+x^2)3/2} = \frac{\mu_o i}{4\sqrt{2}r} \text{ toward } P$$

$$B_Q = \frac{\mu_o 4 i r^2}{2\,(r^2+x^2)3/2} = \frac{\mu_o 4 i}{4\sqrt{2}r} \text{ toward } Q$$

$$B_{\text{Net}} = \frac{\mu_o 3 i}{4\sqrt{2}r} \text{ toward } Q$$

Q. 3 A cubical block of density $\rho_b = 600\ \text{kg/m}^3$ floats in a liquid of density $\rho_e = 900\ \text{kg/m}^3$. If the height of block is $H = 8.0\ \text{cm}$ then height of the submerged part is _____ cm .

Option 1:
4.3

Option 2:
5.3

Option 3:
6.3

Option 4:
7.3

Correct Answer:
5.3

Solution:

$F_B = mg$

$\rho_e V i g = \rho_o \times V_{\text{total}} \times 8$

$\rho_e \times A \times hi = \rho_o \times A \times h_o$

$$hi = \frac{\rho_o}{\rho_e} \times h_o = \frac{900}{600} \times 8\ \text{cm} = 5.3\ \text{cm}$$

Q. 4 In the Young's double slit experiment the intensity produced by each one of the individual slits is I_O. The distance between two slits is 2 mm . The distance of screen from slits is 10 m . The wavelength of light is $6000\AA$. The intensity of light on the screen in front of one of the slits is ____ .

Option 1:
$2I_O$

Option 2:
I_O

Option 3:
$\frac{I_O}{2}$

Option 4:
$4I_O$

Correct Answer:
I_O

Solution:

$$I = 4I_o \cos^2 \frac{\Delta\phi}{2}$$

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$$

$$\Delta x = \frac{dy}{D} = \frac{2mm \times 1mm \times 10^{-6}}{10}$$

$$\Delta\phi = \frac{2\pi}{3}$$

$$I = 4I_o \cos^2\left(\frac{2\pi}{3} \times \frac{1}{2}\right) = I_o$$

Q. 5 The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is ____ .

Option 1:
1

Option 2:
3

Option 3:
4

Option 4:
2

Correct Answer:
2

Solution:

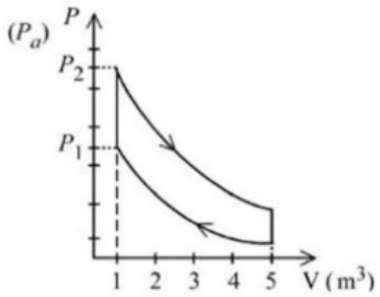
5th harmonic in closed $= \dfrac{5V}{4l_1}$

1st harmonic in open $= \dfrac{V}{2l_2}$

$\dfrac{5V}{24l_1} = \dfrac{V}{xl_2}\dfrac{l_1}{l_2} = \dfrac{5}{2}$

$x = 2$

Q. 6 10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7$ Pa and $P_2 = 30$ Pa, $\mathrm{C}_V = 21$ J/K · mol, R = 8.3 J/mol · K). The value of α is ____



Option 1:
21

Option 2:
24

Option 3:
28

Option 4:
15

Correct Answer:
21

Solution:

$Q = \Delta U + \Delta W$

$Q = \Delta U$

$= nC_V\Delta T \times \frac{R}{R} = \frac{C_v}{R}[P_2V_2 - P_1V_1] = \frac{21}{8-3}[30 \times 1 - 21.3 \times 1] = 21$

Q. 7 A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

Option 1:
decrease by 8 times

Option 2:
increase by 64 times

Option 3:
increase by 8 times

Option 4:
decrease by 4 times

Correct Answer:
decrease by 4 times

Solution:

$$\text{Volume (New)} = \frac{4}{3}\pi r^{13}$$

$$\text{Volume (old)} = \frac{4}{3}\pi r^{3}$$

$$\text{New} = 8 \times \text{old}$$

$$\frac{4}{3}\pi r^{13} = 8 \times \frac{4}{3}\pi r^{3}$$

$$r! = 2r$$

$$\frac{I_{\text{old}} = \frac{p}{4\pi r^2}}{I_{\text{new}} = \frac{p}{4\pi(2r)^2}}$$

$$\frac{I_{\text{old}}}{I_{\text{new}}} = 4$$

$$I_{\text{new}} = \frac{I_{\text{old}}}{4}$$

Q. 8 A regular hexagon is formed by six wires each of resistance $r\Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

Option 1:
$\frac{3}{5}r$

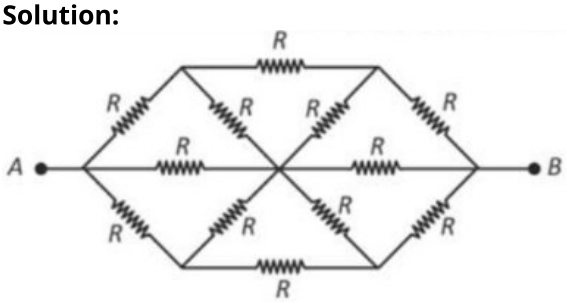
Option 2:
$\frac{4}{5}r$

Option 3:
$\frac{3}{4}r$

Option 4:
$\frac{5}{8}r$

Correct Answer:
$\frac{4}{5}r$

Solution:



Resistance b/w $A\&B$

$R_{AB} = \frac{4r}{5}$

Q. 9 Distance between an object and three times magnified real image is 40 cm . The focal length of the mirror used is _____ cm .

Option 1:
-10

Option 2:
-15

Option 3:
-20

Option 4:
-15/2

Correct Answer:
-15

Solution:

$m = 3 = \frac{v}{u} = \frac{y}{x}$

$y = 3x$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{-y} + \frac{1}{-x} = \frac{1}{f}$

$\frac{1}{-3x} + \frac{1}{-x} = \frac{1}{f}$

$\frac{-4}{3x} = \frac{1}{f} \quad f = \frac{-3x}{4} = \frac{-3}{4}(20) = -15\text{cms}$

Q. 10 The binding energy for the following nuclear reactions are expressed in MeV .

$$2\mathrm{He}^3 + {}_0\mathrm{n}^1 \rightarrow {}_2\mathrm{He}^4 + 20\mathrm{MeV}$$
$$2\mathrm{He}^4 + {}_0\mathrm{n}^1 \rightarrow {}_2\mathrm{He}^5 - 0.9\mathrm{MeV}$$

If $\mathrm{X}_3, \mathrm{X}_4, \mathrm{X}_5$ denote the stability of ${}_2\mathrm{He}^3$, ${}_2\mathrm{He}^4$, and ${}_2\mathrm{He}^5$, respectively, then the order is:

Option 1:
$X_4 < X_5 < X_3$

Option 2:
$X_4 = X_5 = X_3$

Option 3:
$X_4 > X_5 > X_3$

Option 4:
$X_4 > X_5 < X_3$

Correct Answer:
$X_4 > X_5 > X_3$

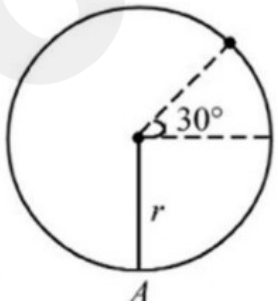
Solution:
${}_2\mu e^3 + {}_0n^1 \rightarrow 2\mu e^4 + 20Mev$

Energy is released, Hence ${}^4_2\mu e$ is more stable than ${}^3_2\mu e$

$2\mu e^4 + {}_0n^1 \rightarrow 2\mu e^5 - 0.9Mev$

Energy is obseryed < hence ${}^4_2\mu e$ is more stable than ${}^5_2\mu e X_4 > X_5 > X_3$
${}^5_2\mu$ is more stable composed to ${}^3_2\mathrm{He}$ as more energy is released in first reason than the energy Absorbed in 2 nd reaction.

Q. 11 In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30 shown in figure, the velocity at the bottom point (A) of the circular path is ($g =$ gravitational acceleration)



Option 1:
$\sqrt{\frac{7}{2}gr}$

Option 2:
$\sqrt{\frac{5}{2}gr}$

Option 3:
$\sqrt{4gr}$

Option 4:
$\sqrt{5gr}$

Correct Answer:
$\sqrt{\frac{7}{2}gr}$

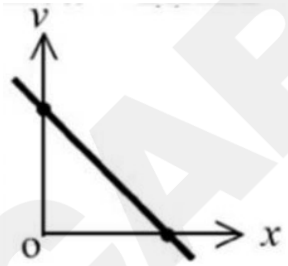
Solution:
Let ' u ' be the velocity at ' A '

$$T + mg\cos bo = \frac{mv^2}{r}$$
$$T = 0$$
$$V^2 = \frac{rg}{2}$$

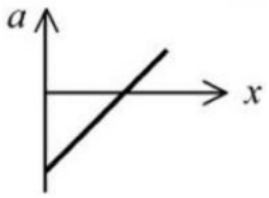
Energy conservation at $A\&B$

$$\frac{1}{2}mu^2 = mg\left(r + \frac{r}{2}\right) + \frac{1}{2}mv^2$$
$$u^2 = 3gr + \frac{rg}{2} \quad u = \sqrt{\frac{7gr}{2}}$$

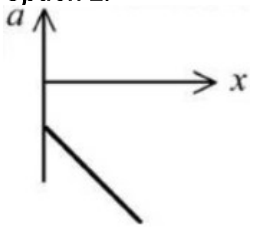
Q. 12 The velocity (v)-Distance (x) graph is shown in figure. Which graph represents acceleration (a) versus distance (x) variation of this system?



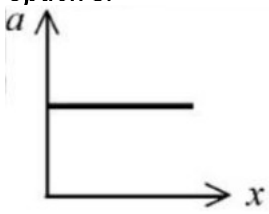
Option 1:



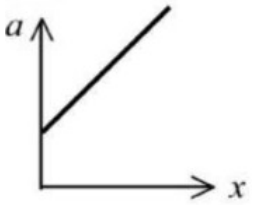
Option 2:



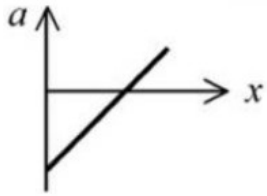
Option 3:



Option 4:



Correct Answer:



Solution:

$v = v_0 - mx$

$$a = (v)\left(\frac{dv}{dx}\right)$$
$$= (v_0 - mx)(-m)$$
$$= -mv_0 + m^2x$$

Q. 13 In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division $= 0.05$ mm, then the least count of the vernier callipers is _____ mm.

Option 1:

0.02

Option 2:

0.002

Option 3:

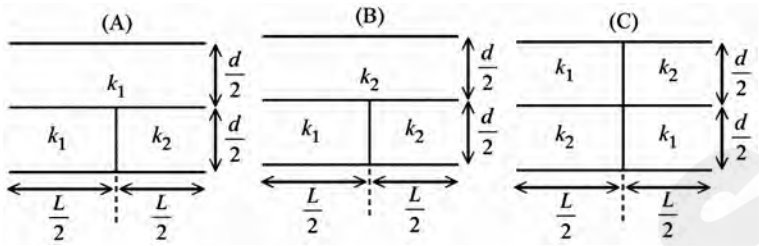
0.005

Option 4:
0.05

Correct Answer:
0.002

Solution:
$50VSD = 48MSD$
$L.C = 1MSD - 1VSD$
$= 1MSD - \frac{48}{50}MSD = \frac{2}{50}MSD = \frac{2}{50} \times 0.05 \text{ mm} = 0.002 \text{ mm}$

Q. 14 Three parallel plate capacitors each with area A and separation d are filled with two dielectric $(k_1$ and $k_2)$ in the following fashion. Which of the following is true? $(k_1 > k_2)$



Option 1:
$C_B > C_C > C_A$

Option 2:
$C_C > C_B > C_A$

Option 3:
$C_C > C_A > C_B$

Option 4:
$C_A > C_C > C_B$

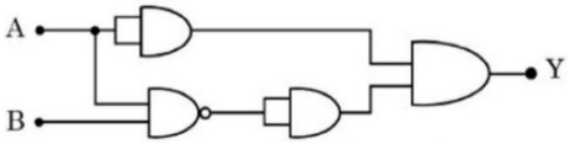
Correct Answer:
$C_A > C_C > C_B$

Solution:
$$\text{Ceq(A)} = \left(\frac{k_1}{2} + \frac{k_1k_2}{k_1+k_2}\right)\frac{\epsilon oA}{d}$$
$$\text{Ceq(B)} = \left(\frac{k_2}{2} + \frac{k_1k_2}{k_1+k_2}\right)\frac{\epsilon_o A}{d}$$
$$\text{Ceq(C)} = \left(\frac{2k_1k_2}{k_1+k_2}\right)\frac{\epsilon oA}{d}$$
$$k_1 > k_2$$

By substituting values for $k_1 \& k_2$

$C_A > C_C > C_B$

Q. 15 Identify the correct truth table of the given logic circuit.



Option 1:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Option 2:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Option 3:

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

CAREERS360

Option 4:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Correct Answer:

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

Solution:

$Y = (\overline{AB})A = (\bar{A} + \bar{B})A = 0 + A \cdot \bar{B}$

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

Q. 16 A moving coil galvanometer of resistance 100Ω shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is _____ Ω

Option 1:
10

Option 2:
0.5

Option 3:
2.5

Option 4:
25

Correct Answer:
25

Solution:
Shunt resistance required is

$$S = \frac{G}{\frac{\text{newrange}}{\text{old range}} - 1} = \frac{100}{\frac{5}{1} - 1} = \frac{100}{4}$$
$$S = 25\Omega$$

Q. 17 Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power $+5D$ as $(25, 96), (30, 62), (35, 37), (45, 35)$ and $(50, 32)$ respectively. Identify correct statement

Option 1:
Readings recorded by P_3 person are incorrect

Option 2:
Readings recorded by P_4 and P_5 persons are incorrect

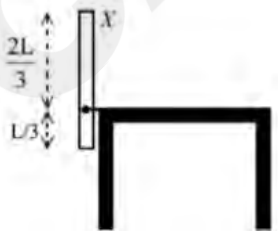
Option 3:
Readings recorded by P_3 and P_2 persons are incorrect

Option 4:
Readings recorded by all persons are correct

Correct Answer:
Readings recorded by P_3 person are incorrect

Solution:
$D = \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{96} + \frac{1}{25} = 0.0104 + 0.04 = 0.0504 \text{ cm}^{-1} = 5D$

Q. 18 A thin uniform rod (X) of mass M and length L is picoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is _____ . (g = gravitational acceleration)



Option 1:
$\sqrt{\frac{3}{2}\frac{g}{L}}$

Option 2:
$\frac{3}{\sqrt{2}}\sqrt{\frac{g}{L}}$

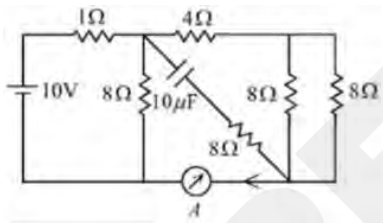
Option 3:
$\sqrt{\frac{3g}{L}}$

Option 4:
$\frac{1}{\sqrt{2}}\sqrt{\frac{g}{L}}$

Correct Answer:
$\sqrt{\frac{3g}{L}}$

Solution:
$$mg\frac{L}{6}=\frac{1}{2}lw^2$$
$$mg\frac{L}{6}=\frac{1}{2}\left\{\frac{mL^2}{12}+m\frac{L^2}{36}\right\}w^2$$
$$w=\sqrt{\frac{3g}{L}}$$

Q. 19 The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is _____ A.



Option 1:
2

Option 2:
1

Option 3:
1/2

Option 4:
0

Correct Answer:
1

Solution:
$I=\frac{10}{5}=2A$

Reading of ammeter $= 1\ \mathrm{A}$

Q. 20 When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V . If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V . The wavelength of first light is _____ m.

$\left(h = 6.63 \times 10^{-34}\ \mathrm{J.s}, \mathrm{e} = 1.6 \times 10^{-19} C, c = 3 \times 10^{8}\ \mathrm{m/s}\right)$

Option 1:
3.1×10^{-7}

Option 2:
2.5×10^{-7}

Option 3:
2.9×10^{-8}

Option 4:
2.2×10^{-8}

Correct Answer:
2.5×10^{-7}

Solution:

$$= \left(\frac{k_2}{2} + \frac{k_1 k_2}{k_1 + k_2}\right)\frac{\epsilon o A}{d}$$

$$e(0.7) = \frac{hc}{2\lambda} - \phi$$

$$2.5 = \frac{hc}{e}\frac{1}{2\lambda}$$

$$\lambda = \frac{12400}{5}$$

$$\lambda = 2480\ \mathrm{A}^0$$

$$\lambda = 2.48 \times 10^{-7}\ \mathrm{m}$$

Q. 21 A soap bubble of surface tension $0.04\ \mathrm{N/m}$ is blown to a diameter of 7 cm . If $(15000 - x)\mu\mathrm{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is _____
$(\pi = 22/7)$

Correct Answer:
11304

Solution:

$w = 8\pi T\left\{r_2^2 - r_1^2\right\}$

$w = 8 \times \frac{22}{7} \times 0.04 \times \frac{1}{4}\left\{\frac{14^2 - 7^2}{10^4}\right\}$

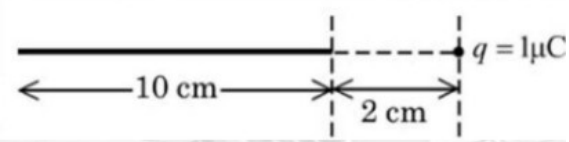
$w = 8 \times \frac{22}{7} \times 147 \times 10^{-6}$

$w = 3696\mu J$

$(15000 - x)\mu J = 3696\mu J = 11304$

Q. 22 A point charge $q = 1\mu C$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\mu C$, distributed uniformly along its length, as shown in figure. Force between q and wire is _____ N.

$\left(\text{Use } : \frac{1}{4\pi\epsilon_o} = 9 \times 10^9 \text{ N.m}^2/\text{C}^2\right)$



Correct Answer:
90

Solution:
$F = 90 \text{ N}$
$= 90 \text{ N}$

Q. 23 In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance lcm from the left. Now when an unknown resistance $x\Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is _____ Ω.

Correct Answer:
6

Solution:
$\frac{2}{3} = \frac{l}{100 - l}$
$l = 40 \text{ cm}$
$\frac{\frac{2}{3x}}{3 + x} = \frac{50}{50} = 1$
$x = 6\Omega$

Q. 24 A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1. A small co-centric cylinder of length $L/2$ and radius $R/3$ carved from this cylinder has moment of inertia about its axis equals to I_2. The ratio I_1/I_2 is _____ .

Correct Answer:
162

Solution:
$I_1 = \frac{MR^2}{2}$

After removing mass $= \frac{m}{18}$

$\text{Rad} = \frac{R}{3} \quad \text{Use } m = v \cdot d$

$I_2 = \frac{1}{2}\left(\frac{M}{18}\right)\left\{\frac{R}{3}\right\}^2$

$I_2 = \frac{MR^2}{324}$

$\frac{I_1}{I_2} = 162$

Q. 25 When 300 J of heat given to an ideal gas with $C_P = \frac{7}{2}R$ its temperature raises from $20C$ to $50C$ keeping its volume constant. The mass of the gas is (approximately) _____ g.
$(R = 8.314\ \text{J/mol. K})$

Correct Answer:
-

Solution:
$Q = nC_v\Delta T$

$300 = n\frac{5R}{2}\{50 - 20\}$

$300 = \frac{\text{mass}}{\text{molar mass}} \times \frac{5R}{2} \times 30$

$300 = \frac{\text{mass}}{\text{molar mass}} \times \frac{5}{2} \times 8.314 \times 30$

Molar mass is not given to calculate mass of gas

Chemistry

Q. 1 At 298 K , the mole percentage of $\mathrm{N}_2(\mathrm{~g})$ in air is 80%. Water is in equilibrium with air at a pressure of 10 atm. What is the mole fraction of $\mathrm{N}_2(\mathrm{~g})$ in water at 298 K ?
(K_H for N_2 is $6.5 \times 10^7 \mathrm{Hg}$)

Option 1:
1.23×10^{-7}

Option 2:
9.35×10^{-5}

Option 3:
1.17×10^{-4}

Option 4:
9.35×10^{-5}

Correct Answer:
9.35×10^{-5}

Solution:

$P = k\mu \times N_2$

$10 = 6.5 \times 10^7 \text{ Xair}$

$10 \times 760 = 6.5 \times 10^7 \text{ Xair}$

$\text{Xair } = \dfrac{10 \times 760}{6.5 \times 10^7} \quad XN_2 = 0.8 \text{ Xair}$

$\dfrac{0.8 \times 10 \times 760}{6.5 \times 10^7} = \dfrac{6080}{6.5} \times 10^{-7} = 9.35 \times 10^{-5}$

Hence, the correct answer is the option (2).

Q. 2 A student has planned to prepare acetanilide from aniline using acetic anhydride. The student has started from 9.3 g of aniline. However, the student has managed to obtain 11 g of dry acetanilide. The % yield of this reaction is :-

Option 1:
97.5%

Option 2:
72.5%

Option 3:
81.5%

Option 4:
59.5%

Correct Answer:
81.5%

Solution:

$\frac{9.3}{93} = 0.1$

$0.1 \times 135 = 13.5g$

$\text{Yield } = \frac{1}{13.5} \times 100$

81.5%

Hence, the correct answer is the option (3).

Q. 3 The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2 , is

Option 1:
$2.28 \times 10^{6}\ \mathrm{cm}$

Option 2:
$1.14 \times 10^{-7}\ \mathrm{cm}$

Option 3:
$1.14 \times 10^{-6}\ \mathrm{cm}$

Option 4:
$2.28 \times 10^{-7}\ \mathrm{cm}$

Correct Answer:
$1.14 \times 10^{-6}\ \mathrm{cm}$

Solution:
$n_2 + n_1 = 4$

$n_2 - n_1 = 2$

$2n_2 = 6$

$n_2 = 3$

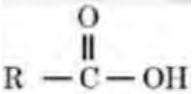
$n_1 = 1$

$\frac{1}{\lambda} = R_H \times z^2 \left(\frac{1}{1} - \frac{1}{9}\right)$

$\frac{1}{\lambda} = 10^7 \times 9 \left(\frac{8}{9}\right)$

$\lambda = \frac{1}{8} \times 10^{-7} = 1.25 \times 10^{-8}\ \mathrm{A} = 1.25 \times 10^{-6}\ \mathrm{cm}$

Hence, the correct answer is the option (3).

Q. 4 Given below are two statements:
Statement I: The dipole moment of R-CN is greater than R-NC and R-NC can undergo hydrolysis under acidic medium to produce 

Statement-II: R-CN hydrolyses under acidic medium to produce a compound which on treatment with $SOCl_2$ followed by the addition of NH_3 gives another compound (x). This compound (x) on treatment with $NaOCl/NaOH$ gives a product, that on treatment with $CHCl_3/KOH/\Delta$ produces R-NC In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are false

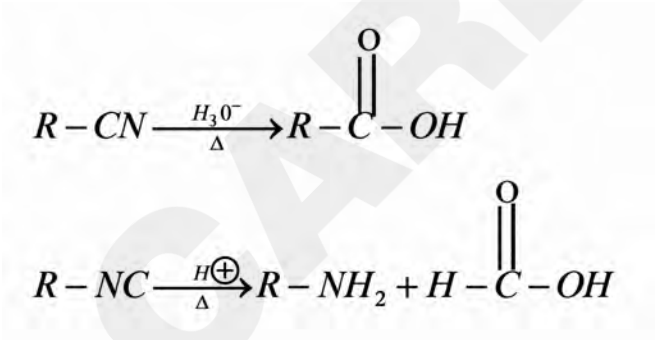
Option 2:
Statement I is true but Statement II is false

Option 3:
Both Statement I and Statement II are true

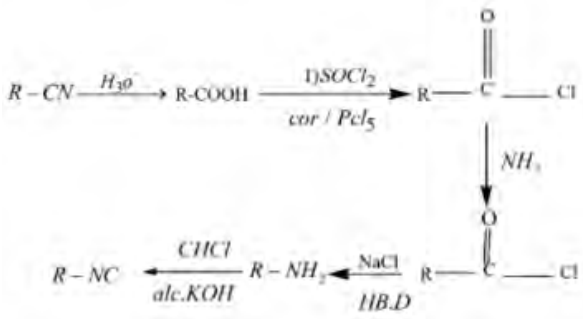
Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
Statement - I



Statement - II



Hence, the correct answer is the option (4).

Q. 5 Pair of species among the following having same bond order as well as paramagnetic character will be

Option 1:
O_2^-, N_2^-

Option 2:
O_2^+, N_2^{2-}

Option 3:
O_2^-, N_2^+

Option 4:
O_2^+, N_2^-

Correct Answer:
O_2^+, N_2^-

Solution:
1. O_2^+

For O_2 : bond order = 2
Removing one electron from antibonding orbital:
Bond order $= 2.5$

It has one unpaired electron → paramagnetic.

2. N_2^-

For N_2 : bond order = 3
Adding one electron to antibonding orbital:
Bond order $= 2.5$
It has one unpaired electron → paramagnetic.

Hence, the correct answer is the option (4).

Q. 6 The wavelength of light absorbed for the following complexes are in the order

$$(I)[Co\,(NH_3)]^{3+}; (II)[Co(H_2O)_6]^{3+}; (III)[Co(CN)_6]^{3+}; (IV)\left[Co(NH_3)_5(H_2O)^{3+}\right]; (V)[CF_6]^{3-}$$

Option 1:
III < I < IV < II < V

Option 2:
III < I < II < IV < V

Option 3:
III < IV < I < II < V

Option 4:
III < I < IV < V < II

Correct Answer:
III < I < IV < II < V

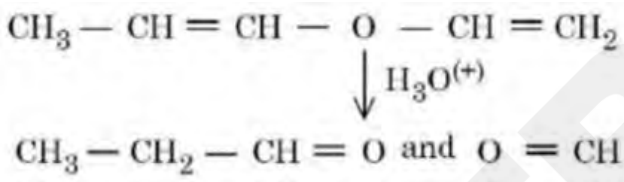
Solution:
Wave length of light absorbed is inversely proportional to ligand strength

Ligand strength:

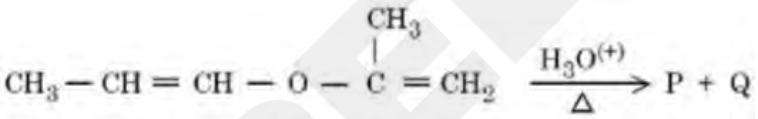
$CN^- > NH_3 > H_2^o > F^-$
Wavelength: $CN^- > NH_3 > H_2 > F^-$

Hence, the correct answer is the option (1).

Q. 7 The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:-



Based on this, predict the solution/reagent that will help to distinguish " P " and " Q " obtained in the following reaction:-



Option 1:
2, 4 - DNP reagent

Option 2:
Lucas reagent

Option 3:
Fehling solution

Option 4:
Saturated $NaHSO_3$ solution

Correct Answer:
Fehling solution

Solution:

$CH_3 - CH = CH - OH + CH_2 = CH - OH$
$CH_3 - CH_2 - CHO \quad CH_3CHO$
$CH_3 - CH_2 - CHO + CH_3 - c^p - CH_3$
$(p) \quad (Q)$

Hence, the correct answer is the option (3).

Q. 8 The heat of atomisation of methane and ethane are ' x' $\mathrm{kJ\ mol^{-1}}$ and ' y'$\mathrm{kJmol^{-1}}$ respectively. The longest wavelength (λ) of light capable of breaking the $\mathrm{C-C}$ bond can be expressed in SI unit as:

Option 1:
$N_A hc\left(y - \frac{6x}{4}\right)^{-1}$

Option 2:
$\frac{N_A hc}{250(y-6x)}$

Option 3:
$\frac{N_A hc}{250(4y-6x)}$

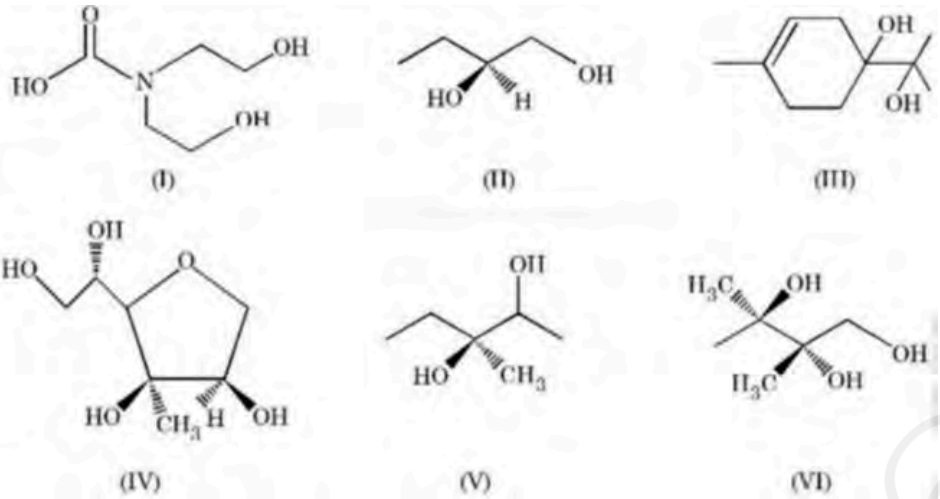
Option 4:
$\frac{hc}{1000}\left(\frac{y-6x}{4}\right)^{-1}$

Correct Answer:
$\frac{N_A hc}{250(4y-6x)}$

Solution:
$\mathrm{CH+ \xrightarrow{\Delta} C + AH}, x\mathrm{kJmol^{-1}}$
$\mathrm{C_2H_6+ \xrightarrow{\Delta} 2C + 6H}, y\mathrm{kJmol^{-1}}$
$\left[\mathrm{C + kH \xrightarrow{\Delta} (H4] \times 2}, -x\ \mathrm{kJ\ mol^{-1}}\right.$
$1000x = 4 \times E_{C-H}$
$1000y = 4 \times E_{C-C} + 6 \times E_{CH}$
$1000y = E_{C-C} + 6 \times \frac{1000}{4}x$
$N_A \frac{hc}{\lambda} = E_{c-c} = \left(1000y - \frac{6000}{4}x\right)$
$\lambda = \frac{N_A hc}{1000(y - 6x/4)}$

Hence, the correct answer is the option (3).

Q. 9 From the following, how many compounds contain at least one secondary alcohol?



Choose the correct answer from the options given below:

Option 1:
Three

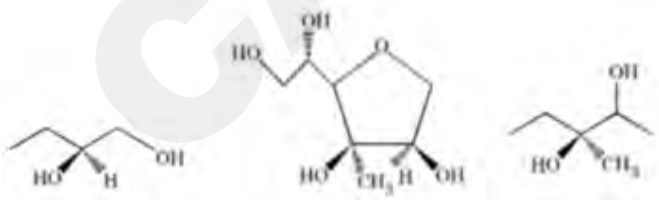
Option 2:
Five

Option 3:
Four

Option 4:
Two

Correct Answer:
Three

Solution:
$2^0 - \mathrm{OH}$ contains molecules are



So final answer is 3

Hence, the correct answer is the option (1).

Q. 10 The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

Option 1:
1. 6

Option 2:
2. 3

Option 3:
3.8

Option 4:
4. 4

Correct Answer:
1. 6

Solution:
Since:

- Amino acids available: alanine (Ala), glycine (Gly), valine (Val)
- Each amino acid can be used only once
- We need tripeptides (3 amino acids)

We are arranging 3 different amino acids in sequence.

Number of sequence= n!=3!=6

$$\begin{aligned}
&\mathrm{Gly}-\mathrm{Ala}-\mathrm{Val}\\
&\mathrm{Gly}-\mathrm{Val}-\mathrm{Ala}\\
&\mathrm{Ala}-\mathrm{Gly}-\mathrm{Val}\\
&\mathrm{Ala}-\mathrm{Val}-\mathrm{Gly}\\
&\mathrm{VAl}-\mathrm{Gly}-\mathrm{Ala}\\
&\mathrm{Val}-\mathrm{Ala}-\mathrm{Gly}
\end{aligned}$$

Hence, the correct answer is the option (1).

Q. 11 One mole of $\mathrm{Cl_2(g)}$ was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of $\mathrm{Cl^-}$, $\mathrm{ClO^-}$ and $\mathrm{OH^-}$ are respectively (assume volume remains constant)

Option 1:
$0.75\mathrm{M}, 0.75\mathrm{M}, 1\mathrm{M}$

Option 2:
$0.5\mathrm{M}, 0.5\mathrm{M}, 1\mathrm{M}$

Option 3:
$1\mathrm{M}, 1\mathrm{M}, 1\mathrm{M}$

Option 4:
$0.5\mathrm{M}, 0.5\mathrm{M}, 0.5\mathrm{M}$

Correct Answer:
$0.5\mathrm{M}, 0.5\mathrm{M}, 1\mathrm{M}$

Solution:
$\mathrm{e}/2 + 20\mathrm{H}^- \rightarrow \mathrm{Cl}^- + \mathrm{clo}^-$
$4\ \mathrm{mol} 1\ \mathrm{mol} 1\ \mathrm{mol}$
$= 0.5\mathrm{M}, 0.5\mathrm{M}, 1\mathrm{M}$

Hence, the correct answer is the option (2).

Q. 12 Two liquids A and B form an ideal solution at temperature T K . At T K , the vapour pressures of pure A and B are 55 and $15\mathrm{kNm}^2$ respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8?

Option 1:
0.340

Option 2:
0.5217

Option 3:
0.663

Option 4:
0.480

Correct Answer:
0.5217

Solution:
$PA = 55\mathrm{kNm}^{-2}$
$P_B = 15\mathrm{kNm}^{-2}$
$Y_A = 0.8$
$P_A = P_A X_B = Y_A P_T$
$P_B = P_B X_B = Y_B P_T$
$\frac{P_A}{P_B} = \frac{P_A X_A}{P_B X_B} = \frac{Y_A}{Y_B} = \frac{0.8}{0.2}$
$\frac{X_A}{X_B} = \frac{0.8}{0.2} \times \frac{P_B}{P} = 4 \times \frac{15}{55} = \frac{60}{55} = \frac{12}{11}$
$\frac{X_A}{(1 - X_A)} = \frac{12}{11}$
$\frac{1}{X_A} - 1 = \frac{11}{12}$
$\frac{1}{X_A} = \frac{11 + 12}{12}$
$X_A = \frac{12}{23} = 0.5217$

Hence, the correct answer is the option (2).

Q. 13 Consider the following gaseous equilibrium in a closed container of volume ' V ' at $T(K)$.

$$P_2(g) + Q_2(g) \quad 2PQ(g)$$

2 moles each of $\mathrm{P}_{2(\mathrm{g})}, \mathrm{Q}_{2(\mathrm{g})}$ and $\mathrm{PQ}_{(\mathrm{g})}$ are present at equilibrium. Now one mole each of ' P_2 ' and ' Q_2 ' are added to the equilibrium keeping the temperature at $\mathrm{T}(\mathrm{K})$. The number of moles of P_2 and Q_2 and PQ at the new equilibrium, respectively, are

Option 1:
1.66, 1.66, 1.66

Option 2:
1.21, 2.24, 1.56

Option 3:
2.67, 2.67, 2.67

Option 4:
2 .56,1.62,2.24

Correct Answer:
2.67, 2.67, 2.67

Solution:

$P_2(9) + Q_2Q1 \rightarrow 2PQ(g)$

$\mathrm{Keg} = \dfrac{(PQ)}{\{P_2\}\{Q_2\}}$

$(3-x)(3-x)(2-x)\mathrm{keg} > Q$

Reaction forward

$keg = \dfrac{(2+2x)^2}{(3-x)^2} = 1$

$P_2 = 3 - x$

$= 3 - \dfrac{1}{3} = 8/3(3-x)^2 = (2+2x)^2$

$Q_2 = 3 - x$

$Q_2 = 3 - \dfrac{1}{3} = 8/3(3-x) = 2 + 2x$

$PQ = 2 + 2x = 2 + 2/3 = 8/3$

$\therefore x = 1/3$

Hence, the correct answer is the option (3).

Q. 14 Choose the INCORRECT statement

Option 1:
Carbon cannot exceed its covalency more than four.

Option 2:
Among the isotopes of carbon, 13_C is a radioactive isotope.

Option 3:
CO_2 is the most acidic oxide among the dioxides of group of 14 elements.

Option 4:
Carbon exhibits negative oxidation states along with +4 and +2 .

Correct Answer:
Among the isotopes of carbon, 13_C is a radioactive isotope.

Solution:
Among the isotopes of carbon

C^{14} is radioactive isotope

Hence, the correct answer is the option (2).

Q. 15 In the Group analysis of cations, $Ba^{2+}\&Ca^{2+}$ are precipitated respectively as

Option 1:
chromate \& sulphide

Option 2:
sulphide \& sulphide

Option 3:
carbonate \& carbonate

Option 4:
hydroxide \& carbonate

Correct Answer:
carbonate \& carbonate

Solution:
For Ba^{+2} and Ca^{+2} group reagent is $(NH_4)_2CO_3$ in the presence of NH_4OH and $NH_4ClBaBa^{+2}$ and Ca^{+2} are precipitated as carbonates

$$Ba^{+2} + Ca^{+2} \rightarrow BaCO_3$$
$$Ca^{+2} + CO^{+2} \rightarrow CaCO_3$$

Hence, the correct answer is the option (3).

Q. 16 Find out the statements which are not true.
A. Resonating structures with more number of covalent bonds and lesser charge separation are more stable.
B. In electromeric effect, an unsaturated system shows +E effect with nucleophile and -E effect with electrophile.
C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.
D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.
E. Stability of carbanion increases with the increase in s - character of the carbon carrying the negative charge.
Choose the correct answer from the options given below:

Option 1:
A, C \& D only

Option 2:
B, D \& E only

Option 3:
B \& D only

Option 4:
A, D \& E only

Correct Answer:
B \& D only

Solution:

- A)True — More covalent bonds & less charge separation → more stable resonance.
- B) False — Correct rule: +E with electrophile, −E with nucleophile.
- C) True — Inductive effect causes polarity → affects dipole moment, bp, mp.
- D) False — More substituted alkene → lower heat of hydrogenation (more stable).
- E) True — Carbanion stability: sp > sp^2 > sp^3 (more s-character → more stable).

Hence, the correct answer is the option (3).

Q. 17 The correct order of $\mathrm{C}, \mathrm{N}, \mathrm{O}$ and F in terms of second ionisation potential is

Option 1:
$F < N < C < O$

Option 2:
$C < F < N < O$

Option 3:
$C < O < N < F$

Option 4:
$C < N < F < O$

Correct Answer:
$C < N < F < O$

Solution:
Given elements, $\mathrm{C, N, O, F}$

$$\begin{array}{cccc} C< & O< & N< & F \\ \downarrow & \downarrow & \downarrow & \downarrow \\ B & N & C & O \end{array}$$

Ionisation energy order of $\mathrm{B, N, C, O}$ is $B < C < O < N$ but they are formed from $C < N < F < O$

Hence, the correct answer is option (4)

Q. 18 " X " is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in the +6 oxidation state in " X ". The color of the potassium salt of X is

Option 1:
green

Option 2:
Purple

Option 3:
Orange

Option 4:
Yellow

Correct Answer:
green

Solution:
The lightest element of group 7 (transition series) is Manganese.
When manganese is in the +6 oxidation state, it forms the oxoanion Manganate ion, $\mathrm{MnO_4^{2-}}$.

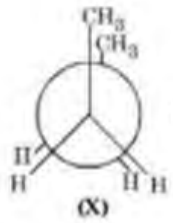
The potassium salt of this ion is Potassium manganate, $\mathrm{K_2MnO_4}$.

Potassium manganate is green in colour.

Hence, the correct answer is option (1)

Q. 19 Given below are two statements:

Statement I: There are several conformers for n-butane. Out of those conformers,



(X)

is the least stable and most stable conformer is



Statement II: As the dihedral angle increases, torsional strain decreases from (X) to (Y).

In the light ofthe above statements, choose the correct answer from the options given below

Option 1:
Statement I is false but Statement II is true

Option 2:
Statement I is true but Statement II is false

Option 3:
Both Statement I and Statement II are true

Option 4:
Both Statement I and Statement II are false

Correct Answer:
Both Statement I and Statement II are true

Solution:
Statement - I is correct because X is an eclipsed confirmation that suffers from a highly eclipsed strain, whereas Y is an anti confirmations there is no eclipsed strain and no steric strain because of groups separated by 180°

Statement -II is also correct because as the dihedral angle changes, potential energy also changes depending on dihydral angle

Hence, the correct answer is option (3)

Q. 20 Given below are two statements:

Statement I: Cross aldol condensation between two different aldehydes will always produce four different products.
Statement II: When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH, it forms a condensation product with acetophenone only.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Statement I is true but Statement II is false

Option 2:
Statement I is false but Statement II is true

Option 3:
Both Statement I and Statement II are true

Option 4:
Both Statement I and Statement II are false

Correct Answer:
Both Statement I and Statement II are false

Solution:
Statement I:
Cross aldol condensation between two different aldehydes does not always produce four different products. If both aldehydes contain α-hydrogen, four products are possible (two self-aldol and two cross-aldol products). However, if one aldehyde lacks an α-hydrogen (e.g., Benzaldehyde), it cannot undergo self-aldol condensation. In such cases, fewer products are formed. Therefore, the statement that it will always produce four products is incorrect.

Statement II:
Semicarbazide reacts with carbonyl compounds to form semicarbazones. Between Benzaldehyde and Acetophenone, aldehydes are generally more reactive than ketones toward nucleophilic addition because they have less steric hindrance and a stronger electrophilic carbonyl carbon. Hence, semicarbazide reacts preferentially with benzaldehyde, not only with acetophenone.

Thus, Statement II is also false.

Hence, the correct answer is option (4)

Q. 21 Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be $1/30$ of the molar conductivity of another weak acid HZ with concentration of 0.02 M. If λ_Q^0 - happened be equal with λ_Z^0-, then the difference of the $\mathrm{pK_a}$, values of the two weak acids $(pK_a(HQ) - pK_a(HZ))$ is ____

(Nearest integer). [Given: degree of dissociation $(\alpha) \ll 1$ for both weak acids λ^0 : limiting molar conductivity of ions]

Correct Answer:
2

Solution:

$$\lambda_m(HQ) = \frac{1}{30} \times \lambda_m(HZ)$$

$$\frac{\alpha_{HQ}}{\alpha_{HZ}} = \frac{\frac{\lambda_m(HQ)}{\lambda_m^{\infty}(HQ)}}{\frac{\lambda_m(HZ)}{\lambda_m^{\infty}(HZ)}} = \frac{1}{30}$$

$$\therefore x = \log\left[\frac{kg(HZ)}{ka(HQ)}\right]$$

$$= \log\left[\frac{0.02 \times \alpha HZ^2}{0.18 \times \alpha HQ^2}\right]$$

$$\therefore x = 2$$

Hence, the answer is 2

Q. 22 The half-life of $^{65}\mathrm{Zn}$ is 245 days After x days, 75% of original activity remained. The value of x in days is. (Nearest integer) (Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

Correct Answer:
102

Solution:

$$k = \left(\frac{\ln 2}{t1/2}\right) = \frac{\ln 2}{245} day - 1$$

$$\ln \frac{[Ao]}{[At]} = kt$$

$$\frac{\ln[Ao]}{0.75[A0]} = \left(\frac{\ln 2}{245}\right)t$$

$$\therefore t = 102 \text{ days}$$

Hence, the correct answer is 102

Q. 23 0.25 g of an organic compound " A " containing carbon, hydrogen, and oxygen was analysed using the combustion method. There was an increase in mass of the $\mathrm{CaCl_2}$ tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g, respectively. The percentage of oxygen in compound A is _____ \%. (Nearest integer)
(Given: molar mass in $\mathrm{gmol^{-1}H:1, C:12, O:16}$)

Correct Answer:
73

Solution:

$$\mathrm{C_xH_yO_z} + \left(x + \tfrac{y}{4} - \tfrac{z}{2}\right)\mathrm{O_2} \rightarrow x\mathrm{CO_2} + \tfrac{y}{2}\mathrm{H_2O}$$

Mass of compound $= 0.25g$
Increase in mass of potash tube = mass of $CO_2 = 0.1837$ g

$$\%\text{ mass of C} = \tfrac{0.1837}{44} \times \tfrac{12\times100}{0.25} = 20.04\%$$

Increase in mass of $CaCl_2$ tube = mass of $H_2O = 0.15$ g

$$\% \text{ mass of H} = \frac{0.15}{18} \times \frac{2 \times 100}{0.25} = 6.66\%$$

$$\% \text{ of oxygen } = 100 - (x + y)$$
$$= 100 - (20.04 + 6.66)$$

$$\%O = 73.3\%$$

Hence, the answer is 73

Q. 24 Grignard reagent $RMgBr(P)$ reacts with water and forms a gas (Q). One gram of Q occupies $1.4 dm^3$ at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z).0.1 mole of (Z) will weigh _____ g. (Nearest integer)

Correct Answer:
6

Solution:
$M_Q = \frac{22.4 \text{ L/mol}}{1.4 \text{ L/g}} = 16 \text{ g/mol}$

A hydrocarbon gas with a molar mass of 16 g/mol is methane (CH_4). Therefore, the R group in the Grignard reagent is a methyl group

First, calculate the molar mass of acetic acid $(C_2H_4O_2)$:

$$MR = (2 \times 12) + (4 \times 1) + (2 \times 16) = 24 + 4 + 32 = 60 \text{ g/mol}$$

Mass = moles × molar mass $= 0.1 \text{ mol} \times 60 \text{ g/mol} = 6$ g

Hence, the answer is 6

Q. 25 A chromium complex with a formula $CrCl_3 \cdot 6H_2O$ has a spin-only magnetic moment value of 3.87 BM, and its solution conductivity corresponds to a $1:2$ electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with an excess of $AgNO_3$. The amount of AgCl formed in the above process is g. (Nearest integer)
[Given: Molar mass in $gmol^{-1} Cr: 52; Cl: 35.5, Ag: 108, O: 16, H: 1$]

Correct Answer:
3

Solution:
$[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O + 2AgNO_3 \rightarrow 2AgCl$

Weight $= 2.75g$
No. of moles $= \frac{2.75}{266.5} = 0.0103$ moles

Moles of $AgCl = 2 \times 0.0103 = 0.0206$ moles
Amount of AgCl formed

$= 0.0206 \times 143.5 = 2.96 \approx 3$

Hence, the answer is 3

Maths

Q. 1 Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$, be 30 . If its eccentricity is the maximum value of the function $f(t) = -\frac{3}{4} + 2t - t^2$, then $\left(a^2 + b^2\right)$ is equal to:

Option 1:
276

Option 2:
256

Option 3:
516

Option 4:
496

Correct Answer:
496

Solution:
$\frac{2b^2}{a} = 30$
$b^2 = 15a$

max value of $f(t) = \frac{-3}{4} + 2t - t^2$ is ' e '

$e = \frac{4(-1)\left(\frac{-3}{4}\right) - 4}{4(-1)}$
$= \frac{-3}{4} + 1 = \frac{1}{4}$
$e = \frac{1}{4}$
$b^2 = a^2\left(1 - e^2\right)$
$= a^2\left(1 - \frac{1}{16}\right)$
$15a = a^2 \cdot \frac{15}{16}$
$a = 16$
$a^2 + b^2 = 256 + 240$
$= 496$

Hence the correct answer is option (4).

Q. 2 Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and $a_1a_2a_3a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

Option 1:
24

Option 2:
27

Option 3:
23

Option 4:
21

Correct Answer:
27

Solution:
Given a_1, a_2, a_3, a_4 are in AP as $a-3d, a-d, a+d, a+3d$
Given $a_1 + a_2 + a_3 + a_4 = 48$

$4a = 48$
$a = 12$

Given $a_1a_2a_3a_4 + l^4 = 361$

$\left(a^1 - 9d^2\right)\left(a^2 - d^2\right) + l^4 = 361$
$\left(144 - 9d^2\right)\left(144 - d^2\right) + l^4 = 361$
$25d^4 - 1440d^2 + 144^2 = 361$
$\left(5d^2 - 144\right) = 19^2$
$5d^2 - 144 = 19 \& -19$
$d^2 = \frac{163}{5} \quad \& \quad \frac{125}{5} = 25$
$d = \frac{\sqrt{163}}{5} \& \frac{125}{5} = 25$
$d = \sqrt{\frac{163}{5}} \quad \& 5$

Largest term $12 + 15 = 27$

Hence the correct answer is option (2)

Q. 3 Let $f(x)=\int \frac{7x^{10}+9x^8}{(1+x^2+2x^9)}dx, x>0, \lim_{x\to 0} f(x)=0 \quad f(1)=\frac{1}{4}$

If $A=\begin{vmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{vmatrix}$ and $B=\text{adj}(\text{adj } A)$ be such that $|B|=81$, then α^2 is equal to:

Option 1:
3

Option 2:
2

Option 3:
4

Option 4:
1

Correct Answer:
4

Solution:

$$f(x)=\int \frac{7x^{10}+9x^8}{(1+x^2+2x^9)^2}dx$$

$$=\int \frac{7x^{10}+9x^8}{x^{18}\left(\frac{1}{x^9}+\frac{1}{x^7}+2\right)^2}dx$$

$$=\int \frac{\frac{7}{x^8}+\frac{9}{x^{10}}}{\left(\frac{1}{x^9}+\frac{1}{x^7}+2\right)^2}dx$$

$$\frac{1}{x^9}+\frac{1}{x^7}+2=t$$

$$\left(-\frac{9}{x^{10}}-\frac{7}{x^8}\right)dx=dt$$

$$-\int \frac{dt}{t^2}=\frac{1}{t}+C$$

$$=\frac{1}{\left(\frac{1}{x^9}+\frac{1}{x^7}+2\right)}+C$$

$$f(x)=\frac{x^9}{1+x^2+2x^9}+C$$

$f(x)=\frac{1}{4} \quad \frac{1}{4}=\frac{1}{4}+C$

$C=0$

$f(x)=\frac{x^9}{1+x^2+2x^9}$

$A=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & 1 & 1\\ \alpha & 4 & 1\end{bmatrix}$

$|A|=1\left(1-\alpha^2\right)$

$B=\text{adj(adjA}$

$|B|=|\text{adj(adjA)}|$

$|B|=|\text{adj(adjA)}|$

$|B|=|A|^4$

$3^4=\left(1-\alpha^2\right)^4$

$1-\alpha^2=\pm 3$

$\alpha^2=4,-2$

$\alpha^2=4$

Hence the correct answer is option (3).

Q. 4 The smallest positive integral value of a , for which all the roots of $x^4-ax^2+9=0$ are real and distinct, is equal to:

Option 1:
9

Option 2:
3

Option 3:
4

Option 4:
7

Correct Answer:
7

Solution:

$f(x)=x^4-ax^2+9$

$x^2=t$

$g(t)=t^2-at+9$

$g(t)$ roots must be + ve

1) $\Delta>0$

2) $\frac{-b}{2a}>0$

1) $a^2-49>0$

$a^2 - 36 > 0$
$a \in (-\infty, -6) \cup (6, \infty)$

2) $\frac{a}{2.1} > 0$

$a > 0$

Hence the correct answer is option (4).

Q. 5 $\left(\frac{1}{3} + \frac{4}{7}\right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3}\right) + \ldots$ upto infinite terms, is equal to:

Option 1:
$\frac{5}{2}$

Option 2:
$\frac{6}{5}$

Option 3:
$\frac{4}{3}$

Option 4:
$\frac{7}{4}$

Correct Answer:
$\frac{5}{2}$

Solution:
$\left(\frac{1}{3} + \frac{4}{3}\right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2}\right) + \left(\frac{1}{3^2} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3}\right)$
$a = \frac{1}{3}, b = \frac{4}{7}$
$S = (a + b) + \left(a^2 + ab + b^2\right) + \left(a^3 + a^2 b + ab^2 + b^3\right)$
$= \frac{1}{(a-b)}\left[a^2 - b^2 + a^3 - b^3 + a^4 - b^4 + - - -\infty\right]$
$= \frac{a^2 + a^3 + - - \infty}{a - b} - \frac{b^2 + b^3 + -\infty}{a + b} = \frac{a^2}{(a-b)(1-a)} - \frac{b^2}{(a-b)(1-b)}$
$= \frac{\frac{1}{9}}{\left(\frac{1}{3} - \frac{4}{7}\right)\left(1 - \frac{1}{3}\right)} - \frac{1}{\left(\frac{1}{3} - \frac{4}{7}\right)\left(1 - \frac{4}{7}\right)}$

Hence the correct answer is option (1).

Q. 6 Let the angles made with the positive x -axis by two straight lines drawn from the point $\mathrm{P}(2, 3)$ and meeting the line $\mathrm{x} + \mathrm{y} = 6$ at distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2. Then the value of $(\theta_1 + \theta_2)$ is:

Option 1:
$\frac{\pi}{12}$

Option 2:
$\frac{\pi}{6}$

Option 3:
$\frac{\pi}{2}$

Option 4:
$\frac{\pi}{3}$

Correct Answer:
$\frac{\pi}{2}$

Solution:

$$\frac{|5-6|}{\sqrt{2}}=\frac{1}{\sqrt{2}}$$

$$\triangle PLA, \cos\theta=\frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{2}}{\sqrt{3}}}=\frac{\sqrt{3}}{2}$$

$$\theta=\frac{\pi}{6}$$

$$\angle APB=\frac{\pi}{3}$$

$\triangle PAB$ is an equilateral triangle

$$\tan 60^{\circ}=\left|\frac{m+1}{1-m}\right|$$

$$\sqrt{3}=\frac{|m+1|}{|1-m|}$$

$$3\left(1+m^2-2m\right)=m^2+1+2m$$

$$2m^2-8m+2=0$$

$$m^2-4m+1=0$$

$$m_1+m_2=4$$

$$m_1m_2=1$$

$$\tan\left(\theta_1+\theta_2\right)=\frac{m_1+m_2}{1-m_1m_2}$$

$$\tan\left(\theta_1+\theta_2\right)=\frac{4}{1-1}=\frac{1}{0}$$

$$\theta_1+\theta_2=\frac{\pi}{2}$$

Hence the correct answer is option (3)

Q. 7 Let f be a function such that $3f(x)+2f\left(\frac{m}{19x}\right)\neq 5x, x\neq 0$, where $m=\sum_{i=1}^{9}(i)^2$. Then $f(5)-f(2)$ is equal to

Option 1:
-9

Option 2:
18

Option 3:
36

Option 4:
9

Correct Answer:
18

Solution:
Given $3f(x)+2f\left(\frac{m}{19x}\right)=5x$

$$\text{Given } m=\sum_{i=1}^{9} i^2=1^2+2^2+\ldots\ldots.9^2=\frac{9.10.19}{6}=19\times 15$$

$$3f(x)+2f\left(\frac{15}{x}\right)=5x$$

$$\left(x \text{ by } \frac{15}{x}\right)3f\left(\frac{15}{x}\right)+2f(x)=\frac{75}{x}$$

$$9f(x)+6f\left(\frac{15}{3}\right)=15x$$

$$6f\left(\frac{15}{x}\right)+4f(x)=\frac{150}{x}$$

$$5f(x)=15x-\frac{150}{x}$$

$$f(x)=3x-\frac{30}{x}$$

$$f(5)-f(2)=\left(15-\frac{30}{5}\right)-\left(6-\frac{30}{2}\right)=(15-6)-(6-15)=9-9=18$$

Hence the correct answer is option (2)

Q. 8 The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

Option 1:
1579

Option 2:
1580

Option 3:
1581

Option 4:
1578

Correct Answer:
1580

Solution:
UDAYPUR

$A---\frac{6!}{2!}$

$D---\frac{6!}{2!}$

$P--\frac{6!}{2!}$

$R----\frac{6!}{2!}$

$UA------5!$

UDAP - - - - - 3!

UDAR -----3!

UDAU -----3!

UDAYPRU ---1

UDAYPUR ---1

$4\times\frac{6!}{2!}+5!+3\times3!+2$

$440+120+18+2=1580$

Hence the correct answer is option (2)

Q. 9 The largest value of n , for which 40^{n} divides $60!$, is

Option 1:
14

Option 2:
13

Option 3:
11

Option 4:
12

Correct Answer:
14

Solution:

$40^n = 2^{3n}5^n$

$E_2(6!) = \left[\frac{60}{2}\right] ..... + \left[\frac{60}{2^5}\right] = 56$

$E_2(60!) = \left[\frac{60}{2}\right] ..... + \left[\frac{60}{2^5}\right] = 14$

$60! = 2^{56} \cdot 5^{14} = 2^{14}\left(2^3 \cdot 5\right)^{14}$

$\therefore$ Max value of n is 14

Hence the correct answer is option (1).

Q. 10 Let $a = 2i - 5j + 5k$ and $b = i - j + 3k$. If C is a vector such that $2(a \times c) + 3(b \times c) = 0 (a - b) \cdot c = -97$, then $|c \times k|^2$ is equal to

Option 1:
205

Option 2:
218

Option 3:
233

Option 4:
193

Correct Answer:
218

Solution:
$2(a \times c) + 3(b \times c) = 0$
$(2a + 3b) \times 5c = 0$
$\therefore c = \lambda(2a + 3b)$
$= \lambda(7i - 13j + 19k)$
$(a - b).c = -97 \quad \therefore C = -7i + 13j + 19k$
$\lambda(7 + 52 + 38) = -97 \quad \therefore |c \times k|^2 = 218$

Hence the correct answer is option (2).

Q. 11 If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2 - 2x - 2}\right)$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

Option 1:
4

Option 2:
5

Option 3:
3

Option 4:
2

Correct Answer:
4

Solution:

$$-1 \leq \frac{1}{x^2-2x-2} \leq 1$$

$$1+\frac{1}{x^2-2x-2} \geq 0$$

$$\frac{x^2-2x-1}{x^2-2x-2} \geq 0$$

$$\frac{(x-1-\sqrt{2})(x-1+\sqrt{2})}{(x-1-\sqrt{3})(x-1+\sqrt{3})} \geq 0$$

$$x \in (-\infty, 1-\sqrt{3}) \cup [1-\sqrt{2}, 1+\sqrt{2}] \cup (1+\sqrt{3}, \infty)$$

$$\frac{1}{x^2-2x-2}-1 \leq 0$$

$$1-\frac{1}{x^2-2x-2} \geq 0$$

$$\frac{x^2-2x-3}{x^2-2x-2} \geq 0$$

$$\frac{(x+1)(x-3)}{(x-1-\sqrt{3})(x-1+\sqrt{3})} \geq 0$$

$$x \in (-\infty, -1] \cup (-\sqrt{3}, \sqrt{3}) \cup [3, \infty)$$

$$\therefore \text{ Domain is } (-\infty, -1] \cup [1-\sqrt{2}, 1+\sqrt{2}] \cup [3, \therefore \infty)$$

$$\therefore \alpha+\beta+\gamma+\delta = 4$$

Hence the correct answer is option (1)

Q. 12 Let $P=[p_{ij}]$ and $Q=[q_{ij}]$ be two square matrices of order 3 such that $q_{ij}=2^{(i+j-1)}p_{ij}$ and $\det(Q)=2^{10}$. Then the value of $\det(\text{adj}(\text{adj } P))$ is:

Option 1:
81

Option 2:
124

Option 3:
32

Option 4:
16

Correct Answer:
16

Solution:
$|Q| = 2^{10}$ and $q_{ij} = 2^{i+j-1}p_{ij}$

$$\begin{vmatrix} 2p_{11} & 2^2p_{12} & 2^3p_{13} \\ 2^2p_{21} & 2^3p_{22} & 2^2p_{23} \\ 2^3p_{37} & 2^4p_{32} & 2^5p_{33} \end{vmatrix} = 2^{10}$$

$$2^9 = \begin{vmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & 2p_{32} & p_{33} \end{vmatrix} = 2^{10}$$

$|p| = 2$

$|\operatorname{adj}(\operatorname{adj}p)| = |p|^{(n-1)^2} = 2^4$

Hence the correct answer is option (4).

Q. 13 Let $a = 2i - j - k, b = i + 3j - k$ and $c = 2i + j + 3k$. Let v be the vector in the plane of the vectors a and b, such that the length of its projection on the vector C is $\frac{1}{\sqrt{14}}$. Then $|v|$ is equal to

Option 1:
13

Option 2:
$\frac{\sqrt{21}}{2}$

Option 3:
7

Option 4:
$\frac{\sqrt{35}}{2}$

Correct Answer:
13

Solution:
Given
$a = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$,
$b = \mathbf{i} + 3\mathbf{j} - \mathbf{k}$,
$c = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$.

Since v lies in the plane of a and b, write
$v = \alpha a + \beta b$.

First compute $a \times b$.

$$a \times b = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & -1\,1 & 3 & -1 \end{vmatrix}$$

$$= \mathbf{i}[(-1)(-1) - (-1)(3)] - \mathbf{j}[2(-1) - (-1)(1)] + \mathbf{k}[2 \cdot 3 - (-1)(1)]$$

$$= 4\mathbf{i} + \mathbf{j} + 7\mathbf{k}.$$

A normal vector to the plane of a and b is
$n = a \times b = (4, 1, 7).$

Any vector in this plane is perpendicular to n, so

$$v \cdot n = 0.$$

The length of projection of v on c is

$$\frac{|v \cdot c|}{|c|} = \frac{1}{\sqrt{14}}.$$

Now

$$|c| = \sqrt{2^2 + 1^2 + 3^2} = \sqrt{14}.$$

Hence

$$|v \cdot c| = 1.$$

Also

$$v = \alpha a + \beta b.$$

So

$$v = \alpha(2, -1, -1) + \beta(1, 3, -1)$$

$$v = (2\alpha + \beta, , -\alpha + 3\beta, , -\alpha - \beta).$$

Now compute $v \cdot c$:

$$v \cdot c = (2\alpha + \beta)2 + (-\alpha + 3\beta)1 + (-\alpha - \beta)3$$

$$= 4\alpha + 2\beta - \alpha + 3\beta - 3\alpha - 3\beta$$

$$= 2\beta.$$

Thus

$$|2\beta| = 1$$

$$\beta = \pm\frac{1}{2}.$$

Now find $|v|$:

$$|v|^2 = (2\alpha + \beta)^2 + (-\alpha + 3\beta)^2 + (-\alpha - \beta)^2.$$

Substitute $\beta = \frac{1}{2}$:

$|v|^2 = (2\alpha + \frac{1}{2})^2 + (-\alpha + \frac{3}{2})^2 + (-\alpha - \frac{1}{2})^2.$

Simplifying,

$|v|^2 = 6\alpha^2 - 2\alpha + \frac{11}{4}.$

Minimizing this quadratic gives

$\alpha = \frac{1}{6}.$

Substitute:

$|v|^2 = 169$

Hence $|v| = 13.$

Therefore the required value is $|v| = 13.$

Hence the correct answer is option (1)

Q. 14 Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in N$. Then $a + b + c$ is equal to

Option 1:
12

Option 2:
6

Option 3:
8

Option 4:
4

Correct Answer:
6

Solution:
Let $p\left(2T, t^2\right)$ on $x^2 = 4y$
Image p in $x - y = 1$ is $Q(h, k)$

$\therefore h = t^2 + 1, k = 2t - 1$

By eliminating t
We get $(k + 1)^2 = 4(h - 1)$

$(y + 1)^2 = 4(x - 1)$
$a = 1, b = 4, c = 1$
$a + b + c = 6$

Hence the correct answer is option (2).

Q. 15 Let $X=\{x \in N: 1 \leq x \leq 19\}$ and for some $a, b \in R, Y=\{a x+b: x \in X\}$. If the mean and variance of the elements of Y are 30 and 750 , respectively, then the sum of all possible values of b is:

Option 1:
100

Option 2:
60

Option 3:
80

Option 4:
20

Correct Answer:
60

Solution:
For X main $x=10; \sigma^2=30$
Variance
For Y main is $ax+b=30$

$\sigma_1^2=750$
$4a^2.30=750$
$a=\pm 5$
$\Rightarrow 10a+b=30$
$\pm 50+b=30$
$\therefore b=-20,80$
$\text{Sum } =80-20$
$=60$

Hence, the correct answer is option (2).

Q. 16 Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x=0, x=1, y^2=x$ and $y=|ax-5|-|1-ax|+ax^2$. Then $(f(0)+f(1))$ is equal to

Option 1:
14

Option 2:
12

Option 3:
7

Option 4:
9

Correct Answer:
7

Solution:

$$f(0) = 4 - \int_0^1 \sqrt{x}dx = \frac{10}{3}$$

$$f(1) = \int_0^1 \left\{\left(x^2+4\right) - \sqrt{x}\right\}dx = \frac{11}{3}$$

$$f(0) + f(1) = \tfrac{10}{3} + \tfrac{11}{3}$$

Hence, the correct answer is option (3).

Q. 17 Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim_{t \to x} \left(\frac{t^2 y(x) - x^2 y(t)}{x - t}\right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to

Option 1:
23

Option 2:
18

Option 3:
27

Option 4:
12

Correct Answer:
23

Solution:

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$$\lim_{t\to x}\frac{t^2y(x)-x^2y(t)}{x-t}=3$$

$$\lim_{t\to x}\frac{t^2\frac{dy}{dx}-2xy(t)}{1}=3\Rightarrow x^2\frac{dy}{dx}-2xy=3$$

$$\frac{dy}{dx}-\frac{2}{x}y=\frac{3}{x^2}$$

$$\text{I.F}=e^{-\int\frac{2}{x}dx}=e^{-2\ln x}=e^{\ln x^{-2}}=\frac{1}{x^2}$$

$$y\cdot\frac{1}{x^2}=\int\frac{3}{x^4}dx=-\frac{1}{x^3}+C$$

$$y=Cx^2-\frac{1}{x}$$

$$y(1)=2,\quad\therefore 2=C-1\Rightarrow C=3$$

$$\therefore y(x)=3x^2-\frac{1}{x}$$

$$y(2)=3(4)-\frac{1}{2}=12-\frac{1}{2}=\frac{23}{2}$$

$$2y(2)=23$$

Hence, the correct answer is option (1).

Q. 18 Consider the following three statements for the function $f:(0,\infty)\to R$ defined by $f(x)=|\log_e x|-|x-1|$:

(I) f is differentiable at all $x>0$
(II) f is increasing in $(0,1)$.
(III) f is decreasing in $(1,\infty)$.

Then,

Option 1:
Only (I) and (III) are TRUE.

Option 2:
Only (I) is TRUE.

Option 3:
Only (II) and (III) are TRUE.

Option 4:
All (I), (II) and (III) are TRUE.

Correct Answer:
Only (I) and (III) are TRUE.

Solution:

$$f(x)=|\log_e x|-|x-1|+5$$

$$f(x)=\begin{cases}\log_e x-(x-1)+5, & x\geq 1\\ -\log_e x+(x-1)+5, & 0<x\leq 1\end{cases}$$

$$f'(x)=\begin{cases}\frac{1}{x}-1, & x\geq 1\\ -\frac{1}{x}+1, & 0<x\leq 1\end{cases}$$

$f'(x)<0$ for $\in(1,\infty)$

$f'(x)<0$ for $\in(0,1)$

$\therefore f(x)$ is decreasing in $(1,\infty)$ and also in $(0,1)$

$\therefore f(x)$ is differentiable in $(0,\infty)$

Hence, the correct answer is option (1).

Q. 19 The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha}=\frac{y-2}{-1}=\frac{z-4}{-\alpha}$ and $\frac{x}{\alpha}=\frac{y-1}{2}=\frac{z-1}{2\alpha}$ is $\sqrt{2}$, is:

Option 1:
-8

Option 2:
8

Option 3:
-6

Option 4:
6

Correct Answer:
-6

Solution:

$$\frac{x+1}{\alpha}=\frac{y-2}{-1}=\frac{z-4}{-\alpha},\frac{x}{\alpha}=\frac{y-1}{2}=\frac{z-1}{2\alpha}$$

$$SD=\frac{\begin{vmatrix}-1 & 1 & 3\\ \alpha & -1 & -\alpha\\ \alpha & 2 & 2\alpha\end{vmatrix}}{\begin{vmatrix}i & j & k\\ \alpha & -1 & -\alpha\\ \alpha & 2 & 2\alpha\end{vmatrix}}=\sqrt{2}$$

$$\Rightarrow\frac{-1(-2\alpha+2\alpha)+1\left(-\alpha^2-2\alpha^2\right)+3(2\alpha+\alpha)}{|i(-2\alpha+2\alpha)+j\left(-\alpha^2-2\alpha^2\right)+k(2\alpha+\alpha)|}=\sqrt{2}$$

$$\Rightarrow\left(-3\alpha^2+9\alpha\right)=\sqrt{2}\sqrt{9\alpha^4+9\alpha^2}$$

$$2\left(9\alpha^4+9\alpha^2\right)=9\alpha^2(-\alpha+3)^2$$

$$\Rightarrow 2\left(\alpha^2+1\right)=\alpha^2-6\alpha+9$$

$$\Rightarrow\alpha^2+6\alpha-7=0$$

$$\Rightarrow(\alpha+7)(\alpha-1)=0$$

$\alpha=-7,1$

Required Sum $=-6$

Hence, the correct answer is option (3).

Q. 20 Let $[t]$ denote the greatest integer less than or equal to t. If the function

$$f(x)=\begin{cases} b^2\sin\left(\frac{\pi}{2}\left[\frac{\pi}{2}(\cos x+\sin x)\cos x\right]\right), & x<0 \\ \frac{\sin x-\frac{1}{2}\sin 2x}{x^3}, & x>0 \\ a & x=0 \end{cases}$$

is continuous at $x=0$, then a^2+b^2 is equal to

Option 1:
$\frac{3}{4}$

Option 2:
$\frac{9}{16}$

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{5}{8}$

Correct Answer:
$\frac{3}{4}$

Solution:

$$\text{R.H.L: } \lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\frac{\sin x-\sin x\cos x}{x^3}$$

$$\lim_{x\to 0^+}\left(\frac{\sin x}{x}\right)\left(\frac{1-\cos x}{x^2}\right)=\frac{1}{2}$$

$$\therefore f(0)=a$$

$$\text{L.H.L: } \lim_{x\to 0^-} f(x)=\lim_{x\to 0^-} b^2\sin\frac{\pi}{2}\left[\frac{\pi}{2}(\sin x+\cos x)\cos x\right]=b^2$$

Since the function $f(x)$ is continuous at $x=0$ $\therefore$ L.H.L = R.H.L = $f(0)$

$\therefore b^2=a=\frac{1}{2}\Rightarrow a^2+b^2=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}$

Hence, the correct answer is option (1)

Q. 21 If $f(x)$ satisfies the relation $f(x)=e^x+\int_0^1 (y+xe^x)f(y)dy$, then $e+f(0)$ is equal to_____

Correct Answer:
2

Solution:

$$f(x) = e^x + \int_0^1 (y + xe^x) f(y) dy$$

$$= e^x + \int_0^1 yf(y)dy + xe^x \int_0^1 f(y)dy$$

$$f(x) = e^x + A + Bxe^x$$

$$A = \int_0^1 yf(y)dy = \int_0^1 y\left(A + e^y + Bye^y\right)dy$$

$$= \left[A \cdot \frac{y^2}{2} + ye^y - e^y + B\left(y^2e^y - 2ye^y + 2e^y\right)\right]_0^1$$

$$= \frac{A}{2} + (0 - 1) + B(e - 2)$$

$$A = \frac{A}{2} + B(e - 2) - 1$$

$$\frac{A}{2} = B(e - 2) - 1 \Rightarrow \frac{A}{2} + B(2 - e) = -1$$

$$B = \int_0^1 (e^y + A + Bye^y)dy$$

$$= \left[e^y + Ay + B\left(ye^y - e^y\right)\right]_0^1 = e - 1 + A + B(0 - (-1))$$

$$B = e - 1 + A + B$$

$$A = 1 - e$$

$$f(x) = e^x + A + Bxe^x$$

$$f(0) = 1 + A \Rightarrow f(0) = 1 + 1 - e$$

$$e + f(0) = 2$$

Hence, the correct answer is 2.

Q. 22 The number of elements in the set
$\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)\}$ is _____

Correct Answer:
4

Solution:

$$\tan(x + 100) = \tan\left(x + 50^0\right)\tan x \tan(x - 50)$$

$$\Rightarrow \frac{\tan(x + 100)}{\tan x} = \tan(x + 50)\tan(x - 50)$$

$$\Rightarrow \frac{\sin(x + 100)\cos x}{\cos(x + 100)\sin x} = \frac{\sin\left(x + 50^0\right)\sin\left(x - 50^0\right)}{\cos\left(x + 50^0\right)\cos\left(x - 50^0\right)}$$

Using Componendo and Dividendo, we get

$$2\sin(2x + 100)\cos 2x + \sin 200 = 0$$

$$\Rightarrow \sin(4x + 100) + \sin 100^0 + \sin(200) = 0$$

$$\Rightarrow \sin(4x + 100) = -2\sin 150 \sin 50$$

$$\sin(4x + 100) = -\sin(50) = \sin\left(-50^0\right)$$

$4x + 100 = n\pi + (-1)^n(-50)$

$x = \dfrac{n\pi + (-1)^{n+1}50 - 100^0}{4}, n \in I$

$x = \dfrac{130}{4}, \dfrac{210}{4}, \dfrac{490}{4}, \dfrac{570}{4}$ in $[0, 180]$

Number of solutions $= 4$

Hence, the correct answer is 4.

Q. 23 Let S be a set of 5 elements and $\mathrm{P(S)}$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $\mathrm{P(S)} \times \mathrm{P(S)}$ such that $A \cap B = \phi$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in N$, then $p + q$ is equal to:

Correct Answer:
15

Solution:
Let $S = \{a, b, c, d, e\}$ The power set $P(S)$ contains $= 2^5 = 32 = 2^{10}$ elements. An ordered pair (A, B) chosen from $P(S) \times P(S)$. The total number of possible pairs $= 32 \times 32$
For condition α each element $x \in S$ has exactly three mutually exclusive Possibilities.
i) $x \in A$ and $x \in B$
ii) $x \notin A$ and $x \in B$
iii) $x \notin A$ and $x \notin B$

Number of favourable pairs for the event $\mathrm{E} = 3^5$
The probability of even $\mathrm{E} = \mathrm{P(E)} = \frac{3^5}{2^{10}} = \frac{3^p}{2^q}$,

$\therefore p = 5, q = 10$
$\therefore p + q = 15$

Hence, the correct answer is 15.

Q. 24 Let $z = (1 + i)(1 + 2i)(1 + 3i) \ldots\ldots (1 + m)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to _____

Correct Answer:
5

Solution:
$z = (1 + i)(1 + 2i)(1 + 3i) \ldots (1 + ni)$
$|z| = \sqrt{1 + 1}\sqrt{1 + 4}\sqrt{1 + 9} \ldots . \sqrt{1 + n^2}$
$|z|^2 = 2.5.10.. \sqrt{n^2 + 1} \ldots\ldots\ldots\ldots\ldots\ldots$

But $|z|^2 = 44200 = 2^3 \cdot 5^2 \cdot 13 \cdot 17$ (given) which can be written as

$= 2.5.(2.5)(17)(2.13)$
$|z|^2 = 2.5.10.17.26 \ldots\ldots$

Comparing (i) and (ii) we get $n^2 + 1 = 26$

$\therefore n = 5$

Hence, the correct answer is 5.

Q. 25 Let (h, k) lie on the circle $C : x^2 + y^2 = 4$ and the point $(2h + 1, 3k + 2)$ lie on an ellipse with eccentricity e. Then the value of $\frac{5}{e^2}$ is equal to

Correct Answer:
9

Solution:
$x^2 + y^2 = 4$

Let $P = (2\cos\theta, 2\sin\theta)$, where $h = 2\cos\theta, k = 2\sin\theta \quad Q = (2h + 1, 3k + 2)$
Coordination of $Q = (4\cos\theta + 1, 6\sin\theta + 2)$
Locus of Q is $\frac{(x-1)^2}{16} + \frac{(y-2)^2}{36} = 1$

$$e^2 = 1 - \frac{16}{36} = \frac{5}{9}$$

$$\frac{5}{e^2} = 9$$

Hence, the correct answer is 9.

Physics

Q. 1 A block of mass 5 kg is moving on an inclined plane which makes an angle of $\mathbf{30}^{\circ}$ with the horizontal. Friction coefficient between the block and the inclined plane surface is $\frac{\sqrt{3}}{2}$. The force to be applied on the block so that the block will move down without acceleration is _____ N. $\left(g = 10\ \mathrm{m/s^2}\right)$.

Option 1:
15

Option 2:
7.5

Option 3:
12.5

Option 4:
25

Correct Answer:
12.5

Solution:
Sol:



$$F = mg(\sin\theta - \mu\cos\theta)$$

$$F = 5 \times 10\left(-\sin 30^{\circ} + \frac{\sqrt{3}}{2}\cos 30^{\circ}\right) = 50\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}\right) = 50\left(-\frac{1}{2} + \frac{3}{4}\right) = 12.5\ \mathrm{N}$$

Q. 2 Given below are two statements:
Statement I: A plane wave after passing through prism remains as plane wave but passing through small pin hole may become spherical wave.
Statement II: The curvature of a spherical wave emerging from a slit will increase for increasing slit width.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Statement I is true but Statement II is false

Option 3:
Statement I is false but Statement II is true

Option 4:
Both Statement I and Statement II are false

Correct Answer:
Statement I is true but Statement II is false

Solution:
Sol: Through plane slit cylindrical wave emerges

Q. 3 The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a planoconcave lens (refractive index $= 1.7$) are same. If the curvature of plano-concave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is _____ .

Option 1:
12:5

Option 2:
5:2

Option 3:
2:5

Option 4:
5:12

Correct Answer:
5:2

Solution:
Sol:

Let biconvex lens radius of curvature be R_1 and $R_2 \cdot P = (\mu_1 - 1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$

Let radius of curvature of plano concave lens $\rightarrow R_2 \cdot P = -\frac{(\mu_2 - 1)}{R_2}$

Since magnitudes are same: $(1.5 - 1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = (1.7 - 1)\left(\frac{1}{R_2}\right)$

$\frac{0.5}{R_1} + \frac{0.5}{R_2} = \frac{0.7}{R_2}$

$\frac{0.5}{R_1} = \frac{0.7 - 0.5}{R_2} \quad \frac{R_1}{R_2} = \frac{0.5}{0.2} = \frac{5}{2}$

Q. 4 In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell is shunted with 12Ω resistance, the balance is shifted to a length of 180 cm . The internal resistance of cell is _____ Ω

Option 1:
3

Option 2:
4

Option 3:
12

Option 4:
6

Correct Answer:
4

Solution:
Sol:Formula: $\frac{R+r}{R}=\frac{l_1}{l_2}$

$$\left(\frac{R_1+r}{R_1}\times\frac{R_2}{R_2+r}\right)=\frac{l_1}{120}\times\frac{180}{l_2}$$
$$\frac{12\times(4+r)}{4\times(12+r)}=\frac{3}{2}$$
$$8+2r=12+r$$
$$2r-r=12-8$$
$$r=4\Omega$$

Q. 5 Water drops fall from a tap on the floor, 5 m below at regular intervals of time. The first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground at the instant when the first drop strikes the ground is _____ m. $\left(g=10\ \mathrm{m/s^2}\right)$.

Option 1:
3.8

Option 2:
4.2

Option 3:
4.0

Option 4:
2.5

Correct Answer:
4.2

Solution:
Sol:

According to Galileo's law of odd numbers

$$\mathrm{d}_{6,5} : \mathrm{d}_{5,4} : \mathrm{d}_{4,3} : \mathrm{d}_{3,2} : \mathrm{d}_{2,1} = 1 : 3 : 5 : 7 : 9$$

Height of 4th drop from ground:

$$= d_{4,3} + d_{3,2} + d_{2,1}$$
$$= \frac{5+3+9}{9+7+5+3+1} \times 5$$
$$= \frac{21}{25} \times 5 = \frac{21}{5} = 4.2 \text{ m}$$

Q. 6 The electric field of an electromagnetic wave travelling through a medium is given by $E(x,t) = 25 \sin\left(2.0 \times 10^{15} t - 10^{7} x\right)\hat{n}$. Then the refractive index of the medium is _____ . (All given measurements are in SI units)

Option 1:
2

Option 2:
1.5

Option 3:
1.7

Option 4:
1.2

Correct Answer:
1.5

Solution:
Sol:

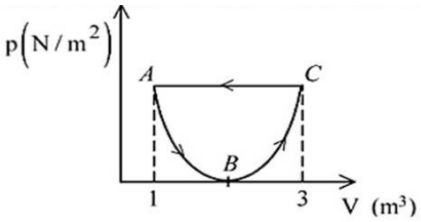
Velocity of wave,
On comparing $E(x,t) = 25 \sin\left(2.0 \times 10^{15} t - 10^{7} x\right)\hat{i}$

$$y = A \sin(\omega t - kx)$$
$$v = \frac{2 \times 10^{15}}{10^{7}} = 2 \times 10^{8}$$

Refractive index,

$\mu = \frac{c}{v} = \frac{3\times10^8}{2\times10^8} = 1.5$

Q. 7 In the following $\mathrm{p} - \mathrm{V}$ diagram the equation of state along the curved path is given by $(V-2)^2 = 4\mathrm{ap}$, where a is a constant. The total work done in the closed path is



Option 1:
$-\frac{1}{3a}$

Option 2:
$-\frac{1}{a}$

Option 3:
$+\frac{1}{2a}$

Option 4:
$+\frac{1}{3a}$

Correct Answer:
$-\frac{1}{3a}$

Solution:
Sol:

$\mathrm{If\ V} = 1 \Rightarrow 4ap_0 = (1-2)^2$

$p_0 = \frac{1}{4a}$

$\mathrm{Work\ Done} = \mathrm{Area\ of\ parabola\ ABC}$

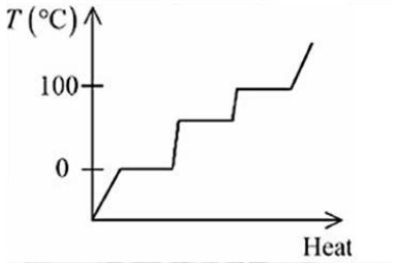
$= \mathrm{Area} = \frac{2}{3} \times \mathrm{Area\ of\ the\ recta\ n\ gle\ ABCD} = \frac{2}{3}p_0(3-1)$

$= \frac{4}{3}p_0 = \frac{4}{3} \times \frac{p_0}{4a}W = -\frac{p_0}{3a}$

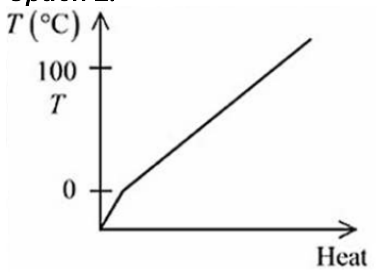
$\therefore\ \mathrm{Work\ done\ by\ gas} = -\frac{1}{3a}$

Q. 8 Which of the following best represents the temperature versus heat supplied graph for water, in the range of $-20^{\circ}\mathrm{C}$ to $120^{\circ}\mathrm{C}$?

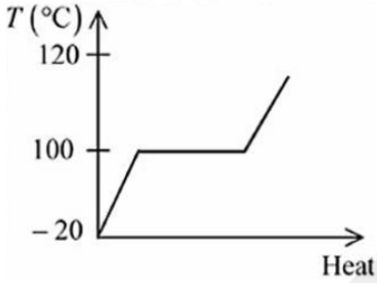
Option 1:



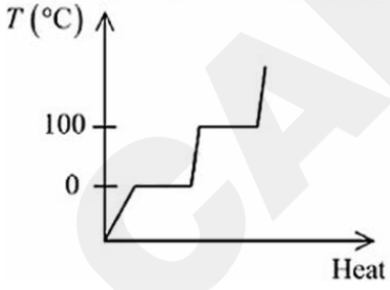
Option 2:



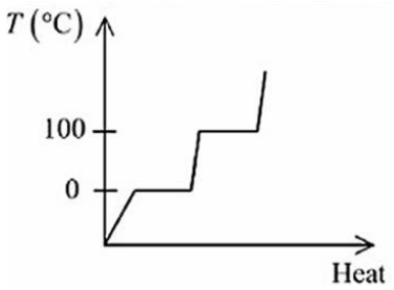
Option 3:



Option 4:



Correct Answer:



Solution:

Sol:

The graph is plotted between temperature (T) versus heat (Q) based on the following steps:
(i) $-20C$ to $0C$ implies Q is directly proportional to T
(ii) At $0C$, Q increases at constant T
(iii) $0C$ to $100C$ implies Q is directly proportional to T
(iv) At $100C$, Q increases at constant T
(v) $100C$ to $120C$ implies Q is directly proportional to T

Q. 9 The magnetic field at the centre of a current carrying circular loop of radius R is 16μ T. The magnetic field at a distance $x = \sqrt{3}R$ on its axis from the centre is ____ μT.

Option 1:
2

Option 2:
4

Option 3:
8

Option 4:
$2\sqrt{2}$

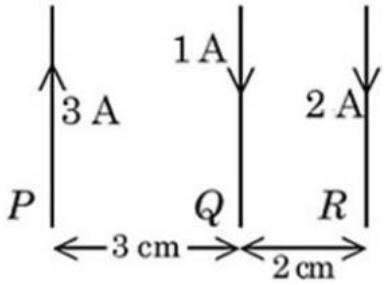
Correct Answer:
2

Solution:
Sol:

$$B = \frac{\mu_0 I R^2}{2(R^2+x^2)^{3/2}} \Rightarrow \frac{B_2}{B_1} = \left(\frac{R^2+x_1^2}{R^2+x_2^2}\right)^{3/2}$$

$$\Rightarrow B_2 = 16\left(\frac{R^2+0^2}{R^2+BR^2}\right)^{3/2} = 16\frac{R}{8R} = 2\mu \text{ T}$$

Q. 10 Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is _____ .



$(\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T.m/A})$

Option 1:
6×10^{-7} N towards R

Option 2:
6×10^{-6} N towards R

Option 3:
6×10^{-7} N towards P

Option 4:
6×10^{-6} N towards P

Correct Answer:
6×10^{-6} N towards R

Solution:
Force on Q = Force on Q due to $\mathrm{A}+$ Force on Q due to R

$$= l\left(\frac{\mu_0}{2\pi}\left(\frac{I_p I_q}{r_{pq}}\right) + \frac{\mu_0}{2\pi} \cdot \frac{I_q I_r}{r_{qr}}\right)$$

$$= \left(\frac{5 \times 10^{-7} \times 2 \times 10^{-7}}{10^{-2}}\right)\left(\frac{3 \times 1}{3} + \frac{1 \times 2}{2}\right) = 30 \times 10^{-7}(1+1)$$

$$= 60 \times 10^{-7} = 6 \times 10^{-6}\ \mathrm{N\ toward\ R}$$

Q. 11 An atom ${}^{8}_{3}\mathrm{X}$ is bombarded by shower of fundamental particles and in 10 s this atom absorbed 10 electrons, 10 protons and 9 neutrons. The percentage growth in the surface area of the nucleons is recorded by:

Option 1:
900%

Option 2:
225%

Option 3:
150%

Option 4:
250%

Correct Answer:
225%

Solution:
Sol:

(a) $A_x = 8, A_y = 8 + 10 + 9 = 27$
(b)Surface Area, $S = 4\pi R^2$

$$\frac{S_y}{S_x} = \left(\frac{R_y}{R_x}\right)^2 = \left(\frac{A_y}{A_x}\right)^{2/3} = \left(\frac{27}{8}\right)^{2/3} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

$\therefore$ Growth in surface area $= \frac{9}{4} \times 100 = 225\%$

Q. 12 The electric current in the circuit is given as $\mathrm{i} = \mathrm{i}_0(\mathrm{t}/\mathrm{T})$. The r.m.s current for the period $\mathrm{t} = 0$ to $\mathrm{t} = \mathrm{T}$ is ____ .

Option 1:
$\frac{i_0}{\sqrt{3}}$

Option 2:
$\frac{i_0}{\sqrt{2}}$

Option 3:
$\frac{i_0}{\sqrt{6}}$

Option 4:
i_0

Correct Answer:
$\frac{i_0}{\sqrt{3}}$

Solution:
Sol:

By definition:

$$i_{\text{rms}}^2 RT = \int_0^T i^2 R dt$$
$$i_{\text{rms}}^2 RT = R\int_0^T \left(\frac{i_0 t}{T}\right)^2 dt$$
$$i_{\text{rms}}^2 RT = R\frac{i_0^2}{T^2}\left[\frac{t^3}{3}\right]_0^T$$
$$i_{\text{rms}}^2 = \frac{i_0^2}{T^3}\frac{1}{3}(T^3 - 0^3)$$
$$i_{\text{rms}}^2 = \frac{i_0^2}{T^3} \times \frac{T^3}{3}$$
$$i_{\text{rms}} = \frac{i_0}{\sqrt{3}}$$

Q. 13 When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4th mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5th division of vernier scale coincides with a main scale division. Measured length of cylinder is _____ mm.
(Least count of Vernier calliper $= 0.1\ \text{mm}$)

Option 1:
15.5

Option 2:
15.1

Option 3:
15.4

Option 4:
15.9

Correct Answer:
15.1

Solution:
Sol:

$$\text{Measured depth of cylinder}$$
$$= \text{MSD} + \text{LC} \times \text{ Vernier coincidence}$$
$$= 15 + 0.1(5-4) = 15 + 0.1 = 15.1 \text{ mm}$$

Q. 14 Two wires A and B made of different materials of lengths 6.0 cm and 5.4 cm , respectively and area of cross sections $3.0 \times 10^{-5}\ \mathrm{m}^2$ and $4.5 \times 10^{-5}\ \mathrm{m}^2$, respectively are stretched by the same magnitude under a given load. The ratio of the Young's modulus of A to that of B is $x : 3$. The value of x is _____ .

Option 1:
1

Option 2:
2

Option 3:
5

Option 4:
4

Correct Answer:
5

Solution:
Sol:

$$l = \frac{mg}{\left(\frac{y_A}{L}\right)}$$
$$\frac{l_A}{l_B} = \frac{L_A Y_B A_B}{Y_A A_B L_B}$$
$$l_A = l_A(\text{ given })$$
$$\frac{l_A}{l_B} = \frac{6 \times y_B \times 4.5 \times 10^{-5}}{5.4 \times 3 \times 10^{-5} \times y_A}$$
$$\frac{y_A}{y_B} = \frac{5}{3}$$

Q. 15 10 kg of ice at $-10^{\circ}\mathrm{C}$ is added to 100 kg of water to lower its temperature from $25^{\circ}\mathrm{C}$. Consider no heat exchange to surroundings. The decrement to the temperature of water is _____ $^{\circ}\mathrm{C}$. (Specific heat of ice $= 2100\ \mathrm{J/Kg.}\ ^{\circ}\mathrm{C}$, specific heat of water $= 4200\ \mathrm{J/Kg.}\ ^{\circ}\mathrm{C}$, latent heat of fusion of ice $= 3.36 \times 10^5\ \mathrm{J/Kg}$)

Option 1:
11.6

Option 2:
10

Option 3:
6.67

Option 4:
15

Correct Answer:
10

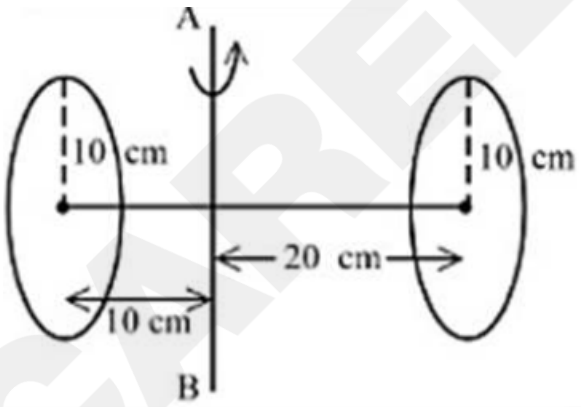
Solution:
Sol:

Heat lost by rice = Heat given by water

$$(10\ \text{kg})(2100\ \text{J/Kg}\cdot\text{C})(10)\text{C} + (10\ \text{kg})\left(3.36\times10^{5}\ \text{J/Kg}\right) + (10\ \text{kg})(4200\ \text{J/Kg}\cdot\text{C})T$$
$$= (10\ \text{kg})(4200\ \text{J/Kg}\cdot\text{C})(25\text{C} - T)$$
$$\text{T} = 15^{\circ}\text{C}\ (\text{final temperature})$$

Decrement $= (25-15)^{\circ}\text{C} = 10^{\circ}\text{C}$

Q. 16 Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure. If the mass of each disc is 600 gm and applied torque between two discs is 43×10^{5} dyne. cm , the angular acceleration of the discs about the given axis AB is $____\ \text{rad/s}^2$.



Option 1:
11

Option 2:
27

Option 3:
22

Option 4:
100

Correct Answer:
11

Solution:
Sol:

$$\text{Torque } = I\alpha$$

$$\alpha = \frac{43 \times 10^5}{\left(\frac{mR^2}{2} + ml_1^2\right) + \left(\frac{mR^2}{2} + ml_2^2\right) + \left(\frac{mR^2}{12} + m(d)^2\right)}$$

$m = 600 \text{ g} = 0.6 \text{ kg}; R = 10 \text{ cm} = 0.1 \text{ m}; l_1 = 10 \text{ cm} = 0.1 \text{ m}; l_2 = 10 \text{ cm} = 0.1 \text{ m}; d = 5 \text{ cm} = 0.05 \text{ m}$

$$\alpha = \frac{43 \times 10^5}{2\left(\frac{(0.6)(0.1)^2}{2} + (0.6)(0.1)^2\right) + \left(\frac{(0.6)(0.1)^2}{12} + (0.6)(0.05)^2\right)}$$

$\alpha \approx 11 \text{ cm}$

Q. 17 For the two cells having same EMF E and internal resistance r, the current passing through the external resistor 6Ω is same when both the cells are connected either in parallel or in series. The value of internal resistance r is _____ Ω.

Option 1:
3

Option 2:
6

Option 3:
4

Option 4:
9

Correct Answer:
6

Solution:
Sol:

$$\frac{2E}{2r+6} = \frac{\left(\frac{\frac{E}{r}}{\frac{1}{r}} + \frac{\frac{E}{r}}{\frac{1}{r}}\right)}{\frac{1}{\frac{1}{r}+\frac{1}{r}} + 6} \Rightarrow \frac{2E}{2r+6} = \frac{2E}{r+12} \Rightarrow r = 6\Omega$$

Q. 18 Two-point charges of 1 nC and 2 nC are placed at the two corners of equilateral triangle of side 3 cm . The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is ____ μJ.

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 N.m^2/C^2$

Option 1:
5.4

Option 2:
2.7

Option 3:
27

Option 4:
3.3

Correct Answer:
2.7

Solution:
Sol:

$$\mathrm{W.D} = \Delta U$$

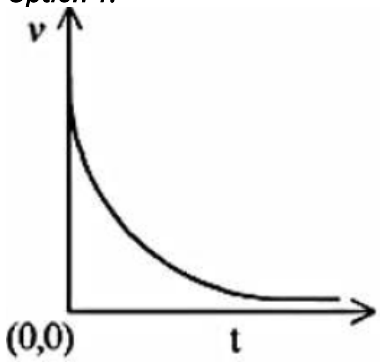
$$W.D = \frac{Kq_1q}{a} + \frac{Kq_2q}{a}$$

$$= \left(\frac{1}{4\pi\epsilon_0}\right)\frac{\left(1\times10^{-9}C\right)\left(3\times10^{-9}C\right)}{3\times10^{-2}m} + \frac{\left(2\times10^{-9}C\right)\left(3\times10^{-9}C\right)}{3\times10^{-2}m}$$

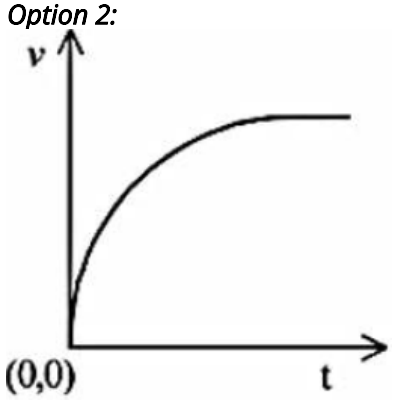
$$= 2.7\times10^{-6}J = 2.7\mu J$$

Q. 19 Particle of mass m falls from rest through a resistive medium having resistive force $\mathrm{F} = -\mathrm{kv}$, where v is the velocity of the particle and k is a constant. Which of the following graphs represents velocity (v) versus time (t)?

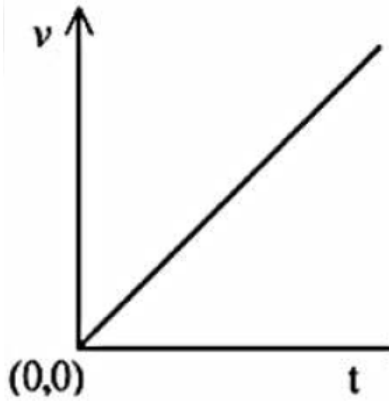
Option 1:



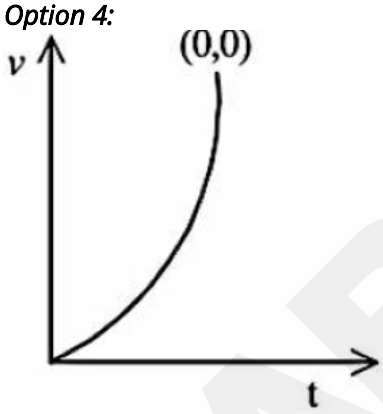
Option 2:



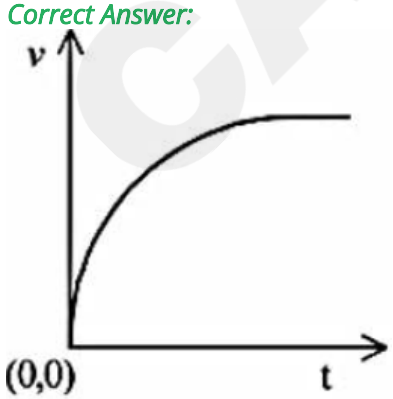
Option 3:



Option 4:



Correct Answer:



Solution:

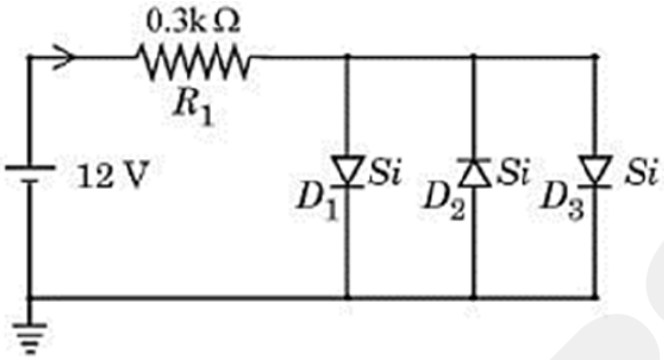
Sol:

$$F_{net} = mg - kv$$

$$m\frac{dv}{dt} = mg - kv$$
$$\int_0^v \frac{dv}{g - \frac{kv}{m}} = \int_0^t dt$$
$$v = \frac{mg}{k}\left(1 - e^{-kt/m}\right)$$

This equation represents exponentially increasing curve that starts from the origin

Q. 20 Assuming in forward bias condition there is a voltage drop of 0.7 V across a silicon diode, the current through diode D_1 in the circuit is _____ mA .
(Assume all diodes in the given circuit are identical)



Option 1:
20.15

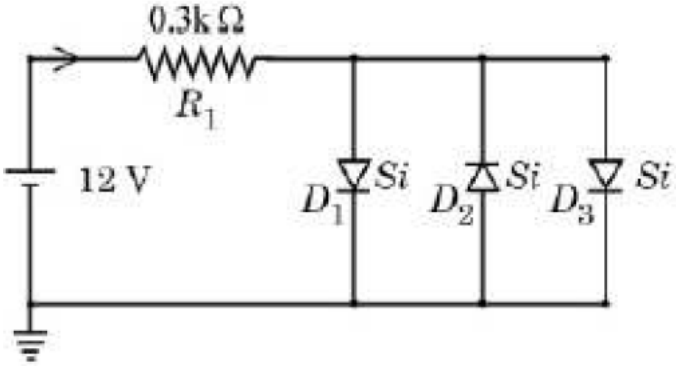
Option 2:
11.7

Option 3:
17.6

Option 4:
18.8

Correct Answer:
18.8

Solution:



$i = \frac{12-0.7}{0.3\times10^{+3}} = 37.6\ \text{mA}$

Current through each diode

$i/2 = 18.8\ \text{mA}$

Q. 21 The displacement of a particle, executing simple harmonic motion with time period T, is expressed as $\mathrm{x(t)} = \mathrm{A}\sin\omega\mathrm{t}$, where A is the amplitude. The maximum value of potential energy of this oscillator is found at $\mathrm{t} = \mathrm{T}/2\beta$. The value of β is _____

Correct Answer:
2

Solution:
Sol:

PE is maximum at extreme position

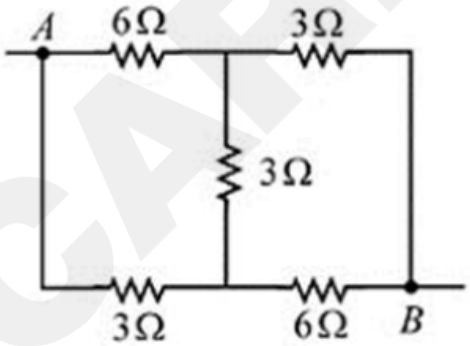
$t = \frac{T}{4}$

$\frac{T}{2\beta} = \frac{T}{4}$

Therefore, the value

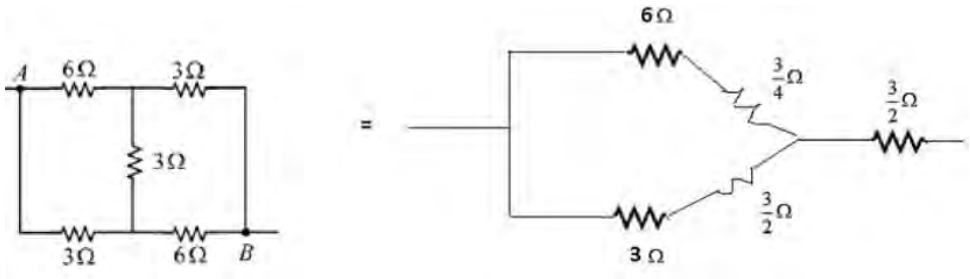
$\beta = 2$

Q. 22 The equivalent resistance between the points A and B in the following circuit is $\frac{\mathrm{x}}{5}\Omega$. The value of x is _____ .



Correct Answer:
21

Solution:
Sol:Given circuit is Unbalanced Wheatstone bridge so applying delta to star conversion



$R_{AB} = \frac{21}{5}\Omega$

Q. 23 The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy 2E, is $n : 1$. The value of n is _____ .
(Assume mass of proton = mass of neutron).

Correct Answer:
2

Solution:
Sol:

DeBrogile wavelength $\lambda = \frac{h}{\sqrt{2mE}}$

$\frac{\lambda_D}{\lambda_\alpha} = \frac{\frac{h}{\sqrt{2\times 2m\times E}}}{\frac{h}{\sqrt{2\times 4m\times 2E}}} = 2$

Q. 24 A solid sphere of radius 10 cm is rotating about an axis which is at a distance 15 cm frol its centre. The radius of gyration about this axis is $\sqrt{n}$ cm. The value of n is _____ .

Correct Answer:
265

Solution:
Sol:

By applying parallel axis theorem

$\frac{2}{5}mR^2 + md^2 = mK^2$
$\frac{2}{5}R^2 + d^2 = K^2$
$\frac{2}{5}(10)^2 + (15)^2 = K^2$
$\frac{200}{5} + 225 = K^2$
$K = \sqrt{265}$ cm

Q. 25 A convex lens of refractive index 1.5 and focal length $\mathrm{f} = 18\ \mathrm{cm}$ is immersed in water. The difference in focal lengths of the given lens when it is in water and in air is $\alpha \times \mathrm{f}$. The value of α is _____ .
(refractive index of water $= 4/3$)

Correct Answer:
3

Solution:
Sol:

In air

$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \frac{1}{18} = (1.5 - 1)\left(\frac{2}{R}\right)$.....(i)

In water $\frac{1}{f} = \left(\frac{1.5}{4/3} - 1\right)\left(\frac{2}{R}\right)$.....(ii)

$(i) \div (ii)$

$\frac{f}{18} = 4 \Rightarrow f = 72 \quad f - 18 = 54 = 3 \times 18$

Chemistry

Q. 1 $20.0\mathrm{dm}^3$ of an ideal gas ' X ' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until the pressure of the gas is 0.2 MPa. Which of the following options is correct?
(Given: $\log 2 = 0.3010$ and $\log 5 = 0.6989$)

Option 1:
$\mathrm{w} = -3.9\ \mathrm{kJ}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0; \mathrm{q} = 3.9\ \mathrm{kJ}$

Option 2:
$\mathrm{w} = +4.1\ \mathrm{kJ}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0; \mathrm{q} = -4.1\ \mathrm{kJ}$

Option 3:
$\mathrm{w} = 9.1\ \mathrm{J}, \Delta\mathrm{U} = 9.1\ \mathrm{J}, \Delta\mathrm{H} = 0; \mathrm{q} = 0$

Option 4:
$\mathrm{w} = -9.1\ \mathrm{J}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0, \mathrm{q} = 9.1\ \mathrm{kJ}$

Correct Answer:
$\mathrm{w} = -9.1\ \mathrm{J}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0, \mathrm{q} = 9.1\ \mathrm{kJ}$

Solution:

Isothermal expansion

$\Delta U = 0 \Rightarrow -W = q = nRT \ln \frac{P_1}{P_2}$

moles of gas $n = \frac{PV}{RT} = \frac{0.5 \times 10^6 \times 9.86 \times 10^{-6}}{0.0821 \times 600} \times 2 = 2$

$-w = 2 \times 2.303 \times 8.314 \times 600 \times \log \frac{0.5}{0.2} = 9.1KJ \therefore +9.1$

$w = -9.1KJ$

$q = 9.1KJ$

Hence, the correct answer is option (4)

Q. 2 Consider a weak base ' B ' of $pK_b = 5.699$. ' x ' mL of 0.02 M HCl and ' y ' mL of 0.02 M weak base ' B ' are mixed to make 100 mL of a buffer of pH 9 at $25C$. The values of ' x ' and ' y ' respectively are:
[Given: $\log 2 = 0.3010, \log 3 = 0.4771, \log 5 = 0.699$]

Option 1:

x	y
11.1	88.9

Option 2:

x	y
14.3	85.7

Option 3:

x	y
42.7	57.3

Option 4:

x	y
85.7	14.3

Correct Answer:

x	y
14.3	85.7

Solution:

$pH = 9 \therefore pOH = 14 - 9 = 5$

$$pOH = pKb + \log \frac{[CA]}{[B]}$$

$[CA] = xml \times 0.02$

$[B] = yml \times 0.02$

Total volume is $100\text{ml} \therefore y = 100 - x$

$$5 = 5.699 + \log\left(\frac{x \times 0.02}{(100 - x) \times 0.02}\right)$$

$x = 16.66 \quad y = 83.34$

Hence, the correct answer is option (2)

Q. 3

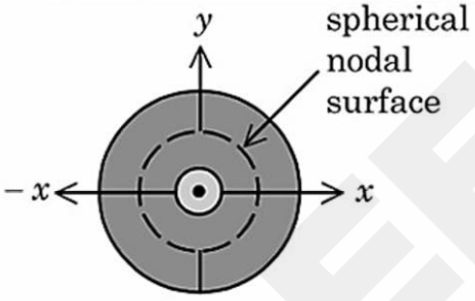


Figure 1. electron probability density for 2s orbital

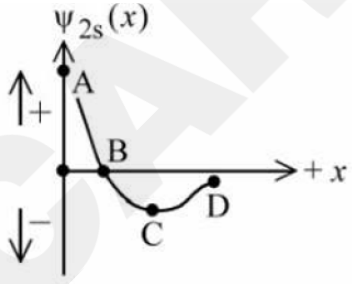


Figure 2. wave function for 2s orbital

Which of the following point in Figure 2 most accurately represents the nodal surface as shown in Figure 1?

Option 1:

C

Option 2:
B

Option 3:
D

Option 4:
A

Correct Answer:
B

Solution:
Sol:

In $\Psi(r) vs r$ plot at radial node Ψ curve intersects the r axis

For hydrogen-like orbitals

Number of radial nodes=n−l−1\text{Number of radial nodes} = n - l - 1Number of radial nodes=n−l−1

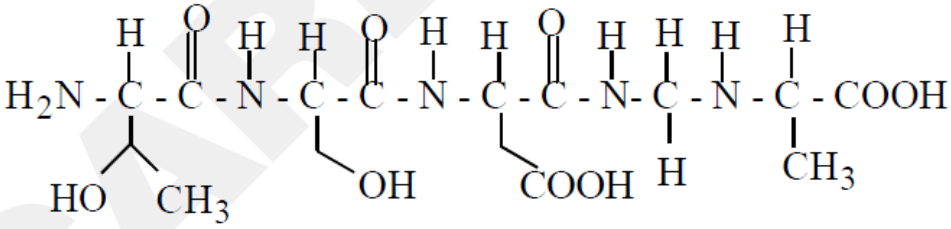
For 2s orbital:

2−0−1=12 - 0 - 1 = 12−0−1=1

So there is one spherical nodal surface, corresponding to point B on the graph.

Hence the correct answer is option (B).

Q. 4 In the given pentapeptide, find out an essential amino acid (Y) and the sequence present in the pentapeptide:



Choose the correct answer from the options given below:

Option 1:

(Y)	(Sequence)
Serine	Thr – Ser – Asp – Ala - Gly

Option 2:

(Y)	(Sequence)
Threonine	Thr – Ser – Asp – Gly - Ala

Option 3:

(Y)	(Sequence)
Threonine	Ser – Thr – Asp – Gly - Ala

Option 4:

(Y)	(Sequence)
Serine	Thr – Ser – Asp – Ala - Gly

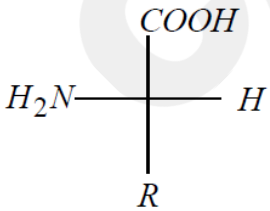
Correct Answer:

(Y)	(Sequence)
Threonine	Thr – Ser – Asp – Gly - Ala

Solution:
Sol:

Thr - Ser - Asp - Gly - Ala

Threonine is essential amino acid
$\therefore$ Natural amino acids general form



$\therefore R \Rightarrow \mathrm{H_3C-CHOH-}$ (Threonine) (essential Amino acid)

$R \Rightarrow \mathrm{HO-CH_2-}$ (Serine)
$R \Rightarrow \mathrm{CH_3-}$ (Alanine)
$R \Rightarrow \mathrm{HS-CH_2-}$ (Cysteine)
$R \Rightarrow \mathrm{HOOC-CH_2-}$ (Aspartic acid)
$R \Rightarrow \mathrm{H-}$ (Glycine)

Hence, the correct answer is option (2)

Q. 5 In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively.

Option 1:
K&Br

Option 2:
Rb&Br

Option 3:
Na&Cl

Option 4:
K&Se

Correct Answer:
K&Br

Solution:
In a given period, the atomic radius decreases from left to right due to an increase in effective nuclear charge. However, the group 17 elements are smaller in size (due tovan-der-Walll forces in Group-18) than the corresponding group 18 elements due to (i) increased cumulative shielding of the electrons that outweighs the nuclear pull.

K	235 pm
Br	114pm(Covalent) 186 (Vander wad)
Kr	116 pm (Covalent) 202pm Calculated

Hence, the correct answer is option (1)

Q. 6 The correct statement among the following is:

Option 1:
$Ni(CO)_4$ is diamagnetic and $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic

Option 2:
$[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic

Option 3:
$Ni(CO)_4$ and $[NiCl_4]^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic

Option 4:
$Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[NiCl_4]^{2-}$ is paramagnetic

Correct Answer:
$Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[NiCl_4]^{2-}$ is paramagnetic

Solution:
For Nickel tetracarbonyl, $Ni(CO)_4$
Nickel is in the 0 oxidation state.
Electronic configuration of Ni:

$[Ar]3d^8 4s^2$
After bonding with CO (strong field ligand), electrons pair up and the complex forms tetrahedral sp^3 hybridisation with all electrons paired.

Hence, $Ni(CO)_4$ is diamagnetic.

For $[Ni(CN)_4]^{2-}$
Oxidation state of Ni :

$$x + 4(-1) = -2$$
$$x = +2$$

Ni^{2+} configuration:

$3d^8$
Cyanide ion is a strong field ligand, so electrons pair up and the complex forms square planar dsp^2 hybridisation. Thus $[Ni(CN)_4]^{2-}$ is diamagnetic.

For $[NiCl_4]^{2-}$
Ni is again $+2\ (3\ d^8)$.

The chloride ion is a weak-field ligand, so electrons do not pair.

The complex forms a tetrahedral geometry and contains two unpaired electrons.
Thus $[NiCl_4]^{2-}$ is paramagnetic.

Hence, the correct answer is option (4)

Q. 7 An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{\text{th}}$ and $\left(\frac{1}{10}\right)^{\text{th}}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$?

$(\log 2 = 0.3)$

Option 1:
0.9

Option 2:
30

Option 3:
3

Option 4:
9

Correct Answer:
9

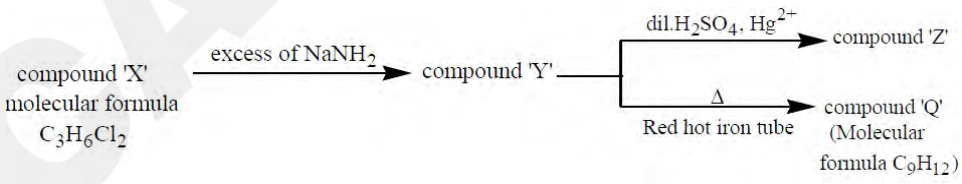
Solution:

$$\frac{t_{1/8}}{t_{1/10}} = \frac{\frac{2.303}{K}\log\frac{[A_O]}{1/8[A_O]}}{\frac{2.303}{K}\log\frac{[A_O]}{1/10[A_O]}} = 0.9$$

$$\frac{t_{1/8}}{t_{1/10}} \times 10 = 9$$

Hence, the correct answer is option (4)

Q. 8 Given below are two statements for the following reaction sequence.



Statement I : Compound 'Z' will give yellow precipitate with NaOI.
Statement II: Compound 'Q' has two different types of 'H' atoms (aromatic : aliphatic) in the ratio 1 : 3
In the light of the above statements, choose the correct answer from the options

Option 1:
Statement I is true but Statement II is false

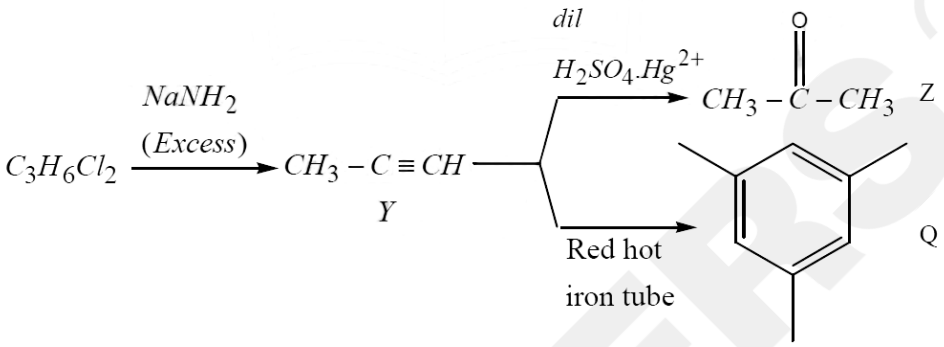
Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is false but Statement II is true

Option 4:
Both Statement I and Statement II are true

Correct Answer:
Both Statement I and Statement II are true

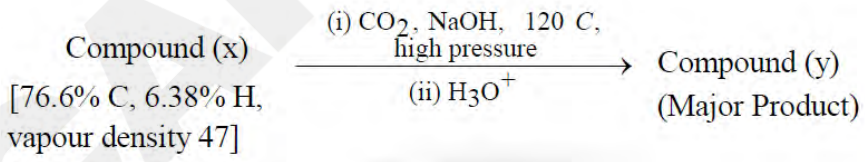
Solution:
Sol:



Z is methyl ketone and hence gives the iodoform test
Compound Q has three aromatic and Nine Aliphatic Hydrogens

Hence, the correct answer is option (4)

Q. 9 Consider the following reaction sequence



Compound (y) develops characteristic colour with neutral FeCl3 solution. Identify the INCORRECT statement from the following for the above sequence.

Option 1:
Both compounds x and y will dissolve in NaOH.

Option 2:
Compound y will dissolve in NaHCO3 and evolve a gas.

Option 3:
Both compounds x and y will burn with sooty flame

Option 4:
Compound x is more acidic than compound y.

Correct Answer:
Compound x is more acidic than compound y.

Solution:
Sol:

$2 \times V \cdot D = \text{ Molar mass}$
$2 \times 47 = 94$
$\text{Gives colour with neutral } FeCl_3 \therefore \text{ Compound is phenol}$

$$\underset{\substack{[76.6\%\ C,\ 6.38\%\ H,\\ \text{vapour density } 47]}}{\text{Compound (x)}} \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CO_2,\ NaOH,\ 120\ C,\ \text{high pressure}} \underset{\text{(Major Product)}}{\text{Compound (y)}}$$

Both $X \& Y$ are soluble in NaOH
Y will dissolve in $NaHCO_3$ and evolve CO_2 gas
Both X and Y being aromatic will burn with sooty flame
However salicylic acid is more acidic than phenol.

Hence, the correct answer is the option (4).

Q. 10 CORRECT order of stability for the following is

$$CH_2 = CH^-, CH_3 - CH_2^-, CH \equiv C$$

Option 1:
$CH_3 - CH_2^- > CH_2 = CH^- > CH \equiv C^-$

Option 2:
$CH_2 = CH^- > CH \equiv C^- > CH_3 - CH_2^-$

Option 3:
$CH \equiv C^- > CH_3 - CH_2^- > CH_2 = CH^-$

Option 4:
$CH \equiv C^- > CH_2 = CH^- > CH_3 - CH_2^-$

Correct Answer:
$\mathrm{CH \equiv C^- > CH_2 = CH^- > CH_3 - CH_2^-}$

Solution:
$\mathrm{CH_3 - CH_2 - H}$ $\quad$ $\mathrm{CH_2 = CH - H}$ $\quad$ $\mathrm{HC \equiv C - H}$
$\mathrm{Sp^3}$ $\quad$ Sp^2 $\quad$ Sp

The higher the 's' nature, the more is the stability of the conjugate base. Hence stability order

$\mathrm{CH_3 - \overline{CH_2} < CH_2 = \overline{CH} < HC \equiv \overline{C}}$

Hence, the correct answer is option (4)

Q. 11 The wave numbers of three spectral lines of the H atom are considered. Identify the set of spectral lines belonging to the Balmer series.
(R = Rydberg constant)

Option 1:
$\frac{3R}{4}, \frac{3R}{16}, \frac{7R}{144}$

Option 2:
$\frac{7R}{144}, \frac{3R}{16}, \frac{16R}{255}$

Option 3:
$\frac{5R}{36}, \frac{8R}{9}, \frac{15R}{16}$

Option 4:
$\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$

Correct Answer:
$\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$

Solution:
For Balmer series $\mathrm{n_i}$(or)$\mathrm{n_f} = 2$
For the absorption spectrum of the H atom

$\mathrm{n_i}$	$\mathrm{n_f}$	
2	3	$\bar{v} = R\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = \frac{5R}{36}$
2	4	$\bar{v} = R\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = \frac{3R}{16}$
2	5	$\bar{v} = R\left[\frac{1}{2^2} - \frac{1}{5^2}\right] = \frac{21R}{100}$

Hence, the correct answer is option (4)

Q. 12 Given below are two statements:

Statement I: The number of pairs, from the following, in which both the ions are coloured in aqueous solutions is 3.

$\left[Sc^{3+}, Ti^{3+}\right], \left[Mn^{2+}, Cr^{2+}\right], \left[Cu^{2+}, Zn^{2+}\right]$ and $\left[Ni^{2+}, Ti^{4+}\right]$

Statement II: Th^{4+} is the strongest reducing agent among Th^{4+}, Ce^{4+}, Gd^{3+} and Eu^{2+} In the light of the above statements, choose the correct answer from the options given below

Option 1:
Statement I is false but Statement II is true

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II are false

Solution:
A transition metal ion is coloured in aqueous solution when it has partially filled d-orbitals (d^1–d^9). Ions with d^0 or d^{10} configuration are colourless.

$\left[Sc^{3+}, Ti^{3+}\right]$

$Sc^{3+} \rightarrow 3\,d^0 \rightarrow$ colourless

$Ti^{3+} \rightarrow 3\,d^1 \rightarrow$ coloured

Both are not coloured → pair not counted.

$\left[Mn^{2+}, Cr^{2+}\right]$

$Mn^{2+} \rightarrow 3\,d^5 \rightarrow$ coloured

$Cr^{2+} \rightarrow 3\,d^4 \rightarrow$ coloured

Both coloured → counted.

$\left[Cu^{2+}, Zn^{2+}\right]$

$Cu^{2+} \rightarrow 3\,d^9 \rightarrow$ coloured

$Zn^{2+} \rightarrow 3\,d^{10} \rightarrow$ colourless

Both not coloured → not counted.
$[\mathrm{Ni}^{2+}, \mathrm{Ti}^{4+}]$
$\mathrm{Ni}^{2+} \rightarrow 3\ \mathrm{d}^{8} \rightarrow$ coloured
$\mathrm{Ti}^{4+} \rightarrow 3\ \mathrm{d}^{0} \rightarrow$ colourless
Both not coloured → not counted.

Thus, only 1 pair has both ions coloured, not 3.
So Statement I is incorrect.

Now check Statement II.

Compare the reducing strength of
$\mathrm{Th}^{4+}, \mathrm{Ce}^{4+}, \mathrm{Gd}^{3+}, \mathrm{Eu}^{2+}$.

A strong reducing agent easily loses electrons to form a more stable state. Eu^{2+} easily oxidizes to Eu^{3+} (stable half-filled $4f^7$ configuration), so Eu^{2+} is a strong reducing agent.

Therefore, Th^{4+} is not the strongest reducing agent.

Hence, the correct answer is option (3)

Q. 13 Given below are two statements:
Statement I: The number of species among BF_4^-, SiF_4, XeF_4 and SF_4, that have unequal E-F bond lengths is two. Here, E is the central atom.
Statement II: Among O_2^-, O_2^{2-}, F_2 and O_2^+, O_2^- has the highest bond order In the light of the above statements, choose the correct answer from the options given below

Option 1:
Statement I is false but Statement II is true

Option 2:
Both Statement I and Statement II are true

Option 3:
Both Statement I and Statement II are false

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II are true

Solution:
Both Statements are false $S-IBF_4^-$, SiF_4 and XeF_4 with Sp^3, Sp^3 and Sp^3d^2 hybridisation. The shapes are tetrahedral, tetrahedral, and square planar. All have equal $E-F$ Bond length $\mathrm{S}-\mathrm{IIO}_2^+$ bond order is 2.5 and $\mathrm{O}_2^- 1.5$

Hence, the correct answer is option (2)

Q. 14 At $\mathrm{T(K)}, 2$ moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of the ideal solution formed is 320 mm Hg. At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg. The vapour pressure in mm Hg) of A and B are respectively:

Option 1:
300,200

Option 2:
600,400

Option 3:
500,200

Option 4:
400,300

Correct Answer:
500,200

Solution:
2 moles of A+ 3 moles of B are mixed

$\mathrm{X_A} = 2/5, \mathrm{X_B} = 3/5$
$\mathrm{Ps} = \mathrm{X_A} P_A^0 + \mathrm{X_B} P_B^0$
$320 = P_A^0 \left(\frac{2}{5}\right) + P_B^0 \left(\frac{3}{5}\right)$
$2P_A^0 + 3P_B^0 = 1600 \ldots\ldots\ldots$

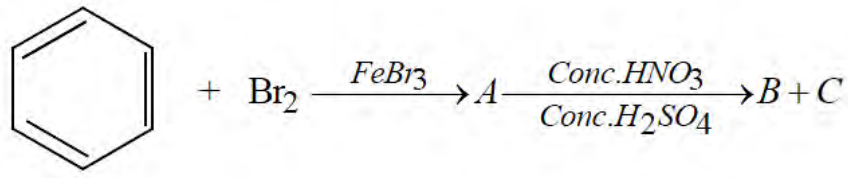
Now, 1 mole of A & 1 mole of B are added

$X'_A = \frac{3}{7}, X'_B = \frac{4}{7}$

$P'_S = 328.6 = P_A^0 \left(\frac{3}{7}\right) + P_B^0 \left(\frac{4}{7}\right)$
$3P_A^0 + 4P_B^0 = 2300 \ldots\ldots\ldots (\mathrm{II})$
$\text{Now eq } (1) \times 3 - \text{eq } (\mathrm{II}) \times 2$
$6P_A^0 + 9P_B^0 = 4800$
$6P_A^0 + 8P_B^0 = 4600$
$P_B^0 = 200 \text{ mm of Hg}$
$P_A^0 = 500 \text{ mm of Hg}$

Hence, the correct answer is option (3)

Q. 15 Method used for separation of mixture of products (B and C) obtained in the following reaction is



Option 1:
Fractional distillation

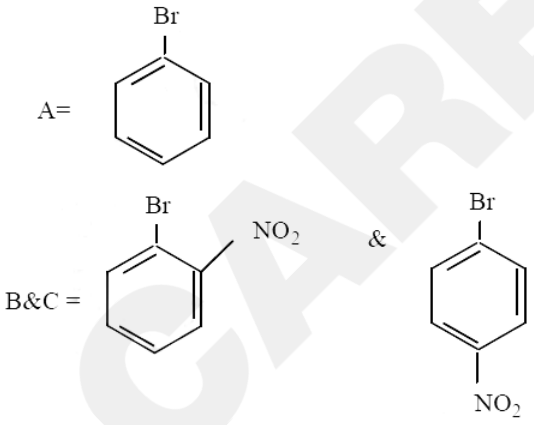
Option 2:
simple distillation

Option 3:
steam distillation

Option 4:
sublimation

Correct Answer:
Fractional distillation

Solution:
Due to their difference in boiling points



The para isomer is more symmetrical, so it packs better in the crystal lattice and has a higher melting point, while the ortho isomer has a lower melting point and remains more soluble.

Therefore, the mixture is separated by fractional crystallization.

Hence, the correct answer is option (1)

Q. 16 Regarding the hydrides of group 15 elements $\mathrm{EH_3}(E = N, P, As, Sb)$, select the correct statement from the following:

A) The stability of hydrides decreases down the group
B) The basicity of hydrides decreases down the group
C) The reducing character increases down the group
D) The boiling point increases down the group

Choose the correct answer from the options given below:

Option 1:
A, B & C only

Option 2:
A, B, C & D

Option 3:
A & D only

Option 4:
B & C only

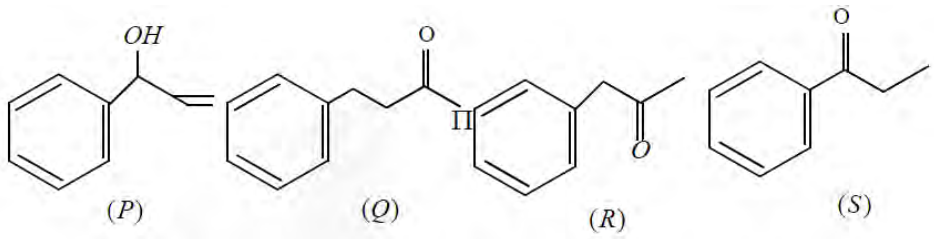
Correct Answer:
A, B & C only

Solution:
Group 15 elements N, P, As, and Sb form covalent hydrides of the type EH_3, namely Ammonia (NH_3), Phosphine (PH_3), Arsine (AsH_3), and Stibine (SbH_3). As we move down the group, the atomic size of the central atom increases and the E–H bond becomes weaker, so the stability of hydrides decreases down the group, making statement A correct. The basicity also decreases down the group because the lone pair present on the central atom becomes less available for donation due to the increase in atomic size and decrease in electronegativity, so statement B is correct. Since stability decreases, these hydrides can release hydrogen more easily, and thus their reducing character increases down the group, making statement C correct. However, statement D is incorrect because the boiling point does not increase regularly down the group; Ammonia has an unusually high boiling point due to hydrogen bonding, which disturbs the expected trend. Therefore, the correct answer is A, B, and C only.

Boiling point order $\mathrm{PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3}$

Hence, the correct answer is option (1)

Q. 17 Given below are the four isomeric compounds (P, Q, R, S)



Identify the correct statements from below.
A. Q, R, and S will give precipitate with 2,4-DNP.
B. P and Q will givea positive Baeyer's test.
C. Q and R will give a sooty flame.
D. R and S will give yellow precipitate with $I_2/NaOH$.
E. Q alone will deposit silver with Tollen's reagent.

Choose the correct option.

Option 1:
A, C, and E only

Option 2:
C and E only

Option 3:
A and E only

Option 4:
A, B, D and E only

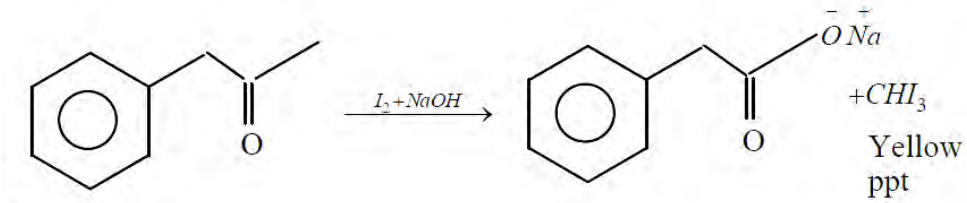
Correct Answer:
A, C, and E only

Solution:
P contains an alcohol group with an alkene, so it does not have a carbonyl group. Q, R, and S contain the carbonyl (>C=O) functional group (aldehyde or ketone). Compounds containing carbonyl groups react with the 2,4-Dinitrophenylhydrazine test to give an orange or yellow precipitate; therefore, Q, R, and S give a 2,4-DNP test, making statement A correct.

In Baeyer's test, dilute alkaline Potassium permanganate detects unsaturation (C=C or C≡C). Only P contains a double bond, while Q does not contain such unsaturation, so statement B is incorrect. Aromatic compounds generally burn with a sooty flame because of their high carbon content; both Q and R contain aromatic rings, so statement C is correct.

The Iodoform reaction is given by compounds containing the CH_3CO- (methyl ketone) group. In the structures shown, this condition is not satisfied simultaneously by R and S, so statement D is incorrect.

Finally, the Tollens' test is given by aldehydes, which reduce the reagent to metallic silver. Among the given compounds, Q is the aldehyde, so only Q gives the silver mirror test, making statement E correct.



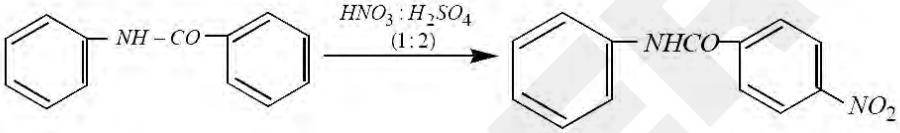
(OR)

Other compounds, P, Q, and S, do not give the Iodoform test.

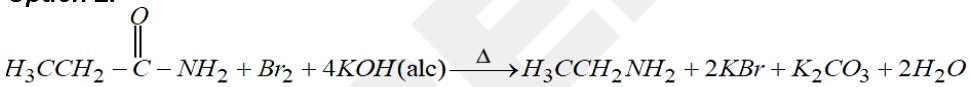
Hence, the correct answer is option (1).

Q. 18 Consider the following reactions, giving the major product. Identify the correct reaction

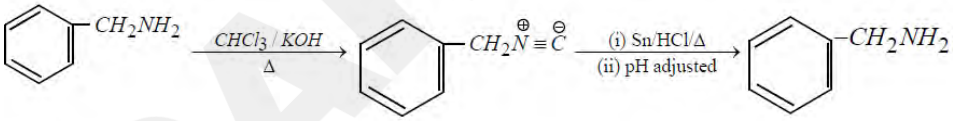
Option 1:



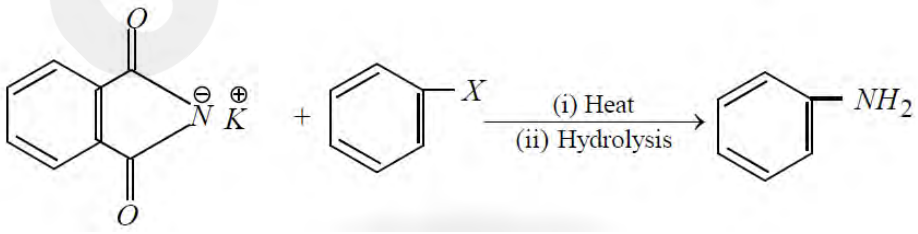
Option 2:



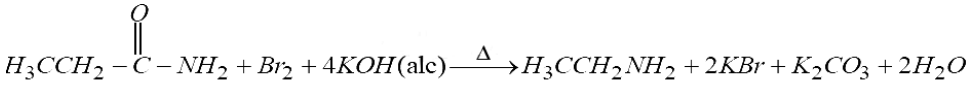
Option 3:



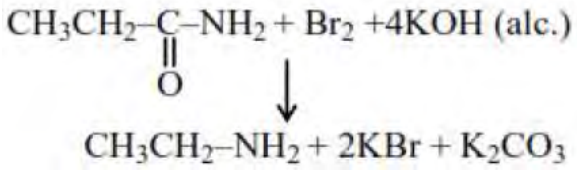
Option 4:



Correct Answer:



Solution:



The Hofmann bromide degradation reaction is correct.

Hence, the correct answer is option (2)

Q. 19 $\mathrm{Ph-CH=CH_2} \xrightarrow[\mathrm{HBr}]{(\mathrm{PhCOO})_2}$ Product

Consider the above reaction
A. The reaction proceeds through a more stable radical intermediate.
B. The role of peroxide is to generate H (hydrogen radical).
C. During this reaction, benzene is formed as a byproduct.
D. 1-Bromo-2-phenylethane is formed as the minor product.
E. The same reaction in the absence of peroxide proceeds via a carbocation intermediate.

Identify the correct statements. Choose the correct answer from the options given below.

Option 1:
C, D & E only

Option 2:
A & E only

Option 3:
A, B & D only

Option 4:
A, C & E only

Correct Answer:
A, C & E only

Solution:
The given reaction is the addition of HBr to styrene (Ph–CH = CH_2) in the presence of benzoyl peroxide $(\mathrm{PhCOO})_2$. In the presence of peroxide, the reaction follows a free radical mechanism rather than the normal ionic pathway. Peroxide decomposes to form radicals, which initiate the chain reaction and generate bromine radicals (Br•). The Br• adds to the double bond such that the more stable benzylic radical intermediate is formed, which is stabilized by resonance with the benzene ring; therefore, Statement A is correct.

During the decomposition of benzoyl peroxide, benzoyloxy radicals can lose CO_2 to form phenyl radicals, which may abstract hydrogen to produce benzene as a by-product. Statement C is also correct. In the absence of peroxide, the reaction proceeds via the normal electrophilic addition mechanism, where the double bond forms a carbocation intermediate following Markovnikov's rule, so Statement E is correct. Thus, the correct statements are A, C, and E.

Hence, the correct answer is option (4)

Q. 20 Given below are two statements:
Statement I: The Griess-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphanilic acid and α-naphthylamine reagent.

Statement II: In the above test, sulphanilic acid is diazotized by the acidified nitrite ion, which, on further coupling with α-naphthylamine, forms an azo-dye.

In light of the above statements, choose the correct answer from the options given below

Option 1:
Statement I is false, but Statement II is true

Option 2:
Both Statement I and Statement II are true

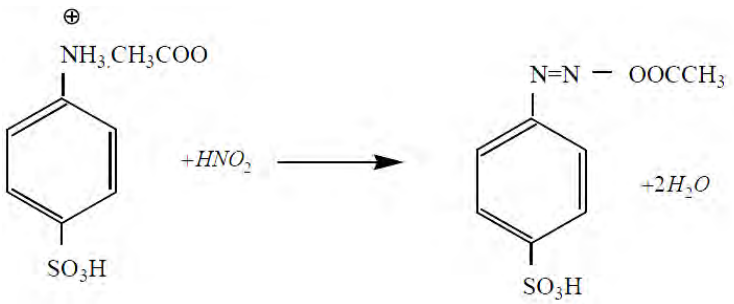
Option 3:
Both Statement I and Statement II are false

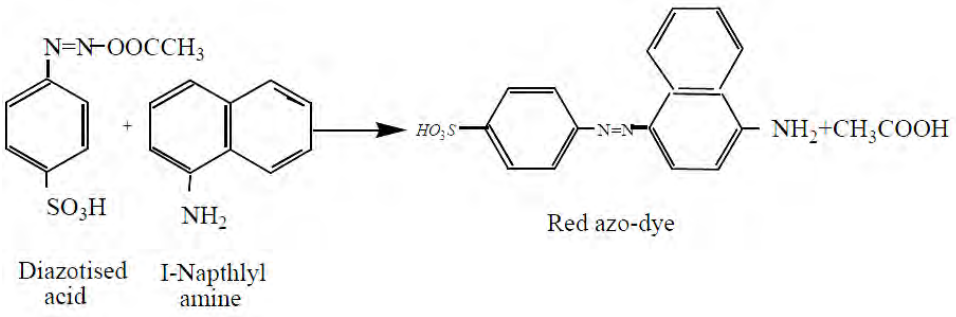
Option 4:
Statement I is true, but Statement II is false

Correct Answer:
Statement I is true, but Statement II is false

Solution:
The Griess-Ilosvay test detects $\left(NO_2^-\right)$ in water, soil, or biological samples by producing a distinct pink-to-red azo dye. It utilizes sulfanilic acid and α-naphthylamine in acetic acid, which diazotizes with nitrite to form a color proportional to the concentration.

$$NO_2^- + CH_3CHOOH \rightarrow HNO_2 + CH_3COO^-$$





Hence, the correct answer is option (4)

Q. 21 Consider the dissociation equilibrium of the following weak acid:

$HA \quad H^+(aq) + A^-(aq)$

If the pKa of the acid is 4, then the pH of a 10 mM HA solution is _____. (Nearest integer)

Given: [The degree of dissociation can be neglected with respect to unity.]

Correct Answer:

3

Solution:

$$pK_a = 4$$

$$C = 10\text{mM} = 0.01M = 10^{-2}M$$

$$pH = \frac{1}{2}[pK_a - \log C]$$

$$pH = \frac{1}{2}\left[4 - \log\left(10^{-2}\right)\right]$$

$$\log\left(10^{-2}\right) = -2$$

$$pH = \frac{1}{2}[4 - (-2)]$$

$$pH = \frac{1}{2}(6)$$

$$pH = 3$$

Hence, the answer is 3

Q. 22 Consider the following redox reaction taking place in an acidic medium:

$$BH_4^-(aq) + ClO_3^-(aq) \rightarrow H_2BO_3^-(aq) + Cl^-(aq)$$

If the Nernst equation for the above balanced reaction is

$$E_{cell} = E_{cell} - \frac{RT}{nF}\ln Q,$$

then the value of n is _____ . (Nearest integer)

Correct Answer:
24

Solution:
First, balance the given redox reaction in an acidic medium:

$$3BH_4^- + 4ClO_3^- \rightarrow 4Cl^- + 3H_2BO_3^- + 3H_2O$$

Now determine the **n**-factor.
In this reaction, boron in BH_4^- gets oxidized to $H_2BO_3^-$.
Change in oxidation state of B:

$$B: -3 \rightarrow +5$$

Increase in oxidation number:

$$(+5 - (-3)) = 8$$

Thus,

$$\text{n-factor of } BH_4^- = 8$$

Number of moles of BH_4^- in the balanced equation:

$$= 3$$

Therefore,

$$n = (\text{ moles }) \times (\text{ n-factor })$$
$$n = 3 \times 8$$
$$n = 24$$

Hence, the answer is (24)

Q. 23 X is the number of geometrical isomers exhibited by $[Pt(NH_3)(H_2O)BrCl]$.
Y is the number of optically inactive isomer(s) exhibited by $[CrCl_2(ox)_2]^{3-}$ Z is the number of geometrical isomers exhibited by $[Co(NH_3)_3(NO_2)_3]$ The value of $X + Y + Z$ is _____ .

Correct Answer:
6

Solution:
Here
$\mathrm{X} = 3$ (Two cis + one trans isomers)
$\mathrm{Y} = 1$ (trans isomer)
$\mathrm{Z} = 2$ (Fac- mer isomer)
$X + Y + Z = 3 + 1 + 2 = 6$

Hence, the answer is 6

Q. 24 0.53 g of an organic compound (x), when heated with an excess of nitric acid (concentrated) and then with silver nitrate, gave 0.75 g of silver bromide precipitate. 1.0 g of (x) gave 1.32 g of CO_2 gas on combustion. The percentage of hydrogen in the compound (x) is _____ %. [Nearest Integer]
[Given: Molar mass in $\mathrm{gmol}^{-1}\mathrm{H}: 1, \mathrm{C}: 12, \mathrm{Br}: 80, \mathrm{Ag}: 108, \mathrm{O}: 16$; Compound (x) $\mathrm{C_xH_yBr_z}$]

Correct Answer:
4

Solution:
Mass of $C = \frac{12}{44} \times 1.32 = 0.36$ g
Percentage of $C = \frac{0.36}{1.0} \times 100 = 36\%$

$$Br = \frac{80}{188} \times 0.75 \approx 0.3191g$$
$$Br = \frac{0.3191}{0.53} \times 100 \approx 60.2\%$$

Percentage of $\mathrm{H} = 100\% - (\%\mathrm{C} + \%\mathrm{Br})$
Percentage of $\mathrm{H} = 100 - (36 + 60.2) = 3.8\% \cong 4$

Hence, the answer is 4

Q. 25 500 mL of 1.2 M KI solution is mixed with 500 mL of $0.2\mathrm{M}\mathrm{KMnO}_4$ solution in a basic medium. The liberated iodine was titrated with standard $0.1\mathrm{M}\mathrm{Na}_2S_2\mathrm{O}_3$ solution in the presence of starch indicator till the blue color disappeared. The volume (in L) of $\mathrm{Na}_2\ \mathrm{S}_2\mathrm{O}_3$ consumed is _____ . (Nearest integer)

Correct Answer:
3

Solution:
$\text{Equivalents of } KMnO_4 = M \times V \times n$

In basic medium, $KMnO_4$ changes from Mn^{7+} to Mn^{4+}, so $\mathbf{n}$-factor $= \mathbf{3}$.

$$\begin{aligned}\text{Equivalents of } KMnO_4 &= 0.2 \times \frac{500}{1000} \times 3 \\ &= 0.2 \times 0.5 \times 3 \\ &= 0.3 \text{ equivalents}\end{aligned}$$

In the titration, the equivalents of oxidizing agent = equivalents of reducing agent.

$$\text{Equivalents of } \mathrm{KMnO_4} = \text{ Equivalents of } \mathrm{Na_2\,S_2O_3}$$
$$0.3 = 0.1 \times V \times 1$$

$$V = \frac{0.3}{0.1} = 3L$$

Hence, the answer is 3

Maths

Q. 1 Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$. If $BC = \begin{pmatrix} 1 & -5 \\ -1 & 2 \end{pmatrix}$ and $CB\binom{x_1}{x_2} = \binom{12}{-6}$, then $x_1 + x_2$ is:

Option 1:
2

Option 2:
-2

Option 3:
4

Option 4:
0

Correct Answer:
0

Solution:
$(I + A)B = I \Rightarrow B + AB = I - - - - - - - - - - (1)$

As well as $B + BA = I$

$A + C = I \Rightarrow A = I - C - - - - - - - - - - -(2)$

Using (1) and (2) $\Rightarrow B + (I - C)B = I \Rightarrow 2B - CB = I$

$$BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix} \Rightarrow B(I-A) = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$$

$$B-(I-B) = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$$

$$2B-I = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$$

$$CB = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} -6 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$x_1 + x_2 = 0$$

Hence, the correct answer is option (4).

Q. 2 The mean and variance of 10 observations are 9 and 34.2 , respectively. If 8 of these observations are $2, 3, 5, 10, 11, 13, 15, 21$, then the mean deviation about the median of all the 10 observations is

Option 1:
7

Option 2:
4

Option 3:
6

Option 4:
5

Correct Answer:
5

Solution:

$$2+3+5+10+11+13+15+21+a+b = 90$$

$$a+b = 90-80$$

$$a+b = 10 \ldots\ldots\ldots\ldots (1)$$

$$4+9+25+100+121+169+225+441+a^2+b^2 = 10(34.2+81)$$

$$1094+a^2+b^2 = 1152$$

$$a^2+b^2 = 58$$

$$(1) \Rightarrow a^2+b^2+2ab = 100$$

$$ab = 21 \text{ then } a = 3, b = 7 \text{ or } a = 7, b = 3$$

$$\text{Numbers} \longrightarrow 2, 3, 3, 5, 7, 10, 11, 13, 15, 21$$

$$\text{Mean} = \frac{7+10}{2} = 8.5$$

$$M.D = \frac{6.5+5.5+5.5+3.5+1.5+1.5+2.5+4.5+6.5+12.5}{10} = \frac{50}{10}$$

$$M.D = 5$$

Hence, the correct answer is option (4).

Q. 3 Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half - plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point $(2, 3)$ and is farthest from the center of C_2, is $x + ay + b = 0$, then $a - b$ is equal to

Option 1:
-2

Option 2:
-6

Option 3:
6

Option 4:
10

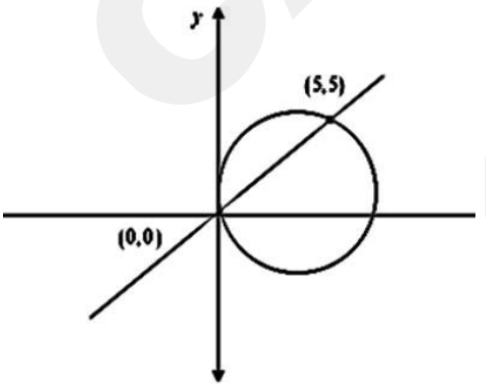
Correct Answer:
-2

Solution:

$C_1 = (x-5)^2 + y^2 = 25$

$C_2 = \left(x - \frac{5}{2}\right)^2 + \left(y - \frac{5}{2}\right)^2 = \frac{25}{4}$

Let centre of C_2 be P



$Q(2, 3)$

Slope of $PQ = \frac{1/2}{-1/2} = -1$

Req $x - y + 1 = 0$

$a - b = -2$

Hence, the correct answer is option **1**.

Q. 4 For three unit vectors a, b, c satisfying $|a-b|^2 + |b-c|^2 + |c-a|^2 = 9$ and $|2a + kb + kc| = 3$, the positive value of k is

Option 1:
3

Option 2:
5

Option 3:
4

Option 4:
6

Correct Answer:
5

Solution:

$6 - 2(\bar{a}\cdot\bar{b} + \bar{b}\cdot\bar{c} + \bar{c}\cdot\bar{a}) = 9$

$\bar{a}\cdot\bar{b} + \bar{b}\cdot\bar{c} + \bar{c}\cdot\bar{a} = \frac{-3}{2}$

$|2\bar{a} + k\bar{b} + k\bar{c}| = 3$

$|2\bar{a} + k(-\bar{a})| = 3$

$|(2-k)||(\bar{a})| = 3$

$2 - k = \pm 3 \Rightarrow k = 5, -1$

Hence, the correct answer is option **2**.

Q. 5 The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{k(k+1)}{k!}\right)$ is

Option 1:
$2/e$

Option 2:
$e/2$

Option 3:
$\sqrt{e}$

Option 4:
$1/e$

Correct Answer:
$1/e$

Solution:

$$\sum_{k=1}^{\infty}(-1)^{k+1}\frac{k(k+1)}{k!}$$

$$\sum_{k=1}^{\infty}(-1)^{k+1}\quad\frac{(k(k-1)+2k)}{k!}$$

$$\sum_{k=2}^{\infty}(-1)^{k+1}\quad\frac{1}{(k-2)!}+\sum_{k=1}^{\infty}\quad\frac{(-1)^{k+1}}{(k-1)!}=-e^{-1}+2e^{-1}=e^{-1}=\frac{1}{e}$$

Hence, the correct answer is option **4**.

Q. 6 Let $\mathrm{S}=\{1,2,3,4,5,6,7,8,9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then:

Option 1:
$45x=7y$

Option 2:
$21x=4y$

Option 3:
$56x=9y$

Option 4:
$29x = 5y$

Correct Answer:
$21x = 4y$

Solution:
$S = \{1, 2, 3, \ldots\ldots 9\}$
$x = 9c_1 \cdot 8c_7 \times \frac{9!}{2} = \frac{9 \times 8 \times 9}{2}$
$y = 9c_2 \cdot {}^7c_5 \times \frac{9!}{2! \times 2!} = \frac{9 \times 8}{2} \times \frac{7 \times 6}{2} \times \frac{9!}{2! \times 2!}$
$\frac{x}{y} = \frac{4}{21}$
$21x = 4y$

Hence, the correct answer is option **2**.

Q. 7 A bag contains 10 balls out of which k are red and $(10 - k)$ are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains 1 red and 9 black balls is:

Option 1:
$\frac{14}{55}$

Option 2:
$\frac{7}{11}$

Option 3:
$\frac{7}{55}$

Option 4:
$\frac{7}{110}$

Correct Answer:
$\frac{14}{55}$

Solution:
$P\left(\frac{1R9B}{3B}\right) = \frac{\frac{9}{10} \times \frac{8}{9} \times \frac{7}{8}}{\sum_{r=3}^{10} \frac{r(r-1)(r-2)}{10 \times 9 \times 8}} = \frac{9 \times 8 \times 7}{1 \times 2 \times 3 \times \sum_{r=3}^{10} r_{c_3}} = \frac{84}{3_{c_3} + 4_{c_3} + \ldots\ldots 10_{c_3}}$
$= \frac{84}{11_{c_4}} = \frac{14}{55}$

Hence, the correct answer is option **1**.

Q. 8 Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x+2\sqrt{2}y=4$. If the co-ordinates of the vertex A are (α,β), then the greatest integer less than or equal to $|\alpha+\sqrt{2}\beta|$ is:

Option 1:
4

Option 2:
2

Option 3:
5

Option 4:
3

Correct Answer:
3

Solution:
A lies on $2\sqrt{2}x-y=10$

$\tan\theta=2\sqrt{2}$

Perpendicular distance from $(0,0)$ on B C $\}P=\frac{4}{\sqrt{1+8}}=\frac{4}{3}\Rightarrow OA=\frac{8}{3}$

$$A=\left(-\frac{8}{3}\cos\theta,-\frac{8}{3}\sin\theta\right)=\left(-\frac{8}{3}\cdot\frac{2\sqrt{2}}{3},-\frac{8}{3}\cdot\left(\frac{1}{3}\right)\right)=\left(\frac{-16\sqrt{2}}{9},-\frac{8}{9}\right)=(\alpha,\beta)$$

$$[\alpha+\sqrt{2}\beta]=\left[\frac{16\sqrt{2}}{9}+\frac{8\sqrt{2}}{9}\right],\left[\frac{24\sqrt{2}}{9}\right]=3$$

Hence, the correct answer is option **4**.

Q. 9 If α,β, where $\alpha<\beta$, are the roots of the equation $\lambda x^2-(\lambda+3)x+3=0$ such that $\frac{1}{\alpha}-\frac{1}{\beta}=\frac{1}{3}$, then the sum of all possible values of λ is

Option 1:
4

Option 2:
8

Option 3:
2

Option 4:
6

Correct Answer:
6

Solution:
$\lambda x^2-(\lambda+3)x+3=0$
$\frac{1}{\alpha}-\frac{1}{\beta}=\frac{1}{3}\Rightarrow\frac{\beta-\alpha}{\alpha\beta}=\frac{1}{3}$
$\frac{\sqrt{(\alpha+\beta)^2-4\alpha\beta}}{\alpha\beta}=\frac{1}{3}$
$\left(\frac{\lambda+3}{\lambda}\right)^2-4\cdot\frac{3}{\lambda}=\frac{1}{3}\times\frac{3}{\lambda}\Rightarrow\lambda^2-6\lambda+8=0$

$\lambda_1+\lambda_2=6$

Hence, the correct answer is option **4**.

Q. 10 If $\int\left(\frac{1-5\cos^2 x}{\sin^5 x\cos^2 x}\right)dx=f(x)+C$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right)-f\left(\frac{\pi}{4}\right)$ is equal to

Option 1:
$\frac{1}{\sqrt{3}}(26-\sqrt{3})$

Option 2:
$\frac{2}{\sqrt{3}}(4+\sqrt{6})$

Option 3:
$\frac{4}{\sqrt{3}}(8-\sqrt{6})$

Option 4:
$\frac{1}{\sqrt{3}}(26+\sqrt{3})$

Correct Answer:
$\frac{4}{\sqrt{3}}(8-\sqrt{6})$

Solution:

$$\int \frac{1-5\cos^2 x}{\sin^2 x \cos^2 x}dx$$
$$\int \frac{\sec^2 x dx}{\sin^5 x} - \int \frac{5}{\sin^5 x}dx$$
$$\int \frac{1}{\sin^5 x}\tan x - \int \frac{-5}{\sin^6 x}\cos x \cdot \tan x dx - \int \frac{5}{\sin^6 x}dx = \frac{\tan x}{\sin^5 x} = f(x)$$
$$f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right) = \frac{32}{\sqrt{3}} - \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^5} = \frac{32}{\sqrt{3}} - 4\sqrt{2} = \frac{4}{\sqrt{3}}(8-\sqrt{6})$$

Hence, the correct answer is option **3**.

Q. 11 Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N \text{ and } a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S, which are divisible by $x^2 + 2$, is

Option 1:
20

Option 2:
6

Option 3:
120

Option 4:
10

Correct Answer:
10

Solution:
$$(x^2+2)(px+q) = x^3 + ax^2 + bx + c$$
$$\Rightarrow px^3 + qx^2 + 2px + 2q = x^3 + ax^2 + bx + c$$
$$p = 1 \Rightarrow x^3 + qx^2 + 2x + 2q = x^3 + ax^2 + bx + c$$
$$\Rightarrow b = 2 \text{ and } c = 2a$$
$$(a, b, c) = (a, 2, 2a) a \in N \text{ and } 1 \leq a \leq 10$$
$$\text{Required polynomials } = 10$$

Hence, the correct answer is option **4**.

Q. 12 Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, x \neq 0$. If $y(2) = 0$, then $\tan(y(1))$ is equal to

Option 1:
$\frac{3}{4}$

Option 2:
$\frac{7}{4}$

Option 3:
$-\frac{7}{4}$

Option 4:
$-\frac{3}{4}$

Correct Answer:
$\frac{7}{4}$

Solution:
$\sec^2 y\frac{dy}{dx} - \frac{2\tan y}{x} = x^3\left(2-x^3\right)$

Put $\tan y = t \Rightarrow \sec^2 y \cdot \frac{dy}{dx} = \frac{dt}{dx}$

$$\frac{dt}{dx} - \frac{2t}{x} = x^3\left(2-x^3\right)$$
$$\text{I.F } = e^{\int -\frac{2}{x}dx} = e^{-2\ln x} = \frac{1}{x^2}$$

$t.\left(\frac{1}{x^2}\right) = \int\left(2-x^3\right)dx + c$

$$\frac{t}{x^2} = 2x - \frac{x^4}{4} + c$$
$$\frac{\tan y}{x^2} = 2x - \frac{x^4}{4} + c$$
$$y(2) = 0 \Rightarrow 0 = 4 - \frac{16}{4} + c \Rightarrow c = 0$$

Now Put $x = 1, \tan y = 2 - \frac{1}{4} = \frac{7}{4}$

$\tan y(1) = \frac{7}{4}$

Hence, the correct answer is option **2**.

Q. 13 The area of the region $R = \left\{(x,y) : xy \le 8, 1 \le y \le x^2, x \ge 0\right\}$ is

Option 1:
$\frac{1}{3}(40\log_e(2)+27)$

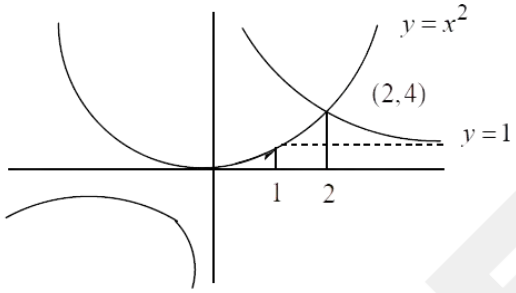
Option 2:
$\frac{1}{3}(49\log_e(2)-15)$

Option 3:
$\frac{2}{3}(24\log_e(2)-7)$

Option 4:
$\frac{2}{3}(20\log_e(2)+9)$

Correct Answer:
$\frac{2}{3}(24\log_e(2)-7)$

Solution:



$$A=\int_1^2(x^2-1)dx+\int_2^8\left(\frac{8}{x}-1\right)dx$$

$$A=8\log_e^4-\frac{14}{3}=16\log_e^2-\frac{14}{3}=\frac{2}{3}(24\log_e(2)-7)$$

Hence, the correct answer is option **3**.

Q. 14 If $g(x)=3x^2+2x-3$, $f(0)=-3$ and $4g(f(x))=3x^2-32x+72$, then $f(g(2))$ is

Option 1:
$-\frac{25}{6}$

Option 2:
$\frac{25}{6}$

Option 3:
$\frac{7}{2}$

Option 4:
$-\frac{7}{2}$

Correct Answer:
$\frac{7}{2}$

Solution:
$g(2)=3(2)^2+4-3=13$
$f(g(2))=f(13)$
$4g(f(x))=3x^2-32x+72$
$4\left(3f^2(x)+2f(x)-3\right)=3x^2-32x+72$
$12f^2(x)+8f(x)-12=3x^2-32x+72$
$\text{Put } x=13$
$12f^2(13)+8f(13)-12=3(13)^2-32(13)+72$
$12f^2(13)+8f(13)-175=0$
$f(13)=\frac{-8\pm 92}{24} \Rightarrow f(13)=\frac{7}{2} \text{ or } f(13)=-\frac{15}{4}(\text{ rejected })$

Hence, the correct answer is option **3**.

Q. 15 The common difference of the A.P.: $a_1, a_2, \ldots\ldots a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots\ldots b_n$. If $b_{31}=-277, b_{43}=-385$, and $a_{78}=327$, then a_1 is equal to:

Option 1:
19

Option 2:
21

Option 3:
16

Option 4:
24

Correct Answer:
19

Solution:
$b_{43}-b_{31}=12d_b=-108$
$\therefore d_b=-9$
$d_a=-9+13=4$
$a_1=a_{78}-77d_b=327-77\times 4=327-308=19$

Hence, the correct answer is option **1**.

Q. 16 The value of $\lim_{x\to 0}\frac{\log_e(\sec(ex)\cdot\sec(e^2x)\ldots\ldots\cdot\sec(e^{10}x))}{e^2-e^{2\cos x}}$

Option 1:
$\frac{(e^{10}-1)}{2e^2(e^2-1)}$

Option 2:
$\frac{(e^{20}-1)}{2e^2(e^2-1)}$

Option 3:
$\frac{(e^{10}-1)}{2(e^2-1)}$

Option 4:
$\frac{(e^{20}-1)}{2(e^2-1)}$

Correct Answer:
$\frac{(e^{20}-1)}{2(e^2-1)}$

Solution:

$$\lim_{x\to 0}\frac{\ln(\sec(ex))+\ln\left(\sec\left(e^2x\right)\right)+\ldots\ldots\ldots+\ln\left(\sec\left(e^{10}x\right)\right)}{e^{2\cos x}\left(\frac{e^{2-2\cos x}-1}{2-2\cos x}\right)\times\frac{2-2\cos x}{x^2}\times x^2}$$

$$\lim_{x\to 0}\frac{\ln(\sec(ex))+\ln\left(\sec\left(e^2x\right)\right)+\ldots\ldots\ldots+\ln\left(\sec\left(e^{10}x\right)\right)}{e^2x^2}$$

Using L-Hopital's Rule

$$\lim_{x\to 0}\frac{e\tan ex+e^2\tan e^2x+\ldots\ldots\ldots+e^{10}\tan e^{10}x}{2e^2x}$$

$$\Rightarrow\frac{1}{2e^2}\left[e^2+e^4+e^6+\ldots\ldots e^{20}\right]$$

$$\Rightarrow\frac{1}{2}\frac{e^2\left(\left(e^2\right)^{10}-1\right)}{e^2\left(e^2-1\right)}=\frac{\left(e^{20}-1\right)}{2\left(e^2-1\right)}$$

Q. 17 If $\frac{\tan(A-B)}{\tan A}+\frac{\sin^2 C}{\sin^2 A}=1, A, B, C\in\left(0,\frac{\pi}{2}\right)$, then

Option 1:
$\tan A, \tan B, \tan$ are in G.P.

Option 2:
$\tan A, \tan B, \tan C$ are in A.P.

Option 3:
$\tan A, \tan C, \tan B$ are in A.P.

Option 4:
$\tan A, \tan C, \tan B$ are in G.P

Correct Answer:
$\tan A, \tan C, \tan B$ are in G.P

Solution:
$\frac{\tan A-\tan B}{\tan A(1+\tan A\tan B)}+\frac{1+\cot^2 A}{1+\cot^2 C}=1$

Let $\tan A=x, \tan B=y, \tan C=z$

$$\frac{x-y}{x(1+xy)}+\frac{1+\frac{1}{x^2}}{1+\frac{1}{z^2}}=1$$

$$\frac{x-y}{x(1+xy)}+\frac{z^2\left(x^2+1\right)}{x^2\cdot\left(z^2+1\right)}=1$$

$$x(x-y)\left(z^2+1\right)+z^2\left(x^2+1\right)(1+xy)=x^2(1+xy)\left(z^2+1\right)$$

After Solving $\Rightarrow \tan^2 C=\tan A\cdot\tan B$

Hence, the correct answer is option **4**.

Q. 18 Let z be a complex number such that $|z-6|=5$ and $|z+2-6i|=5$. Then the value of $z^3+3z^2-15z+141$ is equal to

Option 1:
37

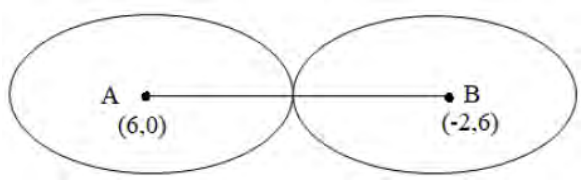
Option 2:
61

Option 3:
50

Option 4:
42

Correct Answer:
50

Solution:
$|z-6|=5$ and $|z-(-2+6i)|=5$



Z is mid point of $A(6,0), B(-2,6)$

$$z=\left(\frac{6-2}{2},\frac{0+6}{2}\right)=(2,3)$$
$$\therefore z=2+3i$$
$$z-2=3i$$

Squaring on both sides

$$z^2+4-4z=-9$$
$$Z^2-4z+13=0$$
$$\text{Let } z^3+3z^2-15z+141$$
$$z^3+3z^2-15z+91+50$$
$$(z+7)\left(z^2-4z+13\right)+50\Rightarrow 0+50=50$$

Hence, the correct answer is option **3**.

Q. 19 If the distances of the point $(1,2,a)$ from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines $L_1:\frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and $L_2:\frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal, then $a+b+c$ is equal to

Option 1:
4

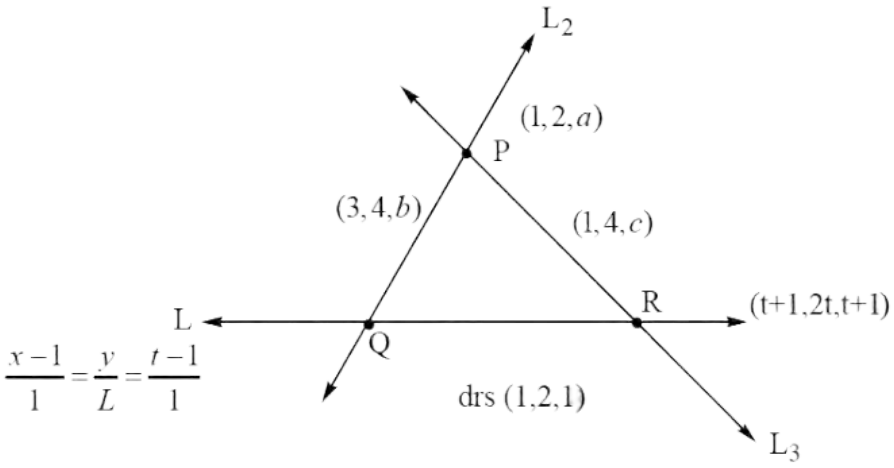
Option 2:
7

Option 3:
5

Option 4:
6

Correct Answer:
7

Solution:



At any point on L is $Q[t+1, 2t, t+1], p(1, 2, a)$
DRs of PQ : $(\mathrm{t}, 2\mathrm{t}-2, \mathrm{t}+1-\mathrm{a})$

$PQ \| L_2$

$$\frac{t}{3} = \frac{2t-2}{4} = \frac{t+1-a}{b} \Rightarrow 4t = 6t - 6$$

$6 = 2t \quad \therefore a + b = 4$

$t = 3$

PQ DRs$(\mathrm{t}, 2\mathrm{t}-2, \mathrm{t}+1-\mathrm{a})$

$RPL_3 \quad$ so $-\frac{\mathrm{t}}{\mathrm{c}} = \frac{2t-2}{4} = \frac{t+1-a}{c}$

$4t = 2t - 2$

$2 = -2t \quad \therefore \quad a - c = 0$

$t = -1$

$\therefore$ point $Q(4, 6, 4), R(0, -2, 0)$

$QP = RP$

$$\sqrt{9 + 16 + (4-a)^2} = \sqrt{1 + 16 + a^2}$$

$25 + 16 + a^2 - 8a = 17 + a^2$

$41 - 17 = 8a$

$a = 3 \quad$ so $c = 3$

$a + b + c = 4 + 3 = 7$

Hence, the correct answer is option **2**.

Q. 20 Let f be a polynomial function such that $f\left(x^2+1\right) = x^4 + 5x^2 + 2$, for all $x \in R$. Then $\int_0^3 f(x)dx$ is equal to

Option 1:

$\frac{33}{2}$

Option 2:
$\frac{27}{2}$

Option 3:
$\frac{41}{3}$

Option 4:
$\frac{5}{3}$

Correct Answer:
$\frac{33}{2}$

Solution:
$f\left(x^2+1\right)=x^4+5x^2+2$

Let $x^2+1=t$

$x^2=t-1$
$f(t)=(t-1)^2+5(t-1)+2$
$f(t)=t^2+3t-2$
$$\int_0^3 f(t)dt=\int_0^3\left(t^2+3t-2\right)dt=\left(\frac{t^3}{3}\right)_0^3+3\left(\frac{t^2}{2}\right)_0^3-(2t)_0^3=\frac{1}{3}(27)+\frac{3}{2}(9)-6=\frac{33}{2}$$

Hence, the correct answer is option **1**.

Q. 21 Let PQR be a triangle such that $PQ=-2i-j+2k$ and $PR=ai+bj-4k, a,b\in Z$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $|PR|=9$ and $PS=i-7j+2k$, then the value of $3a-4b$ is_____

Correct Answer:
5

Solution:
Given
$PQ=(-2,-1,2),;PR=(a,b,-4),;PS=(1,-7,2)$

$|PR|=9\Rightarrow a^2+b^2+16=81$

$a^2+b^2=65$

$Q=(-2,-1,2),;R=(a,b,-4)$

Since S lies on QR,

$S=Q+\lambda(R-Q)$

$R-Q=(a+2,b+1,-6)$

$S = (-2, -1, 2) + \lambda(a + 2, b + 1, -6)$

Equating with $S = (1, -7, 2)$

$1 = -2 + \lambda(a + 2)$

$-7 = -1 + \lambda(b + 1)$

$2 = 2 - 6\lambda$

$\lambda = 0$

$S = Q$

Distance from Q to PQ is 0

Hence distance from Q to PR must also be 0

Thus Q lies on PR

$Q = \mu R$

$(-2, -1, 2) = \mu(a, b, -4)$

$2 = -4\mu \Rightarrow \mu = -\frac{1}{2}$

$-2 = -\frac{1}{2}a \Rightarrow a = 4$

$-1 = -\frac{1}{2}b \Rightarrow b = 2$

But $|PR| = 9$

$a^2 + b^2 = 65$

Possible integer pairs satisfying $a^2 + b^2 = 65$

$(1, 8), (4, 7), (7, 4), (8, 1)$ and their negatives

Using the geometry condition from collinearity requirement gives

$a = 7, b = 4$

$3a - 4b$

$= 3(7) - 4(4)$

$= 21 - 16$

$= 5$

Hence, the correct answer is 5.

Q. 22 If $k=\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right)+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then the number of solutions of the equation $\sin^{-1}(kx-1)=\sin^{-1}x-\cos^{-1}x$ is ____

Correct Answer:
1

Solution:
$K=\tan\theta+\cot\theta=\frac{1}{\sin\theta\cos\theta}=\frac{2}{\sin 2\theta}$

Where $\theta=\frac{1}{2}\sin^{-1}\frac{2}{3}$

$K=\frac{2}{\frac{2}{3}}=3$

$\sin^{-1}(3x-1)=\sin^{-1}x-\cos^{-1}x$

$\sin^{-1}(3x-1)=\dfrac{\pi}{2}-2\cos^{-1}x$

$3x-1=\sin^{-1}\left(\dfrac{\pi}{2}-2\cos^{-1}x\right)$

$3x-1=2x^2-1\Rightarrow x=0,\dfrac{3}{2}$

$x=\dfrac{3}{2}(\text{ Rejected })$

$x=0$ only

$\text{No of solution }=1$

Hence, the correct answer is 1.

Q. 23 In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $R-(a,b)$, then a^2+b^2 is equal to:

Correct Answer:
90

Solution:
Let $\frac{a}{r},a,ar$ first 3 terms of G.P □ Product of them $=27$

$a=3$

$\text{S}=\text{ sum of first 3 terms }=\dfrac{a}{r}+a+ar=a\left(\dfrac{1}{r}+1+r\right)$

$S = 3\left(\frac{1}{r} + r + 1\right)$

If $r < 0$

$\frac{1}{r} + r \leq -2$

if $r > 0$

$\frac{1}{r} + r \geq 2$

$\therefore r + \frac{1}{r} \in (-\infty, -2] \cup [2, \infty)$

$r + \frac{1}{r} + 1 \in (-\infty, -1] \cup [3, \infty)$

$3\left(r + \frac{1}{r} + 1\right) \in (-\infty, -3] \cup [9, \infty)$

$\therefore S \in (-\infty, -3] \cup [9, \infty)$

$S \in R - (-3, 9)$

$a = -3, b = 9$

$a^2 + b^2 = (-3)^2 + 9^2 = 90$

Hence, the correct answer is 90.

Q. 24 The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x|\sin \pi x|dx\right)}\right|\right)$ is ____

Correct Answer:
210

Solution:

$I_r = \int_0^r x|\sin \pi x|dx$

King's Rule $= \int_0^r (r - x)|\sin \pi x|dx$

$2I_r = \int_0^r r|\sin \pi x|dx$

$I_r = \frac{r}{2}\int_0^r |\sin \pi x|dx$

$I_1 = \frac{1}{2}\int_0^1 |\sin \pi x|dx = \frac{1}{2\pi}\int_0^\pi |\sin t|dt = \frac{1}{2\pi} \cdot 2$

$I_2 = \frac{2}{2}\int_0^2 |\sin \pi x|dx = \frac{2}{2\pi}\int_0^{2\pi} |\sin t|dt = \frac{2}{2\pi}.4$

$\therefore S = \sqrt{\pi \cdot \frac{1}{2\pi} \cdot 2} + \sqrt{\pi \cdot \frac{2}{2\pi} \cdot 4} + \sqrt{\pi \cdot \frac{3}{2\pi} \cdot 6 \ldots\ldots\ldots} + \sqrt{\pi \cdot \frac{20}{2\pi} \cdot 2(20)}$

$= 1 + 2 + 3 + \ldots\ldots.20 = \frac{20 \times 21}{2} = 210$

Hence, the correct answer is 210.

Q. 25 For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2 - y^2 \sec^2\theta = 8$ be e_1 and l_1 respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2 \sec^2\theta + y^2 = 6$ be e_2 and l_2 respectively. If $e_1^2 = e_2^2\left(\sec^2\theta + 1\right)$, then $\left(\frac{l_1 l_2}{e_1 e_2}\right)\tan^2\theta$ is equal to _____

Correct Answer:
8

Solution:
Hyperbola $x^2 - y^2\sec^2\theta = 8$

$$\frac{x^2}{8} - \frac{y^2}{8\cos^2\theta} = 1$$

$$e_1 = \sqrt{1 + \frac{8\cos^2\theta}{8}}, l_1 = \frac{2b^2}{a} = \frac{2\cdot\left(8\cos^2\theta\right)}{2\sqrt{2}} = 4\sqrt{2}\cos^2\theta$$

Ellipse $x^2\sec^2\theta + y^2 = 6 \quad e_1^2 = e_2^2\left(\sec^2\theta + 1\right)$

$\frac{x^2}{6\cos^2\theta} + \frac{y^2}{6} = 1$ $\qquad$ $\left(1 + \cos^2\theta\right) = \sin^2\theta\left(\frac{1+\cos^2\theta}{\cos^2\theta}\right)$

$e_2 = \sqrt{1 - \cos^2\theta} = \sin\theta$ $\qquad$ $1 = \tan^2\theta$

$l_2 = \frac{2a^2}{b} = \frac{2\cdot 6\cos^2\theta}{\sqrt{6}} = 2\sqrt{6}\cos^2\theta$ $\qquad$ $\theta \in \left(0, \frac{\pi}{2}\right)$

$\left(\frac{l_1\cdot l_2}{e_1\cdot e_2}\right)\tan^2\theta = \frac{4\sqrt{2}\cos^2\theta\cdot 2\sqrt{6}\cos^2\theta}{\sqrt{1+\cos^2\theta}\cdot\sin\theta}$ $\qquad$ $\because$ so $\theta = \frac{\pi}{4}$

$= \frac{4\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)^2\cdot 2\sqrt{6}\left(\frac{1}{\sqrt{2}}\right)^2}{\sqrt{1+\left(\frac{1}{\sqrt{2}}\right)^2}\cdot\left(\frac{1}{\sqrt{2}}\right)}$ $\qquad$ $= \frac{8\cdot\sqrt{2}\cdot\sqrt{6}\cdot\frac{1}{4}}{\frac{\sqrt{3}}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}} = \frac{2\cdot\sqrt{2}\cdot\sqrt{2}\cdot\sqrt{3}}{\sqrt{3}}\cdot 2 = 8$

Hence, the correct answer is 8.

Physics

Q. 1 The mean free path of a molecule of diameter $5 \times 10^{-10}\ \mathrm{m}$ at the temperature $41^{\circ}\mathrm{C}$ and pressure $1.38 \times 10^{5}\ \mathrm{Pa}$, is given as _____ m .(Given $k_{\beta} = 1.38 \times 10^{-23}\ \mathrm{J/K}$).

Option 1:
$10\sqrt{2} \times 10^{-8}$

Option 2:
$2\sqrt{2} \times 10^{-10}$

Option 3:
$2\sqrt{2} \times 10^{-8}$

Option 4:
2×10^{-8}

Correct Answer:
$2\sqrt{2} \times 10^{-8}$

Solution:

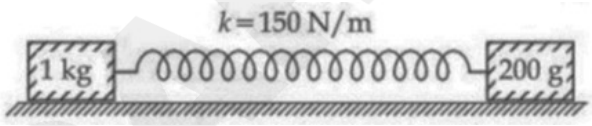
$$\lambda = \frac{1}{\sqrt{2}\pi d^2 \eta} = \frac{k_B T}{\sqrt{2}\pi d^2 p} = \frac{1.38 \times 10^{-23} \times (273+41)}{\sqrt{2} \times \pi \times \left(5 \times 10^{-10}\right)^2 \times 1.38 \times 10^{5}}$$

$$= \frac{314 \times 10^{-21}}{\sqrt{2}\pi 25 \times 10^{-20} \times 10^{5}}$$

$$= \frac{100 \times 10^{-8}}{25\sqrt{2}}$$

$$= 2\sqrt{2} \times 10^{-8}$$

Q. 2 As shown in figure, a spring is kept in a stretched position with some extension by holding the masses 1 kg and 0.2 kg with a separation more than spring natural length and are released. Assuming the horizontal surface to be frictionless, the angular frequency (in SI unit) of the system:



Option 1:
5

Option 2:
30

Option 3:
27

Option 4:
20

Correct Answer:
30

Solution:

$w=\sqrt{\frac{k}{\mu}}=\sqrt{\frac{150}{1/6}}, \mu=\frac{m_1 m_2}{m_1+m_2}=\sqrt{150\times 6}=\sqrt{900}=30$

Q. 3 Identify the correct statements:

A. Effective capacitance of a series combination of capacitors is always smaller than the smallest capacitance of the capacitor in the combination.

B. When a dielectric medium is placed between the charged plates of a capacitor, displacement of charges cannot occur due to insulation property of dielectric.

C . Increasing of area of capacitor plate or decreasing of thickness of dielectric is an alternate method to increase the capacitance.

D. For a point charge, concentric spherical shells centred at the location of the charge are equipotential surfaces.

Choose the correct answer from the options given below:

Option 1:
C and D Only

Option 2:
B and D Only

Option 3:
A,B and C Only

Option 4:
A,C and D Only

Correct Answer:
A,C and D Only

Solution:
Polarisation (movement of charges occur inside dielectric)

Q. 4 Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____

(Given: $h=6.63\times 10^{-34}\ \mathrm{J.S}$ and $c=3\times 10^{8}\ \mathrm{m/s}$)

Option 1:
10×10^{15}

Option 2:
2×10^{16}

Option 3:
5×10^{15}

Option 4:
5×10^{16}

Correct Answer:
2×10^{16}

Solution:

$$p = \frac{n}{t} \times \frac{hc}{\lambda}$$

$$6 \times 10^{-3} = \frac{n}{t} \times \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{6.63 \times 10^{-9} \times 100}$$

$$\frac{n}{t} = \frac{2 \times 10^{-3} \times 10^{-7}}{10^{-26}}$$

$$= 2 \times 10^{16}$$

Q. 5 Which one of the following is not a measurable quantity?

Option 1:
Voltage

Option 2:
Displacement current

Option 3:
Resistance

Option 4:
Voltage difference

Correct Answer:
Voltage

Solution:
We can only measure potential difference

Q. 6 Match List - I with List - II.

List - I	List - II
A. Coefficient of viscosity	I. $\left[ML^{-1}T^{-2}\right]$
B. Surface tension	II. $\left[ML^{2}T^{-2}\right]$
C. Pressure	III. $\left[ML^{0}T^{-2}\right]$
D. Surface energy IV. $\left[ML^{-1}T^{-1}\right]$	

Choose the correct answer from the given options given below:

Option 1:
A-I, B-III, C-II, D-IV

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-I, B-II, C-IV, D-III

Option 4:
A-IV, B-III, C-I, D-II

Correct Answer:
A-IV, B-III, C-I, D-II

Solution:
$F = 6\pi\eta rv$

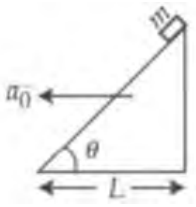
$MLT^{-2} = \eta \times L \times LT^{-1}$

$\eta = M^1L^{-1}T^{-1}$

$S = \dfrac{\text{Force}}{\text{Length}}$

$P = \dfrac{F}{A}$

Q. 7 A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0. The angle between the inclined plane and ground is θ and its base length is L . Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is _____



Option 1:
$\sqrt{\frac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

Option 2:
$\sqrt{\frac{2L}{g\sin\theta - a_0\cos\theta}}$

Option 3:
$\sqrt{\frac{2L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

Option 4:
$\sqrt{\frac{4L}{g\cos^2\theta - a_0\sin\theta\cos\theta}}$

Correct Answer:
$\sqrt{\frac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

Solution:

$$ma_{\text{rel}} = mg\sin\theta - ma_o\cos\theta$$
$$a_{\text{rel}} = g\sin\theta - a_o\cos\theta$$
$$s = 0(t) + \frac{1}{2}a_{\text{rel}}\,t^2$$
$$\frac{L}{\cos\theta} = \frac{1}{2}\times(g\sin\theta - a_o\cos\theta)\times t^2$$
$$t = \sqrt{\frac{2L}{g\left(\sin\theta\cos\theta - a_o\left(\cos^2\theta\right)\right.}}$$
$$t = \sqrt{\frac{4L}{g\left(2\sin\theta\cos\theta - a_o\left(2\cos^2\theta\right)\right.}}$$

Q. 8 A long cylindrical conductor with large cross section carries an electric current distributed uniformly over its cross-section. Magnetic field due to this current is:

A. maximum at either ends of the conductor and minimum at the midpoint

B. maximum at the axis of the conductor

C. minimum at the surface of the conductor

D. minimum at the axis of the conductor

E. same at all points in the cross-section of the conductor

Choose the correct answer from the options given below:

Option 1:
A, D Only

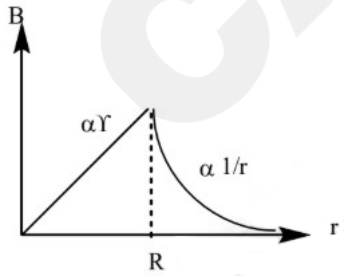
Option 2:
E Only

Option 3:
D Only

Option 4:
B,C Only

Correct Answer:
D Only

Solution:
B due to solid cylindrical wire



Q. 9 A particle starts moving from time $t = 0$ and its coordinate is given as $x(t) = 4t^3 - 3t$

A. The particle returns to its original position (origin) 0.866 units later

B. The particle is 1 unit away from origin at its turning point

C. Acceleration of the particle is non-negative

D. The particle is 0.5 units away from origin at its turning point

E. Particle never turns back as acceleration is non-negative

Choose the correct answer from the options given below:

Option 1:
A, B, C Only

Option 2:
A, C Only

Option 3:
A, C, D Only

Option 4:
C,E Only

Correct Answer:
A, B, C Only

Solution:

$V = \frac{dx}{dt} = 12t^2 - 3$

$V = 0 \Rightarrow 12t^2 = 3$

$t = \frac{1}{4} \Rightarrow t = \pm\frac{1}{2}$

$t = 0.5$

$a = \frac{dv}{dt} = 24t \Rightarrow a = 24\left(\frac{1}{2}\right) = 12 \text{ units / time}^2$

$x = 4\left(\frac{1}{4}\right)^3 - 3\left(\frac{1}{2}\right) = -1$

$x = 0 \Rightarrow 4t^3 = 3t$

$t = \frac{\sqrt{3}}{2} = 0.866$

Q. 10 A plane electromagnetic wave is moving in free space with velocity $c = 3 \times 10^8 \text{ m/s}$ and its electric field is given as $E = 54\sin(kz - wt)jTV/m$, where j is the unit vector along y -axis. The magnetic field vector B of the wave is :

Option 1:
$-1.8 \times 10^{-7}\sin(kz - wt)i \text{ T}$

Option 2:
$1.4 \times 10^{-7}\sin(kz - wt)kT$

Option 3:
$+1.8 \times 10^{-7}\sin(kz - wt)iT$

Option 4:
$1.4 \times 10^{-7} \sin(kz - wt)iT$

Correct Answer:
$-1.8 \times 10^{-7} \sin(kz - wt)i$ T

Solution:
$B_o = \frac{E_0}{c} = 1.8 \times 10^{-7} T$

Magnetic filed along x -axis

$\therefore B = B_0 \sin(kz - wt)i$
$\bar{V} = \bar{E} \times \bar{B}$
$B = -1.8 \times 10^{-7} \sin(kz - wt)i$

Q. 11 When the position $r = xi + yj + zk$ changes sign as $-r$, which one of the following vector will not flip under sign change?

Option 1:
Velocity

Option 2:
Acceleration

Option 3:
Linear momentum

Option 4:
Angular momentum

Correct Answer:
Angular momentum

Solution:
v, mv, a will change it's sign

$L = r \times P$
$L' = -r \times -p = L$

Q. 12 In an experiment, a set of reading are obtained as follows -1.24 mm, 1.25 mm, 1.23 mm, 1.21 mm . The expected least count of the instrument used in recording these readings is
_____ mm

Option 1:
0.1

Option 2:
0.01

Option 3:
0.001

Option 4:
0.05

Correct Answer:
0.01

Solution:
Readings are:

$1.24\ \text{mm}, 1.25\ \text{mm}, 1.23\ \text{mm}$
1.21 mm have two decimal places

This means this instrument can measure upto hundredth place of millimetre. i.e, 0.01 mm

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Q. 13 A nucleus has mass number α and radius R_α. Another nucleus has mass number β and radius R_β. If $\beta = 8\alpha$ then R_α/R_β

Option 1:
2

Option 2:
8

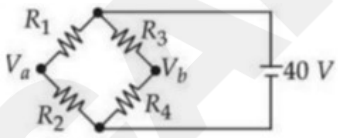
Option 3:
0.5

Option 4:
1

Correct Answer:
0.5

Solution:
$R = R_o A^{1/3}$ $\quad \frac{R_\alpha}{R_\beta} = \frac{R_0\alpha^{1/3}}{R_0\beta y_3} = \left(\frac{\alpha}{\beta}\right)^{1/3} = \left(\frac{\alpha}{\delta\alpha}\right)^{1/3} = \left(\frac{1}{8}\right)^{1/3} = \frac{1}{2}$

Q. 14 A Wheatstone bridge is initially at room temperature and all arms of the bridge have same value of resistances $(R_1 = R_2 = R_3 = R_4)$. When R_3 resistance is heated to some temperature, its resistance value has gone up by 10%. The potential difference $(V_a - V_b)$ (after R_3 is heated) is _____ V



Option 1:
0

Option 2:
1.05

Option 3:
2

Option 4:
0.95

Correct Answer:
0.95

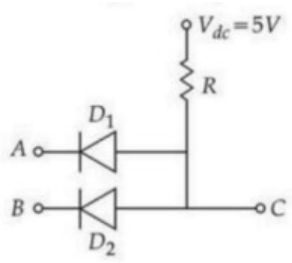
Solution:
R_3 increased by 10%
Initially

$$V_a = 40 \times \frac{R}{R+R} = 20$$
$$\therefore V_6 = 20$$

New Resistance

$$R_3^1 = R_3 + 0.10 \times R_3 = 1.1R$$
$$V_a = 20V\ (R_1 \& R_e \text{ No change })$$
$$V_b^1 = V \times \frac{R_4}{R_3^1 + R_4} = \frac{40}{2.1}V \quad 19.048$$
$$V_a - V_b^1 = 20 - 19.048 \quad 0.95$$

Q. 15 Two p-n junction diodes D_1 and D_2 are connected as shown in figure. A and B are input signal and C is the output. The given circuit will function as a _____



Option 1:
NOR Gate

Option 2:
NAND Gate

Option 3:
AND Gate

Option 4:
OR Gate

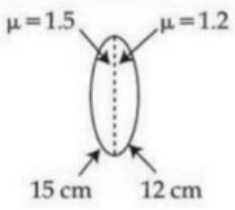
Correct Answer:
AND Gate

Solution:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Configuration of AND Gate

Q. 16 A biconvexlens is formed by using two thin planoconvex lenses, as shown in the figure. The refractive index and radius of curved surfaces are also mentioned in figure. When an object is placed on the left side of lens at a distance of 30 cm from the biconvex lens, the magnification of the image will be :



Option 1:
+2

Option 2:
-2.5

Option 3:
-2

Option 4:
+2.5

Correct Answer:
-2

Solution:

$$\frac{1}{f_{\text{net}}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{15}\right) + \left(\frac{1}{5}\right)\left(\frac{1}{12}\right) = \frac{1}{20}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\text{net}}} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{1}{60}$$

$$\therefore M = \frac{v}{u} = \frac{60}{-30} = -2$$

Q. 17 The time period of simple harmonic oscillator $T = 2\pi\sqrt{\frac{k}{m}}$. Measured value of mass (m) of the object is 10 g with an accuracy of 10 mg and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant (k) is _____ \%

Option 1:
6.76

Option 2:
3.35

Option 3:
3.43

Option 4:
7.60

Correct Answer:
6.76

Solution:
$T = 2\pi\sqrt{\frac{k}{m}} \Rightarrow k = \frac{T^2 m}{4\pi^2} \Rightarrow \frac{\Delta k}{k} = 2\frac{\Delta T}{T} + \frac{\Delta m}{m} = \left[2\left(\frac{2}{60}\right) + \frac{10\times10^{-3}}{10}\right] \times 100\% = 6.76\%$

Q. 18 Identify the correct statements:
A. Electrostatic field lines form closed loops.
B. The electric field lines point radially outward when charge is greater than zero.
C. The Gauss - Law is valid only for inverse - square force.
D. The workdone in moving a charged particle in a static electric field around a closed path is zero.
E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the correct answer from the options given below:

Option 1:
A, C, E Only

Option 2:
B, C, D, E Only

Option 3:
A, B, C, D Only

Option 4:
A, B, D, E Only

Correct Answer:
B, C, D, E Only

Solution:
[Electrostatic lines are not closed]

Q. 19 For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

Option 1:
$\sqrt{2} < n < 2$

Option 2:
$n \geq 2$

Option 3:
$1 < n < 2$

Option 4:
$\sqrt{2} < n < 2\sqrt{2}$

Correct Answer:
$\sqrt{2} < n < 2$

Solution:

$\delta_{\min} = 2i - A = A$

$\Rightarrow i = A$

$\sin i = \mu \sin\left(\frac{A}{2}\right)$

$\Rightarrow 2\sin\left(\frac{A}{2}\right)\cos\left(\frac{A}{2}\right) = \mu \sin\left(\frac{A}{2}\right) \Rightarrow \mu = 2\cos\left(\frac{A}{2}\right)$

But $\cos\left(\frac{A}{2}\right) \leq 1$

$\therefore \mu \leq 2$

Another limit is:

$\sin i = \mu \sin r_1$

if i becomes $\max = 90^\circ$

$r_1 = \frac{1}{2} = \frac{A}{2} = 45^0$

$\therefore \mu \geq \frac{1}{1/\sqrt{2}}$

Q. 20 The speed of a longitudinal wave in a metallic bar is $400\ \mathrm{m/s}$. If the density and Young's modulus of the bar material are increased by 0.5% and 1%, respectively then the speed of the wave is changed approximately to $\mathrm{m/s}$.

Option 1:
402

Option 2:
398

Option 3:
401

Option 4:
399

Correct Answer:
401

Solution:

$v\alpha\sqrt{\frac{Y}{e}} \Rightarrow \frac{\Delta v}{v} = \frac{1}{2}\frac{\Delta^4}{4} - \frac{1}{2}\frac{\Delta^e}{e}$

$= \left(\frac{1}{2} - \frac{1}{4}\right) \times \frac{1}{100} = \frac{1}{400}$

$\therefore \frac{v_2 - v_1}{v_1} = \frac{1}{400} \Rightarrow v_2 = 401$

Q. 21 An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rms) and frequency 50 Hz flows through it. The ratio of inductive reactance to its resistance is. $(\pi = 3.14)$

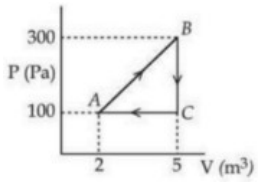
Correct Answer:
314

Solution:
$\frac{1}{2}Li^2 = 16 \Rightarrow \frac{32}{i^2}$

$i^2R = 32 \Rightarrow R = \frac{32}{i^2}$

$\therefore \frac{XL}{R} = \frac{Lw}{R} = w = 2\pi f = 314$

Q. 22 A thermodynamic system is taken through the cyclic process ABC as shown in the figure. The total work done by the system during the cycle ABC is _____ J.



Correct Answer:
300

Solution:
Area of graph = work = $\frac{1}{2} \times 3 \times 200 = 300J$

Q. 23 A fly wheel having mass 3 kg and radius 5 m is free to rotate about a horizontal axis. A string having negligible mass is wound around the wheel and the loose end of the string is connected to 3 kg mass. The mass is kept at rest initially and released. Kinetic energy of the wheel when the mass descends by 3 m is $\mathrm{J}.\left(\mathrm{g} = 10\ \mathrm{m/s^2}\right)$

Correct Answer:
30

Solution:
Decrease in gravitational p.e of connected
Mass=increase in k.e of connected mass + increase in k.e of wheel

$\Rightarrow 3 \times 10 \times 3 = \left(\frac{1}{2} \times 3v^2\right) + \frac{1}{2}\left(\frac{3v^2}{2}\right)$

$\therefore \frac{1}{2}\left(\frac{3v^2}{2}\right) = 30J$

Q. 24 A beam of light consisting of wavelengths 650 nm and 550 nm illuminates the Young's double slits with separation of 2 mm such that the interference fringes are formed on a screen, placed at a distance of 1.2 m from the slits. The least distance of a point from the central maximum, where the bright fringes due to both the wavelengths coincide, is $\times 10^{-5}\ \mathrm{m}$.

Correct Answer:
429

Solution:

$$y_{\text{nthbright}} = \frac{n\lambda D}{d}$$

$$n_1\lambda_1 = n_2\lambda_2$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{11}{13}$$

$$\therefore \text{ Ans } = \frac{11 \times 650 \times 10^{-9} \times 1.2}{2 \times 10^{-3}}$$

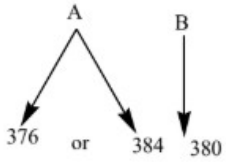
$$= 429 \times 10^{-5} \text{ m}$$

Q. 25 Two tuning forks A and B are sounded together giving rise to 8 beats in 2 s . When fork A is loaded with wax, the beat frequency is reduced to 4 beats in 2 s . If the original frequency of tuning fork B is 380 Hz then the original frequency of tuning fork A is

_____ Hz

Correct Answer:
384

Solution:



∴ From second hint

384 Hz is confirmed

Chemistry

Q. 1 Consider the following reactions

$$\mathrm{Na_2B_4O_7} \xrightarrow{\Delta} 2\mathrm{X} + \mathrm{Y}$$

$$\mathrm{CuSO_4} + \mathrm{Y} \xrightarrow[\text{Non-Luminous flame}]{} \mathrm{Z} + \mathrm{SO_3}$$

$$2\mathrm{Z} + 2\mathrm{X} + \text{ Carbon } \xrightarrow{\text{Luminous flame}} 2\mathrm{Q} + \mathrm{Na_2B_4O_7} + \mathrm{CO}$$

The oxidation states of Cu in Z and Q, respectively, are:

Option 1:
+1 and +1

Option 2:
+1 and +2

Option 3:
+2 and +2

Option 4:
+2 and +1

Correct Answer:
+2 and +1

Solution:

$$Na_2B_4O_7 \cdot 10H_2O \xrightarrow[-10H_2O]{\Delta} Na_2B_4O_7$$

$$\xrightarrow{\Delta} 2NaBO_2 + B_2O_3$$

$$CuSO_4 + B_2O_3 \xrightarrow[\text{Non-Luminoss flaine}]{\Delta} \underset{\substack{(\text{Blue basid})\\(21)}}{+2}\ (BO_2)_2$$

$$\underset{(Z)}{Cu(BO_2)_2} \xrightarrow[\text{Lumin ous Flums}]{\Delta} \underset{\substack{\text{Cololes}\\(s)}}{\overset{+1}{Cu}}\ BO_2$$

Oxidation states of Cu in Z and Q are +2 & +1

Hence, the correct answer is option (4)

Q. 2 Given below are two statements:
Statement I: The increasing order of boiling point of hydrogen halides is

$$HCl < HBr < HI < HF$$

Statement II: The increasing order of melting point of hydrogen halides is

$$HCl < HBr < HF < HI$$

In light of the above statements, choose the correct answer from the options given below:

Option 1:
Statement I is true but Statement II is false

Option 2:
Both Statement I and Statement II are true

Option 3:
Statement I is false but Statement II is true

Option 4:
Both Statement I and Statement II are false

Correct Answer:
Both Statement I and Statement II are true

Solution:
In hydrogen halides, the boiling point generally increases down the group due to an increase in molecular mass and London dispersion forces:

HCl < HBr < HI

However, HF forms strong intermolecular hydrogen bonding, which makes its boiling point much higher than the others.

Thus, the correct order of boiling points is:

HCl < HBr < HI < HF

So, Statement I is correct.

Now consider Statement II (Melting points).

For melting points, the order mainly depends on molecular mass and packing in the crystal lattice. The actual increasing order is:

HCl < HBr < HF < HI

Thus, Statement II is also correct.

Hence, the correct answer is option (2)

Q. 3 The correct increasing order of spin-only magnetic moment values of the complex ions $[MnBr_4]^{2-}(A), [Cu(H_2O)_6]^{2+}(B), [Ni(CN)_4]^{2-}(C)$ and $[Ni(H_2O)_6]^{2+}(D)$ is:

Option 1:
$C = D < B < A$

Option 2:
$A = B < C < D$

Option 3:
$C < B < D < A$

Option 4:
$A = B < D < C$

Correct Answer:
$C < B < D < A$

Solution:
$Mn^{2-}\ 3\ d^5 n = 5$
$Cu^{2-}\ 3\ d^* t_{2g}^{2,2,2} \quad e_g^{2,1} n = 1$

$Ni^{2-}\ 3\ d^z$ square planar $n = 0$
$Ni^{2-}\ 3\ d^8$ tetrahedral $e^{2,2} t_2^{2,1,1} n = 2$

Hence, the correct answer is option (3)

Q. 4 Consider the elements N, P, O, S, Cl and F . The number of valence electrons present in the elements with the most and least metallic character from the above list is, respectively.

Option 1:
5 and 6

Option 2:
5 and 7

Option 3:
6 and 7

Option 4:
7 and 5

Correct Answer:
5 and 7

Solution:
Metallic character increases down a group and decreases from left to right across a period.

Among the given elements:

P and S are in period 3, while N, O, and F are in period 2.

Elements in period 3 are more metallic than those in period 2.

Between P and S, phosphorus is more towards the left, so it is more metallic.

Thus, the most metallic element = P.

Now check the least metallic element. Metallic character decreases towards the top-right of the periodic table. Among the given elements, fluorine (F) lies at the extreme top-right, so it is the least metallic.

Now count their valence electrons:

P (Phosphorus) → Group 15 → 5 valence electrons

F (Fluorine) → Group 17 → 7 valence electrons

Therefore, the number of valence electrons in the elements with the most and least metallic character, respectively, is: 5, 7

Hence, the correct answer is option (2)

Q. 5 The wavelength of photon ' A ' is 400 nm. The frequency of photon ' B ' is $10^{16} s^{-1}$. The wave number of photon ' C ' is $10^4\ \mathrm{cm}^{-1}$. The correct order of energy of these photons is:

Option 1:
$A > C > B$

Option 2:
$A > B > C$

Option 3:
$C > B > A$

Option 4:
$B > A > C$

Correct Answer:
$B > A > C$

Solution:
(1) Wavelength of A=400 nm.

(2) $v = \dfrac{C}{\lambda} \Rightarrow \text{ wavelength of B}(\lambda) = \dfrac{3 \times 10^8}{10^{16}}$
$= 3 \times 10^{-5} = 30 \times 10^{-9} = 30\ \mathrm{nm}$

(3)> Wavelength of $C(\lambda) = \frac{1}{\bar{v}} = \frac{1}{10^4} = 10^{-4}\ \mathrm{cm} = 10^{-6}\ \mathrm{m} = 1000\ \mathrm{nm}$

Here $\lambda_C > \lambda_A > \lambda_B$
$E \propto \frac{1}{\lambda}$
So $E_c < E_A < E_B$

Hence, the correct answer is option (4)

Q. 6 Consider the following aqueous solutions.
I. 2.2 g Glucose in 125 mL of solution.
II. 1.9 g Calcium chloride in 250 mL of solution.
III. 9.0 g Urea in 500 mL of solution.
IV. 20.5 g Aluminium sulphate in 750 mL of solution.

The correct increasing order of boiling point of these solutions will be:
[Given: Molar mass in gmol^{-1} : $\mathrm{H}=1, \mathrm{C}=12,\ \mathrm{N}=14, \mathrm{O}=16, \mathrm{Cl}=35.5, \mathrm{Ca}=40, \mathrm{Al}=27$ and $\mathrm{S}=32$]

Option 1:
I < II < III < IV

Option 2:
III < I < II < IV

Option 3:
II < III < IV < I

Option 4:
II < III < I < IV

Correct Answer:
I < II < III < IV

Solution:
$\Delta T_b = i.k_b.m$

For a dilute solution (M=m)

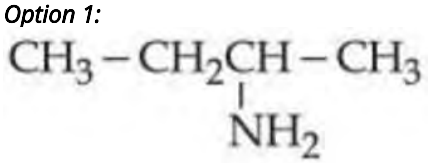
Molarity	i x m
$M_{\text{stucess}} = \frac{2.2}{180} \times \frac{1000}{125} = 0.09$	0.09×1
$\mathrm{M}_{\text{weat}} - \frac{9}{60} \times \frac{1000}{500} = 0.3$	0.03×1
$\mathrm{M}_{\mathrm{CaCl_2}} = \frac{1.9}{111} \times \frac{1000}{250} = 0.068$	0.068×3
$\mathrm{M}_{\mathrm{Al_2(SO_4)3}} = \frac{20.5}{342} \times \frac{1000}{750} \cong 0.08$	0.08×5

Order of $\Delta \mathrm{T_b} = \mathrm{Al_2(SO_4)_3} >$ Urea $> \mathrm{CaCl_2} >$ Glucose

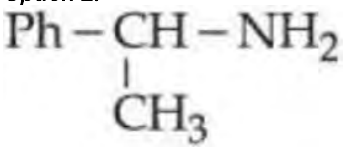
Hence, the correct answer is potion (1)

Q. 7 A student performed an analysis of an aliphatic organic compound ' X ', which on analysis gave
$C = 61.01\%, H = 15.25\%, N = 23.74\%$
This compound, on treatment with $\mathrm{HNO_2/H_2O}$, produced another compound ' Y ' which did not contain any nitrogen atom. However, the compound ' Y ' upon controlled oxidation produced another compound ' Z ' that responded to the Iodoform test. The structure of ' X ' is:

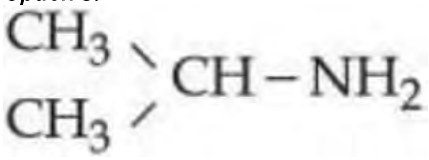
Option 1:



Option 2:



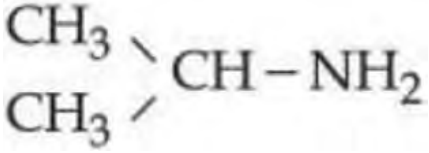
Option 3:



Option 4:

$CH_3CH_2CH_2NH_2$

Correct Answer:

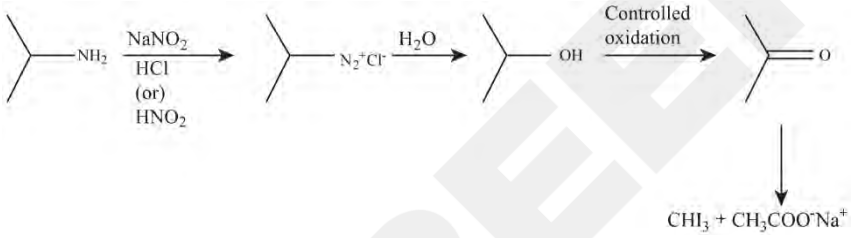


Solution:

$C = \frac{61.01}{12} = 5.08 \quad H = \frac{15.25}{1} = 15.25 \quad N = \frac{23.74}{14} = 1.69$

$\frac{5.08}{1.69} = 3 \quad \frac{15.25}{1.69} = 9 \quad \frac{1.69}{1.69} = 1$

$\text{Emp. For } = C_3H_9N$



Hence, the correct answer is option (3)

Q. 8 Observe the following equilibrium in a 1 L flask.

$A(g) \rightleftharpoons B(g)$

At $\mathrm{T(K)}$, the equilibrium concentrations of A and B are 0.5 M and 0.375 M, respectively. 0.1 moles of A is added into the flask and heated to $\mathrm{T(K)}$ to establish the equilibrium again. The new equilibrium concentrations (in M) of A and B are respectively

Option 1:

$0.367, 0.275$

Option 2:

$0.53, 0.4$

Option 3:

$0.557, 0.418$

Option 4:

$0.742, 0.557$

Correct Answer:
$0.557, 0.418$

Solution:
$\mathrm{A} \rightleftharpoons \mathrm{B}$
$0.50.375$

$$\mathrm{K_{eq}} = \frac{[\mathrm{B}]_{\mathrm{eq}}}{[\mathrm{A}]_{\mathrm{eq}}} = \frac{0.375}{0.5} = 0.75$$

Now 0.1 mole of A is added, so the reaction will move in the forward direction.

$\mathrm{A} \rightleftharpoons \mathrm{B}$
$0.6 - \mathrm{x} \quad 0.375 + \mathrm{x}$

$$\mathrm{K_a} = 0.75 = \frac{0.375 + \mathrm{x}}{0.6 - \mathrm{x}}$$

$0.45 - 0.75\mathrm{x} = 0.375 + \mathrm{x}$
$1.75\mathrm{x} = 0.075$

$$\mathrm{X} = \frac{0.075}{1.75} = \frac{3}{70} = 0.043$$

Moles of $\mathrm{A} = 0.043 = 0.557$

Moles of $\mathrm{B} = 0.418$

Hence, the correct answer is option (3)

Q. 9 A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____. (Given Molar mass in g mol^{-1} $\mathrm{S}: 32, \mathrm{BaSO_4}: 233$)

Option 1:
21.05%

Option 2:
63.15%

Option 3:
48.24%

Option 4:
42.10%

Correct Answer:
21.05%

Solution:
So,

$233\ \mathrm{g BaSO_4} \rightarrow 32\ \mathrm{g\ S}$
Therefore, sulphur present in $0.4813\mathbf{g\ BaSO_4}$ is

$$\text{Mass of S} = \frac{32}{233} \times 0.4813$$

Now calculate the percentage of sulphur in the given organic compound (mass of compound $= 0.314$ g).

$$\%S = \frac{\text{mass of } S}{\text{mass of compound}} \times 100$$
$$\%S = \frac{32 \times 0.4813}{233 \times 0.314} \times 100$$
$$\%S = 21.05\%$$

Thus, the percentage of sulphur in the compound is 21.05%.

Hence, the correct answer is option (1)

Q. 10 For the given reaction;

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

If $90\ g CaCO_3$ is added to 300 mL of HCl, which contains $38.55\% HCl$ by mass and has a density $1.13\ g mL^{-1}$, then which of the following options is correct?
Given the molar mass of H, Cl, Ca and O are 1, 35.5, 40 and $16\ g\ mol^{-1}$ respectively.

Option 1:
32.85 g of $CaCO_3$ remains unreacted

Option 2:
60.32 g of HCl remains unreacted

Option 3:
97.30 g of HCl reacted

Option 4:
64.97 g of HCl remains unreacted

Correct Answer:
97.30 g of HCl reacted

Solution:
$$M_{HCl} = \frac{38.55 \times 1.13 \times 10}{36.5} = 11.93$$
$$\eta_{HCl} = \frac{11.93 \times 300}{1000} = 3.58$$
$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

Initial Molar	0.9	3.58	0	0	0
After Rxn	0	1.78	0.9	0.9	0.9

Unreacted weight $= 1.78 \times 36.5 = 64.97$

Hence, the correct answer is option (3)

Q. 11 Identify the correct statements:

The presence of $-NO_2$ group in benzene ring

A. Activates the ring towards electrophilic substitutions.
B. deactivates the ring towards electrophilic substitutions.
C. activates the ring towards nucleophilic substitutions.
D. deactivates the ring towards nucleophilic substitutions.

Choose the correct answer from the options given below:

Option 1:
A and D only

Option 2:
B and D only

Option 3:
C and A only

Option 4:
B and C only

Correct Answer:
B and C only

Solution:
The $-NO_2$ group is a strong electron-withdrawing group due to both –I (inductive) and –M (resonance) effects. It pulls electron density away from the benzene ring, making the ring electron-deficient.

Because electrophilic substitution reactions require a high electron density in the benzene ring to attract electrophiles, the $-NO_2$ group decreases the electron density of the ring. Hence, the benzene ring becomes less reactive toward electrophilic substitution. Therefore, statement B is correct, and statement A is incorrect.

In contrast, nucleophilic substitution reactions are favored when the aromatic ring is electron-deficient. The presence of the $-NO_2$ group stabilizes the intermediate formed during nucleophilic aromatic substitution by withdrawing electron density through resonance. Thus, it activates the ring toward nucleophilic substitution, making statement C correct and D incorrect.

Hence, the correct answer is option (4)

Q. 12 The plot of $\log_{10} K$ vs $\frac{1}{T}$ gives a straight line. The intercept and slope respectively are (Where K is equilibrium constant).

Option 1:
$-\frac{\Delta S^{\circ} R}{2.303}, \frac{\Delta H^{\circ} R}{2.303}$

Option 2:
$\frac{2.303R}{\Delta H^{\circ}}, \frac{2.303R}{\Delta S^{\circ}}$

Option 3:
$\frac{\Delta S^{\circ}}{2.303R}, -\frac{\Delta H^{\circ}}{2.303R}$

Option 4:
$-\frac{\Delta H^{\circ}}{2.303R}, \frac{\Delta S^{\circ}}{2.303R}$

Correct Answer:
$\frac{\Delta S^{\circ}}{2.303R}, -\frac{\Delta H^{\circ}}{2.303R}$

Solution:

$$\Delta G^{o} = -2.303RT \log_{10} k$$
$$\Delta H^{o} - T.\Delta S^{o} = -2.303RT \log_{10} k$$
$$\frac{T.\Delta S^{o}}{2.303RT^{o}} \frac{-\Delta H^{o}}{2.303RT} = \log_{10} k$$
$$\frac{\Delta S^{o}}{2.303R} - \left(\frac{\Delta H^{o}}{2.303R}\right)\frac{1}{T} = \log k$$
$$\text{Slope } = \frac{-\Delta H^{o}}{2.303R}$$
$$\text{Intercept } = \frac{\Delta S^{o}}{2.303R}$$

Hence, the correct answer is option (3)

Q. 13 The reactions that produce alcohol as the product are:

A. $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$

B. $2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn}$

C. $(CH_3)_3CH \xrightarrow{KMnO_4}$

D. $2CH_4 + O_2 \xrightarrow{Cu/523\ K/100\ atm.}$

E. $CH_3 - CH = CH - CH_3 \xrightarrow{KMnO_4/H^+}$

Choose the correct answer from the options given below:

Option 1:
C and D only

Option 2:
B, D and E only

Option 3:
A and D only

Option 4:
A, C and E only

Correct Answer:
C and D only

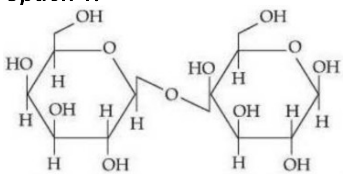
Solution:

$$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} HCHO$$
$$2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn} 2CH_3COOH$$
$$(CH_3)_3CH \xrightarrow{KMnO_4} (CH_3)_3C - OH$$
$$2CH_4 + O_2 \xrightarrow{Cu/523\ K/100\ atm.} 2CH_3OH$$
$$CH_3 - CH = CH - CH_3 \xrightarrow{KMnO_4/H^+} 2CH_3COOH$$

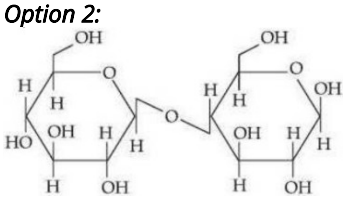
Hence, the correct answer is option (1)

Q. 14 Structures of four disaccharides are given below. Among the given disaccharides, the non-reducing sugar is:

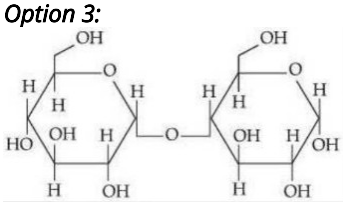
Option 1:



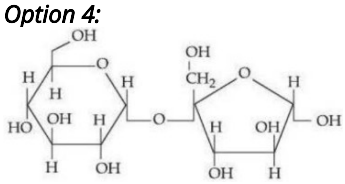
Option 2:



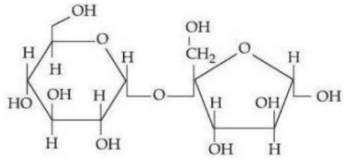
Option 3:



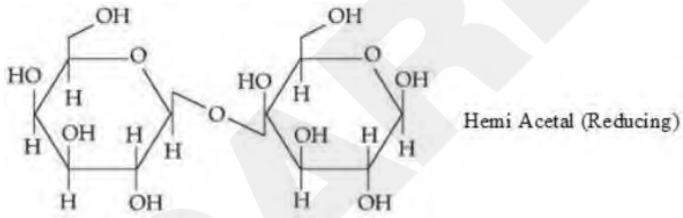
Option 4:



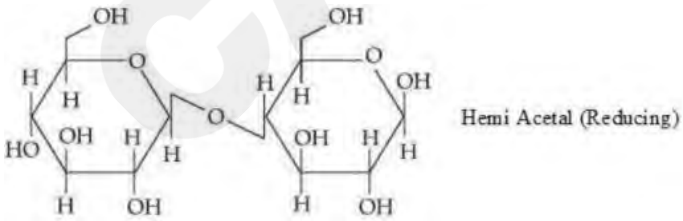
Correct Answer:



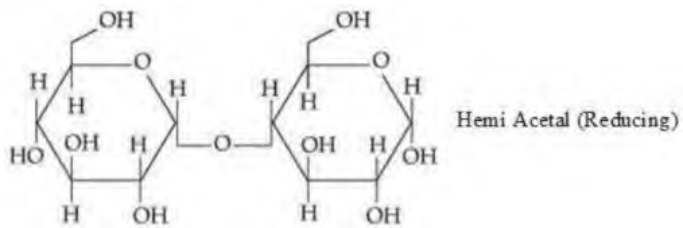
Solution:



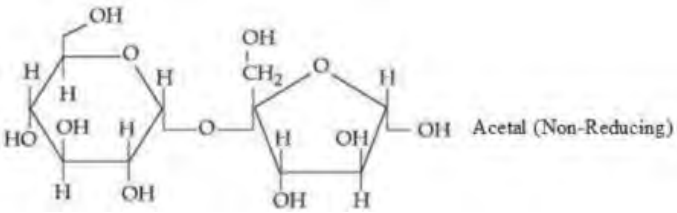
Hemi Acetal (Reducing)



Hemi Acetal (Reducing)



Hemi Acetal (Reducing)



Hence, the correct answer is option (4)

Q. 15 Consider the following statements about manganite and permanganate ions. Identify the correct statements.
A. The geometry of both manganate and permanganate ions is tetrahedral.
B. The oxidation states of Mn in manganate and permanganate are +7 and +6, respectively.
C. Oxidation of $\mathrm{Mn(II)}$ salt by peroxodisulphate gives manganate ion as the final product.
D. Manganate ion is paramagnetic and permanganate ions are diamagnetic.
E. Acidified permanganate ion reduces oxalate, nitrite, and iodide ions.

Choose the correct answer from the options given below:

Option 1:
A, C and D only

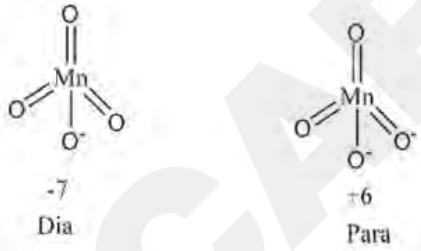
Option 2:
A, D, and E only

Option 3:
A, B and C only

Option 4:
A and D only

Correct Answer:
A, D, and E only

Solution:



Statement 1: Both manganate $\left(\mathrm{MnO_4^{2-}}\right)$ and permanganate $\left(\mathrm{MnO_4^{-}}\right)$ ions have four oxygen atoms bonded to Mn and no lone pair on Mn. According to VSEPR theory, species with four bonding pairs and no lone pairs have a tetrahedral geometry.

Statement 2: For manganate $\left(\mathrm{MnO_4^{2-}}\right)$:

$$x + 4(-2) = -2$$
$$x - 8 = -2$$
$$x = +6$$

For permanganate $\left(\mathrm{MnO_4^{-}}\right)$:

$$x + 4(-2) = -1$$
$$x - 8 = -1$$
$$x = +7$$

The statement says +7 and +6, respectively, which is reversed.

Statement 3: Oxidation of Mn^{2+} by peroxodisulphate $\left(S_2O_8^{2-}\right)$ produces permanganate $\left(MnO_4^-\right)^-$, not manganate.

Statement 4: Electronic configuration:

Mn atomic number $= 25$
Mn^{6+} (in manganate) $\rightarrow 3\,d^1 \rightarrow$ one unpaired electron $\rightarrow$ paramagnetic
Mn^{7+} (in permanganate) $\rightarrow 3\,d^0 \rightarrow$ no unpaired electrons $\rightarrow$ diamagnetic

Statement 5: Acidified permanganate is a strong oxidising agent. It oxidises:

- Oxalate → CO_2
- Nitrite → Nitrate
- Iodide → I_2

Thus, permanganate is reduced while oxidising these species.

The statement says "acidified permanganate reduces oxalate, nitrite, and iodide."
Here, "reduces" means it gets reduced while oxidising them.

Hence, the correct answer is option (2)

Q. 16 The correct order of acidic strength of the major products formed in the given reaction is:

A. $PhNH_2 \xrightarrow{(1)\ NaNO_2+HCl(<5^\circ C)} \xrightarrow{(2)CuCN} \xrightarrow{(3)H_3O^+/\Delta} [A]$

B. $CH_3CH_2CHO \xrightarrow[\Delta]{[Ag(NH_3)_2]^+,OH^-} [B]$

C. $CH_4 + O_2 \xrightarrow[(ii)Na_2Cr_2O_7/H^+]{(i\ Mo_2O_3} [C]$

D. $PhCH_2MgBr + CO_2 \xrightarrow[H_3O^+]{\text{Dry ether}} [D]$

Choose the correct answer from the options given below:

Option 1:
$C > B > A > D$

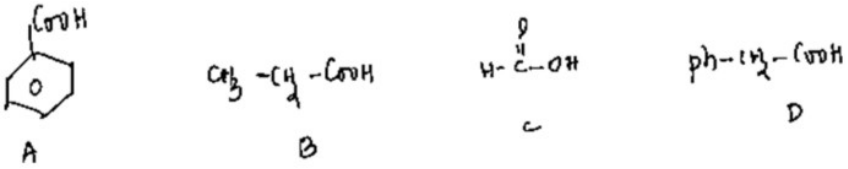
Option 2:
$A > D > C > B$

Option 3:
$C > A > D > B$

Option 4:
$A > D > B > C$

Correct Answer:
$C > A > D > B$

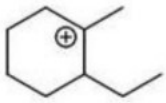
Solution:

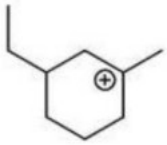


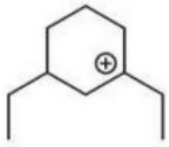
Acidic order: $C > A > D > B$

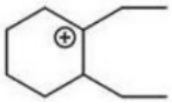
Hence, the correct answer is option (3)

Q. 17 The cyclic cations having the same number of hyperconjugation are:

A. 

B. 

C. 

D. 

Choose the correct answer from the options given below:

Option 1:
A, C and D only

Option 2:
A and C only

Option 3:
B and C only

Option 4:
A and B only

Correct Answer:
A and C only

Solution:
By comparing the substitution pattern around the positively charged carbon in each cyclic cation, options A and C have the same degree of substitution and therefore the same extent of hyperconjugation, while B and D differ.

$A \rightarrow 6\alpha, B \rightarrow 7\alpha, C \rightarrow 6\alpha, D \rightarrow 5\alpha$

Hence, the correct answer is option (2)

Q. 18 Match List – I with List – II according to shape.

List - I	List - II
A. XeO_3	I. BrF_5
B. XeF_2	II. NH_3
C. XeO_2F_2	III. $[I_3]$
D. $XeOF_4$	IV. SF_4

Choose the correct answer from the options given below:

Option 1:
$A-II, B-III, C-IV, D-I$

Option 2:
$A-II, B-III, C-I, D-IV$

Option 3:
$A-II, B-I, C-III, D-IV$

Option 4:
$A-III, B-II, C-IV, D-I$

Correct Answer:
$A-II, B-III, C-IV, D-I$

Solution:
XeO_3, NH_3 } Pyramidal

XeF_2, I_3^- } Linear

XeO_2F_2, SF_4 } Seesaw

$XeOF_4$, BrF_5 } Square pyramidal

Hence, the correct answer is option (1)

Q. 19 Total number of alkali-insoluble solid sulphonamides obtained by reaction of given amines with Hinsberg's reagent is _____.
Aniline, N-Methylaniline, Methanamine, N, N-Dimethylaniline, N-Methyl Methanamine, Phenylmethanamine, N-Propylaniline, N-Phenylaniline, N-Dimethylaniline, Allyl amine, Isopropyl amine

Option 1:
4

Option 2:
8

Option 3:
5

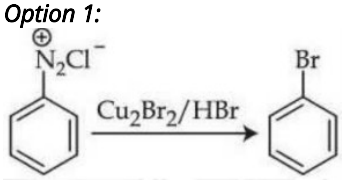
Option 4:
2

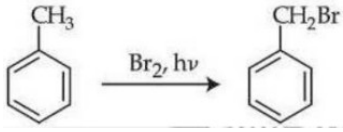
Correct Answer:
4

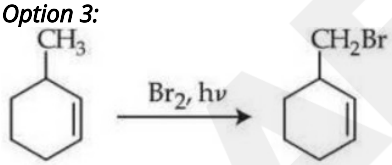
Solution:
$Ph - NH - CH_3$
$CH_3 - NH - CH_3$
$Ph - NH - CH_2 - CH_2 - CH_3$
$Ph - NH - Ph$

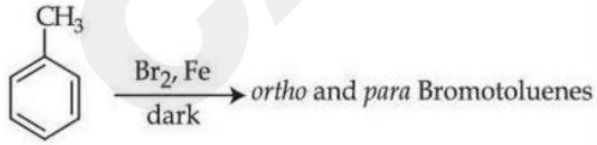
Hence, the correct answer is option (1).

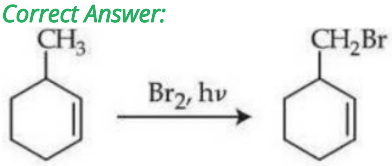
Q. 20 Which of the following reactions is NOT correctly represented?

Option 1:


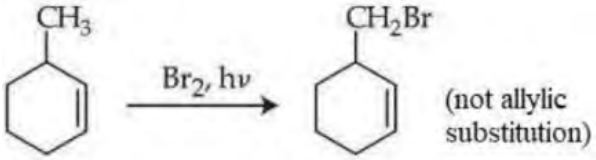
Option 2:


Option 3:


Option 4:


Correct Answer:


Solution:



This reaction involves free radical bromination of an alkene under the conditions Br_2, hv. In the presence of ultraviolet light, bromine undergoes homolytic cleavage to form bromine radicals, and the reaction proceeds via a free radical substitution mechanism. In alkenes, such conditions lead to allylic bromination, meaning substitution occurs at the carbon adjacent to the double bond (allylic position). This happens because the allylic radical formed during the reaction is resonance-stabilized, making it more stable than other possible radicals. In the given reaction, bromination is shown at the methyl group, which is not at the correct allylic position relative to the double bond. Since the substitution does not occur at the resonance-stabilized allylic carbon, the reaction is not correctly represented.

Hence, the correct answer is option (3)

Q. 21 $A \rightarrow B$ (first reaction)
$C \rightarrow D$ (second reaction)
Consider the above two first-order reactions. The rate constant for the first reaction at 500 K is double that at 300 K. At $500\ \mathrm{K}, 50\%$ of the reaction becomes complete in 2 hours. The activation energy of the second reaction is half that of the first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the temperature; then rate constant for the second reaction at 300 K is _____ $\times 10^{-1} h_{our}^{-1}$ (nearest integer).

Correct Answer:
3

Solution:
$A \rightarrow B$
$K_{500} = 2K_{300}$
$t_{50\%} = 2$
$K_{500} = \dfrac{0.693}{2}$
$K_{300} = \dfrac{0.693}{4}$
$C \rightarrow D$
$E_a = E_a = 112.5R$
$\dfrac{A \rightarrow B}{2}$

$K_{500} = 2K_{500} = 2 \times \dfrac{0.693}{2} = 0.693 \log_2 = \dfrac{E_a}{R}\left[\dfrac{200}{300 \times 500}\right]$ Solving $K_{\substack{300 \\ C \rightarrow D}} = 3$

$E_{a \rightarrow B} = 225R$
$E_a = 112.5R$
$A \rightarrow D$
$K_{500} = 2K_{300}$
$t_2 = 2$
500

$K_{A00} = \frac{0.693}{2}$

$K_{A00}^{K_{A\to B}} = \frac{0.693}{4}$

$C \to D$

$E_{C\to D} = E_a = 112.5R$

$K_{500} = 2K_{500} = 2 \times \frac{0.693}{2} = 0.693$

$K_{C\to B}^{300} = ?$

Now apply Arrhenius for $\mathrm{C} \to \mathrm{D}$:

$$\log\left(\frac{K_{500}}{K_{300}}\right) = \frac{112.5R}{2.303R}\left[\frac{1}{300} - \frac{1}{500}\right]$$
$$= \frac{112.5}{2.303} \times \frac{1}{750}$$
$$= \frac{112.5}{1727.25}$$
$$= 0.065$$

So,

$\log\left(\frac{0.693}{K_{300}}\right) = 0.065$

Taking antilog:

$$\frac{0.693}{K_{300}} = 10^{0.065}$$
$$= 1.16$$
$$K_{300} = \frac{0.693}{1.16}$$
$$K_{300} = 0.30$$

$K_{C\to D}^{300} = 0.3 = 3 \times 10^{-1}$

Hence, the correct answer is 3

Q. 22 The number of isoelectronic species among $\mathrm{Sc}^{3+}, \mathrm{Cr}^{2+}, \mathrm{Mn}^{3+}, \mathrm{Co}^{3}$ and Fe^{3+} is ' n '. If ' n ' moles of AgCl is formed during the reaction of the complex with formula $\mathrm{CoCl}_3(en)_2\mathrm{NH}_3$ with an excess of AgNO_3 solution, then the number of electrons present in the t_{2g} orbital of the complex is _____.

Correct Answer:
6

Solution:
Cr^{2-} and Mn^{3+} are isoelectronic

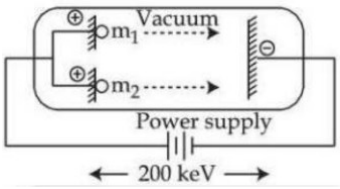
$\mathrm{n} = 2$

Complex is : $[\mathrm{Co(en)_2NH_3Cl}]\mathrm{Cl_2}$

$\Rightarrow \mathrm{Co}^{3+} \quad 3\,\mathrm{d}^6 \quad \mathrm{t}_{2\,\mathrm{g}}^{2,2,2} \quad \mathrm{e}_{\mathrm{g}}^{0,0}$

Hence, the correct answer is 6

Q. 23 Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below.



[Given mass of $m_1 = 1\text{amu}$ and $m_2 = 4\text{amu}$]
The de Broglie wavelength of m_1 will be x times that of m_2. The value of x is _____ (nearest integer)

Correct Answer:
2

Solution:
$\lambda_{\mathrm{d}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mK.E.}}}$

Here KE is same i.e. 200 k eV
So $\lambda_d \propto \frac{1}{\sqrt{m}}$

$$\frac{(\lambda_{\mathrm{d}})_{m_1}}{(\lambda_{\mathrm{d}})_{m_2}} = \sqrt{\frac{m_2}{m_1}} = \sqrt{4} = 2$$
$$(\lambda_{\mathrm{d}})_{m_1} = 2(\lambda_{\mathrm{d}})_{m_2}$$

So $\mathrm{x} = 2$.

Q. 24 For a strong electrolyte Λ_m increases slowly with dilution and can be represented by the equation $\Lambda_m = \Lambda_m^{\circ} - AC^{\frac{1}{2}}$

Molar conductivities of the solutions of strong electrolyte AB at $18^{\circ}\mathrm{C}$ are given below:

$c\left[\mathrm{molL}^{-1}\right]$	0.04	0.09	0.16	0.25
$\Lambda_m\left[\mathrm{S\ cm^2\ mol^{-1}}\right]$	96.1	95.7	95.3	94.9

The value of constant A based on the above data $\left[\text{ in } \mathrm{Scm^2\ mol^{-1}/(mol/L)^{1/2}}\right]$ unit is_____.

Correct Answer:
4

Solution:
$\Lambda_m = \Lambda_m^{\circ} - A\sqrt{C}$
$96.1 = \Lambda_m^{\circ} - A\sqrt{0.04}$
$95.7 = \Lambda_m^{\circ} - A\sqrt{0.09}$
$\text{Solving, A} = 4$

Hence, the correct answer is 4

Q. 25 A volume of x mL of 5 M NaHCO 3 solution was mixed with 10 mL of $2\mathrm{MH_2CO_3}$ solution to make an electrolytic buffer. If the same buffer was used in the following electrochemical cell to record a cell potential of 235.3 mV, then the value of $\mathrm{x} =$
_____ mL (nearest integer).

$$\mathrm{Sn(\ s)\left|Sn(OH)_6^{2-}(0.5M)\right|HSnO_2^-(0.05M)\left|OH^-\right|Bi_2O_3(\ s)\mid Bi(\ s)}$$

Consider upto one place of decimal for intermediate calculations

Given:
$E^{o}_{HSnO_2 \mid Sn(OH)_6^{2-}} = -0.9\ \mathrm{V}$
$E^{o}_{Bi_2O_3|Bi} = -0.44\ \mathrm{V}$
$p\mathrm{Ka}_{(H_2CO_3)} = 6.11$
$\frac{2.303RT}{F} = 0.059\ \mathrm{V}$
$\mathrm{Antilog}\ (1.29) = 19.5$

Correct Answer:
1

Solution:
$E^{o} = +0.9 - 0.44 = 0.46$

$$E_{cell} = E^{o}_{cell} - \frac{0.059}{6}\log\left(\frac{(0.05)^3}{(05)^3(OH^-)^3}\right)$$

$$0.2353 = 0.46 - 0.14 \times 3\log\left(\frac{0.1}{OH^-}\right) \quad -0.2247 = -0.03\log\left(\frac{0.1}{OH^-}\right)$$

$$7.49 = \left(\log(0.1) - \log\left(OH^-\right)\right) \quad 7.49 = \left[-1 + P^{OH}\right]$$

$P^{OH} = 8.49 \quad P^{H} = 14 - 8.49$

$p^{H} = 5.51$

$$p^{H} = P^{Ka} + \log\left[\frac{HCO_3^-}{H_2CO_3}\right]$$

$$5.51 = 6.11 + \log\left[\frac{5x}{20}\right]$$

$$-0.6 = \log\left(\frac{x}{4}\right), -2\log 2 = \log\left(\frac{x}{4}\right)$$

$$\log\left(\frac{1}{4}\right) = \log\left(\frac{x}{4}\right)\frac{1}{4} = \frac{x}{4}, x = 1$$

Hence, the correct answer is 1

Maths

Q. 1 The probability distribution of a random variable X is given below:

X	4k	$\frac{30}{7}k$	$\frac{32}{7}k$	$\frac{34}{7}k$	$\frac{36}{7}k$	$\frac{38}{7}k$	$\frac{40}{7}k$	6k
P(x)	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $E(x) = \frac{263}{15}$, then $p(x < 20)$ is equal to:

Option 1:
$\frac{11}{15}$

Option 2:
$\frac{3}{5}$

Option 3:
$\frac{8}{15}$

Option 4:
$\frac{14}{15}$

Correct Answer:
$\frac{11}{15}$

Solution:

$$\frac{1}{15}\left(8+\frac{30}{7}+\frac{64}{7}+\frac{102}{7}+\frac{36}{7}+\frac{76}{7}+\frac{120}{7}+6\right)k=\frac{263}{15}$$

$$\Rightarrow\left(14+\frac{428}{7}\right)k=263\Rightarrow k=263\times\frac{7}{526}=\frac{7}{2}$$

$$\Rightarrow P(X<20)=\frac{2}{15}+\frac{1}{15}+\frac{2}{15}+\frac{3}{15}+\frac{1}{15}+\frac{2}{15}=\frac{11}{15}$$

Hence, the correct answer is option (1).

Q. 2 Let

$A=\{z\in C:|z-2|\leq 4\}$ and
$B=\{z\in C:|z-2|+|z+2|=5\}$.

Then the max $\{|z_1-z_2|:z_1\in A \text{ and } z_2\in B\}$ is:

Option 1:
9

Option 2:
8

Option 3:
$\frac{15}{2}$

Option 4:
$\frac{17}{2}$

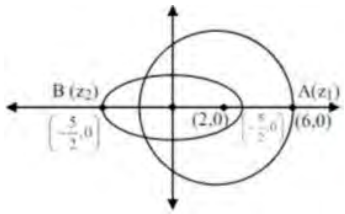
Correct Answer:
$\frac{17}{2}$

Solution:

$$|z-2|=4\Rightarrow(x-2)+y^2=16$$

$$|z-2|+|z+2|=5\Rightarrow\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

$$\Rightarrow\frac{4x^2}{25}+\frac{4y^2}{9}=1$$



Maximum value of $|z_1 - z_2| = 6 + \frac{5}{2} = \frac{17}{2}$

Hence, the correct answer is option (4).

Q. 3 $\frac{6}{3^{26}} + \frac{10.1}{3^{25}} + \frac{10.2}{3^{24}} + \frac{10.2^2}{3^{23}} + \ldots + \frac{10.2^{24}}{3}$ is equal to:

Option 1:
2^{26}

Option 2:
2^{25}

Option 3:
3^{25}

Option 4:
3^{26}

Correct Answer:
2^{26}

Solution:
$S = \frac{6}{3^{26}} + \frac{10}{3^{25}}\left[\frac{(6)^{25}-1}{6-1}\right] \quad S = \frac{6}{3^{26}} + \frac{10}{3^{25}}\left[\frac{6^{25}-1}{5}\right]$
$S = \frac{2}{3^{25}} + 2\left[2^{25} - \frac{1}{3^{25}}\right] \quad S = 2^{26}$

Hence, the correct answer is option 1).

Q. 4 An ellipse has its center at $(1, -2)$ one focus at $(3, -2)$ and one vertex at $(5, -2)$. Then the length of its latus rectum is:

Option 1:
$\frac{16}{\sqrt{3}}$

Option 2:
6

Option 3:
$4\sqrt{3}$

Option 4:
$6\sqrt{3}$

Correct Answer:
6

Solution:

$\Rightarrow ac = 2, a = 4 \Rightarrow b^2 = a^2 - (ac)^2 = 12$

$LR = \dfrac{2 \times 12}{4} = 6$

Hence, the correct answer is option (2).

Q. 5 The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots + x^{1000}$ is:

Option 1:
$1000C_{501}$

Option 2:
$1002C_{501}$

Option 3:
$^{1001}C_{501}$

Option 4:
$^{1002}C_{500}$

Correct Answer:
$^{1002}C_{500}$

Solution:

$S = (1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots . + x^{1000}$

$= (1+x)^{1000} \dfrac{\left(1 - \left(\frac{x}{1+x}\right)^{1001}\right)}{1 - \frac{x}{1+x}} = (1+x)^{1001} - x^{1001}$

$\text{Required sum } = {}^{1001}C_{499} + {}^{1001}C_{500} = {}^{1002}C_{500}$

Hence, the correct answer is option (4).

Q. 6 Given below are two statements:

Statement I: The function $f : R \to R$ defined by $f(x) = \frac{x}{1+|x|}$ is one - one.

Statement II: The function $f : R \to R$ defined by $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many -one.

In the light of above statement, chose the correct answer from the options given below:

Option 1:
Both statement I and Statement II are false

Option 2:
Statement I is false but Statement II is true

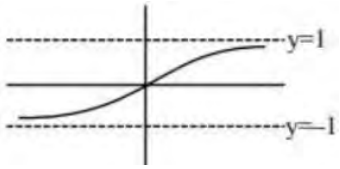
Option 3:
Both Statement I and Statement II are true

Option 4:
Statement I is true but Statement II is false

Correct Answer:
Both Statement I and Statement II are true

Solution:
Statement 1: $f(x)=\frac{x}{1+|x|}$

$$f(x)=\begin{cases}\frac{x}{1+x} & x\geq 0\\ \frac{x}{1-x} & x<0\end{cases}$$



$\mathrm{f(x)}$ is one-one
Statement 2: $f(x)=\frac{x^2+4x-30}{x^2-8x+18}, f(0)=\frac{-30}{18}=\frac{-5}{3}$

$\frac{-5}{3}=\frac{x^2+4x-30}{x^2-8x+18}$

On solving $x=0,>1$

$\Rightarrow f(0)=f(-1)=\frac{-5}{3}$

$\therefore f(x)$ is many-one

Hence, the correct answer is option 3).

Q. 7 Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}be\frac{5}{16}, a>2$. if α is such that $\mathrm{a},4,\alpha,\ \mathrm{b}$ are in A.P., then the equation $\alpha x^2-ax+2(\alpha-2b)=0$ has:

Option 1:
One root in $(1,4)$ and another in $(-2,0)$

Option 2:
One root in $(0,2)$ and another in (-4,-2)

Option 3:
Both roots in the interval (-2,0)

Option 4:
Complex roots of magnitude less than 2

Correct Answer:
One root in $(1,4)$ and another in $(-2,0)$

Solution:
$a=4-d,\alpha=4+d,b=4+2d$
$\Rightarrow (4+d)x^2-(4-d)x+2(4+d-8-4d)=0$
$\Rightarrow (4+d)x^2-(4-d)x+2(-4-3d)=0$
$\text{Also }\frac{\frac{1}{a}+\frac{1}{b}}{2}=\frac{5}{16}$
$\Rightarrow \frac{\frac{1}{4-d}+\frac{1}{4+2d}}{2}=\frac{5}{16}$
$\Rightarrow d=2$

Equation becomes

$6x^2 - 2x - 20 = 0$
$3x^2 - x - 10 = 0$
$x = 2, \frac{-5}{3}$

Hence, the correct answer is option (1).

Q. 8 Let P be a point in the plane vectors $AB = 3\hat{i} + \hat{j} - \hat{k}$ and $AC = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equidistant from the lines AB and AC . If $|AP| = \frac{\sqrt{5}}{2}$, then the area of the triangle ABP is :

Option 1:
$\frac{\sqrt{26}}{4}$

Option 2:
2

Option 3:
$\frac{\sqrt{30}}{4}$

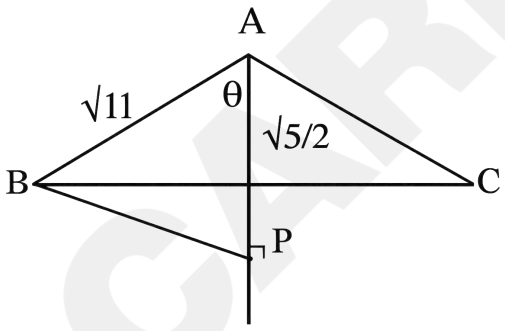
Option 4:
$\frac{3}{2}$

Correct Answer:
$\frac{\sqrt{30}}{4}$

Solution:

$\cos 2\theta = \frac{-1}{11} \Rightarrow \text{Sin}\,\theta = \sqrt{\frac{6}{11}}$

$\text{Area } = \frac{1}{2}\sqrt{11} \times \sqrt{5}/2 \times \sqrt{\frac{6}{11}} = \frac{\sqrt{30}}{4}$



Hence, the correct answer is option (3).

Q. 9 Given below are two Statements:
Statement I: $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7 .
Statement II: The integral part of $(7 + 4\sqrt{3})^{25}$ is an odd number.
In the light of the above statement, choose the correct answer from the options given below:

Option 1:
Statement I is true but statement II is false

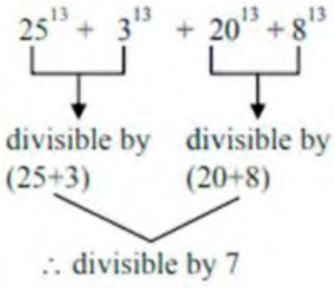
Option 2:
Both statement I and statement II are false

Option 3:
Statement I is false but statement II is true

Option 4:
Both statement I and statement II are true

Correct Answer:
Both statement I and statement II are true

Solution:
Statement - I



$$\text{Statement } -2: R=(7+4\sqrt{3})^{25}=I+f$$
$$R'=(7-4\sqrt{3})^{25}=f'$$
$$\therefore R+R'=2\left[{}^{25}C_0 7^{25}+{}^{25}C_2 7^{23}(4\sqrt{3})^2+\ldots.\right]$$
$$I+f+f'=\text{ even integer}$$
$$\therefore I=\text{ odd integer}$$
$$0<f+f'<2\Rightarrow f+f'=1$$

⇒ Both the statements are correct

Hence, the correct answer is option (4).

Q. 10 Considering the principal values of inverse trigonometric functions, the value of the expression $\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right)-2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to:

Option 1:
$-\frac{33}{56}$

Option 2:
$\frac{16}{63}$

Option 3:
$\frac{33}{56}$

Option 4:
$-\frac{16}{63}$

Correct Answer:
$\frac{33}{56}$

Solution:

$\text{Sin}^{-1}\frac{2}{\sqrt{13}}=\theta, \text{Cos}^{-1}\frac{3}{\sqrt{10}}=\phi, \tan(2\theta-2\phi)=\frac{33}{56}$

Hence, the correct answer is option (3).

Q. 11 Let A be the focus of the parabola $y^2=8x$. Let the line $y=mx+\text{c}$ intersect the parabola at two distinct points B and C . If the centroid of the triangle ABC is $\left(\frac{7}{3},\frac{4}{3}\right)$, then $(BC)^2$ is equal to :

Option 1:
89

Option 2:
32

Option 3:
41

Option 4:
80

Correct Answer:
80

Solution:
$2t_1^2+2t_2^2+2=7 \quad t_1+t_2=1$
$2t_1^2+2\left(1+t_1^2-2t_1\right)+2=7 \quad t_2=1-t_1$
$4t_1^2-4t_1-3=0$
$4t_1^2-6t_1+2t_1-3=0$
$2t_1\left(2t_1-3\right)+1\left(2t_1-3\right)=0$
$t_1=-\frac{1}{2} \quad t_1=\frac{3}{2}$
$t_1=-\frac{1}{2} \quad t_2=1+\frac{1}{2}=\frac{3}{2}$
$B\left(2t_1^2,4t_1\right)=\left(\frac{1}{2},-2\right)$
$C\left(2t_2^2,4t_2\right)=\left(\frac{9}{2},6\right)$
$(BC)^2=\left(\frac{9}{2}\frac{-1}{2}\right)^2+(6+2)^2=16+64=80$

Hence, the correct answer is option (4).

Q. 12 Let Q(a, b, c,) be the image of the point $\text{P}(3,2,1)$ in the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{2}$. Then the distance of Q from the line $\frac{x-9}{3}=\frac{y-9}{2}=\frac{z-5}{-2}$ is:

Option 1:
8

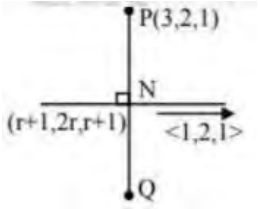
Option 2:
5

Option 3:
6

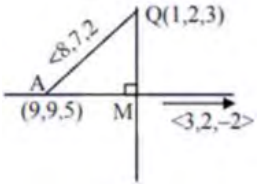
Option 4:
7

Correct Answer:
7

Solution:



$\text{drs of PN} =< r-2, 2r-2, r > 1 \cdot (r-2) + 2(2r-1) + 1 \cdot (r) = 0$
$6r = 6 \Rightarrow r = 1 \quad \therefore N \equiv (2,2,2) \quad \Rightarrow Q \equiv (1,2,3)$



$AQ = \sqrt{64+49+4} = \sqrt{117}$

$AM = \left|\dfrac{24+14-4}{\sqrt{9+4+4}}\right| = \dfrac{34}{\sqrt{17}} = 2\sqrt{17}$
$\therefore QM = \sqrt{117-68} = \sqrt{49} = 7$

Hence, the correct answer is option (4).

Q. 13 Let $P_1 : y = 4x^2$ and $P_2 : y = x^2 + 27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y = \alpha x, \alpha > 0$ and P_1, then α is equal to:

Option 1:
15

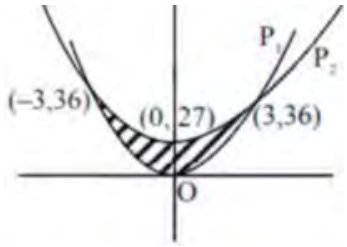
Option 2:
8

Option 3:
6

Option 4:
12

Correct Answer:
12

Solution:



Area bounded between $P_1 \& P_2$ is

$$\int_{-3}^{3}\left(\left(x^2+27\right)-\left(4x^2\right)\right)dx = 2\int_{0}^{3}\left(27-3x^2\right)dx = 2\left[27x-x^3\right]_0^3 = 2[81-27] = 108$$

$\therefore$ Area bounded between $P_1 \& L$ is 18 sq. units
(Area between $x^2 = 4ay$ and line $\mathrm{x} = \mathrm{my}$) is $\frac{8a^2}{3m^3}$
$\therefore$ Area between $x^2 = \frac{y}{4} \& x = \frac{y}{\alpha}$ is
$\frac{8\cdot\left(\frac{1}{16}\right)^2}{3\cdot\left(\frac{1}{\alpha}\right)^3} = 18 \Rightarrow \frac{\frac{8}{16\cdot 16}}{\frac{3}{\alpha^3}} = 18 \Rightarrow \alpha^3 = 2^6\cdot 3^3 \Rightarrow \alpha = 12$

Hence, the correct answer is option (4).

Q. 14 Let $f(x) = \lim_{\theta\to 0}\left(\frac{\cos\pi x - x^{\left(\frac{2}{\theta}\right)}\sin(x-1)}{1+x^{\left(\frac{2}{\theta}\right)}(x-1)}\right), x \in R$. Consider the following two statements:
(I) $f(x)$ is discontinuous at $x = 1$
(II) $f(x)$ is continuous at $x = -1$

Then

Option 1:
Neither (I) nor (II) is true

Option 2:
Both (I) and (II) are true

Option 3:
Only (II) is true

Option 4:
Only (I) is true

Correct Answer:
Neither (I) nor (II) is true

Solution:

$$f(x) = \begin{cases} \cos\pi x & x \to 1^- \\ \frac{-\sin(x-1)}{(x-1)} & x \to 1^+ \end{cases}$$

$$RHL = \lim_{x\to 1}\frac{\sin(x-1)}{(x-1)} = -1$$

$$LHL = \lim_{x\to 1}\cos\pi x = -1$$

$f(x)$ is discontinuous at $\mathrm{x} = 1$

$$f(x) = \begin{cases} \frac{-\sin(x-1)}{-(x-1)} & x \to -1^- \\ \cos\pi x & x \to -1^+ \end{cases}$$

$$LHL = \lim_{x\to 1}\frac{-\sin(x-1)}{-(x-1)} = \frac{\sin 2}{2}$$

$f(x)$ is discontinuous at $x = -1$

Hence, the correct answer is option (1).

Q. 15 Let the circle $x^2 + y^2 = 4$ intersect x -axis at the point $\mathrm{A}(a, 0), a > 0$ and $B(b, 0)$. Let $P(2\cos\alpha, 2\sin\alpha), 0 < \alpha < \frac{\pi}{2}$ and $Q(2\cos\beta, 2\sin\beta)$ be two points such that $(\alpha - \beta) = \frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on :

Option 1:
$x^2 + y^2 - 4x - 4y = 0$

Option 2:
$x^2 + y^2 - 4x - 4 = 0$

Option 3:
$x^2 + y^2 - 4x - 4y - 4 = 0$

Option 4:
$x^2 + y^2 - 4y - 4 = 0$

Correct Answer:
$x^2 + y^2 - 4y - 4 = 0$

Solution:
$x^2 + y^2 = 4$ intersects $x -$ axis at $A(a, 0) a > 0$
$y = 0$
$x^2 = 4$
$x = 2 = a; A(2, 0)$
$x = -2 = b; \quad B(-2, 0)$

$P(2\cos\alpha, 2\sin d) \quad Q(2\cos\beta 2\sin\rho) \quad 0 < \alpha < \dfrac{\pi}{2} \quad \alpha - \beta = \dfrac{\pi}{2} \quad \beta = \alpha - \dfrac{\pi}{2}$
$Q(2\sin\alpha, -2\cos d)$

P I of AQ and BP

$t\frac{\alpha}{2} - t\frac{p}{2} = 1 + t\frac{\alpha}{2}t\frac{p}{2} \quad \dfrac{(y^2 + x^2 - 4)}{4y} = 1$
$x^2 + y^2 - 4y - 4 = 0$

Hence, the correct answer is option (4).

Q. 16 Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2\cot x, x \in (0, \pi)$.If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

Option 1:
3π

Option 2:
π

Option 3:
-3π

Option 4:
$-\pi$.

Correct Answer:
$-\pi$.

Solution:
$xdy - ydx = x^2 \cot xdx$
$x^2 d\left(\frac{y}{x}\right) = x^2 \cot xdx$
$d\left(\frac{y}{x}\right) = \cot \text{xdx}$
$\int d\left(\frac{y}{x}\right) = \int \cot \text{xdx}$
$\frac{y}{x} = \log_e \sin x + C$
$\text{given y}\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$
$\Rightarrow c = 1$

$y = x\left(\log_e \sin x + 1\right)$
$y\left(\frac{\pi}{6}\right) = \frac{\pi}{6}[-\log_e 2 + 1]$
$y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}\left[-\frac{1}{2}\log_e 2 + 1\right]$
$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$
$== \pi\left[(-\log_e 2 + 1) + 2\left(\frac{1}{2}\log_e 2 - 1\right)\right]$
$= \pi[1 - 2] = -\pi$

Hence, the correct answer is option (4).

Q. 17 Let $f(x) = \int \frac{dx}{x^{\left(\frac{2}{3}\right)} + 2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0) = -26 + 24\log_e(2)$. If $f(1) = a + b\log_e(3)$, where $a, b \in z$, then $a + b$ is equal to:

Option 1:
-18

Option 2:
-11

Option 3:
-5

Option 4:
-26

Correct Answer:
-11

Solution:
$f(x) = \int \frac{dx}{x^{2/3} + 2x^{1/2}}$

Put $x = t^6 \Rightarrow dx = 6t^5 dt$

$$= \int \frac{6t^5 dt}{t^4 + 2t^3} = 6\int \frac{(t^2-4)+4}{t+2} dt$$
$$= 6\left[\int (t-2)dt + 4\int \frac{1}{t+2} dt\right]$$
$$= 6\left[\frac{t^2}{2} - 2t + 4\ln(t+2)\right] + C$$
$$= 3x^{1/3} - 12x^{1/6} + 24\left(n\left(x^{16}+2\right)\right) + C$$
$$f(0) = 24(n2 + C = -26 + 24(n2 \text{ (given)}$$
$$\Rightarrow C = -26$$

Now

$$f(1) = -35 + 24(n3 = A + B(n3 \text{ (as given in ques.)}$$
$$\Rightarrow A = -35 \& B = 24$$
$$\Rightarrow A + B = -11$$

Hence, the correct answer is option (2).

Q. 18 Let [.]denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right) dx$ is equal to:

Option 1:
$15\pi + 4$

Option 2:
$12\pi + 5$

Option 3:
$13\pi + 1$

Option 4:
$11\pi + 2$

Correct Answer:
$11\pi + 2$

Solution:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{12(3+[x])dx}{3+[\sin x]+[\cos x]}$$
$$I = \int_{-\frac{\pi}{2}}^{-1} \frac{12(1)dx}{2} + \int_{-1}^{0} \frac{12(2)dx}{2} + \int_{0}^{1} \frac{12(3)dx}{3} + \int_{1}^{\frac{\pi}{2}} \frac{12(4)dx}{3}$$
$$I = 6\left(\frac{\pi}{2} - 1\right) + 12(0+1) + 12(1-0) + 16\left(\frac{\pi}{2} - 1\right)$$
$$I = 3\pi - 6 + 12 + 12 + 8\pi - 16$$
$$I = 11\pi + 2$$

Hence, the correct answer is option (4).

Q. 19 Let the ellipse $E: \frac{x^2}{144} + \frac{y^2}{169} = 1$ and the hyperbola $H: \frac{x^2}{16} - \frac{y^2}{\lambda^2} = -1$ have the same foci. If e and L respectively denote the eccentricity and the length of the latus rectum of H , then the value of $24(e + L)$ is:

Option 1:
296

Option 2:
148

Option 3:
126

Option 4:
67

Correct Answer:
296

Solution:
Equation of hyperbola: $\frac{y^2}{\lambda^2}-\frac{x^2}{16}=1$

Equation of ellipse: $\frac{x^2}{144}+\frac{y^2}{169}=1$

$e'=\sqrt{1-\frac{144}{169}}=\frac{5}{13}$

focus $\Rightarrow (0,5)$

$\Rightarrow \lambda\sqrt{1+\frac{16}{\lambda^2}}=5$
$\Rightarrow \lambda^2+16=25$
$\lambda=3$

Eccentricity of hyperbola $=\sqrt{1+\frac{16}{\lambda^2}}=\frac{5}{3}$
Length of latus rectum of hyperbola $=\frac{2(16)}{3}=\frac{32}{3}$

$24(e+\ell)=24\left[\frac{5}{3}+\frac{32}{3}\right]=8\times 37=296$

Hence, the correct answer is option (1).

Q. 20 The sum of all the elements in the range of

$$f(x)=\operatorname{sgn}(\sin x)+\operatorname{sgn}(\cos x)+\operatorname{sgn}(\tan x)+\operatorname{sgn}(\cot x), x\neq \tfrac{n\pi}{2}, n\in z. \text{ where } \operatorname{sgn}(t)=\begin{cases}1, & \text{if } t>0\\ -1, & \text{if } t<0\end{cases} \text{ is:}$$

Option 1:
0

Option 2:
4

Option 3:
2

Option 4:
-2

Correct Answer:
2

Solution:

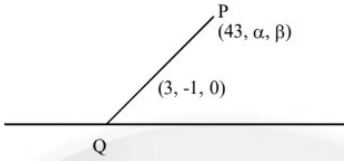
$x \in (0, \pi/2) \Rightarrow y = 1 + 1 + 1 + 1 = 4$
$x \in (\pi/2, \pi) \Rightarrow y = 1 - 1 - 1 - 1 = -2$
$x \in (\pi, 3\pi/2) \Rightarrow y = -1 - 1 + 1 + 1 = 0$
$x \in (3\pi/2, 2\pi) \Rightarrow y = -1 + 1 - 1 - 1 = -2$
$\therefore$ Range of y is $\{-2, 0, 4\}$
Required sum $= -2 + 0 + 4 = 2$

Hence, the correct answer is option (3).

Q. 21 If the distance of the point $P(43, \alpha, \beta), \beta < 0$, from the line $r = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k}), \mu \in R$ along a line with direction ratios $3, -1, 0$ is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to ____

Correct Answer:
170

Solution:



$Q = (43 + 3r, \alpha - r, \beta) = (2s + 4, 0, 3s - 1), (PQ)^2 = 1690 = 10\alpha^2$

Gives $\alpha = 13, \beta = 1$

$(\alpha)^2 + (\beta)^2 = 169 + 1 = 170$

Hence, the correct answer is 170.

Q. 22 Let f be a differentiable function satisfying $f(x) = 1 - 2x + \int_0^x e^{(x-t)} f(t)dt, x \in R$ and Let $g(x) = \int_0^x (f(t) + 2)^{15}(t - 4)^6(t + 12)^{17}dt, x \in R$. if p and q are respectively the points of local minima and local maxima of g , then the value of $|p + q|$ is equal to ____

Correct Answer:
9

Solution:

$f'(x) = 1 - 2x + e^x \int_0^x e^{-1} f(t)dt$

$e^{-x} f(x) = (1 - 2x)e^{-x} + \int_0^x e^{-1} f(t)dt$

$e^{-1} f'(x) - e^{-1} f'(x) = -2e^{-1} + (1 - 2x)e^{-1}(-1) + e^{-1} f(x)$

$f'(x) - 2f(x) = 2x - 3$

$\frac{dy}{dx} - 2y = 2x - 3$

$\Rightarrow y \cdot e^{-2x} = \int e^{-2x}(2x - 3)dx$

On solving we get

$y = 1 - x$

$g'(x) = (3 - x)^{11}(x + 12)^{11}(x - 4)^4$

minima at $x = -12$ & maxima at $x = 3$

$p = -12, q = 3 \Rightarrow |p + q| = 9$

Hence, the correct answer is 9.

Q. 23 Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such that $A^{100} = 100B + I$. Then the sum of all the elements of B^{100} is ____

Correct Answer:
0

Solution:

$$A^2 = A \cdot A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$$

$$A^3 = A^2 \cdot A = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 7 & -12 \\ 3 & -5 \end{pmatrix}$$

$$A^n = \begin{pmatrix} 2n+1 & -4n \\ n & -(2n-1) \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} 201 & -400 \\ 100 & -199 \end{pmatrix}$$

$$A^{100} - I = \begin{pmatrix} 200 & -400 \\ 100 & -200 \end{pmatrix} = 100\,B$$

$$B = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$$

$$B^2 = B \cdot B = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Sum of all elements in B^{100} is zero.

Hence, the correct answer is 0.

Q. 24 If $\sum_{r=1}^{25} \left(\frac{r}{r^4+r^2+1} \right) = \frac{p}{q}$, where p and q are positive integers such that $\gcd(p, q) = 1$, then $p + q$ is equal to ____

Option 1:
976

Option 2:
-

Option 3:
-

Option 4:
-

Correct Answer:
976

Solution:

$$S = \sum \frac{r}{(r^2+r+1)(r^2-r+1)}$$
$$= \frac{1}{2}\sum_{r=1}^{2s}\left(\frac{1}{r^2-r+1} - \frac{1}{r^2+r+1}\right)$$
$$= \frac{1}{2}\left[\left(\frac{1}{1}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{7}\right)+\ldots+\left(\frac{1}{601}-\frac{1}{651}\right)\right]$$
$$= \frac{1}{2}\left[\frac{1}{1}-\frac{1}{651}\right]$$
$$= \frac{1}{2}\left[\frac{650}{651}\right] = \frac{325}{651}$$
$$\frac{p}{q} = \frac{325}{651} \Rightarrow p+q = 976$$

Hence, the correct answer is 976.

Q. 25 Three persons enter in a lift at the ground floor. The lift will go upto 10th floor. the number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floor, is equal to _____

Correct Answer:
210

Solution:
Number of ways $= {}^7C_3 \times 3! = 7 \times 6 \times 5 = 210$

Hence, the correct answer is 210.